# Table of Contents

## Module 18: Routing and Polygons



## Module 19: Output Generation and CAM File Editing



## Module 20: Interfacing to 3D Mechanical CAD

# Module 1: Getting Started With Altium Designer

## Module 1: Getting Started  With Altium Designer





Module Seq = 1

# 1.1 Introduction to Altium Designer

Underlying the Altium Designer environment is a software integration platform that brings together all the tools necessary to create a complete environment for electronic product development, in a single application.

Altium Designer includes tools for all design tasks: from schematic and HDL design capture, circuit simulation, signal integrity analysis, PCB design, and FPGA-based embedded system design and development. In addition, the Altium Designer environment can be customized to meet a wide variety of user requirements.

## 1.1.1 The Altium Designer Integration Platform

When you select **All Programs » Altium Designer Summer 09** from the Windows Start menu to run Altium Designer, you are actually launching DXP.EXE. The DXP platform underlies Altium Designer, supporting each of the editors that you use to create your design.

The application interface is automatically configured to suit the document you are working on. For example, if you open a schematic sheet, appropriate toolbars, menus and shortcut keys are activated. This feature means that you can switch from routing a PCB, to producing a Bill of Materials report, to running a transient circuit analysis, and so on – and the correct menus, toolbars and shortcuts will be readily available.

Also, all toolbars, menus and shortcut keys can also be configured to suit how you like to configure your design environment.


*Figure 1. Altium Designer's software integration architecture*

## 1.2 The Altium Designer environment

The Altium Designer environment consists of two main elements:

- The main document editing area of Altium Designer, shown on the right side in Figure 2.
- The *Workspace Panels*. There are a number of panels in Altium Designer, the default is that some are docked on the left side of the application, some are available in pop-out mode on the right side, some are floating, and others are hidden.

When you open Altium Designer, the most common initial tasks are displayed for easy selection in a special view, called the *Home Page*.

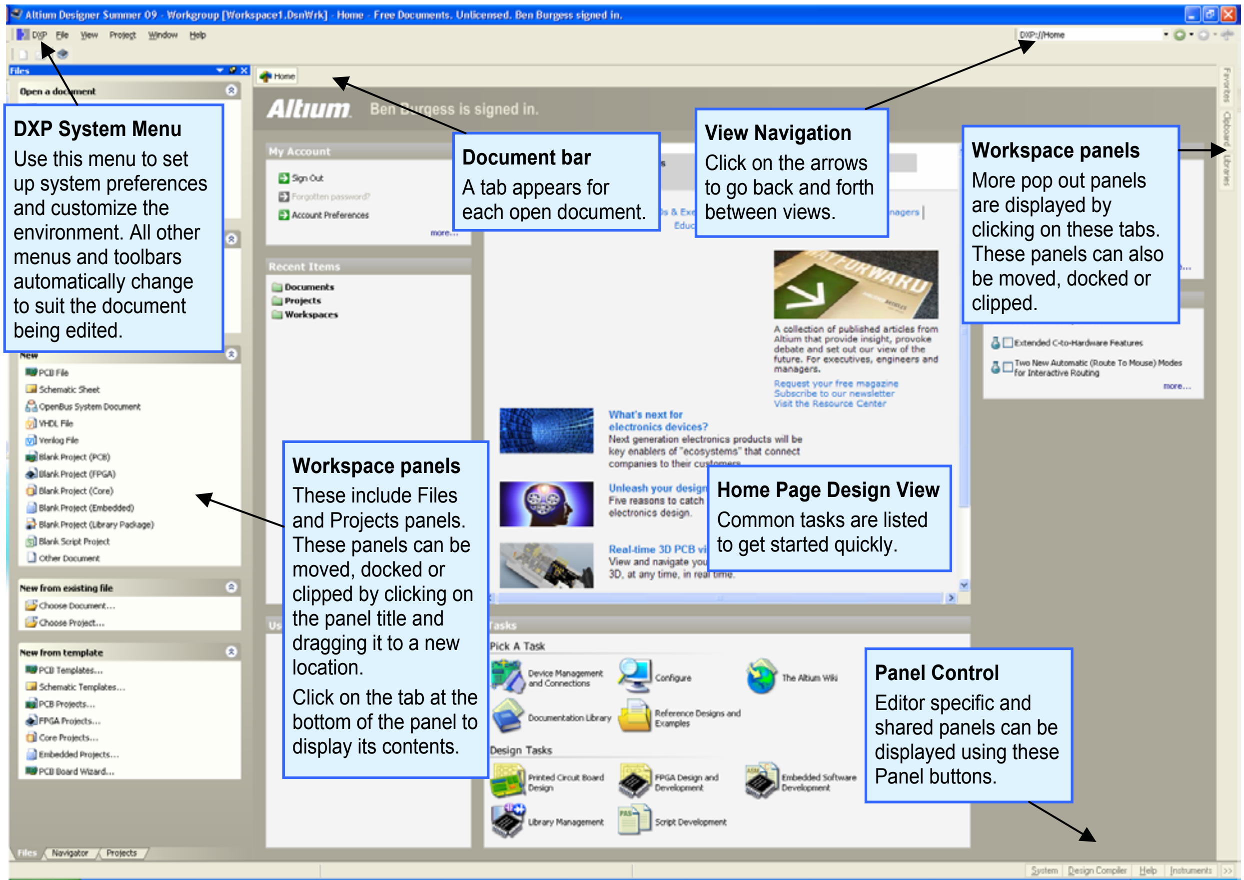



*Figure 2. Altium Designer with the DXP Home Page displayed.*

**Note**: To move an individual panel, click and hold on the panel name. To move a set of panels, click and hold on the panel caption bar away from the panel name. To prevent panels stacking together, hold the CTRL key. To change a docked panel to pop-out mode click the small pin icon at the top of the panel, to change it back to docked click the pin icon again.

**Note**: If you manage to completely ruin your panel layout and wish to revert back to the factory settings, this can be done by going to the **View » Desktop Layouts » Default**. It's best to restart Altium Designer when you run this. To save a custom layout go to **View » Desktop Layouts » Save Layout**. To reload existing layouts go to **View » Desktop Layouts » Load Layout**.

## 1.2.1 The Altium Designer Project

- The basis of every electronic product design is the project.
- The project links the elements of your design together, including the source schematics, the PCB, the netlist, and any libraries or models you want to keep in the project.
- The project also stores the project-level options, such as the error checking settings, the multi-sheet connectivity mode, and the multi-channel annotation scheme.
- There are six project types – PCB projects, FPGA projects, Core Projects, Embedded Projects, Script Projects and Library Packages (the source for an integrated library).
- Altium Designer allows you to access all documents related to a project via the **Projects** panel.
- Related projects can also be linked under a common Workspace, giving easy access to all files related to a particular product your company is developing.
- When you add documents to a project, such as a schematic sheet, a link to each document is entered into the project file. The documents can be stored anywhere on your network; they do not need to be in the same folder as the project file. If they do exist in a directory outside where the project exists or its sub-directories, then a small arrow symbol appears on the document's icon in the **Projects** panel.

## 1.2.2 Demo — Opening an existing Project

1. Select the **File » Open Project** menu to display the *Choose Project to Open* dialog.
2. Navigate to the project folder, `4 Port Serial Interface`, located in the `\Altium Designer Summer 09\Examples\Reference Designs` directory. Locate `4 Port Serial Interface.PRJPCB` (the project file) and double-click on it to open it.
3. The design will now be listed in the navigation tree of the Projects panel.
4. Click on the **–** signs to contract the folders.
5. Click on + (plus) signs to expand folders.
6. Right-click on the project name (4 Port Serial Interface.PrjPcb) to display the context sensitive Projects menu.

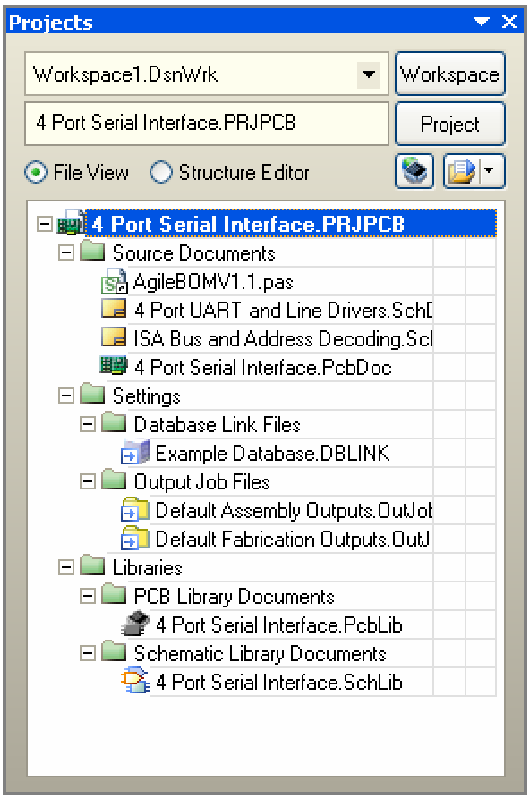



*Figure 3. The open project is displayed in the Projects panel.*

## 1.2.3 Editor View

Each different document kind is edited in an appropriate *Document Editor*, for example the PCB Editor for a PCB document, Schematic Editor for a schematic document, or VHDL Editor for a VHDL document. Figure 4 shows a schematic open for editing in the Schematic Editor.

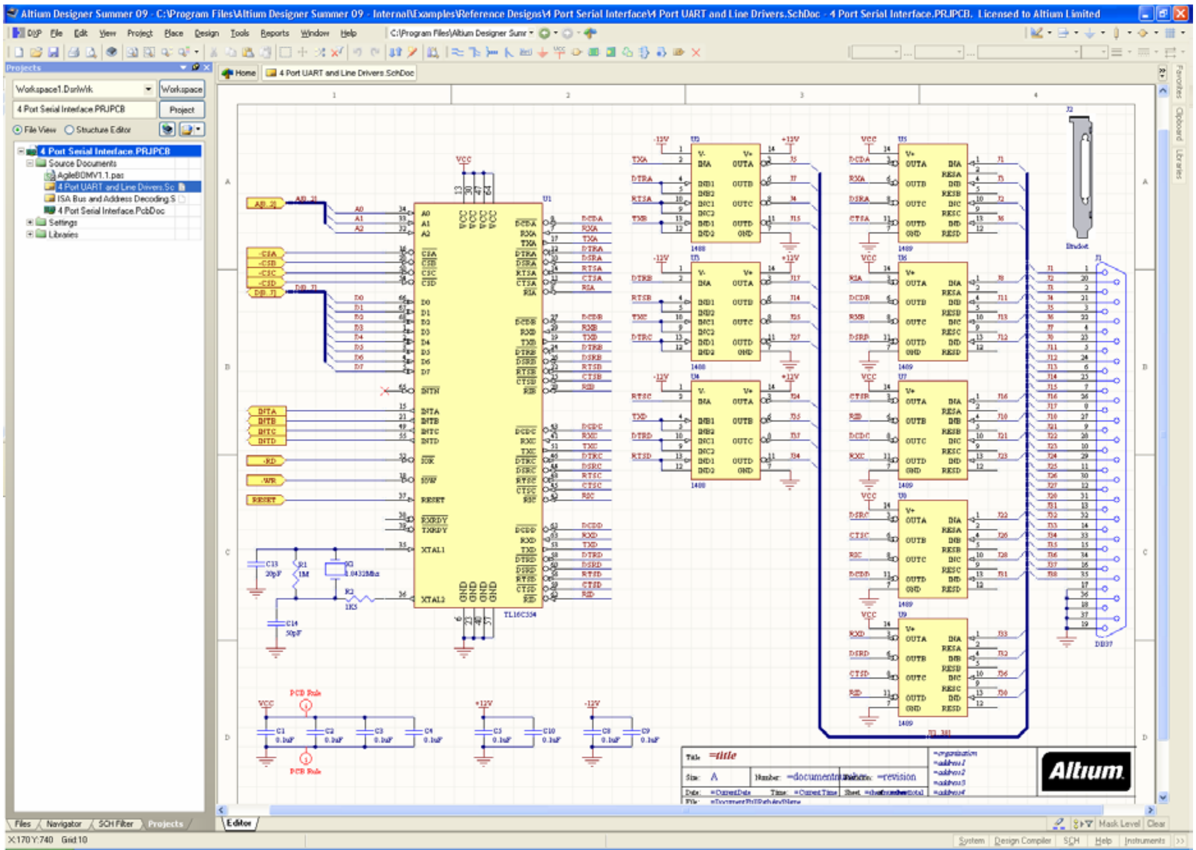


*Figure 4. A schematic open for editing in the Schematic Editor View.*

### 1.2.3.1 Document Tabs in the Documents Bar

Documents that are open are allocated a tab at the top of the application. Click on the relevant tab to display that document and make it the active document for editing. To switch between documents the *Ctrl* + *Tab* shortcut can be used. You can also tweak how *Ctrl* + *Tab* works in the preferences.


*Figure 5. Tabs showing various documents open, note how the PCB tab is highlighted, indicating that it is the document currently being edited.*

**Right-click menu in the Documents Bar**

1. Right-click on any document tab in the Documents bar.
2. Select **Tile All** from the floating menu that appears. All the opened documents are tiled in multiple screen regions.

| **Note**: The number of opened documents determines the number of regions. |
|---|

3. Right-click on a document tab.
4. Select **Close** from the menu.

5. Position the cursor at the point where two regions of a split screen meet and a double-headed arrow will display. Click and drag to resize.

6. Right-click on any one of the tabs in the tiled display and choose **Merge All**. Notice that you have converted a split screen back to a single view.

**Note**: Altium Designer supports multiple monitors. If your PC has multiple monitors you can use the **Open in New Window** command when you right-click on a document, or just drag and drop on to the second monitor and this will cause it to open in a separate Altium Designer application frame.

The right click menu also has options for saving and hiding individual documents as well as groups of documents like groups of schematics.

**Note**: There are a few options that you can tweak to gain more control of how the document bar works in Altium Designer. To do this go select **DXP » Preferences** and open the **System – View** page. At the bottom right is the *Documents Bar* section where things like auto hide, multi-line, ctrl-tab to switch can be set up.

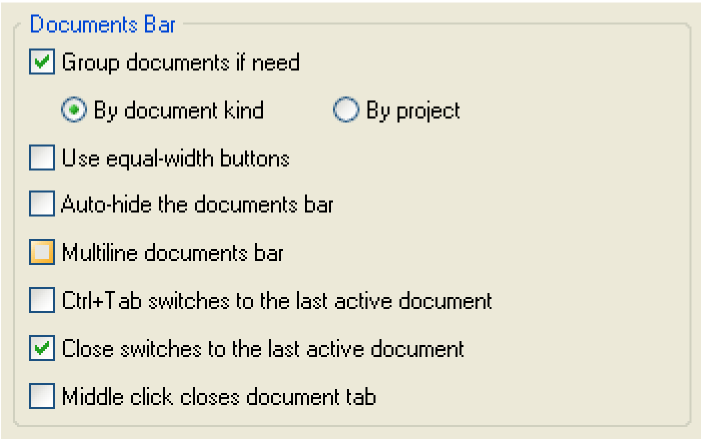



*Figure 6. Document Bar options in Preferences*

## 1.2.4 Exercises — Navigating around Altium Designer

### 1.2.4.1 Using the Projects panel

1. Open `4 Port Serial Interface.PRJPCB`, located in the `\Altium Designer Summer 09\Examples\Reference Designs\4 Port Serial Interface` folder.

2. Expand and then contract the contents of the navigation tree.

3. Double-click on a document in the Projects panel to open it.

4. Double-click on a few more documents in the Projects panel to open them.

5. Right click on the documents bar to see all the options.

6. Tweak some of the settings in preferences and see the results.

# 1.3 Document Editor Overview

To display a document in its editor, double-click on a document icon in the Projects panel. The document will be opened in the appropriate editor, e.g. Schematic Editor, PCB Editor or the Library Editors.

When you create a new document in a design you are required to select a document type, e.g. Schematic or PCB. The document type you select determines which editor is assigned to the document.

## 1.3.1 Working in a document editor

The sections below describe various elements in the user interface of the Altium Designer document editors.

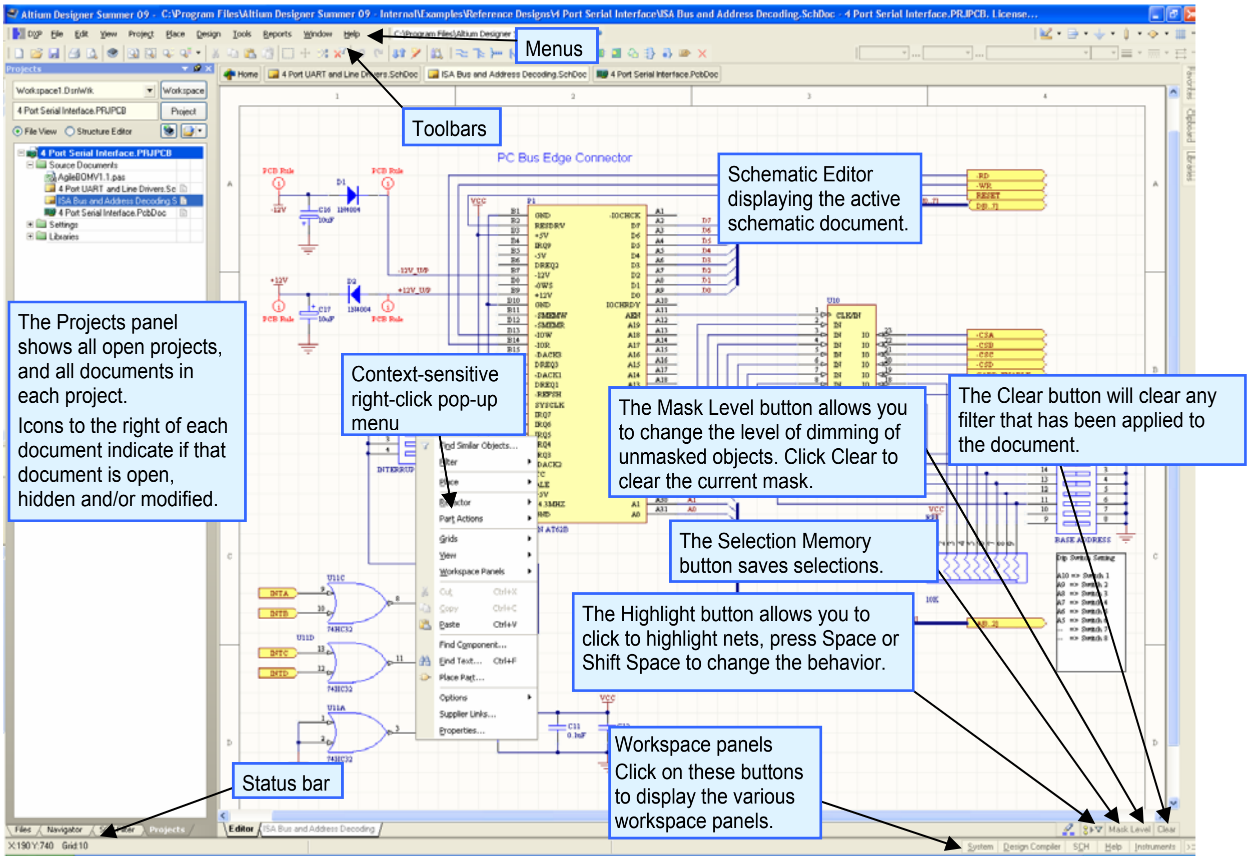



*Figure 7. Schematic Editor Workspace*

### 1.3.1.1 Menus

- Altium Designer menus are similar to standard Windows menus.
- Standard operations, e.g. opening, saving, cut, paste, etc. are consistent across editors.
- Right-click on an empty space on the menu bar or a toolbar caption to open the **Customization Editor** and customize any of the resources for that editor.

### 1.3.1.2 Shortcut keys and pop-up menus

- Menu commands can also be accessed using shortcut keys. The underlined letter indicates the shortcut key for a menu command, e.g. press **F** for the File menu.

- Special shortcut keys give direct access to both menus and sub-menus in the graphical editors, e.g. pressing **F** in the Schematic Editor will pop up the File menu and pressing **S** will pop up the Select sub-menu.

> **Note :** You can gain a list of every shortcut that is available in Altium Designer by looking for a document named **GU0104 Shortcut Keys.PDF** located in the Help directory of the Altium Designer Installation.

#### 1.3.1.3 Toolbars

- Toolbars can be fixed to any side of the workspace or they can be floated.
- Click and drag to move a toolbar. The cursor must be within the toolbar but not actually on a button.
- Toolbars can be reshaped, hold the cursor over the edge of the toolbar and when the resizing cursor appears click and hold to reshape.
- New toolbars can be created and existing toolbars edited.
- Multiple toolbars can be active, right-click on a toolbar to pop up the toolbar display control menu.

#### 1.3.1.4 System and Editor Panels

- Altium Designer uses two types of panels – system-type panels, such as the Files, Messages or Projects panels that are always available, and editor panels, such as the PCB, schematic library or PCB library panels that are only available when a document of that type is active.
- Panels can float, or be docked, on any edge of the Altium Designer workspace. Docked panels can be pinned open, or set to unpinned, where they pop out when their name button is clicked.
- Panels can be clipped together in a set by dragging and dropping one on another, and then dragged around as a set by clicking and dragging on the area of panel title bar that contains no text or icons.
- A panel can be unclipped from a set by clicking and dragging on the panel name.
- Panels can be prevented from docking on particular edges. Right-click on a panel title bar to configure this.

> **Note** : The hide and display speed of unpinned panels is configured in the **System – View** page of the *Preferences* dialog (**DXP » Preferences**). It can be useful to turn off the animation of panels on slower machines.

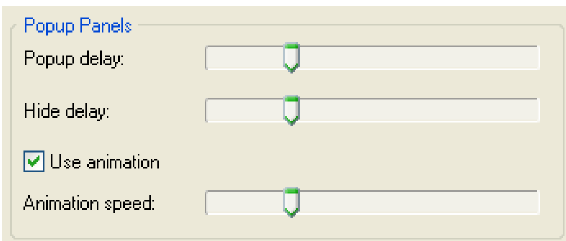



*Figure 8. Configuration for panel control*

#### 1.3.1.5 Status Bar

- The Status Bar is used to display information to the user.
- The Status Bar consists of three display fields divided by separators and a set of panel display buttons. These three display fields are:
  - Cursor position
  - Prompt
  - Options.
- The fields can be re-sized by clicking and dragging on the separators.

- The Status Bar is turned on and off using the menu command **View » Status Bar.**
- The panel display buttons can be added or removed from the Status bar by clicking on the arrow button in the far bottom left.

### 1.3.1.6 Tool Tips

- Tool Tips provide a brief description of how to use a particular function.
- Position the cursor over a toolbar button and leave it stationary for about a second and the Tool Tip will appear.

### 1.3.1.7 Right mouse click context sensitive pop-up menus

- Altium Designer makes extensive use of context sensitive right mouse menus, including in panels and dialogs.
- Right-click anywhere in the environment to pop up a context sensitive menu of commands at the current cursor position. Supported right-click locations include:
  - in a document editor, on an object
  - in a document editor, in free space
  - in the different sections of a panel
  - on the Status bar
  - on a toolbar or menu bar
  - In dialogs, especially those with a grid of information.

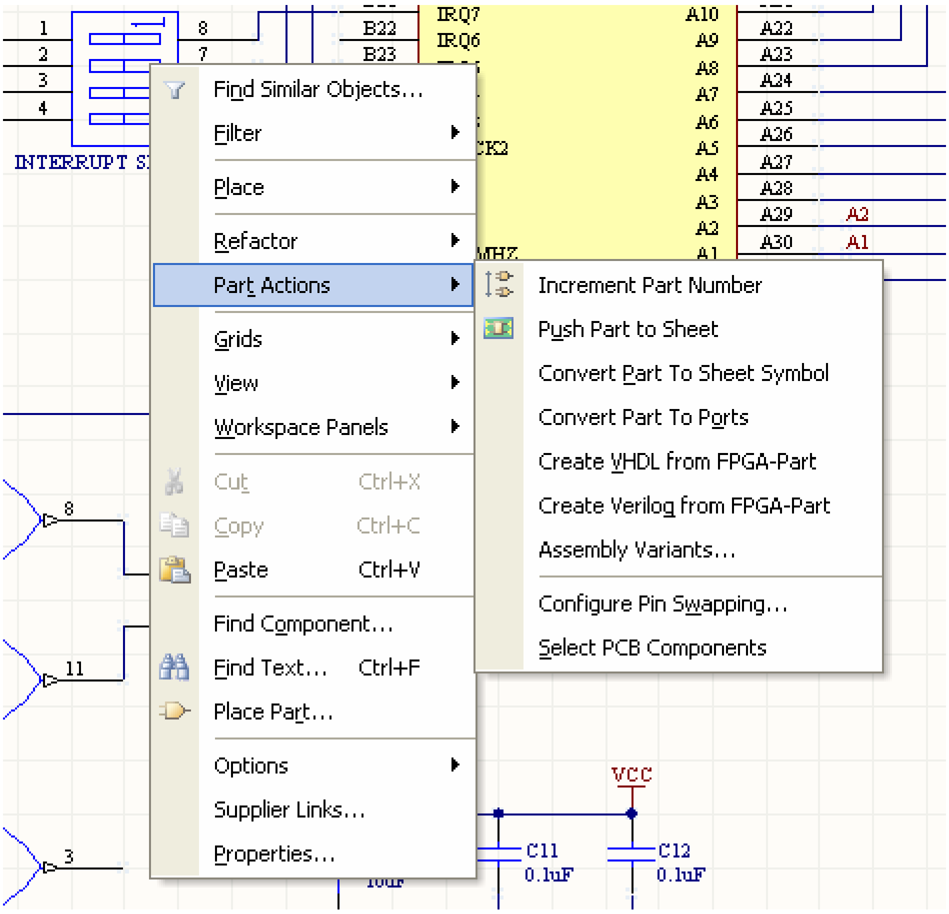



*Figure 9. Context sensitive right mouse menus are available throughout Altium Designer*

### 1.3.1.8 Dialogs

- Dialogs are used to set the parameters for various commands and objects.
- To move from one field to another in a dialog, press the *Tab* key or use the mouse. SHIFT+TAB takes you in the reverse direction.
- Most fields will have an underlined character associated with them that can be pressed (in combination with the ALT key) as an alternative to a mouse click.
- When a field is highlighted, typing can overwrite it.
- You'll find nearly all dialogs will have a question mark icon in the top right hand corner.  Clicking on this icon activates the **What's This Help** (WTH) feature and will display a brief pop-up help message from the next control that you click on. For example, Figure 10 shows the WTH for the **Type** control in the component properties dialog.

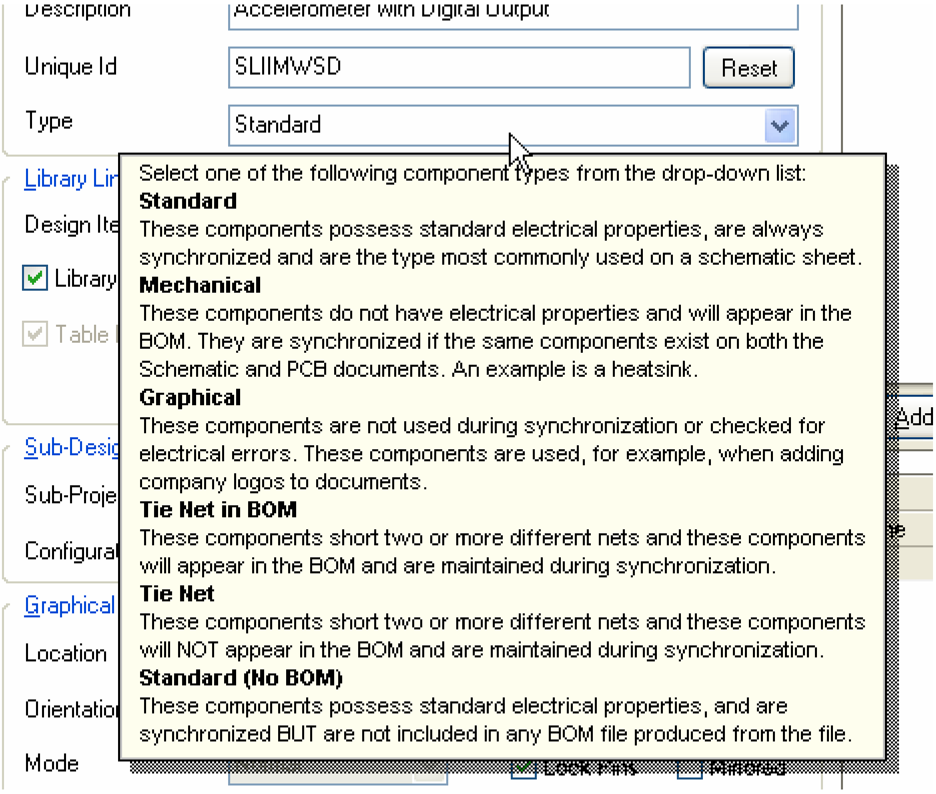



*Figure 10. Using the What's This Help icon to gain help in a dialog*

### 1.3.1.9 Undo/Redo

- Most commands can be undone or then redone using the Undo  and Redo  toolbar buttons. The number of schematic editor and PCB editor undos is set in the *Preferences* dialog (**DXP » Preferences**).
- The shortcut keys for Undo are CTRL+Z or ALT+BACKSPACE, and CTRL+Y or CTRL+BACKSPACE for Redo.

# 1.4 Working with projects and documents

A project is a set of documents that together define all aspects of your design: including schematic sheets, PCB documents, database link definition files, output job definition documents, netlists, and so on. Each project results in a single implementation, for example a PCB project results in one PCB design, and a library package project results in a single integrated library.

Each document in the project is stored as a separate file on the hard drive. The project file itself is also an ASCII document, which includes links to the documents in the project, as well as storing project-level settings.

## 1.4.1 Creating a new project

To create a new PCB project:

1. From the Main Menu, select **File » New » Project » PCB Project**.

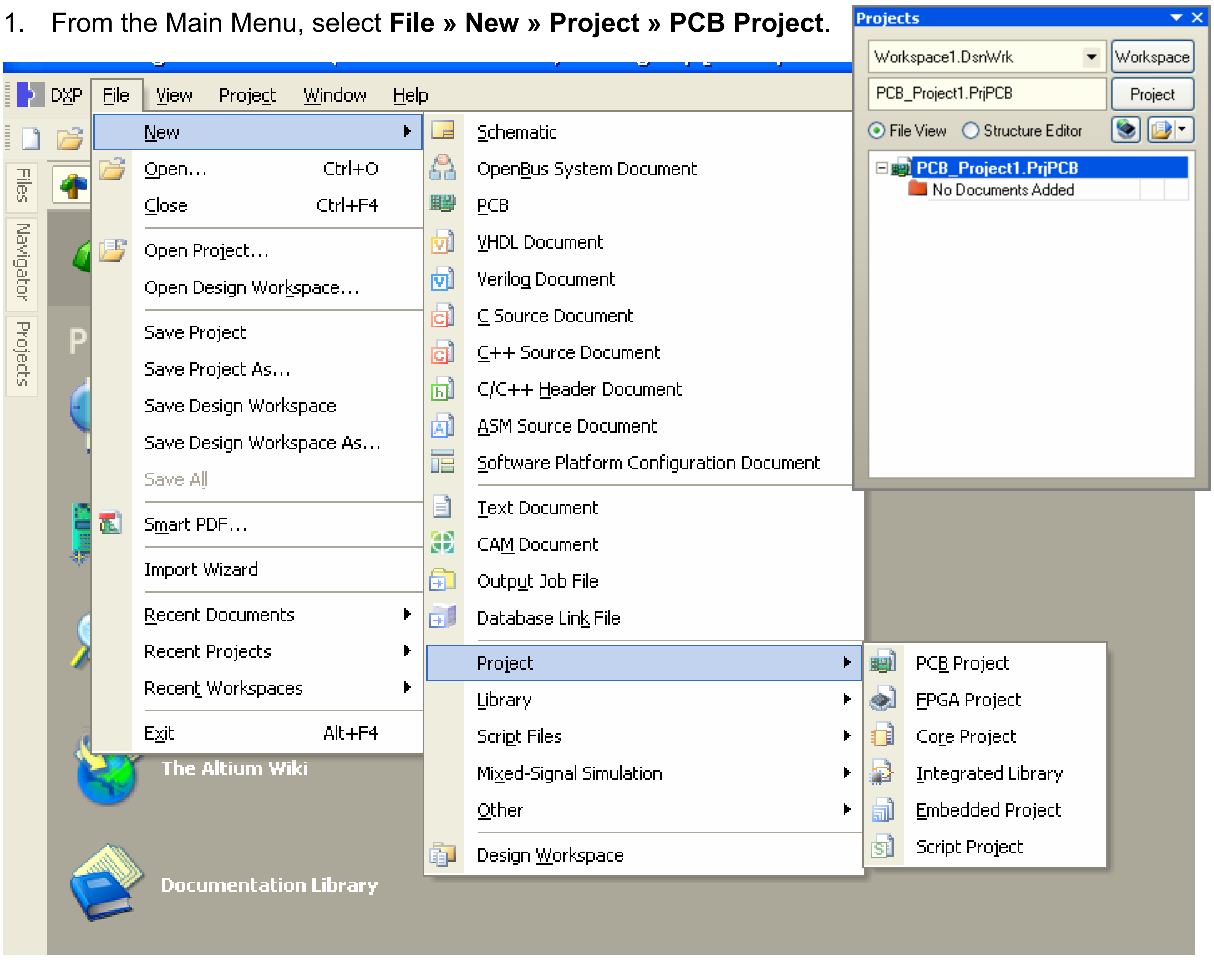



*Figure 11. The new project is displayed in the Projects panel*

2. Select **Save Project As** from the **File** menu to name and save the project document or you can right click on the project in the *Projects Panel* and select **Save Project As**.

3. The new project is ready to add new or existing documents to.

## 1.4.2 Adding a new document to the project

To add a new document to the project:

1. Right-click on the Project name in the Projects panel, and from the **Add New to Project** sub-menu, select the document kind, for example, **Schematic**.
2. Right-click on the new schematic document in the Projects panel and select **Save As** to name and save the schematic.

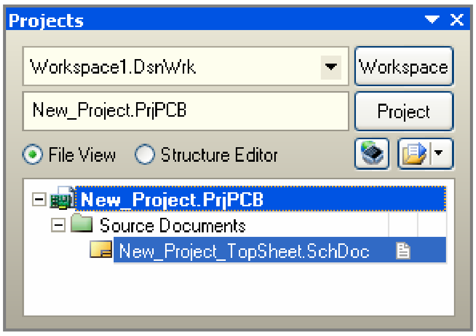



*Figure 12. New schematic added to the*

## 1.4.3 Adding an existing document to a project

To add an existing document to a project:

1. Right-click on the Project name in the Projects panel.
2. Select **Add Existing to Project** in the menu to display the *Choose Document to Add to Project* dialog.
3. Navigate to locate required file and select it.
4. Click on **Open** to add it. The document is added into the currently active project. Note that when you add a document to a project a link is added in the project file to that document. The document can be located anywhere on the hard disk (or network).

The document icon graphic indicates which Editor will be used to edit the document, e.g. a PCB document will have a PCB icon, indicating that it will be opened by the PCB Editor.

**Note:** You can add a document to a project using a two step process. First drag the document from the Windows File Explorer into the Altium Designer Projects panel and then when it appears as a Free Document, click and drag it into the project.

## 1.4.4 Moving or copying a document between projects

1. Since documents are only *linked* into the project, you can easily move a document from one project to another simply by clicking and dragging it.
2. To copy a document to another project, hold the CTRL key as you click and drag.

## 1.4.5 Removing a document from the project

To remove a document from a project, right-click on the document icon in the Project panel and select **Remove from Project**.

**Note:** The document is not deleted from the hard disk, but it is no longer linked into the project.

## 1.4.6 File management with the Storage Manager

The Storage Manager is a system panel that allows you to perform a variety of file management tasks. When you open the **Storage Manager** (**View » Workspace Panels » System » Storage Manager**) it presents a folder/file view of the active project's documents.

The **Storage Manager** can be used for:

- General everyday file management functions such as renaming and deleting files in the project or within the active project's folder structure.
- Management of Altium Designer backups, using the **Local History** feature.
- As a Subversion compliant interface for your Altium Designer projects.

**Note**: Right-click in the different regions of the panel for options.

- As a CVS compliant (Concurrent Versions System) interface for your Altium Designer projects.
- As an SCC (Source Code Control) compliant version control interface for your Altium Designer projects.
- Performing a physical and electrical comparison of any 2 versions in the **Local History**, or the CVS **Revision** list.

The **Folders** view on the left gives access to documents stored in the project folder hierarchy. Next to this the **File** list shows all documents in the selected folder. A number of highlighting modes are used to indicate the state of each document, press F1 when the cursor is over the panel for information on highlighting.

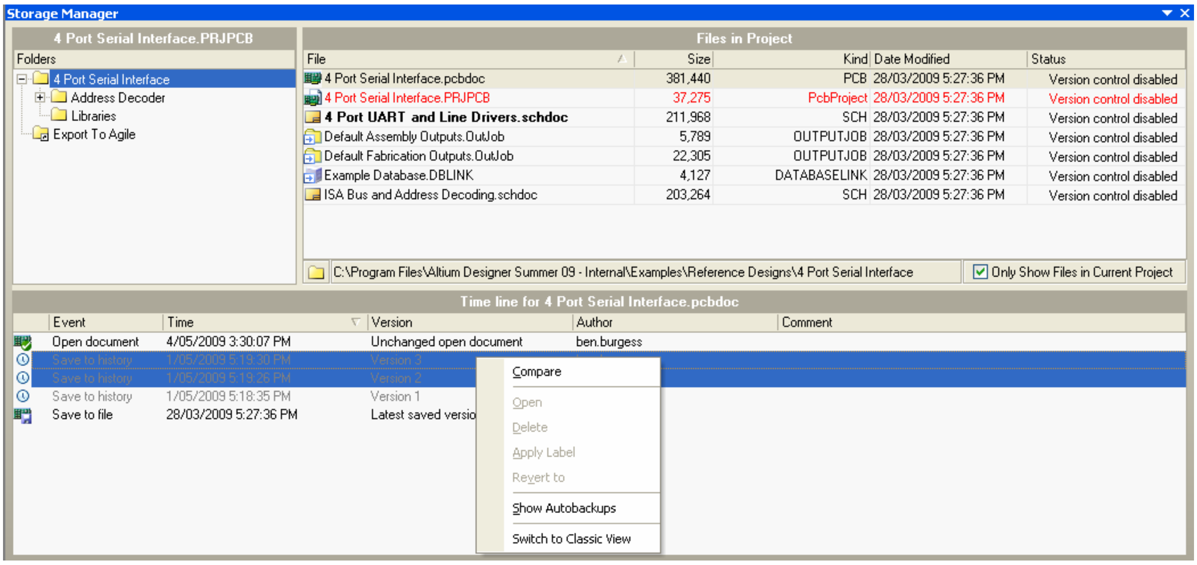


*Figure 13. Use the Storage Manager to manage project files on the hard disk, and to interface to your Version control system.*

**Note**: Press F1 over the panel for access to detailed help.

## 1.4.7 Including other files in the Altium Designer project

- You can include any file in your Altium Designer project, as long as the Microsoft Windows® operating system is aware of the file's associated editor.
- Add it to the project as described in section 1.4.3 (you will need to change the file filter to see non-Altium file types). The file will appear in the Project structure in the the **Projects** panel, under a folder icon titled **Documentation**.

## 1.4.8 Libraries

- Libraries can exist as individual documents, for example, schematic libraries containing schematic symbols, PCB libraries containing PCB footprint models, discrete SPICE models (MDL and CKT), and so on.
- Altium Designer also supports the creation of integrated libraries. An integrated library is the compiled output from a library package. It includes all the schematic libraries in the original library package, plus any referenced models, including footprint, simulation and signal integrity models.
- Most of the supplied libraries are provided as integrated libraries and are stored within the `\Program Files\Altium Designer Summer 09\Library` folder. Integrated libraries can be converted back to their constituent libraries; simply open them in Altium Designer to do this. PCB libraries are also provided in the `\Program Files\Altium Designer Summer 09\Library\Pcb` folder.
- The Schematic Library Editor and PCB Library Editor are covered during the *Schematic Capture* and *PCB Design* training sessions. The basics of creating an integrated library are also covered.

> **Note**: You can use Protel 99 SE libraries directly in Altium Designer. Add them to the Libraries panel to use them without converting them to the Altium Designer format. Note that you will not get all the benefits of the enhanced parameter and model support.

## 1.4.9 Project Packager

An Altium Designer project can include many and varied files - source files, libraries, reports, data sheets, manufacturing files, etc. The Project Packager Wizard simplifies the task of managing and transferring the complete fileset. Guided by the settings you define the Project Packager Wizard gathers and packages the project into a portable time and date stamped ZIP file. The Project Packager supports:

- Any situation where your project must be moved, for example is moving it from one site to another, or backing up your project for secure storage.
- Packaging a complete Altium Designer project tree - ideal for linked PCB + FPGA + Embedded projects.
- Packaging a complete Altium Designer Workspace - ideal for designers that include all the board designs destined for a company product, in a single Workspace.
- Managing how directory paths are handled during packaging.
- Managing how files outside the project folder are handled during packaging.
- Including/excluding Generated files, such as reports, in the project package.
- Including/excluding History files (created by Altium Designer's built-in file history/restore system).

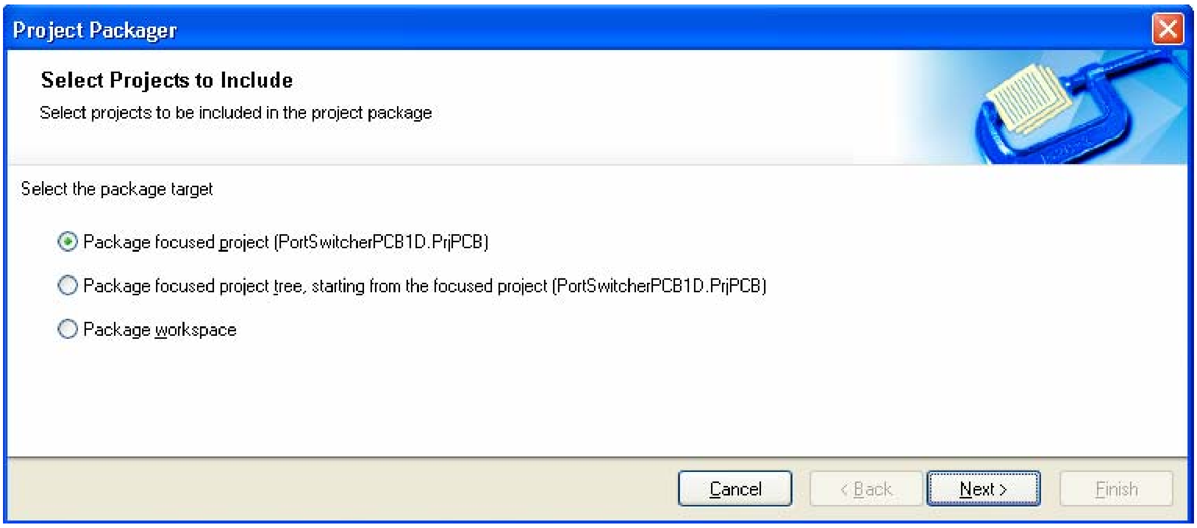



*Figure 14. Project packager project choice page*

## 1.4.10 Exercise – Working with projects and documents

This exercise looks at creating a new project and adding documents to it.

1. Create a new PCB project in the `\Altium Designer Summer 09\Examples\Training\PCB Training\Temperature Sensor` folder and name it `Temperature Sensor.PrjPCB`. We will use this project later during the Schematic Capture training session.
2. Add the following two schematic documents to the project from the `\Altium Designer Summer 09\Examples\Training\PCB Training\Temperature Sensor` folder: `LCD.SchDoc` and `Power.SchDoc`. Use **Add Existing to Project…** command from the right click menu in the **Projects** panel.
3. Save and close the new project `Temperature Sensor.PrjPCB`.
4. Check that the documents exist on the hard drive using the Windows Explorer

# Module 2: Help and DXP system menu

# Module 2: Help and DXP system menu






Module Seq = 2

# 2.1 Using the Help system

## 2.1.1 Dynamic On-line Help

Altium Designer includes a dedicated panel for dynamically displaying context sensitive help as you work. The panel, called the **Knowledge Center** panel in Figure 1 has two sections. The upper section displays help about the current menu entry, toolbar button, selected object, panel, and so on. The lower section is the library navigation area. Here you can browse through the documentation tree to open PDF-based articles, application notes, tutorials and other references.

The help text loads automatically into the upper region of the **Knowledge Center** panel if the **Autoupdate** button is enabled, indicated by the outline around the button. By enabling the Autoupdate feature of the help system, the upper section of the Knowledge Center panel will update to display content sensitive help while you work.  If it is not enabled you can force the context to load by pressing **F1**.

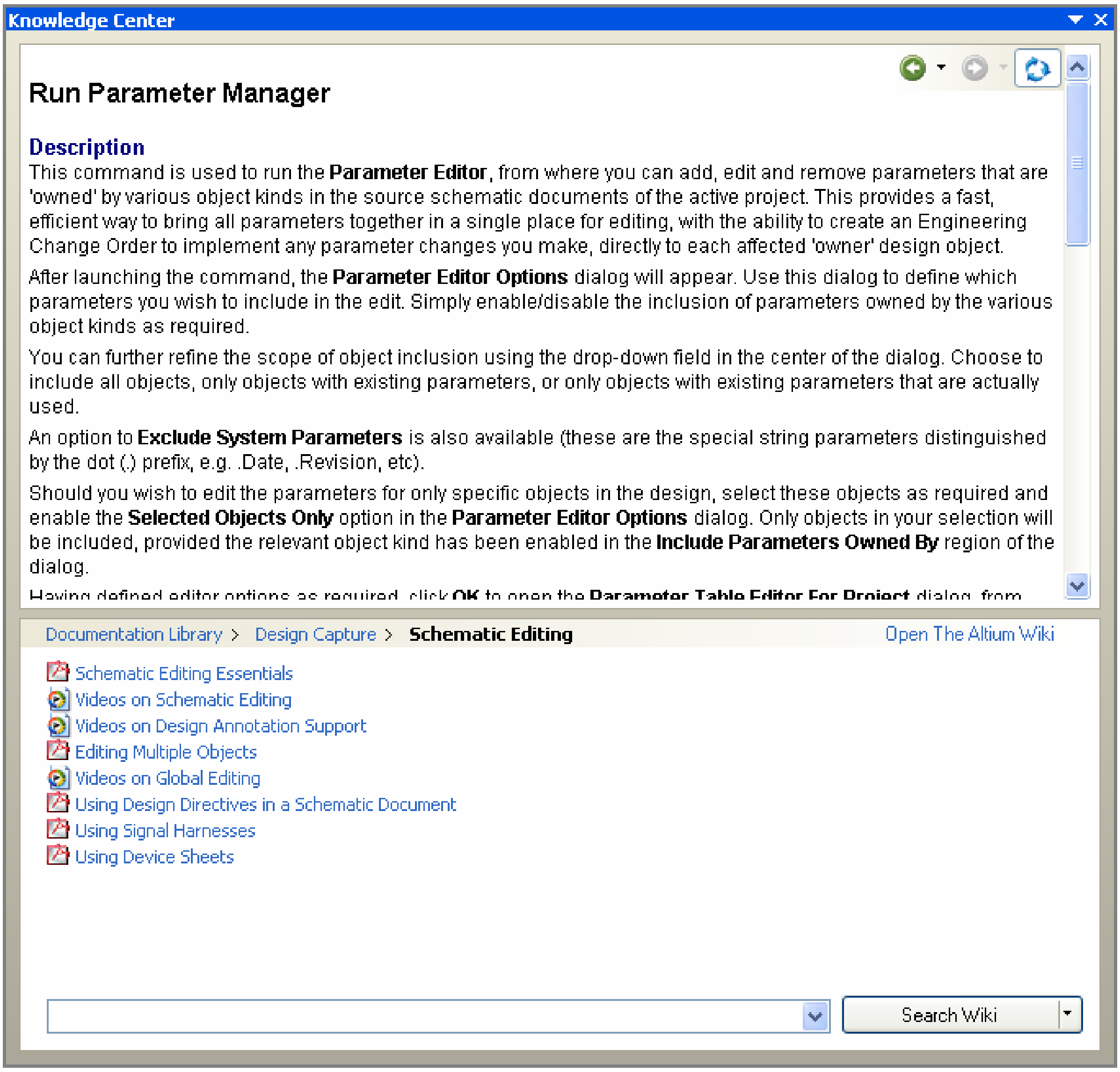



*Figure 1. The Knowledge Center panel is used to access the Documentation Library.*

## 2.1.2 Searching the Documentation Library

Enter a search string in the field at the bottom of the panel and click **Search Wiki** to search the Wiki (http://wiki.altium.com) by default. This opens a web browser window in the software to view the search results. To search via the PDF based documentation in the documentation library click on the drop down arrow on the **search** button and select **Search Local Documents**. Note that the scope or this searching is controlled by your current location in the library, open a specific sub-folder to restrict searching to that topic area, return to the top of the library to search the entire library. Figure 1 shows navigating to the *Design capture, schematic editing section*.

## 2.1.3 Using F1

The Altium Designer environment includes extensive F1 help support. Virtually every aspect of the interface has F1 help support, for example:

- Press **F1** while hovering the mouse cursor over a menu entry, toolbar button or dialog, to directly open the help topic about that command/dialog.
- Press **F1** while hovering the mouse cursor over a panel to obtain detailed help specific to that panel.
- Press **F1** in the Editor environment for help on that editor. If there is a design object under the cursor then you will be presented with help on that object.

## 2.1.4 What's This Help

Use the dialog **What's This Help ?** to gain detailed information about each of the individual options available in a dialog.

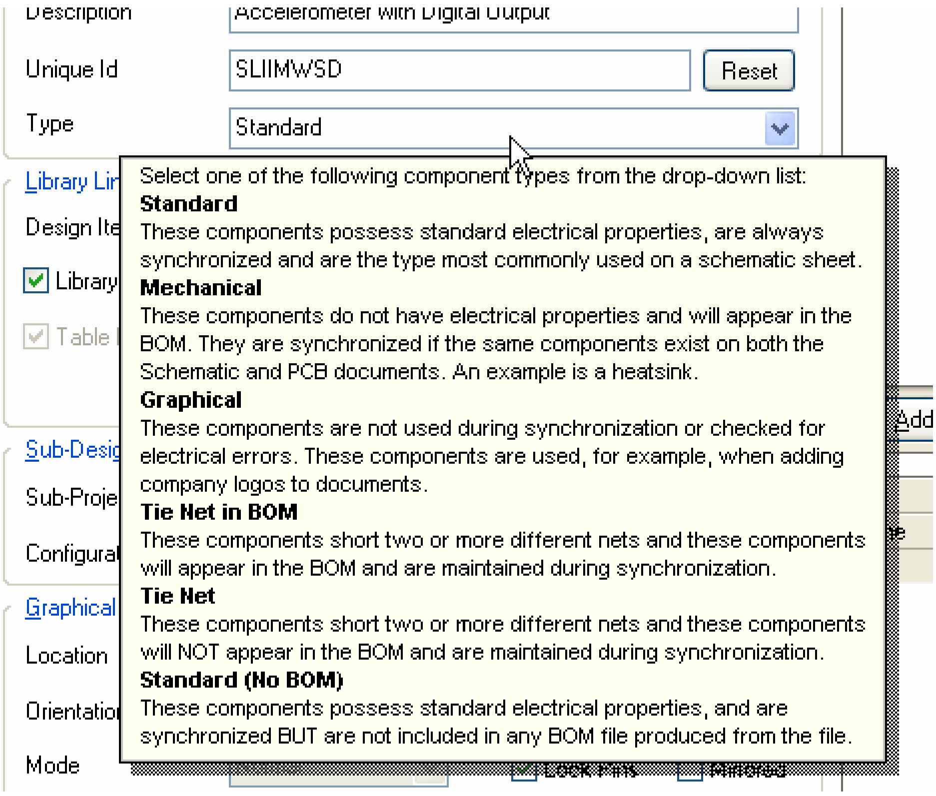



*Figure 2. Using the question mark icon to gain help in a dialog*

# 2.2 Using the Altium website

The Altium website (`www.altium.com`) includes extensive information about Altium's products and services, including access to technical information and Service Packs. It is good practice to regularly visit the website to keep in touch with the latest information.

- Click on the **Community** section at the top the Home Page to access a variety of customer resources.
- The **SUPPORTcenter** and **Learning Guides** are two of the sections available in the **Support** sub-page. Learning guides, such as tutorials, articles and white papers, are in PDF format.
- The Altium Technical Forums can be joined by choosing the **Forums** option in the **Community** menu at the top of the Home Page. The Altium Designer forum is very popular amongst both Altium product users and Altium staff, as a meeting place where they can exchange advice and information.
- The **DEMOcenter** found on the home page to the left side is a useful tool for viewing short videos of new features or selective how-to's in Altium Designer. Whenever a new release comes out with its usual array of new features, videos are created to help customers better utilize these new features in the product.
- The **TRAININGcenter** is another useful resource available in the **Community** section. These videos go into a lot more depth than the **DEMOcenter** movies. Current training material includes Project Management, layout, route and fill to Enterprise Integration videos.

## 2.2.1 SUPPORTcenter

The SUPPORTcenter http://www.altium.com/supportcenter/ is one of the ways users can access high-quality product support and access to the latest product updates service packs.

- Secure log in with username (email address) and password.

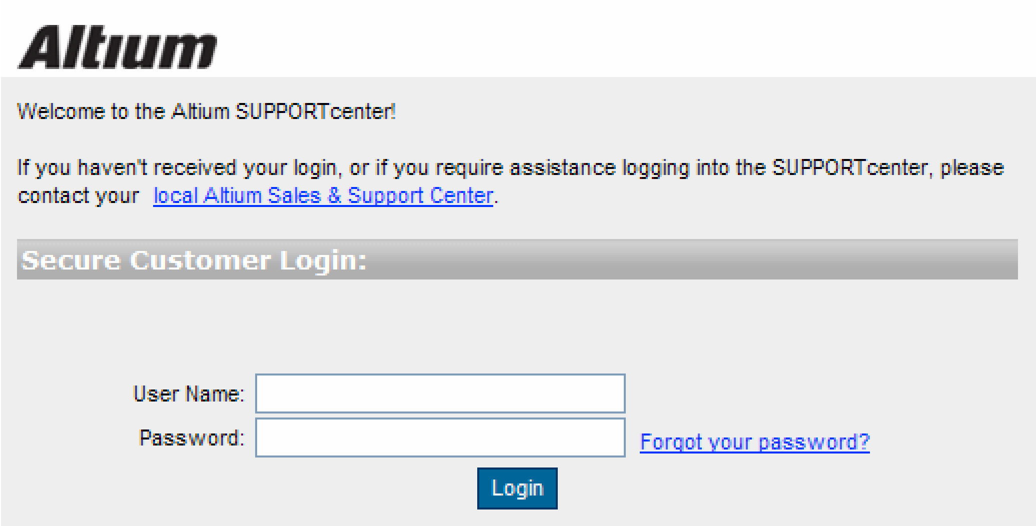



*Figure 3. The login page*

- Once in, it allows you to log, track and search through a solution database
- The Find Solution button searches for possible solutions

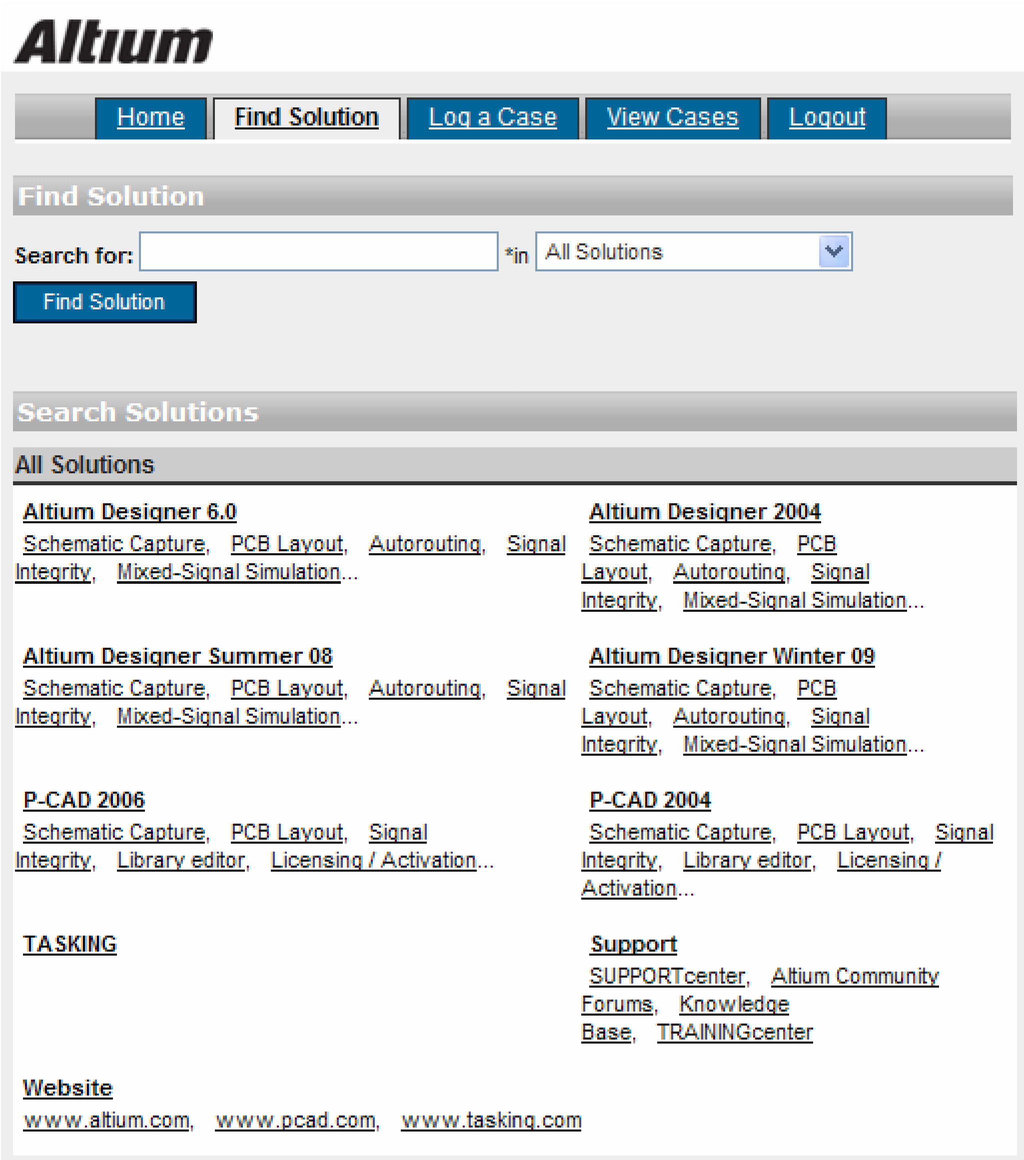



*Figure 4. Search engine for finding solutions*

- Click on the **Log a Case** button and entering in the bug or support query if a valid solution is not available.

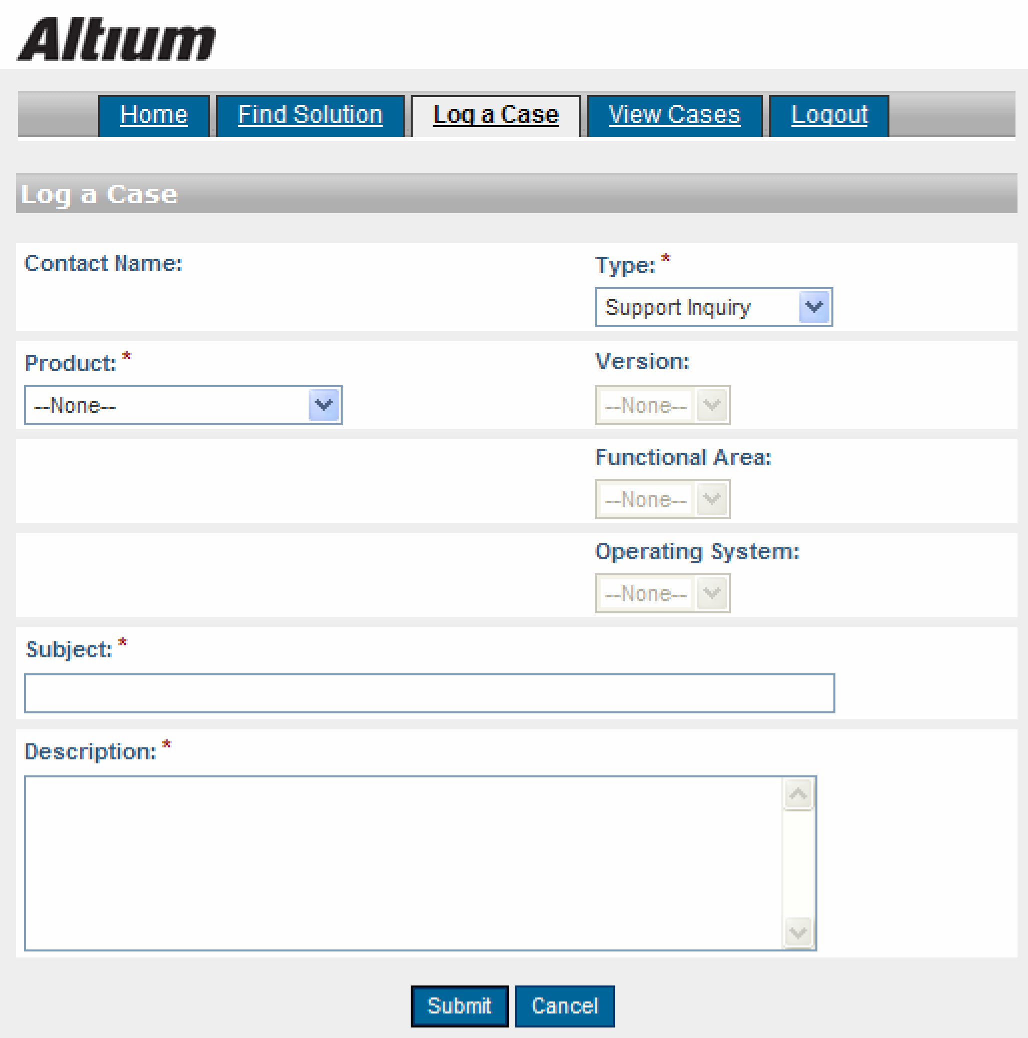



*Figure 5. Logging a case*

- Once the support query is added, click on the **Submit** button to save. The **View Cases** button allows editing, and adding attachments to the original query.

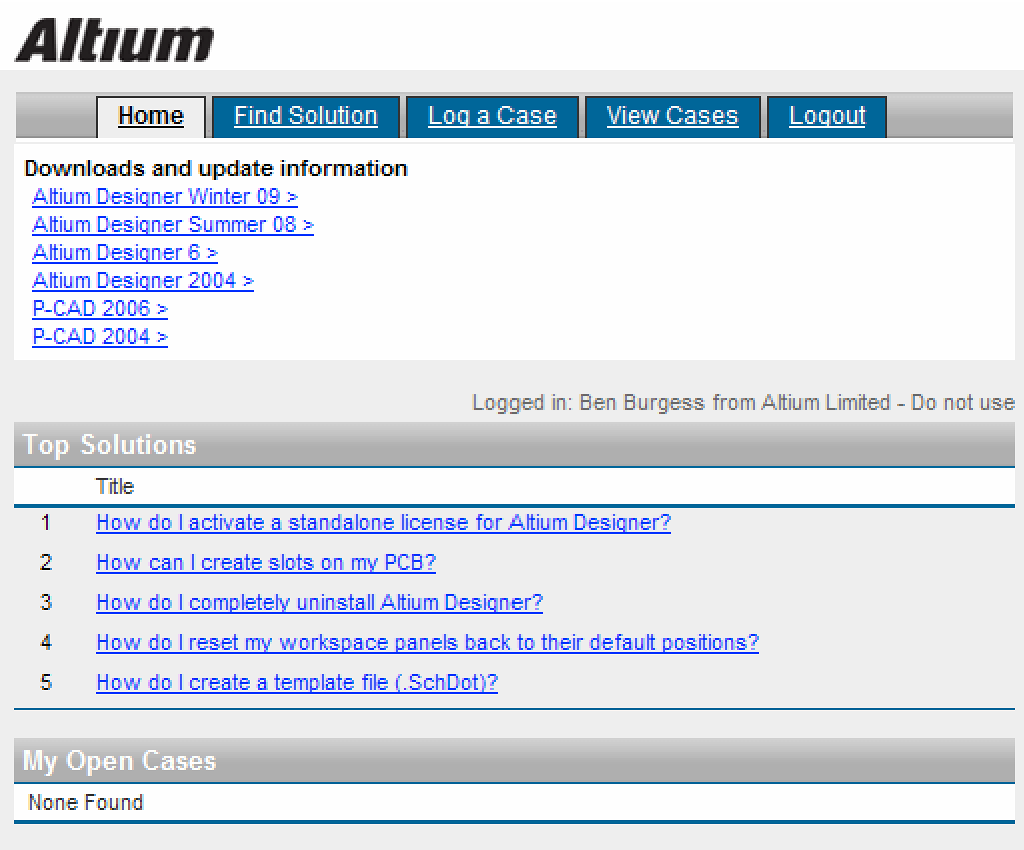



*Figure 6. Home page for downloads*

- The **Download and update information** section is on the main **Home** page of the SupportCenter when you log in.

## 2.2.2 Altium Forums

You can access the forums by going to http://forums.altium.com

- Secure log in with username (email address) and password.
- Access to unique resources developed by users
- Posts replied to by customers or Altium staff including R&D and even the CEO.
- Great 24 hour a day, 7 day a week resource of questions and answers from customers all over the world.

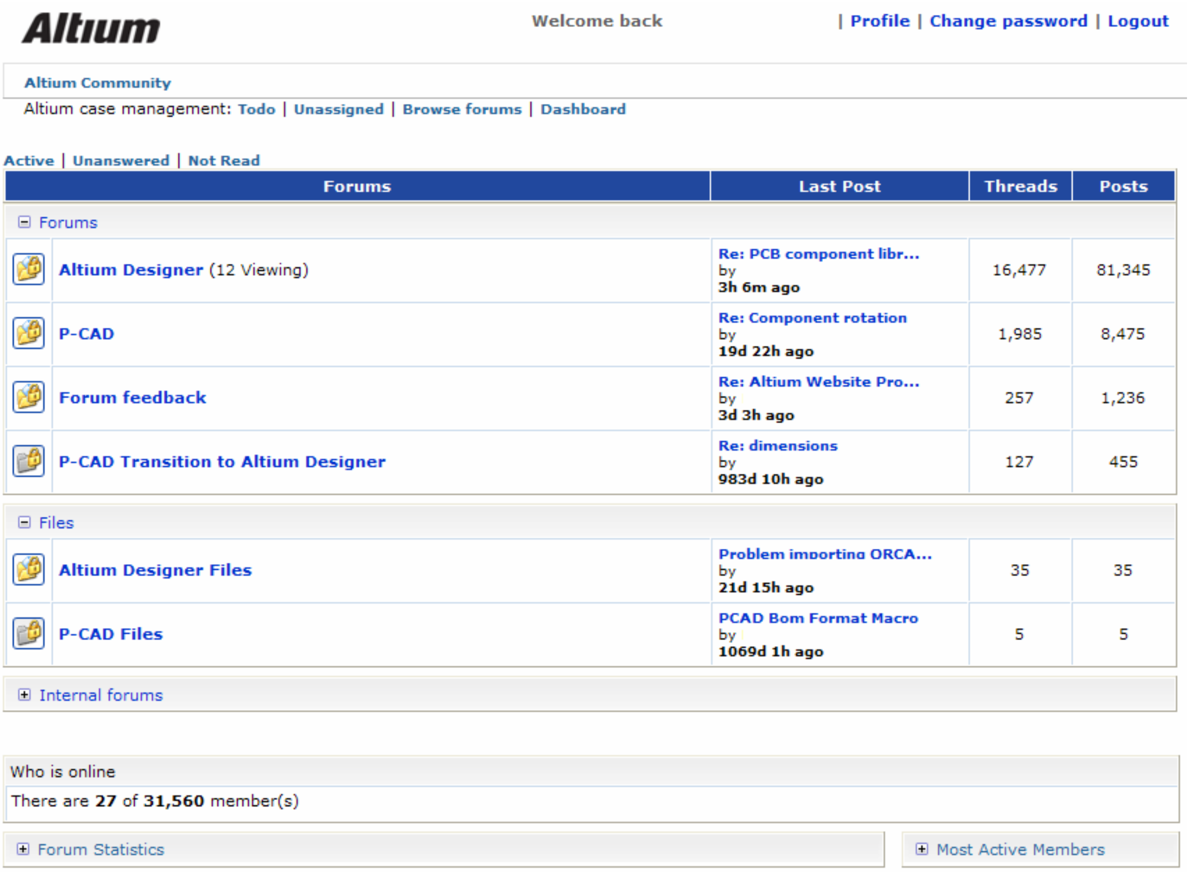



*Figure 7.Altium Community forum home page*

## 2.2.3 DEMOcenter

The DEMOcenter can be accessed from, http://www.altium.com/Evaluate/DemoCenter/ . This resource is constantly updated when new releases come out showing short movies of the new features available.

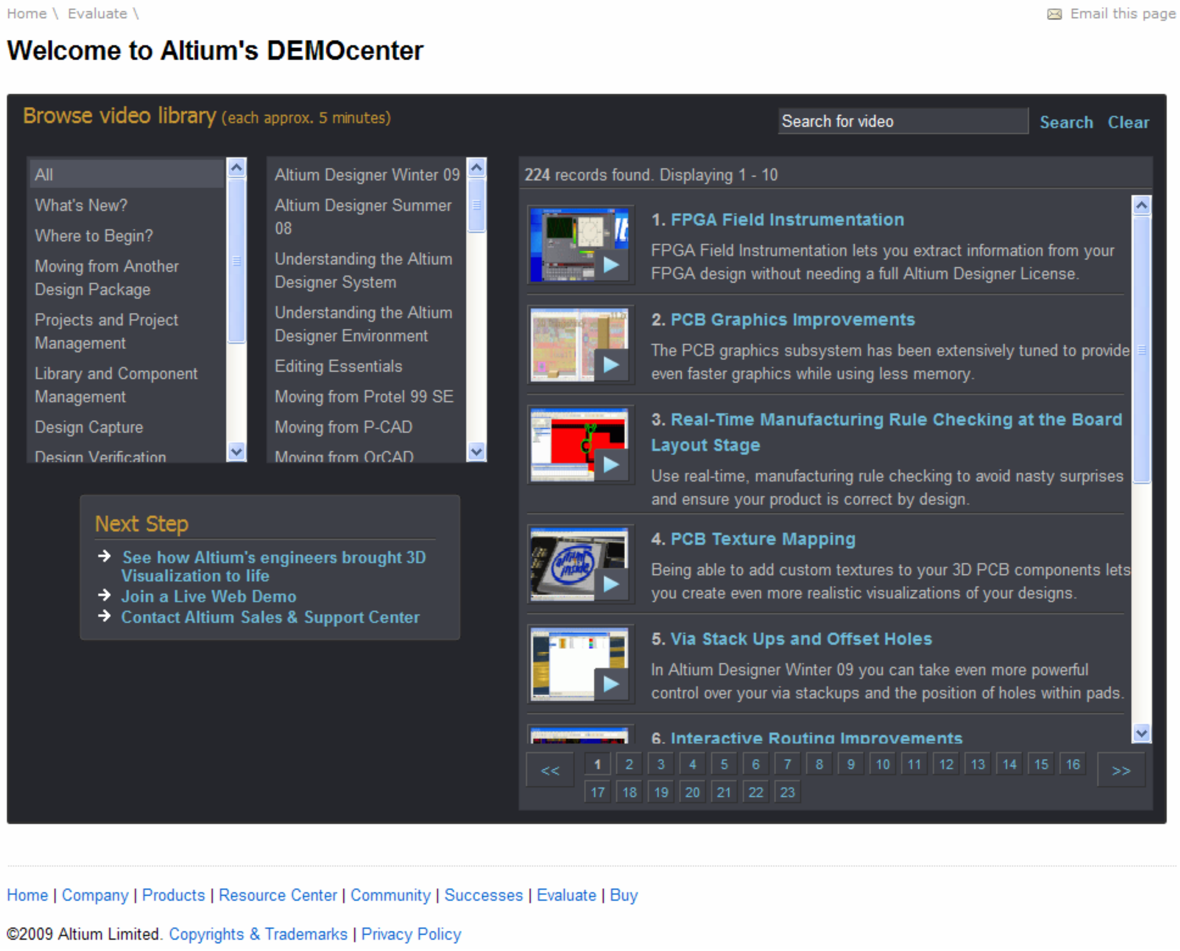



*Figure 8. A wide range of short movies are available through the DEMOcenter.*

## 2.2.4 TRAININGcenter

The TRAININGcenter can be found at, http://www.altium.com/community/trainingcenter/ . These training videos go into great depth into a very particular area of the product. These videos can be used as a resource in keeping your knowledge of the product and how to use it current. It's a great resource if you're completely new to the software or trying to get up to speed again from not using it for some time.

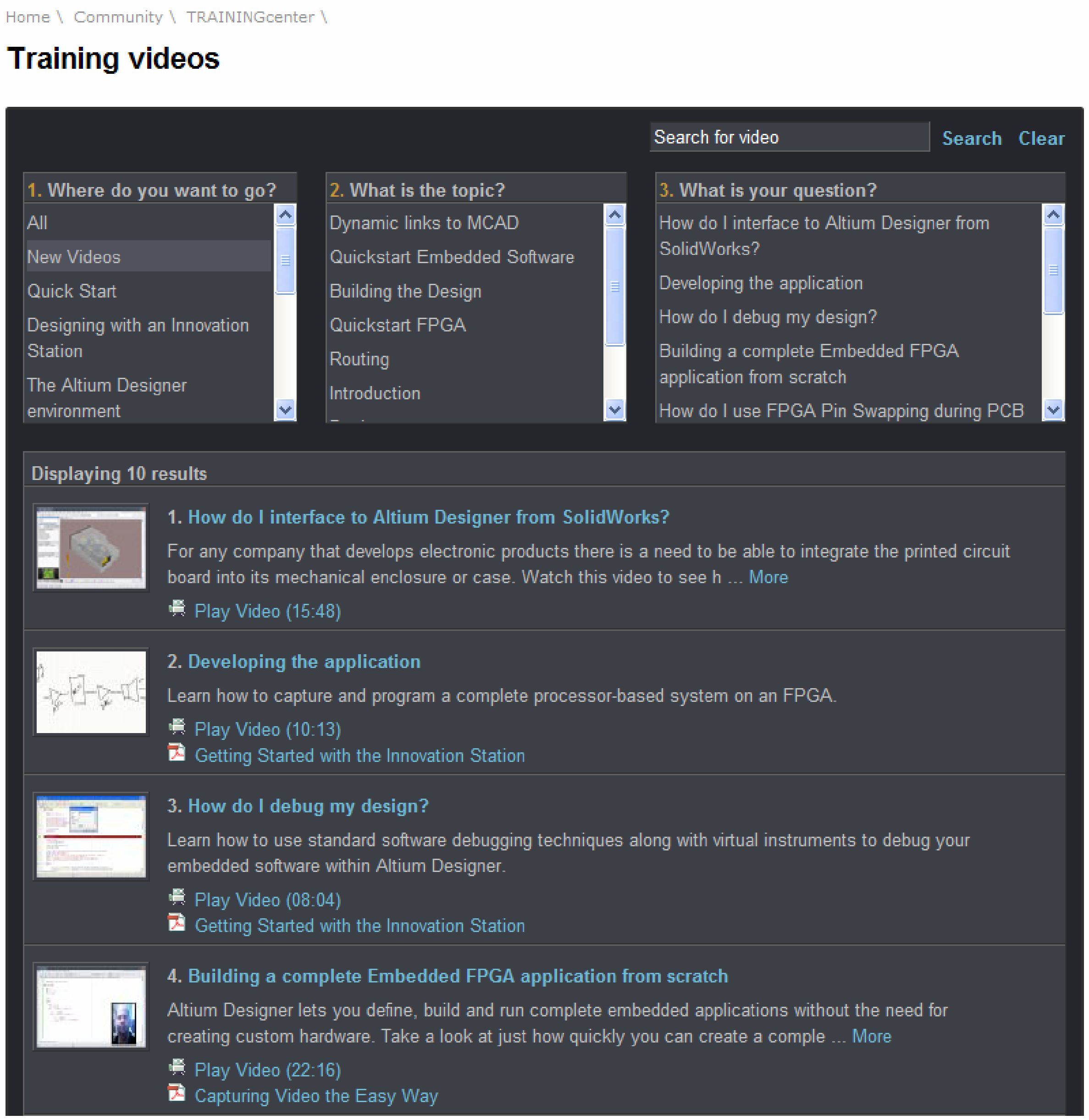



*Figure 9. A list of TRAININGcenter material available – please note new videos will be added to this page regularly.*

## 2.3 DXP System menu

The DXP system menu provides commands for configuring the Altium Designer (DXP) environment. You can access these commands by clicking on the DXP icon located on the left-hand side of the Main menu. The DXP menu is always accessible in Altium Designer, regardless of which editor is currently in use.

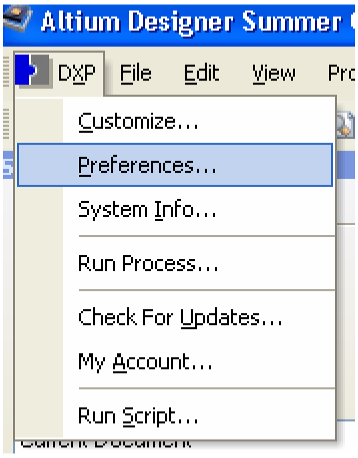



*Figure 10. Configure the environment via the DXP menu*

The following sections describe the entries in the DXP system menu.

### 2.3.1 Customize

The menu enables the management of resources associated with the current editor. For more information on this, the Advanced PCB training course covers this in a lot of detail.

### 2.3.2 Preferences

Various global system preferences can be set for the DXP environment, including file backup and auto-save options, the system font used, the display of the Projects panel, environment view preferences including the popup and hide delay for panels, and enabling the version control interface. You can also access the environment preferences for each of the editors available in Altium Designer, such as the schematic and PCB editors.

To set Altium Designer environment preferences, select **Preferences** from the **DXP** menu. This will open the *Preferences* dialog shown in Figure 11.

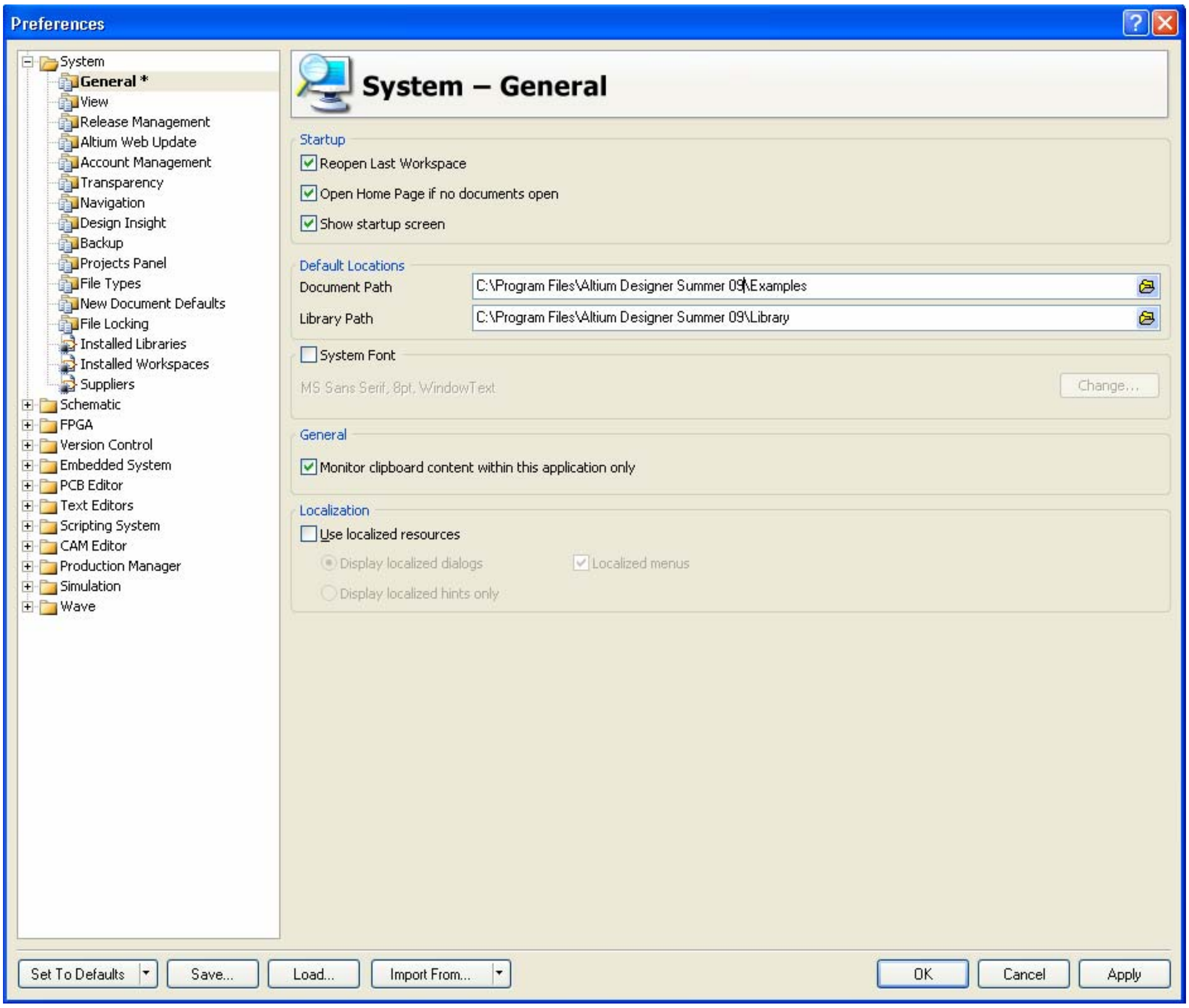



*Figure 11. Preferences dialog, used to configure Altium Designer and all editor preferences.*

#### 2.3.2.1 Autosave and manual save backup options

Altium Designer supports two automatic file backup modes:

1. Backup-on-save – keep a backup whenever a user-initiated save action is performed (on by default). These files are saved in a History folder. The default is to create the History Folder below the active project folder, to configure an alternative central folder open the **Local History** page of the **Version Control** section of the *Preferences* dialog. History files are listed in the **History** section of the **Storage Manager** panel.

2. Timed backup – automatically save a copy of all open documents at a fixed time interval (off by default). Autosave settings, such as number of files and frequency of saves are configured in the **Backup** page of the **DXP System** section of the *Preferences* dialog.

Both backup modes support multiple copies, using the naming convention of:
`OriginalFileName.~(number of save).DocExtension.Zip`

Backup files are automatically compressed to reduce file size.

### 2.3.3 Run Process

Selecting the **Run Process** command from the DXP System menu displays the *Run Process* dialog, which allows you to run any process in the DXP environment.

## 2.3.4 Licensing

Selecting the **My Account** command from the DXP System menu displays the *Licensing* View, where you can select and configure the licensing type – Standalone or (Private Server). With the Summer 09 release there is also now the concept of an On-Demand license.

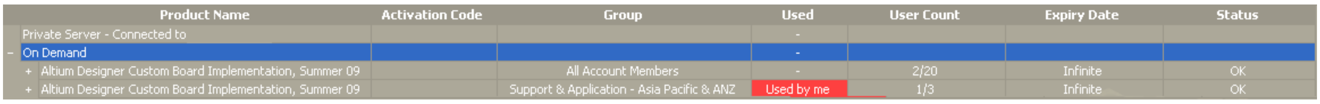



*Figure 12. Connection to Altium based Server*

The On Demand license requires that the user have a SupportCenter login and password in order to see the company or individual license(s) available. This login is entered via the account preferences in **DXP » Preferences » System » Account Management**. If you don't have a SupportCenter login, your local support office can create one for you.

### 2.3.4.1 Roaming License

With the new on-demand license in Altium Designer Summer 09 there is also the concept now of borrowing a license from a pool of company licenses. When a license is borrowed it can be taken offline until the license borrow time comes to pass, at which point the license automatically comes available again in the online pool for other users to use. This function is especially useful for designers that work from home or work on the road.

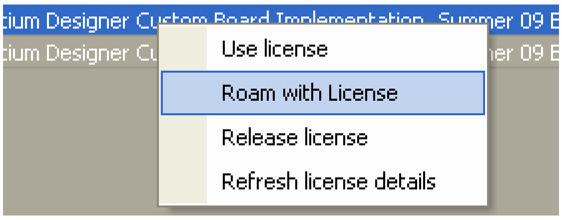



*Figure 13. Right click on the license type to get the roam with license option.*

A dialog appears when Roam with License option is used where the user can enter time in hours and days for the license to be borrowed from the pool. License can be borrowed for an unlimited amount of time in hourly increments.


*Figure 14. Setting the borrowing time for an on demand license.*

More details can be found here.

### 2.3.4.2 Standalone

Altium Designer's Standalone licensing allows you to effectively manage your own license through use of a Standalone licensing file (`*.alf`). This file can be saved, copied and backed-up as required. The `.alf` file is reusable on a home computer (in accordance with the EULA) simply by copying the file to a specific folder on that computer and then adding the file as part of

Standalone License Configuration. You also have the ability to convert a Standalone license to a single-seat On-Demand license.

With a Standalone license, the only time you need to sign in to your Altium account (through the Altium Portal), is when you initially activate your Standalone license, or if you want to convert it to an On-Demand license. Other than that, with this type of license, you really are working in a self-contained, offline fashion – the very definition of standalone. More details can be found here.

#### 2.3.4.3 Private Server

Altium Designer's Private Server licensing offers you floating license capability through implementation of your own dedicated *Private License Server*. Your administrator sets up this central server (also referred to as a network license, or local license server) to access and use Private Server licenses. Once a license is in use on the server, that server can then serve the license to multiple local computers. Users on the local network do not need to sign in to their Altium accounts to acquire the seats from the Private License Server. More details can be found here

### 2.3.5 Altium Account Manager

To complement Altium Designer's licensing system, browser-based license management and reporting is provided, courtesy of the *Altium Account Manager*. The Account Manager allows you to firmly control how the licenses you purchase are assigned within your organization and lets you view license activity.

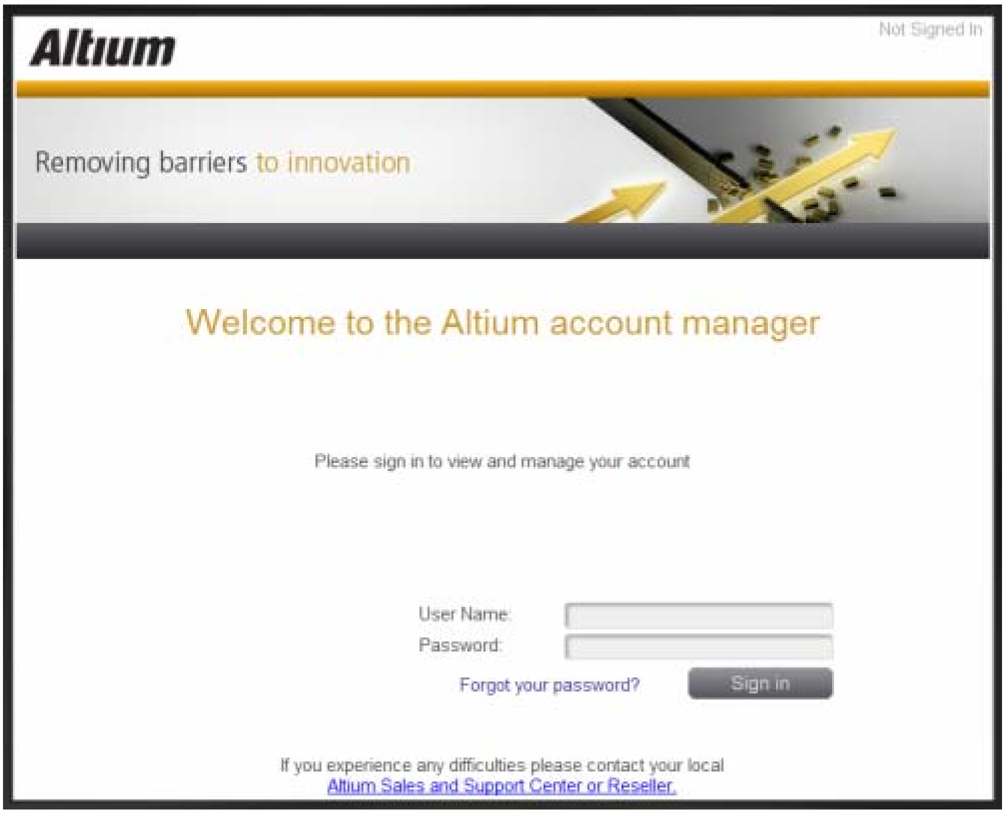



*Figure 15. The browser-based Altium Account Manager.*

#### 2.3.5.1 Accessing the Account Manager

Access the Account Manager – using your favorite web-browser – at myaccount.altium.com. The initial welcome page will load, as pictured below.

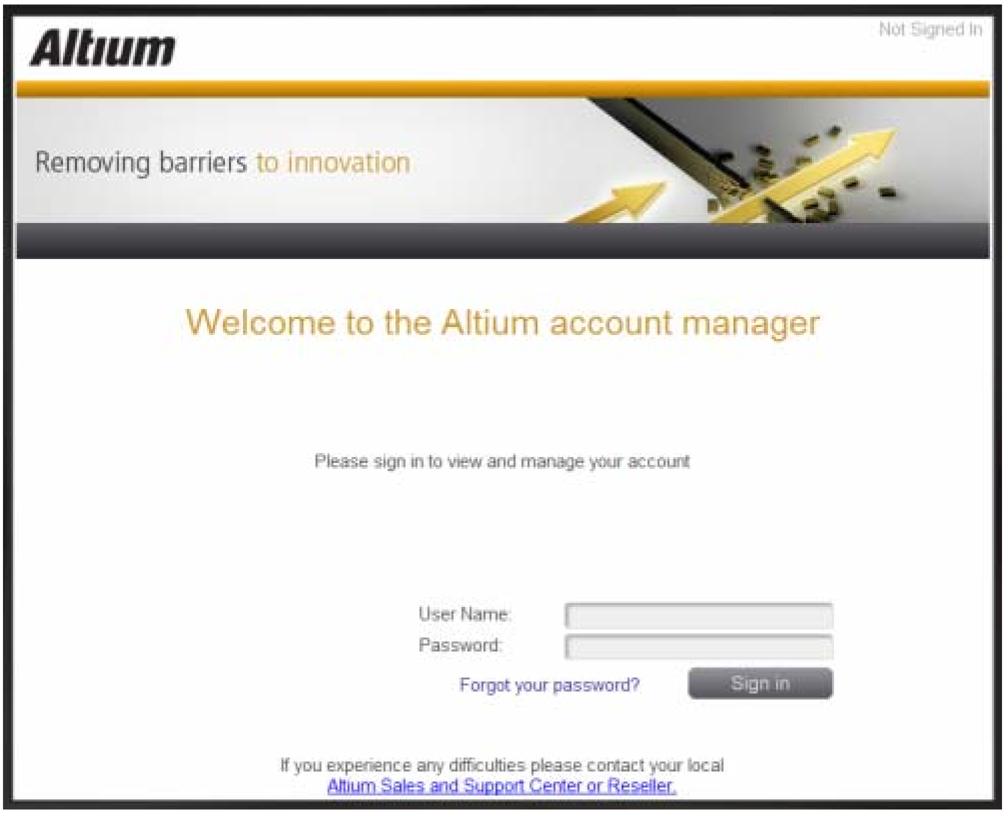



*Figure 16. The browser-based Altium Account Manager.*

To view and manage your account, you will need to sign in to your Altium account through the secure Altium Portal, using your standard login credentials.

The **User Name** and **Password** for the login are the same as those used to access the SUPPORTcenter and Altium Wiki.

### 2.3.5.2 Home Page

Once signed in, you will be taken to the Account Manager's `Home` page, which presents top-level information for the account itself, such as company details, billing address and shipping address. Summary details for the number of users, groups and licenses associated to the account are also presented on this page.

Use the links associated to the summary entries to access the corresponding Users, Groups, and Licenses pages for the account, respectively. Alternatively, click directly on a page heading in the banner area of the page.

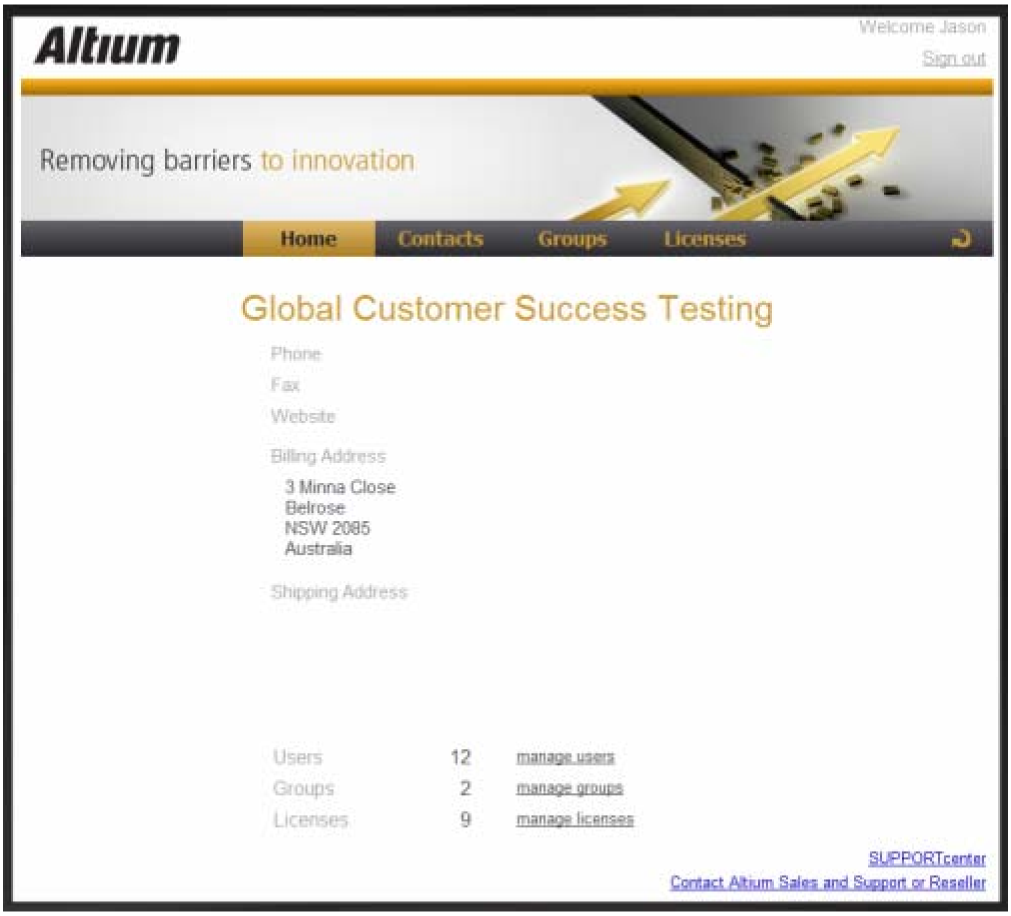



*Figure 17. The Home page presents information for the account, along with controls to access pages on which to manage account users, groups and licenses.*


To edit top-level account information, simply hover the cursor over the area of information you wish to change and click the **change** link that appears. A popup form will appear in which to make the required changes.

*Figure 18. Make changes to account details using the associated popup forms.*

Any changes you make to information on any of the Account Manager pages will refresh in your browser instance. However, it is possible that others, who have administrative access to the account, are also making changes. To refresh the account details – essentially retrieving the latest stored details from Altium, simply click on the button, located at the right of the banner area.

### 2.3.5.3 Users

Management of users for the account is performed from the **Users in account** page. A user is simply a person who it is intended will use a licensed seat of the software.

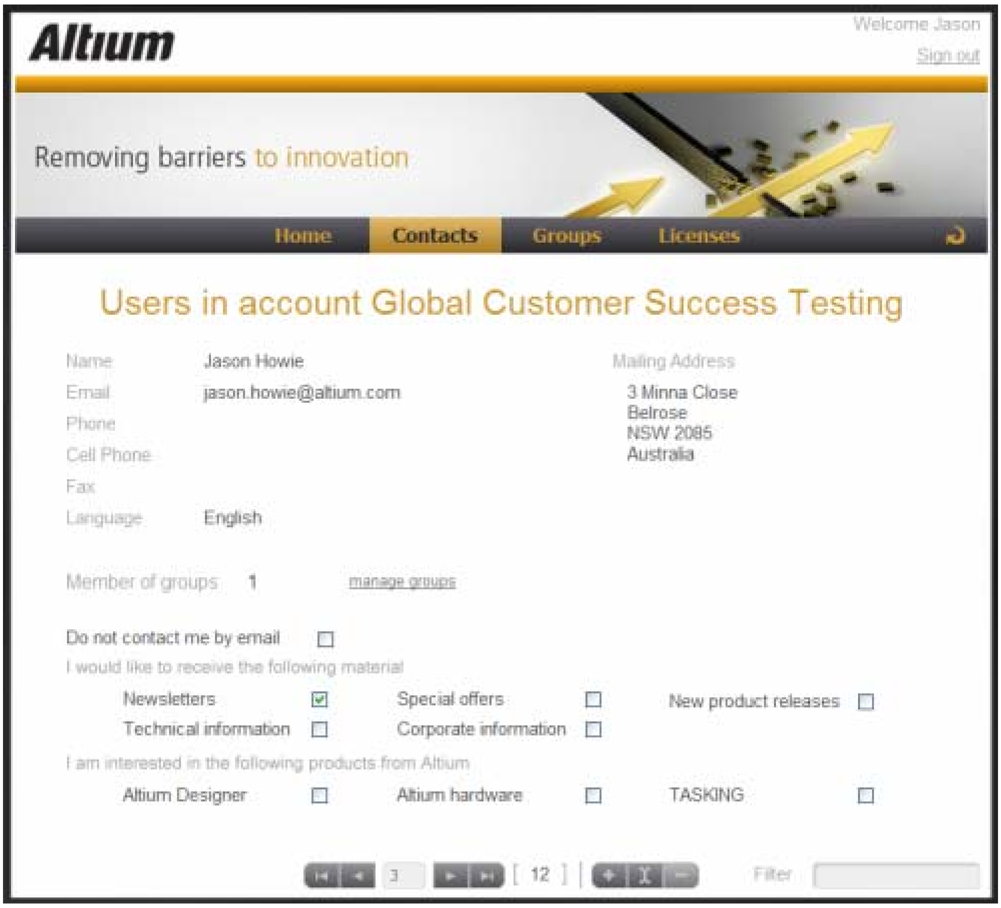



*Figure 19. Create a 'database' of people who are to use the license(s) associated to the account.*

For each user, the page presents:

- Contact information (name, postal address, email address, phone number, cell phone number and fax number. The main language used by the user is also listed).
- How many groups the user belongs to (with a link to the corresponding group management page for that user).
- Whether or not the user can be contacted via email and, if so:
  - What type of material the user has elected to receive from Altium.
  - Which Altium solutions the user is interested in.

#### 2.3.5.4 Groups

Management of groups for the account is performed from the **Groups in account** page. Groups allow you to further organize your users according to, for example, the particular section of the company in which they are involved, or the design team they are in. Groups make assignment of licenses more streamlined.

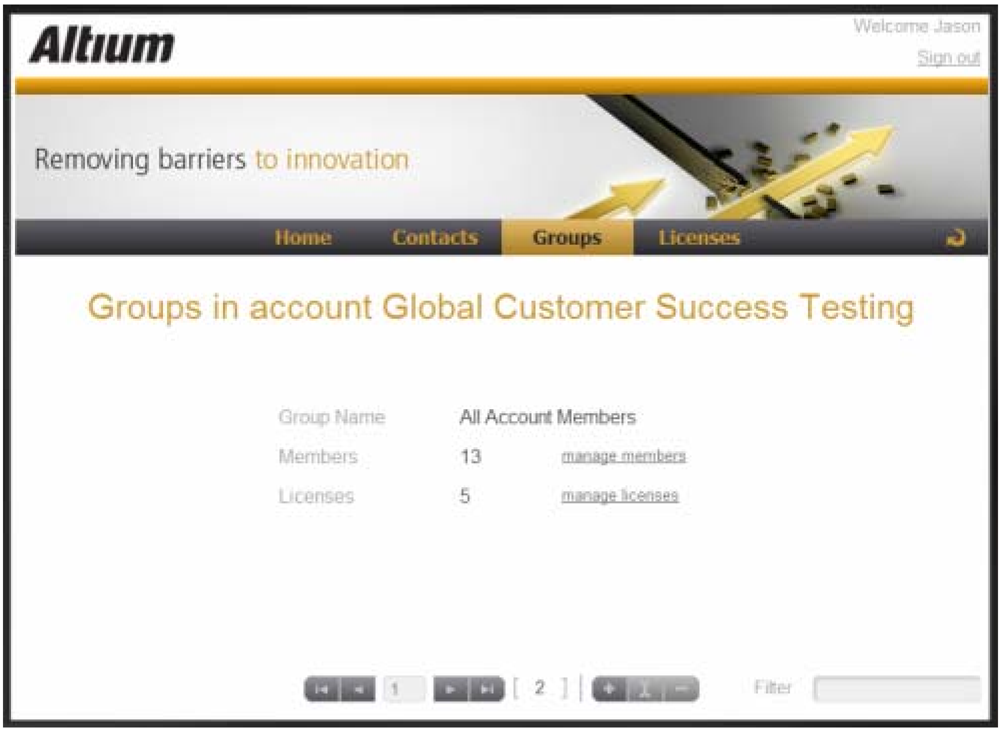



*Figure 20. Create specific groupings (or 'memberships') of users and then assign licenses to those groupings.*

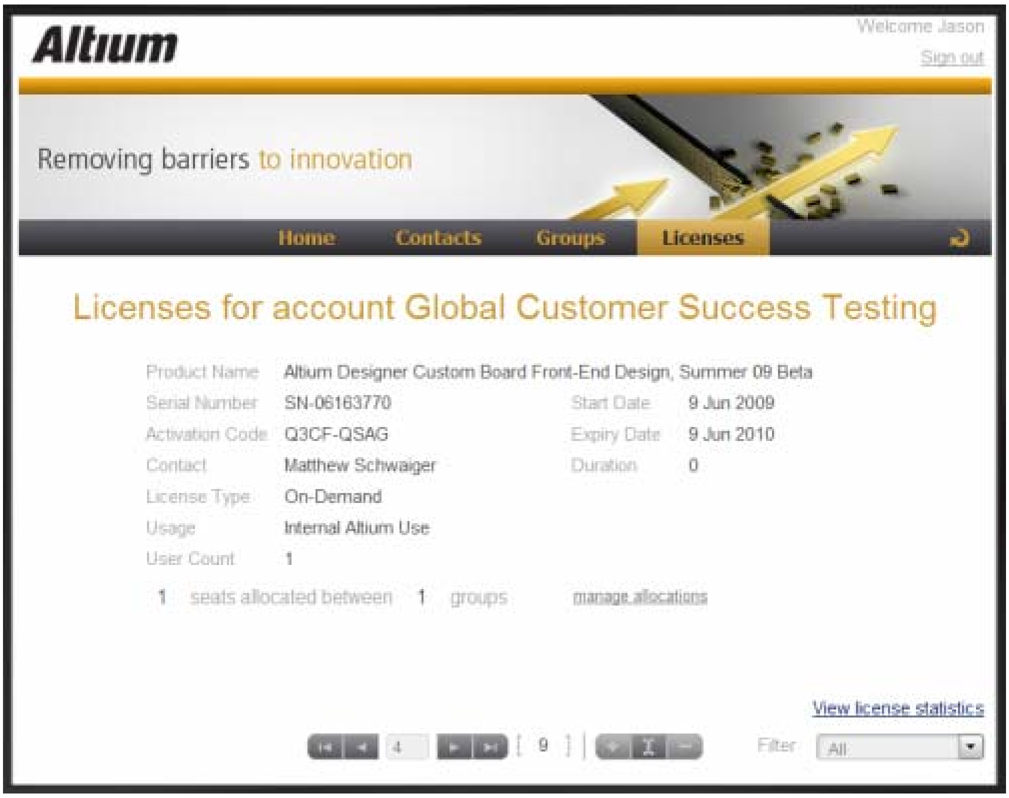
There are two system groups, defined by default for an account. These are:

- *All Account Members* – this group automatically includes every user in the account. So if a license is assigned to this group, then any user in the account can use that license.
- *Group Administrators* – this group gives administrative privileges to the members of the group. Anyone who is a member of this group can make changes in the myaccount site, but anyone who is not a member cannot.

These two system groups cannot be edited or deleted. You can, however, manage members and license assignment for these groups, as with any user-created group.

#### 2.3.5.5 Licenses

Management of licenses for the account is performed from the **Licenses for account** page. From here, you can view all licenses that you have purchased from Altium.



*Create specific groupings (or 'memberships') of users and then assign licenses to those groupings.*

For an On-Demand license, you can:

- Allocate the license to one or more defined groups of users
- Determine how many seats of the license are assigned to each group (for a multi-seat license)
- Determine, on a group basis, whether or not the licenses can be 'roamed', and for how long.

For each license in the account, you can also view license statistics – showing weekly or monthly usage graphs and on which day the maximum number of concurrent seats for the license was used. This is most useful for On-Demand licenses, and can give you an indication of whether a license could be put to better use by simple reassignment between the groups that use it.

## 2.3.6 Run Script and Run Script Debugger

Altium Designer includes a powerful scripting system, supporting the built-in DelphiScript language, as well as popular Windows scripting languages, including VisualBasic Script and JavaScript.

- The built-in scripting language, DelphiScript, is a Pascal-like language. There is also a complete Form design interface, allowing dialogs to be quickly created.
- Selecting the **Run Script** menu entry will pop up the *Select Item to Run* dialog, click on the script name to execute that script on the current document.
- Selecting the **Run Script Debugger** menu entry will open the Script Debugger, where you can set break points, single step through the script, and so on.
- There is more information on scripting in the **Scripting** section of the Documentation Library. Browse to it in the **Configuring the System** folder in the lower section of the **Knowledge Center** panel.
- Two useful scripts you can try are located in the directory `\Altium Designer Summer 09\Examples\Scripts\Delphiscript Scripts\Pcb\`. The names of the scripts are

**CreateRegionsFromBitmap** and **PCB Logo Creator** and they create regions from a bitmap and create tracks and arcs from a bitmap respectively. To use these scripts open the document with the ***.PRJSCR** extension, then open up a PCB document and use **DXP » Run Script**.

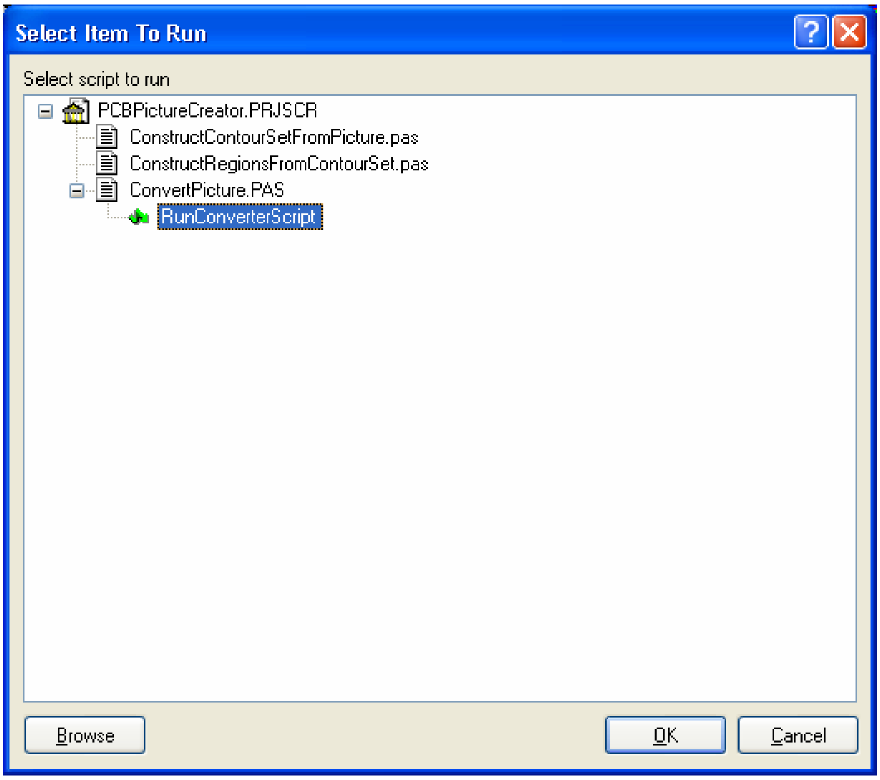



*Figure 21. Executing a script*

Click **Ok,** then load up a Bitmap file. You should see what's in Figure 22.

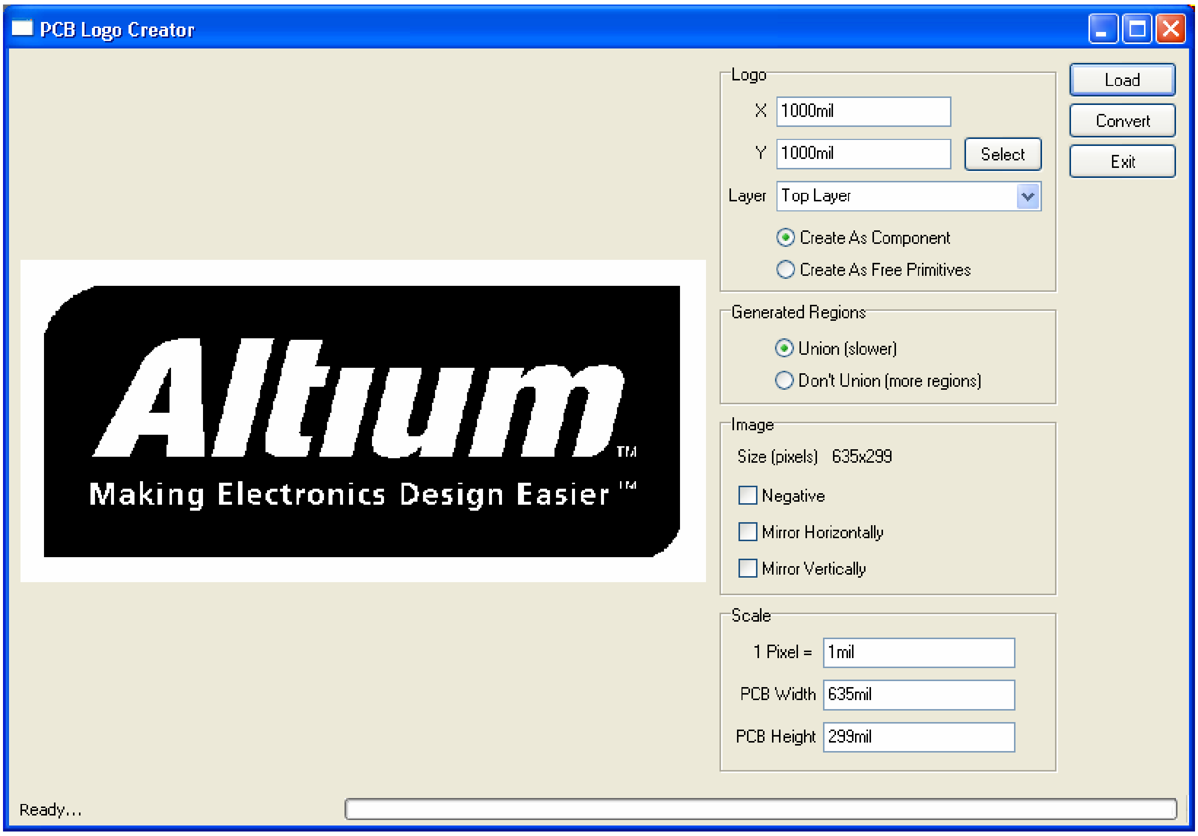



*Figure 22. Create regions from bitmap using Altium logo.*

It is also possible to customize Altium Designer, and add a script to a menu item or toolbar, so that it doesn't need to be opened each time. This functionality is covered in the *Altium Designer Advanced Schematic Capture and PCB Editing* training course.

### 2.3.7 Exercise – Configuring Altium Designer System Preferences

1. Open the DXP *Preferences* dialog and click on different nodes in the tree on the left of the dialog to get an idea of what options can be set – the options for the schematic and PCB editors will be covered later in the course.
2. In the **System » View** page of the *Preferences* dialog disable the **Use animation** option, and reduce the **Hide delay** option.
3. Close the *Preferences* dialog.
4. Hover the cursor over the **Libraries** tab on the right-hand side of the workspace to see how the popup of the panel is affected and then move the cursor away from the **Libraries** panel to check the hide delays. If you don't have the libraries panel loaded go to the panel control down the bottom right of your screen and click **System » Libraries**.

TOP SECRET

Altium Designer

# Module 3: Schematic Editor Basics

## Module 3: Schematic Editor Basics





Module Seq = 3

# 3.1 Schematic Editor basics

The Schematic Editor opens when you open an existing schematic document or create a new one. This editor makes use of all the workspace features in the Altium Designer environment. This includes multiple toolbars, resource editing, right-click menu, shortcut keys and Tool Tips.

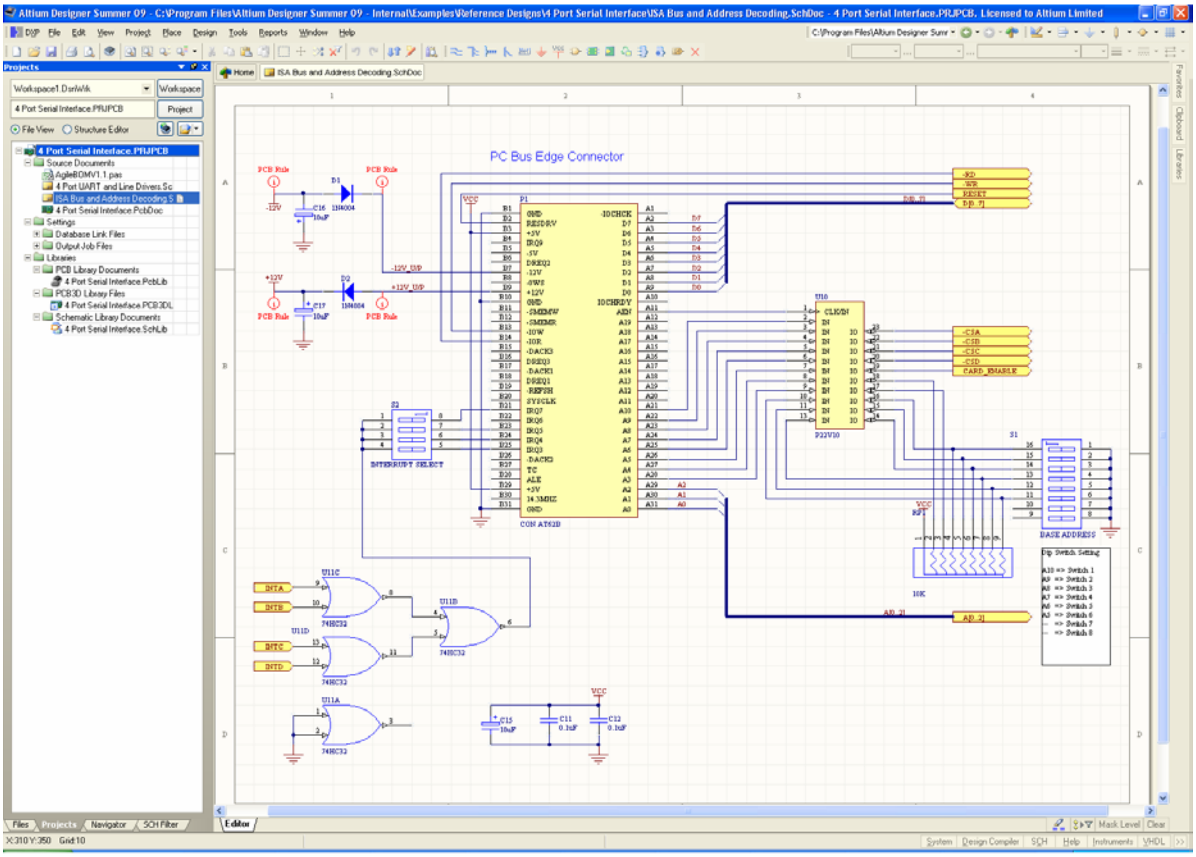



*Figure 1. Schematic Editor Workspace*

In this section, we will explore the basics of working in the Schematic Editor.

- If not already open, open the following project: `4 Port Serial Interface.PrjPcb`, found in the `\Altium Designer Summer 09\Examples\Reference Designs\4 Port Serial Interface` folder (as shown above in Figure 1), and then open the schematic sheet, `ISA Bus and Address Decoding.SchDoc` by double-clicking on the document name in the Projects panel.

## 3.1.1 View Commands

The View commands can be accessed from the View menu and are listed below.

| Command | Toolbar | Shortcut Key | Description |
|---|---|---|---|
| Fit Document | | VD | Display entire document |
| Fit All Objects | | VF | Fits all objects in the current document window |
| Area | | VA | Display a rectangular area of document by selecting diagonal vertices of the rectangle |
| Around Point | | VP | Display a rectangular area of document by selecting the centre and one vertex of the rectangle |
| Selected Objects | | VE | Fits all selected objects in the current document window |
| 50% | | V5 | Set display magnification to 50% |
| 100% | | V1 | Set display magnification to 100% |
| 200% | | V2 | Set display magnification to 200% |
| 400% | | V4 | Set display magnification to 400% |
| Zoom In | | VI | Zoom In around current cursor position |
| Zoom Out | | VO | Zoom Out around current cursor position |
| Pan | | VN | Re-centre the screen around current cursor position |
| Refresh | | VR | Update (redraw) the screen display |

*Table 1. View command summary*

While executing a command (when a crosshair is attached to the cursor), auto panning becomes active by touching any edge of the Design Window. Press the SHIFT key while auto panning to increase the panning speed. Auto panning speed is configured in the Auto Pan Options section of the **Graphical Editing** page in the *Preferences* dialog **(Tools » Schematic Preferences).** Auto panning can also be turned off here.

**Note** : The auto pan options apply in both the schematic and the schematic library editors. Since you work at a much higher zoom level in the library editor window area, you may find that auto panning moves the view too quickly. If this is the case you may prefer to disable auto panning and pan with the right mouse button (refer to 3.1.1.2).

The following shortcut keys provide a useful alternative for manipulating the view of the workspace. These shortcut keys can be used while executing commands.

| Keystroke | Function |
|---|---|
| END | Redraws the view |
| PAGE DOWN | Zoom out (holds the current cursor position) |
| PAGE UP | Zoom in (holds the current cursor position) |
| CTRL+PAGE DOWN | View Document |
| HOME | View pan (pan to centre the current cursor position) |
| SPACEBAR | Stops screen redraw |
| ARROW KEYS | Moves the cursor by one snap grid point in direction of the arrow |
| SHIFT+ARROW KEY | Moves the cursor by 10 snap grid points in the direction of the arrow |

*Table 2. Shortcut keys for view manipulation*

#### 3.1.1.1 Using the mouse wheel to pan & zoom

The mouse wheel can also be used to pan and zoom when in a design document.

**Panning**

Roll the mouse wheel upwards to pan upwards, and downwards to pan downwards.

Press SHIFT and roll the mouse wheel downwards to pan to the right.

Press SHIFT and roll the mouse wheel upwards to pan to the left.

**Zoom In**

Press CTRL and roll the mouse wheel upwards to zoom in.

**Zoom Out**

Press CTRL and roll the mouse wheel downwards to zoom out.

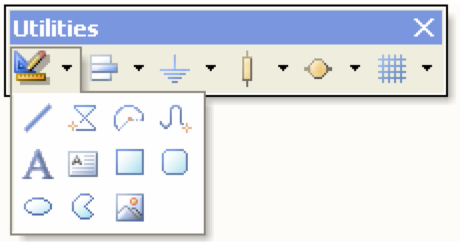
#### 3.1.1.2 Using the right mouse button to pan

Click and hold down the right mouse button (RMB) and move the cursor to pan in a design document. The hand-shaped cursor indicates you are in panning mode. Release the right mouse button to stop panning.

## 3.2 Schematic graphical objects

### 3.2.1 General

- Use the Drawing Tools available on the Utilities toolbar to place the graphical objects. Turn the Utilities toolbar on and off by selecting **View » Toolbars » Utilities**.
- Drawing toolbar functions can also be accessed through the **Place » Drawing Tools** menu, except for Paste Array (**Edit » Smart Paste**).




- When placing an item, press the tab key to edit its properties. Double-click on a placed object to modify its properties.
- When an object is selected, its handles are displayed.
- While in a command, you can select another command, without quitting the first command, provided you use a shortcut key. This powerful feature, called *re-entrant editing*, will considerably enhance your productivity.

### 3.2.2 Drawing schematic graphical objects

For an example of each graphical object, open `Graphical Objects.SchDoc` found in the `Altium Designer Summer 09\Examples\Training\PCB Training\Practice Documents` folder.

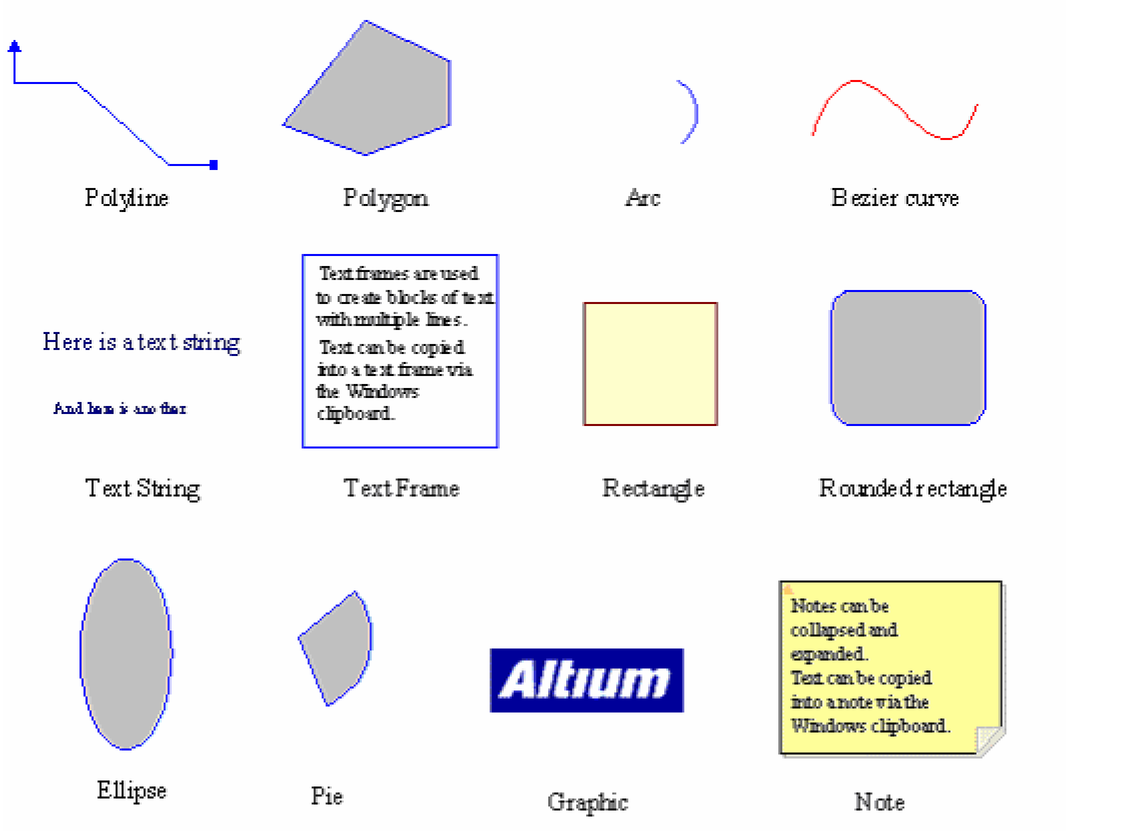



*Figure 2. Schematic graphical objects*

The placement of each of these objects is described in the following sections.

#### 3.2.2.1 Lines

To draw a line:

1. Select the **Place Line** toolbar button or **Place » Drawing Tools » Line**.
2. Click once to start the line.
3. Click to place each vertex. The BACKSPACE key deletes the last vertex placed.
4. Right-click once to end the line.
5. Right-click again to end the command.

#### 3.2.2.2 Polygons

To draw a polygon:

1. Select the **Place Polygon** toolbar button or **Place » Drawing Tools » Polygon**.
2. Click to place each vertex.
3. Right-click to end the polygon.
4. Right-click again to end the command.
5. Turn the **Draw Solid** option off in the *Polygon* dialog to draw a polygon that is not filled.

**Note**: The fill color and border color of polygons are independent.

#### 3.2.2.3 Arcs

To place a circular arc:

1. Select the Place » Drawing Tools » Arc menu command.
2. Click to place the arc centre.
3. Click to determine the arc radius.
4. Click to place the start of the arc and click to place the end of the arc.
5. Right-click to end the command.

#### 3.2.2.4 Elliptical arcs

To place an elliptical arc:

1. Select the **Place Elliptical Arc** toolbar button or **Place » Drawing Tools » Elliptical Arc**.
2. Click to place the arc centre.
3. Click to determine the arc X-radius.
4. Click to determine the arc Y-radius.
5. Click to place the first end of the arc and click to place the second end of the arc.
6. Right-click to end the command.

#### 3.2.2.5 Bezier curves

A Bezier curve is a curve of best fit between points defined by mouse clicks.

To draw a Bezier curve:

1. Select the **Place Bezier Curve** toolbar button or **Place » Drawing Tools » Bezier**.
2. Click once to place the first control point at the start of the curve.
3. Click to place the second control point.
4. Click to place the third and fourth control points.
5. Continue to click to place further control points.
6. Right-click to end the command.
7. To reshape the curve, click on one end of the curve and then move, add (INSERT key) or delete new control points (handles).

#### 3.2.2.6 Annotation (Text)

To place a line of text:

1. Select the **Place Annotation** toolbar button or **Place » Annotation**.
2. Press **Tab** to edit the contents and the font of the text. You can add special strings from the Text drop-down list as well, such as the date and document information. This topic is covered in more detail in the *Schematic Capture* training session.
3. Click to position the text.
4. Right-click to end the command.

Text strings can also be edited by selecting the string and clicking again to highlight the text.

#### 3.2.2.7 Text frames

Text frames are used to place paragraphs of text on the sheet.

To place a text frame:

5. Select the **Place Text Frame** toolbar button or **Place » Text Frame**.
6. Press **Tab** to edit the contents and properties for the text frame and click **OK**.
7. Click to position the top left corner of the frame and then click to position the bottom right corner of the frame.
8. Right-click to stop placing text frames.

The following keys apply when entering text into the frame:

| Action | Keystroke |
|---|---|
| Insert a tab | CTRL+TAB |
| Cut | SHIFT+DELETE or CTRL+X |
| Copy | CTRL+INS or CTRL+C |
| Paste | SHIFT+INS or CTRL+V |

*Table 3. Text Frame action summary*

The Cut, Copy and Paste commands apply to the Windows clipboard. The clipboard can also be used to bring text in from other applications.

#### 3.2.2.8 Notes

Notes are used to place paragraphs of text on the sheet that can be collapsed and expanded at will.

To place a Note:

1. Select **Place » Notes » Note** or **PEO** shortcut.
2. Press **Tab** to edit the contents and properties for the Note and click **OK**.
3. Click to position the top left corner of the Note and then click to position the bottom right corner of the frame.
4. Right-click to stop placing Notes.

The following keys apply when entering text into the frame:

| Action | Keystroke |
|---|---|
| Insert a tab | CTRL+TAB |
| Cut | SHIFT+DELETE or CTRL+X |
| Copy | CTRL+INS or CTRL+C |
| Paste | SHIFT+INS or CTRL+V |

*Table 4. Note action summary*

The Cut, Copy and Paste commands apply to the Windows clipboard. The clipboard can also be used to bring text in from other applications.

5. Notes can also be collapsed and expanded by clicking on the small up arrow that is in the top left hand corner of the note.

#### 3.2.2.9 Rectangles

To place a rectangle:

1. Select the **Place Rectangle** toolbar button  or **Place » Drawing Tools » Rectangle**.
2. Click to place top left corner.
3. Click to place bottom right corner.
4. Right-click to end the command.

#### 3.2.2.10 Rounded rectangles

Rounded rectangles are rectangles with rounded corners. The radius of the arcs at the rectangle corners is set in the X-Radius and Y-Radius fields in the *Round Rectangle* dialog.

To place a rounded rectangle:

1. Select the **Place Rounded Rectangle** toolbar button  or **Place » Drawing Tools » Rounded Rectangle**.
2. Press **Tab** to set the corner radii and click **OK**.
3. Click to place top left corner and click to place bottom right corner.
4. Right-click to end the command.

#### 3.2.2.11 Ellipses

Use this command to draw circles as well. To place an ellipse:

1. Select the **Place Ellipse** toolbar button or **Place » Drawing Tools » Ellipse**.
2. Click to place the ellipse centre.
3. Click to determine the ellipse X-radius.
4. Click to determine the ellipse Y-radius.
5. Right-click to end the command.

### 3.2.2.12 Pie charts

To place a pie shape:

1. Select the **Place Pie Chart** toolbar button or **Place » Drawing Tools » Pie Chart**.
2. Click to place the pie centre.
3. Click to determine the pie radius.
4. Click to place the first edge of the pie and click to place the second edge.
5. Right-click to end the command.

### 3.2.2.13 Graphic images

Graphic images with the following formats can be added to your schematic:

- .bmp, .rle, .dib
- .jpg, .tif (uncompressed)
- .wmf, .pcx, .dcx, .tga.

The file containing the graphical image can be embedded into the sheet or linked. If the image file is linked it must be transferred with the schematic file when moving the schematic from one location to another.

To place a graphic image:

1. Select the **Place Graphic Image** toolbar button or **Place » Drawing Tools » Graphic**.
2. Click to place the top left corner of the image and click to place the bottom right corner of the image.
3. Locate the file that contains the image and click **OK**.
4. To embed the image, double click on it to open the *Graphic* dialog and enable the **Embedded** option.

> **Note**: The advantage of embedding the image into the schematic or schematic template is that the image will still be visible if the file is moved. The disadvantage is that the schematic file size will be larger.

# 3.3 Schematic electrical objects

## 3.3.1 General

Schematic electrical design objects define the physical circuit you are capturing. Electrical objects include components (parts) and connective elements, such as wires, buses and ports. These objects are used to create a netlist from the schematic, which is then used to transfer circuit and connection information between design tools.

- Use the Wiring toolbar to place electrical objects.

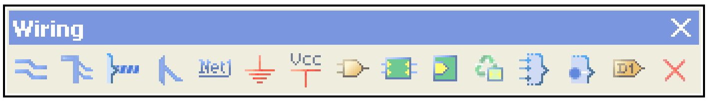


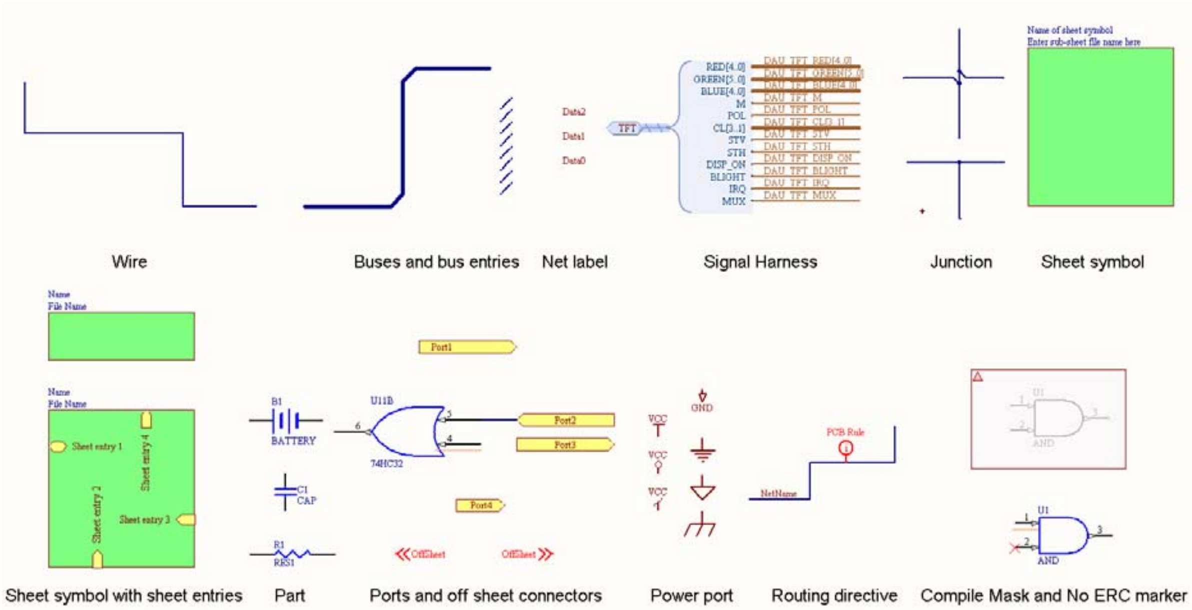

*Figure 3. Schematic electrical objects*

- All Wiring Tools toolbar functions can be accessed through the **Place** menu.
- Text in electrical objects can be over scored, typically to indicate an active low signal, by adding '\' after the character, e.g. R\ESET would display 'R' as over scored text. To overscore the entire word with a single '\' character, enable the **Single '\' Negation** option in the **Schematic – Graphical Editing** page of the *Preferences* dialog.

In the following sections, the use of each electrical object is explained.

## 3.3.2 Summary of Electrical Objects

For an example of each electrical object, open `Electrical Objects.SchDoc` found in the `Altium Designer Summer 09\Examples\Training\PCB Training\Practice Documents` folder.



#### 3.3.2.1 Wires

- Select the **Place Wire** toolbar button or **Place » Wire**.
- Wires are used to represent an electrical connection between points.

> **Note**: Be careful to use the **Place » Wire** command, and not use the **Line** command by mistake.

- Press the *shift* + *SPACEBAR* to change the placement mode. There are Four placement modes as follows:
  - 90 degree
  - 45 degree
  - any angle
  - auto wire.
- The *BACKSPACE* key deletes the last vertex placed.
- The *SPACEBAR* is used to change the current placement modes start and end point. This only works for 90 and 45 degrees modes.
- A wire end must fall on the connection point of an electrical object to be connected to it. For example, the end of a wire must fall on the hot end of a pin to connect.
- Wires have the Auto Junction feature, which automatically inserts a Junction object if a wire starts or ends on another wire or runs across a pin.
- To add more vertices click and hold the left mouse button on a wire and press the *INSERT* Key.
- To remove a vertex, select the vertex, right click and select **Edit Wire Vertex NO.** Wire properties dialog comes up in vertices mode and click **Remove**.

#### 3.3.2.2 Buses

- Buses are used to graphically represent how a group of related signals, such as a data bus, is connected on a sheet. They are also used to collect together all the signals belonging to a bus on a sheet and connecting them to a port to enter or leave a sheet. In this instance, they must have a net label of this format: D[0..7].
- Select the **Place Bus** toolbar button or **Place » Bus**. Place a bus line in the same manner as placing wires, i.e. press SPACEBAR to change placement mode and press the BACKSPACE key to delete the last vertex placed.
- Buses can only represent connections to ports and sheet entries and only at their end points.
- The same shortcut commands used for wire mode apply to the Bus placement mode.

#### 3.3.2.3 Bus Entries

Bus entries are used to represent a connection between a wire and a bus.

To place a bus entry:

1. Ensure that an appropriate snap grid is set so that connections will be made.
2. Select the **Bus Entry** toolbar button or **Place » Bus Entry**.
3. Press the SPACEBAR to rotate the bus entry.
4. Click once to position the bus entry.
5. Right-click to stop placing bus entries.

> **Note**: The use of bus entries is optional. Many users prefer to place a 45-degree wire.

### 3.3.2.4 Signal Harness

Wires are used to represent an electrical connection between points. Buses are used to represent a group of related signals, which are identified by a specific naming convention.

Signal Harnesses allow the logical grouping of multiple signals, including wires, buses and other signal harnesses. This group can be treated as a single entity, which can be used across the entire project.

There are 4 key elements to a Signal Harness system:

- Signal Harness
- Harness Connector
- Harness Entry
- Harness Definition File

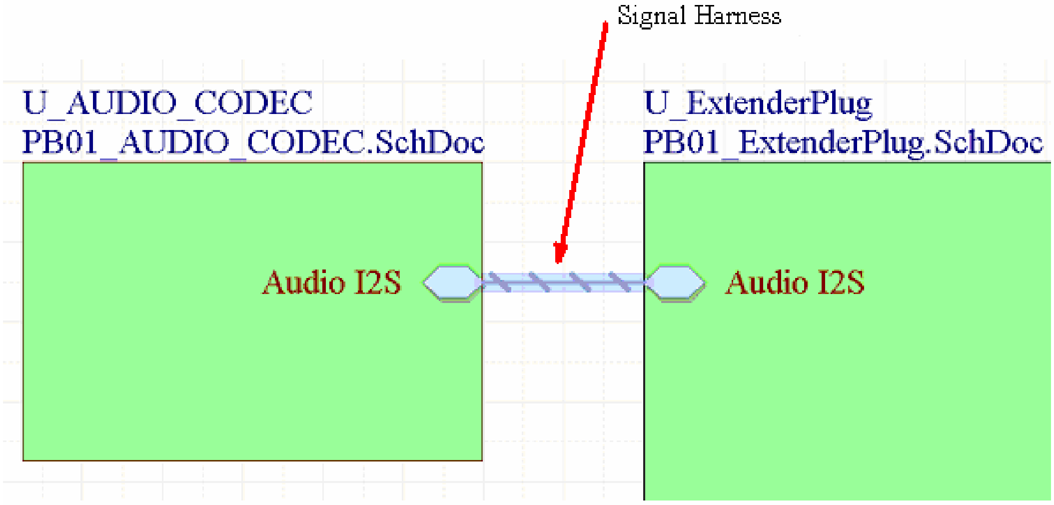



*Figure 4. A Signal Harness can carry any combination of wires, buses and other signal harnesses.*

**Note**: Hover the mouse over a harness to detail all contained nets, buses and sub-harnesses.

### 3.3.2.5 Harness Connector

This object is used to group together the various signals that form a Signal Harness. The harness connector will include a harness entry for each net, bus and sub-harness being bundled into the Signal Harness.

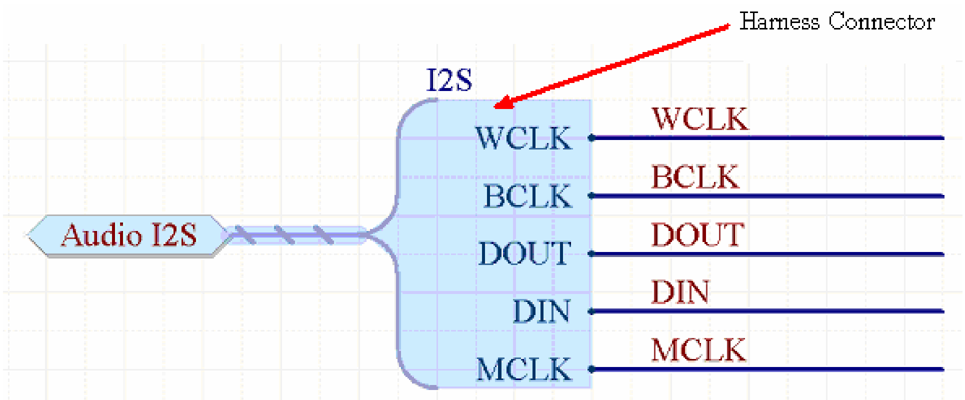



*Figure 5. The Harness Connector groups the signals in the Signal Harness. It is defined by the Harness Type, which in this example is* `I2S`

**Note**: Use the Smart Paste command to convert selected Net Labels into Harness Connectors.

### 3.3.2.6 Harness Entry

For a net, bus or sub-harness to be included in a Signal Harness, it must connect to a Harness Entry in a Harness Connector.

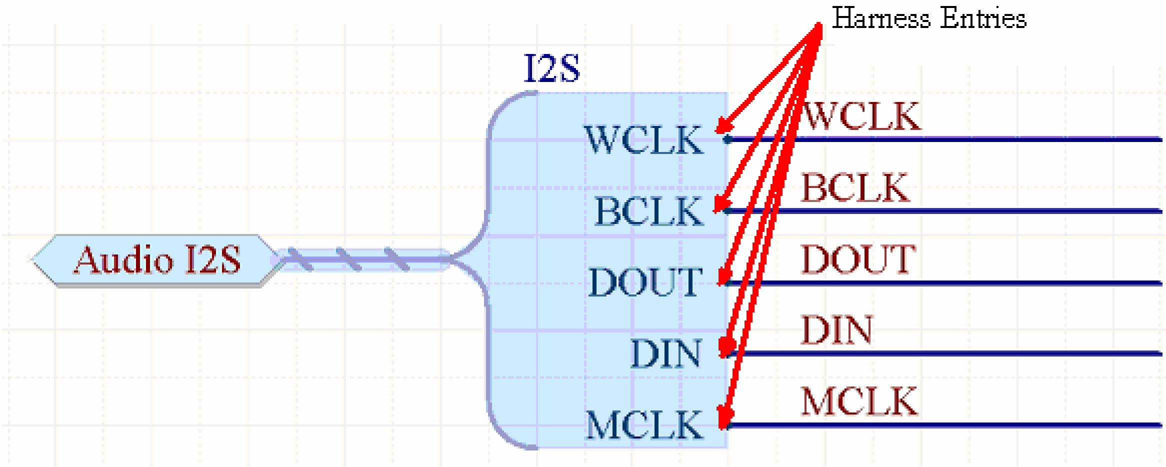



*Figure 6. The Harness Entries provide the connection point for each of the nets feeding into the Signal Harness. In this case, the Harness Entries are WCLK, BCLK, DOUT, DIN and MCLK*

### 3.3.2.7 Harness Definition

For each Signal Harness included in a design, there is a Harness Definition file. Each Harness Definition comprises of a Harness Type (I2S in the image below), and the related Harness Entries (in this case: BCLK, DIN, DOUT, MCLK, WCLK).

The Harness Definition files (*.Harness) are created and managed automatically by Altium Designer. Harness Definition Files appear in the Projects panel under the `Settings\Harness Definition Files` sub-folder.

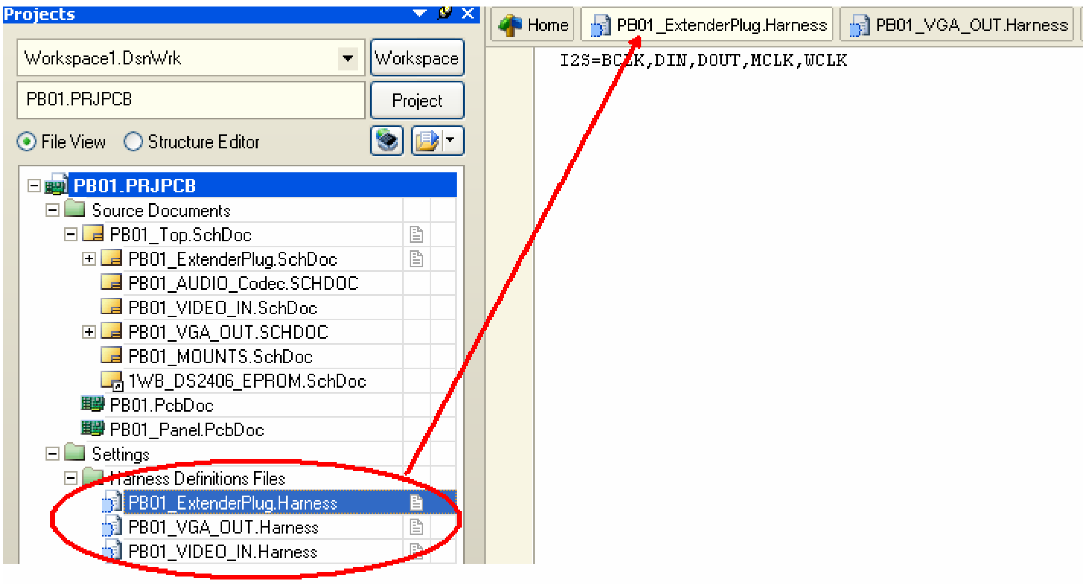



*Figure 7. The Harness Definition Files are text files that are located under Settings in the Projects Panel. These contain textual representations of Signal Harnesses and their respective Harness Entries. They can be identified by their extension* `*.Harness`

**Note**: It is possible to create a Signal Harness without the Harness connector. In this case the Harness Definition file must be created and managed manually.

### 3.3.2.8 Net Labels

- A net label is used to make a net easily identifiable and also provides a method of connecting pins belonging to the same net without placing a wire.
- A connection is made between all wires with identical net labels on a sheet. In some cases, all wires with identical net labels in a project will be connected together. Hierarchies will be explored in more detail during the *Schematic Capture* training session.
- All net labels on a net must be identical.
- The net list generator will convert all net labels to upper case.
- To associate a net label with a wire, place it so that its reference point (bottom left corner) falls on the wire.
- The electrical grid is active when placing net labels.
- If the last character in a net label is a number, it will increment when subsequent net labels are placed.

To place a net label:

1. Ensure that an appropriate snap grid is set so that connections will be made.
2. Select the **Place Net Label** toolbar button  or **Place » Net Label**.
3. Press **Tab** to edit the net label text. The *Net Label* dialog displays.

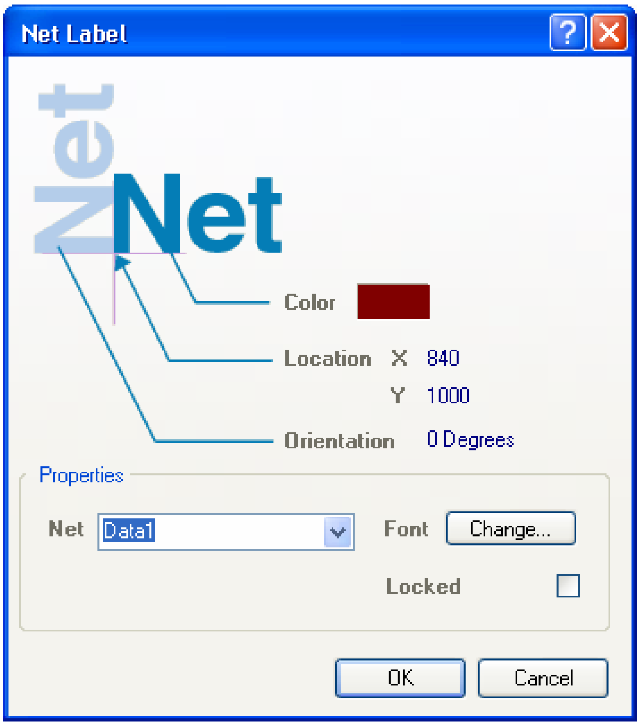



*Figure 8. Net Label dialog*

4. Click on the down arrow in the Net field to display the names of nets already defined on the sheet, or type in the new net name. Click OK.
5. Press spacebar to rotate the net label.
6. Click once to position the net label.
7. Right-click to stop placing net labels.

**Note** : During net label placement, hover the mouse over existing text on a schematic and press the **Insert** shortcut key, the net label will pick up that string value.

### 3.3.2.9 Power Ports

- All power ports with the same Net property in a project will be connected.
- To connect to a power port, make sure that a wire falls on the end of the power port pin.
- The style of the power port only changes its appearance. It does not affect the connectivity as this is established through the Net property.
- Power ports will connect to hidden pins with the same name throughout the design, regardless of the net identifier scope used.
- The Power Port buttons on the Wiring toolbar will only place a single power port. To change this behavior and place multiple ports, edit the button and add the parameter Repeat=True.

To place a power port:

1. Select either the GND  or VCC  **Power Port** toolbar buttons, or **Place » Power Port**.
2. Press TAB to edit the power port properties for a net name other than GND or VCC.

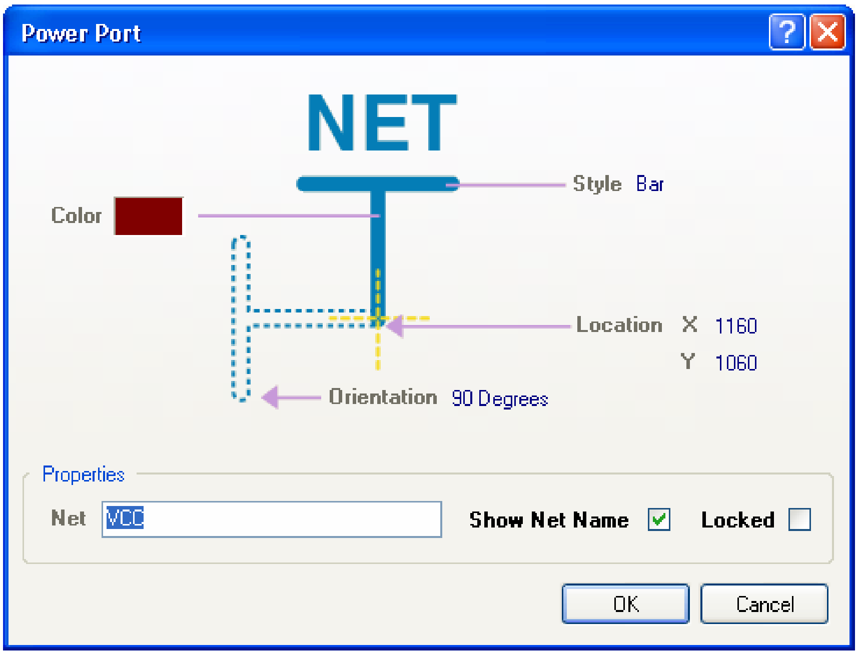



*Figure 9. Power Port dialog*

3. Click to position the port. Right-click to stop placing power ports.

### 3.3.2.10 Parts

- When **Place » Part** (PP) is selected or you click on the Place Part toolbar button , the *Place Part* dialog is displayed. You can enter the name of the component in the Lib Ref field or you can click on the **Browse** button (…) to locate the part by browsing and adding the required library.

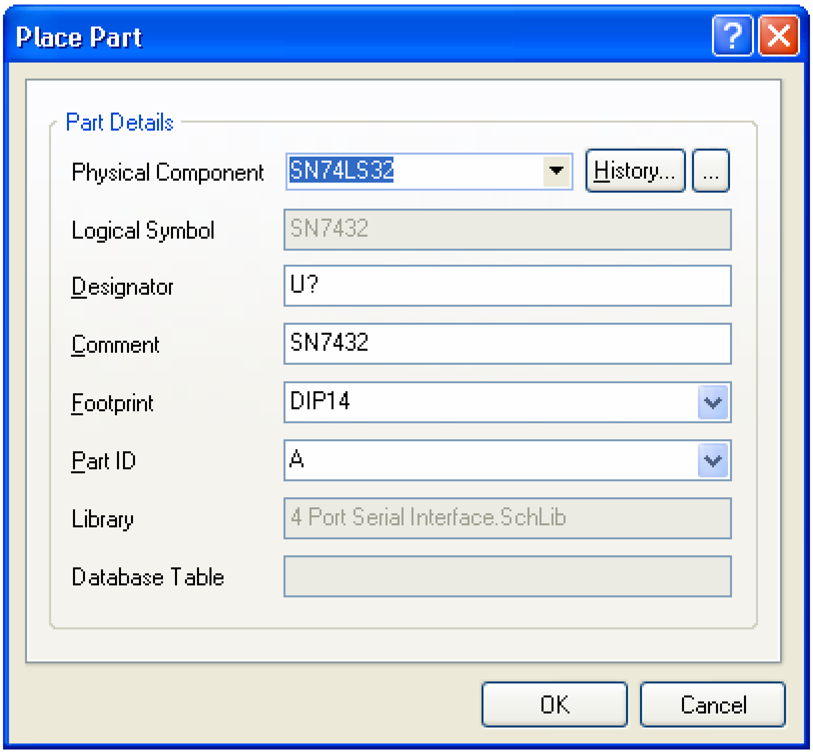
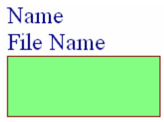



*Figure 10. Place Part dialog*

- Parts can also be placed using the **Place** button in the Schematic Library Editor.
- When placing parts, use a snap grid that will cause the pin ends to fall on a grid point, e.g. 10. Press **G** to cycle through the snap grid settings of 1, 5 and 10.

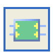
### 3.3.2.11 Sheet Symbols

- Sheet symbols are used when you wish to break the design into a number of sheets.
- A sheet symbol must be placed for each schematic document in the project.




- The sheet symbol name is a descriptive name for the sheet.
- The sheet symbol filename must be the document name of the schematic document it represents. All sheets in a project should be in the same directory.
- When changing the size of the sheet symbol, make sure the edges of the sheet symbol fall on the snap grid to ensure connection between wires and sheet entries.

To place a sheet symbol:

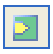
1. Select the Sheet Symbol toolbar button  or Place » Sheet Symbol (PS).
2. Press **Tab** to edit the sheet symbol name and sheet symbol file name.
3. Click to place the top left corner.
4. Click to place the bottom right corner.
5. Right-click to stop placing sheet symbols.

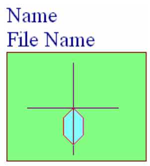
**Note** : Sheet symbols, complete with sheet entries, can be created using the command **Design » Create Sheet Symbol From Sheet or HDL**. The **Synchronize Sheet Entries and Ports** command (Design menu, or right-click on Sheet Symbol menu) can also be used to synchronize the entries on a sheet symbol to the ports on the sheet below. This is covered in *Module 5 - Multi-Sheet Design*.

### 3.3.2.12 Sheet Entries

- Sheet entries are used in the sheet symbols if you are doing the design in a true hierarchical structure, with nets interconnecting the sheet symbols.
- Each sheet entry needs a matching port on the sub-sheet.
- Use **Sheet Symbols / Port Connections** as the Net Identifier Scope when creating netlists or running the Electrical Rules Checker.

To place a sheet entry:

1. Select the Place Sheet Entry toolbar button  or Place » Add Sheet Entry (PA).
2. Click on the sheet symbol that the sheet entry is for and the sheet entry symbol appears within the sheet symbol box.




3. Press **Tab** to edit the sheet entry properties.
4. Click on the down arrow in the name field to list all the Sheet Entry names used on the current sheet.
5. Position the sheet entry on any side of the sheet symbol and click.
6. Right-click to stop placing sheet entries.

> **Note**: Altium Designer can insert sheet entries automatically, by enabling the **Place Sheet Entries automatically** option in the **Graphical Editing** page of the *Preferences* dialog. An entry is placed automatically when the wire is terminated on the edge of a sheet symbol, if the net currently being wired can be identified.

### 3.3.2.13 Ports

- Ports provide a method of forming connections from one sheet to another sheet.
- Click on the down arrow in the Name field to list all the Port names defined on the sheet.
- The port I/O Type is used by the ERC when checking for connection errors.
- The port style only changes the appearance of the port.

To place a port:

1. Select the Port toolbar button  or Place » Port.
2. Press **Tab** to edit the port properties.

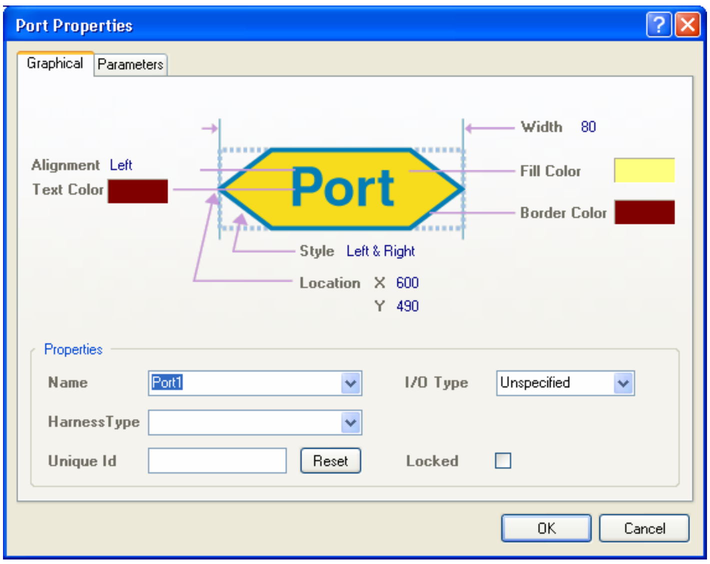



*Figure 11. Port Properties dialog*

3. Press the Spacebar to rotate or X and Y to flip.
4. Click to position one end of the port. Drag the mouse to set the port length and click to finish the port.
5. Right-click to stop placing ports.

> **Note** : The Port Direction can be set automatically, based on the Port I/O type and where the wire touches the port. Enable the **Port Direction** option in the **General** page of the *Preferences* dialog.

### 3.3.2.14 Off Sheet Connectors

Off Sheet Connectors are used to connect nets across multiple schematic sheets that are descended from sheet entries of the same parent sheet symbol. To successfully connect a particular net across two or more sheets, the Off Sheet Connectors on each sheet must be assigned to the same net.

1. Select **Place » Off Sheet Connector** (**PC**).
2. Press **Tab** to edit the Off Sheet Connector properties.
3. Click to place the Off Sheet Connector. Right-click to exit placement mode. 


> **Note** : The primary function of Offsheet connectors is for translation to and from Orcad Schematic Capture. If you intend to use Altium Designer to draw up a schematic, but the end schematic is required in Orcad format, then Offsheet connectors can be used.

### 3.3.2.15 Junctions

- The software automatically adds an auto-junction at valid connection points, including T joins, and when a wire crosses the end of a pin. Auto-Junctions are not added at crossovers.
- Manual junctions can be used to force a junction at a crossover, select **Place » Manual Junction** (**PJ**). The crosshair cursor appears with a junction marker (red dot) on it. Click to place the junction marker.
- The Auto-Junction display is set in the **Compiler** page of the *Preferences* dialog.

### 3.3.2.16 Compile Mask

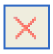
- A Compile Mask hides, or masks, all components and wiring underneath it, from the Altium Designer compiler. Clicking the small triangle in the upper left will collapse the Mask, returning those components and wiring to the circuit. Because they are masked from the compiler, those components are ignored during an update to PCB. Use Compile Masks to mask out sections of the design that are work in progress, to perform partial updates to the PCB, or to mask simulation sources (which are needed for simulation, but do not exist on the PCB).

To place a compile mask:

1. **Place » Directives » Compile Mask.**
2. Press Tab to edit the compile mask.
3. Click to place the top left corner.
4. Click to place the bottom right corner.
5. Right-click to stop placing compile masks.

### 3.3.2.17 No ERC Marker

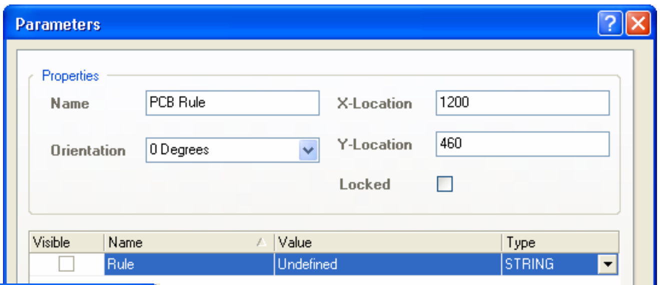
- Placing a No ERC symbol on a node in the circuit suppresses any report warnings and errors that may be generated when compiling the schematic. These markers can also be suppressed for printing.

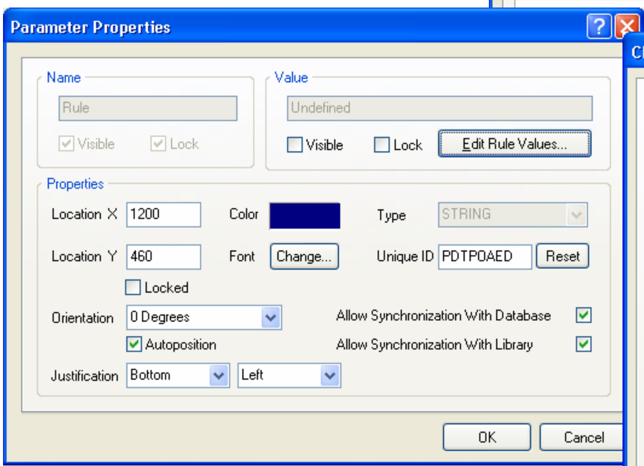
- Select the **Place No ERC** toolbar button  or **Place » Directives » No ERC.** Click to place the No ERC marker on a pin or existing ERC marker. Right-click to exit placement mode.

### 3.3.2.18 Parameter Sets

Some objects do not support directly specifying parameters, such as wires and buses. To add a parameter to these attach a Parameter set object. PCB design rules can also be added to wires (nets) or buses using parameter set objects, these rule specifications are then passed to the PCB during synchronization.


1. Select **Place » Directives » PCB Layout**. The cursor appears with a directive symbol attached.
2. Press TAB to edit the PCB layout directive in the *Parameters* dialog. Edit in the parameters in the *Parameters Properties* dialog by clicking on the **Edit** button.
3. Click on the **Edit Rule Values** button.
4. Select a rule, click **OK** and set the values accordingly, and press **OK**. Keep pressing **OK** until all dialogs are gone.
5. Position the directives symbol so that its hot point (the end of the stem) touches the wire or bus. Click to place it.
6. Right-click to stop placing routing directives.
7. To add more than 1 rule to a directive, edit the directive and use the **Add As Rule** button.



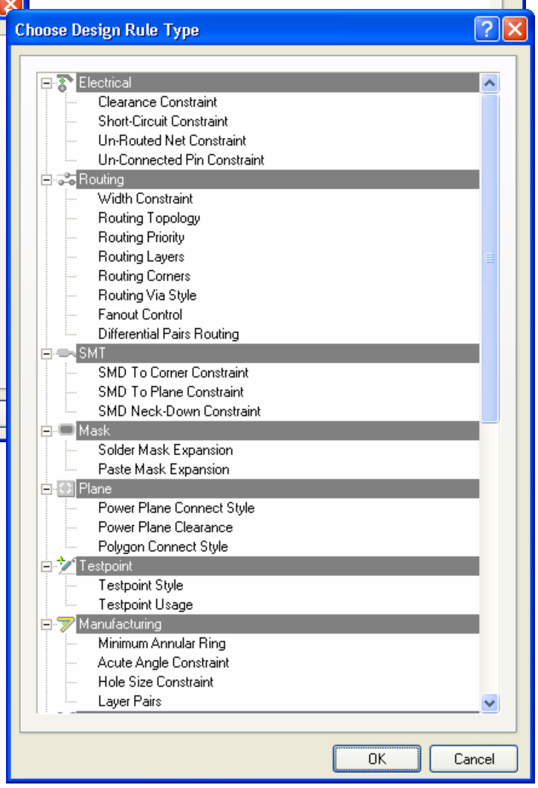




*Figure 12. The parameter set object is a container for adding parameters to objects, including PCB design rule definitions.*

**Note**: PCB routing directives are Parameter Set objects. The separate menu entry to place PCB Routing Directives has been retained for user compatibility with earlier versions.

## 3.3.3 Exercise – Schematic graphical objects

1. Open `Graphical Objects.SchDoc` found in the `Altium Designer Summer 09\Examples\Training\PCB Training\ Practice Documents` folder and experiment with placing each of the drawing objects in the space provided.
2. Select each object and observe the handles.
3. Investigate the effect of moving handles.
4. Insert a new vertex into a polyline object, and then remove it.
5. Double-click on some of the objects to display and modify their properties.
6. Close the sheet without saving.

## 3.3.4 Exercise – Schematic electrical objects

1. Open `Electrical Objects.SchDoc` found in the `Altium Designer Summer 09\Examples\Training\PCB Training\ Practice Documents` folder and experiment with placing each of the schematic electrical objects.
2. Select each object and observe the effect of moving the handles.
3. Double-click on some of the objects to display and modify their properties.
4. Close the sheet without saving.

## 3.3.5 Favorites Panel

Like a web browser, a list of favorite documents can be stored in this panel for future reference. A thumbnail of the view as well as title and comment is stored. For Altium Designer documents the zoom level and location is included.

Favourites can be tied to the project itself making it a useful mark up tool for design collaboration. Project favourites are stored in a 'ViewsOf' folder in the same folder as the project file.

- Open the Favourites panel from the panel controls down the bottom right by clicking on **System » Favorites**.
- The contents may be divided into folders. A new folder can be created from the right click menu
- To add the current view to a folder use Add Current Document View from the right click menu.
- To recall a view simply double click the entry in the list
- The size of the thumbnails is configured in the System preferences in the View section.

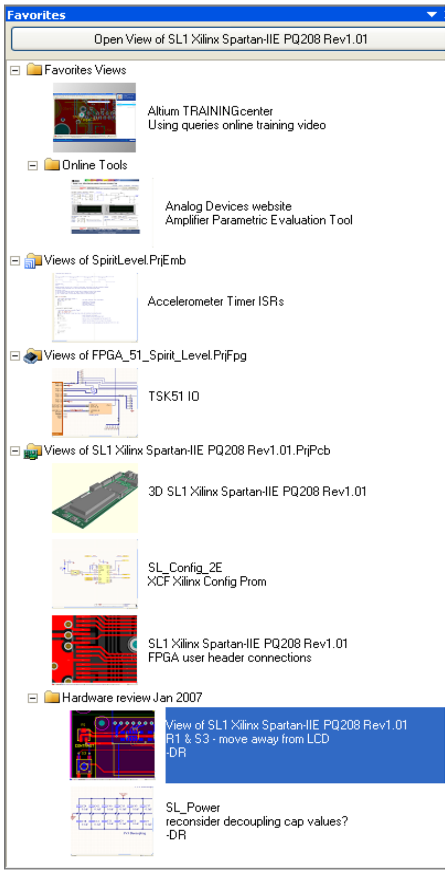



*Figure 13. Favorites panel*

## 3.3.6 Snippets Panel

The Snippets Panel provides a way to store portions of a design for later reuse. The panel will store sections of schematic, PCB layout and source code.

- Open the Snippets panel from the panel controls down the bottom right by clicking on **System » Snippets**
- The contents can be divided into folders. These are just regular Windows folders and the location can be configured from the Snippets Folders button. Multiple folders can be defined; using a shared network resource will let you share a snippets library amongst an entire design team.
- To create a snippet select the objects in the PCB, schematic, or code editor and then from the right click menu select **Snippets » Create Snippet from selection**. File the snippet away with a title and comments.
- To Place a snippet select it in the panel and then click the Place button at the top.
- If you reset selected component designators before using them to create a snippet, you can avoid duplicate designators when the snippet is placed.

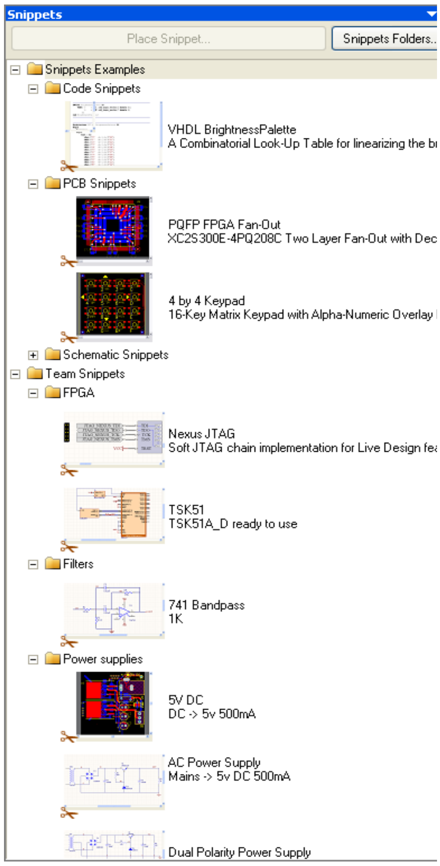



*Figure 14. Snippets panel*

## 3.3.7 Clipboard panel

The Clipboard Panel provides a way of recalling earlier Copy actions. The panel stores copied sections of schematic, PCB layout and source code.

- Open the Clipboard panel from the panel controls down the bottom right by clicking on **System » Clipboard**
- The clipboard panel has the added advantage of being able to read the Windows clipboard, allowing data to be transferred from other programs into Altium Designer. To enable this, turn off the **Monitor clipboard content within this application only** option in the **Systems – General** page of the *Preferences* dialog.
- Clipboard panel data is only available for the current editing session, unlike the Snippets panel which stores data on the hard drive.

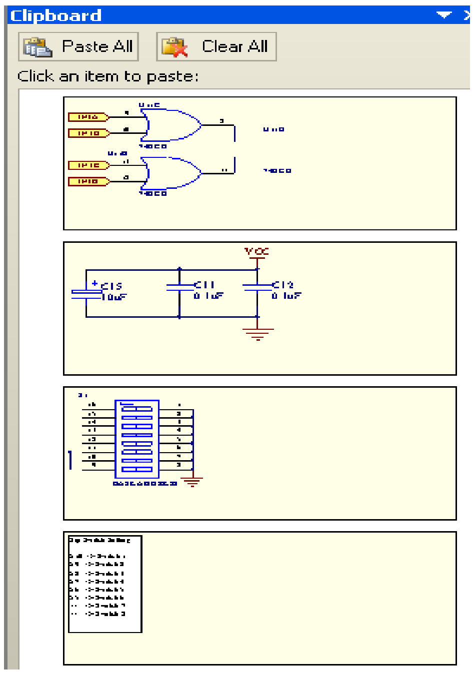



*Figure 15. Clipboard Panel*

## 3.3.8 Selection

The Schematic Editor provides selection capabilities that are similar, although not identical, to selection in other Windows applications.

Below are some key points about selection in the Schematic Editor:

- The main use of selection is to nominate objects for a clipboard operation, i.e. which objects will be moved or copied to the clipboard when the Cut or Copy commands are invoked.
- Once objects are on the clipboard, they can then be pasted elsewhere onto the current schematic or into another schematic, or to another Windows application which supports the Windows clipboard.
- Selection is not cumulative. The selected object deselects when you click on another object.
- Hold the SHIFT key to select multiple objects.
- Press DELETE to delete all selected objects.

To select an object you can use:

| Keystroke | Function |
|---|---|
| Click and drag | Select all objects enclosed by drag area |
| SHIFT+click on object | Select an object (on a selected object, this will de-select it) |
| **Edit » Select** menu (**S**) | Select Inside Area, Outside Area, All, Net or Connection |
| Select Inside Area | button on the Main toolbar |

*Table 5. Select command summary*

Selected objects can be:

| Function | Keystroke |
|---|---|
| Cut, copied, pasted or cleared | Using the Edit menu commands |
| Moved | Click-and-hold on any selected object |
| Moved or dragged | Using the **Edit » Move** menu commands (M) |
| Aligned | Using the **Edit » Align** menu commands (A) |
| Deleted | Using DELETE |

*Table 6. Selected object command summary*

**Note:** Selection in earlier versions of Altium software differed from other Windows applications in that selection was persistent – selected objects always remained selected until you deliberately de-selected them. Altium Designer includes an option to mimic that behavior, if you disable the **Click Clears Selection** option in the **Schematic – Graphical Editing** page selected objects will remain selected until you deliberately clear the selection. It is recommended you try the standard behavior first, and if you need to 'hold' the selection state of a set of objects, use the Selection Memory feature (see 3.3.8.2).

**Note:** If you find that you keep inadvertently selecting certain objects, you can make them harder to select by enabling the **Shift Click to Select** option in the **Schematic – Graphical Editing** page of the *Preferences* dialog. Click the **Primitives** button to configure which objects require Shift to be held during selection.

#### 3.3.8.1 Selection hints

- Before starting a selection, it is a good idea to de-select all objects first.
- Only items that fall completely inside the selection area are selected.
- The selection color is set in the **Graphical Editing** page of the *Preferences* dialog.
- The Move menu allows you to move selections:
  - without maintaining connectivity (move)
  - Maintaining connectivity (drag).
- The S key pops up the Select menu.
- The X key pops up the Deselect menu.

#### 3.3.8.2 Selection memory

Eight selection memories are available in the Schematic and PCB editors, which can be used to store and recall the selection state of up to eight sets of objects on the schematic or PCB. Select the objects you want to remember using any of the methods described above in Table 5, and then store them for quick recall later.

The following selection memory options are available:

- Store in memory (CTRL + number 1 to 8)
- Add to memory (SHIFT + number 1 to 8)
- Recall from memory (ALT + number 1 to 8)
- Recall and Add from memory (SHIFT + ALT + number 1 to 8)
- Apply memory as a workspace filter (SHIFT + CTRL + number 1 to 8).

You can also access the selection memories using the **Edit » Selection Memory** sub-menu.

Alternatively, use the Selection Memory control panel that is opened by clicking the  button next to the Mask Level button ( bottom right of the workspace), or pressing **CTRL+Q**. Click on a **STO** button to store a selection or **RCL** to recall a selection. The filtering options at the bottom of the control panel will determine how the selection is displayed.

To prevent accidentally overwriting a selection memory, enable the Confirm Selection Memory Clear option in the **Schematic – Graphical Editing** page of the *Preferences* dialog. Selection Memory locations can be locked from being overwritten by checking the **Lock** checkbox associated with that selection memory.

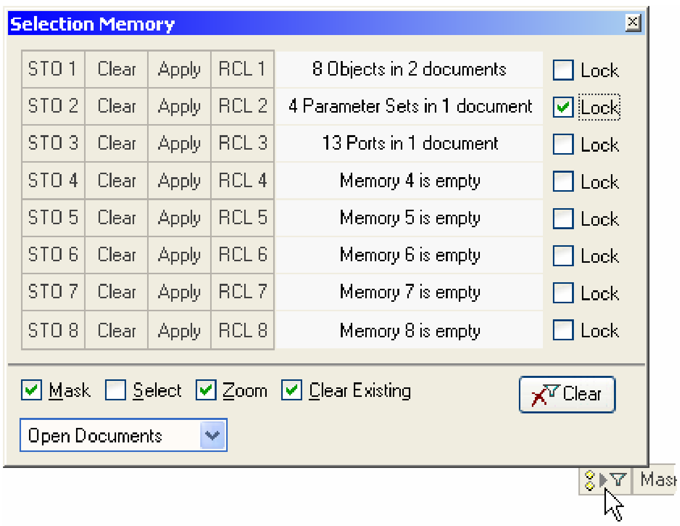



*Figure 16. Selection Memory control panel*

## 3.3.9 Masking

As well as regular selection in Altium Designer, there is also the concept of masking were objects are masked out from view and manual selection. There are many ways to apply a mask to Schematic objects which includes, Find similar objects, navigator panel, messages panel, select connection or Autofocus.

When a mask is set the schematic objects appear in a transparent like mode as shown in Figure 17. When in this mode, you can run **Edit » Select » All** and only the objects that are not masked out will be selected.

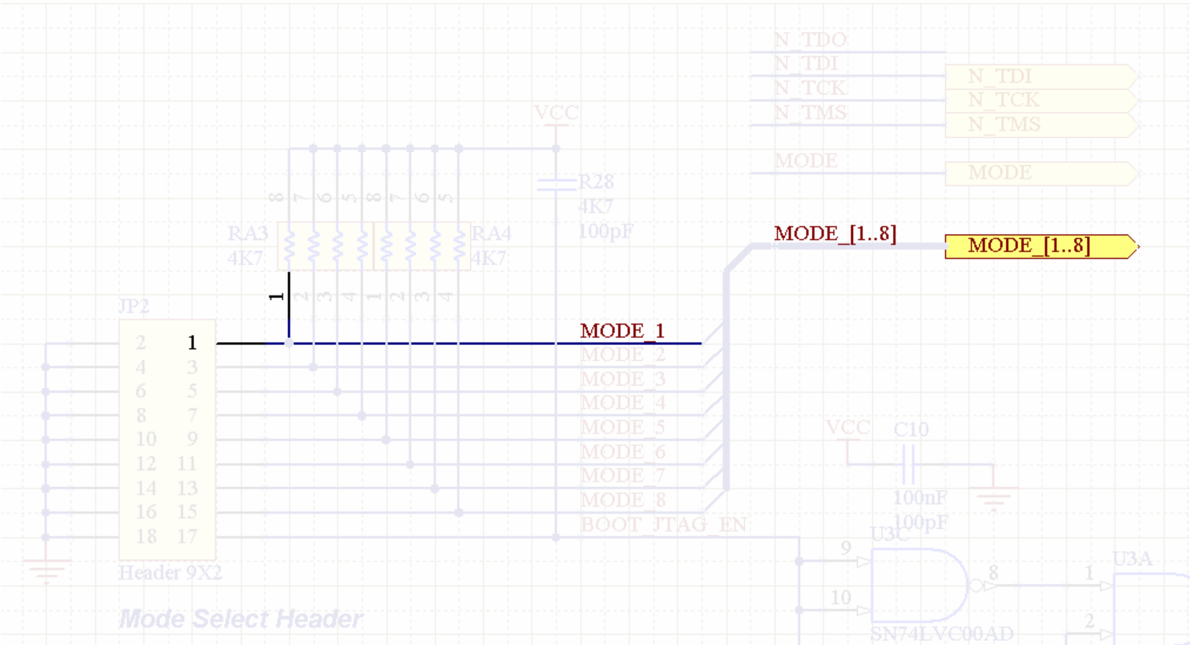


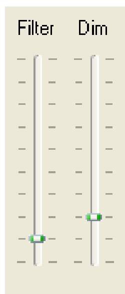

*Figure 17. A schematic showing masked mode.*

### 3.3.9.1 Clearing a mask

To clear a mask in schematic click on the *Clear* Button 

located in the bottom right of the Altium Designer screen.

### 3.3.9.2 Changing the mask level




To change the mask level click on the *Mask Level* Button 

located in the bottom right of the Altium Designer screen. Once clicked, a small popup appears were the mask level is set with sliders as shown to the right. Move the filter slider down to mask more or up to mask less.

## 3.3.10 Other mouse actions

The mouse operations listed below are universal throughout the Schematic Editor and should be used in preference to menu commands:

| Keystroke | Function |
|---|---|
| Click-and-hold on object | Move an object |
| CTRL+click on object | Drag an object whilst maintaining connectivity. Press the SPACEBAR to change mode. |
| Double-click on object | Edit an object's properties |
| Left-click | ENTER |
| Right-click | ESC |

*Table 7. General mouse shortcut summary*

While an object is on the cursor, the following keystrokes can be used:

- SPACEBAR to rotate
- X key to flip around the vertical axis
- Y key to flip around the horizontal axis.

## 3.3.11 Multiple objects at the same location

When working in the Schematic Editor, the situation sometimes occurs where a click to perform an operation is made where there are multiple objects. In this situation, the Schematic Editor pops up a menu listing all the objects it has detected at the location of the click. You can then select the object you wish to operate on from this menu.

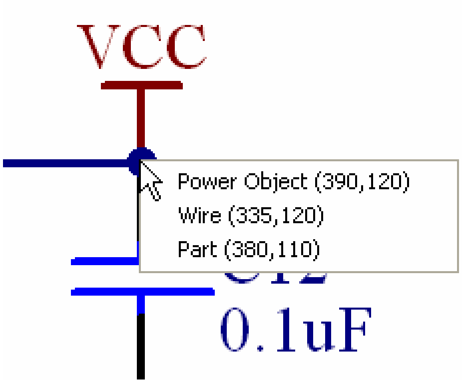



*Figure 18. Menu listing objects at mouse click point*

## 3.3.12 Smart Paste

The Schematic Editor's Smart Paste feature allows you to transform the copy of the selected objects into other objects as you paste them. For example you could copy a selection of Net Labels, and Smart Paste them as Ports, or the selected Sheet Entries could be pasted as Ports+Wires+Net Labels, all in a single paste action.

- Create a set of selected objects in the normal way, for example net labels, then copy them to the clipboard (**Ctrl+C**).
- Choose **Edit » Smart Paste** from the menus (**Ctrl+Shift+V**), to display the Smart Paste dialog, as shown in Figure 19.

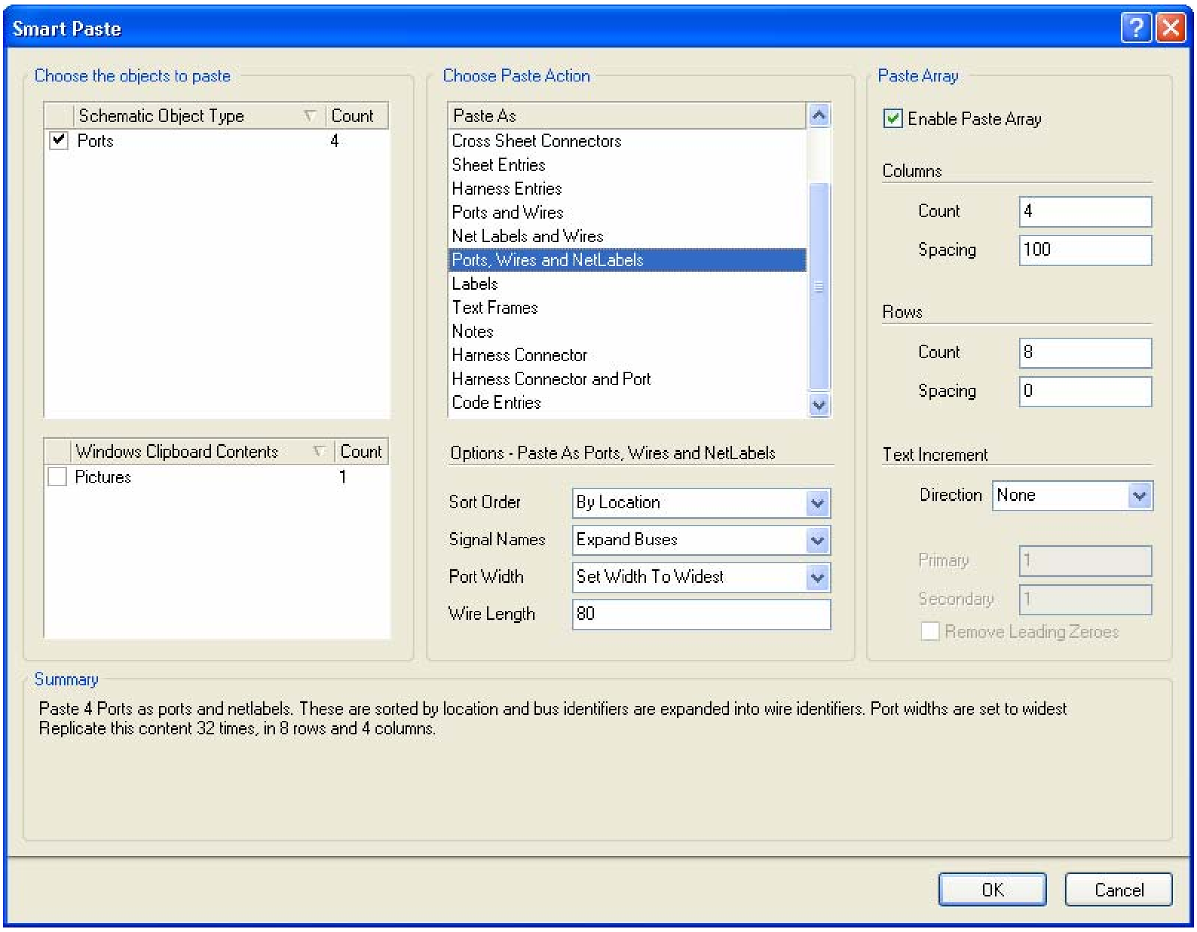



*Figure 19 Smart Paste dialog*

### 3.3.12.1 Choose the objects to paste section

This section displays a list of all the objects in the clipboard, grouped by their type. The check box allows you to control which set of objects you would like to paste. Before you can paste you also need to select a **Paste Action**, this determines how your selected objects will be placed onto your schematic sheet.

### 3.3.12.2 Choose Paste Action section

Before you can paste you also need to select a **Paste Action**. This determines how your selected objects will be transformed as they are placed onto your schematic sheet. The **Paste As** action called **Themselves** is a standard paste operation. The other options allow you to transform the source object into a different object, or collection of objects, when pasting.

The possible transformations include:

- Ports, Sheet Entries or Net Labels can be transformed into equivalent ports, sheet entries, net labels, or one text frame/note or a port and net label set per object (with wires).
- Label, Text Frame or Notes can be transformed into Label, Text Frame or Note.
- Windows Clipboard Text can be transformed into net labels, ports, sheet entries, labels, text frames, notes, or a port and net label set per object (with wires).
- Windows Clipboard Graphics can be transformed into an image.

#### 3.3.12.3 Paste Array section

Enable this option to copy your selected objects as a two-dimensional array. The total number of copies you will create are the number of columns times the number of rows. On clicking Ok, you will be prompted to select a start location on the document, where the array will be inserted. Simply position the cursor at the desired location and click.

**Columns**

This specifies the number of columns you want in your paste array. Each column will be separated by the Column Spacing setting. Enter positive or negative values for spacing, to determine whether the array will be pasted to the right or left respectively for horizontal placement, or upwards or downwards respectively for vertical placement.

**Rows**

This specifies the number of rows you want in your paste array. Each row will be separated by the Row Spacing setting. Enter positive or negative values for spacing, to determine whether the array will be pasted to the right or left respectively for horizontal placement, or upwards or downwards respectively for vertical placement.

**Text Increment**

Select what method you would like to use to increment strings (such as designators) on the copies you are pasting. You can select from the following options:

- **Direction**
  - **None –** do not increment, meaning each copy will have the same strings
  - **Horizontal First –** this will increment strings increasing the value of a string from its predecessor by the Primary amount. The successor string to increment is found by finding the next string in the sequence immediately to the right. Once a row has been re-sequenced, move to the start of the next row above. Pins can also be incremented using the Secondary setting.
  - **Vertical First –** this will increment strings increasing the value of a string from its predecessor by the Primary amount. The successor string to increment is found by finding the next string in the sequence immediately above. Once a column has been re-sequenced, move to the start of the next column to the right. Pins can also be incremented using the Secondary setting.
- **Primary**
  - Strings are incremented/decremented from its predecessor by the Primary amount. Pins can also be changed using the Secondary setting.
- **Secondary**
  - Strings are incremented/decremented from its predecessor by the Primary amount. Pins can also be changed using the Secondary setting.

### 3.3.13 Modifying Polylines

All line objects that have multiple segments are also referred to as polylines – this includes lines, wires and buses. Techniques for modifying a polyline include:

- **Adding or removing a vertex –** To add a new vertex, click once to select the polyline object and display the existing vertices, click and hold anywhere along a segment (the cursor will be a double arrow), press the INSERT key, then move the mouse to position the new vertex. To remove a vertex click and hold on the vertex, and press the DELETE key.
- **Moving a segment in the polyline –** Click once to select the polyline, click and hold on the segment, and move it to the new location.

- **Moving a vertex –** click once to select the polyline, then click and hold on the vertex to move it. Note that when you move an end vertex you can also move the cursor to add a new segment. To prevent this occurring hold the ALT key as you move the end vertex.
- **Move an entire polyline –** while the polyline is *not* selected, click and hold on it and move the mouse to relocate it.

## 3.3.14 Font Management

Fonts are controlled via the *Font* dialog.

- If an object supports direct font editing, you will be able to access the *Font* dialog when you double-click on the string. This dialog is displayed whenever you edit text and click the font **Change** button. The default font for each object-kind is set in the **Default Primitives** page of the *Preferences* dialog.
- Changing the font for text that cannot be edited directly, such as pin names, port names and sheet text, is done via the **Change System Font** button in the *Document Options* dialog (**Design » Document Options**). This changes the system font for the active document only.

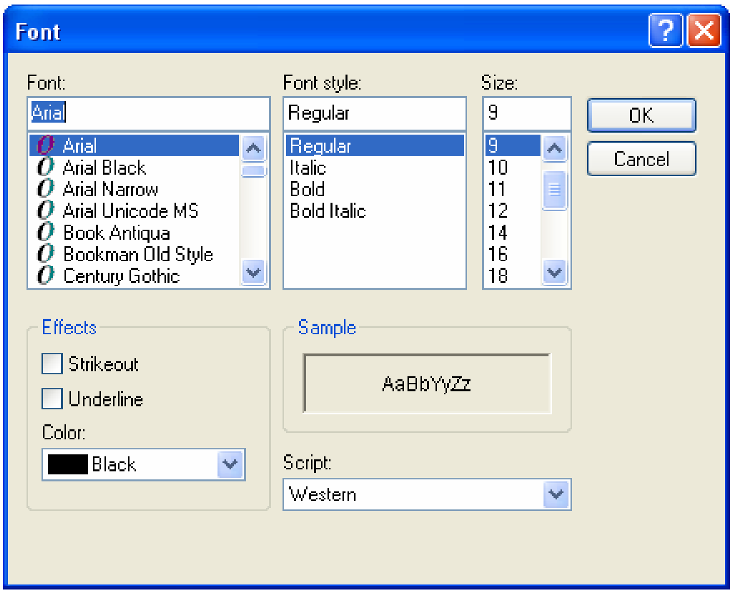



*Figure 20. Font dialog*

## 3.3.15 Exercises – Schematic Editor basics

### Zooming and panning

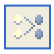
1. Open the schematic sheet, `ISA Bus and Address Decoding.SchDoc`, found in the `\Altium Designer Summer 09\Examples\Reference Designs\4 Port Serial Interface` folder.
2. Experiment with each of the display commands listed in Table 1 using the View menu, shortcut keys and the Main toolbar.
3. Use the mouse wheel to pan and zoom.
4. Select the menu command **Place » Text String** (**PT**) and experiment with the shortcut keys listed in Table 2 in conjunction with the display commands you have just mastered. To exit the command, press the ESC key or right-click.

5. Now try auto panning. Select the menu command **Place » Text String** again, then move the cursor to an edge of the window. The display will start panning. Hold down the SHIFT key while the display is panning. Note the crosshair cursor displaying while the **Place » Text String** command is active.

## Favorites Panel

1. Bring up the Favourites panel if you don't already have it on screen by going to the bottom right of Altium Designer and clicking on **System » Favourites**.
2. Zoom into the area of the schematic you want to create a favourite for and in the favourite's panel right click on it and select **Add current document view**.
3. Now close that schematic and from the favourite's panel double click on the saved favourite and it'll open the schematic and zoom in how you had it.

## Selection and mouse actions

1. Click on a component, e.g. P1. Observe the dashed box indicating it is the selected object.
2. Click on another component, e.g. a capacitor. It will now be the selected object.
3. Click somewhere on the sheet where there are no parts. Nothing will be selected now.
4. Click on the wire to select it. Notice the handles are now displayed.
5. With a wire selected, experiment with moving a vertex and moving a segment (a length of line between two vertices). Add a vertex by clicking and holding on the wire where you want the new vertex, pressing INSERT and then moving the new vertex to its new location. Delete the new vertex by clicking on it and pressing DELETE.
6. Make sure all objects on the sheet are not selected using **Edit » DeSelect » All** (X, A) or  on the main toolbar.
7. Using the click and drag selection feature, select a section of the circuit. Using the **Edit » Copy** menu command, copy the items to the clipboard.
8. Open a new sheet and paste the clipboard contents onto it. De-select the pasted objects.
9. Close the new sheet (no need to save it).
10. Try moving the selected objects on the original sheet using the **Edit » Move** menu commands. Deselect all objects.
11. While holding the CTRL key, click on the component U10. You can now drag it around and still maintain connectivity.
12. Click and hold on capacitor C12 and start to move it. While moving it press the ALT key, noting how the movement is now constrained to the horizontal or vertical direction only. The choice between constraining horizontal or vertical is defined by the proximity of the cursor to the object – simply push the object in the desired direction to see the effect.
13. Double-click on one of the capacitors. The *Component Properties* dialog displays. You can now edit any of the device's properties.
14. Close the schematic without saving any changes.

TOP SECRET

Altium Designer

# Module 4: Schematic Capture

## Module 4: Schematic Capture





Module Seq = 4

# 4.1 Introduction to Schematic Capture

The Schematic Capture training session covers how to create single sheet schematics and multi-sheet hierarchical projects from initial setup through to component placement, wiring, design verification and printing. The functionality of the Schematic Editor will be explored and a series of exercises will show you how to capture a design as a schematic, ready for PCB design.

Figure 1 outlines the workflow to be followed when creating a schematic in Altium Designer.

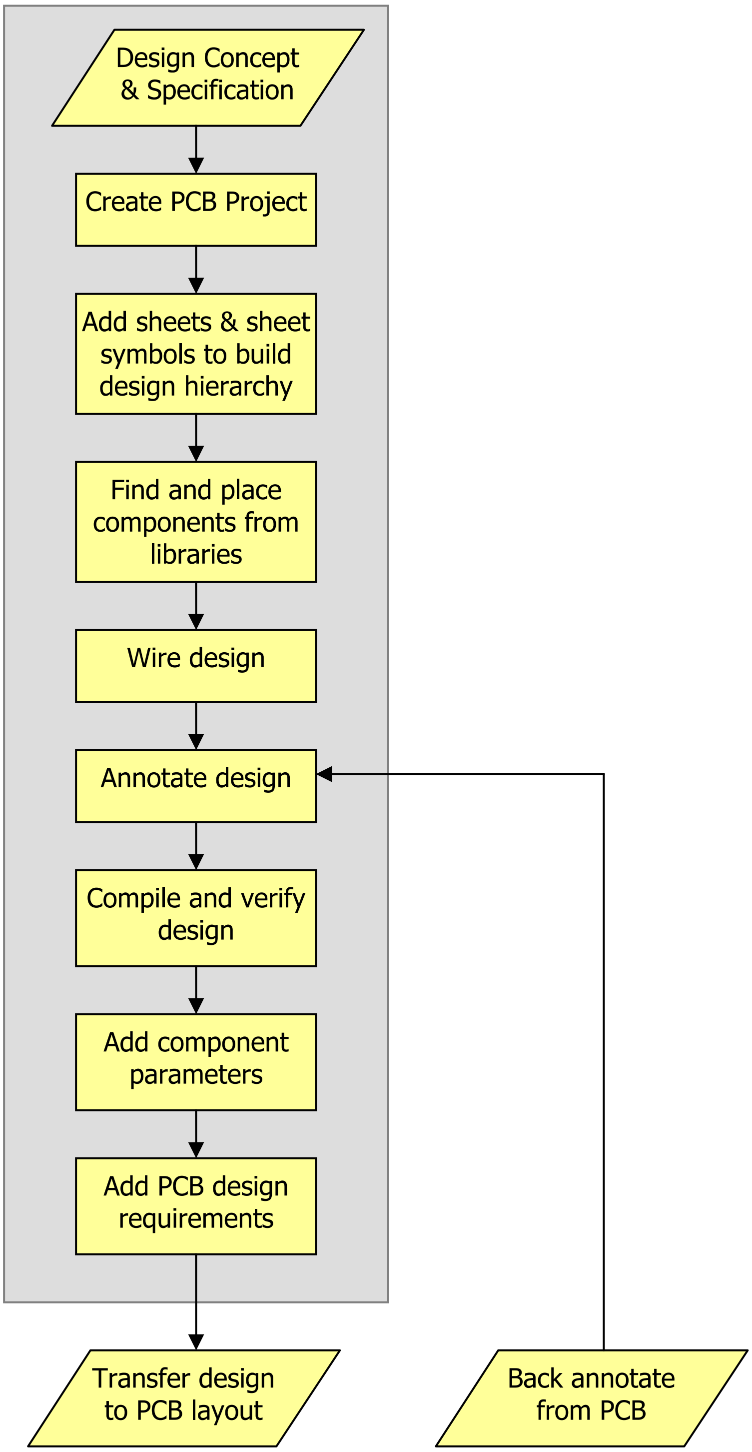



*Figure 1. The Altium Designer schematic capture workflow*

# 4.2 The Schematic Editor workspace

This section describes how to set up and browse the Schematic Editor workspace, done via the *Document Options* and *Preferences* dialogs. Sheet options, such as grids and templates, as well as preferences and defaults can be set through these dialogs.

To open the Schematic Editor, simply create a new schematic document (**File » New » Schematic**) or open an existing `.SchDoc` document in Altium Designer.

## 4.2.1 Document Options

The *Document Options* dialog allows you to:

- Set parameters relating to *individual schematic files.*
- The settings in this dialog are saved with that schematic file.
- The *Document Options* dialog is displayed by double-clicking on the sheet border, or by choosing the **Design » Document Options** menu command.

The tabs of the *Document Options* dialog are described in the following sections.

### 4.2.1.1 Sheet Options tab

The **Sheet Options** tab of the *Document Options* dialog is shown in Figure 2. The options in each of the sections are explained below.

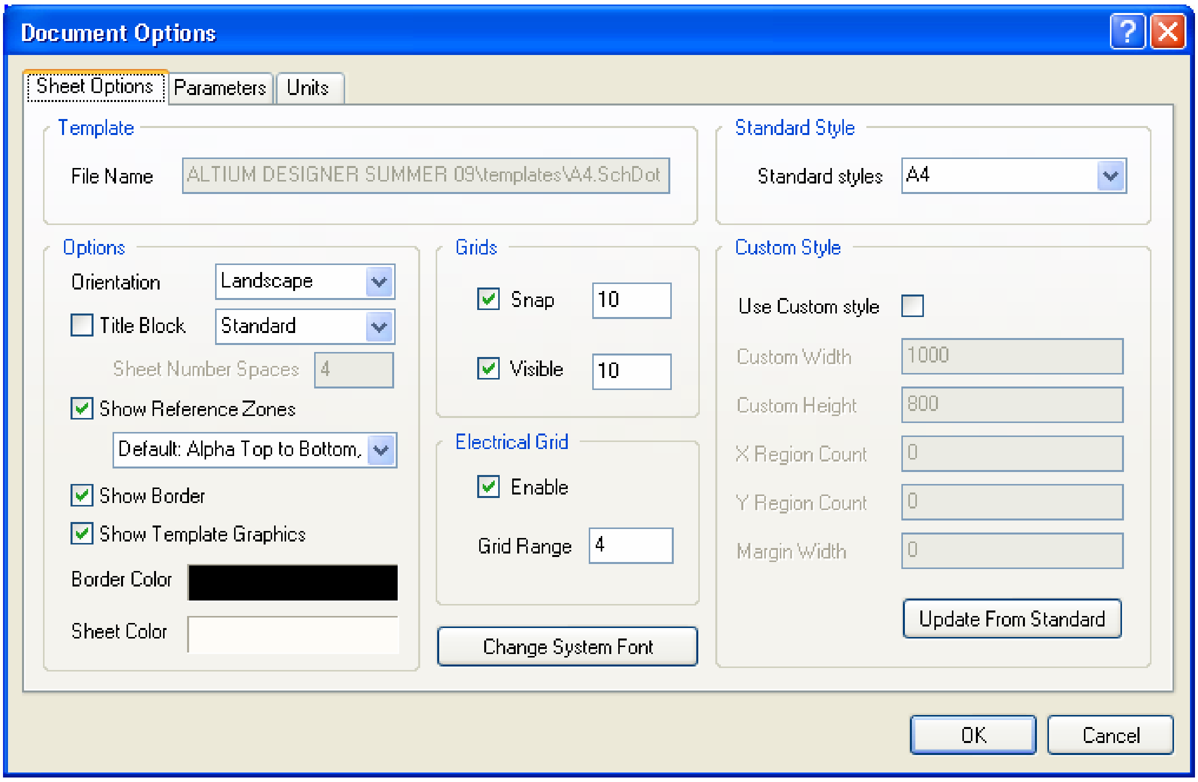



*Figure 2. Sheet Options tab of the Document Options dialog*

## Template section

Displays the filename of the associated template, if any. Use the **Template** options in the **Design** menu to apply, update or remove the associated template. Set the default template in the **DXP » Preferences » Schematic » General**.

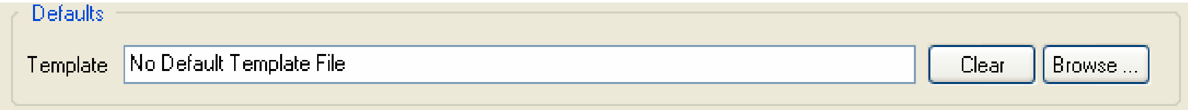



*Figure 3. Setting a default template to use when creating a new schematic.*

## Options section

### Orientation

Sets the sheet orientation to Landscape or Portrait.

### Title Block

When checked, a standard title block is attached to the sheet. The format of that title block is set using the drop-down list next to this option. Note that this is typically only used when there is no associated template.

### Show Reference Zones

When checked, the sheet has a reference grid defined in its border. A new option has also been added to have the zoning as per ASME Y14.1 standard.

### Show Border

When checked the sheet border is displayed.

### Show Template Graphics

When checked, any objects placed in the template file defined for the sheet will be displayed in the sheet. This is typically used to display a non-standard title block, in which case you would uncheck the Title Block option.

### Border Color

Allows you to set the border color from the *Choose Color* dialog.

### Sheet Color

Allows you to set the background color of the sheet.

## Standard Style section

Allows you to select the size of the sheet from a number of standard sizes e.g. A4, A3.

## Custom Style section

Allows you to define a custom sheet size and border. Use this option if you want a sheet size not covered in the Standard Style section.

## Change System Font

This button allows you to change the font used to display pin numbers, pin names, port text, power port text and sheet border text.

## Grids section

Grids Options allow you to set the size and turn on or off the Snap Grid and the Visible Grid.

#### SnapOn

The Snap Grid forces the mouse click location to the closest snap grid point. The Snap Grid is set and can be turned on or off in the *Document Options* dialog. You can also cycle though three predefined grids by pressing the **G** shortcut key at any time.

#### Visible

The Visible Grid displays a grid when turned on. This is independent of the Snap Grid. The Visible Grid can also be turned on or off in the **View** menu (**VV**).

### Electrical Grid section

The Electrical Grid can be turned on or off and the Electrical Grid Range can be set in the *Document Options* dialog. It can also be turned on or off in the **View** menu (**VE**).

When the Electrical Grid is turned on and you are executing a command that supports the electrical grid, the cursor overrides the Snap Grid and jumps to key points on objects.

For example, if you are using the **Place » Wire** command and move the cursor to a certain distance within the Electrical Grid Range of a pin, the cursor will jump to the pin.

### 4.2.1.2 Parameters tab

The **Parameters** tab is used as a convenient method of editing sheet-level text. Each parameter is automatically linked to a text string on the sheet, where the text string is the same as the parameter name, except that it is preceded by an equals sign.

For example, the Parameter *Address1* is automatically linked to the text string =*Address1*. The equals sign is an instruction to the schematic editor to automatically replace the text string on the sheet with the value of a parameter with a name of *Address1*. Any number of these parameters can be added to a document, either a schematic template or a schematic sheet. Using these special strings allows template text properties, such as font, size and color, to be predefined in the template, while the actual text string value is defined when that template is applied to a schematic.

This replacement occurs automatically during printing, it can also be performed on screen by enabling the **Convert Special Strings** option in the **Graphical Editing** tab of the *Preferences* dialog (**Tools » Schematic Preferences**).

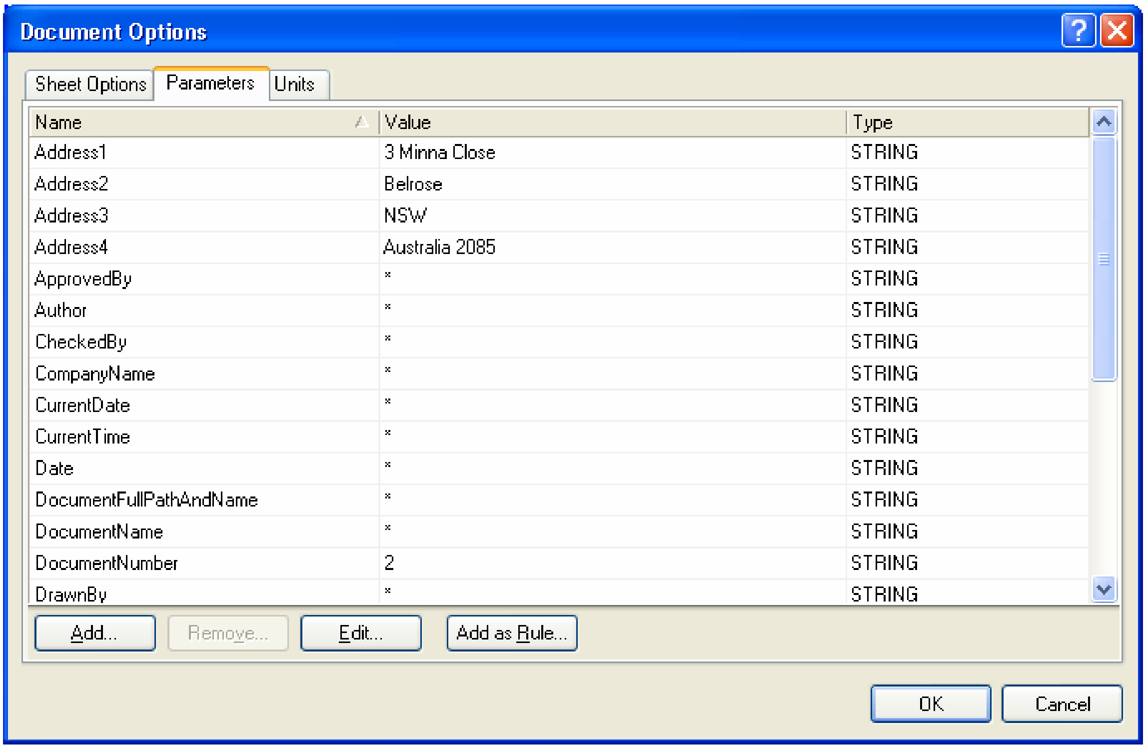



*Figure 4. Parameters tab of the Document Options dialog*

The default special strings are listed in the table below, but you can create custom parameters to suit your document and design requirements.

| Special String | Description | Special String | Description |
|---|---|---|---|
| =Address1 | Line of an address | =Engineer | Engineer's name |
| =Address2 | Line of an address | =ImagePath | Path to image file |
| =Address3 | Line of an address | =Modified Date | Computer system date of last modification to file (value entered automatically) |
| =Address4 | Line of an address | =Organization | Organization name |
| =ApprovedBy | Approver's name | =Revision | Revision number |
| =Author | Author's name | =Rule | Rule description if added using Add as Rule option |
| =Checked By | Checker's name | =SheetNumber | Schematic sheet number |
| =CompanyName | Company name | =SheetTotal | Total number of sheets in the project |
| =CurrentDate | Computer system date (value entered automatically) | =Time | Time (not automatically updated) |
| =CurrentTime | Computer system time (value entered automatically) | =Title | Title of schematic sheet |
| =Date | Date (not automatically updated) | =Engineer | Engineer's name |
| =DocumentFullPath AndName | Filename with full path of the schematic sheet (value entered automatically) | =ImagePath | Path to image file |
| =DocumentName | Filename without the path (value entered automatically) | =Modified Date | Computer system date of last modification to file (value entered automatically) |
| =DocumentNumber | Document number | | |
| =DrawnBy | Draftsperson's name | | |

*Table 1. List of default parameters when creating a new schematic*

Figure 5 shows how Special Strings are entered in a title block. Text entered as the value of a parameter in the **Parameter** tab will display where the special string is placed. The properties of the special strings (i.e. font, color) determine the properties of the text that is displayed.

You place special strings by selecting **Place » Text String** and then pressing the TAB key. The *Annotation* dialog displays. Clicking on the down arrow in the name field lists a special string for each of the parameters defined. Click on the string required and place it. Special strings display their content when the **Convert Special Strings** option is selected in the **Graphical Editing** tab of the *Preferences* dialog (**Tools » Schematic Preferences**), or when the schematic is printed or plotted.

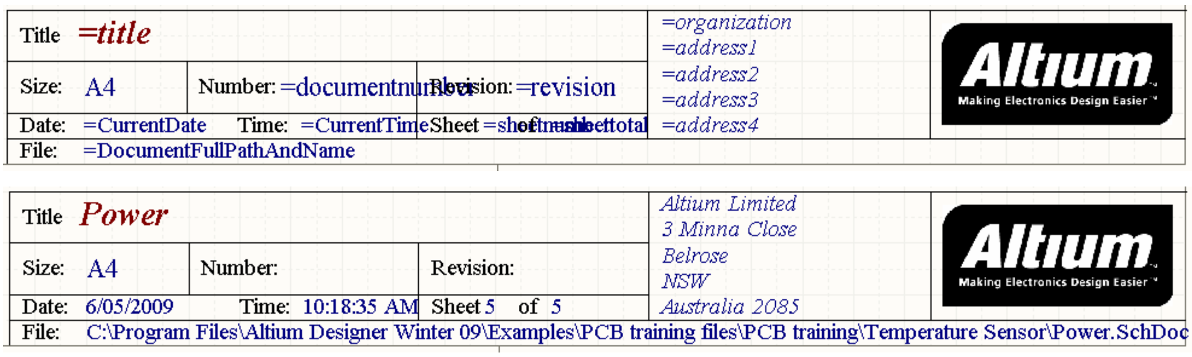



*Figure 5. Special strings in a title block, with and without the Convert Special Strings option enabled*

### 4.2.1.3 Units tab

The **Units** tab allows you to define the various units used in the Schematic Editor. Grids will be a multiples of these units so for example, using the DXP Default Units in which each *unit* is equal to 10 mils, a Snap Grid value of 5 (or 5 units) would equate to 50 mils, 10 units = 100 mils, etc. when printed actual size or 1:1.

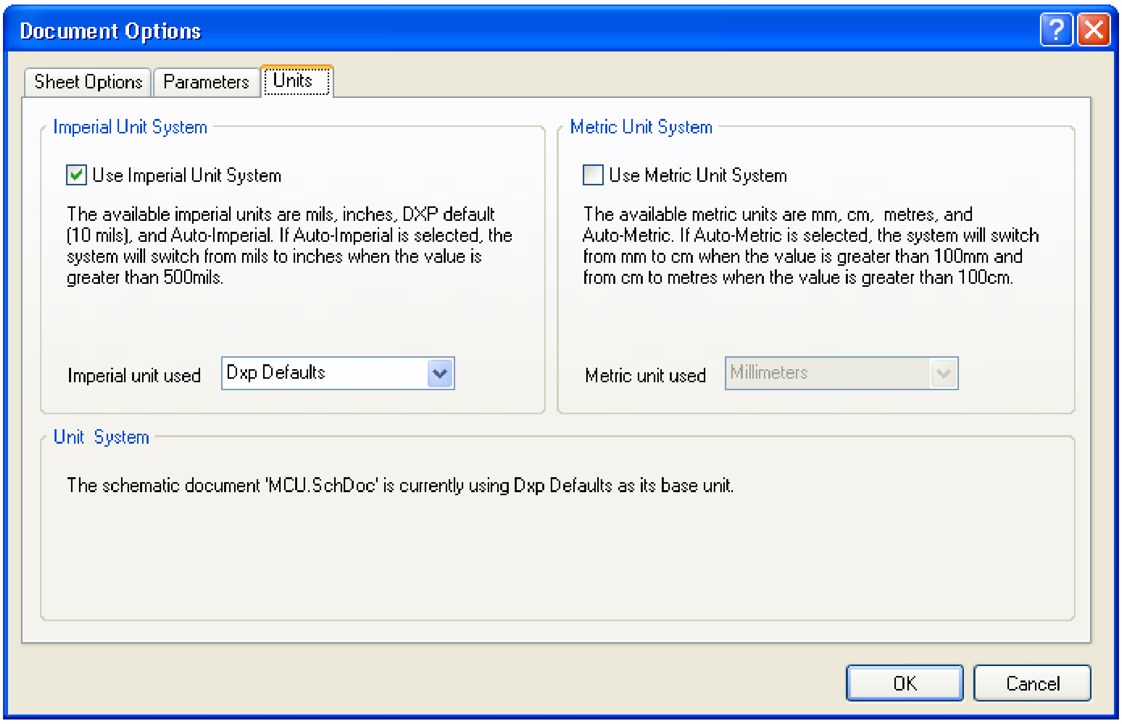



*Figure 6. Units tab of the Document Options dialog*

## Imperial Unit System Section

### Use Imperial Unit System

Check this option to use imperial units in your schematic(s). You will also want to specify which imperial units to be used (DXP defaults - 10 mil, mils, inches and auto imperial units) in the Imperial unit used drop down list.

### Imperial unit used

This option is used to select from one of the available imperial units; mils, inches, DXP default units (10 mils) and auto imperial. If the Auto-Imperial unit is selected, the system will switch from mils to inches when the value is greater than 500 mils.

## Metric Unit System Section

### Use Metric Unit System

Check this option to use metric units in your schematic(s). You will also want to specify which metric units to be used (millimeters, centimeters, meters, and Auto-Metric units) in the Metric unit used drop down list.

### Metric unit used

This option is used to select from one of the available metric units; millimeters, centimeters, meters, and Auto-Metric. If the Auto-Metric unit is selected, the system will switch from millimeters to centimeters when the value is greater than 100 centimeters.

#### 4.2.1.4 Using templates

Standard sheet templates (`*.SchDot`) are supplied with Altium Designer and are accessible in the \`Altium Designer Summer 09\Templates\`folder. You can also create your own templates and store them anywhere.

- Select **Design » Template » Set Template File Name**. This option removes any existing template and uses the one you choose.
- Select **Design » Template » Update**. Use this command when a template is modified and you need to refresh the sheets which use it.
- Select **Design » Template » Remove Current Template**. This option removes the template but retains the old sheet size from the old template.

For each of these commands you will be prompted to indicate if the change is to apply to the current sheet, all open, or all in project.

### 4.2.2 Preferences

The **Schematic** section of the *Preferences* dialog allows you to set up parameters relating to the Schematic Editor workspace. This dialog is displayed using the **Tools » Schematic Preferences** menu command. Settings in this dialog are saved in the Altium Designer environment and therefore remain the same when you change active schematic documents. For help with the options available in the preferences, use the ? button in the preferences dialog; then click on the option more information is required on.

# 4.3 Libraries and components

This section explores the Altium Designer libraries and how to find schematic components within them.

## 4.3.1 Locating and loading libraries

The supplied components are stored within a set of Integrated Libraries. An integrated library includes the schematic symbols, plus it can also include all associated models, such as footprints, spice models, signal integrity models, and so on. Most of the supplied integrated libraries are manufacturer-specific.

Integrated libraries are compiled from separate source schematic libraries, PCB footprint libraries, etc. The components in an integrated library cannot be edited, to change a component the source library is edited and recompiled to produce an updated integrated library.

There are a number of other special purpose integrated libraries, e.g. special function simulation components.

Components can also be placed directly from schematic symbol libraries if this is preferred to integrated libraries, and you can also place them from Protel 99 SE format schematic symbol libraries.

Available components are listed in the **Libraries** panel. The libraries presented in this panel include:

- Libraries in the active project. If the project that the currently active document belongs to includes any libraries, they are automatically listed.

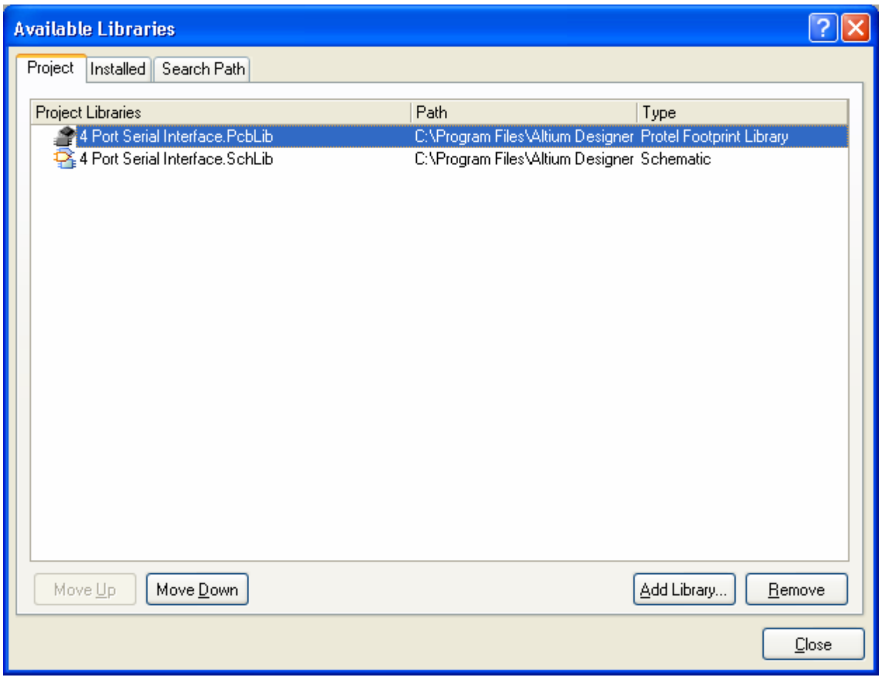



*Figure 7. Available Libraries dialog, showing the project tab which is libraries in the current project*

- Installed libraries. Installed libraries are those that have been made available in the environment. Use this option for company libraries that are used across different projects.

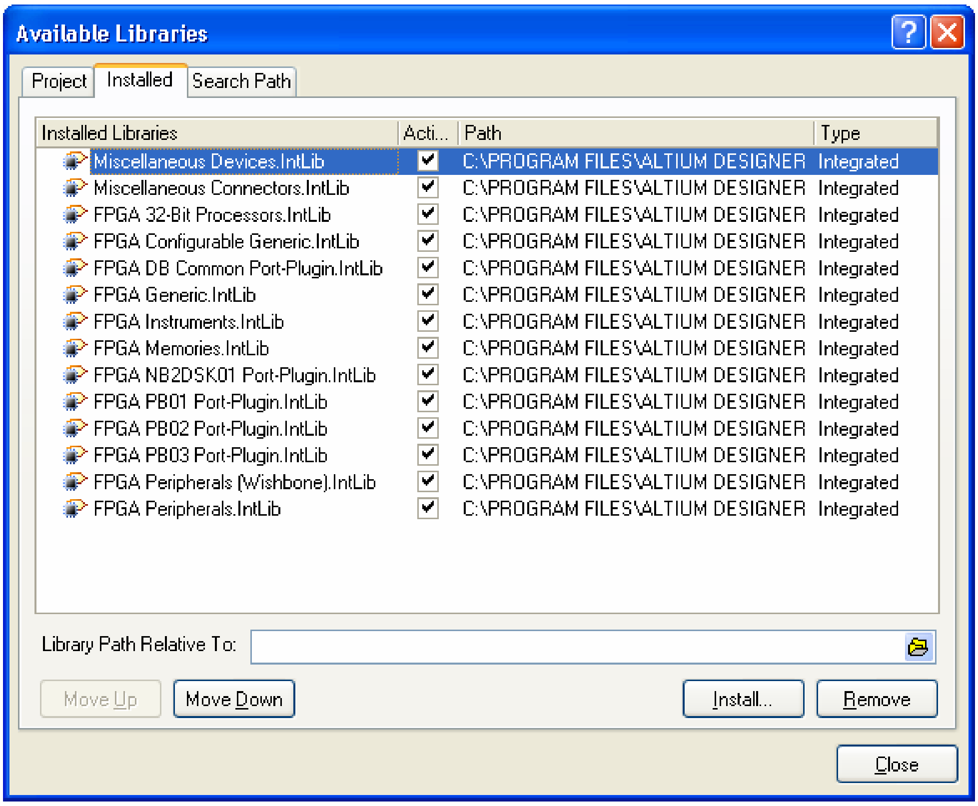



*Figure 8. Available Libraries dialog, showing the installed tab which is available to any open project.*

- Libraries found down the defined project search path. This option is particularly useful for accessing simulation models. Search paths are defined in the *Project Options* dialog.

### 4.3.1.1 Adding a library to make its components available

1. To add a library, press the **Libraries** button in the Libraries panel or select **Design » Add/Remove Library.** The *Available Libraries* dialog displays.
2. Click on the **Install** button at the bottom of the **Installed** Tab of the dialog.
3. Navigate to the required libraries directory and click on a library to select it. The library you selected will now be listed in the **Installed Libraries** list in the dialog.
4. Click **Close** when you have installed the libraries you need.

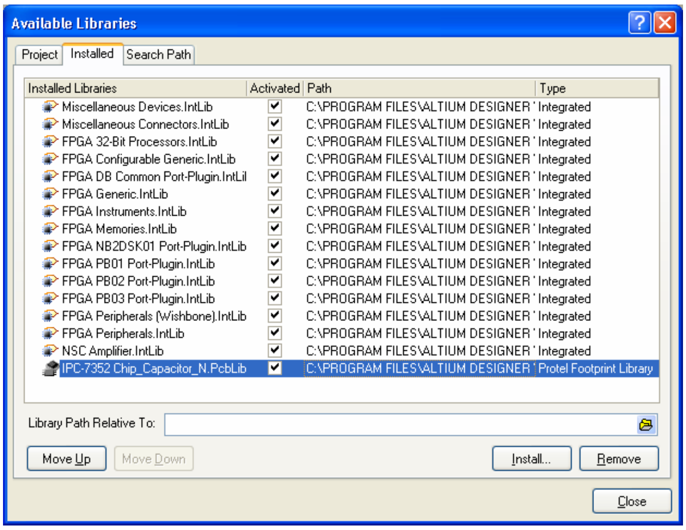



*Figure 9. Available Libraries dialog, use the Installed tab to install or*

| |
|---|
| **Note:** The supplied integrated libraries are located in the: `\Program Files\Altium Designer Summer 09\Library\` folder. |

## 4.3.2 Locating components

When the location of the library isn't known, use the **Search** button in the Libraries panel or the **Tools » Find Component** menu command. The *Libraries Search* dialog displays. The first time this dialog is executed, the dialog shown in Figure 10. is displayed. This initial mode is referred to as **Simple** mode. In **Simple** mode each searchable field has to be entered either manually or from a list of drop down items found within the currently installed libraries. After the field data is entered such as Name for example, an operator and value needs to be entered. The operator can be selected from a drop down list of Equals, contains, starts with and ends with. The value field can also be selected via a drop down of values found within the currently installed libraries or typed manually.

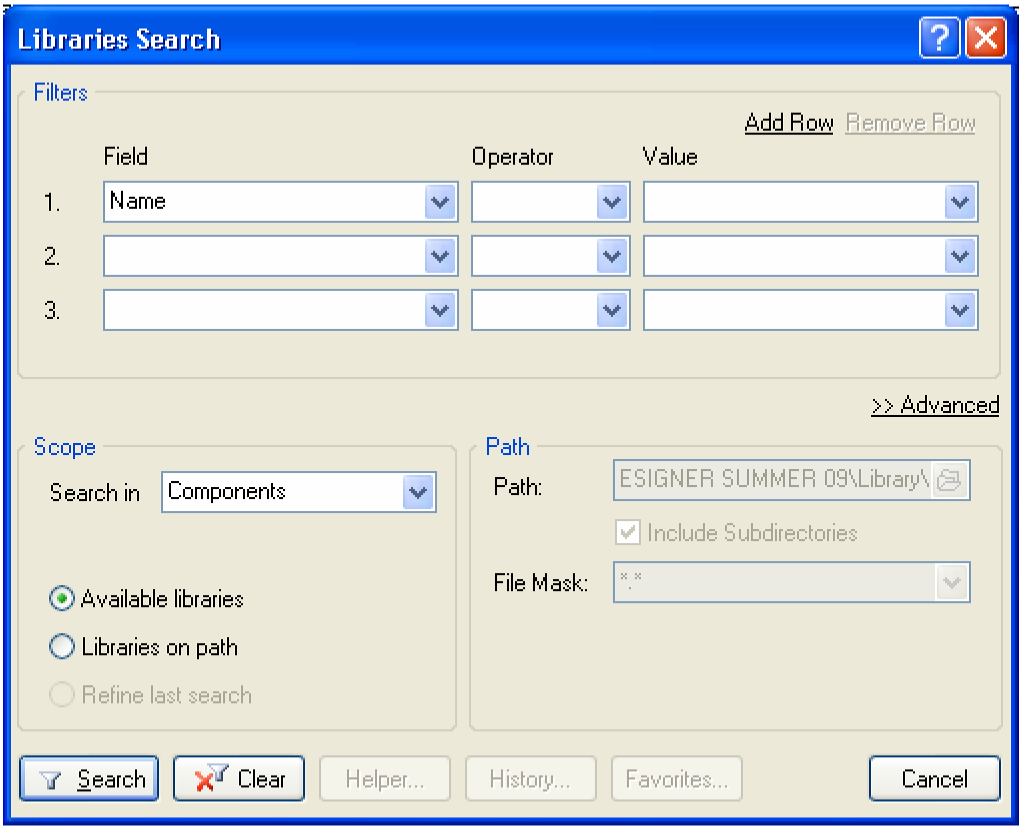



*Figure 10. Simple view for library search dialog*

Once data is entered, press the search button and a search starts. Going back to the search dialog and press the **Advanced** hyperlink button, the query string for the filter created in **Simple** mode is displayed.

Depending on current knowledge of the query language in Altium Designer, it is recommended, new uses use the Simple mode and users familiar with the query language use the advanced mode to manually type the exact search query.

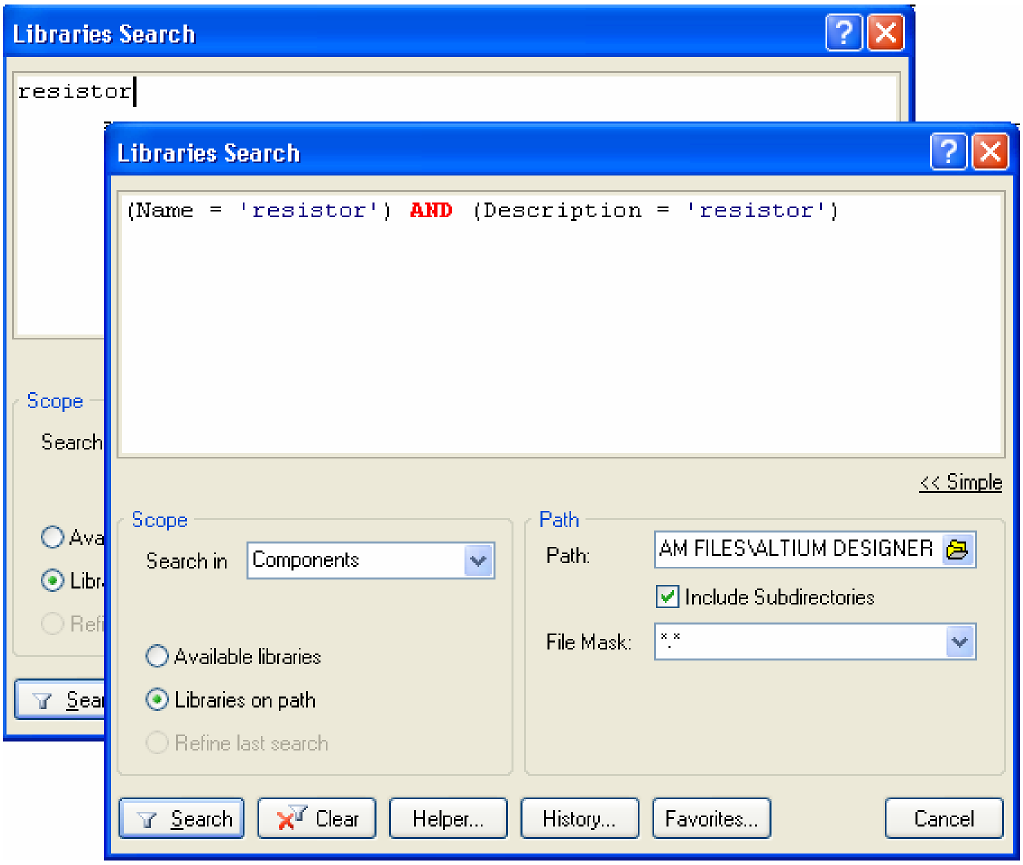
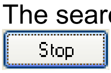



*Figure 11. Advanced view for Search Libraries dialog*

Tips for finding components:

- The search dialog uses a standard query to search the libraries – if the string you type does not include a query keyword it is assumed that the text is either part of the component *Name* or *Description* and a query is built automatically, as shown in Figure 11.
- The default search Scope is **Available libraries** that are those libraries currently listed in the **Libraries** panel. Change this to **Libraries on path** to search across all the supplied libraries.
- Search results are presented in the **Libraries** panel – note that the drop down where you select the current library will change to **Query Results**.
- If you attempt to place a component in the query results from a library that is not currently installed you will be asked if you wish to install that library now, you can still place without installing the library if you wish.
- The search can be terminated as soon as an instance of the part is found by clicking the 

 button on the **Libraries** panel.
- If your search does not produce results, check that the search path is correct. Also, try searching for a component you know exists in a library to check that everything is set correctly

**Note**: When you are searching libraries, check that the Filter field at the top of the Libraries panel has nothing in it. If there is any thing other than a *, then the search results will be further filtered, as shown in Figure 12.

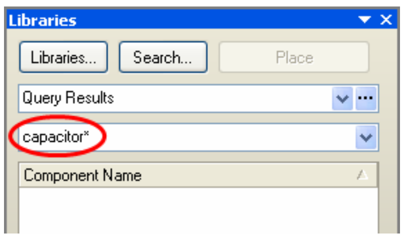



*Figure 12. Ensure that the filter field is clear during searching.*

## 4.3.3 Browsing libraries

The **Libraries** panel gives access to all components that are currently available to be placed.

- In the Panel control, bottom right, select **System » Libraries**.
- Click the **Libraries** button to display the *Available Libraries* dialog. This dialog displays all components currently available to be placed in the active project. Select the **Installed** tab, and click the **Install** button to add libraries to the library list.
- Components contained in the selected library are listed in the box below the Filter field. The Filter allows you to control what component names are listed, e.g. `RES*` will display only component names starting with RES.
- You can also type directly in the list of components, the type-ahead feature will automatically jump through the list as you type. Press ESC to stop performing a type-ahead action.
- Clicking on the name of a component will: display that component symbol in the viewer in the middle of the panel, list the associated models below that, and show the selected footprint model below that.
- The **Place** button places the component currently selected. Double-clicking on the name of a component also achieves this.
- The **Search** button is a powerful searching tool, allowing you to search through libraries for parts. Clicking this button pops up the *Libraries Search* dialog.

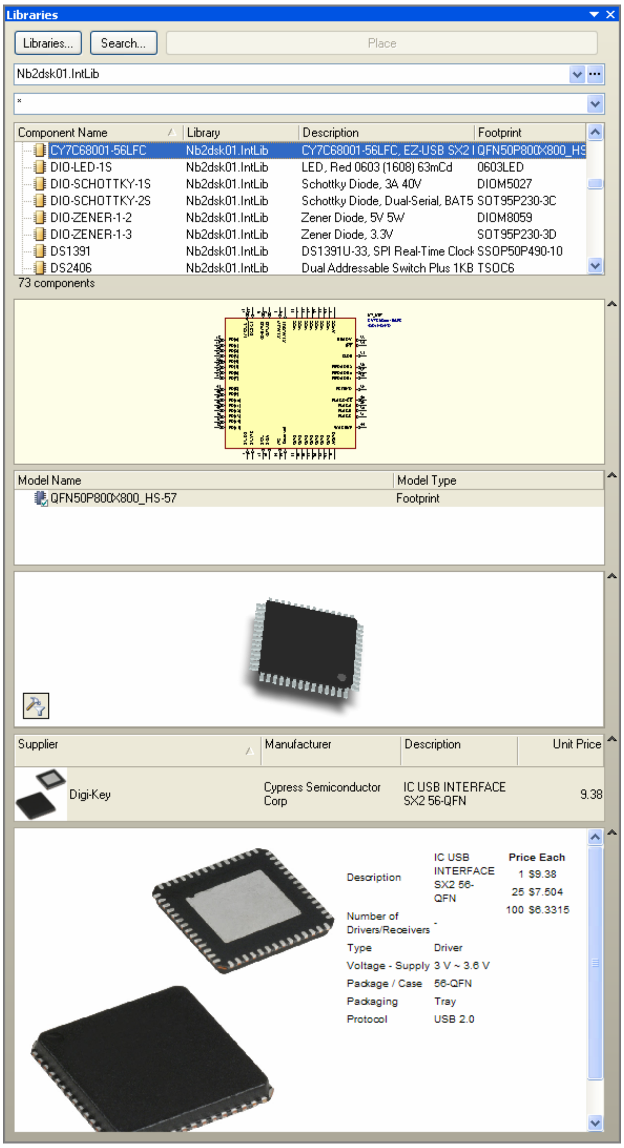



*Figure 13. Browsing libraries with the Schematic Editor Panel*

- If a component has several parts, the sub-parts will be shown in the symbol mini-viewer.
- You can control what columns are displayed in the component or model lists, right-click and choose **Select Columns** to do this. Using this with database libraries is especially useful as there is the ability to sort by a particular column like shown in Figure 14. Database libraries are covered in more detail in the Advanced PCB training course.

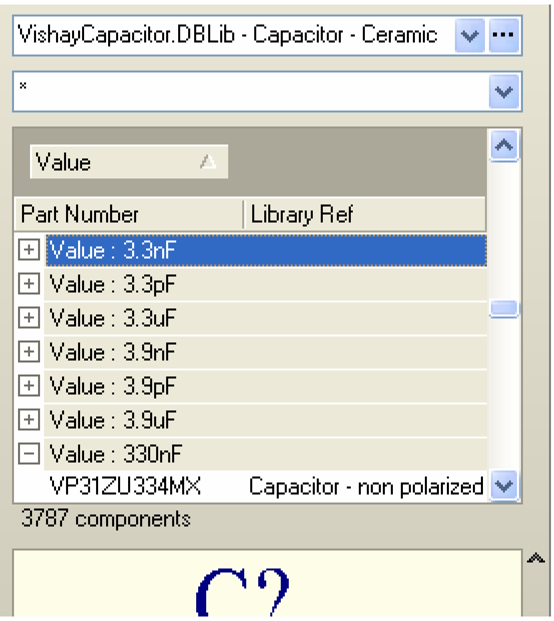



*Figure 14. Using a database library and the sort by column feature to sort by Value.*

## 4.3.4 Exercises – Libraries and components

### 4.3.4.1 Locating and loading libraries when the required library is known

The training design is a microcontroller driven temperature sensor. To install one of the supplied libraries and see if it includes a PIC microcontroller library, complete the following steps:

1. Open a schematic document to activate the Schematic Editor.
2. Click the **Libraries** button on the Libraries panel to display the *Available Libraries* dialog.
3. Select the **Installed** tab of dialog, then click the **Install** button and navigate to the `\Program Files\Altium Designer Summer 09\Library\`directory. This directory contains sub directories containing the integrated libraries supplied with Altium Designer's Schematic Editor.
4. Scroll down through the library directories. Open the `Microchip` folder, select and add the `Microchip Microcontroller 8-Bit PIC16.IntLib`.
5. Click the **Close** button to close the *Available Libraries* dialog.
6. Select this Microchip library in the list of libraries at the top of the Libraries panel. The library's contents will be displayed in the box below the Filter field section. Confirm that the library includes a PIC16C773/SO.

### 4.3.4.2 Finding components when their library is unknown

Often you will want to locate a component but do not know which library it is in, or you may want to see what family types are available in the libraries. To search for components, we use the **Search** button or the **Tools » Find Component** menu command.

1. Click on the **Search** button and the *Libraries Search* dialog will appear.
2. For this exercise, **Advanced** mode is going to be used on the Library search dialog. Click on the **Advanced** Hyperlink button.
3. Set the **Scope** to **Libraries on path** and set the search **Path** to `C:\Program Files\Altium Designer Summer 09\Library` (the **Include subdirectories** option should be on).

4. The power supply in the training design uses a LM317MSTT3 adjustable regulator. To search the supplied libraries for a suitable device type the string `LM317` in the Search field at the top of the dialog and click the **Search** button.
5. Note that the library currently being searched is listed in the **Libraries** panel. Depending on the speed of the PC it will take a few minutes to search the entire 90,000+ components for the required part.
6. The result set should include components in the `ON Semi Power Mgt Voltage Regulator.IntLib`, confirm that the LM317MSTT3 part is listed.
7. To install this library so that component will be available later you can either right-click in the result list and select **Add or Remove Libraries** (this will simply open the *Available Libraries* dialog), or you can double-click on the component name in the list to place it (you can easily delete it if it is the wrong sheet), when you do the Confirm dialog will appear, giving you an opportunity to Install the library.

### 4.3.4.3 Locating components within an open library

1. Select the library `Miscellaneous Devices.IntLib` in the Libraries panel.

   This library is one of two PCB libraries installed by default when the software is installed. It includes a variety of discrete components, including resistors, capacitors, diodes, etc.
2. Type `cap` into the Filter field. Notice that only the capacitor-type components are listed.
3. Try `diode` in the Filter field. The only components listed now are the diodes whose library reference starts with the string diode.
4. Now try `*diode`, this time components that include the word diode anywhere in their name or description will be listed.

### 4.3.4.4 Finding footprints when their library is unknown

1. Footprints can be searched for in the same way as component symbols; the only difference is that you need to set the **Search in,** in the *Libraries Search* dialog to **Footprints** before pressing the **Search** button.
2. Set the Search Path to `C:\Program Files\Altium Designer Summer 09\Library\Pcb\IPC-7350 Series\`.

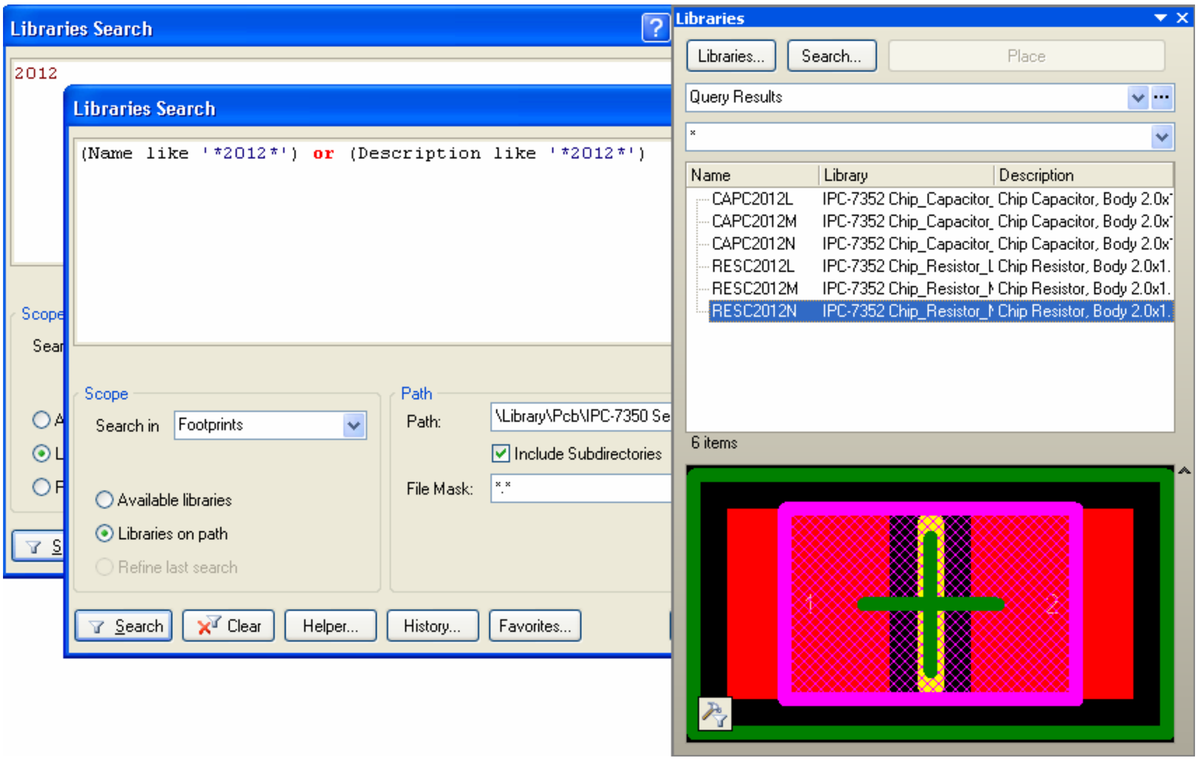


*Figure 15. Searching for a footprint*

3. Enter the string `2012` and click **Search**. The search results will include a number of libraries, including *IPC-7352 Chip_Resistor_N.PcbLib.*
4. Double-click on the `RESC2012N` footprint in the query results, a dialog will appear letting you know that the library is not currently installed, click **Yes** to install the library.

#### 4.3.4.5 Setting the library search order

When you type in a component name, for example in the *Place Part* dialog, or when you type in a footprint name in the *Footprint Model* dialog, the available libraries are searched in a defined order. This search order is the order that the libraries are listed in the *Available Libraries* dialog. To configure the search order:

1. Click the **Libraries** button in the Libraries panel to display the *Available Libraries* dialog.
2. Click on the **Installed** tab, then in the list of Installed Libraries click to select the *IPC-7352 Chip_Resistor_N.PcbLib* to highlight it, and then click the **Move Up** button to move it to the top of the list.
3. Close the *Available Libraries* dialog.

You now have all the components and footprints required to complete the training design.

| |
|---|
| **Note:** Refer to the PDF, *AR0104 Component, Model and Library Concepts* article in the online documentation for further information on definitions, library search order and component to model linking. |

## 4.4 Placing and wiring

This section looks at how to place components and then wire them together. The exercise takes you through the creation of a complete schematic sheet.

### 4.4.1 Placing components

- To place a component, double-click on its name in the Libraries panel.
- To edit a component's properties before you place it, press the TAB key. The *Component Properties* dialog displays. To step through the fields in the dialog press TAB (down), or SHIFT+TAB (up).
- New text will overwrite text that is selected.
- If you set the component designator before placing the component, then subsequent components will be automatically designated with the next designator value.
- You can also use the **Place » Part** menu command if you know the name of a component. When you select this command, you are prompted for the name of the component. Once you type the component name in, the open libraries are searched and if the component is located, it becomes attached to the cursor for placement.

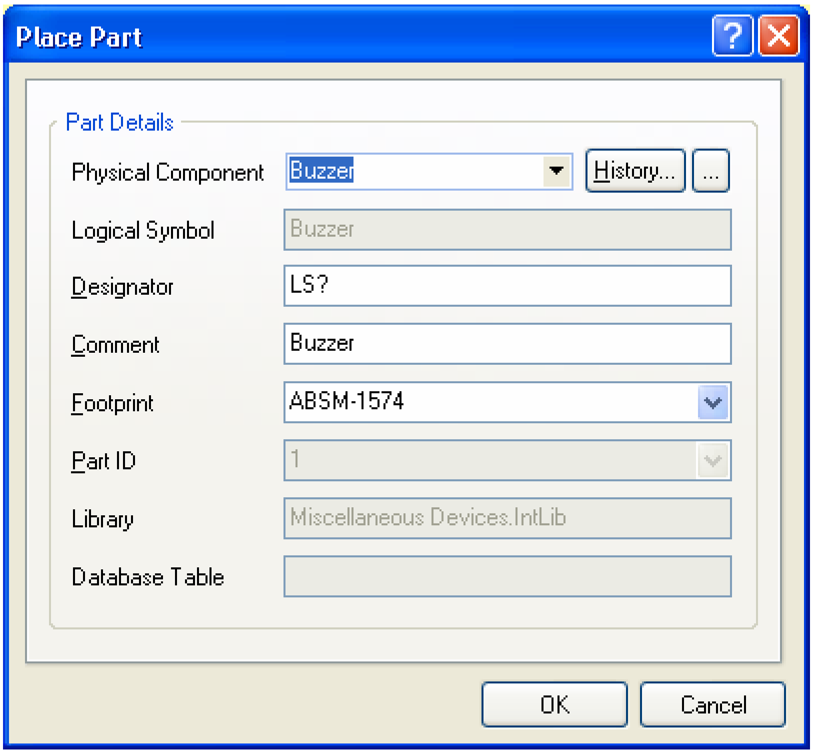



*Figure 16. Place part dialog*

#### 4.4.1.1 Auto-incrementing designators

When placing a component, if the initial designator is set before placing, its designator will be assigned by incrementing the designator of the last component placed. This will only occur for subsequent parts placed after the TAB key was pressed to assign the initial designator. Once you stop placing this type of part, the next designator in the sequence is no longer remembered.

Generally it is easier to leave the annotation of designators until the design is complete to allow the designators to be assigned in a logical and controlled manner on each sheet. Annotation is covered in detail later in *Module 6 - Building the Project.*

## 4.4.2 Pin-to-pin wiring

- Wires are used to create an electrical connection between points.
- Be careful to use **Place » Wire** and not place lines by mistake.
- Press SHIFT+SPACEBAR to change the wire placement mode. Press SPACEBAR to toggle between start and end corner modes.
- Press BACKSPACE to delete the last vertex placed.
- A point on a wire must touch on the connection point of an electrical object to be connected to it, e.g. the wire must touch on the hot end of a pin to connect to it.
- Use buses to graphically represent how a group of related signals, such as a data bus, are connected on a sheet. Also, use buses to connect related signals to ports and sheet entries.
- Buses must use the bus name / bus element referencing system as shown in Figure 20, and must include the individual net labels and the bus net label.
- The bus range can increment [0..7], or decrement [7..0].
- To move a component on the schematic and maintain the wiring (referred to as dragging), hold the CTRL key as you click, hold and move the mouse (release the CTRL key once you start dragging). Press the SPACEBAR or SHIFT+SPACEBAR while dragging to change the wiring mode. Press the M shortcut to drag a selection.
- If you are having trouble seeing the grid in schematic goto **Tools » Schematic Preferences » Schematic » Grids** and change the *Grid Color* to one that is not so white.

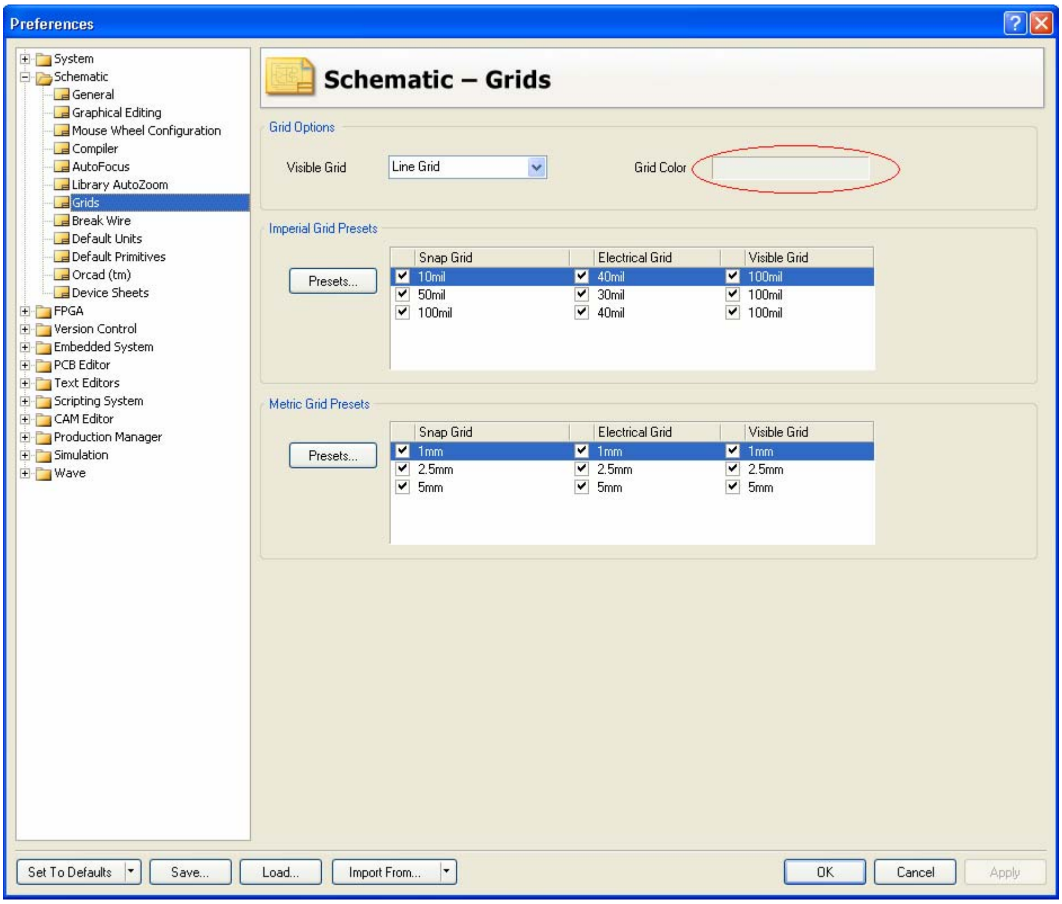



*Figure 17. Grids setting in preferences*

## 4.4.3 Exercise – Drawing the schematic

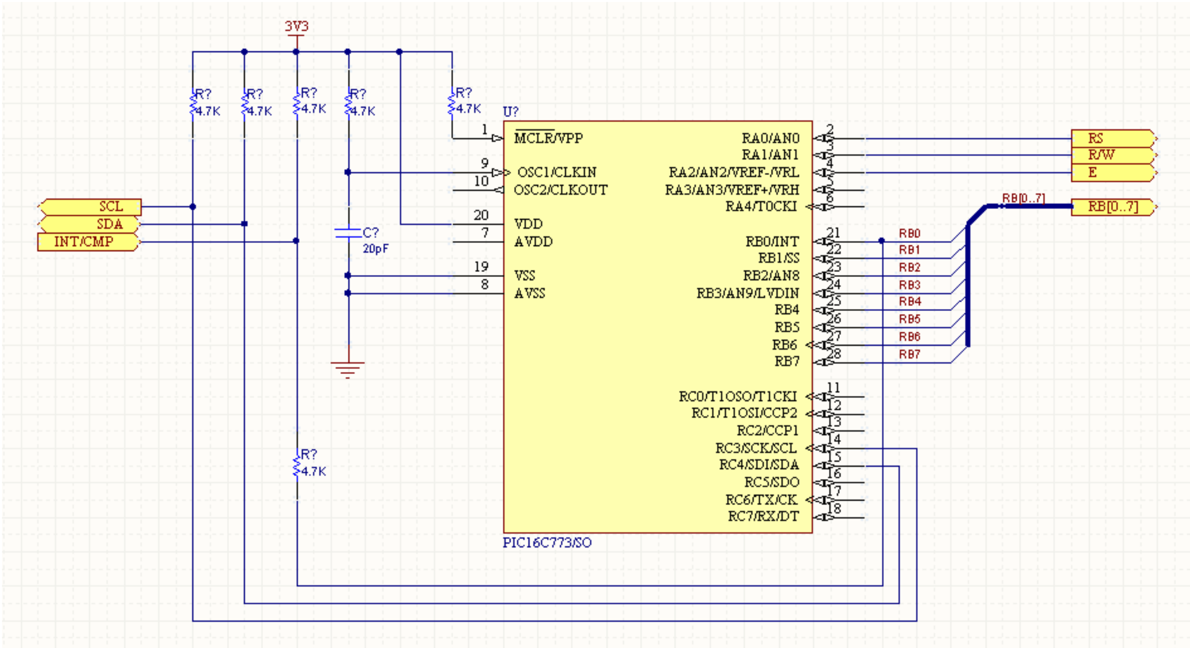



*Figure 18. MCU schematic example*

1. If it is not already open, re-open the project created during the *Module 1 - Getting Started With Altium Designer*, `\Program Files\Altium Designer Summer 09\Examples\Training\PCB Training\Temperature Sensor\Temperature Sensor.PrjPcb`.
2. Add a new schematic document to the project, to do this right-click on the project file name in the **Projects** panel and select **Add New to Project » Schematic**.
3. Right-click on the new schematic sheet in the Projects panel, and select **Save As** from the context menu. Save the schematic as `MCU.SchDoc` in the `\Program Files\Altium Designer Summer 09\Examples\Training\PCB Training\`Temperature Sensor folder.
4. Set the template for your schematic to `A4.SchDot` by choosing **Design » Template » Set Template File Name** and choosing the A4 size template from `\Program Files\Altium Designer Summer 09\Templates` folder.
5. Verify that the electrical grid is on and set to 4 and that the snap grip is on and set to 10 before placing any objects (double-click in the sheet border to open the *Document Options* dialog).
6. Draw up the schematic shown in Figure 18 above. When placing the components, press TAB to define the **Designator** and **Comment** (component value) *before* placing the component.

| Component | Library Reference |
|---|---|
| Microcontroller | PIC16C773/SO |
| Resistors | Res1 |
| Capacitor | Cap |

7. To rotate a component press the SPACEBAR, press the **Y** key to flip it vertically, and the **X** key to flip it horizontally.
8. Set the **Port I/O Type** to match their display **Style**. Set the Ground **Style** power port net attribute to GND.
9. Set the bus name and port name to `RB[0..7]` so as to connect nets RB0 through to RB7 into a bus.

10. To build up the nets in the bus, first create the port on the right. Copy the port and run **Edit » Smart Paste**. Select **Ports** from the left side and select **net labels and wires** on the right side. On the bottom section of the dialog select **expand buses** for signal names. You can also set the wire length to suit. Note that the spacing between the wires well be as per the current grid setting which should be on 10 so the wires and netlabels line up to the pins of the component.

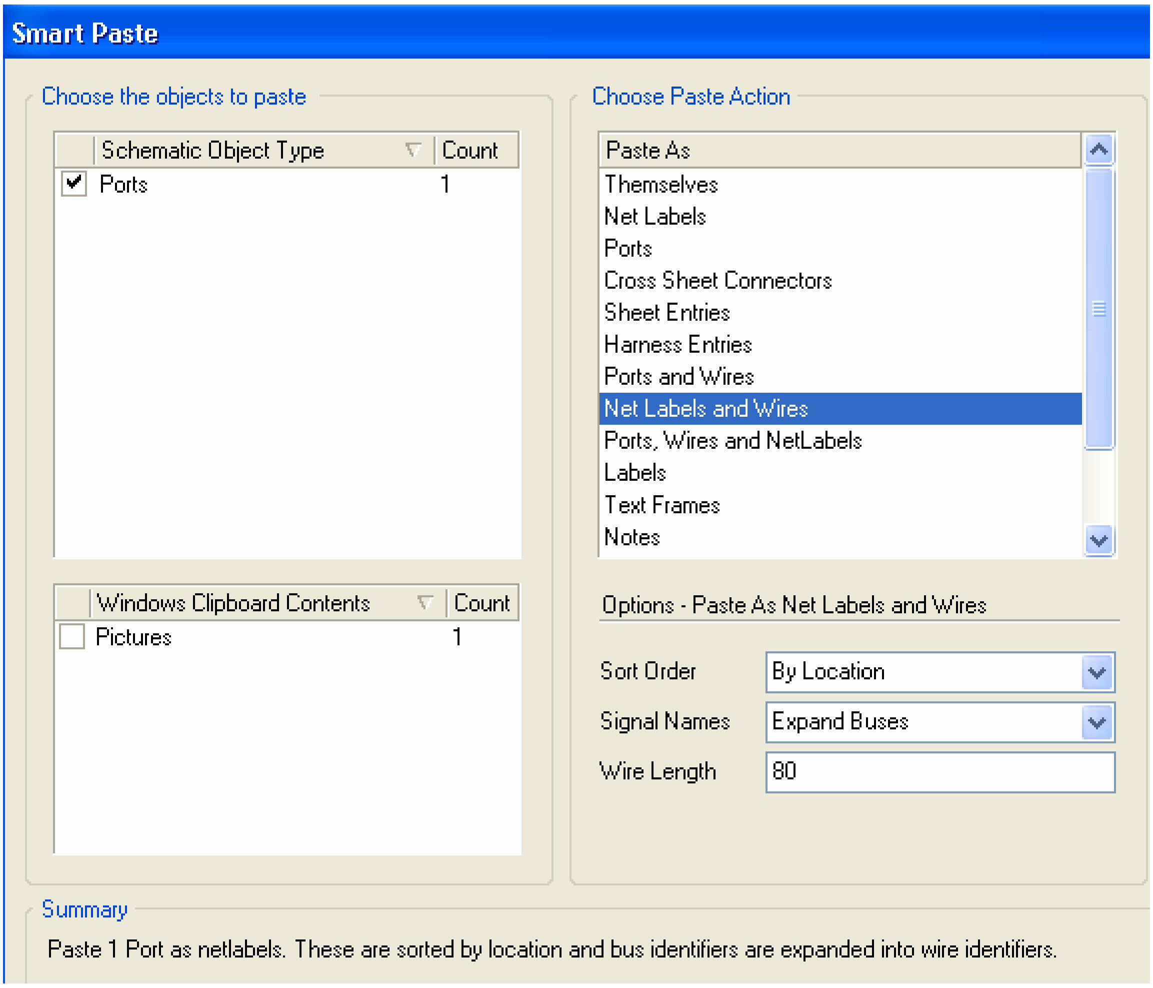



*Figure 19. Smart paste dialog to create the netlabels and wires that make up the signals on a bus.*

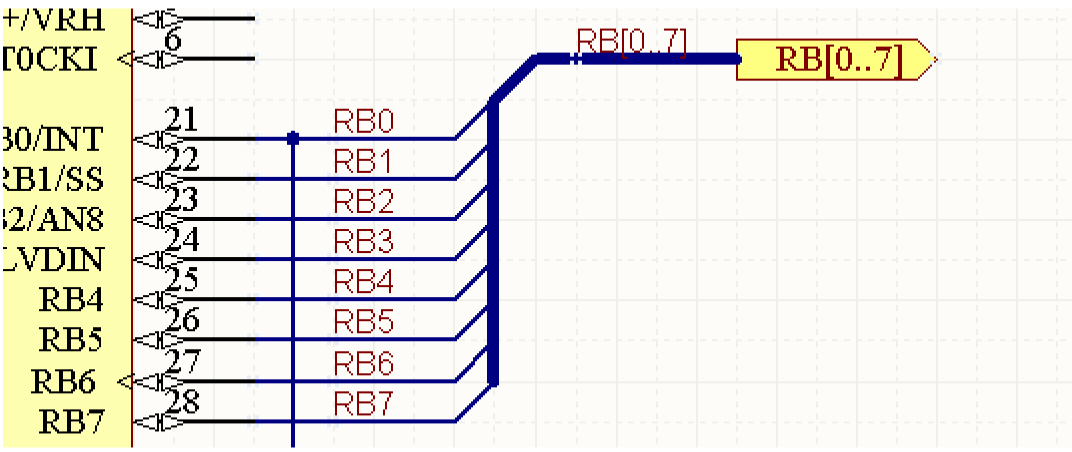



*Figure 20. Buses are defined using the referencing system shown.*

11. While placing the resistors and capacitors you can use the component cut wires feature to quickly cut into the wires rather than wiring around them.
12. While placing some of the ports and power ports hold down the ctrl key and drag to extend the wires with the object. This can save a lot of time when getting to the wiring stage.
13. Enter the necessary document information in the **Parameters** tab of the *Document Options* dialog. Enter the title as `PIC Microcontroller.` Don't worry about setting the sheet/Document No. at this time as there is a feature that can do this automatically over your whole project.

## 4.4.4 Exercise – The Sensor schematic

At this stage in the training, your Temperature Sensor project should look like Figure 21. However, the `Sensor.SchDoc` and the project libraries are not there as yet. To complete it:

1. Create a new schematic and call it **Sensor.schdoc** if you haven't done so already.
2. Save and close the `Sensor.SchDoc` sheet. Note that this sheet is just a blank schematic for now. It'll be completed later in the training.
3. Also you'll need to add the library files as shown in Figure 21 to the project and save the project PCB.

At this point we are only going to create a blank sheet. In *Module 15 - Schematic Library Editor,* we will create the sensor sheet after creating a new component.

The last step to complete the sensor design is to add the top schematic sheet which is done in the next module *Module 5 - Multi-Sheet Design.*

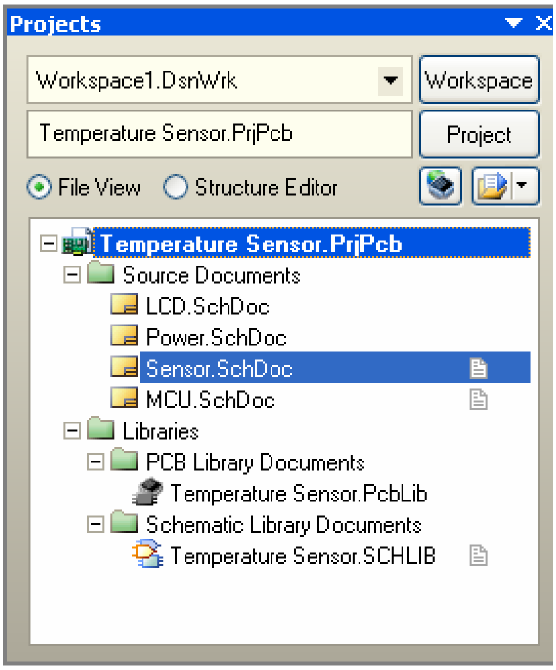



*Figure 21. Project structure after completing the MCU schematic.*

TOP SECRET

# Module 5: Multi-Sheet Design

# Module 5: Multi-Sheet Design






Module Seq = 5

# 5.1 Multi-Sheet Design

## 5.1.1 Structuring a multi-sheet design

All but the smallest designs will need to be laid out over multiple schematic sheets. There are essentially two approaches to structuring a multi-sheet design, either flat, or hierarchical. A flat design is one where the connectivity between nets that span sheets is directly from one sheet to the other or potential to many others.

While a flat design is acceptable for a design with a small number of sheets and nets, perhaps 6 sheets, it becomes unwieldy when the design is larger. Since a net can go to any of the other sheets, a larger flat design needs navigation instructions to guide the reader as they attempt to find that net on the other sheets. The advantage of the flat design is that there are normally fewer sheets, and less wiring to draw.

A hierarchical design is one where the structure – or sheet-to-sheet relationships – in the design is represented. This is done by symbols, known as sheet symbols, which represent lower sheets in the design hierarchy. The symbol represents the sheet below, and the sheet entries in it represent (or connect to) the ports on the sheet below. The advantage of the hierarchical design is that it shows the reader the structure of design, and that the connectivity is completely predicable and easily traced, since it is always from the child sheet up to the sheet symbol on the parent sheet.

The diagram below shows the top sheet for the Temperature Sensor project. Each sheet symbol represents a child schematic in the design.

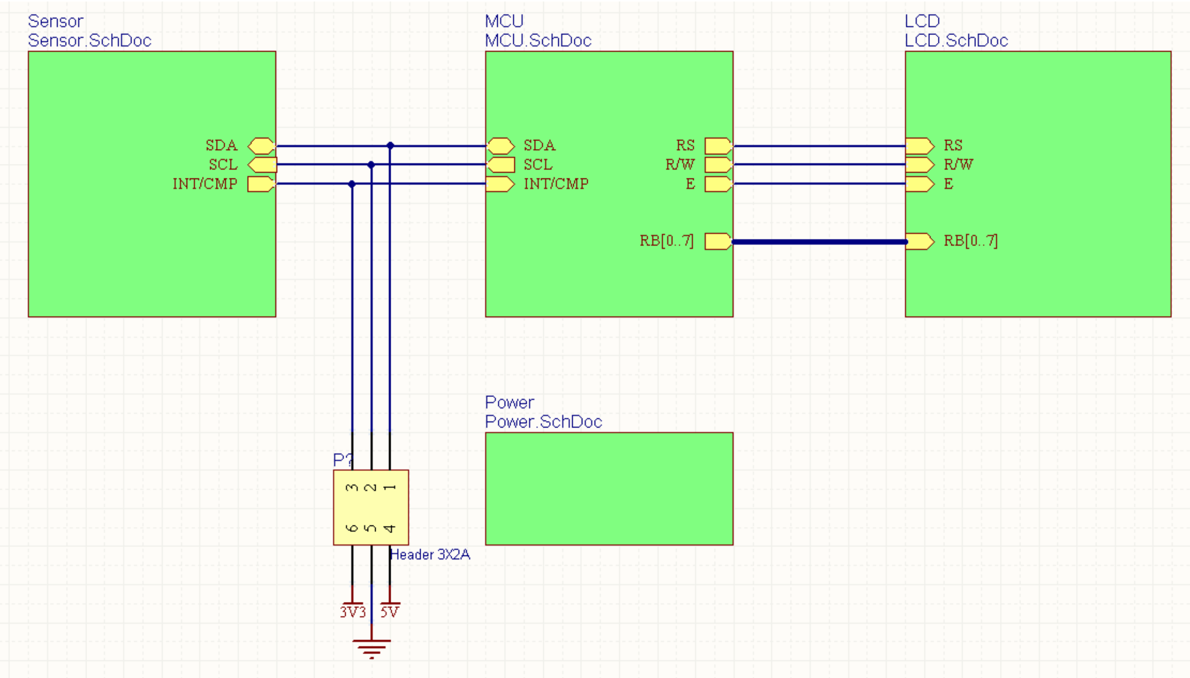



*Figure 1. Temperature Sensor top sheet*

## 5.1.2 Multi-sheet design connectivity

Multi-sheet designs are also defined at the electrical (or connective) level by *net identifiers* which provide the ‘glue’ between nets in schematic sheets.

### 5.1.2.1 Net identifiers

Net identifiers create logical connections between points in the same net. This can be within a sheet, or across multiple sheets. Physical connections exist when one object is attached directly to another electrical object by a wire. Logical connections are created when 2 net identifiers of the same type (eg, two net labels) have the same **Net** property. Note that logical connections are not created between different net identifiers, for example a port and a net label. The only exception to this is when a port connects to a sheet entry of the same name, in the sheet symbol that represents the sheet the port is on (more on this later).

Net identifiers include:

- **Net Label –** Use a net label to uniquely identify a net. This net will connect to other nets of the same name on the same sheet, and can also connect to nets of the same name on different sheets, depending on the connectivity mode defined for the design (referred to as the net identifier scope). Net labels are attached to individual wires, part pins and buses.
- **Port –** Depending on the method of connectivity, a port can connect horizontally to other ports with the same name, or vertically to a sheet entry with the same name.
- **Sheet Entry –** When the connectivity is vertical, you can use a sheet entry to connect to a port of the same name on the sheet below. A sheet entry is added to a sheet using the **Place » Add Sheet Entry** command.
- **Power Port –** All power ports with the same name are connected throughout the entire design.
- **Hidden Pin –** Hidden pins behave like power ports, connecting globally to nets of the same name throughout the entire design.

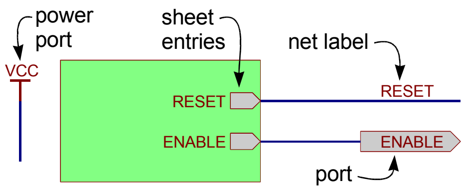



*Figure 2. Net identifiers*

### 5.1.2.2 Net identifier scope

When you create a connective model of a design, you must define how you want these net identifiers to connect to each other – this is known as setting the *Net Identifier Scope*. The *scope of net identifiers* is specified in the **Options** tab of the *Project Options* dialog. The scope of net identifiers should be determined at the beginning of the design process.

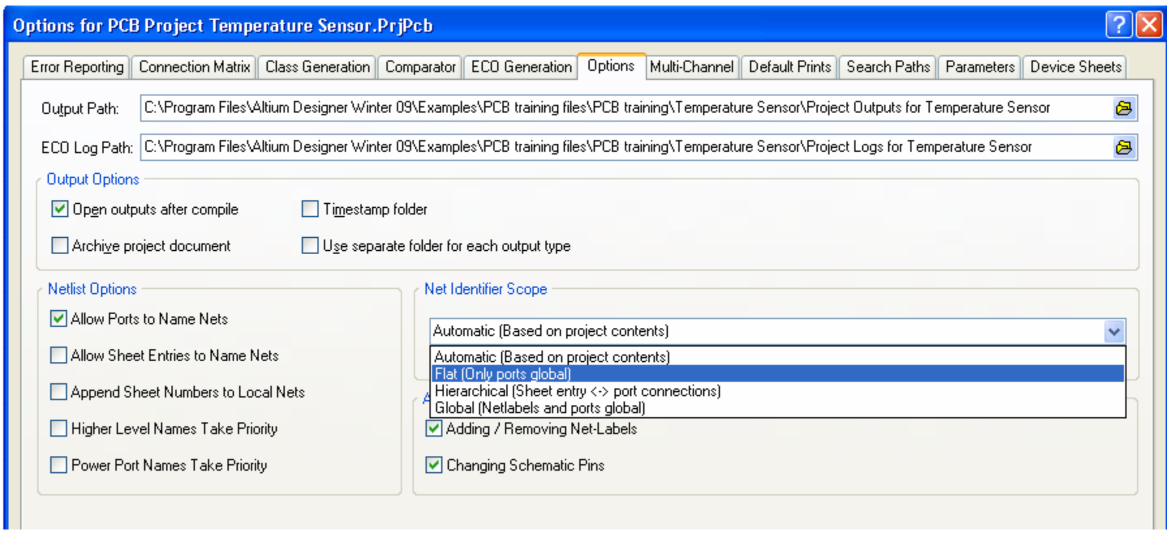



*Figure 3. Net identifier scope and netlist options*

There are essentially two ways of connecting sheets in a multi-sheet design: either *horizontally*, directly from one sheet, to another sheet, to another sheet, and so on; or *vertically*, from a sub-sheet to the sheet symbol that represents it on the parent sheet. In horizontal connectivity, the connections are from port to port (net label to net label is also available). In vertical connectivity, the connections are from sheet entry to port.

The **Net Identifier Scope** specifies how you want the net identifiers to connect:

- **flat** – ports connect globally across all sheets throughout the design. With this option, net labels are local to each sheet; they will not connect across sheets. All ports with the same name will be connected, on all sheets. This option can be used for flat multi-sheet designs. It is not recommended for large designs as it can be difficult to trace a net through the sheets.
- **global** – ports and net labels connect across all sheets throughout the design. With this option, all nets with the same net label will be connected together, on all sheets. Also, all ports with the same name will be connected, on all sheets. If a net connected to a port also has a net label, its net name will be the name of the net label. This option can also be used for flat multi-sheet designs, however it is difficult to trace from one sheet to another, since visually locating net names on the schematic is not always easy.
- **hierarchical (sheet entry/port connections)** – connect vertically between a port and the matching sheet entry. This option makes inter-sheet connections only through sheet symbol entries and matching sub-sheet ports. It uses ports on sheets to take nets or buses up to sheet entries in corresponding sheet symbols on the top sheet. Ports without a matching sheet entry will not be connected, even if a port with the same name exits on another sheet. Net labels are local to each sheet; they will not connect across sheets. This option can be used to create designs of any depth or hierarchy and allows a net to be traced throughout a design on the printed schematic.
- The **automatic** mode automatically selects which of the three net identifier modes to use, based on the following criteria: if there are sheet entries on the top sheet, then *Hierarchical* is used; if there are no sheet entries, but there are ports present, then *Flat* is used; if there are no sheet entries and no ports, then *Global* is used.

> **Note:** Two special net identifier objects are always deemed to be global: power ports and hidden pins.

**Summary**

- If you are using sheet symbols with sheet entries, the net identifier scope should be set to *Sheet Entries/Port Connections.* If this mode is chosen, the top sheet must be wired.
- If you are not, connectivity can be established via Ports and/or Net labels, so you will use one of the other two net identifier scopes.
- Net labels do not connect to ports of the same name.

## 5.1.3 Constructing the top sheet

The process of creating a top sheet can be done in a manual fashion, where the sheet symbols are placed, the filename attribute for each is set to point to the correct sub-sheet and the sheet entries are added to correspond to each port on the sub-sheet.

There are also commands to speed the process of creating a multi-sheet design.

The **Create Sheet from Symbol** command is for top-down design. Once the top sheet is fully defined, this command creates the sub-sheet for the chosen sheet symbol and places matching ports on it.

The **Create Symbol from Sheet** command is for bottom-up design, creating a sheet symbol with sheet entries based on the chosen sub-sheet. This is the mode we will use now.

### 5.1.3.1 Exercise – creating the top sheet for the Temperature Sensor project

Refer to Figure 1 to complete this exercise.

1. To create the top sheet, add a new schematic document to the Temperature Sensor project, set the template to A4 and save it as `\Program Files\Altium Designer Summer 09\Training\PCB Training\Temperature Sensor\Temperature Sensor.SchDoc`.
2. Rather than manually placing sheet symbols and editing them to reference the lower sheets, we will use the **Design » Create Sheet Symbol from Sheet or HDL** command. Select this command from the menus.
3. In the *Choose Document to Place* dialog, select `Sensor.SchDoc`.
4. The sheet symbol will appear floating on the cursor. Place the sheet symbol in an appropriate position on the sheet, as shown in Figure 1.
5. Another important point about sheet entries, their I/O type is an independent attribute from their style (the direction they point), unless you have the auto **Sheet Entry Direction** option enabled in the Schematic tab of the *Preferences* dialog. The SCL sheet entry was pointing inward when it was on the left, now that it is on the right it will be pointing out if the **Sheet Entry Direction** option is currently disabled. Open the *Preferences* dialog and confirm that the option is enabled.
6. Repeat this process of creating symbols for the MCU, LCD and Power sub-sheets. As you place each sheet symbol, you will notice that the sheet entries will appear on both sides of the symbol, depending on their I/O type: Inputs and Bidirectionals on the left, Outputs on the right. Drag the sheet entries onto the correct sides of each symbol, according to Figure 1.

**Note**: Sheets with no ports and even blank sheets if they are to be included in the hierarchy require having a sheet symbol created; otherwise the compile process fails to see these sheets.

7. Place the connector. It is a Header 3X2A, which can be found in the `Miscellaneous Connectors.IntLib` (one of the two integrated libraries installed by default).
8. Enable the **Place Sheet Entries Automatically** option in the **Schematic – Graphical Editing** page of the *Preferences* dialog. Use this feature to automatically place the sheet entries on the sensor schematic sheet symbol.

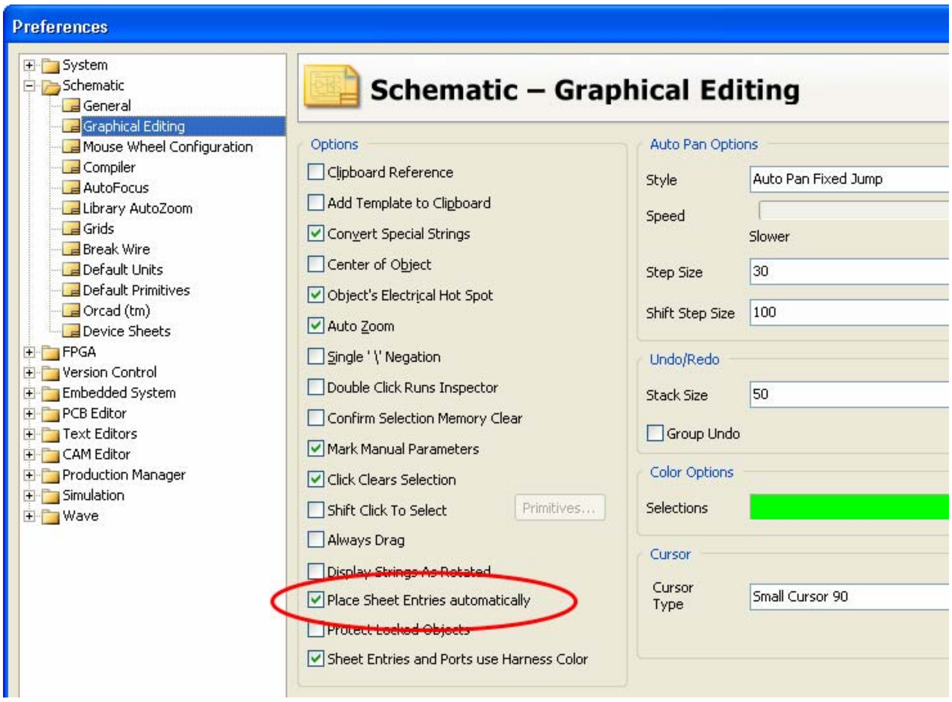



*Figure 4. Sheet entries can be placed automatically*

9. Wire the top sheet as shown in Figure 1.
10. Save the top sheet.

    This completes the capture phase of the design process. To confirm that the project hierarchy is correct we will now compile the design. This is covered in detail in section *Module 6 – Building the Project*, for now we will simply compile to show the correct structure of the design in the **Projects** panel.

11. To compile the project, select **Project » Compile PCB Project Temperature Sensor.PrjPcb**, or you can run the shortcut **CC**. Make sure you run the correct compile command, as there are two in the **Project** menu. One compiles the current schematic document, the other compiles the whole project. We want to compile the project.
12. Save the Project (right-click on the project in the **Projects** panel)

The design is now complete. However, before it can be transferred to PCB layout there are a few other tasks to complete, these include:

- Assigning the sheet numbers for each sheet in the hierarchy
- Assigning the designators
- Checking the design for errors

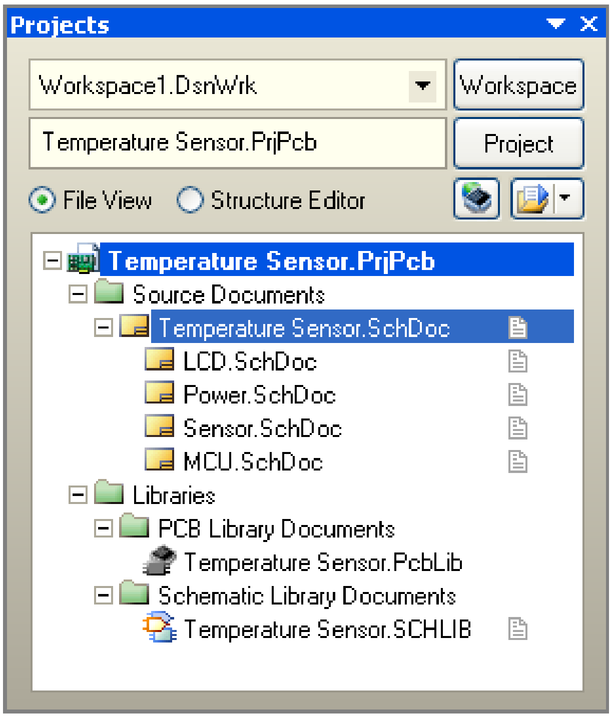



*Figure 5. The project hierarchy is displayed once the project has been compiled.*

## 5.1.4 Assigning the sheet numbers and total number of sheets

Sheet numbering is performed using documents parameters, linked to special strings placed on the schematics, as described earlier in *Module 4 - Schematic Capture*. Sheets can be automatically numbered by selecting the **Tools » Number Schematic Sheets** command.

- The *Sheet Numbering* dialog can be used to
  - number the sheets (**SheetNumber** parameter),
  - set the document number (**DocumentNumber** parameter),
  - set the total number of sheets (**SheetTotal** parameter).
- Click in the column to be edited to access the commands to edit that column.
- The sheets and documents can be numbered in a variety of ways, to do this click in the **SheetNumber** column, then click the **Auto Sheet Number** button.
- Cells can be edited manually, select the target cell(s), then right-click and select edit (or press the SPACEBAR). Alternatively, use the **Move Up** and **Move down** buttons, the number the sheets based on the **Display Order**.

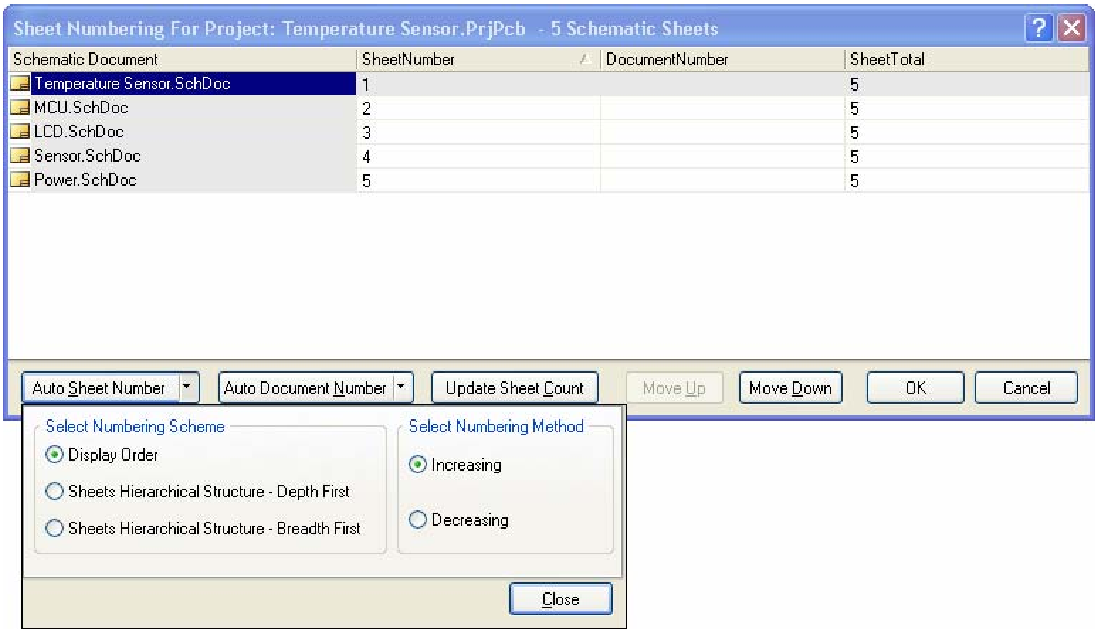



*Figure 6. Use the Sheet Numbering feature to review and update sheet numbers.*

> **Note:** Schematics appear in the Projects panel in the order they were added to the project. You can change this order if you want, simply click, drag and drop to re-order them within the Projects Panel.

### 5.1.4.1 Exercise – Number the sheets

1. Select the **Tools » Number Schematic Sheets** command in the schematic editor.
2. Sort the order of the schematics to match that of Figure 6 using the **Move up** and **Move down** buttons.
3. Press the buttons **Auto Sheet Number**, **Auto Document Number** and **Update Sheet Count**.
4. Press **OK** to save the changes.

> **Note:** Remember that these values are presented on the schematic sheets using parameters and matching special text strings. To display the values instead of the strings, enable the **Convert Special Strings** option in the **Schematic – Graphical Editing** page of the *Preferences* dialog.

### 5.1.5 Checking sheet symbol to sub-sheet synchronization

Typically the design hierarchy is not developed in a purely top-down or bottom-up fashion, the reality is that the design will evolve. This means that there will be modifications to the design that affect the net connectivity established between the sheet entries in the sheet symbol and the ports on the sub sheet below.

To manage the sheet entry to port relationships, use the *Synchronize Ports to Sheet Entries* dialog. Select **Design » Synchronize Sheet Entries and Ports** to display the dialog.

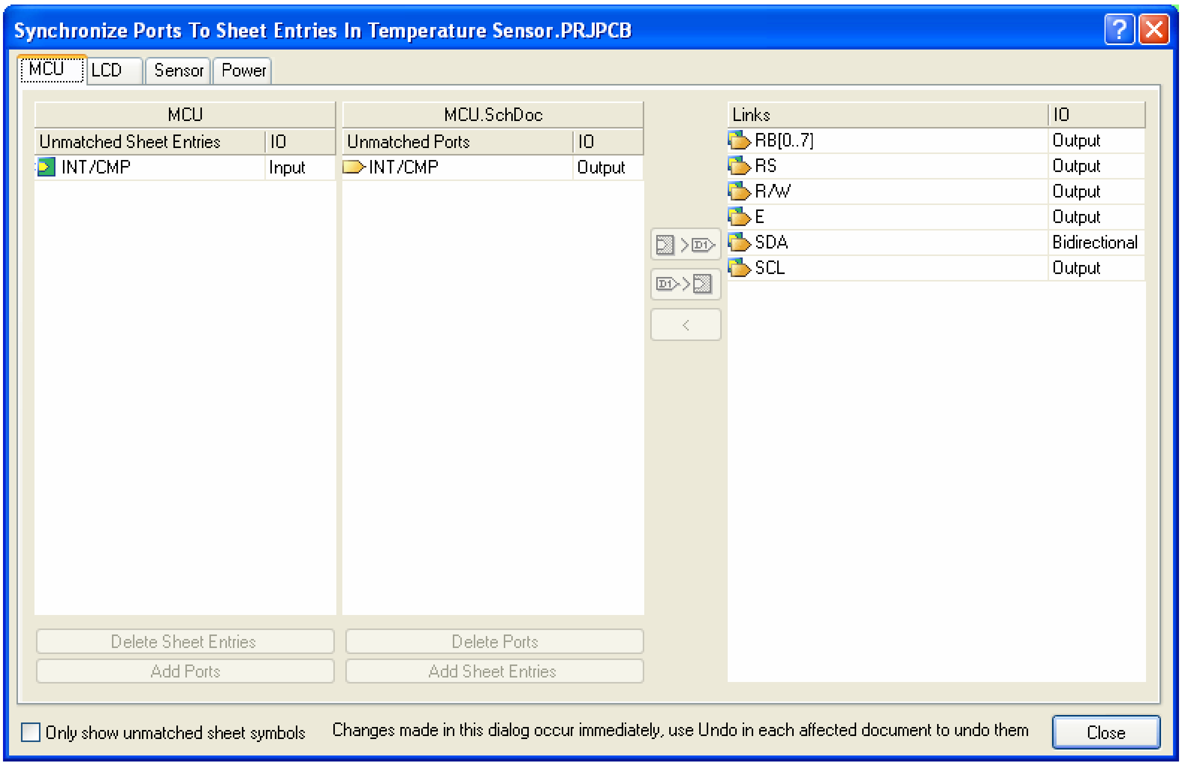



*Figure 7. Use the Synchronize dialog to ensure that sheet entries match with ports. Uncheck the checkbox down the bottom left to show all sub-sheets in the entire design.*

The Synchronize dialog can be used to:

- Match any selected Entry to any selected Port (name and IO type will be changed).
- Add or remove Entries or Ports to either the sheet symbol or the sub sheet.
- Edit the name or IO direction of a matched Entry / Port (done in the Links column on the right).

**Note:** Remember that changes made in the *Synchronize Ports to Sheet Entries* dialog are performed immediately; use the Undo command on each affected sheet to undo any updates.

### 5.1.5.1 Exercise – Synchronize the sheet entries and Ports

1. From any schematic in the temperature sensor project, run the command **Design » Synchronize Sheet Entries and Ports**.
2. You may see a problem like the one shown in Figure 7, where the I/O direction of the sheet entry (Input) does not match the I/O direction of the port (Output). You may also see a problem with the 3 sheet entries on the sensor sheet symbol, not having matching ports on the sensor schematic. Both of these can be resolved in the *Synchronize Ports to Sheet Entries* dialog.
3. To resolve the first problem, we need to determine which direction is correct. Once the correct direction has been determined, select each entry in the columns on the left, and then click on the  to drive the sheet entry direction onto the port, or click on  to drive the port direction onto the sheet entry.
4. The second problem is that the sheet entries exist, but they have no matching ports. Select all 3 sheet entries as shown in Figure 8, once you do you will have the option of either deleting the sheet entries, or adding ports. Click **Add Ports**, when you do the dialog will temporarily disappear, the Sensor schematic will become active, and the cursor will have 3 ports on it. Place them anywhere for now, then close the dialog and return to the Sensor schematic and correctly position the ports onto their wires.
5. Save your design.

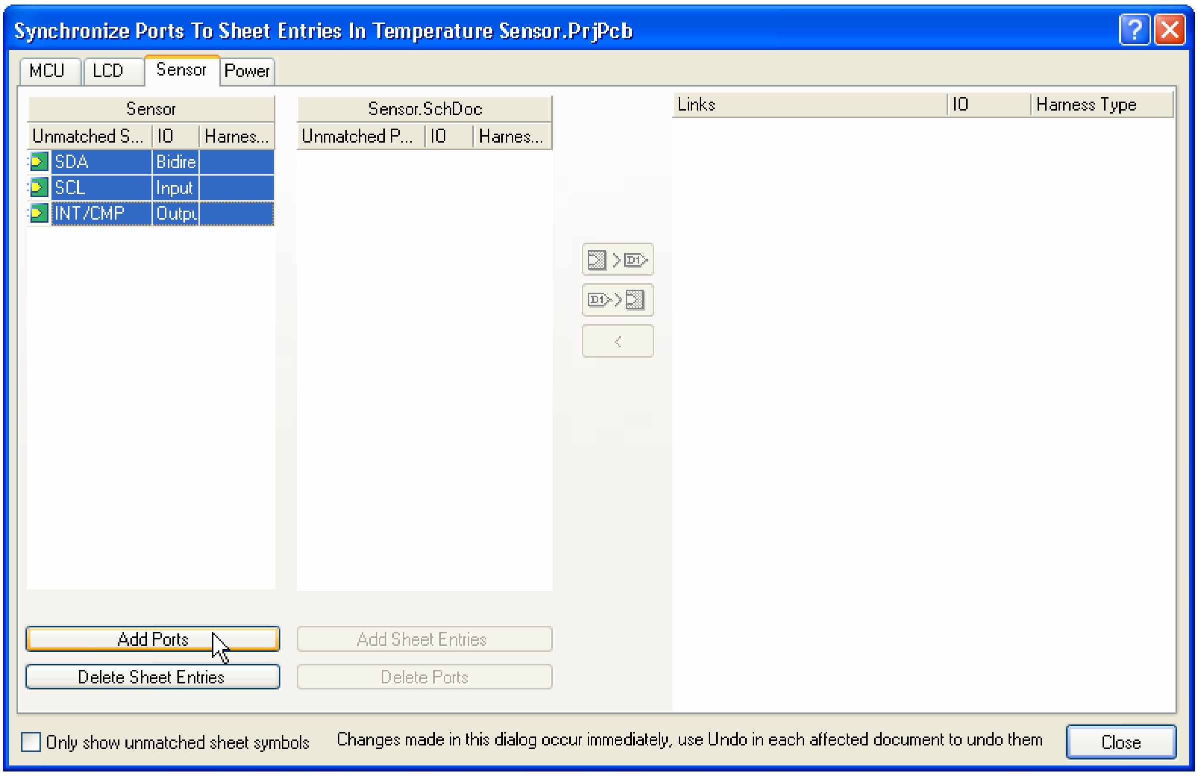



*Figure 8. Problems found - missing ports on a sheet*

TOP SECRET

Altium Designer

# Module 6: Building the Project

## Module 6: Building the Project





Module Seq = 6

# 6.1 Assigning designators

The Schematic Editor includes a positional-based re-annotation tool for allocating component designators.

## 6.1.1 Using Annotate to assign designators

The Schematic Editor provides an automated method of assigning designators. This is the Annotate command. This will take any component which has '?' appended to its designator and allocates a unique designator to those parts.

The order in which designators are assigned is based on the components' position on the sheet. The *Annotate* dialog allows you to set one of four positional annotation options. The annotation grid is based on the sheet border reference, so change the number of regions in the border reference to control the annotation grid.

To run Annotate, choose the **Tools » Annotate Schematic** menu command. This displays the *Annotate* dialog shown in Figure 1.

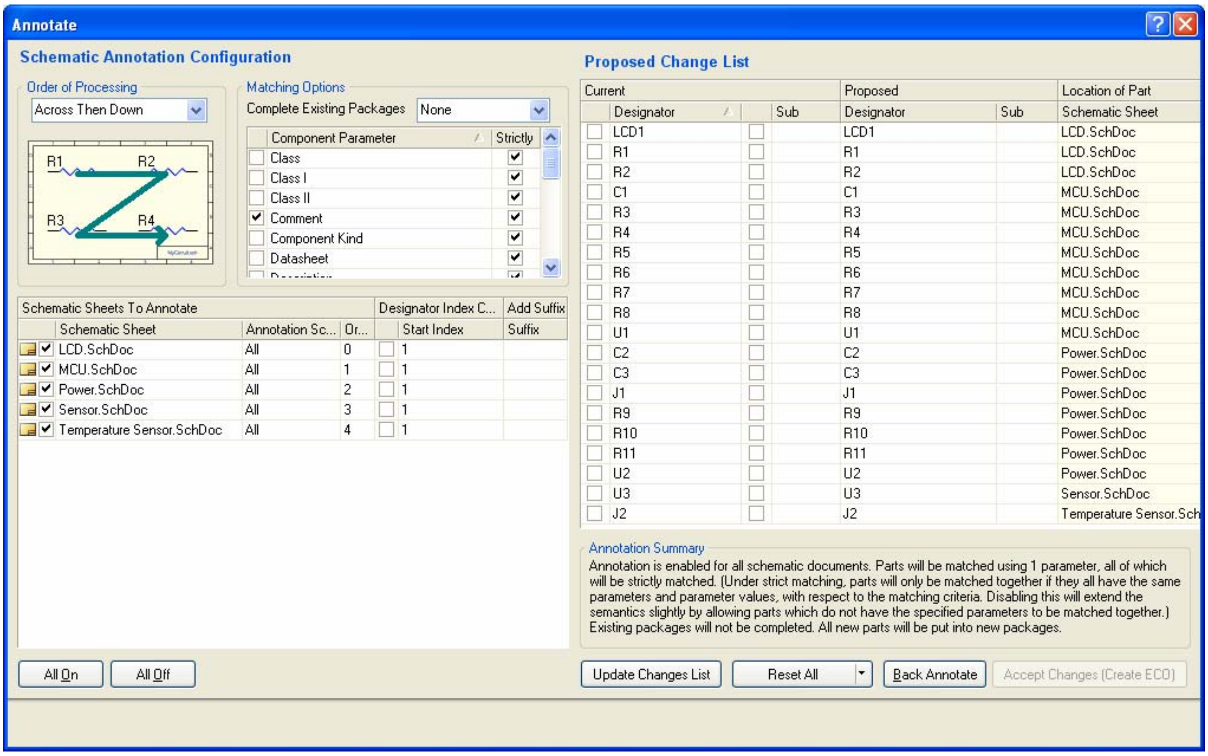



*Figure 1. Annotate dialog*

The Annotate options include:

- **Update Change List –** this button will reassign all designators that are not currently assigned (their designator currently ends in a ?).
- **Reset All –** use the Reset All button to reset all designators so that they end in a ?. You can also limit this to resetting only duplicates.
- **Order of Processing –** there are four directional options available. Select the preferred one at the top left of the dialog. This uses the sheet grid to define the across/down increments.
- **Matching Options –** enable the parameters to be used to package parts of a multi-part component. Typically, this is based on the component comment. If there are particular parts that must be packaged together, give both a common parameter and enable this parameter

in the **Component Parameter** list (e.g. filter-stage1). Note that the **Annotation Summary** down the bottom right of the dialog gives information about the matching behavior.

- **Schematic Sheets to Annotate –** this section of the dialog gives sheet-by-sheet control of the annotation, sheets can be excluded from the process and you can also control the annotation starting number for each sheet.
- **Back annotate –** click this to load a Was/Is file. This is only required if the board is not being designed in Altium Designer. If you are designing the board in Altium Designer you can back annotate directly from the PCB to the schematic by selecting the **Design » Update** menu option.
- Whenever an **Update** or **Reset** is performed an *Information* dialog will appear. This dialog details how many changes have been made from the previous state (since the last Update or Reset) and the information *dialog* also lists the changes from the original state (since the *Annotate* dialog was opened).
- Once you are happy with the designator assignments, click the **Accept Changes** button to generate an ECO. From the *ECO* dialog you can update the schematic.

**Note:** To prevent a component from having its designator changed by the Annotation process, enable the **Locked** checkbox adjacent to the **Designator** in that component's *Component Properties* dialog.

### 6.1.2 Designators on multi-part components

The suffix for multi-part components can be either Alpha or Numeric, depending on the Alpha Numeric Suffix option in the *Preferences* dialog. This is an environment setting and will apply to all open schematic sheets.

You can change parts within a component using the **Edit » Increment Part Number** command. Select this command and then click on the part of interest.

**Note:** To prevent multi-part component parts being swapped during the annotation process enable the **Locked** checkbox adjacent to the Part selector in the *Component Properties* dialog.

### 6.1.3 Project Order

The project order of a project is the order the documents appear in the projects panel. However when dealing with hierarchical designs, due to the top sheet becoming the root schematic, the project order can be difficult to determine. To make this process easier an option in the preferences can be turned on to see the project order numbered position. To turn this on goto **DXP » Preferences » System » Projects Panel**. In the General category, tick on the option **Show document Position in project**. Click **OK**, once ticked.

The picture in Figure 2 shows the numbering that appears when the position is shown. Once this data appears, the order can be changed by dragging and dropping the documents in the project panel to change the project order.

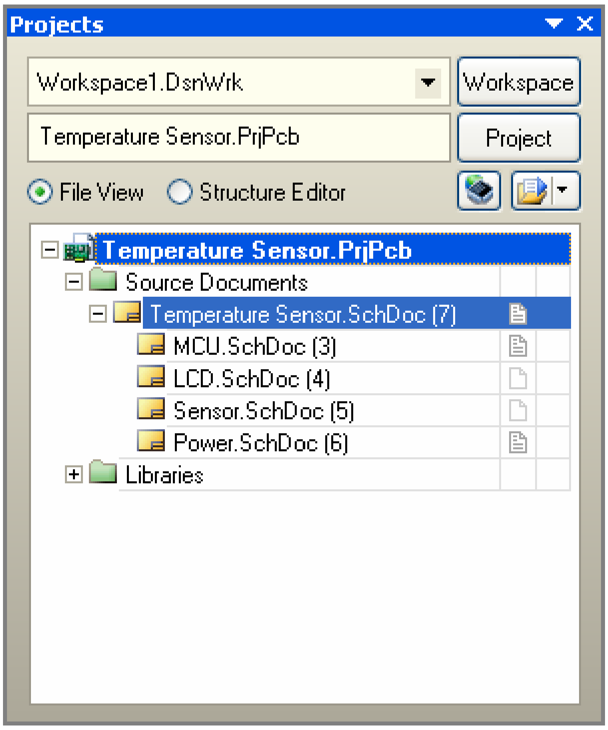



*Figure 2. Project panel with project order showing*

## 6.1.4 Exercise – Annotating the design

1. Select **Tools » Annotate Schematics** from the menus.
2. In the *Annotate* dialog, click the **Reset All** button, then click **OK** in the *Information* dialog that appears. Note that the **Proposed Designator** column in the dialog now shows all designators having a ? as their annotation index.
3. The Order the schematics will be annotated is configured in the **Schematic Sheets to Annotate** section of the dialog. You can change the Order manually, or you can right click and select **Order Alphabetically** or **Order By Project Order**, as shown in Figure 3. To check each schematic's position in the project order, enable the **Show Document Order** in Project option in the **System – Project Panel** page of the *Preferences* dialog and the number will be displayed in the Projects panel.

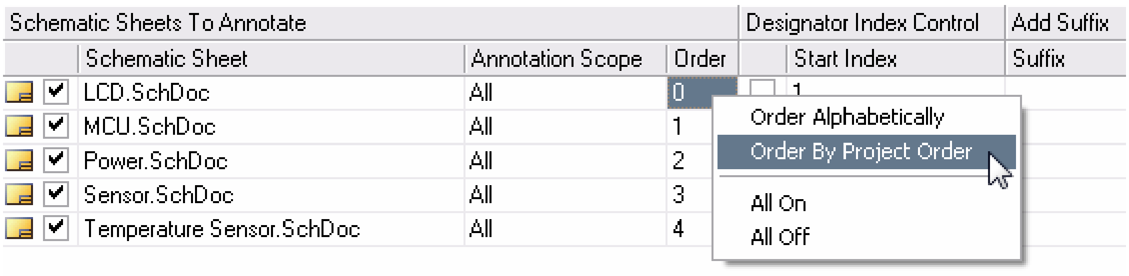
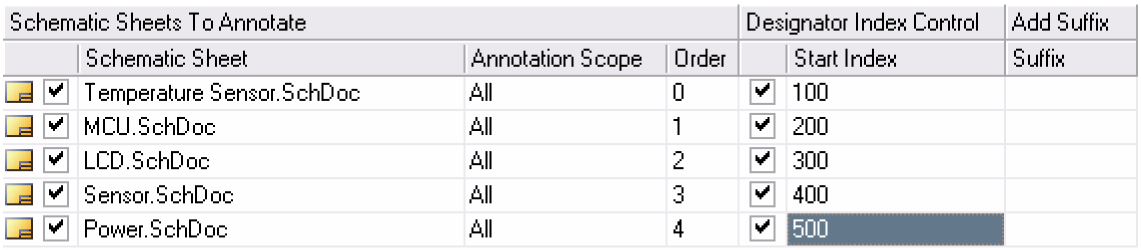



*Figure 3. Setting the sheet numbering order to Order By Project Order.*

4. Altium Designer supports annotating each sheet from a fixed starting index, to use this feature tick all the **Designator Index Control tick** boxes, as shown in Figure 4.

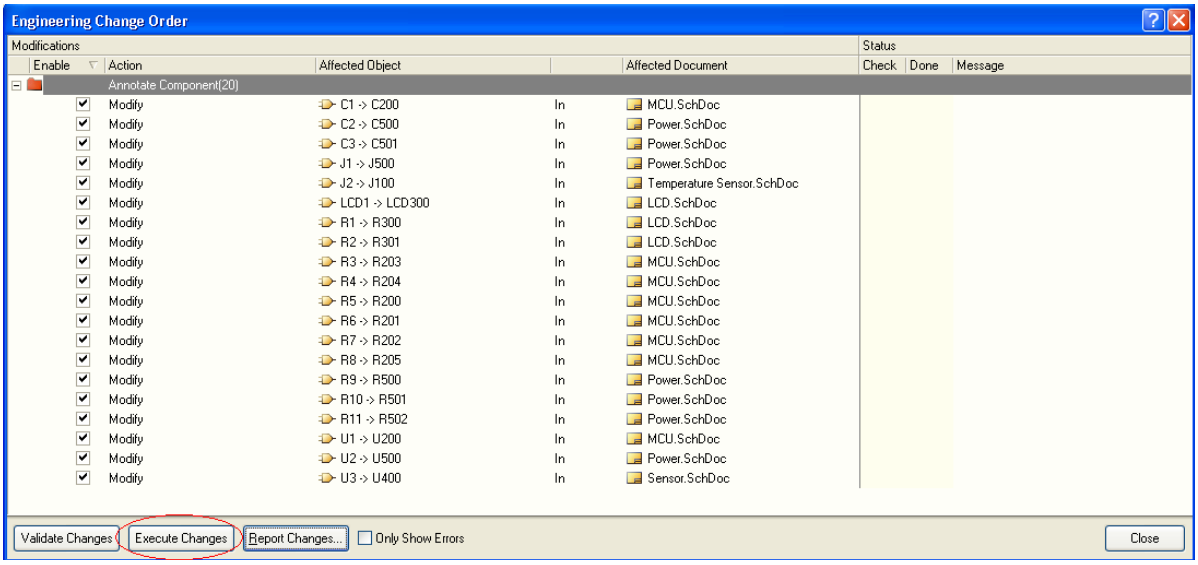


| Schematic Sheets To Annotate | | | | | Designator Index Control | Add Suffix |
|---|---|---|---|---|---|---|
| | Schematic Sheet | Annotation Scope | Order | | Start Index | Suffix |
| ✔ | Temperature Sensor.SchDoc | All | 0 | ✔ | 100 | |
| ✔ | MCU.SchDoc | All | 1 | ✔ | 200 | |
| ✔ | LCD.SchDoc | All | 2 | ✔ | 300 | |
| ✔ | Sensor.SchDoc | All | 3 | ✔ | 400 | |
| ✔ | Power.SchDoc | All | 4 | ✔ | 500 | |

*Figure 4. Set the Starting Index for each sheet*

5. Now set the **Start Index** for each sheet, as shown in Figure 4.
6. Click the **Update Changes List** button to assign a unique designator to each component. The components are annotated positionally, according to the direction setting selected at the top left of the **Annotate** dialog. The *Information* dialog that appears indicates how many designators have changed from their original state.
7. Repeat the process of resetting and assigning, trying different direction options (the actual designators on the schematic are not being changed as you do this). Finish with a direction option that you prefer.
8. To commit the changes and update the components, click the **Accept Changes** button to generate an ECO. Click **Execute Changes** in the *ECO* dialog, then close the *ECO* and the *Annotate* dialogs.

**Note**: Changes are only made if the ECO is executed.



Engineering Change Order

| Modifications | | | | | Status | | |
|---|---|---|---|---|---|---|---|
| Enable | Action | Affected Object | | Affected Document | Check | Done | Message |
| | Annotate Component(20) | | | | | | |
| ✔ | Modify | C1 -> C200 | In | MCU.SchDoc | | | |
| ✔ | Modify | C2 -> C500 | In | Power.SchDoc | | | |
| ✔ | Modify | C3 -> C501 | In | Power.SchDoc | | | |
| ✔ | Modify | J1 -> J500 | In | Power.SchDoc | | | |
| ✔ | Modify | J2 -> J100 | In | Temperature Sensor.SchDoc | | | |
| ✔ | Modify | LCD1 -> LCD300 | In | LCD.SchDoc | | | |
| ✔ | Modify | R1 -> R300 | In | LCD.SchDoc | | | |
| ✔ | Modify | R2 -> R301 | In | LCD.SchDoc | | | |
| ✔ | Modify | R3 -> R203 | In | MCU.SchDoc | | | |
| ✔ | Modify | R4 -> R204 | In | MCU.SchDoc | | | |
| ✔ | Modify | R5 -> R200 | In | MCU.SchDoc | | | |
| ✔ | Modify | R6 -> R201 | In | MCU.SchDoc | | | |
| ✔ | Modify | R7 -> R202 | In | MCU.SchDoc | | | |
| ✔ | Modify | R8 -> R205 | In | MCU.SchDoc | | | |
| ✔ | Modify | R9 -> R500 | In | Power.SchDoc | | | |
| ✔ | Modify | R10 -> R501 | In | Power.SchDoc | | | |
| ✔ | Modify | R11 -> R502 | In | Power.SchDoc | | | |
| ✔ | Modify | U1 -> U200 | In | MCU.SchDoc | | | |
| ✔ | Modify | U2 -> U500 | In | Power.SchDoc | | | |
| ✔ | Modify | U3 -> U400 | In | Sensor.SchDoc | | | |



*Figure 5. Engineering change order dialog for annotation changes,*

9. Note that each document that has been affected by the changes has an * next to its name on the document tab at the top of the window. Save all documents in the project.

# 6.2 Compiling and verifying the project

This section looks at how to verify a design, an essential step before transferring to PCB layout. In Altium Designer, checking the design is done by *compiling* the design which checks for logical, electrical and drawing errors.

To compile your design, select **Project » Compile PCB Project** or use the **CC** shortcut.

Compiled results are displayed in the **Messages** panel; from here, you can double-click to jump to an error or warning. The **Messages** panel will only open automatically if there are errors, if it is not visible click on the **System » Messages** button at the bottom of the workspace to display the panel.

> **Note**: The default error checking options are on the cautious side, you should always review the settings in the *Project Options* dialog and adjust them to suit your project and design requirements.

Once the design has been compiled, it can be navigated in the **Navigator** panel.

## 6.2.1 Setting up to compile the design

When you compile the design, DXP builds a connective model of the design – you can think of it as an internal netlist. The presence of the internal netlist allows you to navigate or browse the connective structure of the design.

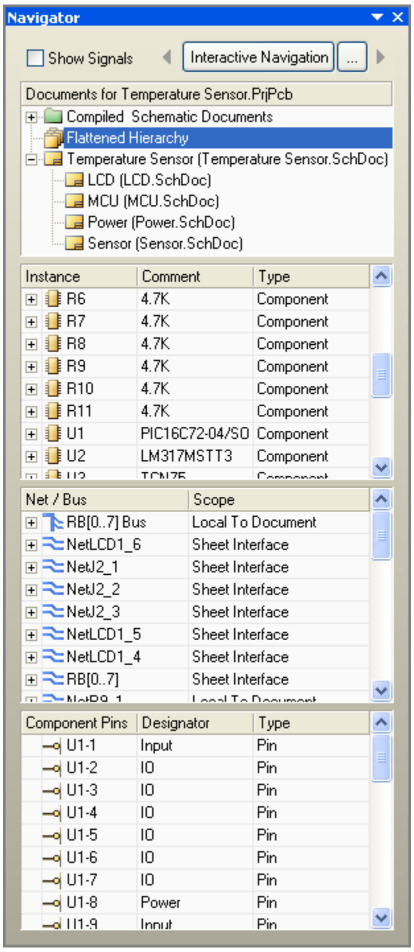



*Figure 6. Use the Navigator to check the design connectivity*

### 6.2.1.1 Compiler options

- Before the design can be compiled, the project options must be configured. This is done in **Options** tab of the *Options for Project* dialog (**Project » Project Options**).
- The **Net Identifier Scope** must be appropriate for the structure of the design. This was covered in *Module 5 - Multi-Sheet Design*.
- When the design is compiled, it can be navigated using the Navigator panel. The Navigator can be brought up by going to the panel control in the bottom right and selecting **Design Compiler » Navigator**. Select the **Flattened Hierarchy** at the top of the **Navigator**. When you click on a component or a net, that component or net will be displayed in the workspace.
- Expand the component or net using the small + sign to access all pins in the component or all pins/net identifiers in the net.
- Click the ... button to the right of the Interactive Navigation button to configure options that control how the workspace will be displayed.
  - **Zoom:** jump to the sheet and zoom in on the object of interest.
  - **Select:** select the objects of interest.
  - **Mask:** fade all objects except those of interest. Control the mask fade level using the **Mask Level** button at the lower right of the screen. Clear the Mask using the **Shift+C** shortcut.

- **Connective Graph:** show the connective relationship with either red (for net objects) or green (components) graph lines.

- The Navigate button in the panel allows you to navigate spatially. Click it to get a crosshair cursor, then click on an electrical object in the workspace, such as a wire, net label, port etc, to highlight all electrical connected objects.
- To move Up/Down in the hierarchy hold down the **CTRL** key and double click on either a sheet entry, Sheet symbol or Port to navigate the design.

### 6.2.1.2 Error Reporting options

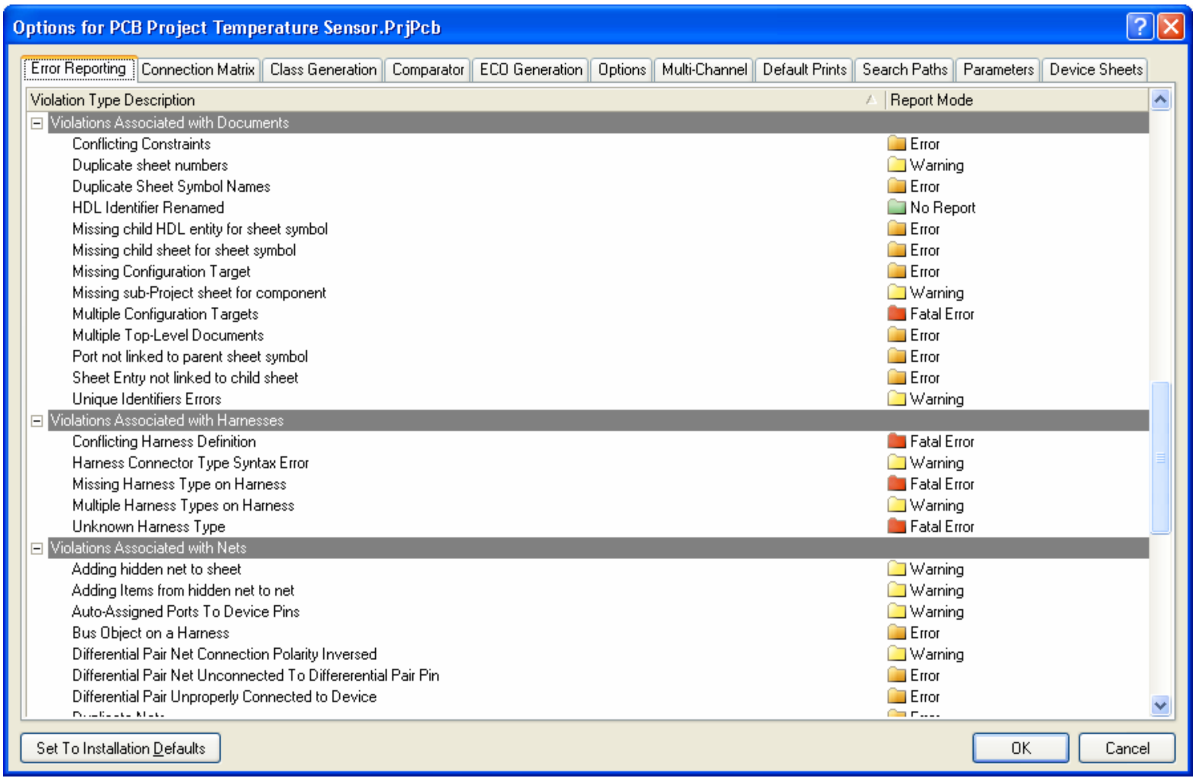



*Figure 7. Setup for Error Reporting*

- Error reporting options are configured in the **Error Reporting** tab and the **Connection Matrix** tab.
- There is an extensive array of error reporting options which have default settings that are on the cautious side. Generally, it is better to compile the design and then if there are warnings that are not an issue for your design, change the reporting level.
- One option of interest is **Nets with only one pin**. This can be used to find single node nets, where a pin has been connected to a port or Netlabel, but does not connect to another pin. This is set to **No Report** by default.

### 6.2.1.3 Connection Matrix

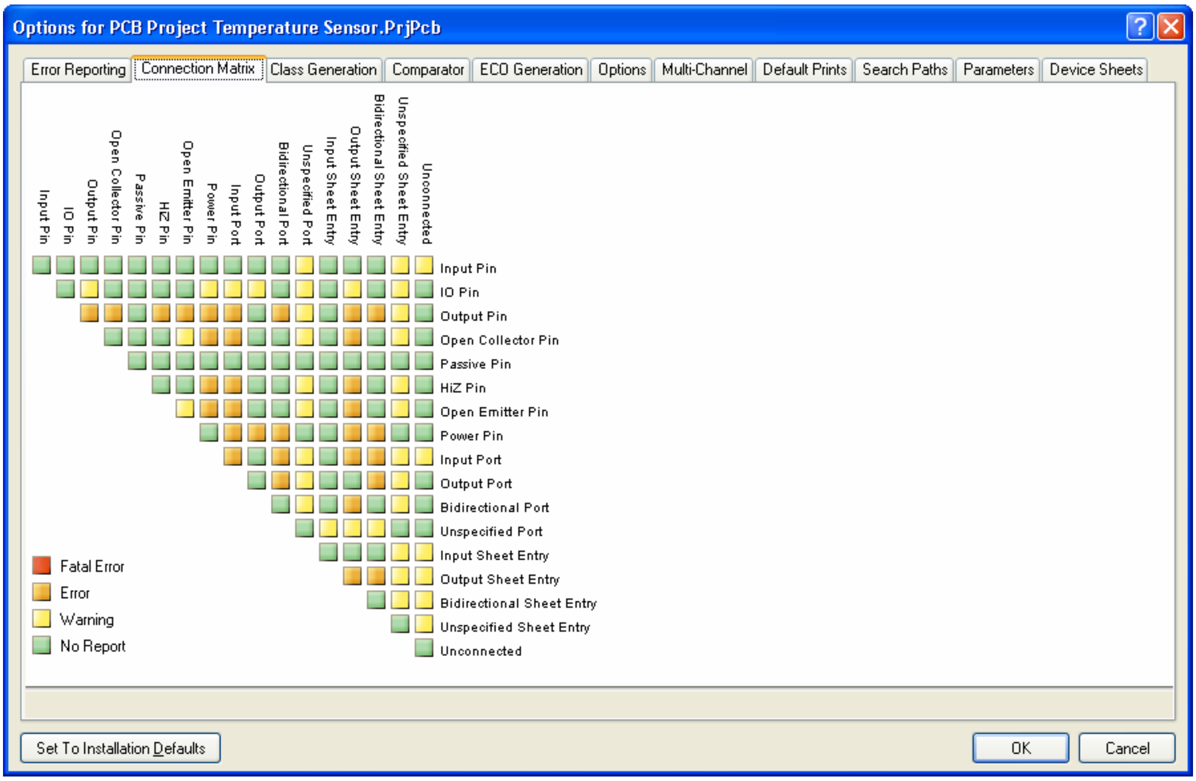



*Figure 8. ERC Rule Matrix tab*

- The **Connection Matrix** tab in the *Options for Project* dialog is shown in Figure 8. This matrix provides a mechanism to establish connectivity rules between component pins and net identifiers. It defines the logical or electrical conditions that are reported as warnings or errors.
- For example, an input pin connected to an input pin would not normally be regarded as an error condition, but connected output pins would not. This is reflected in the table.
- Rules can be changed by clicking on the appropriate square in the matrix, causing it to cycle through the available options.

## 6.2.2 Interpreting the messages and locating the errors

- When you compile the project, any conditions which generate a warning or error will be listed in the **Messages** panel. Note that the **Messages** panel will only open automatically if there is an error condition.
- Double-click on a warning/error to pop up the **Compile Errors** panel, then double-click on an object in that list to jump to it on the schematic.
- Right-click in the **Messages** panel to clear messages. Click on the column headings to sort by that column. Double-click on a message to display the **Compile Errors** panel in which you can double-click to cross probe to that object.
- Subsequent compilations will remove warning/error messages once the error conditions have been corrected.
- It is important to examine each warning/error and resolve them, change the error checking Report Mode, or mark them with a No ERC marker. This should always be done prior to transferring the design to PCB layout.

#### 6.2.2.1 Exercise – Configuring the project options

1. Select **Project » Project Options** to display the *Options for Project* dialog and click on the **Options** tab.
2. For this project, the **Net Identifier Scope** can be left on automatic. Enable only the **Allow Ports to Name Nets** in the **Netlist Options**.
3. Click **OK** to save the settings. Note that the Project PCB file has been modified. You should save the Project PCB file.

#### 6.2.2.2 Exercise – Design verification

1. Check your design by compiling your design and checking any errors or warnings.
2. Resolve any errors. Note that **Nets with no driving source** reports any net that does not contain at least one pin of the following electrical types: IO, Output, OpenCollector, HiZ, Emitter or Power.
3. If you have any remaining warnings that will not affect your design, you can consider turning that warning type to *No Report* in the **Error Reporting** tab of the *Options for Project* dialog.

### Some tips

- Examine each of the objects associated with the error.
- Enable the Graph option to examine the connectivity of a net. Once a net is selected in the Navigator panel, it is highlighted throughout the design. You can also ALT+click on a net to highlight it on the current sheet.
- Errors with input pins are often due to problems with their source. If the input looks OK, trace the signal back to the source (output pin / port).

| **Note:** To open a sub-sheet, hold CTRL as you double-click on the sheet symbol. |
|---|

# Module 7: Setting Up for Transfer to PCB and Importing Data

TOP SECRET

# Module 7: Setting Up for Transfer to PCB and Importing Data





Module Seq = 7

# 7.1 Setting Up for Transfer to PCB

This section outlines how to transfer a schematic design to the PCB Editor using the Synchronizer or a netlist.

## 7.1.1 Setting the relevant project options

There are a number of settings that control what data is transferred between the schematic design and PCB layout. Select **Project » Project Options** to display the *Options for Project* dialog and click on the **Comparator** tab.

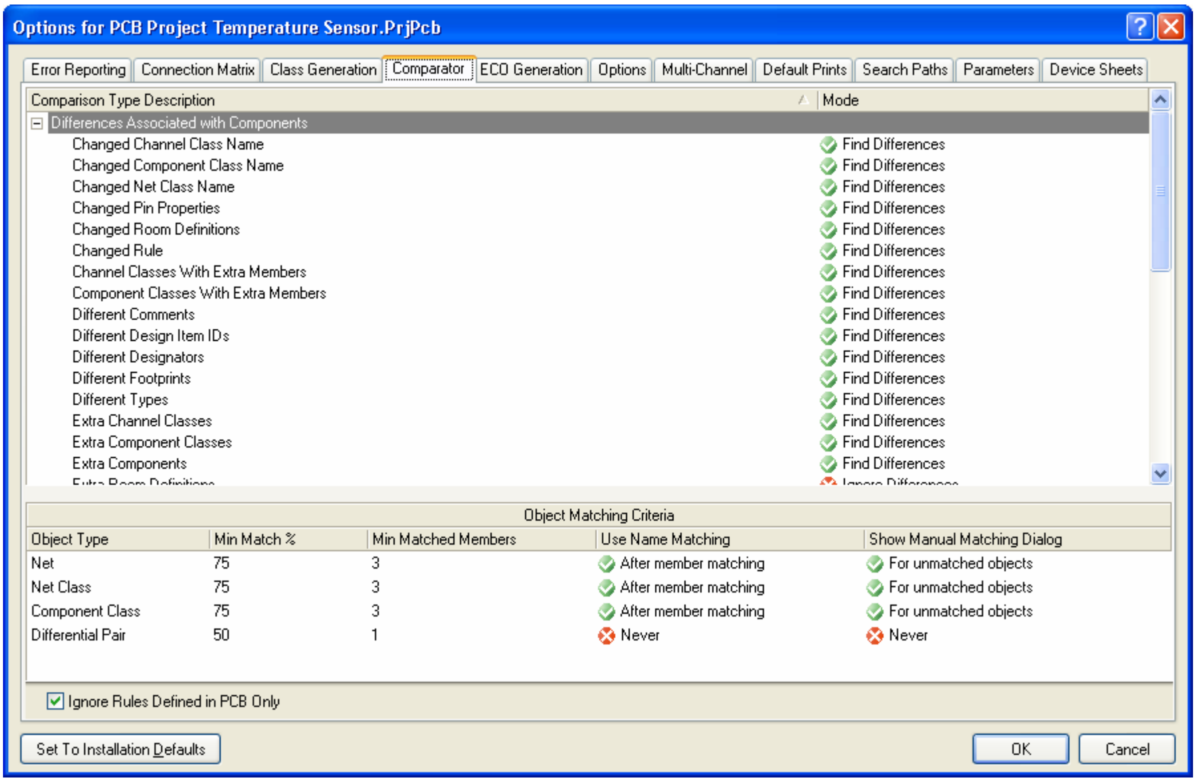



*Figure 1. The Comparator options define what information is transferred to PCB.*

By default, all options are on. For a simple design such as the training design, you might not want Placement Rooms to be created for each schematic sheet.

## 7.1.2 Transferring a design to the PCB Editor

- If you are using Altium Designer's PCB Editor to do the board layout, the best method of transferring design information between the schematic and the PCB (and from the PCB back to the schematic) is the Design Synchronizer. Using the Synchronizer, there is no need to create a netlist in the schematic and load that netlist into the PCB. Selecting **Design » Update PCB** will start the synchronization process.
- When you have a PCB and select this command, the *Engineering Change Order* dialog will be displayed. This lists all the changes that must be made to the PCB to get it to match the schematic. This process will be covered in detail during the PCB training module.
- You can also transfer the design using the **Project » Show Differences** command. This uses the design synchronizer, but gives more comprehensive control of the transfer process.

## 7.1.3 Netlist formats

A netlist is an ASCII file that contains the component and connectivity information defined in the schematic. The netlist can be used to transfer component and connectivity information to other design tools, including PCB Design packages from other vendors. Note that you can still use it to transfer to Altium Designer's PCB editor, but since it does not include unique component ID information it is an inferior method of design transfer.

Netlists are generated by using the **Design » Netlist for Project** menu. By default, there are 25 netlist formats in the menu, including EDIF, Xspice and Multiwire.

## 7.1.4 Design transfer using a netlist

For most situations, the Synchronizer has superseded netlist loading. In cases where the PCB is being designed from a schematic drawn on another EDA vendor's schematic editor, a netlist can be used.

Using the difference engine, the component and connectivity information in the netlist can be compared to the PCB.

Using a netlist is not as powerful as direct synchronization since during direct synchronization components on both the schematic and PCB is issued with a unique ID (UID). By using UIDs, the designators are not required as the synchronization link and can be changed at will on both sides.

### 7.1.4.1 Loading a netlist

To load a netlist:

- Select the **Project » Show Differences** menu command. This displays the *Choose Documents to Compare* dialog.
- Enable the **Advanced** check box, as shown in Figure 2.

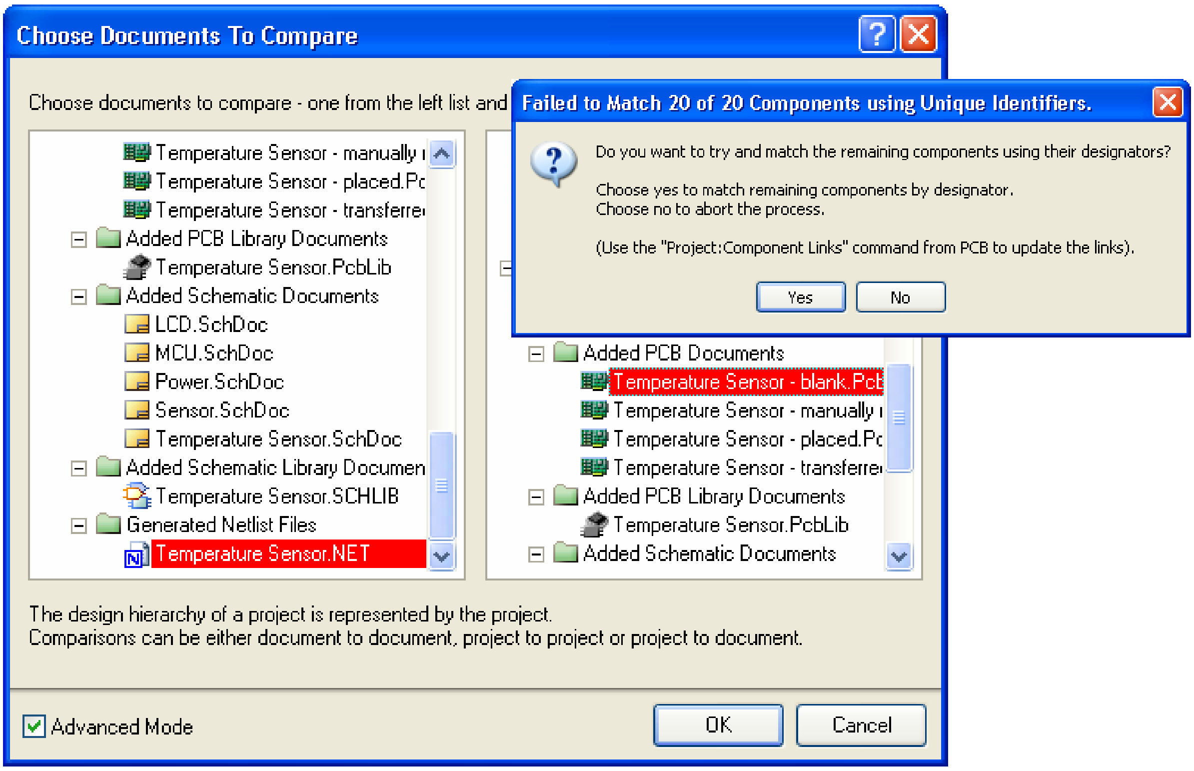



*Figure 2. Advanced mode chosen in the Choose Documents to Compare dialog*

- Select the required Netlist on one side and the PCB on the other. The Netlist must either be open in Altium Designer or included in the Project.
- When you click **OK**, the *Confirm* dialog will indicate that it is unable to match using UIDs. Click **Yes** to proceed using designators to match by.
- The *Difference* dialog will appear from where the process is the same as direct synchronization.

### 7.1.5 Exercise – setting project options for design transfer

1. Open the *Options for Project* dialog, and display the **Comparator** tab.
2. Set the Extra Room Definitions option to **Ignore Differences**.
3. Close the dialog and save the project.

TOP SECRET

Altium Designer

# Module 8: PCB Editor Basics

# Module 8: PCB Editor Basics





Module Seq = 8

# 8.1 PCB Editor Basics

The PCB Editor opens when you open or create a PCB document. It shares all the workspace features offered by the Altium Designer environment.

## 8.1.1 PCB Editor User Interface

Use of the PCB Editor is consistent with the Schematic Editor, with additional features that are detailed in the following sections.

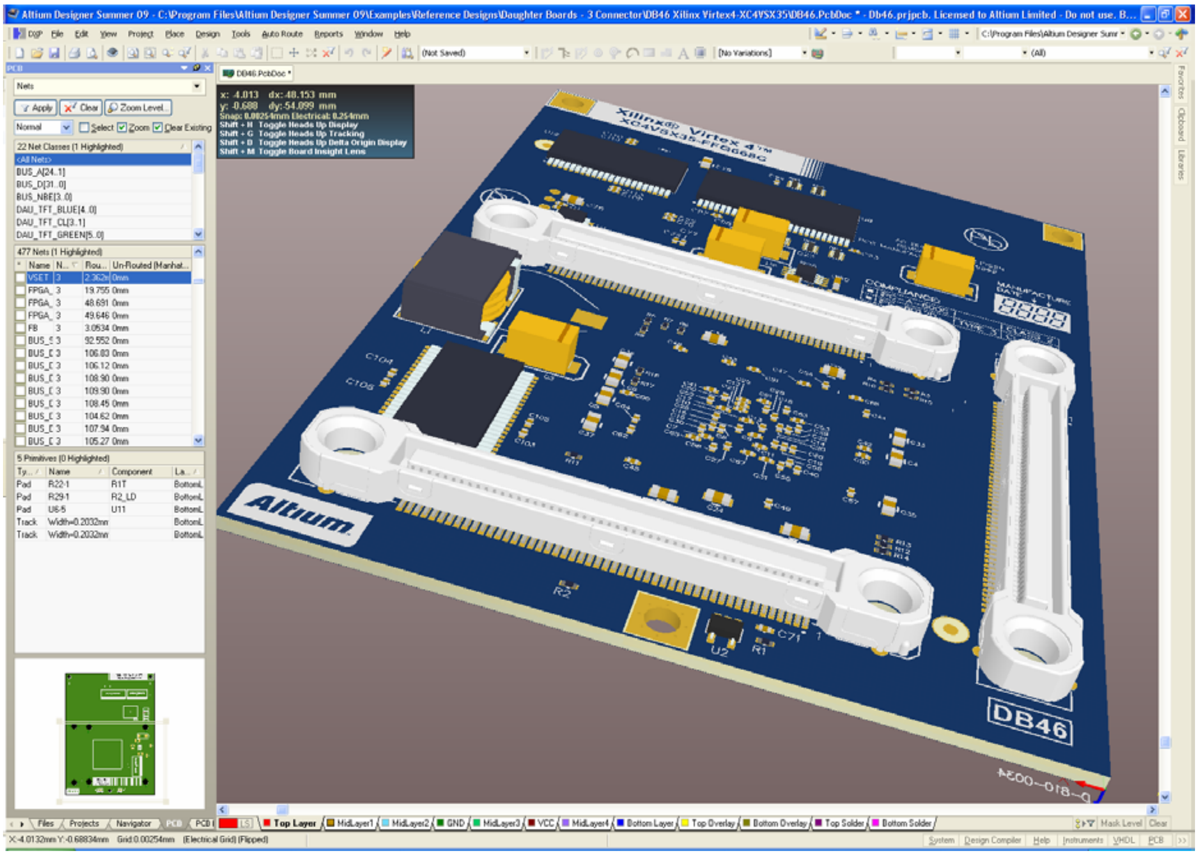


*Figure 1. PCB Editor workspace*

### 8.1.1.1 Layer tabs

A PCB is fabricated as a series of layers, including copper electrical, insulation, protective masking, text and graphic overlay layers. The tabs associated with each layer are located along the bottom edge of the PCB Editor design window. They allow you to switch the current layer and give a visual indication of which layers are currently being displayed and which is the current layer (the highlighted tab). If there are more layer tabs than can be displayed at one time, use the arrows to scroll through the tabs. Layer colors will be displayed to the left of the layer tabs and clicking the layer color will launch the Board Layers and Colors dialog.


**Note:** To switch between signal layers use the numerical pad * key and for all layers use the numerical pad **+/-** keys.

## 8.1.2 View Commands

The **View** commands can be accessed in the **View** menu or the Main toolbar. The table below lists the main display commands.

| Menu Command | Toolbar | Shortcut | Description |
|---|---|---|---|
| Fit Document |  | **VD** | Fits all objects in the current document window |
| Fit Board | | **VF** | Fits all objects located on signal layers in the current document window |
| Area |  | **VA** | Display a rectangular area of document by selecting diagonal vertices of the rectangle |
| Around Point | | **VP** | Display a rectangular area of document by selecting the centre and then a vertex of the rectangle |
| View Selected Objects |  | **VE** | Fits selected objects in the current document window |
| View Filtered Objects |  | **VE** | Fits filtered objects in the current document window |
| Zoom In | | **VI** | Zooms in on cursor position |
| Zoom Out | | **VO** | Zooms out from cursor position |
| Zoom Last | | **VZ** | Returns display to its state before the last view command |
| Refresh | | **VR** | Updates (redraws) the screen |

*Table 1. View command summary*

The following shortcut keys are very useful for manipulating the view of the document window. These shortcut keys can be used at any time, i.e. even when executing commands.

| Keystroke | Function |
|---|---|
| END | Redraws the view |
| ALT+END | Redraw Current layer |
| PAGE DOWN | Zoom out (holds the current cursor position) |
| PAGE UP | Zoom in (holds the current cursor position) |
| CTRL+PAGE DOWN | View Document |
| CTRL+PAGE UP | Massive Zoom In around the current cursor position |
| HOME | View pan (pan to centre the current cursor position) |
| ARROW KEYS | Moves the cursor by one snap grid point in the direction of arrow |
| SHIFT+ARROW KEY | Moves the cursor by 10 snap grid points in the direction of arrow |

*Table 2. Shortcut keys for PCB view manipulation*

### 8.1.2.1 Autopanning

Autopanning becomes active when executing commands, i.e. when the cursor appears as a crosshair. When in this state, touching any edge of the document window will initiate autopanning.

The autopanning speed is controlled via Autopan Options section of the **PCB Editor – General** page of the *Preferences* dialog (**Tools » Preferences**). Autopanning can also be turned off here.

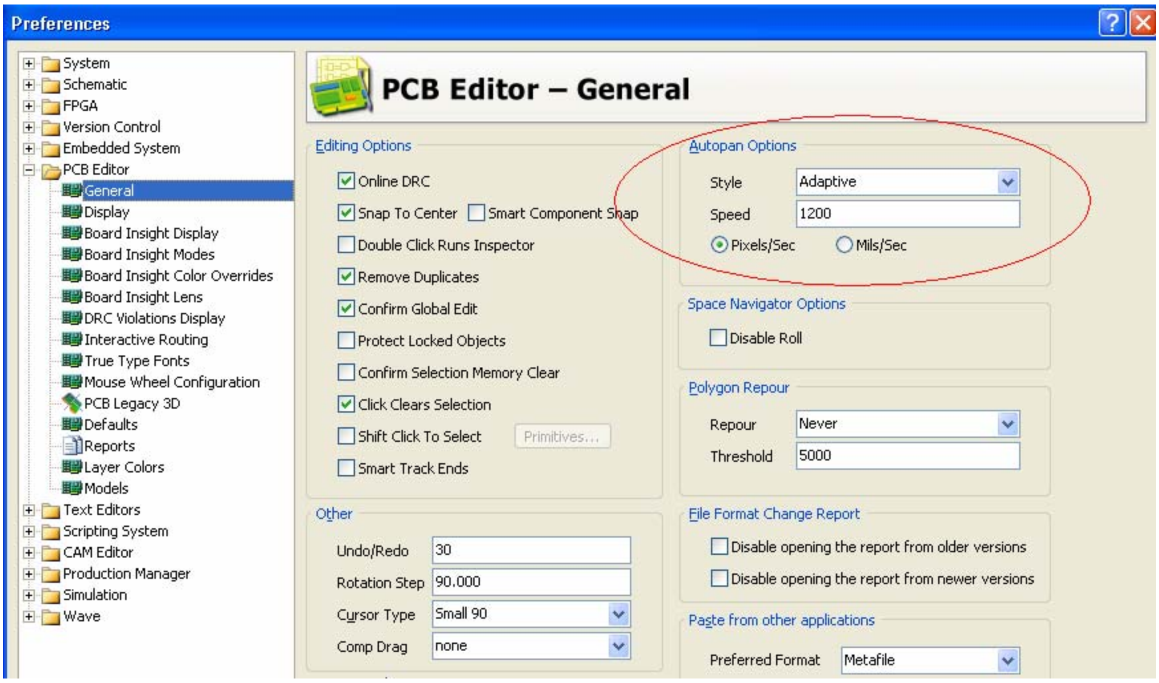



*Figure 2. Autopan settings in preferences*

#### 8.1.2.2 Right mouse panning

You can also use the Right Mouse pan feature to pan across your PCB document.

1. Place the cursor in the PCB Editor workspace.
2. Right-click and hold. A hand symbol displays on the cursor.
3. Move the cursor in the desired direction to pan.

**Note**: Once the cursor is off the sheet, the panning will stop and you will need to release the right button and repeat the process.

#### 8.1.2.3 Displaying connection lines

The **View » Connections** menu command displays a menu that allows displaying or not displaying of connection lines either by net, component net or the whole board.

# 8.2 PCB design objects

## 8.2.1 General

A variety of objects are available for use in designing a PCB. Most objects placed in a PCB document will define copper areas or voids. This applies to both electrical objects, such as tracks and pads, and non-electrical objects, such as text and dimensioning. It is therefore important to keep in mind the width of the lines used to define each object and the layer on which the object is placed.

Most of the PCB design objects are also referred to as *primitives* that can be edited in the PCB Editor. Components are made up of a variety of primitive objects and are editable only in the PCB Library Editor. Placing components, polygon planes, split planes and rooms will be covered in detail in up and coming modules.

For an example of each PCB design object, open `PCB Objects.PcbDoc` found in the `Practice Documents` folder in `\Altium Designer Summer 09\Examples\Training`.

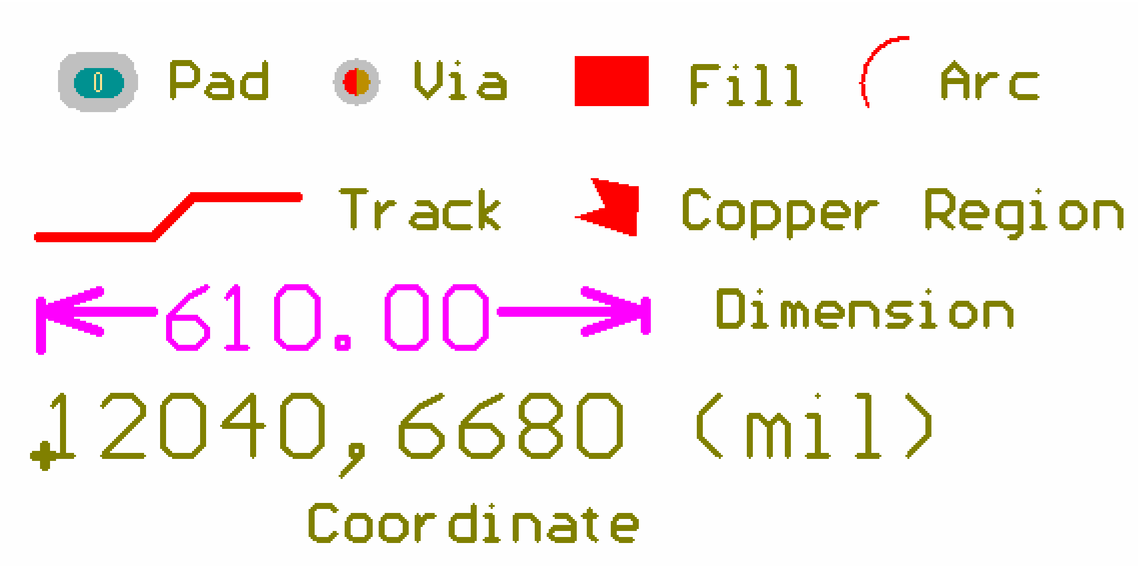
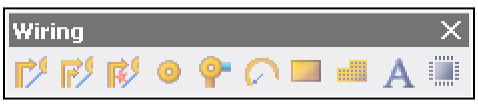



*Figure 3. The PCB Editor primitive objects*

- The object placement commands are selected using either the **Place** menu or the **Wiring** and **Utilities** toolbars.




- To set the properties of an object while placing it, press the TAB key and the *Properties* dialog for that object will be displayed.
- Once an object is placed, you can change its properties by double-clicking on it to display the *Properties* dialog for that object. Alternatively, you can click once to select an object, and then edit the properties in the Inspector panel (**F11** to open).
- Set the default properties for each object type in the **Defaults** section of the *Preferences* dialog (**Tools » Preferences**).

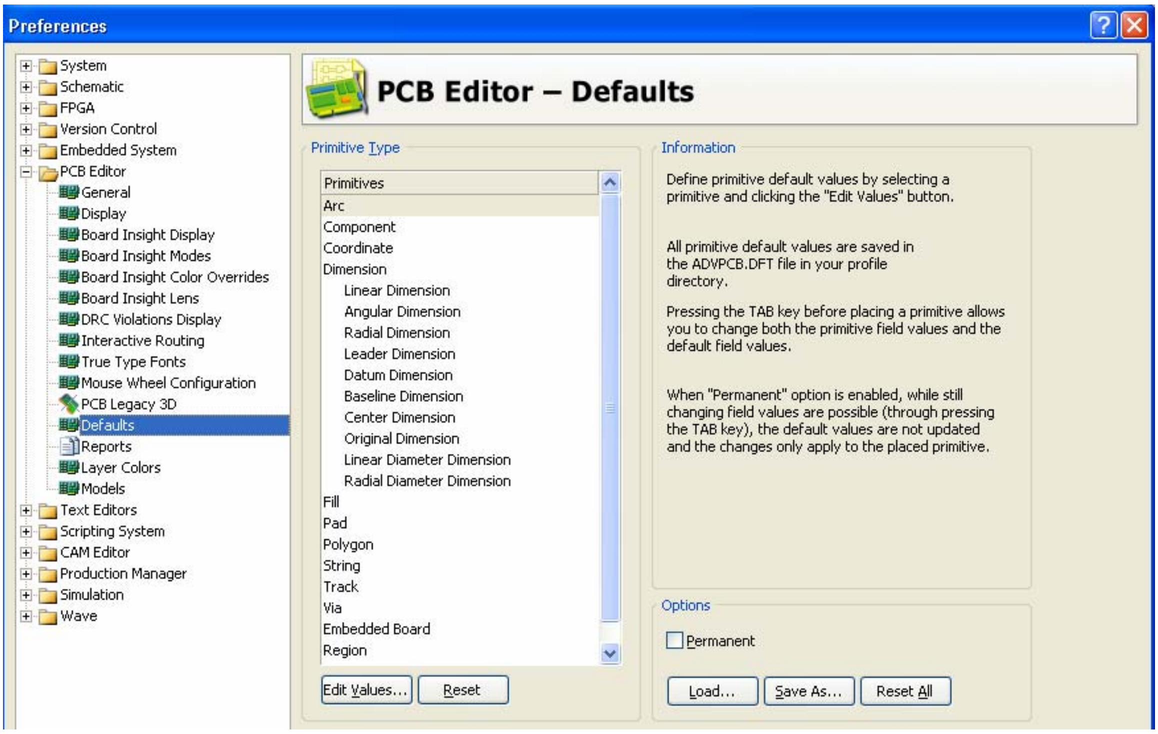



*Figure 4. Default Primitives in PCB preferences*

- The current layer determines the layer on which the object is placed.

## 8.2.2 Tracks

The Interactive Routing command is used to place tracks with associated net information.

To start Interactive Routing, select the  toolbar button or **Place » Interactive Routing** (**PT**). Click where you wish to begin the first track and then use the track placement and start/end modes detailed below.

Pressing TAB during interactive routing will display the *Interactive Routing* dialog where you can set widths, sizes and related design rules.

You can change the signal layer that you are routing on by pressing the * (asterisk) shortcut key on the keypad and a via will be automatically added.

### Track Placement modes

Once you are in the interactive routing command and have clicked to start the first track, press SHIFT +SPACEBAR to change the placement mode. Each mode defines a different corner style. Check the status bar to see which mode is active.

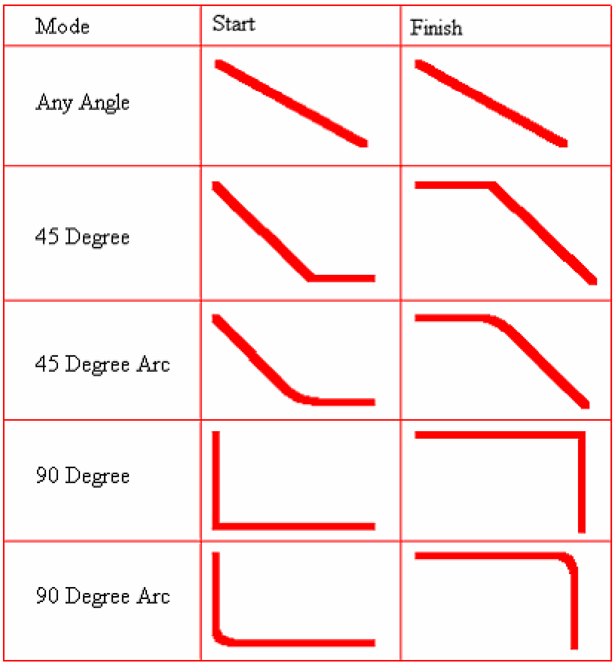



*Figure 5. Track Placement modes*

There are five track placement modes:

1. Any angle
2. 45 degree
3. 45 degree with arc - 45 degree line with rounded corner.
4. 90 degree (horizontal and vertical).
5. 90 degree with arc - horizontal and vertical orientation with rounded corner.

> **Note**: The two arc in corner modes use the Corner Style design rule to define the arc size. If the rule includes a range in the setback size then you can adjust the arc *within* this range during track placement by holding the comma key (,) to make it smaller, or the full stop key (.) to make it bigger.

**Start and Finish modes**

In addition, the track placement modes are supplemented with a Start Mode and a Finish Mode (see image Track Placement Modes above). After you have selected the Track Placement mode, you can press the SPACEBAR to toggle between the Start Mode option and the Finish Mode option.

If a track starts at an object with a net assigned to it, the track will also be assigned to the net. The interactive routing command will adhere to any rules assigned to that net.

A routed net can be highlighted by holding down the CTRL key as you click on it. Use SHIFT+CTRL+CLICK to highlight multiple nets.

### 8.2.2.1 Graphically modifying tracks

When a track segment is selected, three handles appear — one at each end of the segment and one in the middle. Below are the actions that can be performed.

**To re-position a segment end**

1. Place the cursor on one of the end handles.
2. Click and hold the left mouse button.
3. Move cursor (and the attached vertex) to new location. Altium Designer will add track segments to maintain orthogonal/diagonal patterns in the tracks.

**Breaking a track segment in the middle**

1. Place the cursor on the middle handle.
2. Click and hold the left mouse button.
3. Move the cursor. Altium Designer will add track segments to maintain orthogonal/diagonal patterns in the tracks.

**Drag the track segment away from other track segments**

1. Deselect all track segments.
2. Click and hold on the track segment.
3. Drag the segment to a new location.

> Note: Older versions of Altium Designer and Protel did not support maintaining neat orthogonal/diagonal patterns in the tracks when tracks were dragged – in earlier versions the handle remained attached to the cursor and it was up to the designer to position it to maintain neat track placement. This mode is still available, hold down the **Alt** key when starting to drag, and release **Alt** once you are moving. This is discussed in more detail in *Module 18 - Routing and Polygons*.

### 8.2.3 Lines

The Place Line command is provided for placing lines other than tracks, such as the board outline or keepout boundaries on non-electrical layers. Line placement behaves exactly the same as track placement during interactive routing; however, lines have no nets associated with them. When placed on non-electrical layers, lines are not constrained by the design rules.

Pressing TAB when placing lines displays the *Line Constraints* dialog. Note, however, that when you double-click on a line to edit its properties, the *Track* dialog displays.

To draw lines, select the toolbar button or **Place » Line**.

### 8.2.4 Pads

- Place pads using the **Place » Pad** command or the **Place Pad** toolbar button .
- Pads are mainly used as part of components but can be used as individual objects, such as testpoints or mounting holes.

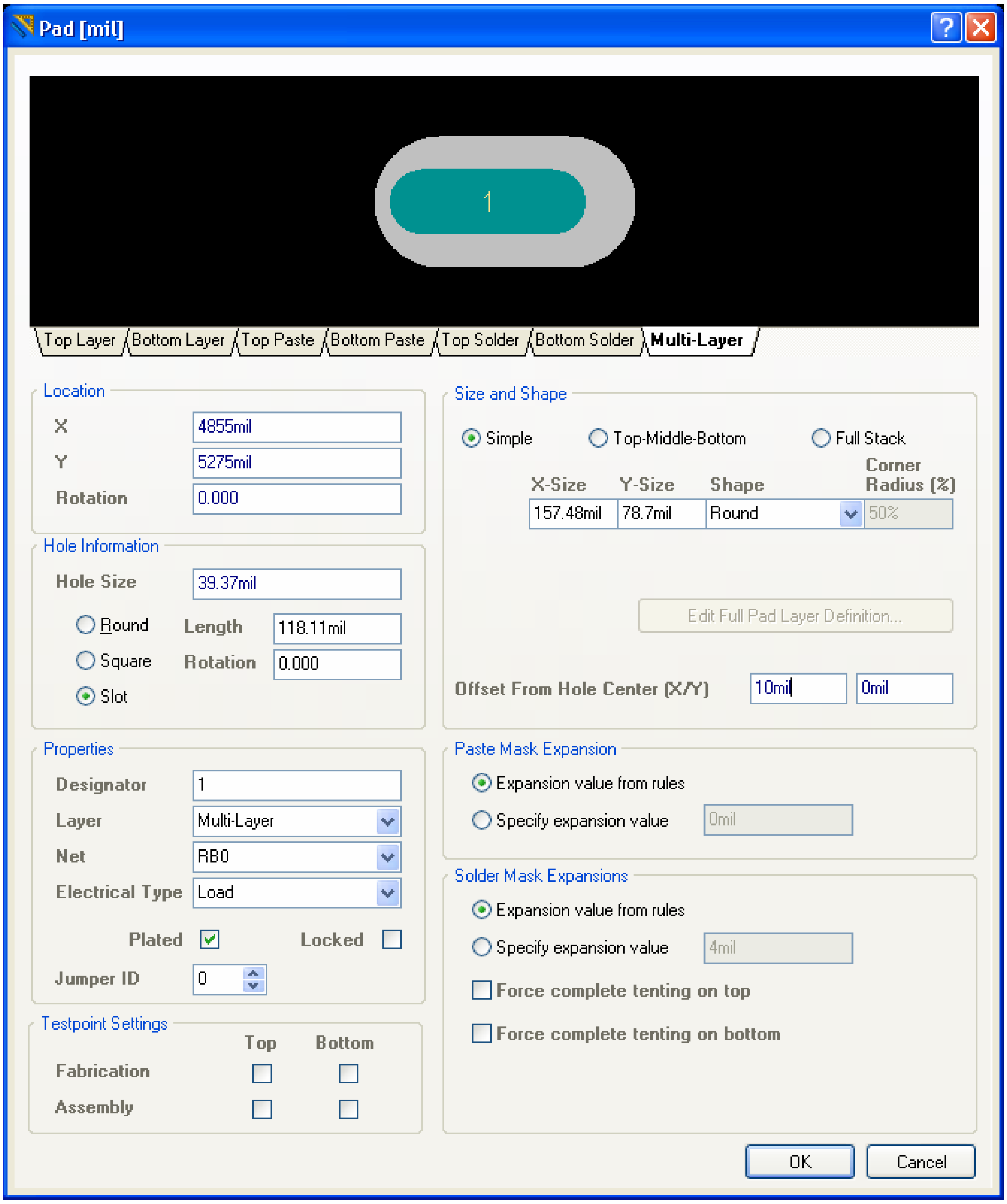



*Figure 6. Pad Properties dialog*

- Pad properties are set in the *Pad* dialog that is displayed by pressing the TAB key while placing the pad or double-clicking on a placed pad.
- If a pad is to have different sizes on the mid layers or bottom layer, check **Top-Middle-Bottom** in the Size and Shape section. Click on **Full Stack** and then **Edit Full Pad Layer Definition** to edit more complicated stack ups.
- Assign a net to the pad, define the pad's electrical type (i.e. load, terminator or source) and set whether or not the pad's hole is plated. The NC drilling software selects separate drill tools for plated and non-plated holes.
- Pads can be assigned as Top and/or Bottom Layer Testpoints on both Fabrication and/or Assembly.
- Pads can be set to Slotted or Square holes
- Pads holes can be set as offset from center
- A preview is created at the top of the dialog that changes in realtime.
- A new shape of rounded rectangle has been added

- Paste and Solder Mask expansion can be set at the pad level bypassing what ever is set in the design rules.

## 8.2.5 Vias

- Vias can be placed using the **Place » Via** command or the **Place Via** toolbar button , but they are normally placed automatically when you change layers while placing a track. The Autorouter also places vias.
- Via properties are set in the *Via* dialog which is displayed by pressing the TAB key while placing a via, or by double-clicking on a placed via. The via diameter, hole size, net and Start and Finish layers are set in the *Via* dialog.
- Setting the Start and Finish layers to any layers other than Top Layer and Bottom Layer automatically assign the via as a *blind* or *buried* via. Blind and buried vias can be easily identified as their hole is displayed as two half circles with different colors.
- Vias can be assigned as Top and/or Bottom Layer testpoints on both Fabrication and/or Assembly.
- If a net being manually routed is to connect to an internal power plane, press the / (forward slash) key on the numeric keypad to place a via connecting to the appropriate power plane. This will work in all track placement modes except 'any angle' mode.

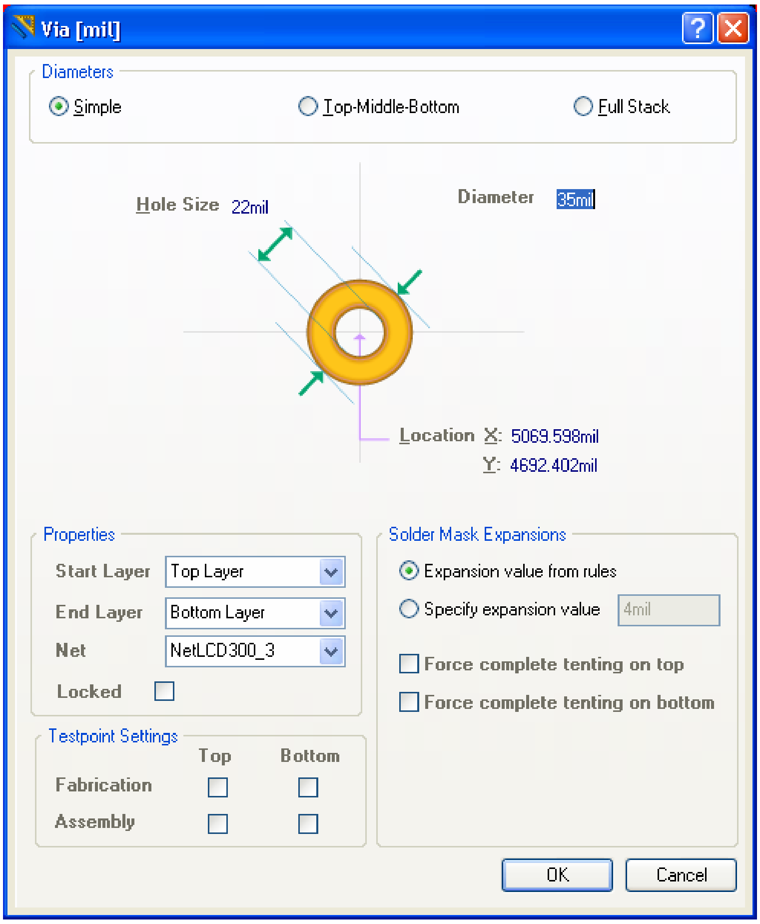



*Figure 7. Via Properties dialog*

- If a via is to have different sizes on the mid layers or bottom layer, check **Top-Middle-Bottom** in the Size and Shape section. Click on **Full Stack** and then **Edit Full Pad Layer Definition** to edit more complicated stack ups.

**Solder Mask Expansions**

Checking the **Specify expansions value** check box allows you to override the Solder Mask setting in the design rules by filling in the required expansion in the field provided.

**Tenting**

Checking the Tenting check boxes causes any Solder Mask settings in the design rules to be ignored and results in no opening in the solder mask for this via.

## 8.2.6 Arcs

The table below lists the arc placement options:

| Place Menu Command | Placement Toolbar |
|---|---|
| Arc (Edge) |  |
| Arc (Centre) |  |
| Arc (Any Angle) |  |
| Full Circle |  |

*Table 3. Arc Placement commands*

- All of the above commands result in an arc object being placed.
- An arc can be placed on any layer.
- Arc properties are set in the *Arc* dialog that is displayed by pressing the TAB key while placing an arc, or double-clicking on a placed arc.

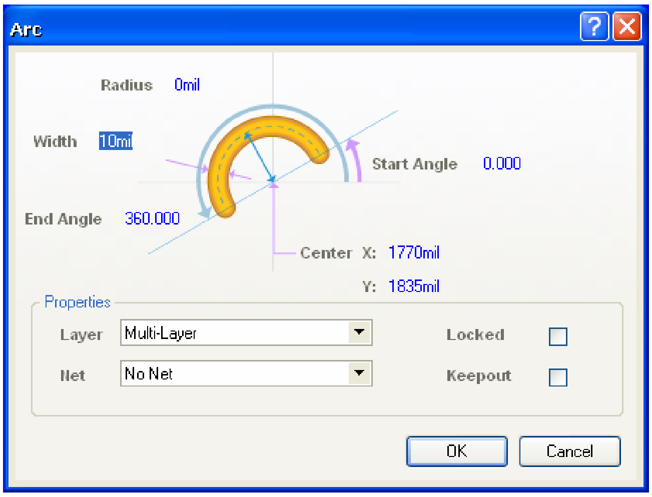



*Figure 8. Arc Properties dialog*

## 8.2.7 Strings

- A string is a single line of text that is placed using the **Place » String** command or the **Place String** toolbar button .
- String properties are set in the *String* dialog that is displayed by pressing the TAB key while placing a string, or double-clicking on a placed string. The actual text string to be placed is entered in the Text field.

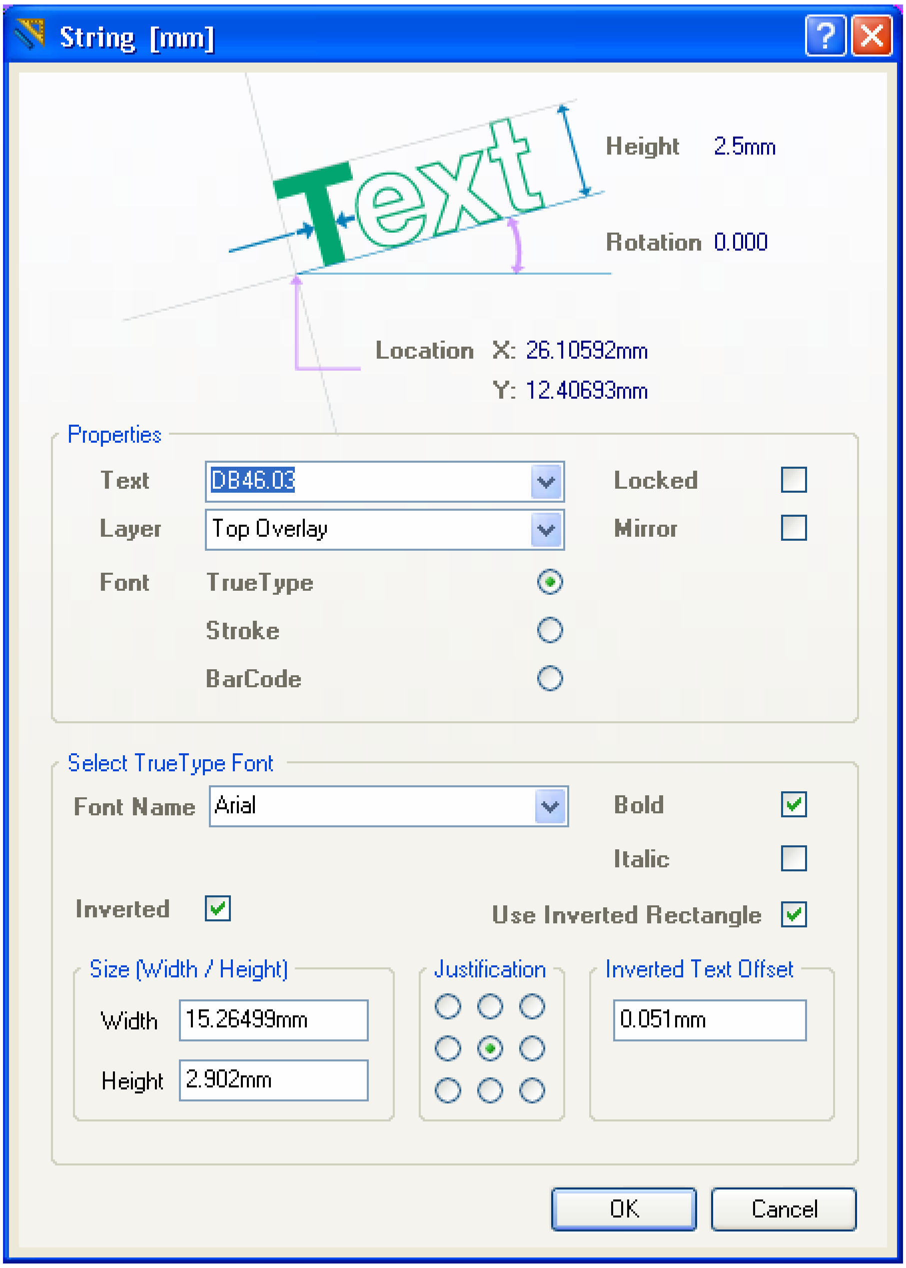



*Figure 9. String Properties dialog*

### 8.2.7.1 Special Strings

To assist in producing manufacturing documentation, special strings are provided. These include strings, such as .Arc_Count and .Component _Count, that display the number of objects in the PCB file when the PCB document is printed or plotted. Other special strings relate to layer names, file names and printing options. The .Comment and .Designator strings are used when creating component footprints. The .Legend string shows a drill symbol legend when the string is placed on the Drill Guide layer.

While most special strings are only converted during printing or plotting, .Layer_Name, .Pcb_File_Name and .Pcb_File_Name_No_Path can be viewed on screen. To see the values of these special strings placed on a PCB, select **Convert Special Strings** in the **Display** tab of the *Preferences* dialog (**Tools » Preferences**). For example, the special string .Layer_Name placed on the Top Layer of a PCB document would now display on the screen as TopLayer.

You place a special string using the **Place » String** command, but instead of filling in the **Text** field in the *String* dialog, use the drop-down list to display the special strings (see Figure 10). Select the desired special string, press OK and click to place it.

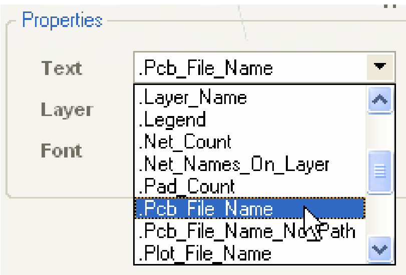



*Figure 10. String dialog showing special strings*

### 8.2.7.2 True Type Fonts

True type fonts can now by used on strings placed in PCB or the PCB library editor. The true type fonts used can be any font that is installed on windows. Using true type fonts gives the extra option of **Inverted**. Using true type fonts also enables the use of Unicode character sets, like Asian character sets or even ROHS character sets.

One draw back to using true type fonts is if it's a unique front set and you move the PCB to another machine that doesn't have it, then that machine is going to read the preference set in **Tools » Preferences » PCB Editor » True Type Fonts**. There is the option in here of setting the font to embedded which converts it to copper regions and the substitution font.

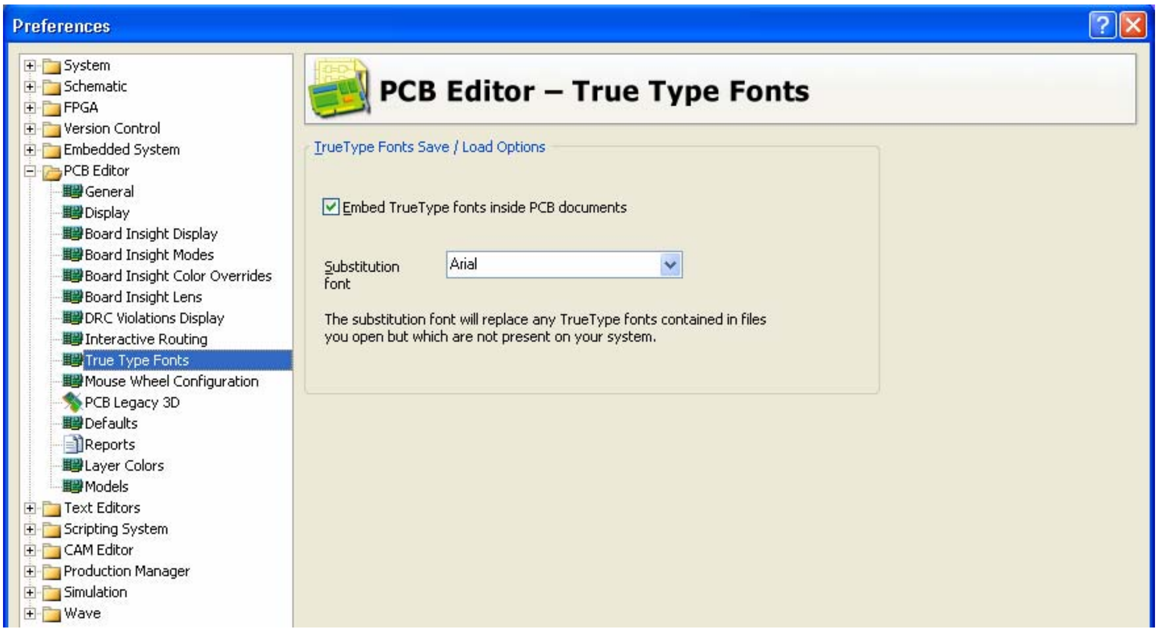


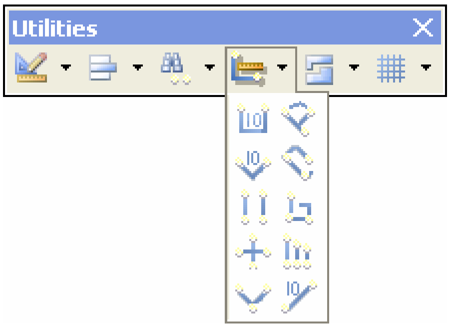

*Figure 11. Setting the substitution font and embedding the true type font*

## 8.2.8 Dimensions and coordinates

Dimensions and coordinates can be added to the current layer. All measurements and cursor positions are displayed relative to the current origin. The absolute origin (0, 0) for a PCB document is the lower left corner of the design area.

You can set the current origin to be any point in the PCB workspace by selecting **Edit » Origin » Set.** Click where you want to set the new current origin. To set the current origin back to the absolute origin, select **Edit » Origin » Reset**.

### 8.2.8.1 Placing dimensions




Dimensions can be added to the current layer by selecting from the **Dimension** tools on the **Utilities** toolbar (**View » Toolbars » Utilities)** or the **Place » Dimension** (**PD**) submenu. Click to define the start and end points. Watch the Status bar for instructions on placing the dimension. Press TAB to set the properties, such as the text height and width. Right-click or press ESC to exit the command.

The dimension value automatically updates as you move the start or end points.

### 8.2.8.2 Placing coordinates

A coordinate object places X, Y coordinate information measured as the horizontal (X) and vertical (Y) distance of the coordinate marker from the current origin. Select the **Place Coordinate** toolbar button  or **Place » Coordinate** (**PO**). Click to place the coordinate. Right-click or press ESC to exit the command. The position values are automatically updated when you move a coordinate object.

## 8.2.9 Fills

- The Fill object is a solid rectangle and can be placed on any layer. A fill is placed using the **Place » Fill** command or the **Place Fill** toolbar button .
- To place a fill, the first click defines a corner of the fill and then the next click defines the opposite corner of the fill. Fill properties are set in the *Fill* dialog that is displayed by pressing the TAB key while placing a fill, or double-clicking on a placed fill.
- When a fill is selected, you can change its size by clicking and dragging its handles and you can rotate it by clicking on the small circle.

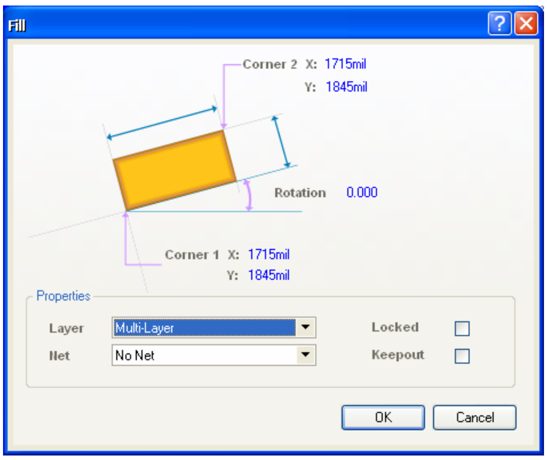



*Figure 12. Fill dialog*

## 8.2.10 Copper Region

- The Copper Region object is a multi-sided solid object. Although it is referred to as a Copper Region it can be placed on any design layer, including mechanical, mask, plane, or silkscreen layers.
- A region is placed using the **Place » Copper Region** command, or the **Place Copper Region** toolbar button .
- To place a region, click to define each vertex on the multi-sided object, when finished right-click to drop out of vertex placement mode. Region properties are set in the *Region* dialog that is displayed by pressing the TAB key while placing a region, or double-clicking on a placed region.
- A region can also be used to create a void in a solid or hatched polygon pour.
- A region can be used to create a board cutout region where a mechanical object may require room within a board space.

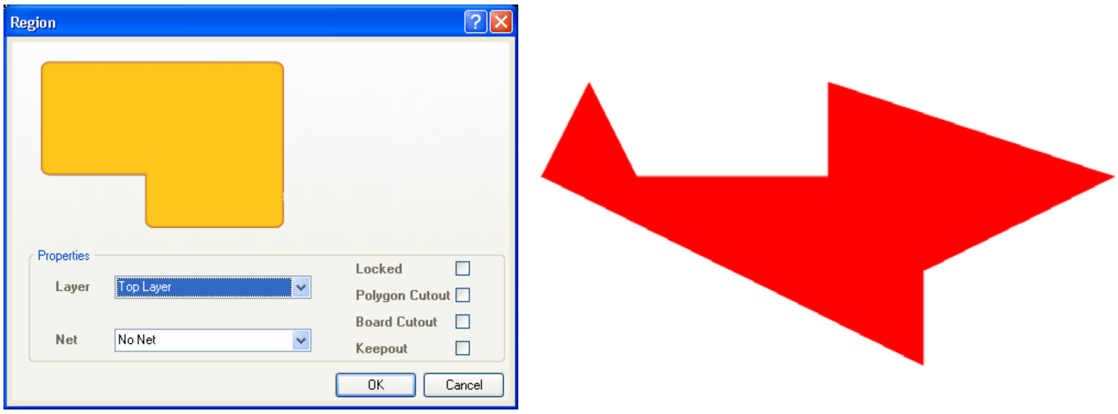



*Figure 13. Region dialog, and an example of a region*

## 8.2.11 Keepout objects

Tracks, fills and arcs can be used to assign an area on a specific electrical layer to act as a routing barrier. Objects defined as keepouts are ignored by output operations, such as photo plotting and printing.

A keepout can be defined using the commands in the **Place » Keepout** sub-menu (**PK)**. Existing tracks, fills and arcs can be defined as layer-specific keepouts by selecting the Keepout option in the object's *Properties* dialog.

## 8.2.12 Paste commands

There is an additional paste command in the PCB Editor — **Edit » Paste Special**. This command can be used for panelizing an entire PCB design or pasting multiple copies of selected objects. However, this does create things like duplicate nets, and large numbers of primitives on the board. A much better way to do this is to use embedded board arrays, which is covered in the *Altium Designer Advanced Schematic Capture and PCB Editing* training course.

Before using this command, copy selected objects to the clipboard using **Edit » Copy** (**EC**) or **Edit » Cut**. Click to select a reference point, i.e. the point used to hold the selection while positioning it during the Paste operation.

From the *Paste Special* dialog, you can choose to paste objects on the current layer (selected option) or retain their original layers (deselected). Clicking on **Keep Net Name** retains the original net names of pasted objects. If this option is not selected, the pasted object's net attribute is set to 'No net'.


*Figure 14. Paste Special dialog*

If components have been copied, the other options will become selectable. The **Duplicate Designator** option should be selected when panelizing an entire design to keep the designator names the same on each panel. Otherwise, generic default designator names are used.

Select the **Add to Component Class** option to make sure pasted components are added to the same class as the components from which they were copied.

## 8.2.13 Exercise – PCB design objects

1. Open `PCB Objects.PcbDoc` found in the `\Altium Designer Summer 09\Examples\Training\PCB Training\Practice Documents` folder. Experiment with placing each of the PCB design objects in the spaces provided.
2. Place a few pads and then connect them by placing tracks, using the various track placement modes.
3. Select each object and observe the effect of moving the handles.
4. Double-click on some of the objects to display and modify their properties.
5. Close the PCB document without saving.

## 8.2.14 Favorites Panel

Like a web browser a list of favourite documents can be stored in this panel for future reference. A thumbnail of the view as well as title and comment is stored. For Altium Designer documents the zoom level and location is included.

Favourites can be tied to the project itself making it a useful mark up tool for design collaboration. Project favourites are stored in a 'ViewsOf' folder in the same folder as the project file.

- Open the Favourites panel from the panel control in the bottom right by going to **System » Favorites**.
- The contents may be divided into folders. A new folder can be created from the right click menu
- To add the current view to a folder use Add Current Document View from the right click menu.
- To recall a view simply double click the entry in the list
- The size of the thumbnails is configured in the System preferences in the View section.

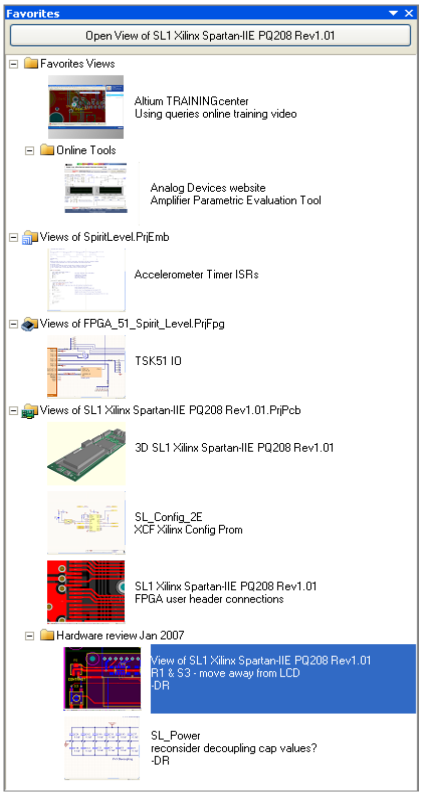



*Figure 15. Favorite's panel.*

## 8.2.15 Snippets Panel

The Snippets Panel provides a way to store portions of a design for later reuse. The panel will store sections of schematic, PCB layout and source code.

- Open the Snippets panel from the panel control in the bottom right by going to **System » Snippets**
- The contents can be divided into folders. These are just regular Windows folders and the location can be configured from the Snippets Folders button. Multiple folders can be defined; using a shared network resource will let you share a snippets library amongst an entire design team.
- To create a snippet select the objects in the PCB, schematic, or code editor and then from the right click menu select **Snippets » Create Snippet from selection**. File the snippet away with a title and comments.
- To Place a snippet select it in the panel and then click the Place button at the top.
- Ideally reset component designators before using them to create a snippet to avoid duplication when they are placed.

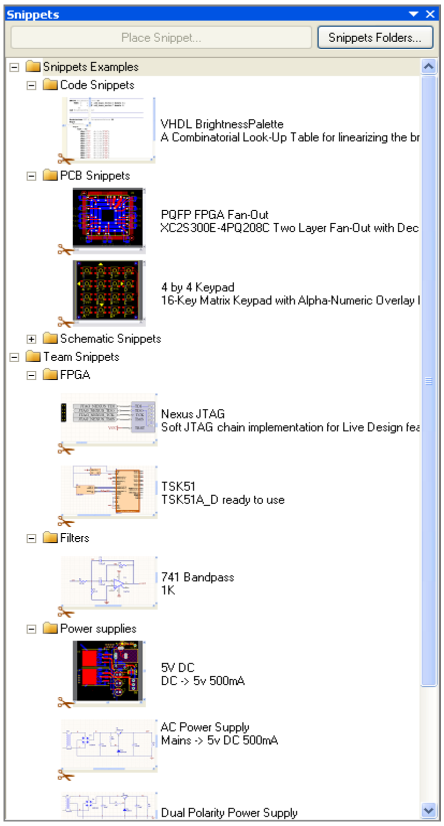



*Figure 16. Snippets panel*

## 8.2.16 Clipboard panel

The Clipboard Panel provides a way to store portions of a design for later reuse. The panel will store sections of schematic, PCB layout and source code. The only limitation of this panel is the data is only available per session rather than all the time like the snippets panel.

- Open the Clipboard panel from the panel control in the bottom right by going to **System » Clipboard**
- The clipboard panel has the added advantage of being able to read the windows clipboard so data can be transferred from other programs to Altium Designer. This option needs to be enabled in **DXP » Preferences » Systems » General**. Turn off the option of **Monitor clipboard content within this application only**.

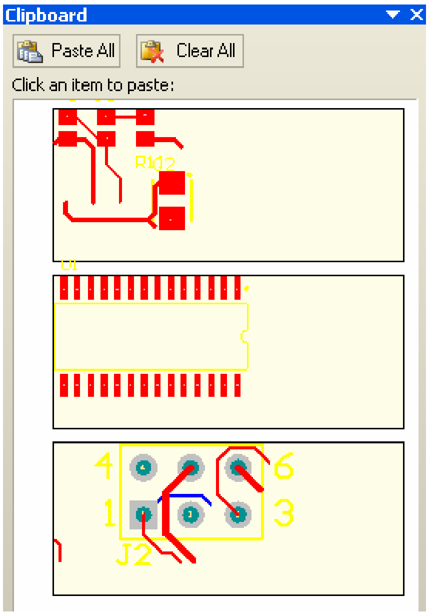



*Figure 17. Clipboard Panel*

## 8.2.17 Selection

Use the Select function to graphically edit an object. Below are some key points about using select:

- An object becomes selected when you click on it with the left mouse button.
- Clicking on an object that is selected allows you to move it.
- When selected, handles appear at key points on the object. The method for editing objects varies between objects, but typically, a click on a handle enables you to move the handle.
- When placing objects, the last object placed remains selected.
- To de-select an object, simply click in an area of the workspace where there are no objects.

> **Note:** PCB components cannot be graphically edited unless you unlock the component primitives. Component footprints are normally only edited in the PCB footprint library.

The PCB Editor provides selection capabilities that are similar, although not identical, to selection in other Windows applications. Below are some key points about selection in the PCB Editor:

- Selected objects can be cut or copied to the clipboard. They can then be pasted elsewhere onto the current PCB file or into another PCB file.
- There are a number of PCB Editor commands that operate on the selected group of objects, e.g. the **Tools » Interactive Placement** commands.
- The PCB Editor uses a special proprietary clipboard that supports PCB data such as connectivity and layer properties of primitives. When a copy action is performed a graphical metafile representation is also placed on the Windows clipboard, ready for pasting into another Windows application.

To select objects, you can use the following methods.

| Method | Function |
|---|---|
| Click and drag box around | Select all objects enclosed by drag area |
| SHIFT+ click | Select several objects (on a selected object this will de-select it). |
| **Edit » Select** menu (**S**) | Select Inside Area, Outside Area, Touching Rectangle, Touching Line, All, Board, Net, Connected Copper, Physical Connection, Component Connections, Component Nets, Room Connections, All on Layer, Free Objects, All Locked, Off Grid Pads or Toggle selection. |
| Select Inside Area | This button on main toolbar |

*Table 4. Select command summary*

Once objects have been selected, you can:

| Function | Menu command | Shortcut keys |
|---|---|---|
| Cut | **Edit » Cut** | CTRL+X |
| Copy | **Edit » Copy** | CTRL+C |
| Paste | **Edit » Paste** | CTRL+V |
| Delete | **Edit » Clear** | CTRL+DELETE |
| Move | **Edit » Move » Move Selection** | Click-and-hold, or **M, S** |
| Rotate | **Edit » Move » Rotate Selection** | SPACEBAR |
| Flip | **Edit » Move » Flip Selection** | **X** or **Y** |
| Align | **Edit » Align** | **A** (Align submenu) |
| Jump to | **Edit » Jump » Selection** | **J** (Jump submenu) |
| View | **View » Selected Objects** | **V** (View submenu) |
| Convert | **Tools » Convert** | **TV** (Tools submenu) |

*Table 5. Selected object command summary*

To de-select objects, use the **Edit » DeSelect** menu (X) commands or the **DeSelect All** button on the Main toolbar.

**Note:** Selection in earlier versions of Altium software differed from other Windows applications in that selection was persistent – selected objects always remained selected until you deliberately de-selected them. Altium Designer includes an option to mimic that behavior, if you disable the **Click Clears Selection** option in the **PCB Editor – General** page selected objects will remain selected until you deliberately clear the selection. It is recommended you try the standard behavior first, and if you need to 'hold' the selection state of a set of objects, use the Selection Memory feature.

**Note:** If you find that you keep inadvertently selecting certain objects, you can make them harder to select by enabling the **Shift Click to Select** option in the **PCB Editor – General** page of the *Preferences* dialog. Click the **Primitives** button to configure which objects require Shift to be held during selection.

#### 8.2.17.1 Selection hints

- Before starting a selection, it is a good idea to de-select all objects first.
- Only items that fall completely inside the selection area will be selected.
- The selection color is set in the *Board Layers & Colors* dialog (**Design » Board Layers & Colors**) or use the **L** shortcut.
- Pressing the S key pops up the Select menu.
- Pressing the X key pops up the DeSelect menu.
- Eight selection memories are available in the PCB editor which can be used to store and recall the selection state of up to eight sets of objects on the PCB.

> **Note:** The PCB editor includes a number of extra selection modes, including **Select Touching Line**, **Select Touching Rectangle** and **Select Connected Copper**. Press S to pop up the Select menu and access these commands.

### 8.2.18 Masking & Dimming

As well as regular selection in Altium Designer, there are also ways of hiding certain objects from view. Two methods of doing this exist. The first is dimming, which dims out any objects that are not currently of interest. The second is masking, which is similar, but prevents the user from accidentally selecting or changing objects unintentionally. There are many ways to apply a mask/dim effect to PCB objects, including: Find similar objects, navigator panel, messages panel, Using CTRL key or Autofocus.

When a mask or dim effect is set the PCB objects appear darker as shown in Figure 18.

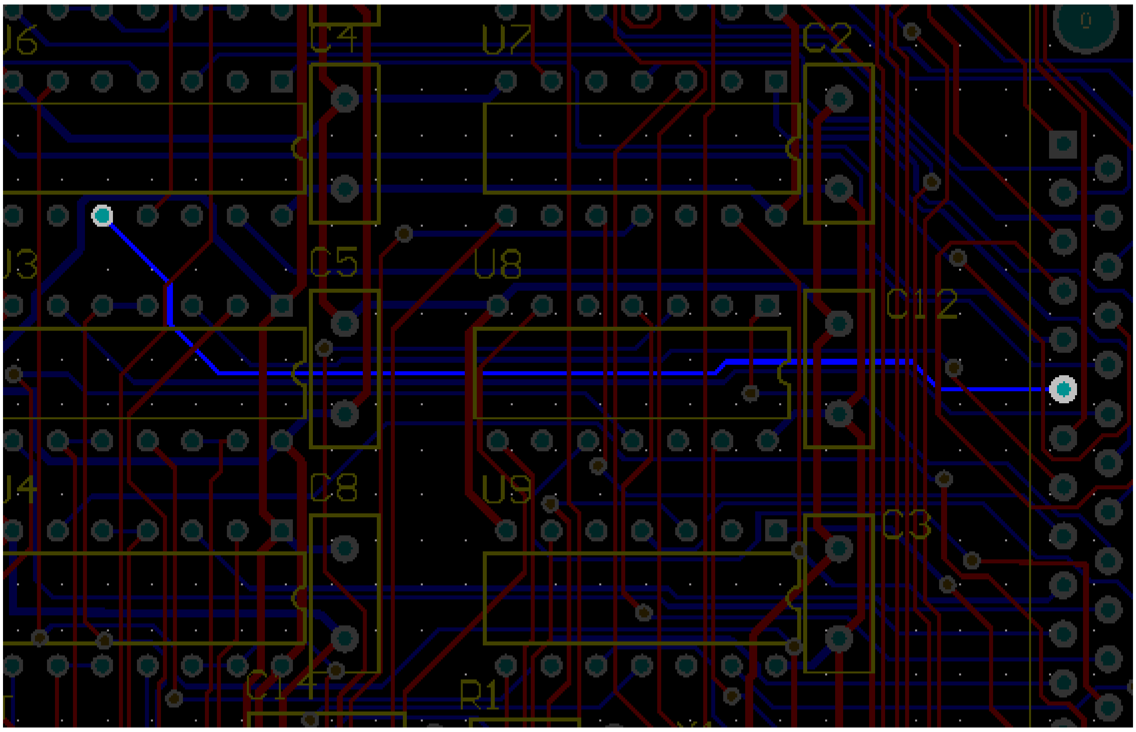



*Figure 18. A PCB showing masked mode.*

### 8.2.18.1 Clearing a Mask/Dim Effect

To clear a mask or dim effect in PCB click on the *Clear* Button Clear located in the bottom right of the Altium Designer screen. The **SHIFT + C** shortcut can also be used.

### 8.2.18.2 Changing the Mask/Dim Level

To change the mask or dim level click on the *Mask Level* Button Mask Level located in the bottom right of the Altium Designer screen. Once clicked, a small popup appears with the mask and dim levels, as shown below. Move the filter slider to the left to mask more or right to mask less.

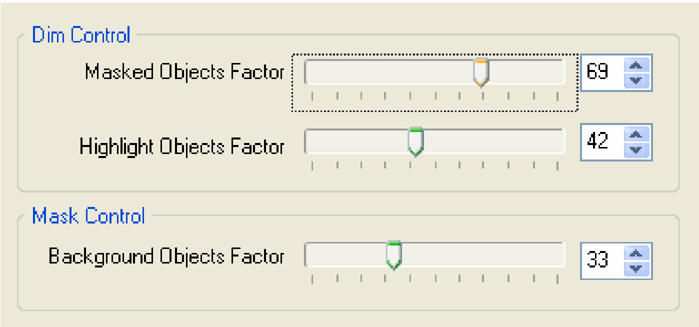



*Figure 19. The mask Level setup in PCB*

## 8.2.19 Other mouse operations

The mouse operations listed below are universal throughout the PCB Editor and should be used in preference to menu commands.

| Mouse Operation | Function |
|---|---|
| Double-click | Change an object |
| Click | ENTER |
| Right-click | ESCAPE |

*Table 6. General mouse shortcut summary*

## 8.2.20 Multiple objects at the same location

When working in the PCB Editor, the situation often occurs where a click to perform an operation is made where there are multiple objects. In this situation, the PCB Editor displays a menu listing all the objects it has detected at the location of the click, with a small preview of the object currently chosen in the menu. You can then select the required object off this menu.

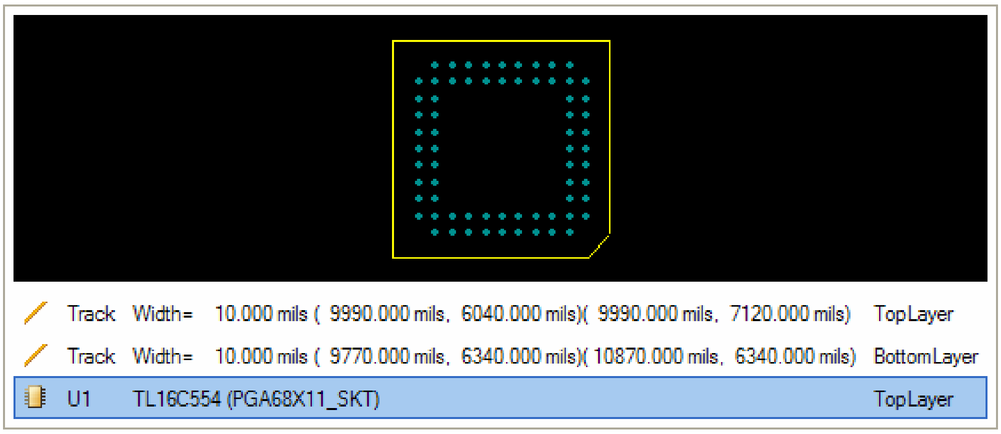



*Figure 20. Menu listing objects at mouse click point*

As well as the above there is also the shortcut **SHIFT + X** which gives a similar popup but it shows you what the object is part. For example a track and what net it's part of and what the whole net looks like.

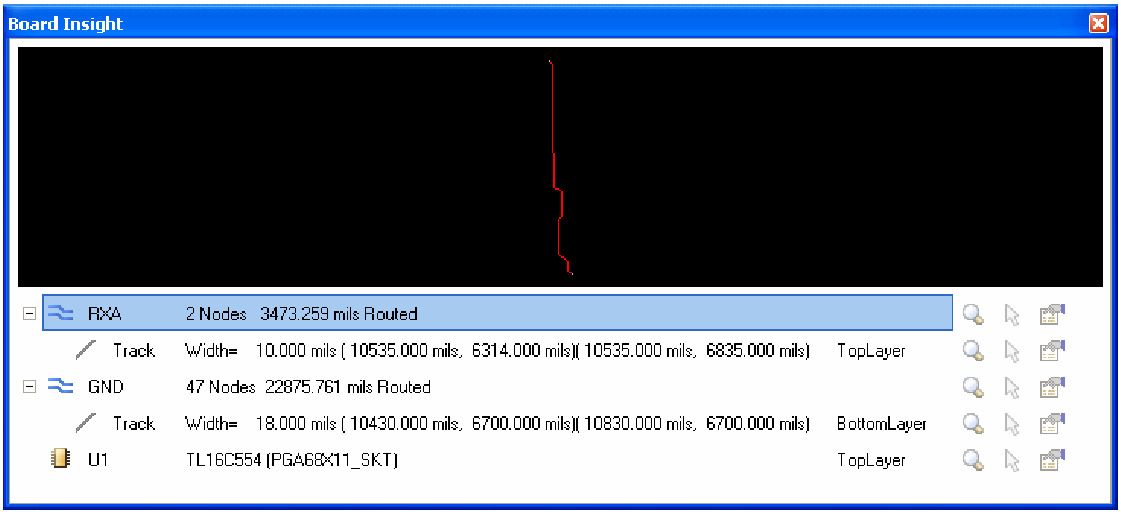



*Figure 21. Menu listing objects when using shift + x shortcut*

## 8.2.21 Jump menu

The Jump menu commands provide you with a number of commands for positioning the cursor. The Jump sub-menu commands are described as follows:

| Menu Command | Shortcut | Description |
|---|---|---|
| Absolute Origin | **JA** | Positions the cursor at the Absolute Origin. CTRL+HOME also does this. |
| Current Origin | **JO** | Positions the cursor at the Origin. ctrl+end also does this. |
| New Location | **JL** | Positions the cursor at a specified coordinate. |
| Component | **JC** | Positions the cursor over the specified component. |
| Net | **JN** | Positions the cursor over a pad assigned to the specified net. |
| Pad | **JP** | Positions the cursor over the specified pad. |
| String | **JS** | Positions the cursor over the specified text string in the PCB file. |
| Error Marker | **JE** | Positions the cursor over the next DRC error marker. |
| Selection | **JT** | Zooms in on the selected group. |

*Table 7. Jump menu commands*

If a Jump command does not appear to jump to the correct location, zoom in to display the correct coordinates.

## 8.2.22 Exercise — PCB basics

1. Open `4 Port Serial Interface.PcbDoc`, located in the `\Altium Designer Summer 09\Examples\Reference Designs\4 Port Serial Interface` folder.
2. Work through some of the commands in Table 1 to Table 7 in this section to get familiar with the PCB display and selection commands listed. Try using the commands from the toolbar and using shortcut keys.
3. Go to the menu **Design » Board Layers and Colors** (**L**) and turn off the Visible Grid 2. Go to **Design » Board Options** and set Visible Grid 1 to 50 mil and set the Snap Grid to 25 mil.
4. Place a Solid Region using the **Place » Solid Region** menu command. Observe that when you exit this command the solid region is selected. Move the handles by clicking on them. Move the solid region by clicking on the object. De-select the object by clicking at a point away from any object.
5. Perform the **View » Fit Document** (**VD**) command on your PCB file.
6. Move a component by clicking and holding on it.
7. While you are moving the component, press the SPACEBAR to rotate it (SHIFT+SPACEBAR for clockwise rotation) and press the **L** key to flip the component to the other side of the board (you may need to enable layers to see all the component primitives when it is on the bottom layer).
8. Click another component and start to move it. While moving it, hold the **Alt** key. Note that this will constrain the movement to a vertical, horizontal, or diagonal line from the starting point. The choice between directions is defined by the proximity of the cursor to the object – simply push the object in the desired direction to see the effect. This feature is particularly useful if you want to move a component and maintain its alignment.
9. Select a group of components (click-and-hold and then drag the cursor over the components).
10. Select the **Edit » Copy** menu command to copy the selected group to the Altium Designer clipboard. Don't forget to give the reference location.
11. Select the **Edit » Paste** menu command. The contents of the clipboard will now be moving with the cursor. Rotate and flip the group as you did when moving a component. Place the group of components by clicking at the required location.
12. Close the document without saving the changes.

TOP SECRET

Altium Designer

# Module 9: Setting up the PCB

## Module 9: Setting up the PCB





Module Seq = 9

# 9.1 Setting up the PCB

## 9.1.1 Board Options dialog

The *Board Options* dialog allows you to set parameters relating to individual PCB documents. Select **Design » Board Options** (**DO**) from the menus to open the dialog. The settings in this dialog are saved with the PCB file.

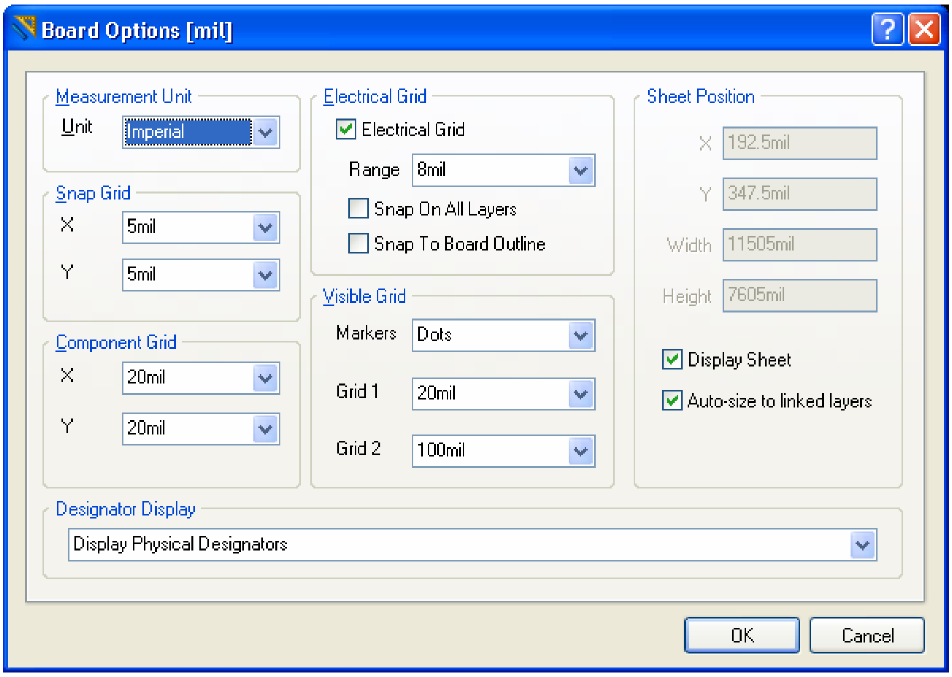



*Figure 1. Set grid options in the Board Options dialog.*

#### Measurement Unit

Sets the coordinate system to either metric or imperial.

| | |
|---|---|
| Snap X | X value for the snap grid |
| Snap Y | Y value for the snap grid |
| Component X | X value for the component grid |
| Component Y | Y value for the component grid. |

#### Electrical Grid

When the electrical grid is enabled and you are executing a command which supports the electrical grid and you move the cursor within the Grid Range value of an object assigned to a net, the cursor will jump to that object.

#### Visible Grid

Sets the size and style of the visible grids.

#### Sheet Position

The sheet is a calculated object, drawn to represent the printed page. The sheet size can either be defined by the **Size** and **Location** settings in this dialog, or it can be linked to the contents of mechanical layer(s). If it is linked to the contents of mechanical layer(s), you can use the **Design**

**» Board Shape » Auto-position Sheet** command to recalculate it when the contents of the linked mechanical layers change.

Typically, the linked mechanical layers would be used for drawing detail that is required on the printout. Another advantage of linking the sheet to mechanical layers is that both the sheet and the mechanical layers can be hidden by disabling the **Display Sheet** option.

> **Note:** It's a good idea to turn off the sheet when attempting to place the components on the PCB after transferring the design from schematic. The reason for this is generally the sheet color is white and the selection color is also white, making it hard to see the components when they are picked up and moved using the cursor.

### Designator Display

The designator display can be either the logical designator shown on the schematic or the physical designator assigned when the design is compiled. Normally, these are the same except in a multi-channel design when the physical designator includes channel identifier information.

## 9.1.2 View Configurations

- This dialog is used to set the display state and color of each layer in the PCB in 2D view mode, and the colors and transparency in 3D view mode (**L** shortcut key to open the dialog).
- It is also used to configure other view related information, such as the display of each object-kind, and the display of net names on pads.
- View configurations can be saved and reloaded, with the last-used view configuration being automatically applied when the board is re-opened.

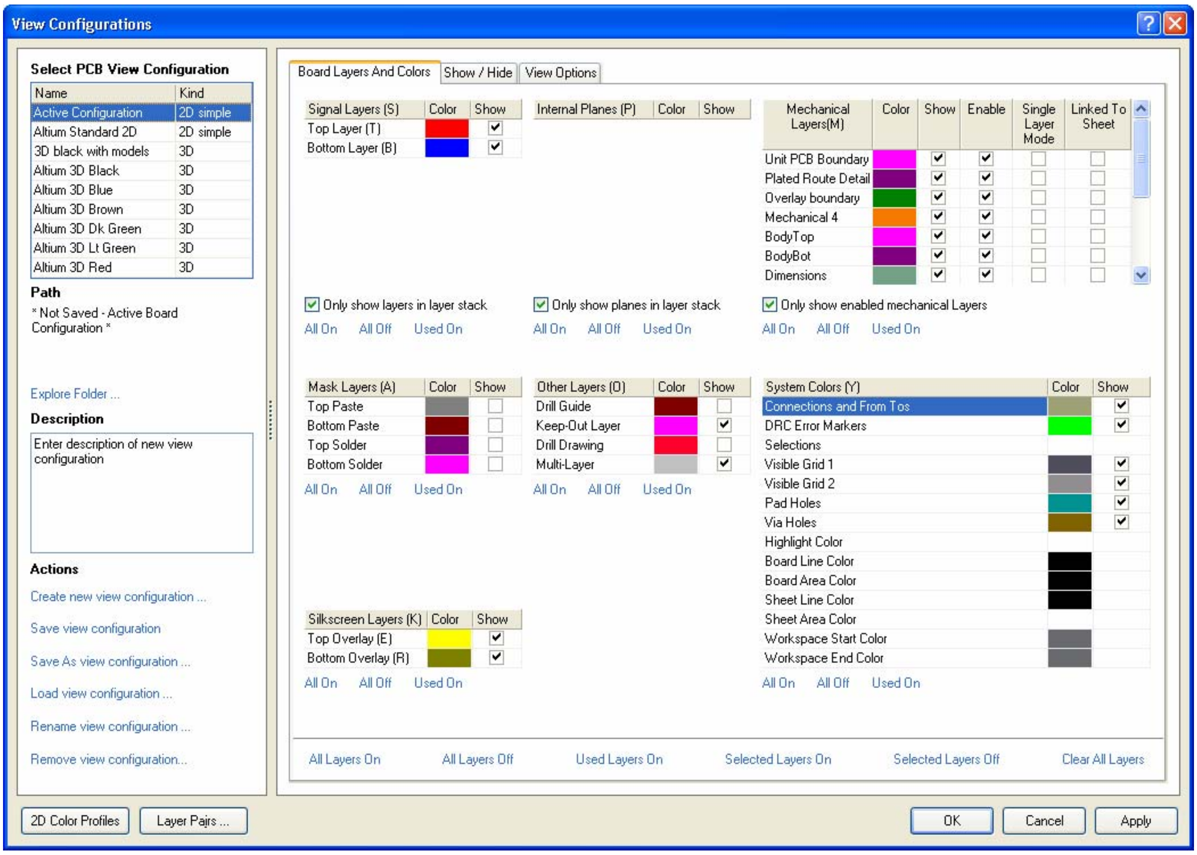



*Figure 2 View Configurations dialog*

**Signal Layers and Internal Planes**

These layers are added too and removed from the PCB in the Layer Stack Manager. Their color and display state is controlled in this dialog.

**Note:** Press the accelerator key in brackets () next to the layer name to toggle that layers show property while in this dialog

**Mechanical Layers**

There are 32 mechanical layers, disable the **Only Show Enabled** option to display the entire set and enable a new mechanical layer for this PCB. Press **F2** to edit the name of a mechanical layer.

**Layer Pairs**

Layer pairs are mechanical layers that have been associated to handle layer-specific component data. For example, if you have component footprints that require glue information, define this on a mechanical layer in the Library Editor, then pair this mechanical layer with another. When the footprint is flipped to the bottom of the board, the information on the first mechanical layer is automatically transferred to the paired mechanical layer.

**Color Sets**

The **Default Color Set** button sets the colors to the default settings with a pale yellow background. Default colors cannot be used if the **Transparent Layers** option (**Display** tab) is selected. The **Classic Color Set** button sets the colors to the traditional black background setting.

**Keep-Out Layer**

The keep out layer is a special layer. Objects placed on the keep out layer act as an obstacle or boundary to an object placed on any signal layer. The keep out layer is typically used to define regions such as the board routing and placement boundary, or areas of the board that must be kept free of components and routing. The keep out layer is discussed more in section 9.2.

## 9.1.3 The PCB coordinate system

The PCB Editor has a coordinate system with the origin located in the bottom left hand corner of the workspace. This point has the coordinates of (0,0) and is known as the Absolute Origin. The workspace size is 100 inches by 100 inches. The reference point of the coordinate system can be re-defined at any time using the **Edit » Origin » Set** menu command and this sets what is known as the relative Origin. The coordinate readout in the status bar references this relative Origin. The **Edit » Origin » Reset** menu command sets the relative Origin back to the Absolute Origin.

An Origin Marker shows the location of the relative Origin. This is displayed by checking the **Display Origin Marker** check box in the **Display** tab of the *Preferences* dialog.

The coordinate system units can be either metric or imperial. The **View » Toggle Units** menu command or the **Q** shortcut key toggles the co-ordinate system between metric and imperial.

## 9.1.4 Grids

### 9.1.4.1 Snap Grid

The Snap Grid ensures accurate movement and placement of objects. The Snap Grid causes the coordinates of a mouse click to snap to the nearest snap grid point. The Snap Grid has X and Y values and is set in the *Board Options* dialog. Press **G** or CTRL+G shortcuts to change the grid.

### 9.1.4.2 Component Grid

The Component Grid is similar to the Snap Grid except that it is only active when placing or moving components. The Component Grid has X and Y values and is set in the *Board Options* dialog.

### 9.1.4.3 Visible Grid

The Visible Grids either display as lines or dots when turned on. They are independent of the Snap Grid. The PCB Editor has two visual grids that you can set in the *Board Options* dialog and display independently.

### 9.1.4.4 Electrical Grid

The Electrical Grid can be thought of as *a range of attraction*. During interactive editing the cursor will jump to any existing electrical object when the cursor falls within the range of the electrical grid setting.

When the Electrical Grid overrides the Snap Grid an octagon displays on the cursor when the hot-spot (or electrical centre-point) is under the cursor. When you see that octagon, you know that the cursor is precisely located on the object it has jumped to.

The Electrical Grid is set and turned on or off in the *Board Options* dialog. You can also toggle the Electrical Grid on and off using the SHIFT+E shortcut, or disable it temporarily during an edit-type operation (such as interactive routing) by holding down the CTRL key.

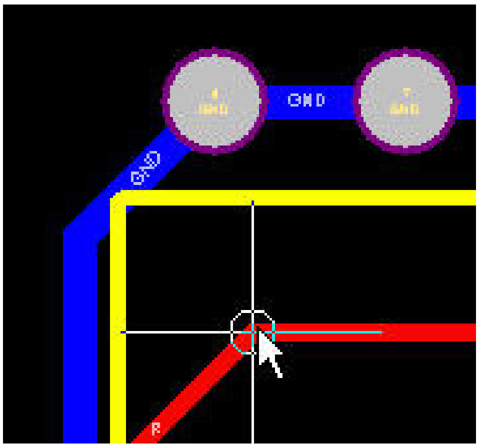



*Figure 3. Cross hair indicating the electrical grid*

**Shortcut keys for setup options**

Pressing the **O** shortcut key displays a menu that provides a quick way of accessing the setup dialogs. Combine this shortcut with the underlined letter in the menu options, e.g. **OB** to display the board options. The options in this menu are described below.

> **Note:** Right click on a layers tab at the bottom of the PCB workspace to access layer related dialogs and commands.

| Option | Dialog displayed | Shortcut |
|---|---|---|
| Board Options | *Board Options* dialog | **OB** |
| Board Layers | *View Configurations* dialog | **L** or **OY** |
| Layer Stack Manager | *Layer Stack Manager* dialog | **OK** |
| Classes | *Object Classes* dialog | **OC** |
| Preferences | **General** tab of *Preferences* dialog | **OP** |
| Display | **Display** tab of *Preferences* dialog | **OI** |
| Show/Hide | **Show/Hide** tab of *View Configurations* dialog | **OD** |
| Defaults | **Defaults** tab of *Preferences* dialog | **OU** |

*Table 1. Dialogs and preferences in PCB*

### 9.1.4.5 Exercise – Exploring document and environment options

Use this exercise to experiment with document and environment options.

1. Open the document `4 Port Serial Interface.PcbDoc` located in the `\Altium Designer Summer 09\Examples\Reference Designs\4 Port Serial Interface` folder.
2. Experiment with the **Used On**, **All On** and **All Off** buttons and with turning on and off individual layers in the *Board Layers & Colors* dialog.
3. Observe the display change when the **Display Sheet** option is toggled in the *Board Options* dialog.
4. Experiment with changing the colors of various layers.
5. Now, experiment with changing the various grid settings to see changes in the grid display and object movement in the *Board Options* dialog.
6. In the **PCB Editor » Defaults** page of the *Preferences* dialog, select **Component** and click on the **Edit Values** button. In the **Comment** section of the *Component* dialog, make sure the **Hide** option is enabled. Also check the Autoposition option is set to **Left-Above** in the **Designator** section.

# 9.2 Creating a new PCB

This section looks at how to create a new PCB using the Board Wizard.

## 9.2.1 Creating the Blank PCB

There are three ways to create a new PCB:

- Select **File » New » PCB** from the menus. This creates an empty PCB workspace, with a 6in by 4in board shape.
- In the **New from Template** region of the **Files** panel, select **PCB Templates**. This opens the *Choose Existing Document* dialog where you can select from an array of template files. The template name indicates the sheet size and each template file also includes a default board shape, typically 6in by 4in.
- Using the Board Wizard. This is launched from the bottom of the **Files** panel. The Wizard can be used to select from a pre-defined list of industry standard board shapes or generate a simple board outline.

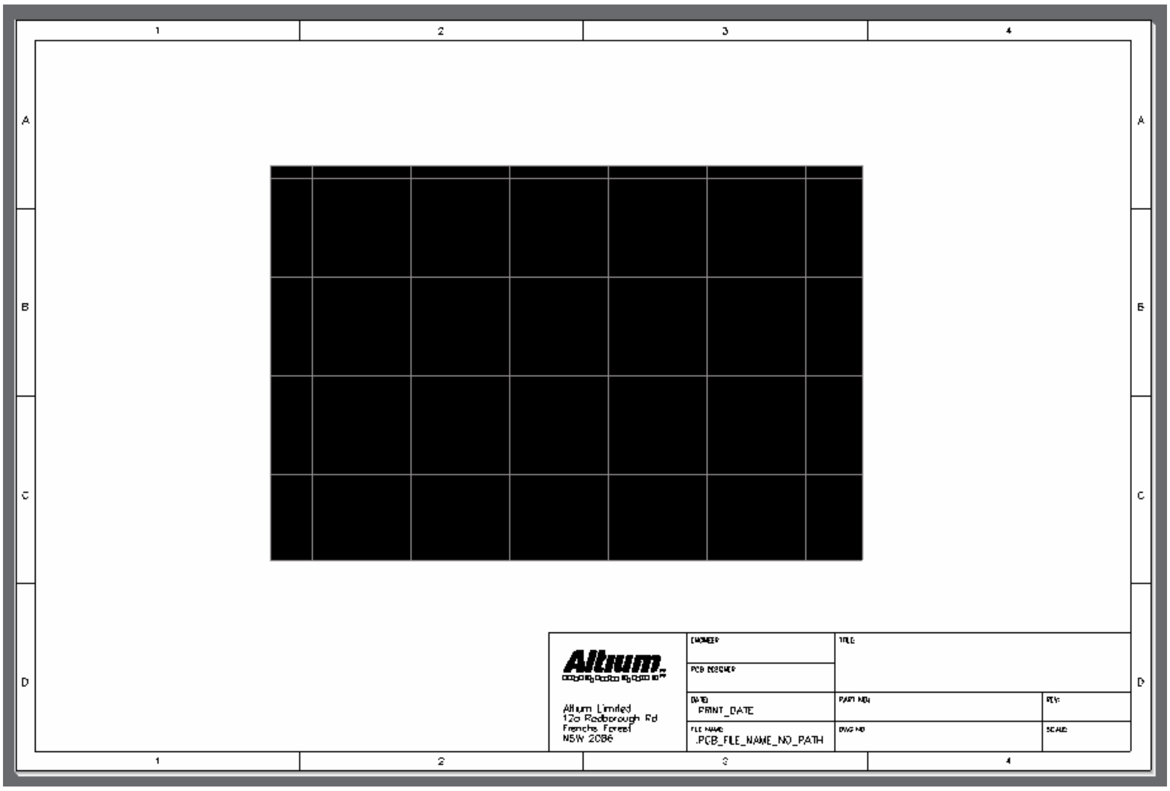



*Figure 4. A new PCB created by using the New from Template option.*

## 9.2.2 Defining a sheet template

The PCB sheet template is simply a display feature that is linked to mechanical layers in the PCB design. In the *View Configurations* dialog there is a checkbox next to each mechanical layer, titled **Linked to Sheet**. Any layer with this enabled is used by the software to calculate the size of the white sheet region.

- Define a template on a mechanical layer using the standard design objects, enable the **Linked to Sheet** checkbox, and enable the display of the sheet in the *Board Options* dialog. If you change the shape or size of the template, select **Design » Board Shape » Auto**

**Position Sheet** from the menus to automatically resize the white sheet region to just enclose all objects on the linked mechanical layers.

- There are a number of pre-defined PCB sheet templates in the `\Altium Designer Summer 09\Templates` folder, open the required size and copy the contents of Mechanical 16 into your own PCB to create a sheet template.

### 9.2.3 Defining the Board Shape, and Placement / Routing Boundary

Once the blank board has been created the next step is to define the shape of the board (typically this is the final finished board shape), and the routing and placement boundary.

- The board shape can be defined manually using the commands in the **Board Shape** sub-menu, or by getting the software to define it automatically from a set of selected objects. Defining it from selected objects is typically done when you have imported a board shape definition from another tool, such as a mechanical CAD package.
- The placement and routing boundary is defined by placing a continuous barrier on the Keep out layer. Any object placed on the keep out layer is considered an obstacle to objects on all the signal layers. Typically the keep out boundary is defined along, or slightly in from the board outline, taking into consideration any mechanical clearance requirements, such as brackets, card guides, and so on.

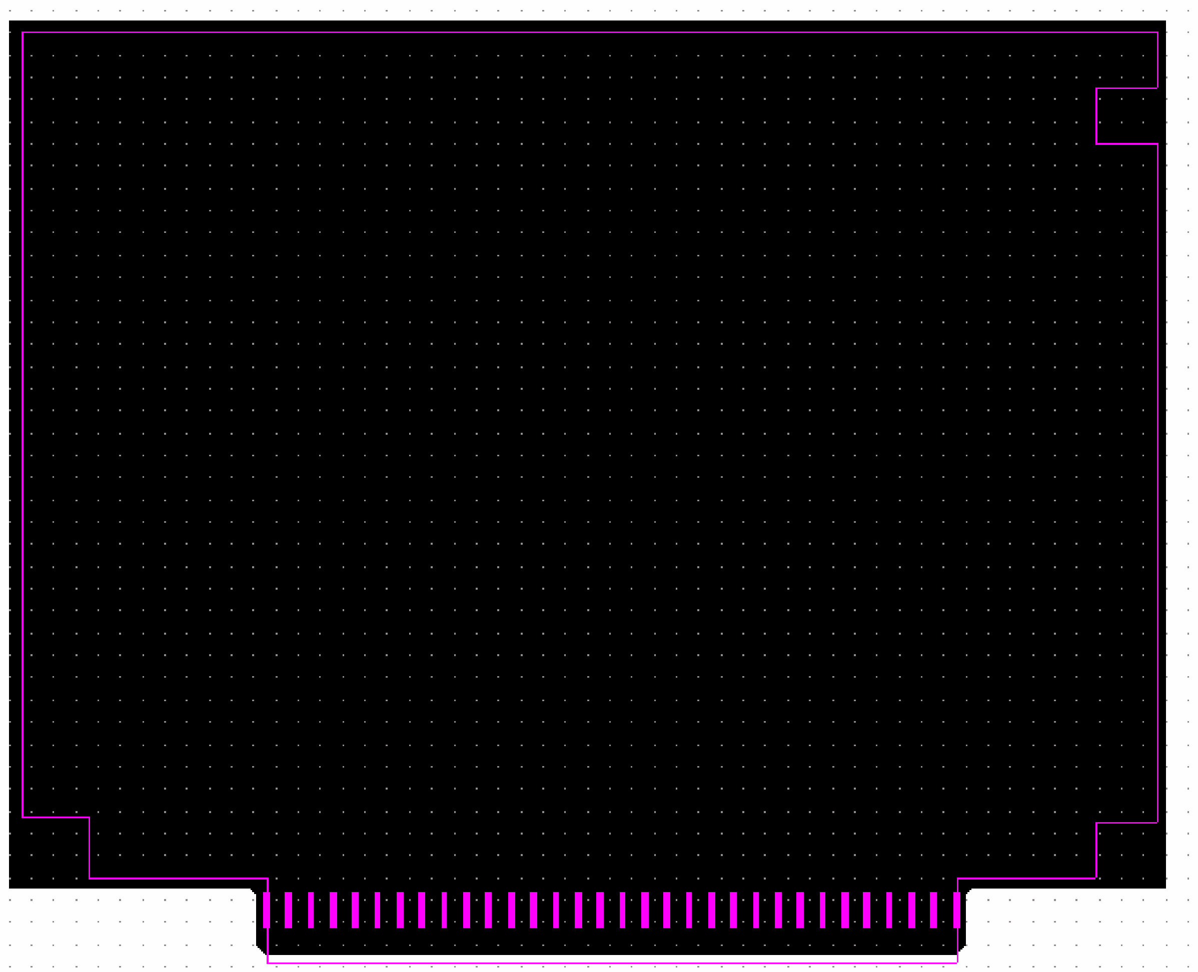


*Figure 5. Board shape (black region) and keep out boundary for the 4 Port Serial Interface example PCB. The row of small fills is there to prevent routing between the contacts of the edge connector.*

## 9.2.4 Exercise – Creating a board outline & placement / routing boundary

This exercise creates a new board outline for the training example.

1. Display the Files panel (**System » Files** from panel control) and click on the **PCB Templates** option in the **New from template** section.
2. Choose `A4.pcbdoc` in the *Choose Existing Document* dialog. The new blank PCB will open, as shown in Figure 4, where the black region on the sheet represents the board shape. We will now redefine it based on data in a DXF mechanical file.
3. Select **File » Import** to display the *Import File* dialog.
4. Set the Files of Type option to **AutoCAD (*.DXF, *.DWG)**
5. Browse and locate the file `\Altium Designer Summer 09\Examples\Training\PCB Training\Temperature Sensor\Outline.DXF` and open it.

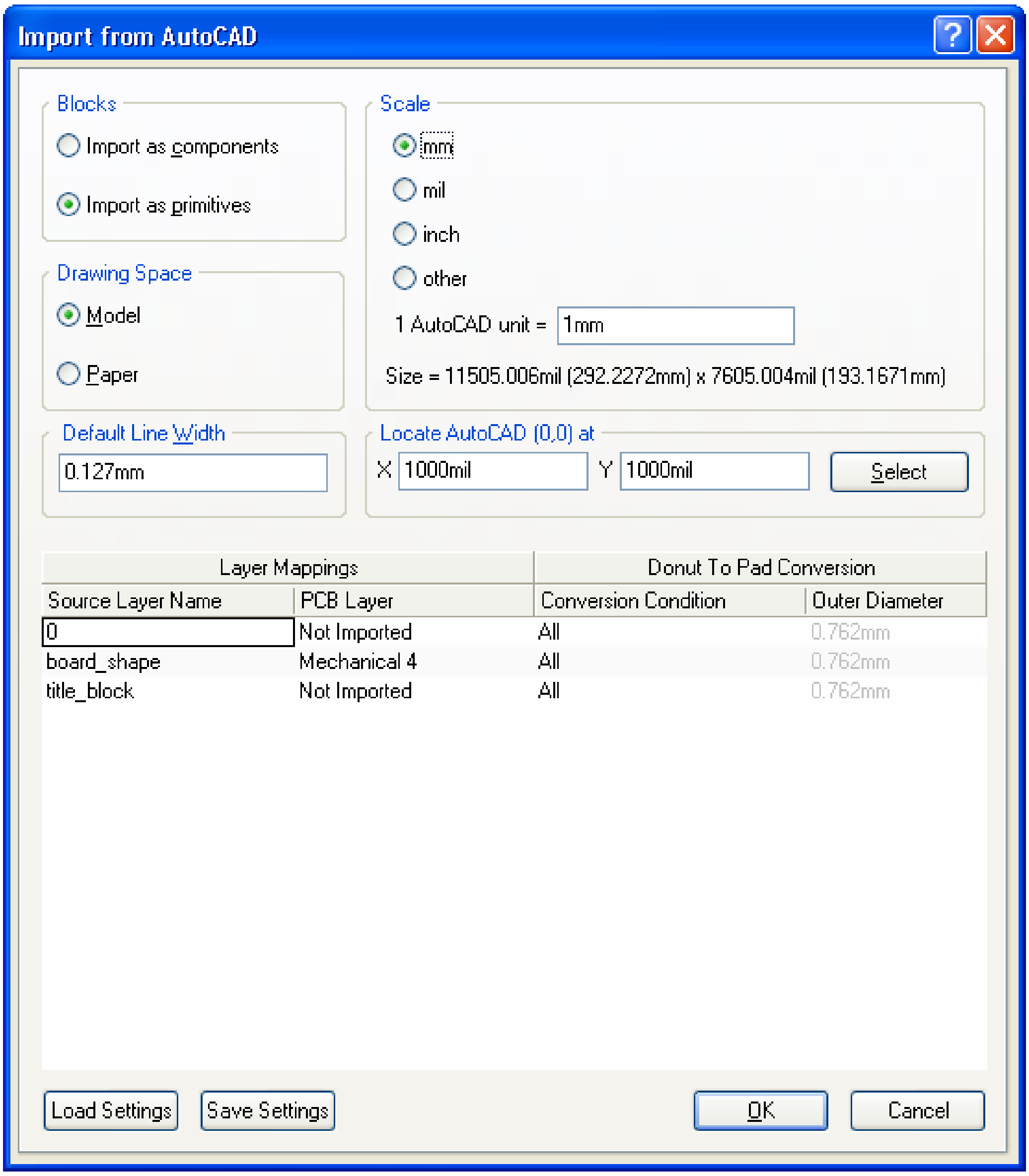



*Figure 6. Import the board shape from a DXF file.*

6. When the *Import from AutoCAD* dialog appears, set the following:
7. Set the **Scale** to mm (the imported shape should be approximately 2021mil x 2755mil)

8. In the **Layer Mapping**, map the source DXF layer to mechanical layer 4
9. Set the **Insertion Point** to something sensible, for example X=1000, Y=1000. The value is not crucial, as you will move it after importing.
10. leave other options at their defaults
11. When the **OK** button is clicked, a shape, forming a rectangle like shape, will appear on Mechanical layer 4.
12. We will now redefine the board shape to match this shape. Select the shape segments (drag a rectangle around them).
13. Select **Design » Board Shape » Define from selected objects**. The black board shape will redefine to match the imported tracks.
14. To move the new board shape to the centre of the sheet, drag a rectangle to select the board shape and the mechanical layer tracks, press the **M** key to display the **Move** submenu and select **Move Selection**. Click somewhere on the selection to define the point where it will be held, then move the board outline and mechanical layer tracks approximately to the centre of the sheet, and click to place them.

    **Note:** To ensure that objects remain on your preferred working grid it is generally better to select a meaningful point when moving or copying & pasting objects, in this case the point at the bottom left of the rectangle where the vertical and horizontal tracks meet would be suitable. If you want to set your reference point based on an object, make the layer that the object is on the active layer – that way the electrical grid will pull the cursor to a meaningful point on the object. Alternatively, press Shift+E to toggle to the Electrical Grid (All Layers) mode.

15. Change the Visible grid 2 to 100 mils in the *Board Options* dialog.
16. To define the placement / routing boundary first deselect all. The easiest way to select all the tracks on Mechanical layer 4 is to use the select on current layer command. To do this, make the Mechanical layer the active layer (use the layer tabs at the bottom of the PCB workspace), press **S** for select, then **Y** to select all on the current layer.
17. Choose **Edit » Copy** from the menus, choosing an appropriate reference point to hold the selection by when prompted (such as one of the corners).
18. Make the Keep out layer the current layer. If the Keep out layer is not currently enabled, press **L** to display the *View Configurations* dialog and enable it.
19. You are now going to paste the selection onto the current layer (the Keep out layer). To do this select **Edit » Paste Special** from the menus, enable the **Paste on Current Layer** option in the *Paste Special* dialog, and click **Paste** to return to the workspace where you can paste the tracks onto the keep out layer.
20. Save the new PCB as `\Altium Designer Summer 09\Examples\Training\PCB Training\Temperature Sensor\Temperature Sensor.PcbDoc`.
21. Check in the **Projects** panel If the board is part of the Temperature Sensor project. If it is not, click and drag the board, dropping it on the project name.
22. Right-click on the project name and choose **Save Project** from the floating menu.

# 9.3 Setting up the PCB layers

## 9.3.1 Enabling Layers

The PCB Editor has a concept of design layers to represent the various physical layers created to fabricate a printed circuit board. When placing objects using the PCB Editor, you need to consider which layer they are to be placed on. Objects are placed on the current layer, shown as the active layer tab at the bottom of the PCB workspace.

- Electrical layers are added in the *Layer Stack Manager* dialog (**Design » Layer Stack Manager**).

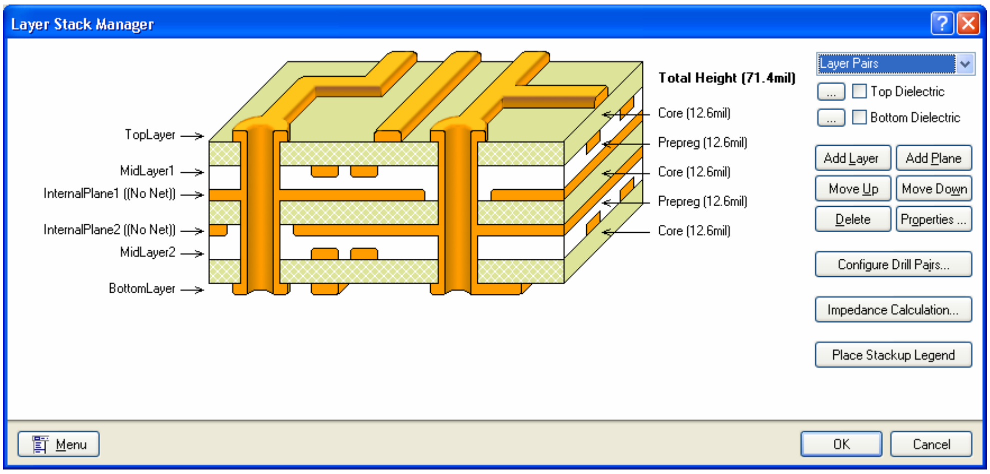



*Figure 7. Define the required electrical layers in the Layer Stack Manager dialog.*

- Layer display and the control of other non-electrical layers is done in the *View Configurations* dialog (**Design » Board Layers & Colors**).

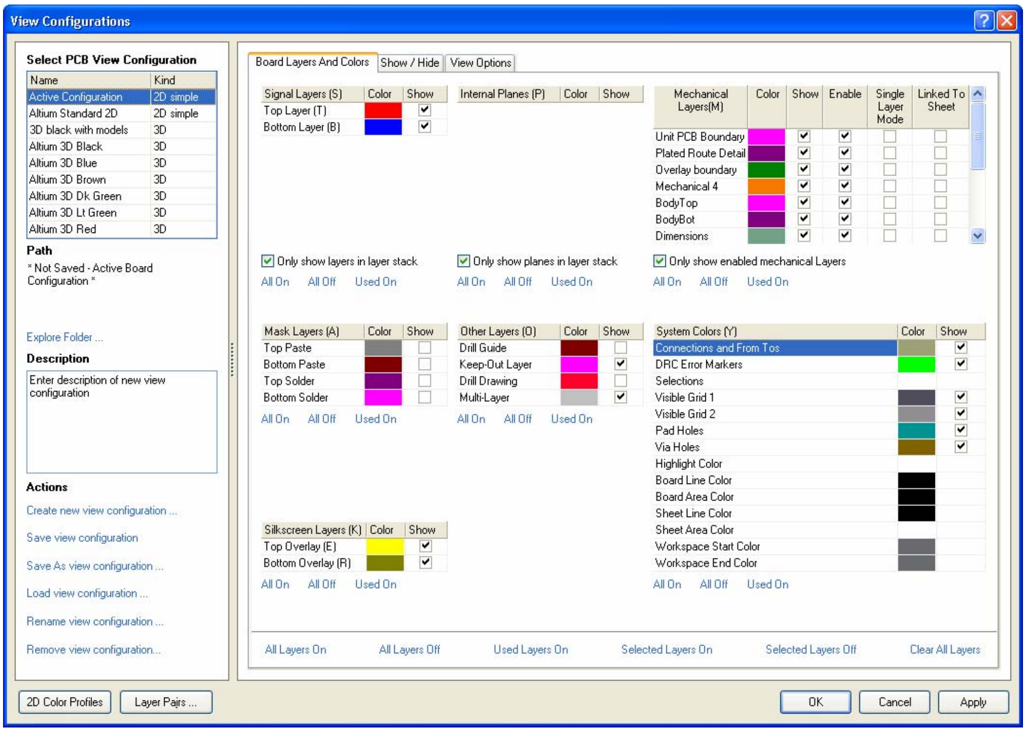



*Figure 8. Control the display of layers in the View Configurations dialog.*

- The current layer (the layer you are placing on) is set by any of the following:
- Clicking on the appropriate Layer tab at the bottom of the workspace,
- Pressing the * key to toggle to the next copper layer,
- Pressing the **+** or **–** keys on the numeric pad to move up or down to the next layer.

## 9.3.2 Layer definitions

Each of the PCB Editor layers is described below.

### Signal Layers

There are 32 signal layers that can be used for track placement. Anything placed on these layers will be plotted as solid (copper) areas on the PCB. As well as tracks, other objects (e.g. fills, text, polygons, etc.) can be placed on these layers. The signal layers are named as follows:

| | |
|---|---|
| Top Layer | Top signal layer |
| MidLayer1 to MidLayer30 | Inner signal layers |
| Bottom Layer | Bottom signal layer |

Signal layer names are user-definable.

### Internal Planes

Sixteen layers (named Internal Plane 1–16) are available for use as power planes. Nets can be assigned to these layers and multi-layer pads and vias automatically connect to these planes. Plane layers can be split into any number of regions, with each region being assigned to a different net. Nested split planes are supported. Internal Plane layer names are user-definable. Internal planes are designed and output in the negative, objects that are placed on the plane define regions of no copper.

### Silkscreen layers

Top and Bottom Overlay (silkscreen) layers are typically used to display component outlines and component text (designator and comment fields that are part of the component description).

### Mechanical layers

Thirty two mechanical drawing layers are provided for fabrication and assembly details, such as dimensions, alignment targets, annotation or other details. Mechanical layer items can be automatically added to other layers when printing or plotting artwork. Mechanical layer names are user-definable. Mechanical layers can also be paired; use this when creating library components that require side-of-board layer-related information, such as glue dots.

### Solder Mask

Top and bottom Solder Mask layers are provided for creating the artwork used to make the solder masks. These automatically generated layers are used to create masks for soldering, usually covering everything except component pins and vias. You can control the expansions for these masks when printing/plotting by including a Solder Mask Expansion rule, or the manual override feature in the pad/via dialogs. Refer to the *Design Rules* section for more information on the Solder Mask Expansion rule. User-defined openings in the mask can also be created by placing design objects directly on the mask layer. These layers are designed in the negative, the visible objects become openings in the mask.

### Paste Masks

Top and bottom Paste Mask layers are provided to generate the artwork which is used to manufacture stencils to deposit solder paste onto surface mount pads on PCBs with surface mount devices (SMDs). The size of the paste deposit is controlled by Paste Mask Expansion rule, refer to the *Design Rules* section for further information. It can also be defined using the manual override in the pad/via dialog, or by placing objects manually on the paste mask layer.

### Drill Drawing

Coded plots of board hole locations are typically used to create a drilling drawing that shows a unique symbol for each hole size at each hole location. Individual layer pair plots are provided when blind/buried vias are specified. Three symbol styles are available: coded symbol; alphabetical codes (A, B, C etc.) or the assigned size.

### Drill Guide

A drill guide plots all holes in the layout. Drill guides are sometimes called *pad masters*. Individual layer pair plots are provided when blind/buried vias are specified. These plots include all pads and vias with holes greater than zero (0) size.

### Keep Out layer

This layer is used to define the regions where components and routes can validly be placed. For example, the board boundary can be defined by placing a perimeter of tracks and arcs, defining the region within which all components and tracks must be placed. No-go areas for components and tracks can be created inside this boundary by blocking off regions with tracks, arcs and fills. Keepouts apply to all copper layers. The basic rule is that components cannot be placed over an object on the Keep Out layer and routes cannot cross an object on the Keep Out layer.

Note that there are also layer-specific keepouts, each standard design object has a keepout attribute, and when this is enabled the object behaves as a layer-specific keepout and is automatically excluded from Gerber and ODB++ output generation.

### Multi-layer

Objects placed on this layer will appear on all copper layers. This is typically used for through-hole pads and vias, but other objects can be placed on this layer.

### System section

The options described below cannot have objects placed on them but they are turned on or off in the **System Colors** section of the *Board Layers & Colors* dialog.

#### DRC Errors

This option controls the display of the Design Rule Check (DRC) error marker.

#### Connections

This option controls the display of the connection lines. The PCB Editor displays connection lines wherever it locates part of a net that is unrouted.

#### Pad and Via Holes

Controls the display of pad and via holes. To be able to distinguish pads from vias in draft mode, pad holes are outlined in the current Pad Holes color.

#### Visible Grids

Controls the display of the two visible grids.

## 9.3.3 Exercise – Configuring the layer display

To confirm that the required layers are displayed:

1. Press the **L** shortcut key to display the *View Configurations* dialog.
2. Click the **Used Layers On** link at the bottom of the dialog, to display all layers that have objects on them.
3. Confirm that the **Connections and From Tos** check box is enabled.
4. Note that mechanical layer 16 is linked to the sheet. This layer contains all the objects used to create the sheet template.

## 9.3.4 Defining the Electrical Layer Stackup

The number and order of electrical layers is defined in the *Layer Stack Manager* dialog.

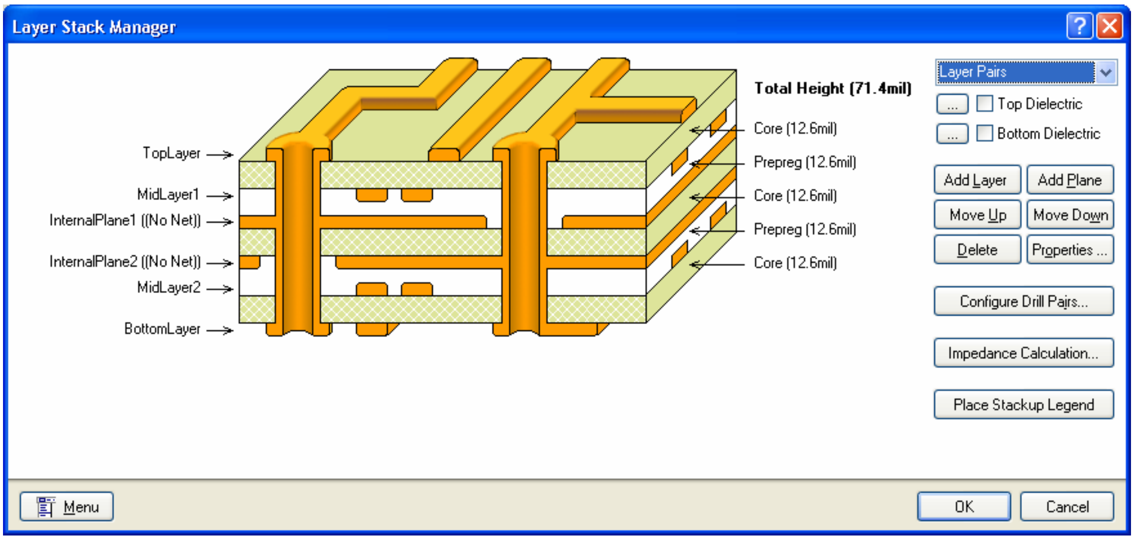



*Figure 9 Layer Stack Manager dialog*

The Layer Stack Manager allows you to visualize the 'stack up' of your PCB, i.e. the relationship between copper, substrate and Prepreg. A picture of your layer stack can be copied to the Windows clipboard and pasted into project documentation by right-clicking and selecting **Copy to Clipboard**.

### 9.3.4.1 Adding layers

#### Adding a Signal or Plane layer

Use the buttons on the right to add signal and plane layers to the board. The new layer is added below the layer selected in the dialog (unless the selected layer is the Bottom Layer). You can also right-click to add new layers. Typically PCBs are fabricated from an even number of layers; these can be any mix of signal and plane layers. Double-click on the layer name to define the layer name, the copper thickness and assign the net name for plane layers.

#### Adding Insulation layers

As additional layers are added to the PCB, insulation layers are automatically added. The insulation layer can be either Core or Prepreg and this is determined by the Stack Up style setting.

### 9.3.4.2 Working with layers

#### Editing layer properties

Double click on a layer name to edit the layer properties, including the name and the physical properties.

#### Deleting a layer

To delete a layer, click on the name text of an existing layer and then click on the **Delete** button, or right-click and choose **Delete** from the right-click menu.

#### Editing the Stack Up order

To change the order in which layers are defined in your PCB, click on the name of the layer and click on the **Move Up** or **Move Down** buttons, or right-click and choose **Move Up** or **Move Down**.

**Editing the Stack Up style**

The Stack Up style defines the order in which the PCB substrate, copper and prepreg insulation layers are fabricated as well as the finish on the PCB. The style is selected in drop down list in the top right corner of the Layer Stack Manager. The choices are:

- Layer Pairs
- Internal Layer Pairs
- Build Up.

The board finish is defined by selecting the buttons next to the Top and Bottom Dielectric check boxes. Click on these to set the material, thickness and dielectric constant for the finish.

#### 9.3.4.3 Where the physical properties are used

The physical properties that are defined in the different layer dialogs, including insulation type, thickness and dielectric constant, and the copper thickness, is used by the signal integrity analysis feature.

### 9.3.5 Layer Sets

To define a layer set there is now a tool in PCB that has predefined layer sets or you can create your own. This is located in **Design » Manage Layer Sets » Board Layer Sets**.

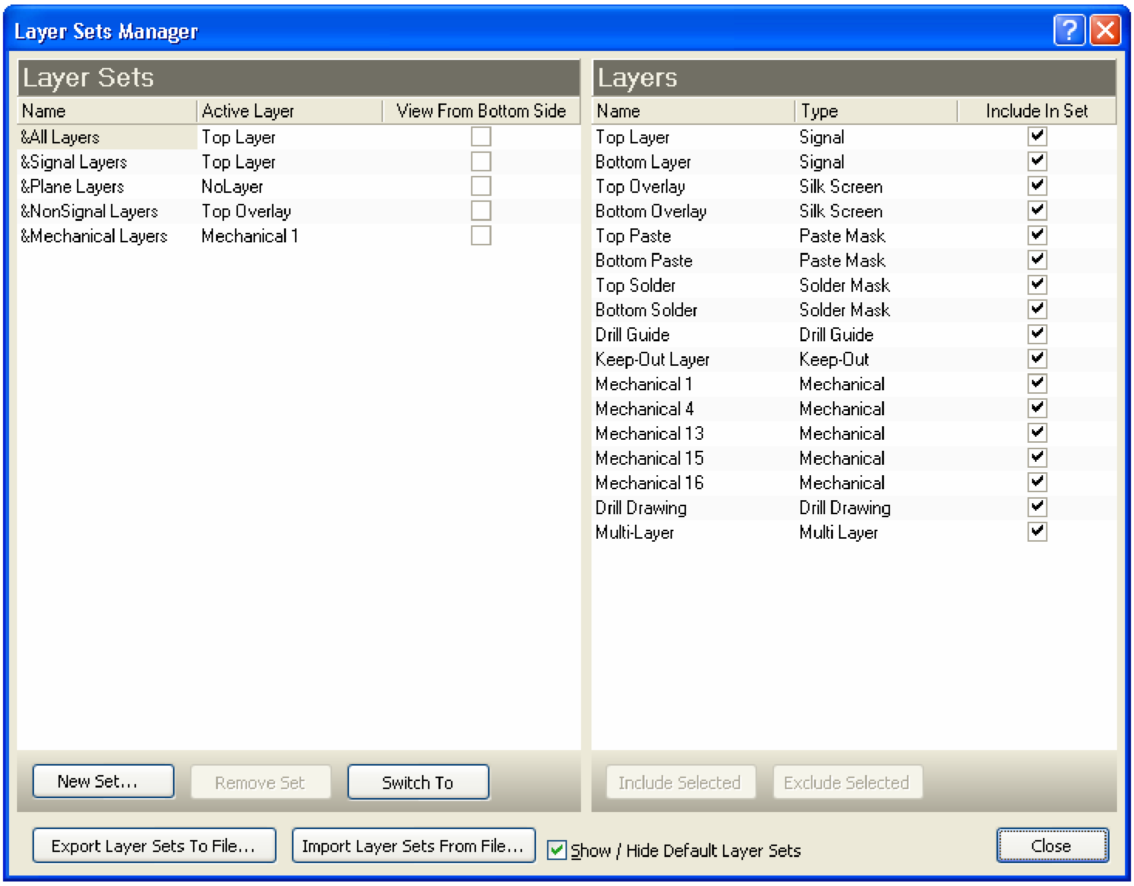



*Figure 10 Defining new layer sets*

To create a new layer set simply click on **New Set**, give the set a name and pick the layers you wish to include in the set. Layer sets can also be imported and exported to and from PCB designs.

## 9.3.6 Drill pairs

The term *drill pairs* refers to the two layers that a drilling operation starts from and finishes at. By default, one Top-Bottom drill pair is defined. If blind or buried vias are to be used on your PCB, layer pairs must be defined for these. Click on the **Drill Pairs** button in the Layer Stack Manager to display the Drill Pair Manager.

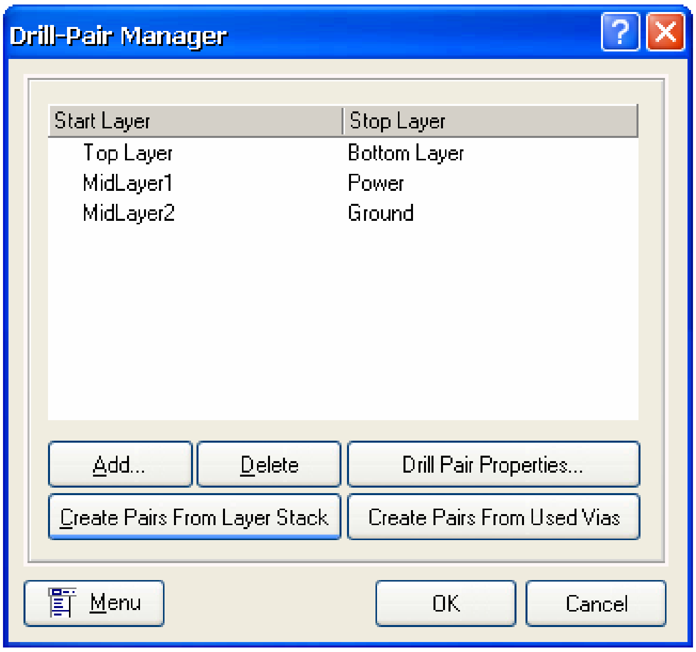



*Figure 11 Define the drill pairs if the board uses blind/buried vias*

## 9.3.7 Placing a Stackup Legend

A legend that details the current layer stackup and individual layer properties can be placed anywhere in the workspace.

- Place the legend via the **Place Stackup Legend** button in *Layer Stackup* dialog.
- After clicking once to define the first corner of the legend, press Tab to configure which detail should be included.
- The detail in the legend is automatically determined from the current board configuration, if the configuration changes you will need to re-place the legend.

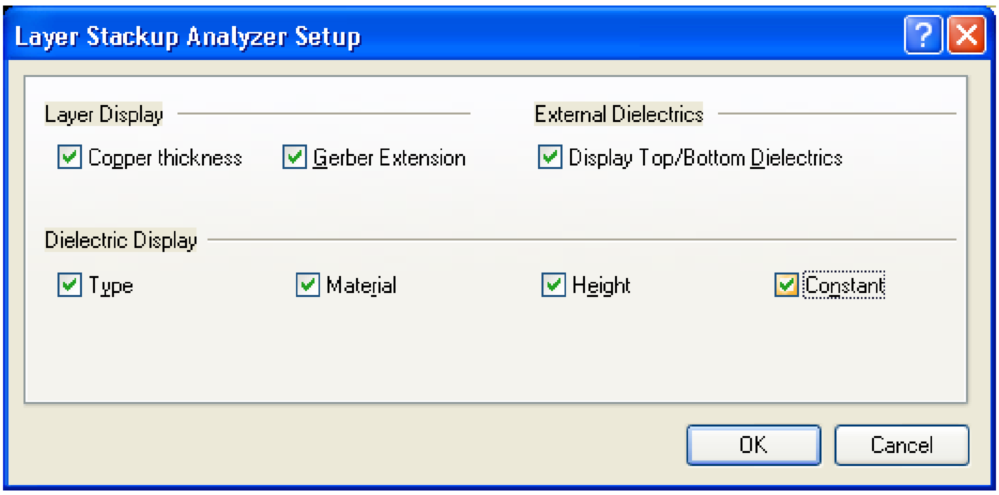



*Figure 12. Layer stackup legend setup dialog.*

Layer Stack Up Detail for: Temperature Sensor - blank.PcbDoc

| Layer Name | Gerber Document | Copper Thickness | Dielectric Height | Dielectric Material | Dielectric Constant | Dielectric Type |
|---|---|---|---|---|---|---|
| Top Solder Mask | (.GTS) | | 0.4mil | Solder Resist | 3.50 | |
| Top Layer | (.GTL) | 1.4mil | | | | |
| | | | 12.6mil | FR-4 | 4.80 | Core |
| MidLayer1 | (.G1) | 1.4mil | | | | |
| | | | 12.6mil | FR-4 | 4.80 | PrePreg |
| InternalPlane1 | (.GP1) | 1.4mil | | | | |
| | | | 12.6mil | FR-4 | 4.80 | Core |
| InternalPlane2 | (.GP2) | 1.4mil | | | | |
| | | | 12.6mil | FR-4 | 4.80 | PrePreg |
| MidLayer2 | (.G2) | 1.4mil | | | | |
| | | | 12.6mil | FR-4 | 4.80 | Core |
| Bottom Layer | (.GBL) | 1.4mil | | | | |
| Bottom Solder Mask | (.GBS) | | 0.4mil | Solder Resist | 3.50 | |

*Figure 13. Stackup legend placed on a mechanical layer*

## 9.3.8 Defining Mechanical layers

Mechanical layers are added to the PCB workspace in the *View Configurations* dialog. Before a Mechanical layer can be used, it must be enabled.

- To enable a new layer first disable the **Only show enabled mechanical layers** check box. This will result in all layers being listed. Enable the new layer, then turn the **Only show enabled mechanical layers** on again.
- To edit a mechanical layer name, click to select the name and press **F2** to edit it.

| Mechanical Layers(M) | Color | Show | Enable | Single Layer Mode | Linked To Sheet |
|---|---|---|---|---|---|
| Mechanical 1 | | ✓ | ✓ | ☐ | ☐ |
| Mechanical 4 | | ✓ | ✓ | ☐ | ☐ |
| Mechanical 13 | | ✓ | ✓ | ☐ | ☐ |
| Mechanical 15 | | ✓ | ✓ | ☐ | ☐ |
| Mechanical 16 | | ✓ | ✓ | ☐ | ☐ |
| Mechanical 31 | | ✓ | ✓ | ☐ | ☐ |
| Mechanical 32 | | ✓ | ✓ | ☐ | ☐ |

*Figure 14. Setting up Mechanical Layers in the Board Layers & Colors dialog.*

- The **Show** check box allows you to control the display of a mechanical layer.

- When checked, the **Display In Single Layer Mode** check box causes that layer to be displayed when Single Layer Mode is invoked (SHIFT+S).
- Check the **Linked to Sheet** check box to relate a mechanical layer to the white sheet object. Related mechanical layers are then hidden when the Display Sheet option is disabled (*Board Options* dialog). They are also used to determine the extents of the sheet when the **Auto-position sheet** option is chosen in the **Board Shape** sub-menu.

### 9.3.8.1 Mechanical Layer Pairs

Mechanical layers can be paired. Used this feature when you need extra detail in the component footprint, and that detail needs to change to another layer when the component is moved from the top side of the board to the bottom side. An example of this is glue dot information.

- Configure Mechanical pairs by clicking the **Layer pairs** button at the bottom of the *View Configurations* dialog, or right clicking on the layer tabs and selecting **Configure Mechanical Pairs**.
- The objects placed on a mechanical layer in the PCB library will flip to its paired layer when the component is switched to the bottom of the board.
- Click the **Add**, **Delete** or **Mechanical Pair Properties** buttons in the *Mechanical Layer Pairs* dialog to modify the mechanical layer pairs.

## 9.3.9 Internal power planes

The PCB Editor supports up to sixteen power planes. These planes are defined in the negative – objects placed on a plane layer become regions of no copper.

### 9.3.9.1 Defining an internal power plane

- An internal power plane is added, named and assigned to a net using the *Layer Stack Manager*. When a net has been assigned to an internal plane layer, pins in that net automatically connect to that plane layer using thermal relief connections.
- Double-click on the plane in the Layer Stack Manager, or in the workspace to assign the net. The PCB Editor automatically connects pins that belong to the power plane net and isolates all other pins from the plane.
- The style of plane connections is defined in the **Power Plane Connect Style** design rule. Nets that are not connected to the plane are isolated from it by a clearance that is defined in the **Power Plane Clearance** rule.
- The pullback, or region of no-copper required around the edge of the PCB, is defined in the *Edit Layer* dialog. Double-click on the plane in the Layer Stack Manager to display this dialog.

### 9.3.9.2 Defining a split power plane

- Internal power planes can be split and shared amongst multiple nets.
- A plane is split by placing objects (typically lines) to divide it into separate regions (select **Place » Line**). As soon as you stop placing lines on a plane the layer is analyzed and each separate split region detected.
- The width of the placed lines defines the clearance between the split regions. Press the TAB key during line placement to change this width.
- Double-click on a split region to assign it to a net. Alternatively, set the display mode of the PCB panel to **Split Plane Editor**.
- Splits can be created completely within another split region.

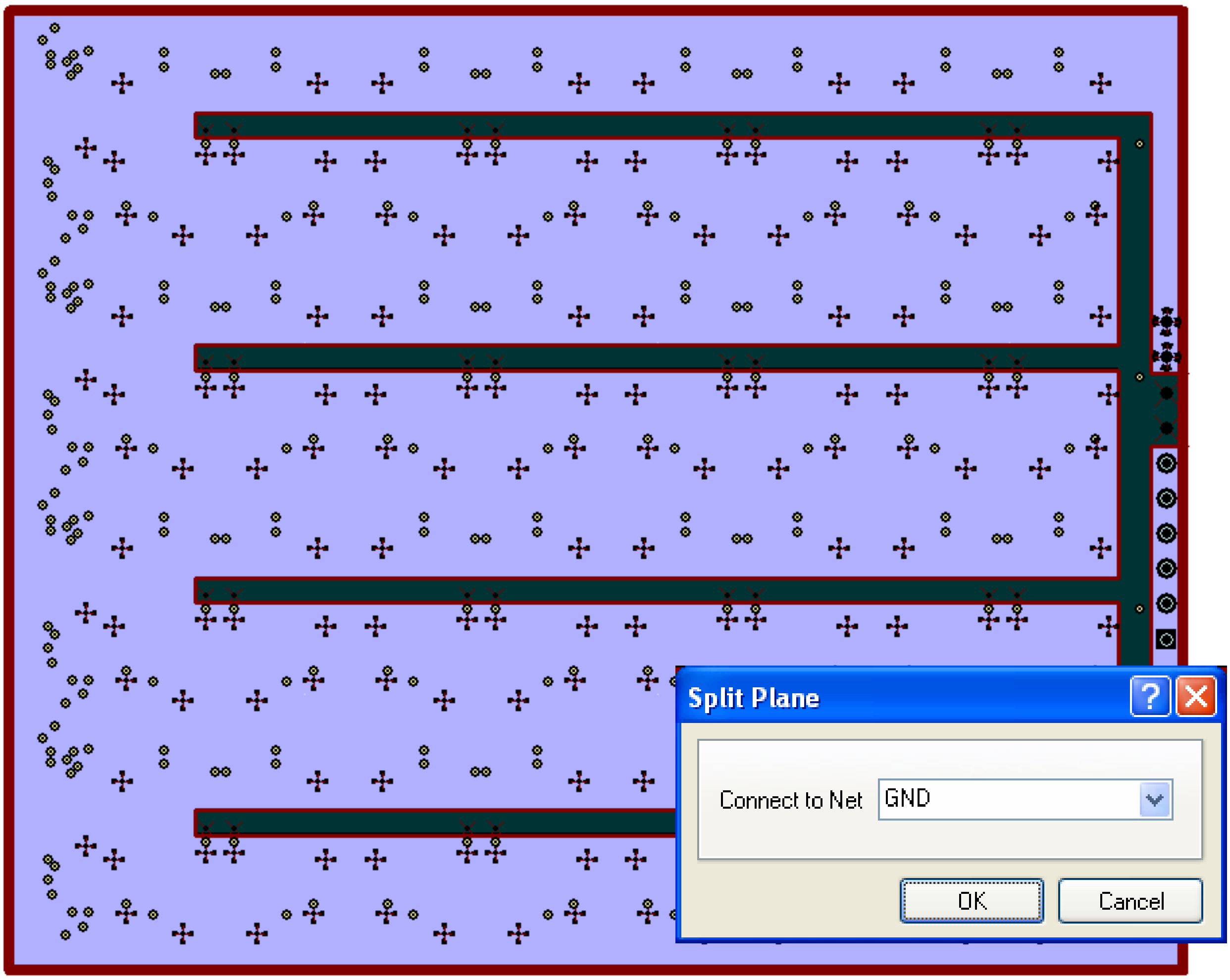



*Figure 15. Split planes on an Internal plane layer with the Split Plane dialog showing the net assignment for the large split region (Peak Detector With Banking.PcbDoc).*

#### 9.3.9.3 Re-defining a split plane

A split plane is defined by the set of objects that make up its boundary. Move and modify these to redefine the split plane.

#### 9.3.9.4 Deleting a split plane

Delete the split boundary lines to delete a split plane.

### 9.3.10 Exercise – Setting up layers

1. Set up the layers in the Layer Stack Manager. Select layer names, right-click and set the properties, i.e. names and copper thickness. Note that you can use the buttons to add and delete layers and move them up and down in the stack.
2. Open the *View Configurations* dialog and select the layers you need to show in the design window, e.g. Top and Bottom layers, Keep-Out Layer, Drill Drawing, Multi Layer and Top Overlay.
3. Show and enable Mechanical layers 1, 4 and 16. Make sure the **Only Show enabled mechanical layers** are deselected first to show all mechanical layers available. Then turn this option on again when you have set up the layers you wish to use. Link Mechanical 16 to the sheet so that the title block of the template will appear on this layer.

TOP SECRET

Altium Designer

# Module 10: Global Editing

## Module 10: Global Editing





Module Seq = 10

# 10.1 Editing Multiple Text Objects

One of the powerful features of Altium Designer is the data editing system. The Schematic Editor includes a feature that provides a very efficient mechanism to edit text strings in your schematics. The reason this is powerful is that the same process using find similar objects would take three separate operations on netlabels, ports and sheet entries.

## 10.1.1 Find and Replace Text

You can perform complete or partial substitutions on text using the following methods.

- To target a section of a string, include the * or ? wildcards as appropriate. In Figure 1, the combination of the Text to Find, the Sheet Scope and the Restrict to net Identifiers option will result in any net label, port, sheet entry or power port whose net attribute starts with the letters RB being found.
- If you wish to replace the entire contents of a text field with a new value, simply enter the new value in the Replace With field.
- Partial string substitutions can be performed using the syntax {oldtext=newtext}. This means you can change a portion of the current string (oldtext) to a new string (newtext). In Figure 1, the letters RB will be replaced with LCD. Any other characters in each found net identifier will remain the same. For example, the following changes would occur:

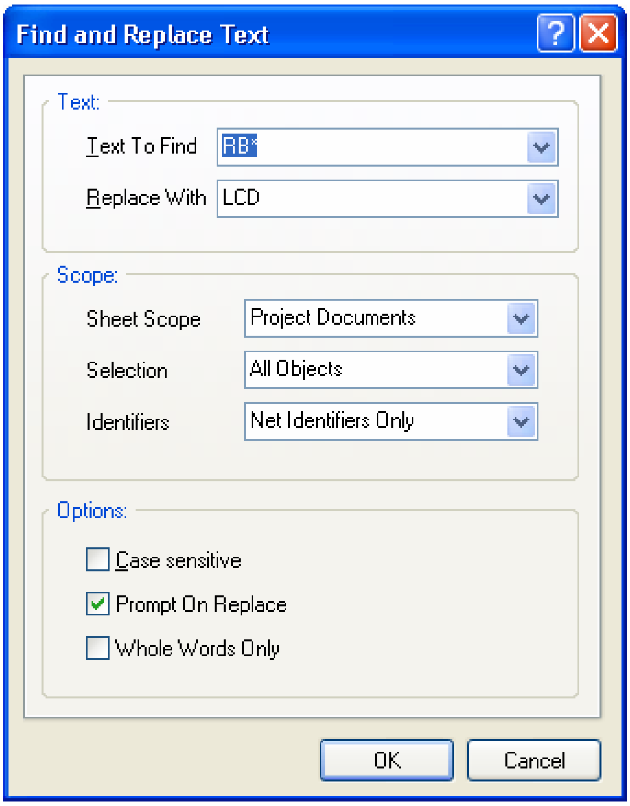
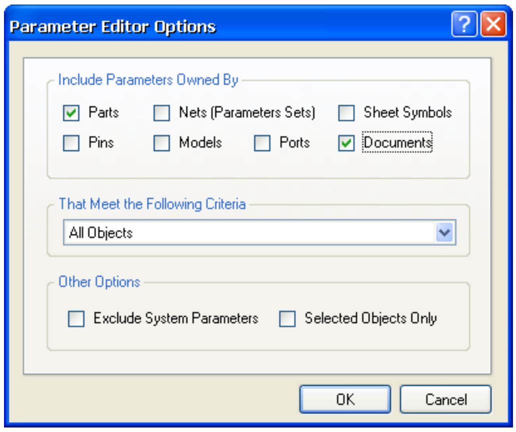



*Figure 1. Performing a partial string substitution*

| Before Find and Replace | After Find and Replace |
|---|---|
| RB1 | LCD1 |
| RB200 | LCD200 |
| RBout | LCDout |
| RB_CLK | LCD_CLK |
| RB[0..7] | LCD[0..7] |

## 10.1.2 The Parameter Manager

- The Parameter Manager allows you to control all your parameters in one single editor. Open the Parameter Manager by selecting **Tools » Parameter Manager** from the menus. User-defined parameters can be added, removed or renamed in the Parameter Manager. You can modify the values of system-level parameters but these cannot be added, removed or renamed.
- You can select which parameters will be included in the *Parameter Table Editor* by limiting the types of parameters you wish to use in the *Parameter Editor Options* dialog. For example, you can exclude all system parameters, or only use document-level or part parameters.




- Changes to the values or names of parameters are made in the *Parameter Table Editor* and then an ECO is generated to execute the changes in the design or schematic library.

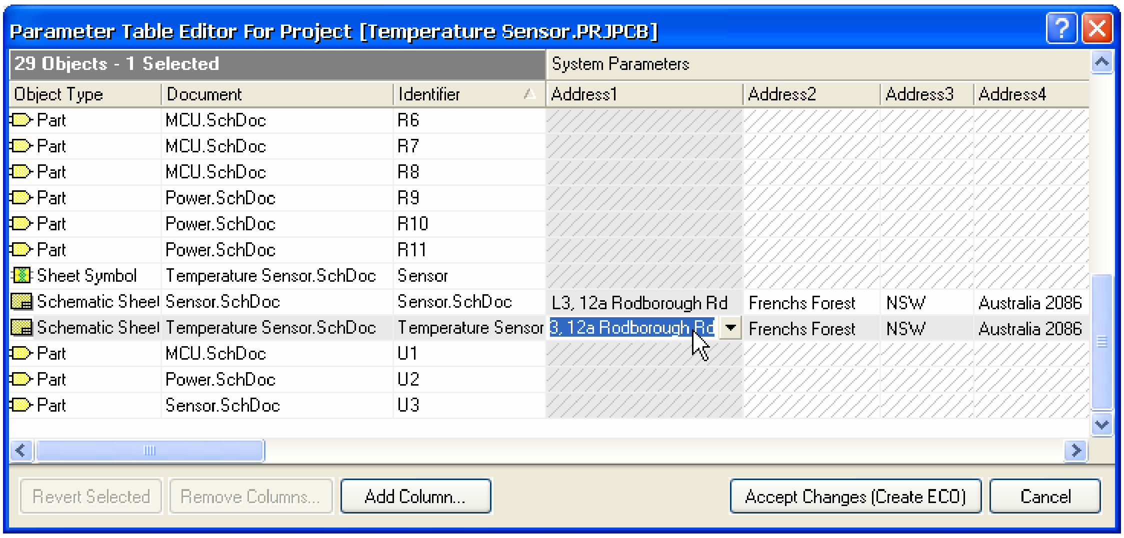


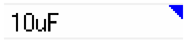

*Figure 2. Parameters can be modified using the Parameter Table Editor.*

### Tips for using the Parameter Manager

- Editing in the *Parameter Table Editor* is similar to editing in an Excel spreadsheet. For example, press **F2** or SPACEBAR to edit, type in the value or select it from a drop-down list, if available, and then press **Enter**. Use the arrow keys to move through the spreadsheet.
- You can edit multiple instances of the same parameter value by selecting the cells, right-clicking and selecting **Edit** for the drop-down menu. Type in the new value and press **Enter**. Right-click and choose **Revert** to undo changes to selected cells.
- Cells are highlighted in the *Parameter Table Editor* according to whether the parameter exists or has current values.

| | |
|---|---|
|   | the object possesses the parameter and the string entry in the field is its value. |
|  | the object possesses the parameter, but it currently has no value |
|  | the object does not possess the parameter. |

- When you modify a parameter, markers in the right-hand top corner of the cell indicates what changes will be made.

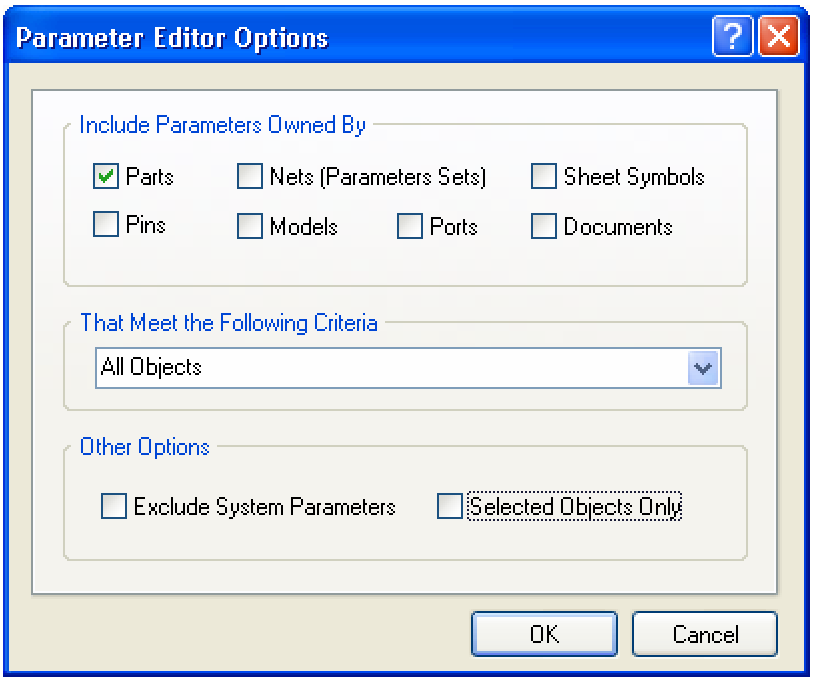
| | |
|---|---|
|   | the value assigned to the parameter will be changed. |
|  | the parameter will be added to the object but, in this case, no value will be assigned. |
|  | the parameter will be removed from the object. |

- Note that any changes made within the table are virtual changes that will not be implemented until the execution of an Engineering Change Order.
- Press **F1** in the Parameter Manager dialogs for more information.

## 10.1.3 Exercises – Using the Parameter Manager

### 10.1.3.1 Adding new parameters using the Parameter Manager

1. This exercise adds a new parameter, named Part Number, to all components in the design. If it is not open, re-open the 4 Port Serial Interface example project in the Reference Designs.
2. In the Schematic Editor with the required schematic documents open, select **Tools » Parameter Manager** to display the *Parameter Editor Options* dialog.
3. In the **Include Parameters Owned By** section of the dialog clear the **Documents** checkbox, enable the **Parts** checkbox, then select **All Objects** as the criteria. Click **OK**.




4. In the *Parameter Table Editor* dialog, click on **Add Column**. The *Add Parameter* dialog displays. Type in a new parameter name, e.g. `Part Number`, and enable **Add to all objects**. Click **OK** to create the new parameter column. Enter values in the new Part Number column as required. Click on **Accept Changes (Create ECO)**.
5. Click on **Validate Changes**. If the validation is successful, click on **Execute Changes**.
6. When the changes have been executed, click **Close**. The new parameters are added to the components in the schematic. These can be checked by double-clicking on the components in the schematic document to display the *Component Properties* dialog. The new parameter is added to the Parameters list.

## 10.2 The Data Editing System

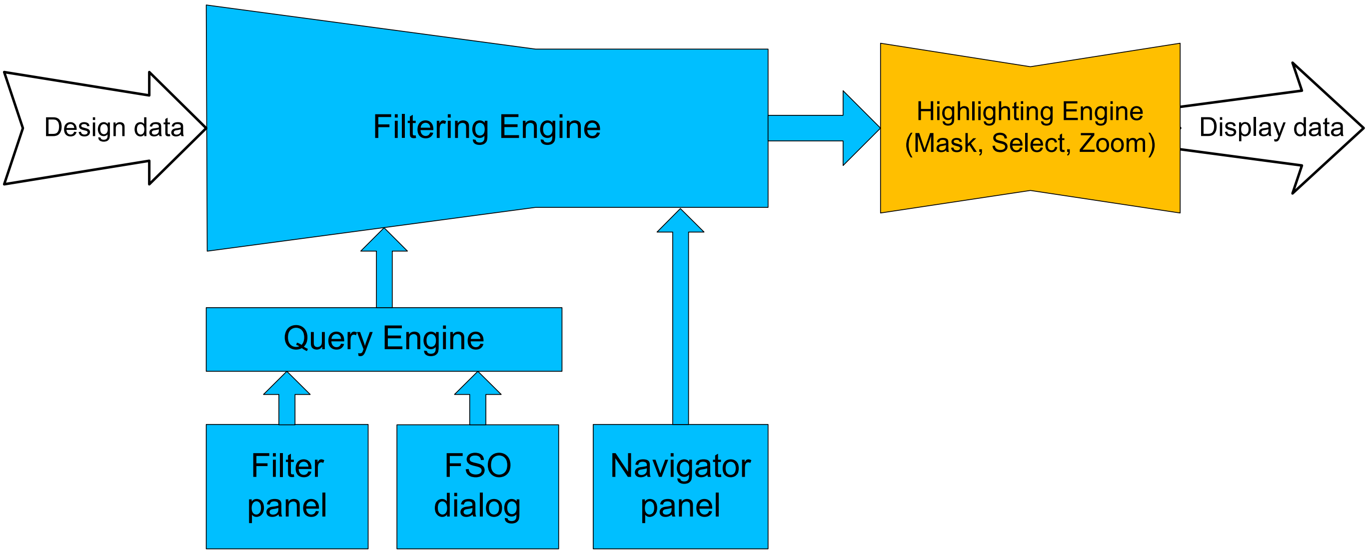



*Figure 3. Diagram of the filtering/highlighting system*

One of the greatest challenges you face as a designer is managing the large amounts of design data that is created during the design process. To facilitate this, Altium Designer has a powerful data editing system. This system allows you to manage, find and edit design data in a variety of ways.

To provide flexible and appropriate methods of editing data, three alternate views of the data can be used to access and edit design objects:

- The traditional graphical view
- The **Inspector** panel (press **F11** to toggle it on and off)
- The **List** panel (press **Shift+F12** to toggle it on and off)
- The **Filter** panel (press **F12** to toggle it on and off)

The **Inspector** displays the attributes of the currently selected object(s), with the total number selected being listed at the bottom. Note that the **Inspector** can be used to edit different kinds of objects simultaneously.

The **List** panel gives a spreadsheet-like, or tabular list of objects in the schematic sheet or PCB workspace. Individual or multiple cells can be edited in the **List** panel.

A powerful filtering engine is used to control the amount of data that is presented for editing in all three views. Data can be filtered using the *Find Similar Objects* dialog, the PCB editor panel, or by writing a query in the **Filter** panel. Figure 3 shows a diagram of the data editing system.

The **Filter** panel is used to type in a query that filters the entire data set, reducing both the graphical display and the **List** panel to display only those objects that satisfy the query. In the graphical display this can be shown by the fading of objects that have been filtered out (and are no longer editable).

One of the powerful features of this data editing system is the ability to edit multiple objects simultaneously. The basic approach to use the data editing system is to:

***Select*** the required objects for editing

***Inspect*** the objects

***Edit*** the object attribute(s).

## 10.2.1 Finding and Selecting Objects

### 10.2.1.1 Using the PCB Panel

The **PCB** panel can be used for browsing objects in a PCB. To open the panel click **PCB » PCB** in the panel control buttons down the bottom right of the workspace.

- The options at the top of the panel control how the chosen object(s) will be highlighted:
  - **Mask:** this option fades all objects except those of interest. While masked objects are still visible, they can not be edited. Use the Mask Level control at the bottom right to control the amount of fading.
  - **Select:** select the target object(s). Selected objects are highlighted using various white shading techniques.
  - **Zoom:** zoom in to fit the highlighted objects in.
  - **Clear existing:** enable this to automatically clear any existing highlighting whenever you choose another object.
- Use the CTRL+Click combination to highlight multiple objects.
- Right-click in the panel to control which primitive kinds are displayed, this is very handy for excluding certain object kinds.
- Click the **Clear** button Clear to clear all masking and selections.
- Highlighting results are displayed in all three views – graphical, List and Inspector (if Selection is enabled).

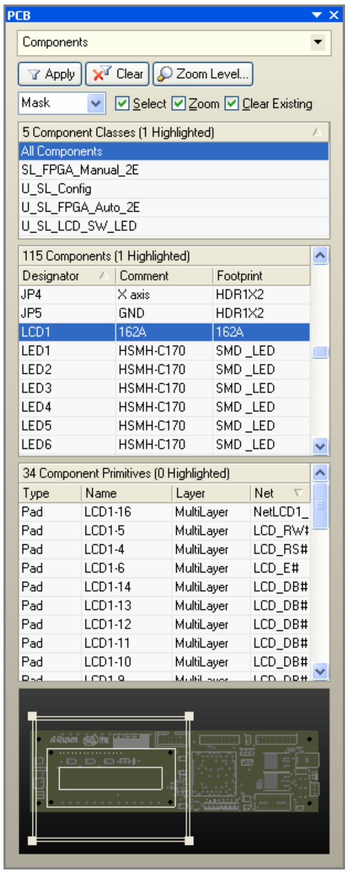



*Figure 4. Use the PCB panel to find and select objects.*

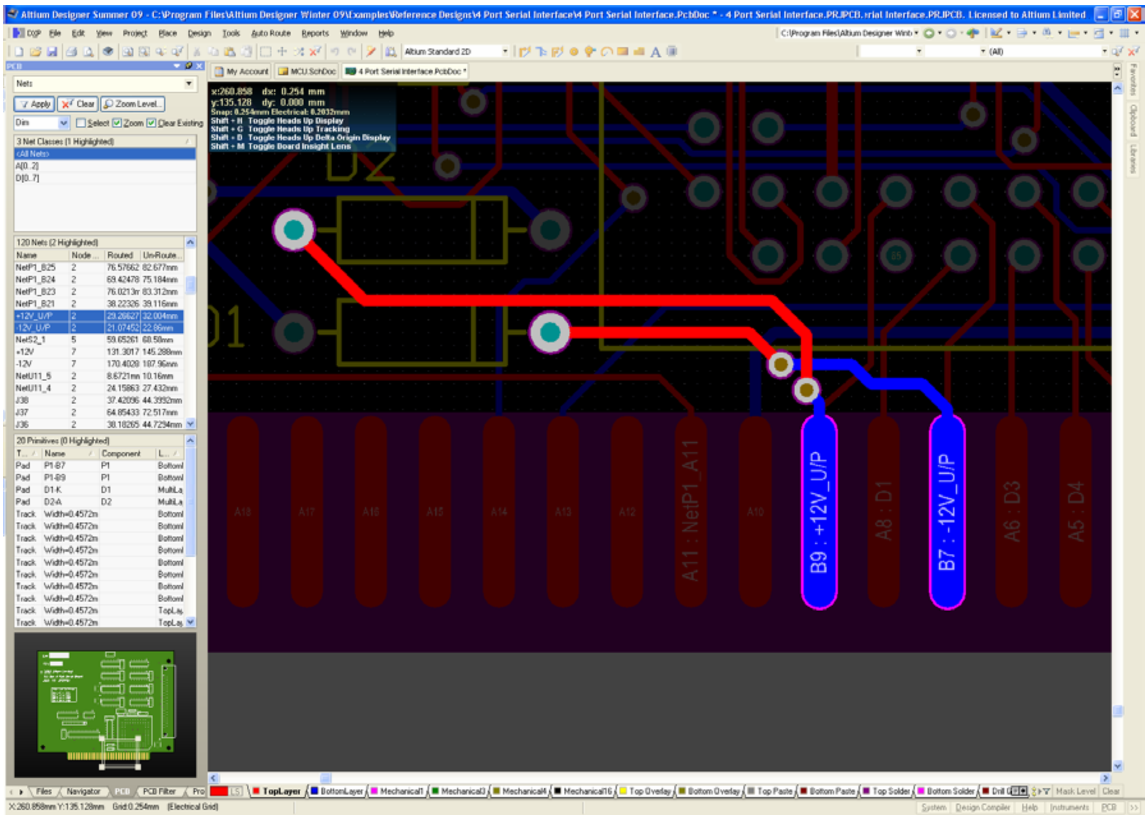



*Figure 5. Using the panel to highlight two nets. Note that all other objects have been faded (masked).*

### 10.2.1.2 Using the *Find Similar Objects* dialog

The panel is ideal when working with group-type objects like components and nets. When you are working at a primitive object level, it can be more efficient to use the *Find Similar Objects* dialog. This dialog works in both PCB and Schematic.

- To launch the *Find Similar Objects* dialog, right-click on an object of interest, and select **Find Similar** from the floating menu.
- The dialog will appear, listing the attributes of that object. Next to each attribute is a drop down. Set this if you wish to use it as a matching criteria.
- Clicking **Apply** will run the search for matching objects but will leave the dialog open. Clicking **OK** will close the dialog and run the search.
- This will select all objects that match the find criteria. Figure 6 shows the *Find Similar Objects* dialog configured to find and select all PCB text strings that are component comments.
- Enable the **Run Inspector** check box to automatically launch the Inspector, where you can edit any attribute(s) of the found and selected objects.

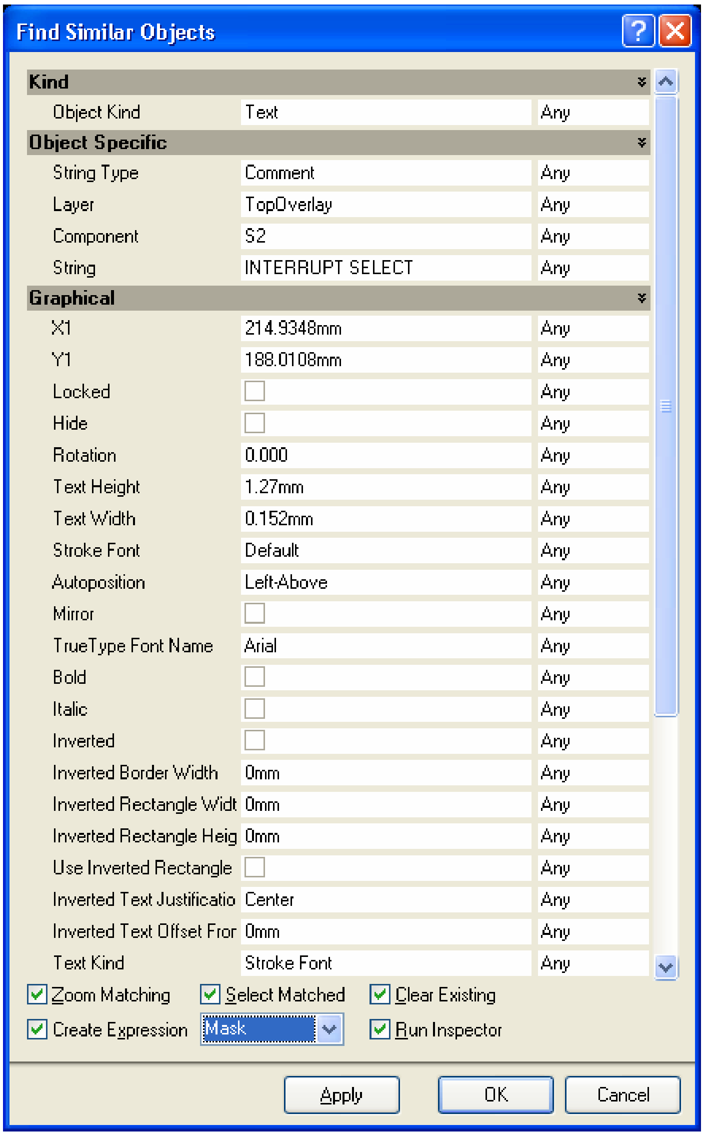



*Figure 6. Using the Find Similar Objects dialog to highlight all component comments.*

### 10.2.1.3 Using the Filter Panel

Underlying the techniques for finding objects described so far is a powerful data filtering engine. You can also access this engine directly by writing a query to describe the objects that you wish to target.

- Press **F12** to toggle the **Filter** panel on, where you write the query.
- A Query is an instruction, written using query language keywords. For example, entering the query `IsComment or IsDesignator` in the PCB editor List panel will reduce the contents of both the graphical display and the List panel to only display the component designator and comment strings on the PCB.
- For a complete list of query keywords, click the Helper button or refer to the document *TR0110 Query Language Reference.PDF* in the Altium Designer help directory. When the cursor is on a keyword, press the **F1** key in the Query Helper dialog for a complete description of that keyword. Press **F1** when the cursor is within an arithmetic operator for information on the operators.
- Refer to the document, *AR0129 An Insiders Guide to the Query Language.PDF* for detailed information on writing queries.

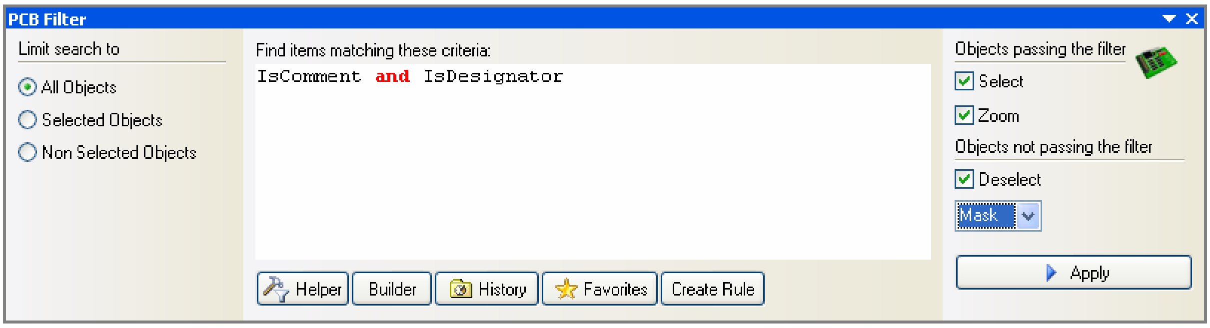
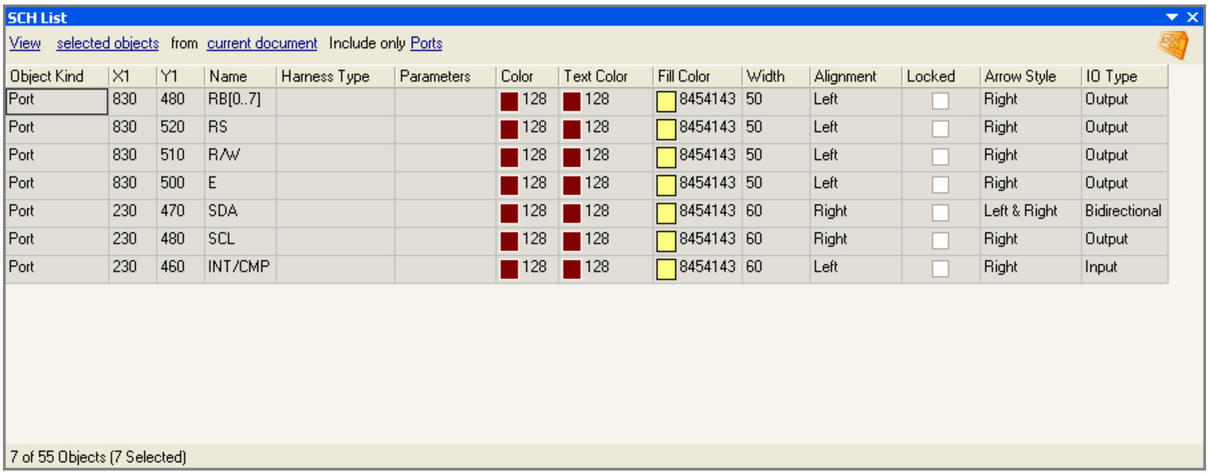



*Figure 7. Use the Filter panel to query the design data and access specific objects.*

### 10.2.1.4 Using the List Panel

The List panel can also be used to examine and edit the properties of objects. Use the List panel when you want to examine/compare attributes, edit only some of the objects or even paste in text from external editors like Excel.

- Press **Shift+F12** to toggle the **List** panel on/off.
- An individual cell in the List can be edited, press the SPACEBAR or right-click and select **Edit**.
- Multiple cells can be edited simultaneously, select them, press the SPACEBAR, type in the new value and press ENTER on the keyboard.
- Blocks of cell data can be copied and pasted to/from a spreadsheet.
- For group-type components, such as components or nets, you can include their primitive parts (child objects) by right-clicking and choosing the appropriate **Show Children** option.
- When there are multiple object types displayed, only attributes that are common to all are displayed. You can remove objects from display in the List panel, select those you wish to keep, right-click and choose **Remove Non-Selected** from the menu.
- Column display is managed by right-clicking on the column headers and selecting **Choose Columns**.




| Object Kind | X1 | Y1 | Name | Harness Type | Parameters | Color | Text Color | Fill Color | Width | Alignment | Locked | Arrow Style | IO Type |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Port | 830 | 480 | RB[0..7] | | | 128 | 128 | 8454143 | 50 | Left | | Right | Output |
| Port | 830 | 520 | RS | | | 128 | 128 | 8454143 | 50 | Left | | Right | Output |
| Port | 830 | 510 | R/W | | | 128 | 128 | 8454143 | 50 | Left | | Right | Output |
| Port | 830 | 500 | E | | | 128 | 128 | 8454143 | 50 | Left | | Right | Output |
| Port | 230 | 470 | SDA | | | 128 | 128 | 8454143 | 60 | Right | | Left & Right | Bidirectional |
| Port | 230 | 480 | SCL | | | 128 | 128 | 8454143 | 60 | Right | | Right | Output |
| Port | 230 | 460 | INT/CMP | | | 128 | 128 | 8454143 | 60 | Left | | Right | Input |

*Figure 8. Using the List to examine / edit all designator and comment strings.*

**Note:** Make sure you set the list panel to edit mode. By default it's in view mode as shown in Figure 8. This is located in the top left corner of the list panel

#### 10.2.1.5 Inspecting and editing the selected objects

The Inspector panel is used to examine the properties of and edit the currently selected objects. Use the Inspector when you want to apply an edit to all the selected objects.

- Press **F11** to toggle the Inspector panel on/off.
- The Inspector can be used to examine 1 or many objects.
- The set of selected objects can be built up in many ways, including; manually, by writing a query, or using the Find Similar dialog.
- Dissimilar objects can be selected and edited, only their common attributes will be available for editing.
- After changing a value in the Inspector, press ENTER to apply it.
- String substitutions can be performed in the Inspector panel.

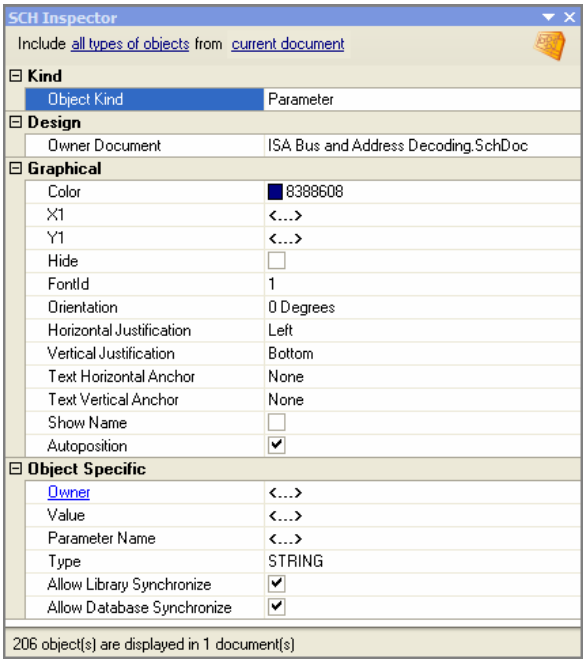



*Figure 9. The Inspector displaying the properties of the selected designator and comment strings.*

### 10.2.2 Exercises – editing objects

This exercise will demonstrate different ways of changing the width of the component overlay tracks and arcs and the height of component text.

#### 10.2.2.1 Changing the visibility of the component Comment strings

1. Locate component `S1` on the 4 Port Serial Interface PCB.
2. Right-click on the comment string and select **Find Similar** from the floating menu.
3. The *Find Similar* dialog appears, presenting the attributes of the object clicked on. Note that the **String Type** attribute has a value of Comment. Set the match by setting for this attribute to **Same** (as shown in Figure 6).
4. Enable the **Run Inspector** check box, clear the **Create Expression** check box and click **OK**.
5. The Inspector will appear with 37 objects selected. Click on the **Hide** attribute, clear the checkbox and press ENTER to apply the change.

All the component comment strings will now be visible on the board.

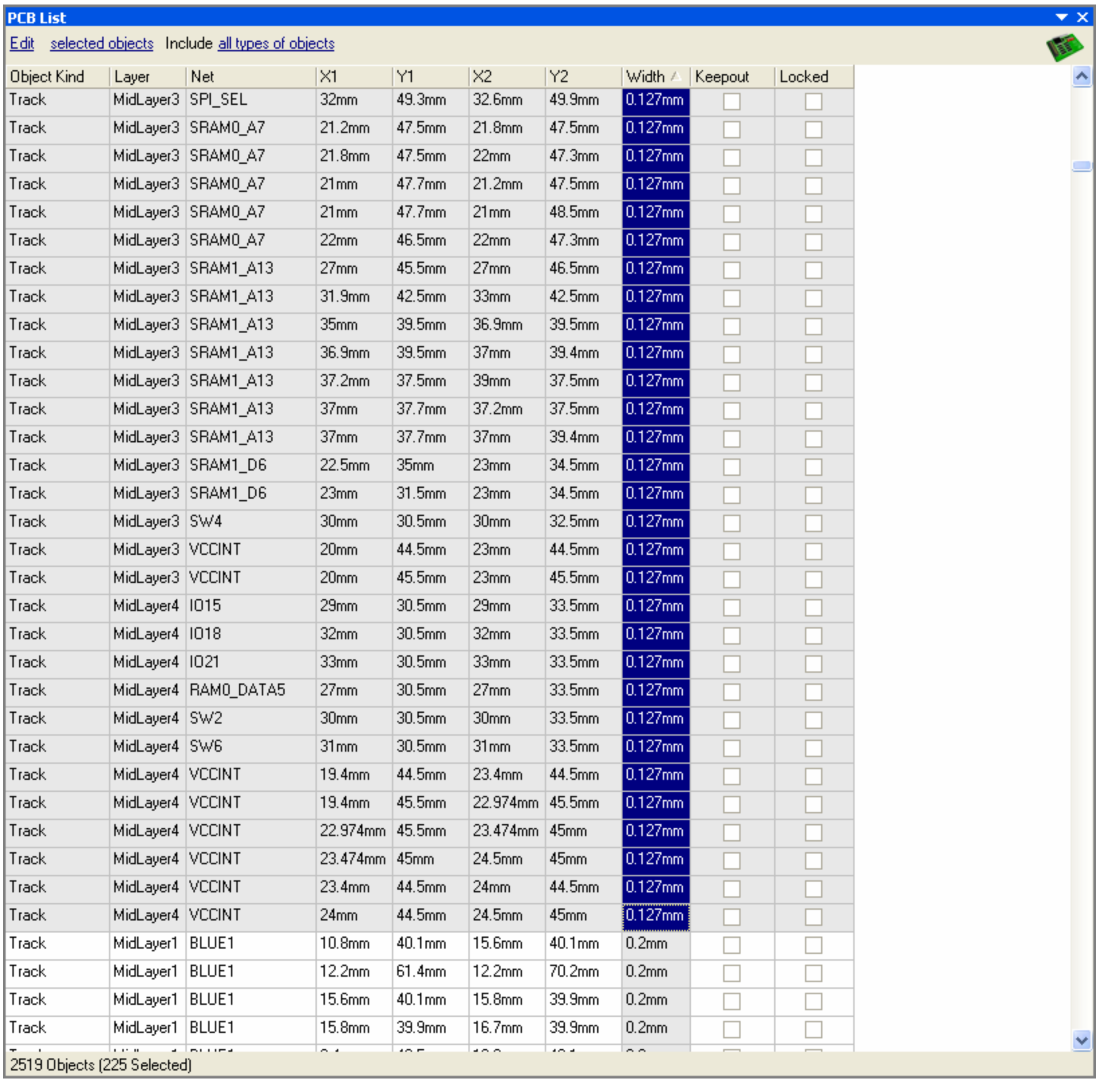
#### 10.2.2.2 Changing the height of designator and comment strings

1. In the editing region of the Filter panel, type the query `IsDesignator or IsComment` and then click the **Apply** button.
2. Open the Inspector panel, select the Text Height field, type in a new value of `40` and press ENTER on the keyboard to apply the change.

> **Note:** For more examples on editing multiple objects, refer to the tutorial *Editing Multiple Objects.*

### 10.2.2.3 Editing the width of tracks and arcs on midlayers

1. Open the **`NBP8 Xilinx Virtex-II Pro BGA456 Rev1.01`** example project, from *C:\Program Files\Altium Designer Summer 09\Examples\Reference Designs\Daughter Boards - 2 Connector\NBP8 Xilinx Virtex-II Pro BGA456 Rev1.01* directory then open the PCB.
2. Set the PCB panel to browse by *Nets*, enable the **Select**, **Zoom** and **Clear Existing** checkbox options at the top of the panel, and set the display mode dropdown to **Mask**.
3. Left-click on the All Nets option in the top section of the panel. This will populate the second section with all the nets. Right click in this second section and click **Select All**
4. Right-click in the Primitives region and set it so only **Tracks** and **Arcs** are displayed.
5. With only Tracks and Arcs displayed, sort the list of primitives by the **Layer** Column, by clicking on it. Once sorted find all the tracks and arcs that exist on the 4 midlayers and select them. There should be 2519 objects selected.
6. Open the PCB **List** panel. Set the options at the top of the panel to **Edit**, **Selected Objects** and **all types of objects** by clicking and selecting from the drop downs.
7. Sort by the **Width** column and select all the 0.127mm width tracks using the shift key. Once selected press F2 key, type in the new width of 0.2mm and press Enter to apply the change. All selected widths (225 in total, as shown at the bottom of the List panel) are changed to 0.2mm. View Figure 10 for details of the selection and what to set in the list panel.



PCB List

Edit selected objects Include all types of objects

| Object Kind | Layer | Net | X1 | Y1 | X2 | Y2 | Width | Keepout | Locked |
|---|---|---|---|---|---|---|---|---|---|
| Track | MidLayer3 | SPI_SEL | 32mm | 49.3mm | 32.6mm | 49.9mm | 0.127mm | | |
| Track | MidLayer3 | SRAM0_A7 | 21.2mm | 47.5mm | 21.8mm | 47.5mm | 0.127mm | | |
| Track | MidLayer3 | SRAM0_A7 | 21.8mm | 47.5mm | 22mm | 47.3mm | 0.127mm | | |
| Track | MidLayer3 | SRAM0_A7 | 21mm | 47.7mm | 21.2mm | 47.5mm | 0.127mm | | |
| Track | MidLayer3 | SRAM0_A7 | 21mm | 47.7mm | 21mm | 48.5mm | 0.127mm | | |
| Track | MidLayer3 | SRAM0_A7 | 22mm | 46.5mm | 22mm | 47.3mm | 0.127mm | | |
| Track | MidLayer3 | SRAM1_A13 | 27mm | 45.5mm | 27mm | 46.5mm | 0.127mm | | |
| Track | MidLayer3 | SRAM1_A13 | 31.9mm | 42.5mm | 33mm | 42.5mm | 0.127mm | | |
| Track | MidLayer3 | SRAM1_A13 | 35mm | 39.5mm | 36.9mm | 39.5mm | 0.127mm | | |
| Track | MidLayer3 | SRAM1_A13 | 36.9mm | 39.5mm | 37mm | 39.4mm | 0.127mm | | |
| Track | MidLayer3 | SRAM1_A13 | 37.2mm | 37.5mm | 39mm | 37.5mm | 0.127mm | | |
| Track | MidLayer3 | SRAM1_A13 | 37mm | 37.7mm | 37.2mm | 37.5mm | 0.127mm | | |
| Track | MidLayer3 | SRAM1_A13 | 37mm | 37.7mm | 37mm | 39.4mm | 0.127mm | | |
| Track | MidLayer3 | SRAM1_D6 | 22.5mm | 35mm | 23mm | 34.5mm | 0.127mm | | |
| Track | MidLayer3 | SRAM1_D6 | 23mm | 31.5mm | 23mm | 34.5mm | 0.127mm | | |
| Track | MidLayer3 | SW4 | 30mm | 30.5mm | 30mm | 32.5mm | 0.127mm | | |
| Track | MidLayer3 | VCCINT | 20mm | 44.5mm | 23mm | 44.5mm | 0.127mm | | |
| Track | MidLayer3 | VCCINT | 20mm | 45.5mm | 23mm | 45.5mm | 0.127mm | | |
| Track | MidLayer4 | IO15 | 29mm | 30.5mm | 29mm | 33.5mm | 0.127mm | | |
| Track | MidLayer4 | IO18 | 32mm | 30.5mm | 32mm | 33.5mm | 0.127mm | | |
| Track | MidLayer4 | IO21 | 33mm | 30.5mm | 33mm | 33.5mm | 0.127mm | | |
| Track | MidLayer4 | RAM0_DATA5 | 27mm | 30.5mm | 27mm | 33.5mm | 0.127mm | | |
| Track | MidLayer4 | SW2 | 30mm | 30.5mm | 30mm | 33.5mm | 0.127mm | | |
| Track | MidLayer4 | SW6 | 31mm | 30.5mm | 31mm | 33.5mm | 0.127mm | | |
| Track | MidLayer4 | VCCINT | 19.4mm | 44.5mm | 23.4mm | 44.5mm | 0.127mm | | |
| Track | MidLayer4 | VCCINT | 19.4mm | 45.5mm | 22.974mm | 45.5mm | 0.127mm | | |
| Track | MidLayer4 | VCCINT | 22.974mm | 45.5mm | 23.474mm | 45mm | 0.127mm | | |
| Track | MidLayer4 | VCCINT | 23.474mm | 45mm | 24.5mm | 45mm | 0.127mm | | |
| Track | MidLayer4 | VCCINT | 23.4mm | 44.5mm | 24mm | 44.5mm | 0.127mm | | |
| Track | MidLayer4 | VCCINT | 24mm | 44.5mm | 24.5mm | 45mm | 0.127mm | | |
| Track | MidLayer1 | BLUE1 | 10.8mm | 40.1mm | 15.6mm | 40.1mm | 0.2mm | | |
| Track | MidLayer1 | BLUE1 | 12.2mm | 61.4mm | 12.2mm | 70.2mm | 0.2mm | | |
| Track | MidLayer1 | BLUE1 | 15.6mm | 40.1mm | 15.8mm | 39.9mm | 0.2mm | | |
| Track | MidLayer1 | BLUE1 | 15.8mm | 39.9mm | 16.7mm | 39.9mm | 0.2mm | | |

2519 Objects (225 Selected)

*Figure 10. The list panel with the mid layer tracks we are after selected.*

8. Click the **Clear** button (SHIFT+C) at the bottom right to remove all masks and selections.

> **Note:** A number of useful queries have been stored in the query **Filter** menu. Press **Y** while your document is active to pop up this menu, then choose **Examples**. It includes an option to filter out all objects except the component tracks and arcs on the overlay. Your own Favorite queries are automatically added to the **Filter** popup menu, add these via the **Filter** panel.

### 10.2.2.4 Importing component pins with the List panel

1. Open the 4 port serial interface example project and open the schematic library that is contained within this project.
2. Click on **SCH Panel Button » SCH Library** in the panel control on the bottom right to open up the library panel.
3. Goto **Tools » New Component** and give the component a name of **EP2C5**.
4. Set the current grid to 10. You can see the current grid in the bottom left of the screen. Use the G shortcut to change the grid on the fly.
5. In Windows open the Excel file TestBook.xls from C:\Program Files\Altium Designer Summer 09\Examples\Training\PCB Training\Practice Documents\ directory.
6. In the spreadsheet there is a group of columns starting with **Object Kind** and ending with **Designator**. This group of columns has been created by copying and pasting sections from

the Altera Cyclone II datasheet and also by using Excels easy to use mathematical functions to increment. Copy these columns, including the column names, to the clipboard.

7. Back in Altium Designer, click on **SCH » SCHLIB List** in the panel control on the bottom right to open up the library list panel.
8. At the top of the List panel, select **Edit** mode, and **current component**.
9. Right click in the list panel itself, and select **Smart Grid Insert** from the floating menu.

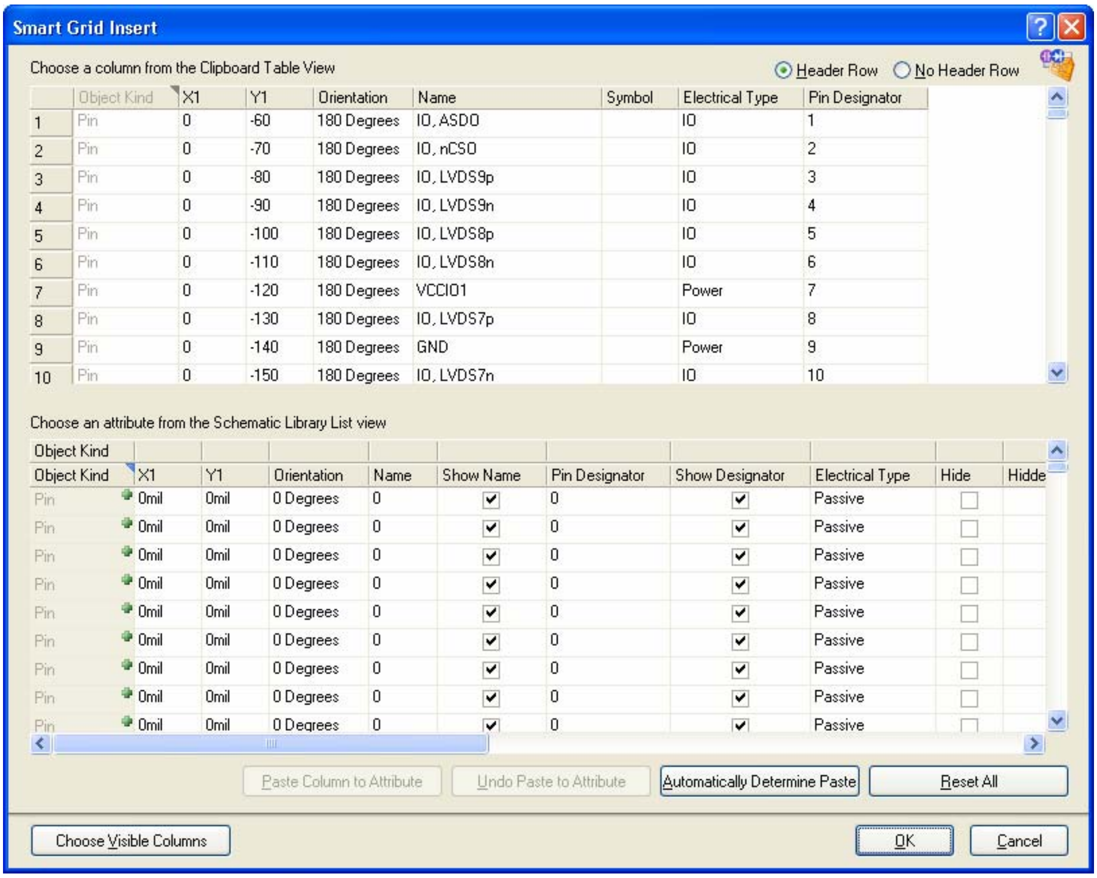



*Figure 11. Smart grid insert dialog that comes up upon pasting.*

10. The Smart Grid Insert dialog will open (Figure 11). This dialog is used to map spreadsheet columns to Altium Designer pin properties. Since the column names have been pre-defined in Excel and were copied onto the clipboard, you can use the **Automatically Determine Paste** button. If the column names are not known, individual columns can be mapped one-by-one, using the **Paste Column to Attribute** button to link them.
11. Once you are satisfied with the column mapping, click **Ok**. The pins for the schematic symbol will be created based on the information in the dialog.

| |
|---|
| **Note:** If you find that all the pins are placed on top of each other, make sure your grid has been set to 10. |

## 10.2.3 The Footprint Manager

Component footprints across the entire project can be managed in the Footprint Manager. Use the Footprint Manager to preview, validate, add, remove or edit footprint associations. Once changes are made the ECO system updates both the schematic and the PCB if required.

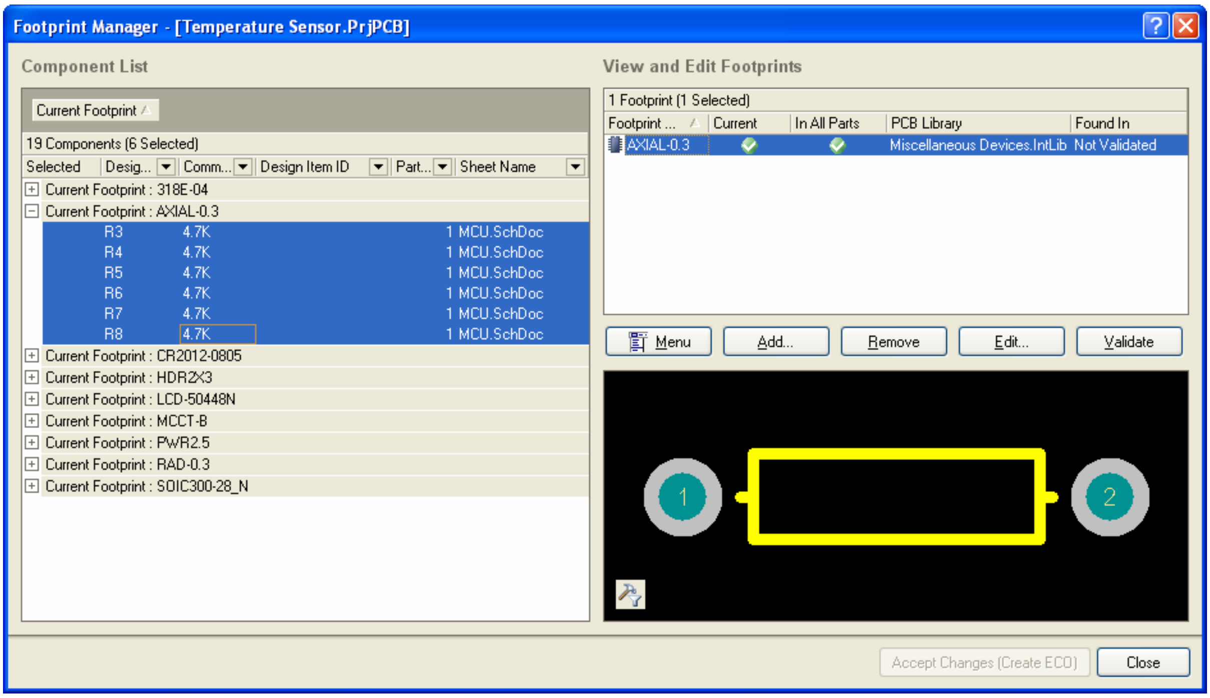



*Figure 12. The Footprint Manager, with all components that use an Axial-0.3 type footprint selected.*

- In the Schematic Editor, select **Tools » Footprint Manager**, to open the *Footprint Manager* dialog.
- The dialog will initially list every component individually, drag a column heading to the top left region of the dialog to group by that column. For example dragging **Current Footprint** will cluster all components that share the same footprint.

#### 10.2.3.1 Exercise – Setting the component's footprint

The footprints currently assigned to the resistor and capacitor components are not surface mount and are not suitable for the board design. To use the Footprint Manager to change these:

1. With any of the schematics in the Temperature Sensor project active, select **Tools » Footprint Manager**.
2. Group by the **Current Footprint** column, and select the components that have the footprint **AXIAL-0.3**, as shown in Figure 12.
3. With the components selected, the top right hand side of the dialog will show all footprints that are common to all of the selected components. At this stage we can either edit the existing **AXIAL-0.3** footprint and change it to an SMD type footprint, or we can add a new footprint to the selected components. For this exercise we'll add a new footprint.
4. Click on the **Add** button, the *PCB Model* dialog will open. Before locating the new footprint, enable the **Library name** option. Doing this ensures that the selected footprint will come from the selected library.
5. Now click the **Browse** button at the top of the *PCB Model* dialog. In the *Browse Libraries* dialog that opens, click the **Find** button to search for a suitable footprint.

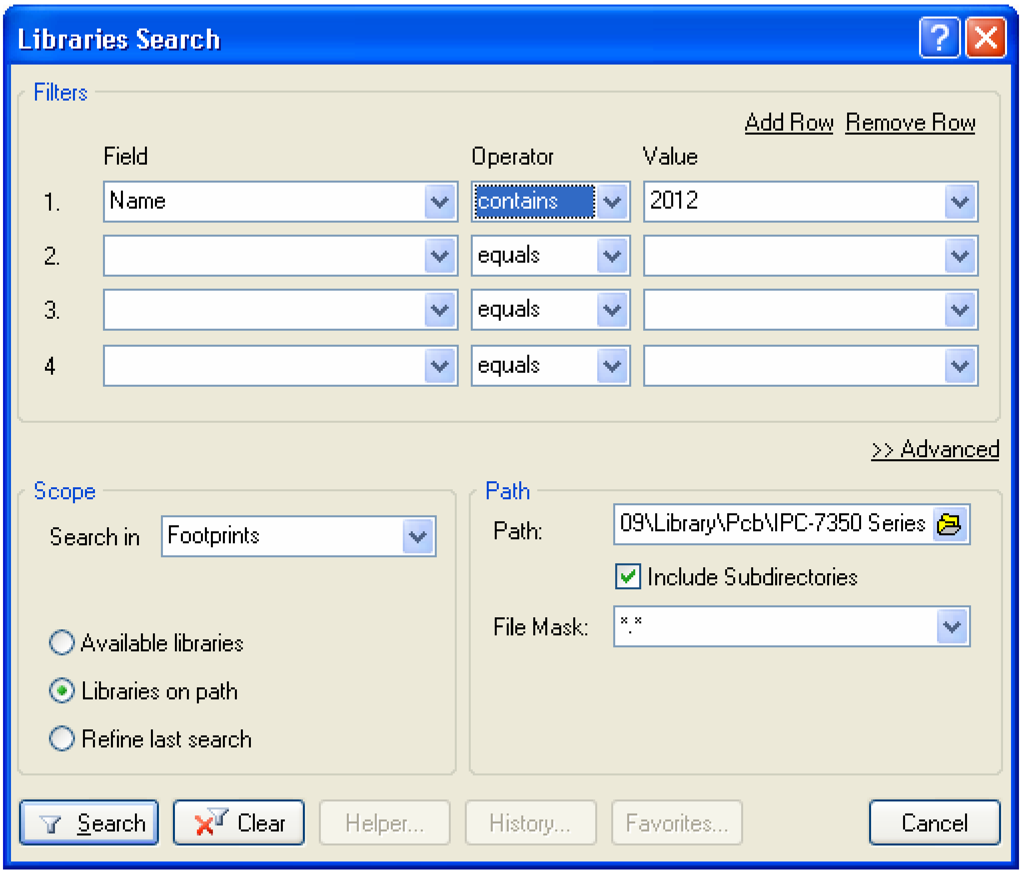



*Figure 13. The Libraries Search dialog.*

6. In the *Libraries Search* dialog set the Scope to **Libraries on Path**, the path to `C:\Program Files\Altium Designer Summer 09\Library\Pcb\IPC-7350 Series`, and enter the search string `2012`, as shown in Figure 13.
7. The search should return a number of matching footprints, such as **RESC2012N**. Select this footprint in the library *IPC-7352 Chip_Resistor_N.PcbLib.*

8. Whenever you choose a component or footprint from a search that is in a library that is not installed, you will be prompted to install the library. Click **Yes** to do this. Tick the **Don't ask again** checkbox if you always want the chosen library to always be installed.
9. The *PCB Model* dialog is now configured for the new footprint, as shown in Figure 14. Click **OK** to close the dialog and Add this footprint to the selected components.
10. At the moment, the **AXIAL-0.3** footprint is still assigned as the current footprint for the selected components, to change this right click on the newly added RESC2012N footprint in the top right hand section of the *Footprint Manager*, and select **Set as Current**, as shown in Figure 15.

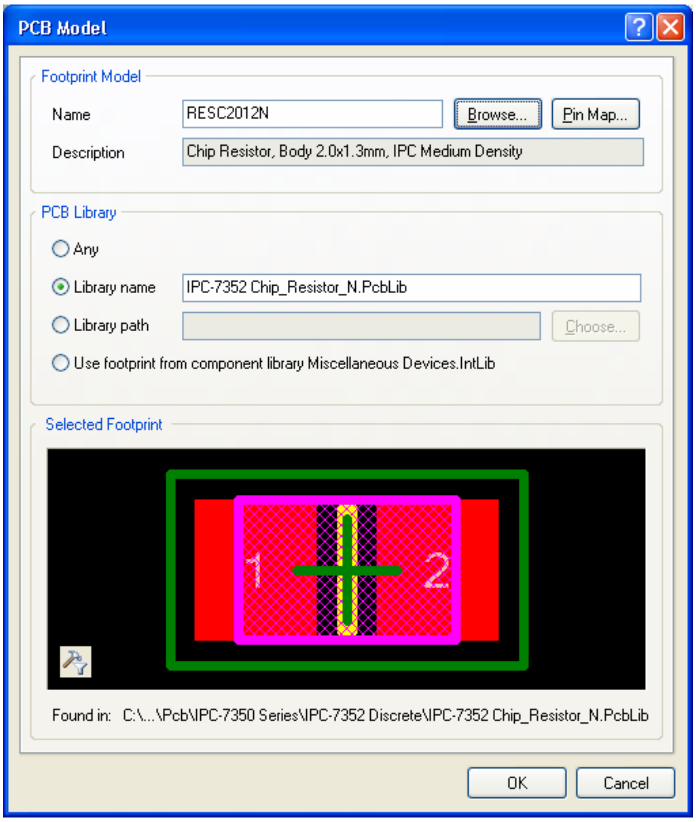



*Figure 14. Configure the PCB Model dialog for the new resistor footprint.*

11. There is also a capacitor using an inappropriate thru hole footprint, RAD-0.3. Repeat the process of locating a new footprint called **CAPC2012N**, using steps 4, 5 and 6 again. The library containing this footprint is called *IPC-7352 Chip_Capacitor_N.PcbLib*. Note you will get a confirmation prompting you to add the IPC-7352 Chip_Capacitor_N.PcbLib confirm with Yes.
12. For the capacitor, set the **CAPC2012N** footprint as the current footprint in the *Footprint Manager*.
13. Repeat for capacitors 1812.
14. The actual components on the schematic sheets have not been modified yet. To update them click on **Accept Changes (Create ECO)**, then **Execute** in the *Engineering Change Order* dialog to update the schematics.

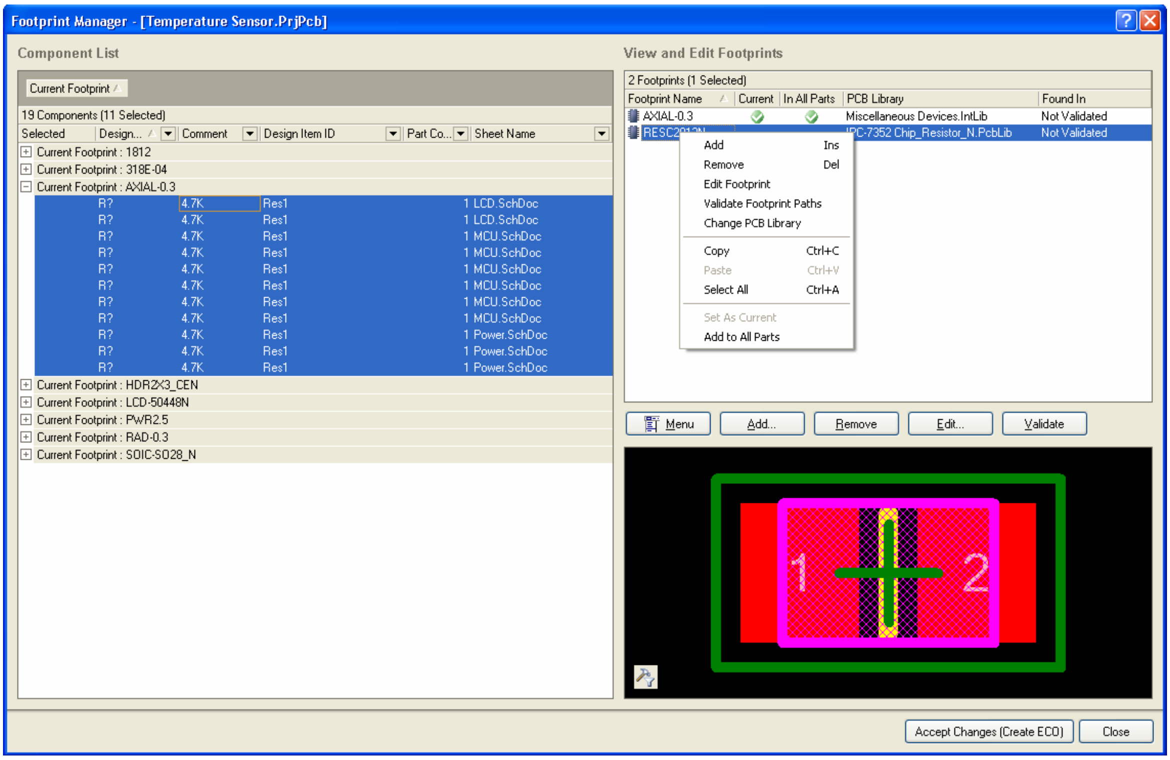



*Figure 15. Set the Current Footprint for the selected components to the 0805 SMD footprint.*

**Note:** As well as being able to change multiple footprints, you can also validate all the footprints, checking to see if they can be found in the available libraries.

TOP SECRET

# Module 11: PCB Design Flow, Transferring a Design and Navigation

## Module 11: PCB Design Flow, Transferring a Design and Navigation





Module Seq = 11

# 11.1 PCB design process

The PCB Design training covers how to use the PCB Editor to create a PCB from setup, through component placement, routing, design rule checking and CAM output. We first look at the overall PCB design process.

The diagram below shows an overview of the PCB design process from schematic entry through to PCB design completion.

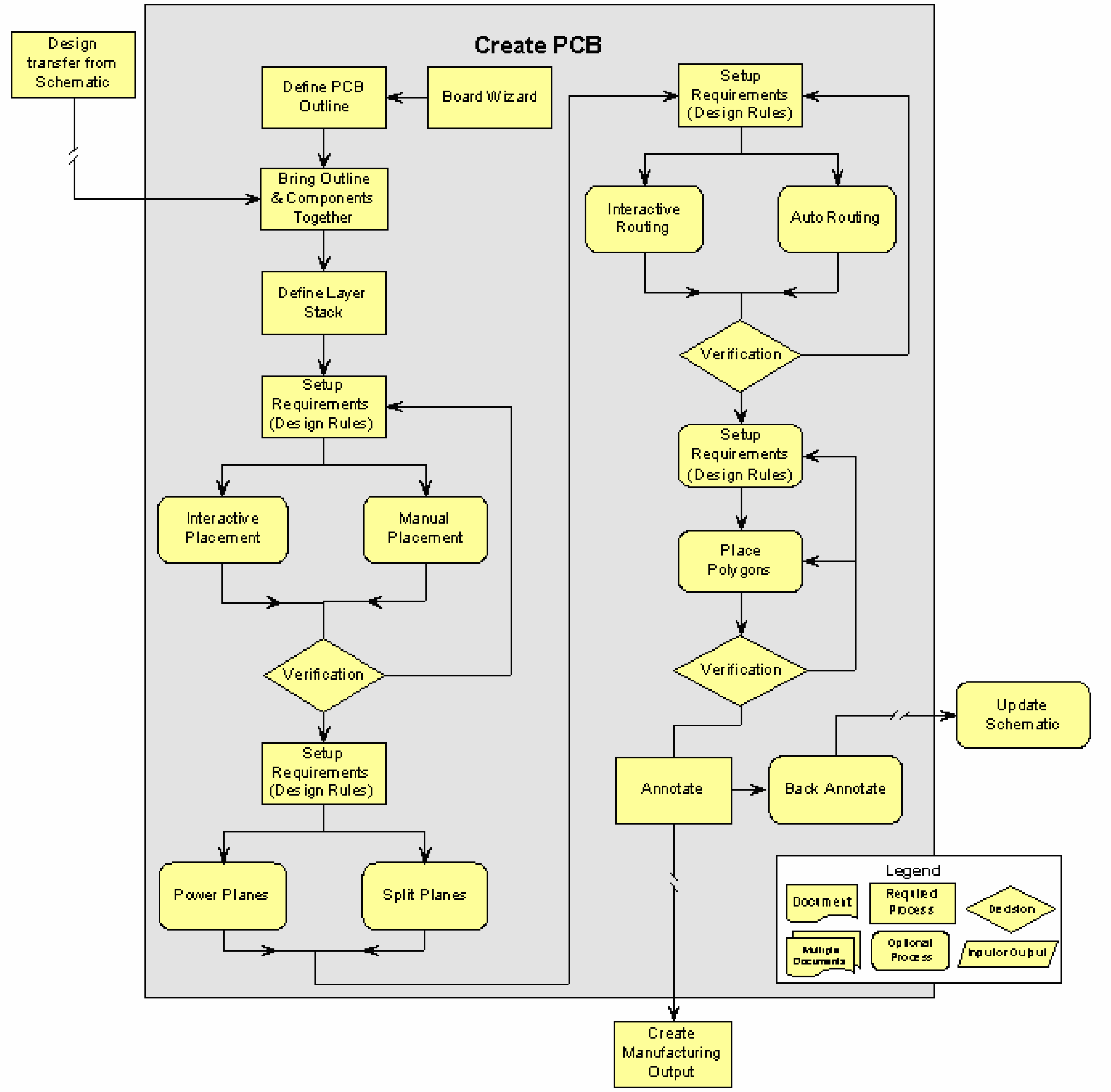



*Figure 1. Overview of the PCB Design Process*

Once the PCB design is completed and verified, the Create Manufacturing Output process is used to generate the PCB output files. This process is outlined below in Figure 2.

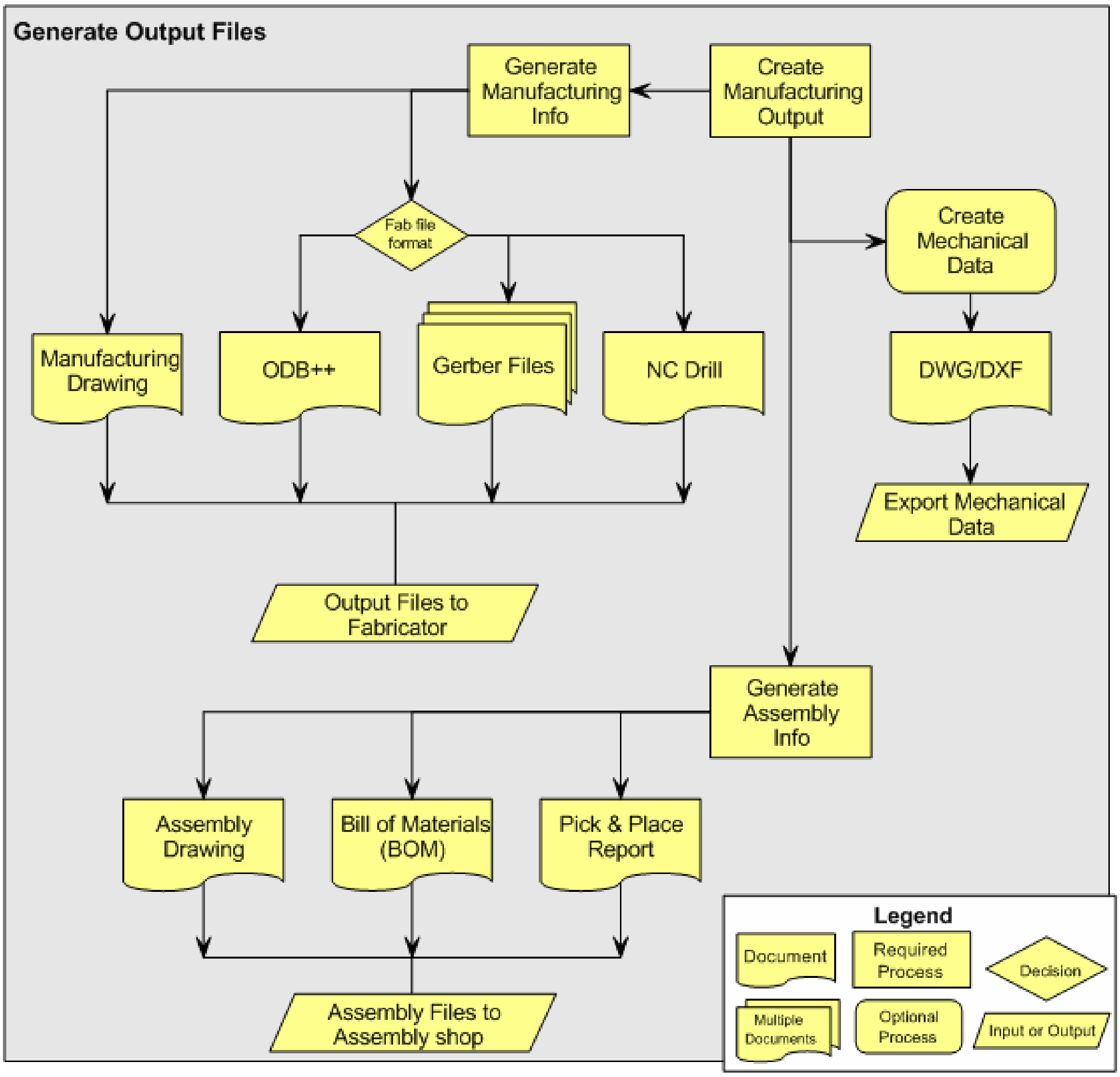



*Figure 2. Work flow for generating PCB output files*

# 11.2 Transferring design information to the PCB

Rather than using an intermediate netlist file to transfer design changes from the schematic to the PCB, Altium Designer has a powerful design synchronization feature.

## 11.2.1 Design synchronization

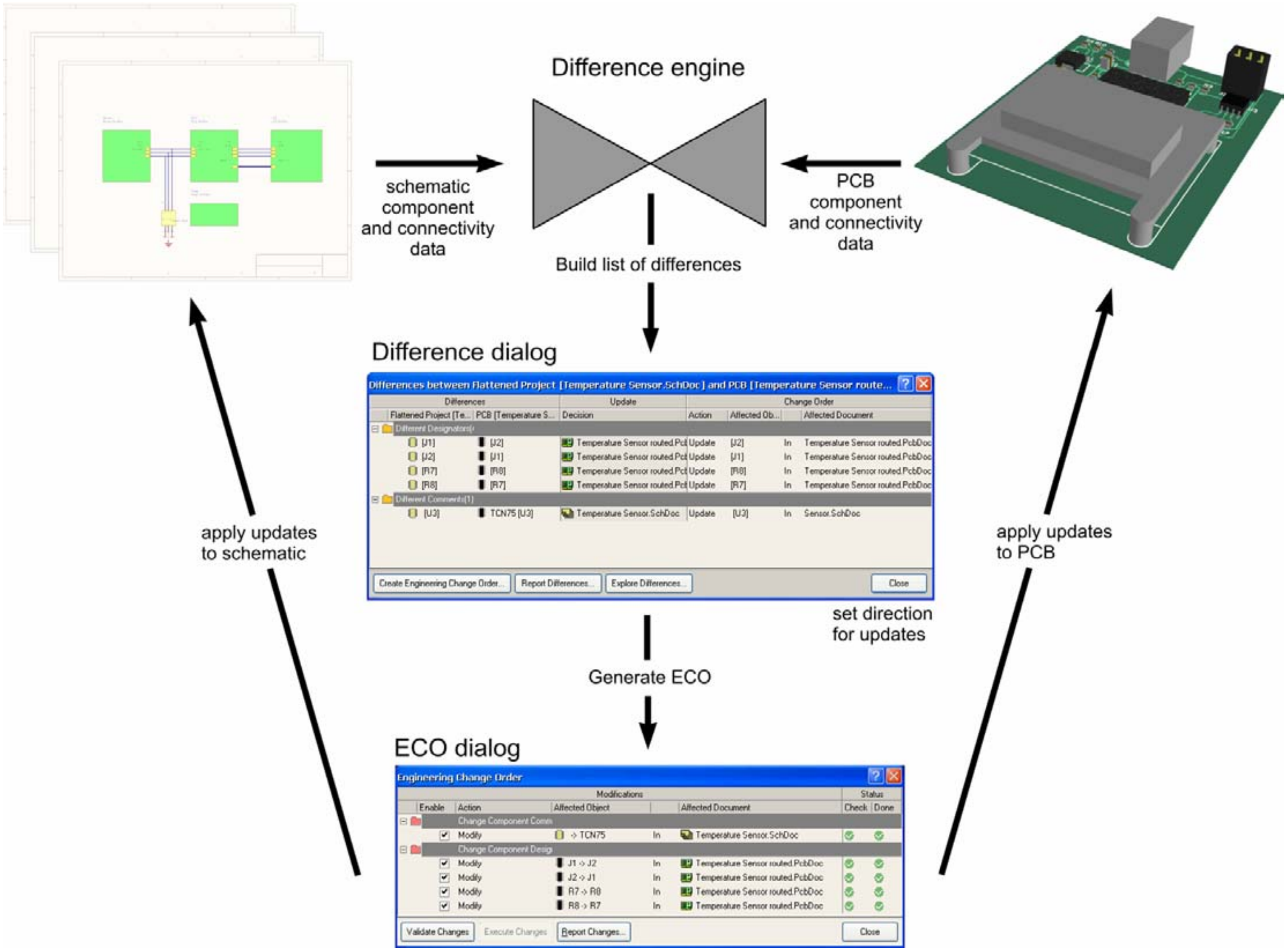



*Figure 3. Design synchronization flow*

The core features of the synchroniser are:

- ***Difference engine*** – compares the schematic project to the PCB. The difference engine can compare the component and connective information between almost all kinds of documents. It can compare a schematic project to a PCB, one PCB to another PCB, a netlist to a PCB, a netlist to a netlist, and so on. The differences found by the difference engine are listed in the difference dialog.
- ***Difference dialog*** – lists all differences detected between the compared documents. You can then define which document should be updated to synchronize the documents. This approach allows you to make changes in both directions in a single update process, giving your bi-directional synchronization. Right-click in the dialog for direction options. This dialog is usually only seen when using the process **Project » Show Differences**.

- ***Engineering Change Order dialog –*** Once the direction of update for the differences has been defined, a list of engineering change orders is generated. A report of these can be generated.

There are two approaches to performing an update:

- Select **Design » Update** to push all changes from schematic to PCB (or PCB to schematic). If you choose this option, you have indicated the direction to use, so you go straight to the *ECO* dialog.
- Select **Project » Show Differences** if you need selective control of the direction. You also use this option if you wish to compare any other document kinds, for example, to compare a netlist to a PCB (also referred to as loading a netlist into a PCB).

## 11.2.2 Resolving synchronization errors

Most problems with synchronizing a design generally fall into two categories:

- Missing component footprints. This occurs when:
  - A footprint is missing from the component information in the schematic.
  - You have forgotten to add the required PCB libraries to the currently available libraries.
  - The footprint in the schematic does not match any PCB library component.
- Footprint pin numbers not matched to schematic pin numbers. Altium Designer supports user-definable pin-to-pad mapping, the default behavior is to expect the same number/letter on both sides. Pin-to-pad mapping is defined in the *PCB Model* dialog (edit the schematic symbol, select the footprint in the Model region of the dialog, and click **Edit**).
- Components not matching (Figure 4). By default Altium Designer attempts to match the component on the schematic with the component on the PCB, using the Unique ID (UID). If there are any miss-matches the *Failed to Match* dialog will open, offering to attempt to match by designator instead. Generally it is better to resolve the UID mismatches, rather than matching by designator. To check and resolve unmatched UIDs, select **Project » Component Links** when the PCB is the active document.

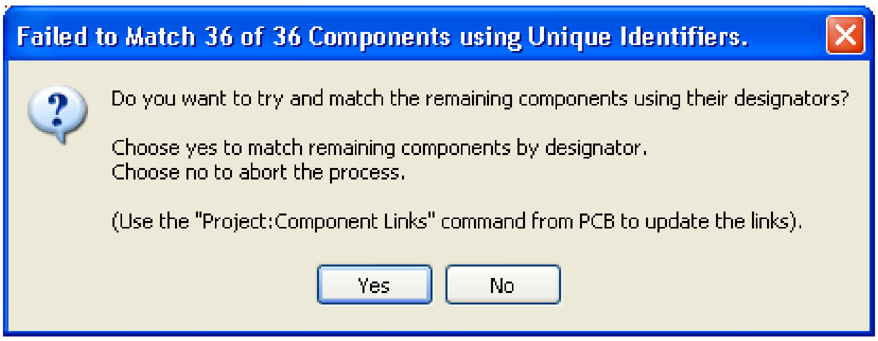



*Figure 4. Warning of component unique ID's not matching.*

- To resolve errors, perform a **Show Differences**, then in the *Differences* dialog click the **Explore Differences** button. The Differences panel will appear – as well as information on what the problem is. This panel lists the objects in question on both the schematic and PCB. Click on an object to display it.

**Note:** If there are large scale net connectivity changes it can be easier to clear the netlist in the PCB editor, the synchronisation process will reload them all. You will then need to reapply the net information to any routing, to do this use the **Update Free Primitives from Component Pads** command (**Design » Netlist**). Be careful using this feature as it creates connections on connected copper, so if shorts exists then these shorts are propagated to the net naming.

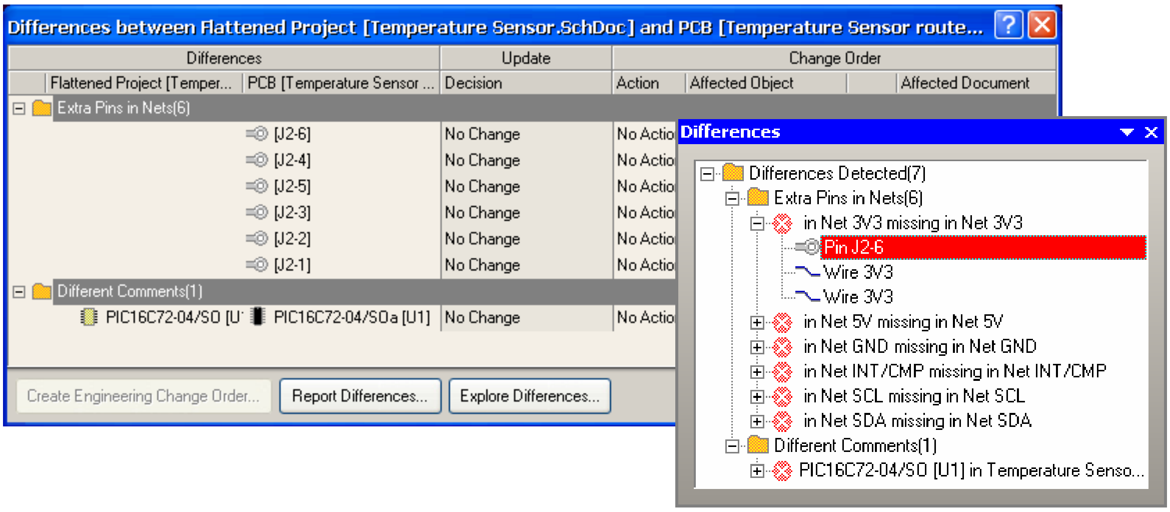



*Figure 5. Differences dialog and using cross probe to mask out problem areas.*

## 11.2.3 Exercise – Transferring the design

In this exercise, you will transfer the design data from the schematic into the new PCB that you have created. This means that all required footprints must be present in available libraries. Keep these points in mind:

- Footprints that are in your project PcbLib are automatically available
- For components placed from an integrated library, such as the PIC Microcontroller, the default state is to only look for the footprint in that integrated library, so it must be available during design transfer.

To transfer the design:

1. In the Libraries panel, click the Libraries... button to open the *Available Libraries* dialog. This dialog shows all libraries that are currently available to you.
2. Confirm that the Temperature `Sensor.PcbLib` is listed in the **Projects** tab.
3. In the **Installed** tab, confirm that the following libraries are installed:
   - Microchip Microcontroller 8-Bit PIC16.IntLib
   - ON Semi Power Mgt Voltage Regulator.IntLib.
   - IPC-7352 Chip_Resistor_N.PcbLib
   - IPC-7352 Chip_Capacitor_N.PcbLib

4. The 2 default libraries must also be installed, `Miscellaneous Devices.IntLib` and `Miscellaneous Connectors.IntLib`. If these have been uninstalled, they can be found in the root of the `\Altium Designer Summer 09\Library` folder.
5. Select **Design » Import Changes from Temperature Sensor.PrjPCB** from the PCB editor menus. The *ECO* dialog displays, listing all the changes that must be made to the PCB so that it matches the schematic. Note that you do not need to open the schematic sheets, this is handled automatically.
6. Scroll down through the list of changes, they should include adding 19 components, 22 nets, 4 component classes, 1 net class and 3 design rules. Click on **Validate Changes** to check the changes are valid.
7. Click on **Execute Changes** to transfer the design data. Close the *ECO* dialog.
8. The components will be placed on the new PCB, positioned to the right of the board outline.
9. Save the board.

**Note:** If you did **not** complete the exercises in the previous modules, you can copy the following project and schematic documents (located in the `Training\Backup` folder) to the `Temperature Sensor` folder and then complete this exercise:

- `Temperature Sensor.PRJPCB`
- `Temperature Sensor.SchDoc`
- `MCU.SchDoc`
- `Sensor.SchDoc`

# 11.3 Using the PCB Panel

This section investigates how to browse through a PCB design. The PCB panel is loaded by going to the panel control and clicking **PCB » PCB**.

## 11.3.1 PCB Panel

The PCB panel provides a powerful method of examining the contents of the PCB workspace.

Clicking on an entry in the panel will filter the workspace to highlight that object – the highlighting will depend on the settings of the options at the top of the panel. To begin with, enable all the options.

### 11.3.1.1 Browse mode selection list

The drop down list at the top of the panel allows you to list, locate or edit the following PCB object types in the active PCB document:

- Components (and then Component Classes)
- Nets (and then Net Classes)
- From-Tos
- Split planes
- Differential pairs
- Polygons
- Hole sizes

When you select an object in the panel, it will be highlighted in the workspace, according to the options at the top of the panel. Each Browse function is described in the following pages.

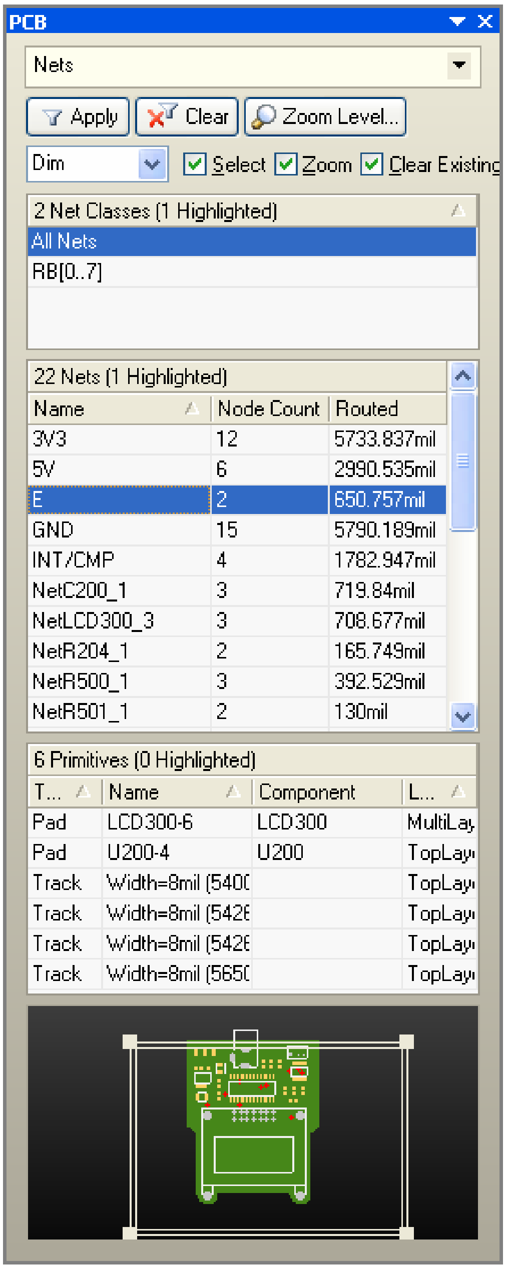



*Figure 6. PCB Editor panel*

### 11.3.1.2 Browsing nets and net classes

- To browse nets, select **Nets** from the drop-down list in the **PCB** panel.
- Click on **All Nets** in the Net Classes region of the dialog to browse all nets on the PCB. The nets are listed in the region below and they are also highlighted on the PCB.
- If the design includes Net Classes these are also listed. Net classes such as D[0..7] have been generated automatically from busses in the design.
- Click on a net name in the Nets region to choose it – all the objects that belong to that net are listed in the Net Items region. Also, the net is highlighted on the PCB.
- Click on an item in the Net Items region and note that it is highlighted on the PCB. Also note that the object that you clicked on is selected.
- Multi-select keys are supported. Hold SHIFT or CTRL as you click on entries in the list.
- Right-click in the Net Items section and note that you can control which net items are displayed.
- Double-click on a net name to open the *Edit Net* dialog. Here you can change the net name, add or remove nodes from the net and define the color of the connection lines for this net.
- The Nets and the Net Items region have multiple columns. Note that you can control the sorting by clicking the heading on a column.
- Type-ahead is supported. You can type on the keyboard to jump through the lists. Press **Esc** to abort the current type-ahead search and start another.

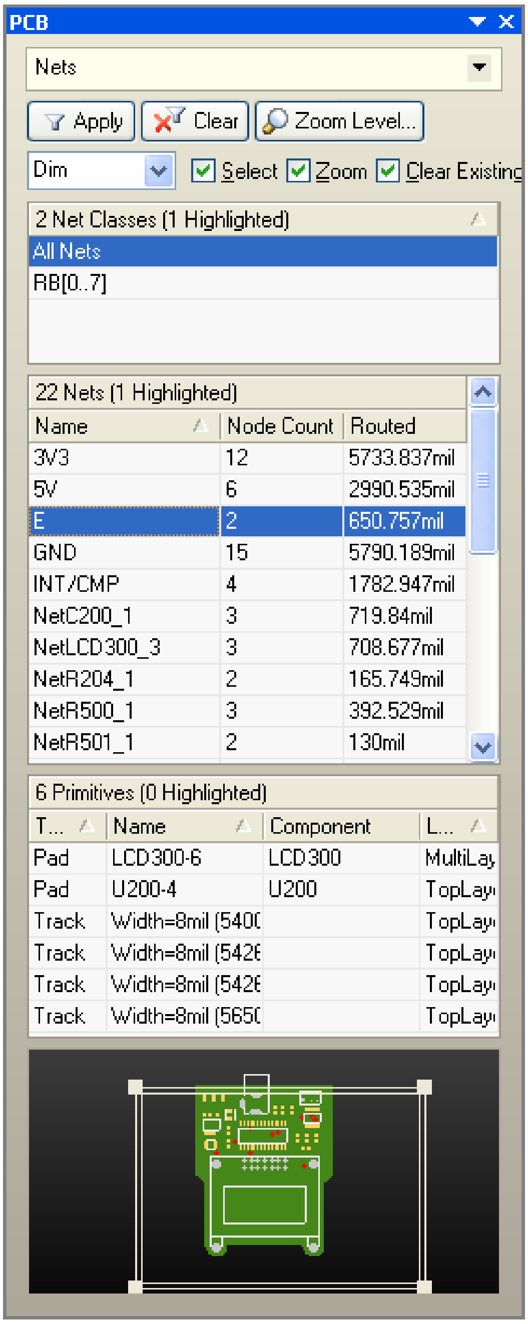



*Figure 7. Browsing nets from the PCB panel*

### 11.3.1.3 Browsing components and component classes

- To browse components, select **Components** from the drop-down list.
- When the panel is being used to filter (highlight) components, you might find it better to have the **Select** option at the top of the panel switched off.
- Click on **All Components** in the Components Class region to browse all components on the PCB. The components are listed in the Components region, as well as being highlighted on the display.
- If the design includes component classes, these are listed too, when you click on a component class only the components in that class are listed and highlighted.
- Click on a component name in the Components region to choose it. All the objects that belong to that component are listed in the Component Primitives region. Also, the component is highlighted on the PCB.
- Click on an item in the Component Items region, Note that it is highlighted on the PCB. Also note that the object that you clicked on is selected.
- Multi-select keys are supported. Hold SHIFT or CTRL as you click on entries in the list.
- Right-click in the Component Items section. Note that you can control which component primitives are displayed.
- Double-click on a component name to open the *Component* dialog where you can modify any attribute of the component.
- The Components and the Component Items region have multiple columns. Note that you can control the sorting by clicking the heading on a column.
- The order of the columns can also be changed; click and drag a column to change the column order. This is handy when you wish to use the type-ahead feature on a different column.
- Type-ahead is supported. You can type on the keyboard to jump through the lists. Press ESC to abort the current type-ahead search and start another. The type-ahead is always performed on the left-most column, so drag any column to make it the left-most.

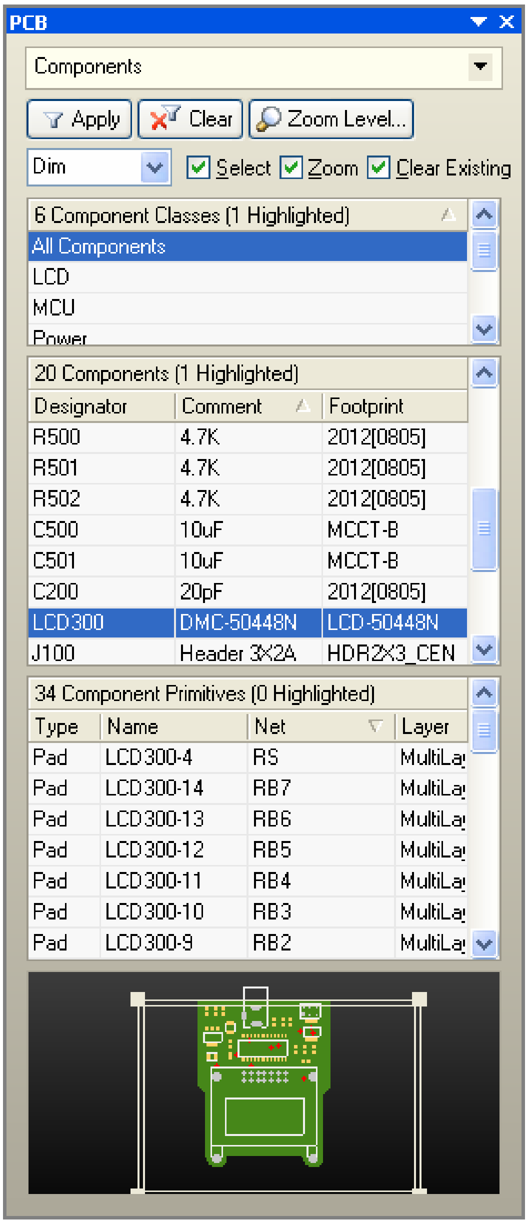



*Figure 8. Browsing components from the PCB panel*

### 11.3.1.4 From-To editor

- Choose **From-To Editor** from the drop-down field at the top of the PCB panel. The top list section of the panel will fill with all nets currently defined for the design.
- As you click on a net entry, all of the nodes on that net will be loaded into the middle list section of the panel. Filtering will be applied and a mask automatically used in order to leave just the nodes (pads) on the net fully visible. All other objects are dimmed.
- Double-click on a net entry to open the *Edit Net* dialog where you can edit the properties of the net.
- To add a new from-to, select the Nodes on Nets to which you want to add the from-to and click the **Add From To** button. The new from-to appears in the From-Tos on Net section.  Click on the from-to in the From-Tos on Net section and click on **Generate** and select a from-to topology, e.g. Shortest, Daisy varieties or Starburst.
- The From-To editor can only be used to create from-tos. To browse for existing from-tos, create a query in the Filter panel using the `IsFromto` keyword.
- Note that all connection lines, other than those that have been defined as From-Tos on the currently selected net, will remain dimmed. Switch the panel back to Nets to restore the display of connection lines.

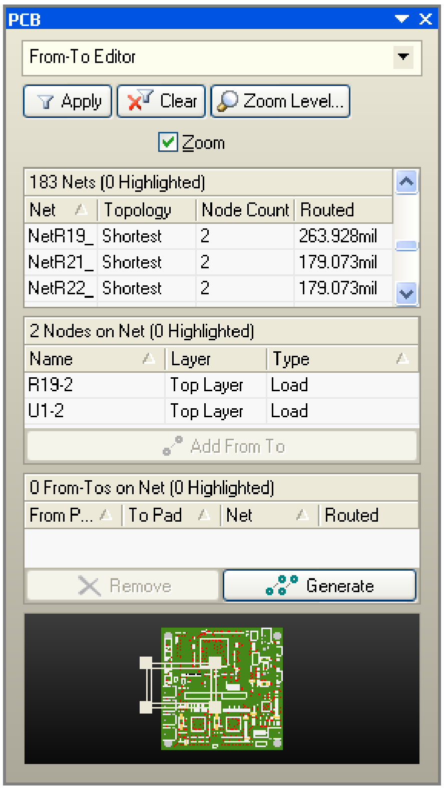



*Figure 9. The From-To Editor in the PCB panel*

### 11.3.1.5 Split Plane editor

- You can review and edit split planes in the PCB panel by selecting the Split Plane Editor from the drop-down list at the top of the panel.
- Select the plane you want to display by clicking on the Plane name. The split planes and their nets on that power plane are listed.
- Click on a split plane name in the Split Planes and Nets section to show the pads and vias on that split plane.
- Double-click on a split plane name to edit the net associated with the split plane.
- Right-click on a split plane name to select an option from the menu.

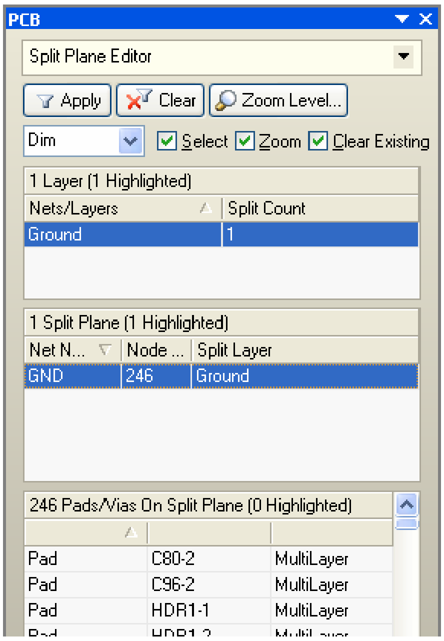



*Figure 10. Use the Split Plane Editor to display and edit split planes*

### 11.3.1.6 Differential Pairs Editor

- You can review and edit Differential Pairs in the PCB panel by selecting the Differential Pairs Editor from the drop-down list at the top of the panel.
- Select the Differential Pair Class you want to display by clicking on the Differential Pair Class name. The Differential Pair Designators will then be listed.
- Click on a Differential Pair name in the Differential Pair section to show the constituent nets of the pair, both positive and negative.
- Double-click on a Differential Pair name to edit the nets associated with the Pair and view the options.
- Right-click on any Differential Pair Class listing (Excepting the default class of All Differential Pairs) and the Object Class Explorer dialog will open allowing you to modify your Classes.

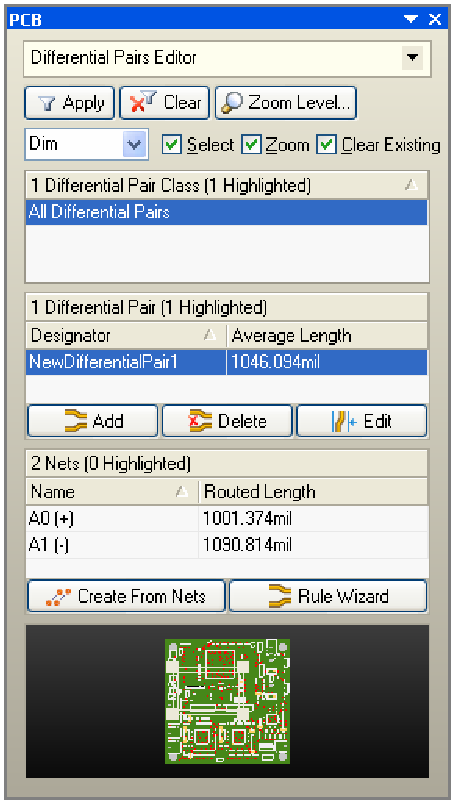



*Figure 11. Use the Differential Pairs Editor to display Differential Pairs.*

### 11.3.1.7 Browsing Polygons

- You can review and edit Polygons in the PCB panel by selecting the Polygons from the drop-down list at the top of the panel.
- Select the Polygon Class you want to display by clicking on the Polygon Class name. The Polygons in this class will then be listed.
- Click on a Polygon name in the Polygon section to show the objects that make up the polygon.
- Double-click on a Polygon name to open the Polygon properties dialog.

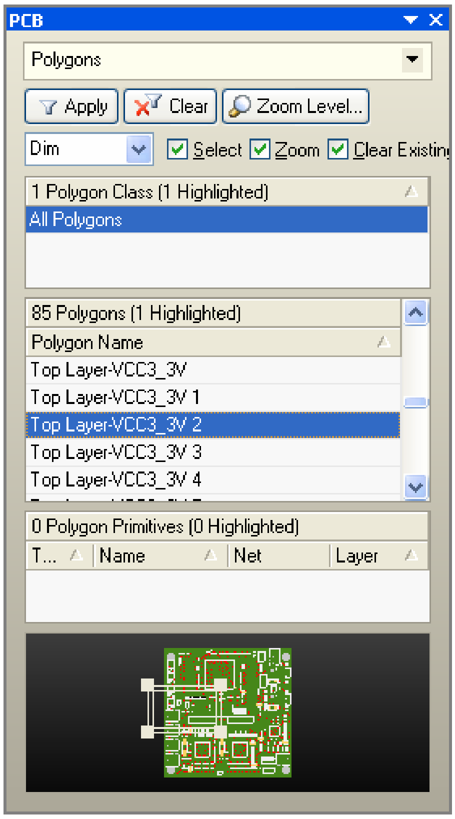



*Figure 12. Use the Polygons drop down to display all polygons and all the primitives that make up the polygons for selection.*

### 11.3.1.8 Hole Size Editor

Use the **Hole Size Editor** mode of the **PCB** panel to review and manage the different drill sizes on your board. Various criteria can be defined, allowing you to locate and display only holes of interest. Criteria includes:

- Pad and/or via holes.
- Plated and/or non-plated holes.
- Free or component pad/via holes.
- Type of hole (all, round, square or slotted holes).
- Selected and/or unselected holes.
- Only the layer-pairs of interest.

Once the criteria defined, the panel lists all unique hole definitions and clusters the pads and/or vias associated with each.

- Click on a **Unique Hole** entry to view all of the instances of that hole size in the design, in accordance with your current highlighting options (Zoom, Mask, and Select).
- Edit the selected holes directly in the panel (ToolSize, Length, Type, Plated).
- Individual pad or via properties can be edited directly – double-click on an entry in the lower list to open the pad/via properties dialog.

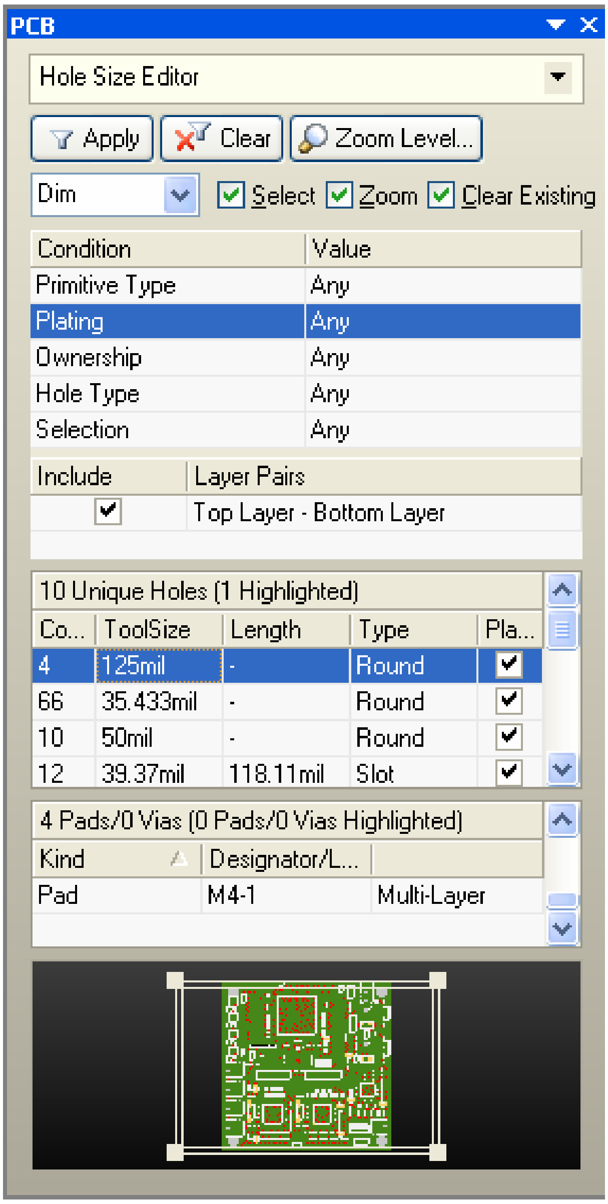



*Figure 13. Use the Hole size editor to find and filter particular holes on a PCB, displaying the count and the primitives.*

### 11.3.1.9 3D Models

Use the **3D Models** mode of the PCB panel to review 3D models used in the board design.

- Click on a component to locate that component on the board and examine the 3D model.
- The panel lists each component individually, click the **Footprint** heading to sort by footprint kind and examine multiple components that have the same 3D model.
- Use the upper **Highlighted Models** dropdown to change the opacity of the currently selected 3D model.
- If the footprint includes both STEP and 3D bodies, use the check boxes to control which is currently displayed, or the lower **Highlighted Models** dropdown to control the opacity of one or the other types of 3D models.
- Elements that are not part of the actual board design, such as the product case, can be placed in the workspace for interference checking (**Place » 3D Body**). Use the ***Free Models** option at the top of the component list to examine and change the opacity of these elements.

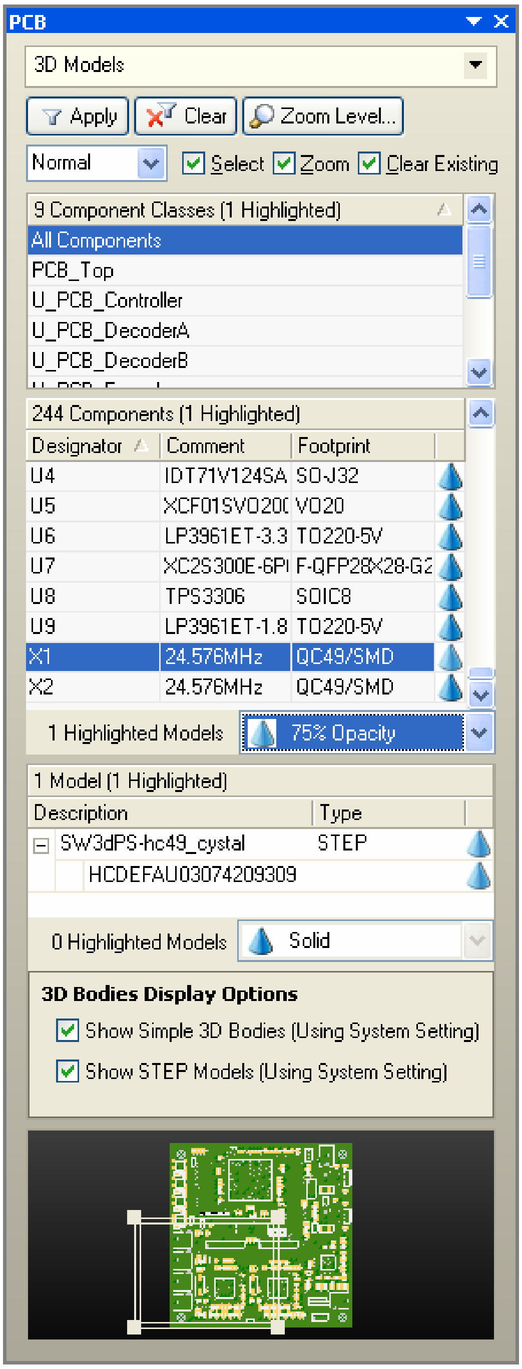



*Figure 14. Review and manage 3D models from the PCB panel.*

## 11.3.2 PCB Rules and Violations

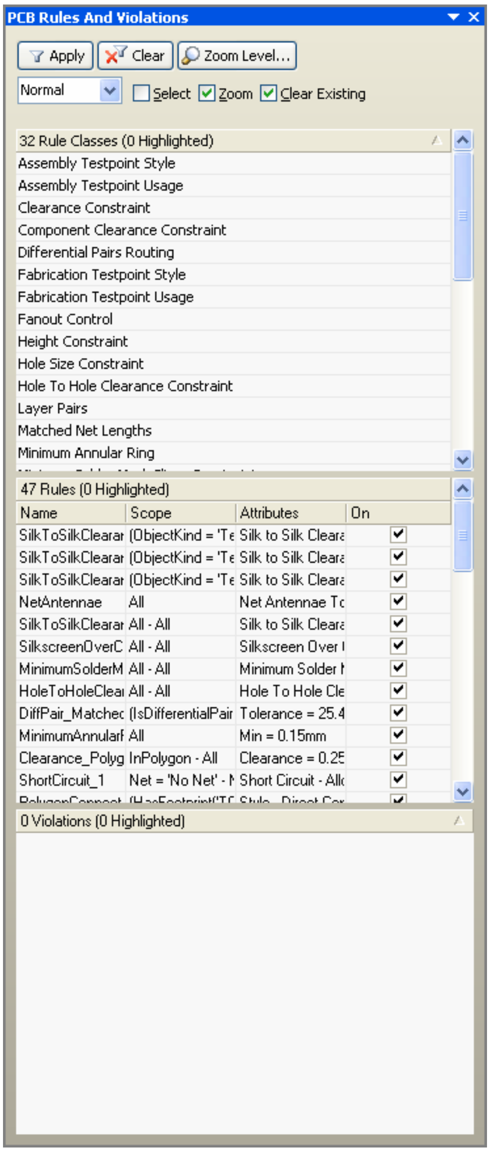



*Figure 15. PCB Rules and Violations Panel*

To browse design rules, Goto PCB » PCB Rules and Violations panel. All Rules classes are listed.

- Click on a Rule Class and all rules defined for that class are listed in the Rules list.
- Click on a rule in the Rules list to highlight all objects targeted by that rule.
- Double-click on the rule to display a dialog to edit that rule.
- If the selected rule is in violation, all violating objects are listed in the Violations region. To check all rules for violations, select [All Rules] in the Rule Classes section.
- Click on a violation to highlight the object causing the violation.
- Double-click on a violation to display the *Violation Details* dialog which details the rule that is being violated and the parameters of the primitive that is causing the violation.
- For more information about design rule checking and violations, refer to *Module 12 - Design Rules*.

You can also view violations by using the shortcut **SHIFT + V** while hovering the cursor over a violation in green.

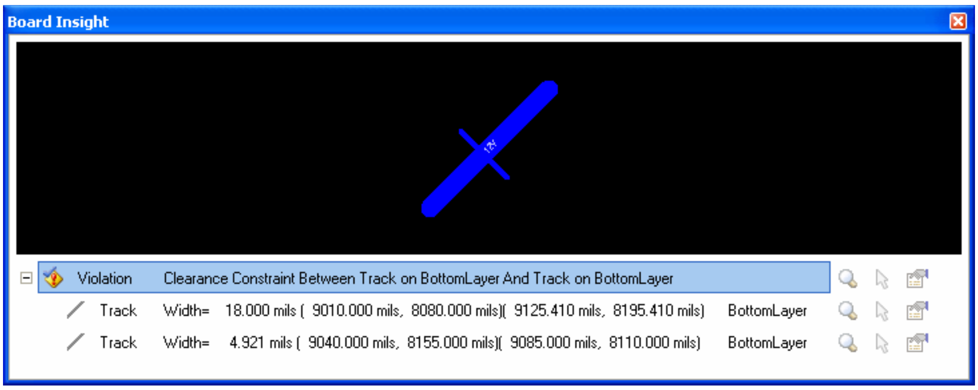



*Figure 16. The board insight for Violations*

### 11.3.3 Exercise – Browsing a PCB document

In this exercise, you will examine the various ways to browse through a PCB document.

1. Open the document `4 Port Serial Interface.PcbDoc` located in the `\Altium Designer Summer 09\Examples\Reference Designs\4 Port Serial Interface` folder.
2. Choose the **Fit Board** view command. Try the other view control options in the **View** menu.
3. Use the MiniViewer to move around the board.
4. Browse each object type and observe how the display changes as you click in the different sections of the panel. As you do, try the Mask, Select and Zoom options.

# 11.4 Project Navigation and Cross Probing

## 11.4.1 Compiling the PCB project

Compiling means creating a connective model (internal netlist) which converts a set of drawings into an electrically wired project. Design navigation is also enabled by compiling the design. To compile a PCB project, select **Project » Compile PCB Project**. (**CC**).

## 11.4.2 Navigating

The DXP **Navigator** panel supports the traditional click-to-highlight style of browsing the design. As you click, the selected object(s) is presented on screen. You can also analyze and trace the connectivity in the design – either spatially in the actual workspace, or in the **Navigator** panel.

- The **Navigator** panel can be used to browse and cross probe to documents, components, buses, nets and pins. A single click on an entry in the panel will browse to that object in the source schematics and VHDL documents.
- Hold the **Alt** key as you click to simultaneously cross probe to the same object(s) on the PCB. The current document remains active, so both must be displayed for this to have any visible effect.

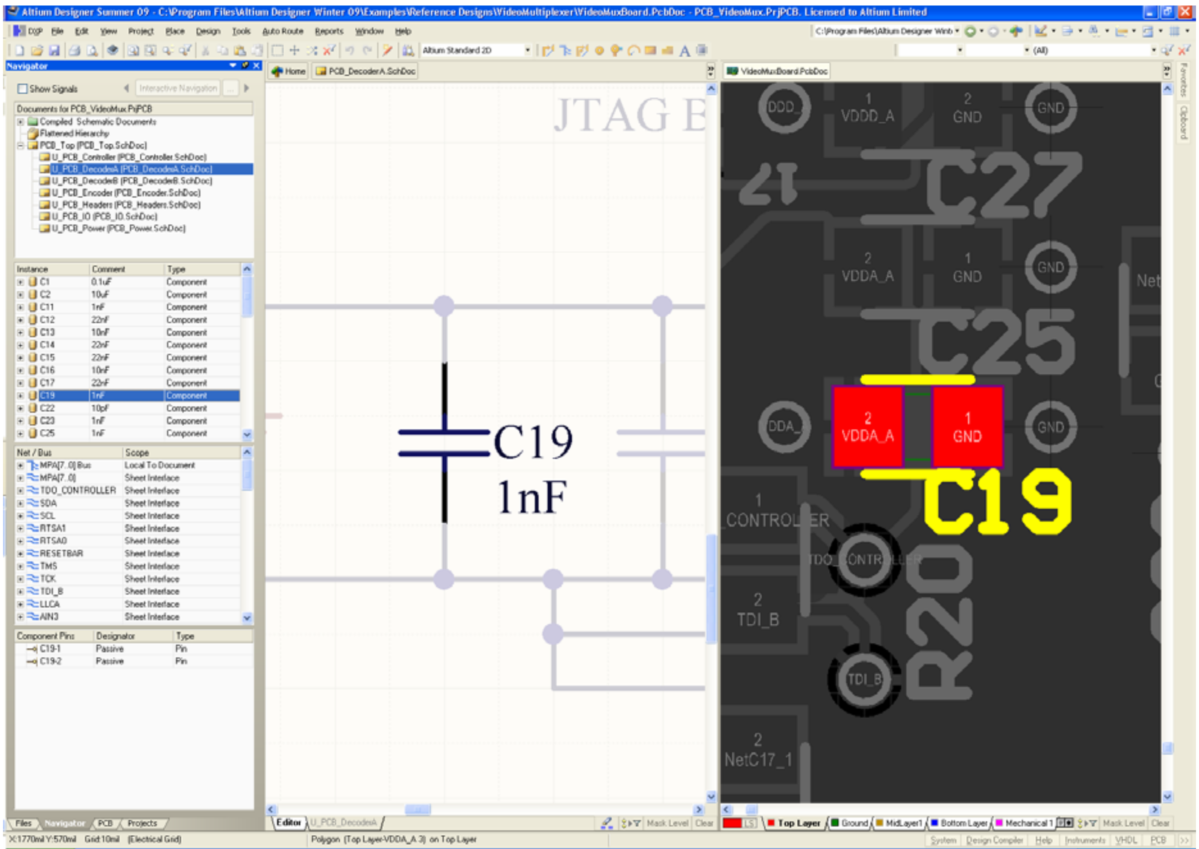


*Figure 17. Holding down the Alt key as you click in the Navigator panel will highlight corresponding objects in both schematic and PCB documents.*

- Navigation highlighting options are controlled from the **System – Navigation** page of the *Preferences* dialog (**DXP » Preferences)**. Alternately this dialog can be accessed by clicking the **…** button to the right of the Interactive Navigation button.

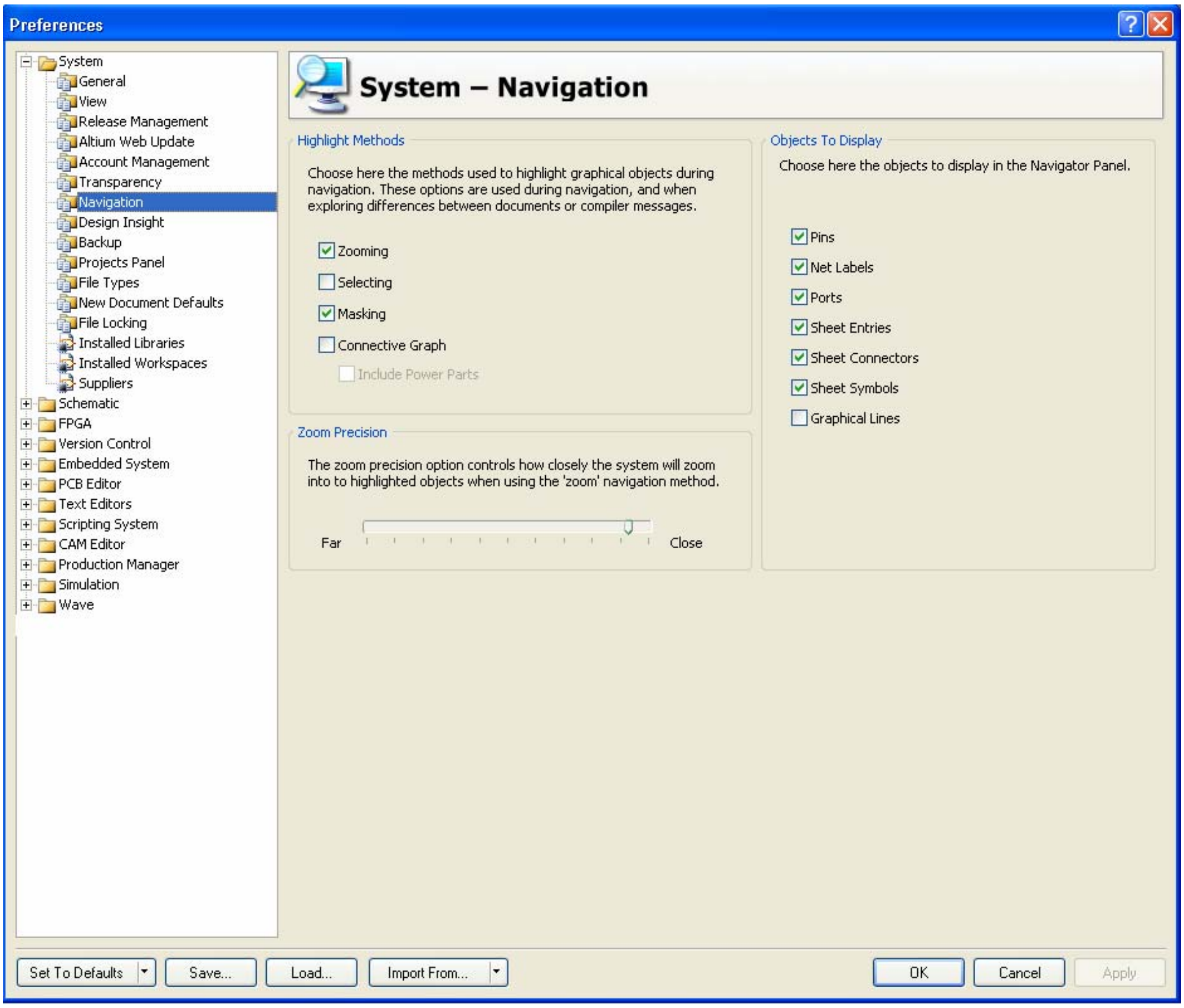



*Figure 18. Preference dialog controlling the highlighting navigation options.*

- The Connective Graph option is useful for showing the connection relationships between different components (green links) and Nets (red links).
- Pressing the Interactive Navigation button causes the component instance information to be updated in the Navigator panel when design elements are selected in the schematic sheet.
- The **Navigator** panel lets you view components and nets by individual sheets or hierarchical groups. Use the flattened hierarchy to see all the components and nets in your design.

## 11.4.3 Cross probing from the schematic to the PCB

Cross Probing is a powerful searching tool to help you locate objects in other editors by selecting the object in the current editor.

- Often when you are analyzing/debugging your design you will want to cross probe from the schematic to the PCB. Full cross probing support is provided, for nets, pins and components.
- You can also cross probe all nets in a bus, and the contents of an entire sheet.

- Use the **Cross Probe** button to be able to click on an object in one view (say the schematic) and display the same object in another view (say the PCB).
- The default behavior is to find the object in the target document then return to the source document. Hold the CTRL key as you cross probe to jump to the target document.
- You can also cross probe using the Navigator panel. Hold the ALT key as you click on something in the panel to highlight it in both the schematic and the PCB. This can be a pin, a component, a net, bus, or a sheet. This works well if you split the view to display both the schematic and the PCB.

### 11.4.4 Exercise — Navigation and Cross Probing

1. Open the project `4 Port Serial Interface.PRJPCB`, found in the `\Altium Designer Summer 09\Examples\Reference Designs\4 Port Serial Interface` folder.
2. Open the schematic, `ISA Bus and Address Decoding.SchDoc` and the PCB document, `4 Port Serial Interface.PcbDoc`. Tile these windows vertically.
3. Make `ISA Bus and Address Decoding.SchDoc` the active document. Use the Navigation panel to highlight components and nets in the schematic.
4. Hold down the **Alt** key when selecting a component or net from the Navigation panel to cross probe to the PCB.
5. Make `4 Port Serial Interface.PcbDoc` the active document and click on the **Cross Probe** toolbar button.
6. Click on component `S1`. The Schematic Editor opens the related schematic document and displays the component S1 centred in the Design window.
7. Click on the **Cross Probe** toolbar button in the Schematic Editor and click on `D1`. The PCB document displays zoomed in on the component.
8. Now try cross probing nets and pads/pins between the open editors.
9. Close all open documents without saving any changes.

TOP SECRET

# Module 12: Design Rules

# Module 12: Design Rules






Module Seq = 12

# 12.1 Design rules and design rule checking

In Altium Designer, design rules are used to define the requirements of your design. These rules cover every aspect of the design – from routing widths, clearances, plane connection styles, routing via styles, and so on. Rules can be monitored as you work and you can also run a batch test at any time and produce a DRC report.

Altium Designer design rules are not attributes of the objects; they are defined independently of the objects. Each rule has a scope that defines which objects it must target.

Rules are applied in a hierarchical fashion, for example, there is a clearance rule for the entire board, then perhaps a clearance rule for a class of nets, then perhaps another for one of the pads in a class. Using the rule priority and the scope, the PCB Editor can determine which rule applies to each object in the design.

This section describes how design rules are defined and how to check for design rule violations.

## 12.1.1 Adding design rules

Design rules are defined in the *PCB Rules and Constraints Editor* dialog that is displayed by selecting **Design » Rules**.

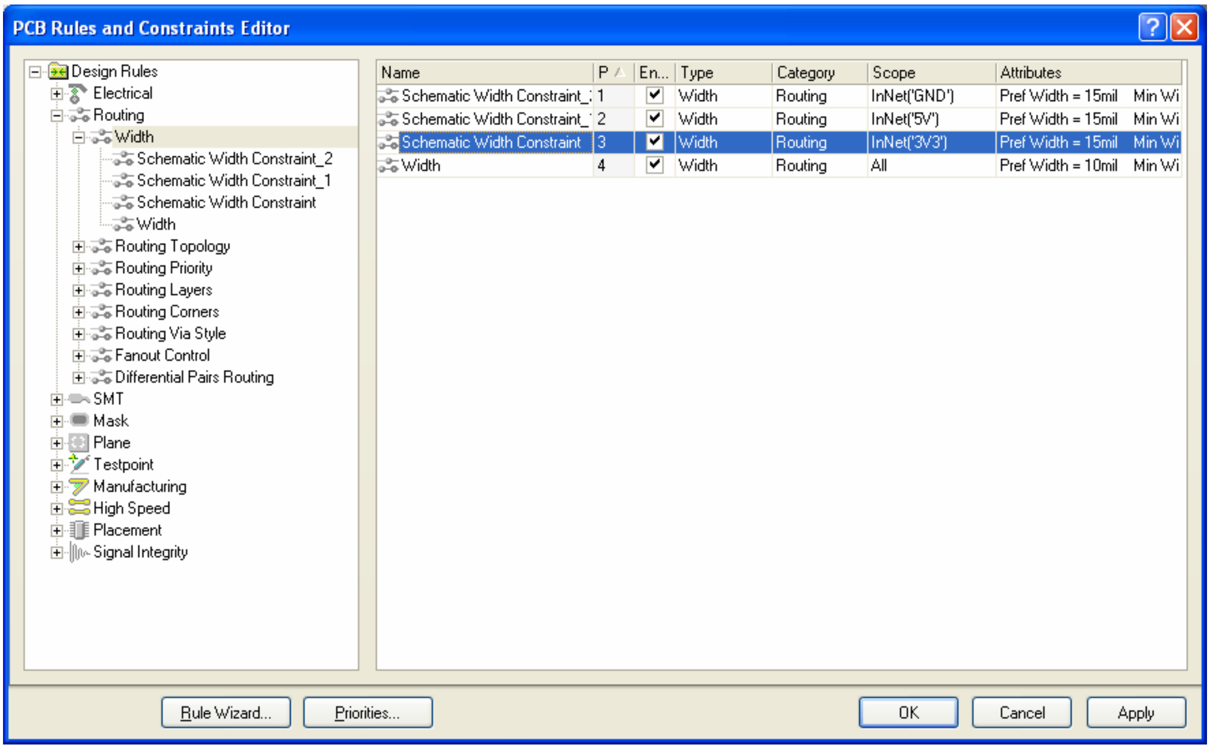



*Figure 1. PCB Rules and Constraints Editor dialog.*

To set up a design rule:

1. Click on the ⊞ to expand the required rule category in the tree on the left.
2. Click on the ⊞ next to the rule kind to display the rules of that kind that have been defined. Notice how in Figure 1 the tree is expanded to show the four Width rules.
3. Click on a specific rule to display the properties of that rule.
4. Right-click on a rule kind to add a new rule of that kind.

- You can use the PCB Rules and Violations panel to see the objects targeted by a rule.
- Alternatively, right-click on an object in the workspace and select **Applicable Unary Rules** or **Applicable Binary Rules** to work out what rules are being applied to an object(s).


*Figure 2. Unary rules dialog showing what's applied to a component*

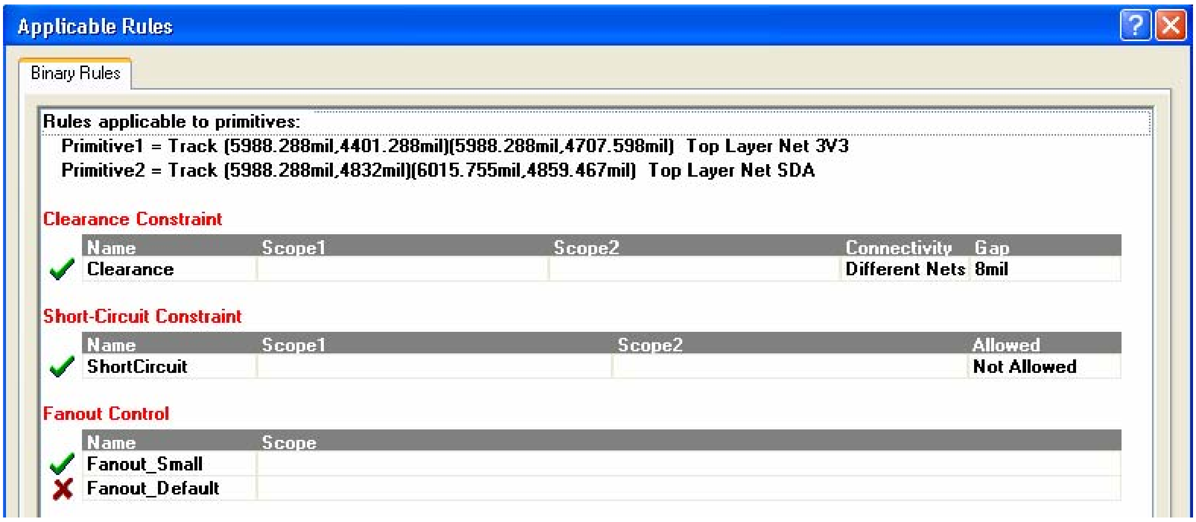



*Figure 3. Binary rules dialog showing what's applied between two nets*

## 12.1.2 Design rules concepts

To effectively apply the design rules, the concepts of rule type, object set, query and priority need to be understood.

### 12.1.2.1 Rule type

There are two types of design rules – unary and binary.

**Unary design rules**

These apply to one object, or each object in a set of objects. For example, Width Constraint.

**Binary design rules**

These apply between any object in the first set to any object in the second set. Binary rules have two object set sections that must be configured. An example of a binary rule is the Clearance rule – it defines the clearance required between any copper object in the first set and any copper object in the second set, as identified by the two rule queries.

### 12.1.2.2 Object set

This refers to the group of objects that the rule applies to. The scope of the object set is determined by the rule Query.

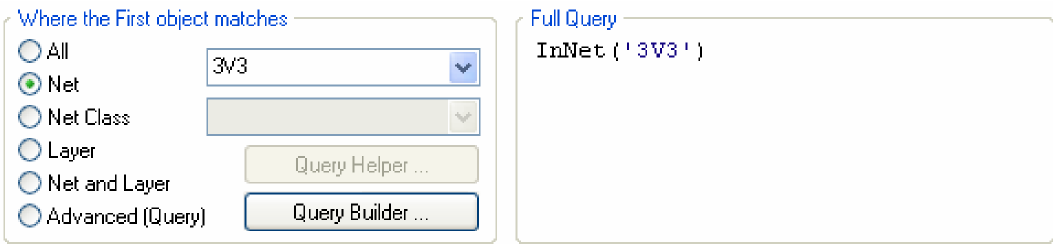



*Figure 4. The scope of the rule defines the objects it targets. This rule targets the 3V3 net.*

### 12.1.2.3 Rule Query

The Query is a description of the objects that this rule applies to. The Query can be typed in directly, it can be constructed automatically using the controls on the left of the Full Query edit field, or it can be constructed using the Query Builder.

For more information on queries, refer to the article, *An Insiders Guide to the Query Language.*

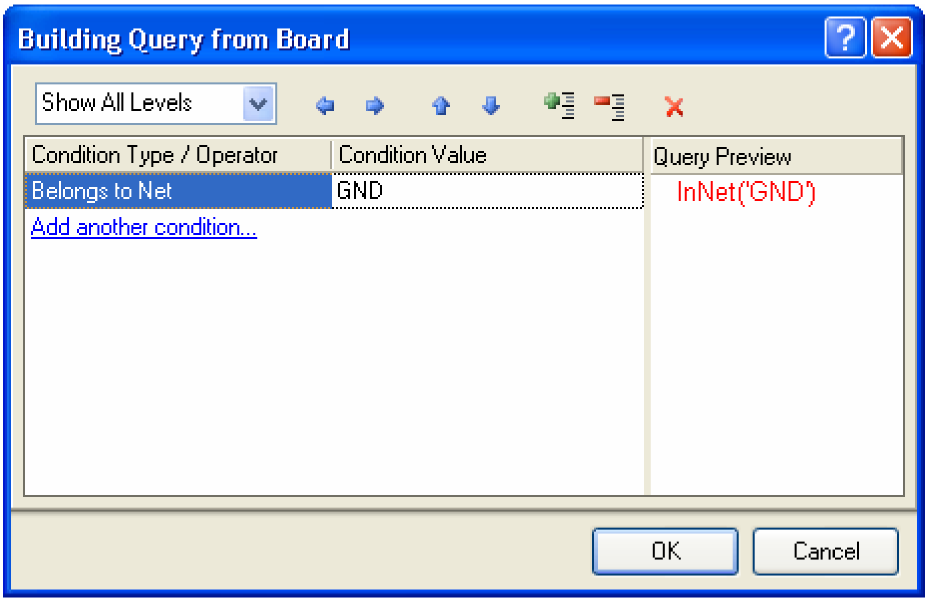



*Figure 5. Use the Query Builder to construct the rule query.*

### 12.1.2.4 Query errors

If you are typing the query in and you make a mistake, for example, you leave off a bracket, a message will appear warning that there are errors when you attempt to close the Rules dialog. It is important to resolve these errors, as if you do not, the on-line DRC can become very slow. Rules that have a query error have their name displayed in red in the tree on the left of the dialog.

### 12.1.2.5 Setting the rule priority

The priority, or order that the rules are tested to determine the applicable rule, is user-defined. When a new rule is added it is automatically set to the highest priority for rules of that kind. It is essential that the priority is set appropriately for them to be applied correctly. To get to the Priorities dialog, click on the **Priorities** Button. Before pressing this button you need to set the area of design rules you wish to alter, like for example clearance rules, width rules to only display rules of this type.

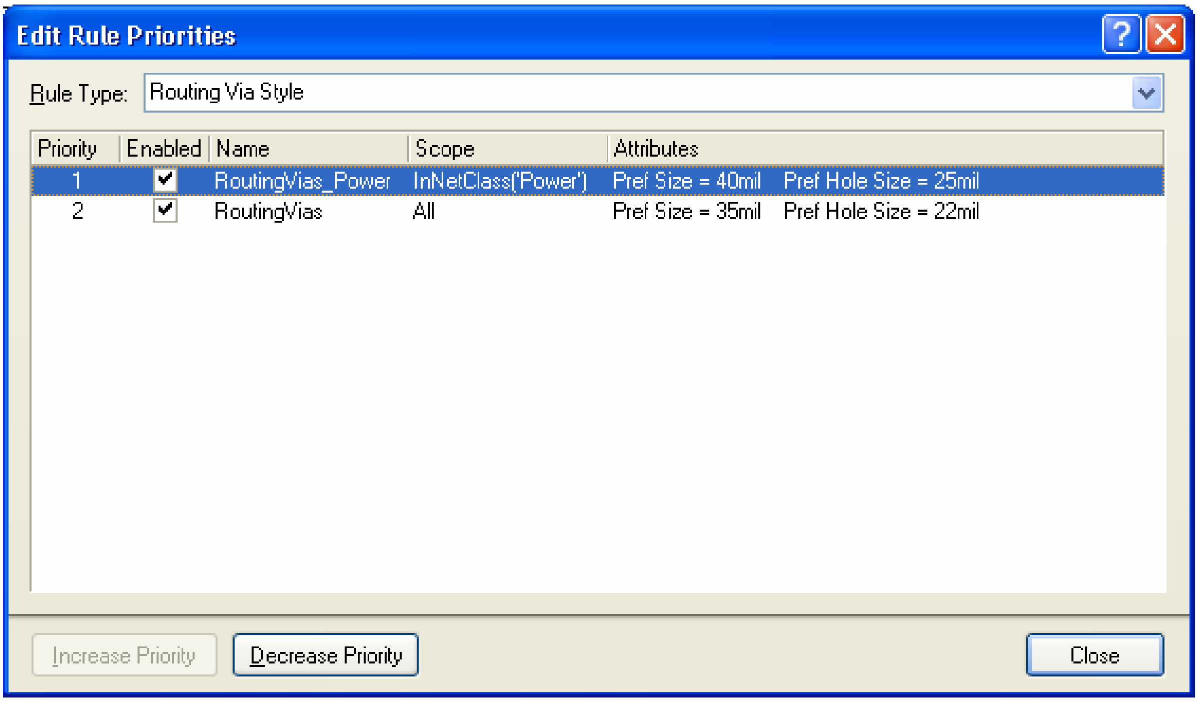



*Figure 6 After adding a rule, make sure that the priority is appropriate*

In Figure 6 a routing via style rule for the NetClass Power exists. Note that it has a rule priority of 1 (the highest priority). If it had a priority lower than the RoutingVias rule, which has a scope of All, it would never be applied.

## 12.1.3 How rules are checked

Design rules are checked by the Design Rule Checker (DRC) either online as you work, or as a batch process (with an optional report). The batch mode can be run at any time, and it is good design practice to run it as a final verification check when the board is completed.

### 12.1.3.1 Online DRC

If the Online DRC option is turned on, all DRC violations are marked as you create them. This is especially helpful when manually routing to immediately highlight clearance, width and parallel segment violations.

Checking the **Online DRC** check box in the **General** page of the *Preferences* dialog (**Tools » Preferences**) turns on the Online DRC.

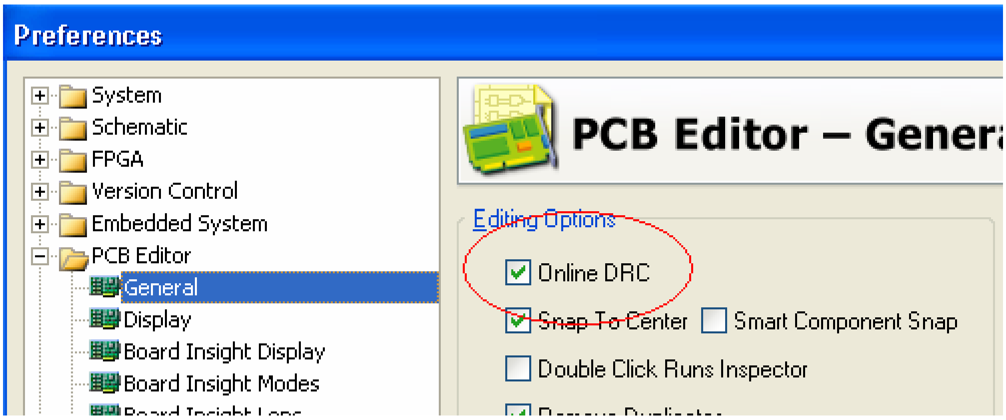



*Figure 7. Online DRC tick box in the preferences*

Each rule is then individually enabled for online and/or batch checking in the **Rules to Check** page of the *Design Rule Checker* dialog, as shown in Figure 8 (select **Tools » Design Rule Check** from the menus). Enable the online checkbox for each rule that you want to have automatically monitored as you work.

DRC errors display in the color chosen in the **Board Layers and Colors** tab of the *View Configurations* dialog, when the **Show** checkbox is enabled. Also DRC markers can be set via **Tools » Preferences » DRC Violations Display**. From here there are new styles and display options to be set to show an icon based view of a DRC violation.

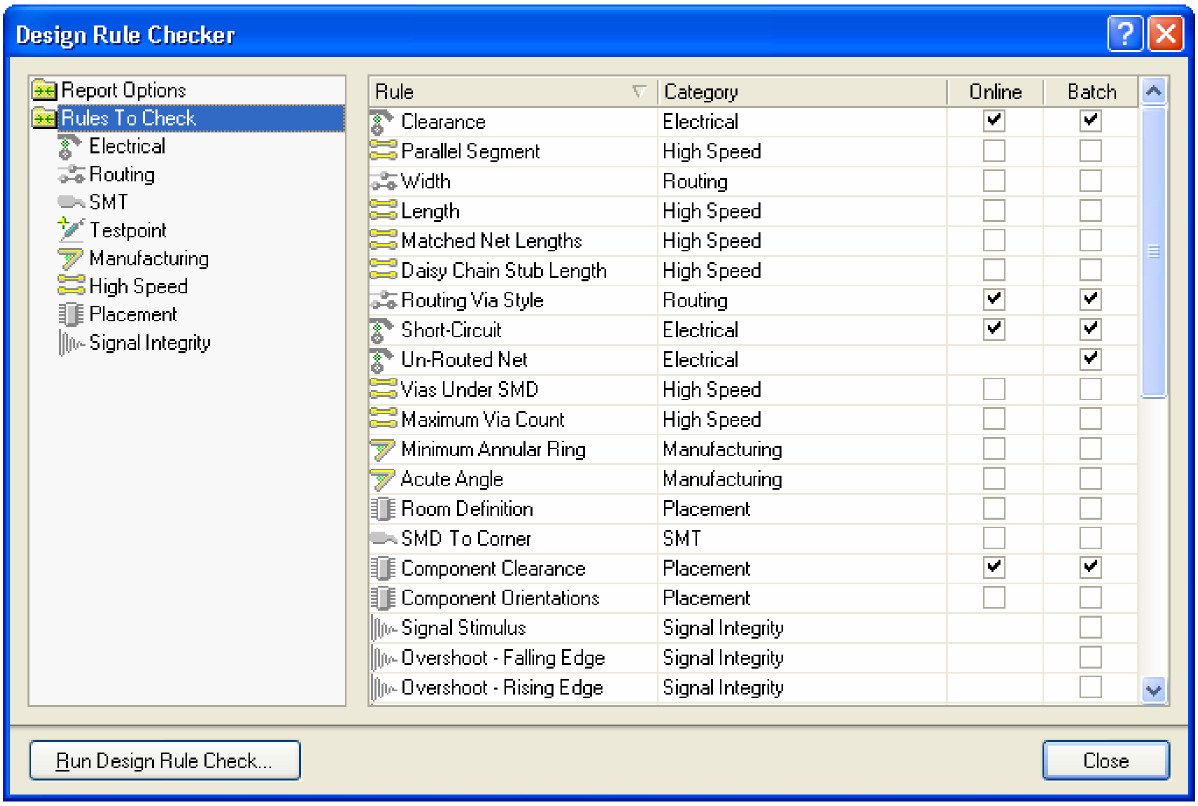



*Figure 8. Configure when each rule is to be checked in the Design Rule Checker dialog.*

## 12.1.4 Where rules apply

### 12.1.4.1 Routing rules

| Rule Class | Manual Route | Auto Route | Online DRC | Batch DRC | Other |
|---|---|---|---|---|---|
| Clearance Constraint | Y | Y | Y | Y | Place Polygon |
| Routing Corners | Y | | | | Specctra DSN export |
| Routing Layers | | Y | | | |
| Routing Priority | | Y | | | |
| Routing Topology | | Y | | | |
| Routing Via Style | Y | Y | | | |
| SMD Neckdown Constraint | | Y | Y | Y | |
| SMD To Corner Constraint | | | Y | Y | |
| SMD To Plane Constraint | | | Y | Y | |
| Width Constraint | Y | Y | Y | Y | Physical connected copper |

*Table 1. Routing rules*

### 12.1.4.2 Manufacturing rules

| Rule Class | Auto Route | Online DRC | Batch DRC | Output Generation | Other |
|---|---|---|---|---|---|
| Acute Angle Constraint | | Y | Y | | |
| Hole Size Constraint | | Y | Y | | |
| Layer Pairs | | Y | Y | | Manual route |
| Minimum Annular Ring | | Y | Y | | |
| Paste Mask Exp | | | | Y | |
| Polygon Connect Style | | | | | Place Polygon |
| Power Plane Clearance | Y | | | Y | Internal Planes |
| Power Plane Connect Style | Y | | | Y | Internal Planes |
| Solder Mask Exp | | | | Y | |
| Testpoint Style | Y | Y | Y | Y | Find Testpoint |
| Testpoint Usage | Y | Y | Y | Y | Find Testpoint |
| Hole to Hole clearance | | Y | Y | | |
| Minimum Solder Mask Silver | | Y | Y | | |
| Silkscreen Over Component Pads | | Y | Y | | |
| Net Antennae | | Y | Y | | |
| Silk to Silk Clearance | | Y | Y | | |

*Table 2. Manufacturing Rules*

### 12.1.4.3 High Speed rules

| Rule Class | Auto Route | Online DRC | Batch DRC | Output Generation | Other |
|---|---|---|---|---|---|
| Daisy Chain Stub Length | | Y | Y | | |
| Length Constraint | | Y | Y | | |
| Matched Length Nets | | Y | Y | | Interactive length tuning tool |
| Differential Pairs | | Y | Y | | Interactive Differential pair length tuning tool |
| Maximum Via Count | | Y | Y | | |
| Parallel Segment | | Y | Y | | |
| Vias Under SMD | | Y | Y | | |

*Table 3. High Speed Rules*

### 12.1.4.4 Placement rules

| Rule Class | Auto Route | Online DRC | Batch DRC | Output Generation | Other |
|---|---|---|---|---|---|
| Component Clearance Constraint | | Y | Y | | Cluster Auto Placer |
| Component Orientation | | | | | Cluster Auto Placer |
| Nets To Ignore | | | | | Cluster Auto Placer |
| Permitted Layers | | | | | Cluster Auto Placer |
| Room Definition | | Y | Y | | Arrange within room |

*Table 4. Placement Rules*

### 12.1.4.5 Signal Integrity rules

All Signal Integrity rules apply only to Signal Integrity Analysis and Batch DRC.

### 12.1.4.6 Other design rules

| Rule Class | Auto Route | Online DRC | Batch DRC | Output Generation | Other |
|---|---|---|---|---|---|
| Short Circuit Constraint | | Y | Y | | |
| Unconnected pin Constraint | | Y | | | |
| Unrouted Net Constraint | | Y | Y | | |

*Table 5. Other Design Rules*

## 12.1.5 From-tos

The PCB Editor allows commands to operate on a particular pin-to-pin connection in a net, in a different manner to the rest of the net. A specific pin-to-pin connection is defined as a *from-to* Commands will operate on a from-to if a design rule for that from-to has been defined.

From-tos are created using the From-To Editor. Select **From-To Editor** in the PCB panel to display this editor.

The top region of the panel lists all nets in the design. Click on a net to list that nets nodes in the Nodes on Net region of the panel. When you click on any two nodes in the net (use CTRL+Click to multi-select), the **Add From To** button will be enabled. When this is clicked, the new from-to will appear in the From-Tos on Net section of the panel.

The **Generate** button allows you to create from-tos for a complete net in the pattern of the selected topology.

## 12.1.6 Exercise – Setting up the design rules

This exercise looks at setting up the required design rules.

1. Using the Temperature sensor project PCB document, confirm that the basic (The ALL rule) clearance constraint design rule is set to 8mils.
2. Add a clearance constraint to keep polygons at least 15mils from other copper objects. To do this:
   - add a second clearance constraint rule
   - for the First Object Matches query, type in the query **`InPolygon`**
   - leave the Second Object Matches query as **`All`**
   - set the minimum clearance to 15mils
   - set the rule name to `Clearance_Polygon`.
3. Confirm that basic (The ALL rule) Board scope width constraint is set to 8 mils (all three settings).
4. For the three power nets on the schematic included parameter set objects that defined the width rule required for these nets. Confirm that a width constraint has been created for each of these nets with a width of 15 mils.
5. Edit the Routing Via Style design rule, setting the via diameter to 35 and the hole size to 22 (all three settings).
6. Save the board.

## 12.1.7 Design Rule Checking

- The Design Rules Checking (DRC) functions are provided to check that your design conforms to the design rules.
- There are both Online and Batch DRC functions.
- A design should only be submitted for manufacturing when all DRC violations have been resolved.
- DRC violations can be located using its own PCB Rules and Violations panel.

### 12.1.7.1 Design Rules Check report

The DRC report is often referred to as the Batch DRC. This performs design rules checks based on the options selected and marks any violations found. Selecting the **Tools » Design Rule Check** menu command runs the DRC. This displays the *Design Rule Checker* dialog shown in Figure 9.

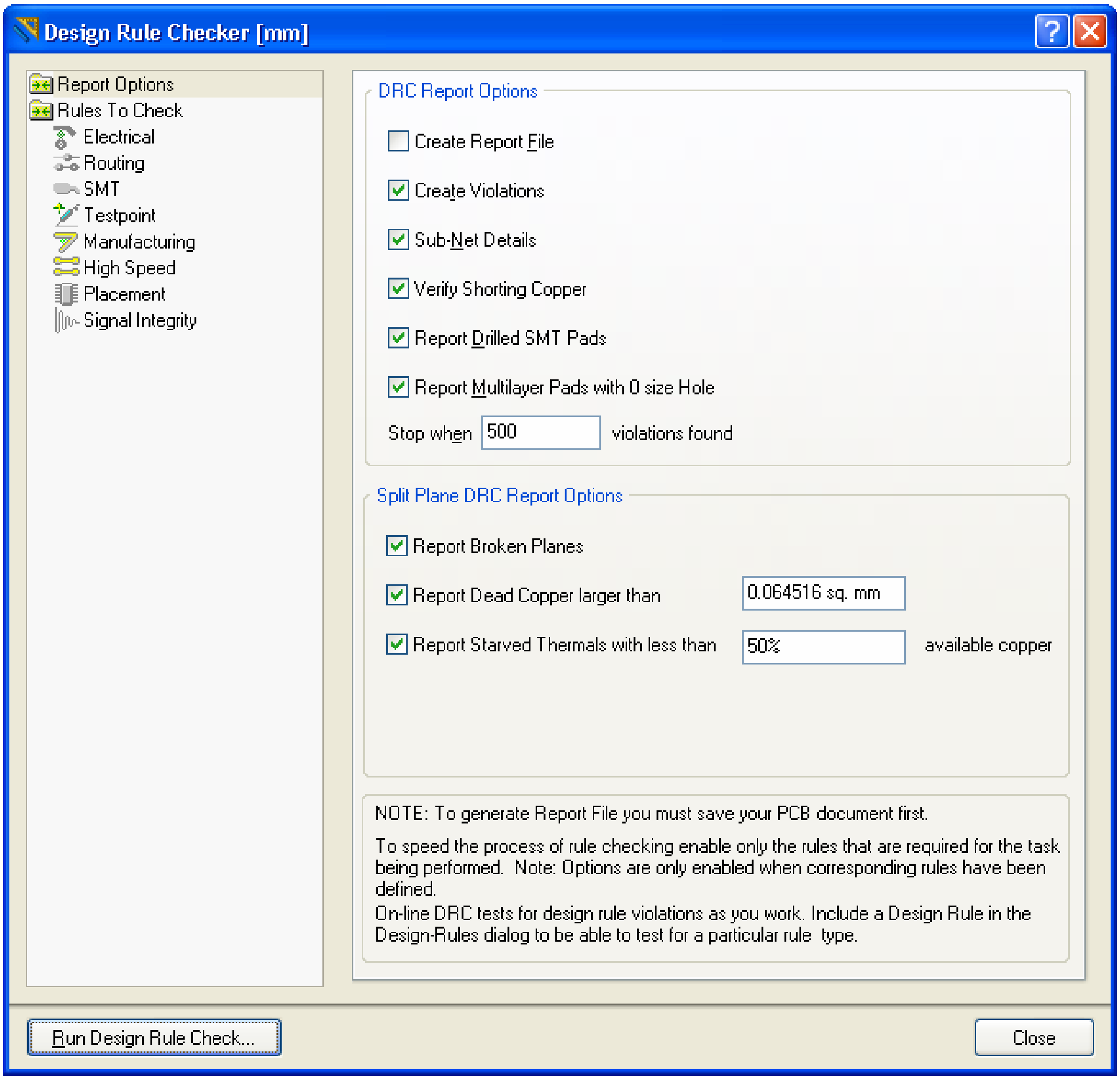



*Figure 9. Report Options in the Design Rules Checker dialog*

The Rules to Check sections of this dialog enables you to select which design rules the DRC will check for violations. Click on the **Run Design Rule Check** button to start a DRC check on the

PCB. A report (.DRC) is generated and displays in the Text Editor if the Create Report File option is enabled.

#### 12.1.7.2 Locating design rule violations

The following features are provided to locate and interpret DRC violations:

- PCB Rules and Violations panel. Select [All Rules] in the Rule Class section of the panel to list all violations. Click once on a violation to display it (and mask all other objects). Double-click to open the *Violations Details* dialog.
- The Message panel. This panel lists all violations detected in the design. Double-clicking on most message types will jump you to the violation (but will not mask like using the panel).
- The DRC report. This report is generated if the **Create Report File** option is enabled in the *Design Rule Checker* dialog.
- The right-click Violations menu entry. Right-click on a violation and select Violation to display information about the violations on that object, select a violation entry to open the *Violation Details* dialog.
- **Shift + V** shortcut while hovering the mouse over the top of a violation.

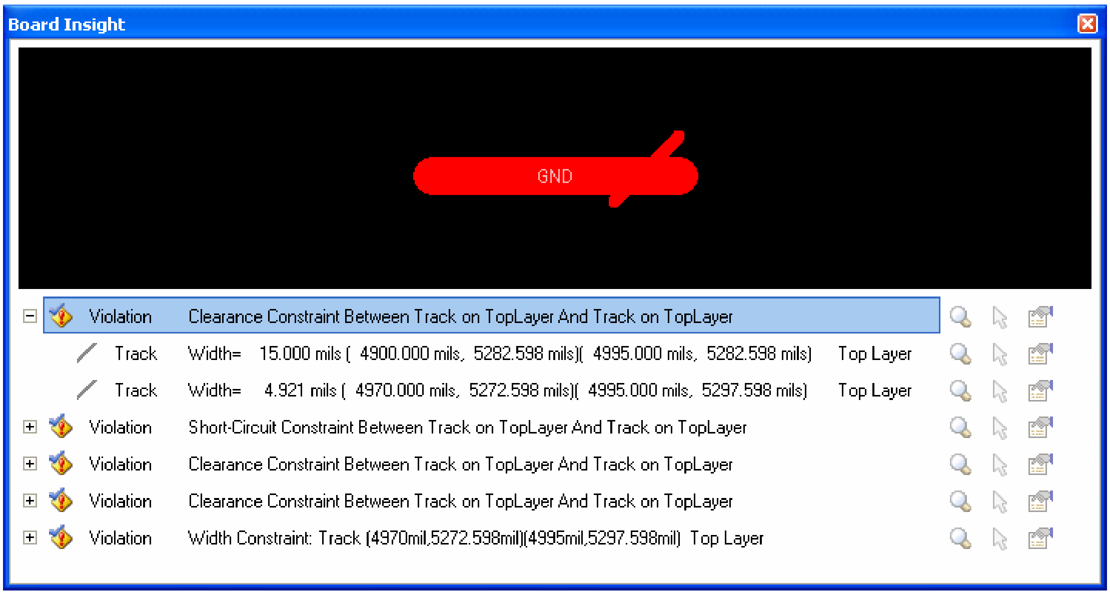



*Figure 10. Using shift + v to display violation in board insight*

### 12.1.8 Exercise – Running a DRC

In this exercise, you will run a Design Rule Check (DRC) to check for PCB design violations.

1. Run a DRC and review the violations in the PCB Rules and Violations Panel. There should be at least three violations as the pads in J200, the power connector, have holes that are larger than the maximum permitted by the default hole size constraint rule.
2. Change the rules to suit the requirements of the connector. A suggestion for an effective rule in this case would be to use a rule that targets the footprint name. The query to use in this case would be, **HasFootprint('PWR2.5')**
3. Create a new rule for Hole size in the manufacturing section and set to the query in step 2. Make sure the priorities are setup correctly after creating this rule. Also make sure there are two rules at this stage. One should be set to All scope.
4. Note that the Unrouted Net design rule is used to check for nets that have not been completely routed, if your board is not routed yet you should disable checking of this rule in the *Design Rule Checker* dialog.

5. On top of above the is also some manufacturing rules checked that at this stage are causing errors that we'll fix later on. These include the silk to silk clearance, silk over component pads and min solder mask sliver
6. To make sure the rule added in step two is working correctly right click on the offending pad found in step 1 and select **Applicable Unary Rules**. It should be using the new rule for the offending pads and the original rule for any other pads.
7. Save the board.

> **Note:** Make sure that all used layers are on when you are trying to resolve design rule violations. Keep in mind that with the default settings, the DRC stops after detecting 500 errors (configured in the *Design Rule Checker* dialog).

TOP SECRET

Altium Designer

# Module 13: Classes and Rooms

# Module 13: Classes and Rooms






Module Seq = 13

# 13.1 Object classes

## 13.1.1 Defining Classes

Classes are provided to enable various commands to operate on sub-sets of object types, for example a group of components or a group of nets. Note that an object can belong to more than one class.

Commands will operate on a class if a design rule for that class has been defined.

Classes can be created for:

- nets
- components
- pads
- from-tos
- differential pairs
- layers.
- design channels
- polygons

To create an object class, select **Design » Classes**. This displays the *Object Class Explorer* dialog shown in Figure 1 below.

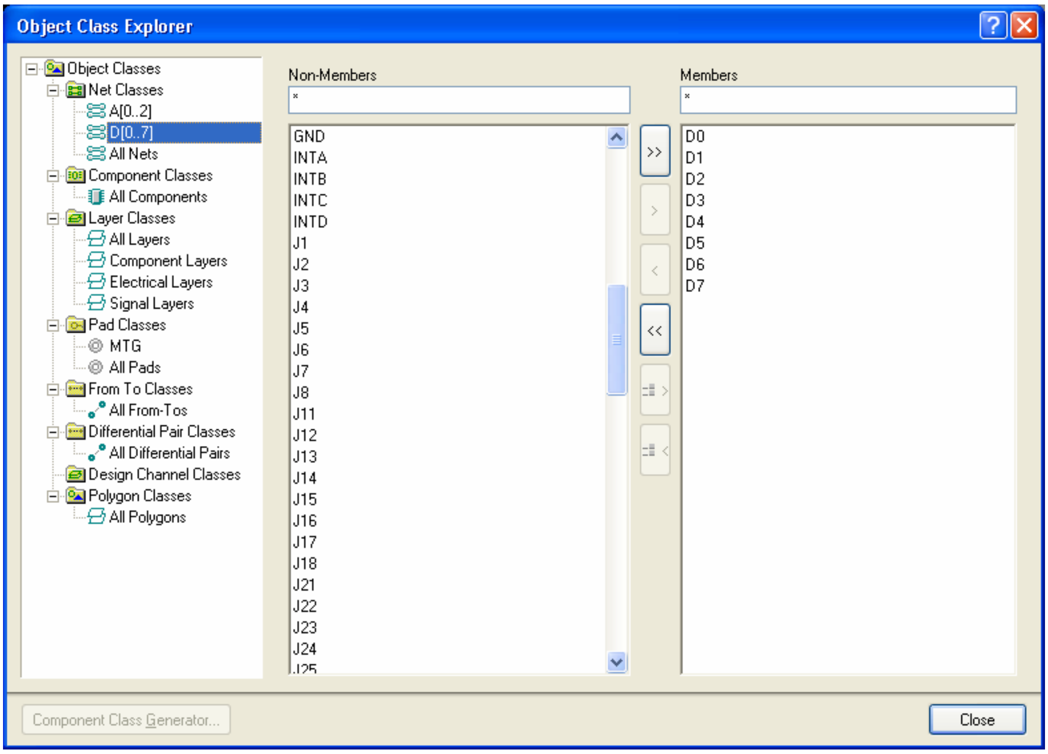



*Figure 1. Use the Object Class Explorer to create and manage Object Classes.*

- Click on the class type of the class you want to create, right-click and select **Add Class**. A new class will appear in the list with the default name of **New Class.** Click on the class name to edit the class and add the members, right-click on the class name and select **Rename Class** to rename it.

- Note that there are transfer buttons for selected objects; often it is easier to select the objects in the workspace first, then use these transfer selected buttons to build the class. If you have created a selection before hand using *Find Similar objects*, *PCB panel*, *list panel* etc, then the button shown in Figure 2 should enable. Pressing this button shifts the selection from the non-members column to the members column.

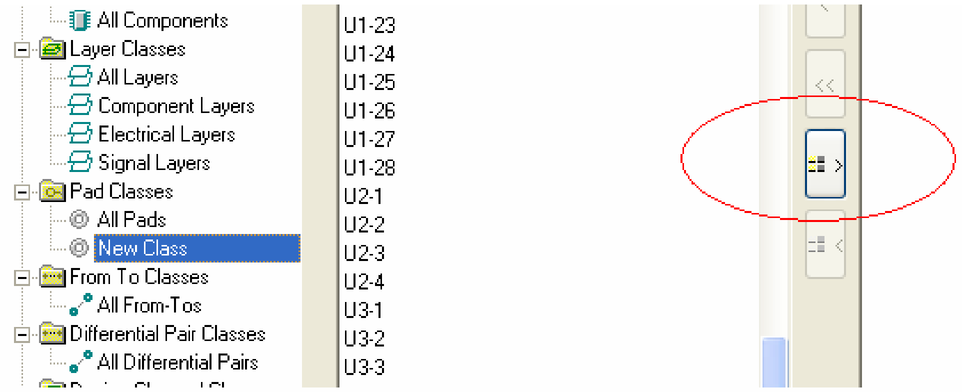



*Figure 2. The selection button enabled.*

## 13.1.2 Component Class Generator

The *Edit Component Class* dialog includes the **Class Generator** button, which, when clicked, displays the *Component Class Generator* dialog. This allows you to quickly create a component class containing components based on selected properties.

## 13.1.3 Exercise – Using classes in design rules

This exercise looks to continue with some of the rule changes we have made in the previous module.

1. Using the Temperature sensor project PCB document, create a Net Class called `Power`, which includes the following nets: 3V3, 5V and GND. To do this:
   - Select **Design » Classes**
   - Right-click on **Net Classes** in the tree on the left and select **Add Class**.
   - Click on the New Class entry that is added to the list, and press **F2** to rename the class to `Power`.
   - Add the class members 3V3, 5V, GND and close the dialog.
2. In the **Design Rules** dialog, add a new routing via style rule that targets the Power net class, with settings of Via diameter = 40 and a hole size of 25. Name this rule `RoutingVias_Power`

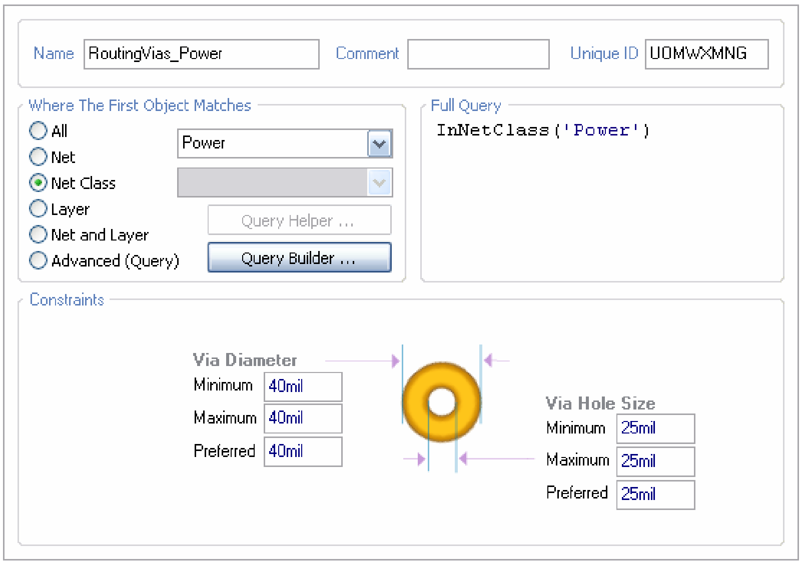



*Figure 3. Set the rule to target the Power net class*

3. Save the board.

# 13.2 Rooms

A room is a region that defines an area where components can either be kept within or kept out.

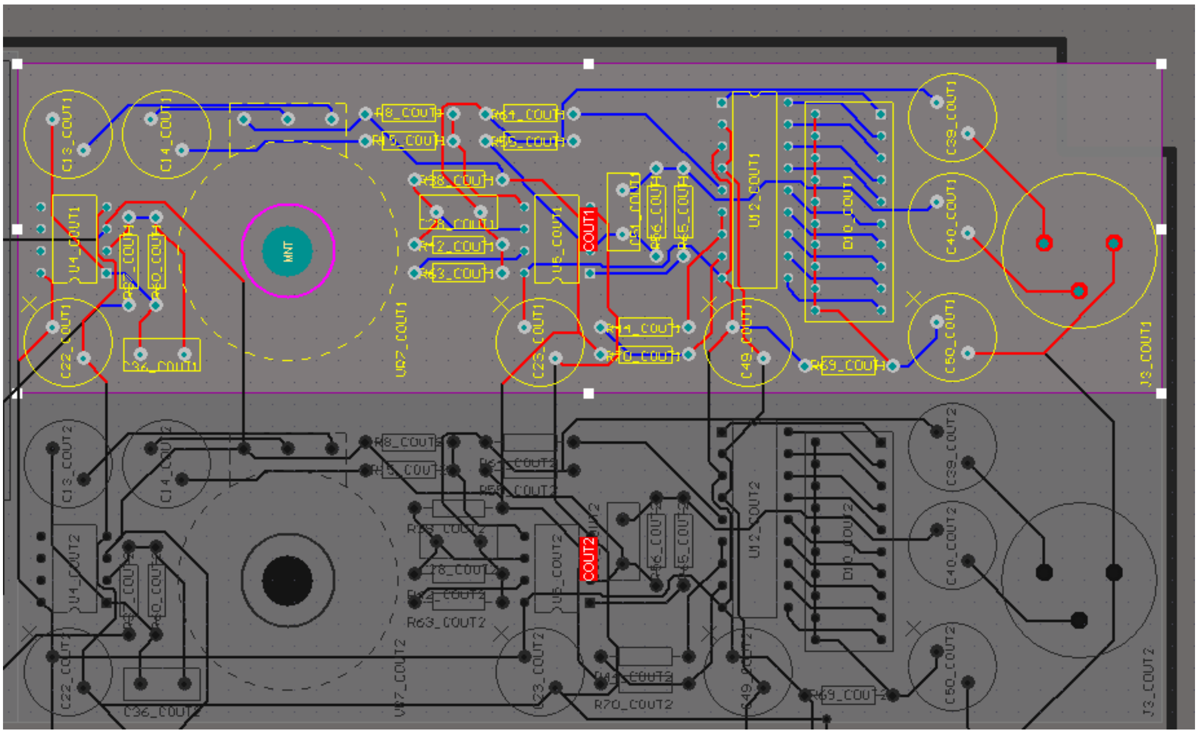


*Figure 4. Components and routing in the room COUT1 highlighted, all other components and routing on the board are masked..*

## 13.2.1 Defining Rooms

- Rooms are placed using the commands in the **Design » Rooms** sub-menu, or using the Room tools on the Utilities toolbar.
- A Room Definition design rule is created for each room that is placed.
- Once a room is placed you define the components associated with it (and whether they are to be kept in or kept out), as the scope, or query, for that rule.
- To define the components associated with a room, double-click on the room to display the *Room Definition* dialog. This dialog can also be accessed in the **Placement** region of the *Rules* dialog. Set the scope of the rule to the required component, component class or footprint.

### 13.2.1.1 Moving components into a room

- Components that have been assigned to a room can be automatically moved into it by selecting the **Tools » Component Placement » Arrange Within Room** command, or clicking the **Arrange Components Within Room** button in the Alignment tools in the Utilities toolbar. You will be prompted to click on the room.

### 13.2.1.2 Moving rooms

- Once component(s) have been assigned to a room, they move when the room is moved. To move a room without moving the components, temporarily disable the Room Definition rule in the **Placement** section of the *PCB Rules& Constraints* dialog.

- If a component is moved such that it is in violation of the Room Definition rule, it is displayed with a Design Rule Check (DRC) error marker.

### 13.2.1.3 Using a Room to scope another Rule

- Rooms have a dual nature in that they are defined as a rule themselves, but they can also be used as the scope of other design rules.
- To use a room as the scope of another rule, for example to define a region where you require larger or tighter routing clearances, you first set the Room rule to target nothing by setting its rule Query to something like: **Not IsComponent**.
- You can then define a Routing Clearance design rule that uses a Query like **WithinRoom(MyRoomDefinition)**.
- Examples of where you might use the query WithinRoom include Width, Clearance and Via Style design rules, defining exactly the area of the PCB were that rule is to be applied.

## 13.2.2 Copying Room Formats

The placement and routing of one room, can be applied to other rooms. Use this capability in multi-channel designs, where the component placement and routing can be identical in all channels.

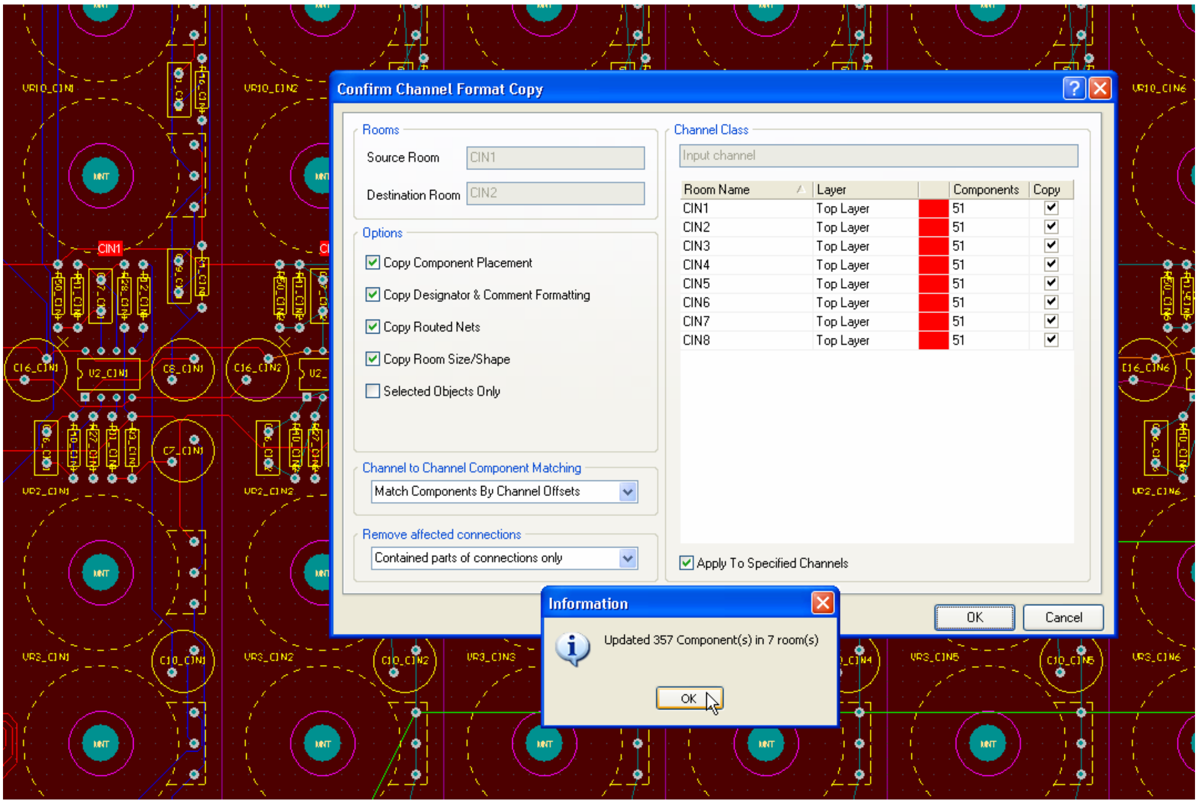



*Figure 5. Component placement and routing being copied from one room, to seven other identical rooms. This 'step and repeat' capability is an excellent time-saver for a multi-channel PCB.*

- To copy the placement and routing of one room onto others, run the command **Design » Rooms » Copy Room Formats**. You will be prompted to choose the source room, click anywhere in that room. You will then be prompted to choose a destination room, click anywhere within one of the target rooms.
- The **Confirm Channel Format Copy** dialog will appear. Use this dialog to define the copy options as required. The options relate to component placement, designator and comment formatting, net routing and room size/shape.

- There is also an option to limit the copy to only those objects that are selected in the source room.
- If the source room is part of a defined channel class, the class and all of its members will be listed on the right-hand side of the dialog. This section is, by default, unavailable until you enable the **Apply To Specified Channels** option. For each room in the class, its name, layer and the amount of components it contains, is listed. There is also a check box to define whether a room should be included in the copy.
- Use the **Channel to Channel Component Matching** field to determine the technique used to match components between the source and destination room(s). Matching can be carried out either by use of the **Channel Offset** for a component - stored in the **Schematic Reference Information** region of the **Component** dialog - or by source designator for the component, i.e. the logical designator, as used in the source schematics.
- After defining the criteria for the copy as required, pressing **OK** will close the dialog and proceed with the copy. An information dialog will appear, listing how many components were updated and in how many rooms.

## 13.2.3 Polygon Rooms

As well as rectangular shaped rooms, polygonal rooms can also be placed (**Design » Rooms » Place Polygon Room**).

- As with rectangular rooms, the room can either be placed empty and components associated at a later stage, or it can be placed around components in the design, automatically associating them to the room.
- While defining the shape of the room, use the **SPACEBAR** to cycle through various corner modes. Modes available are: 90 Degrees, 45 Degrees and Any Angle.
- Press **TAB** during placement to open the **Edit Room Definition** dialog, from where you can define the scope and constraints of the room.
- Double-click on a placed room to open the **Edit Room Definition** dialog, in order to edit its rule properties. Alternatively, use the **PCB Rules and Violation panel** dialog to gain access to a particular room definition.
- To copy an existing rectangle room format to a polygon shape room format you can use the copy room format command found in Section 13.2.2 Copying Room Formats for more detail.

## 13.2.4 Exercise – Using a polygon room and the WithinRoom Query with a BGA

1. Open up the Project `\Altium Designer Summer 09\Examples\Reference Designs\ Daughter Boards - 2 Connector\NBP8 Xilinx Virtex-II Pro BGA456 Rev1.01\ NBP8 Xilinx Virtex-II Pro BGA456 Rev1.01.PrjPcb` . Once the project is open, open the PCB in the project.
2. Select **Tools » Polygon Pours » Shelve polygons**. There should be 16 polygons to shelve. The shelving process hides the polygons from view.
3. Select **Tools » Unroute » All**. You can also do this by using the shortcut **UA**. Click **OK** to allow unrouting of locked primitives if asked.
4. Zoom in to clearly show the BGA. Select **Design » Rooms » Place Polygon Room**. Place the room so it goes around the outline (top overlay) of the BGA component. It does not need to be exact for the exercise.
5. Once placed a room definition will be added to the design rules. Select **Design » Rules** to open the rules dialog.

6. Display the **Placement** section, and then the **Room Definition** section. There should be a room called *RoomDefinition,* change its name to *BGAwidth*, and copy that name so it can be used in a moment.
7. Now go to the **Routing** section of the rules dialog, and display all the **Width** rules.
8. To add a new width design rule, right click on the word **Width** and select **New Rule** from the pop up menu.
9. Rename the new rule to *BGA routing width*, and set the Query to **WithinRoom('***BGAwidth')*,
10. In the Constraint section of the rule, set the **Min**, **Preferred** and **Max** width settings to 0.05mm for all three widths. This defines a width rule for routing within the area of the polygonal room.
11. Click on the **Priorities** button at the bottom of the rules dialog, and check that the new *BGA routing width* has the highest priority.
12. Once all the settings are defined click **Apply** to ensure that there are no problems with the rule settings (the rule name will go red if there are), then click **OK** if there are no errors.
13. Run the command **PT** or **Place » Interactive Routing**, and start routing off one of the pads on the BGA. After you place the first corner outside of the room, the routing width will become wider (as per the width specified by the next applicable width design rule).

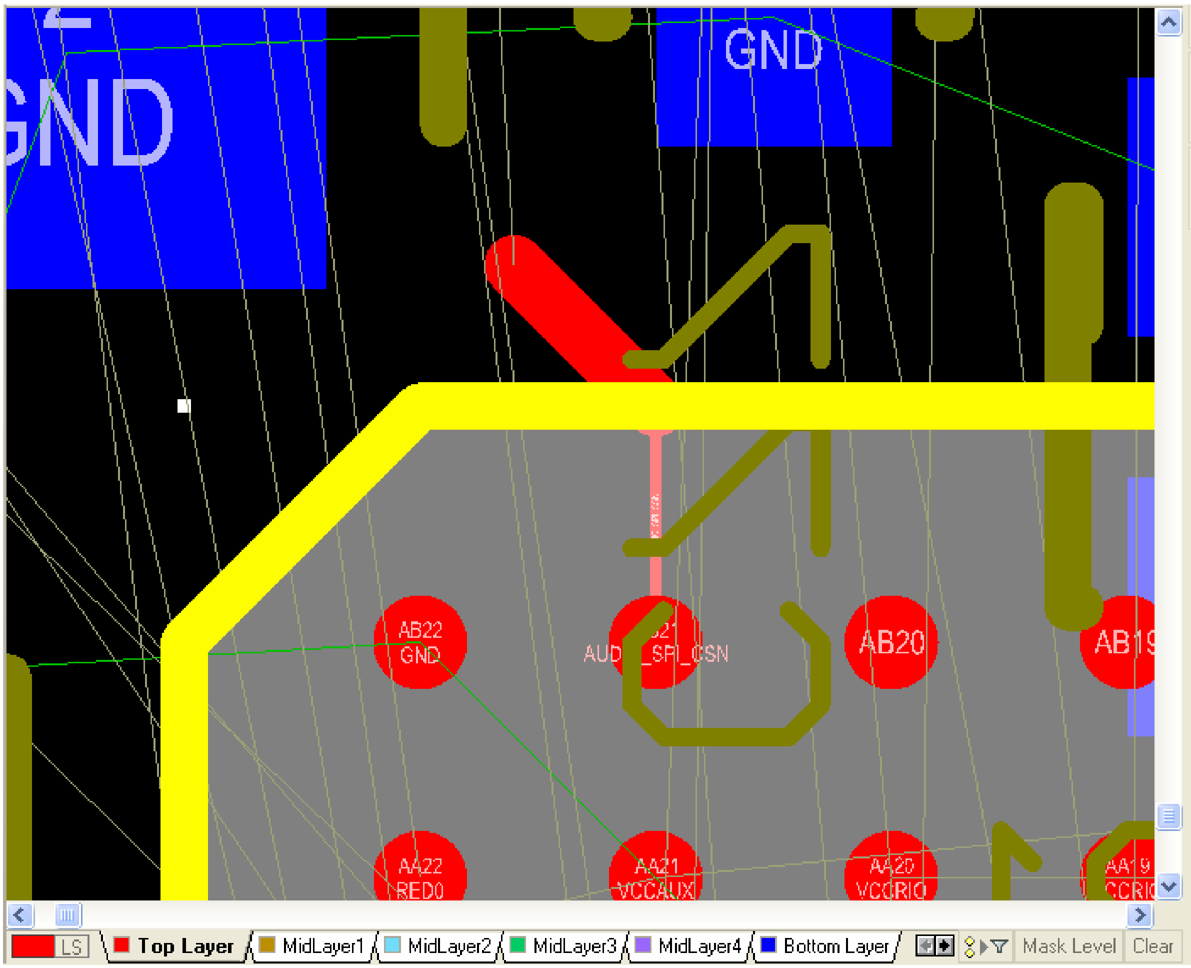



*Figure 6. The width change of the track as it leaves the defined polygon room area.*

> **Note**: You can also use the WithinRoom query to define rules for clearance and via style for the BGA.

TOP SECRET

Altium Designer

# Module 14: Placement and Re-annotation

## Module 14: Placement and Re-annotation





Module Seq = 14

# 14.1 Component Placement tools

## 14.1.1 Placing Components

Component footprints can be placed on a PCB board manually from the PCB libraries. Alternatively, they are placed to the side of the board when the Synchronizer is run from a schematic document, ready for moving to their correct locations.

### 14.1.1.1 Adding libraries

- For component footprints to be placed, they must be available in a library. Footprint libraries can be made available by including them in the project, installing them in the Libraries panel, or defining a search path to their location. Libraries are searched in the order just mentioned. Installed and search path libraries can have their search order defined.
- Click the **Libraries** button at the top of the Libraries panel to install a footprint library.
- Search paths are defined in the *Project Options* dialog.
- Footprint libraries included with Altium Designer are located in `\Program Files\Altium Designer Summer 09\Library\Pcb`.

### 14.1.1.2 Placing a Component

- Component footprints can be placed in a PCB document from any open footprint library by double-clicking on the name in the Libraries panel, using the **Place** button on the panel, or using the **Place » Component** command. If you use the **Place » Component** command, the footprint name you type in must be in an available library.
- The *Place Component* dialog appears. Enter the designator and comment as required.
- During placement, the component may be moved, rotated (press SPACEBAR) or swapped to the bottom layer (press **L**).

## 14.1.2 Finding Components for placement

- If you can visually locate components that you are positioning on the board, click and hold to move them.
- Otherwise, select **Edit » Move » Component** (**M C**) and click where there are no objects. This displays the *Choose Component* dialog.
- From this list, you can select the component to be placed.
- You also select the behavior you would like – to move the cursor to the component, the component to the cursor or no special action.
- You can also browse for a component in both the schematic and the PCB, by holding the ALT key as you click on the component in the **Navigator** panel (note the project must be compiled).
- Another technique to finding component footprints is to use the schematic as a reference. Select the required component(s) on the schematics and select **Tools » Select PCB Components** from the menus.

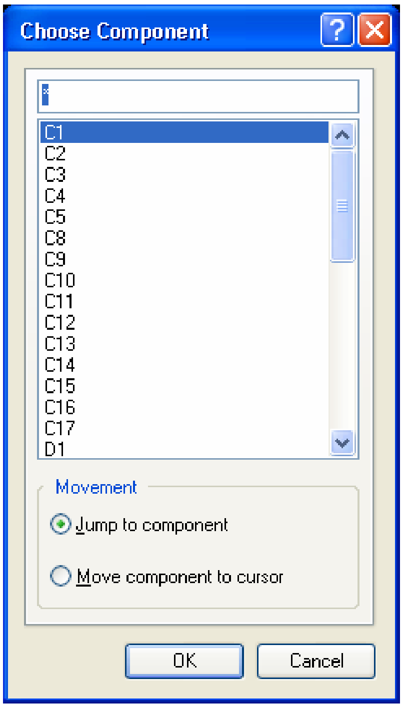



*Figure 1. Choose Component dialog*

- Alternatively, once the components are selected on the schematic, switch to the PCB and use the **Reposition Selected Components** command (**Tools » Component Placement** menu) to position them in the order you selected them on the schematic.

## 14.1.3 Moving Components

- Click and hold on a component to move it. While you are moving the component the connection lines directly connected to it will drag with it while all other connection lines are not displayed.
- As you move the component, connection lines are dynamically optimized so that every connection line is following the shortest path to any other object with the same net name.
- Also, while you are moving a component, pressing the **N** key will toggle the display of connections.
- Pressing the **L** key while moving a component toggles the component between the top and the bottom layer of the PCB.

### 14.1.3.1 Component unions

The Union feature allows you to group components together so that they can be moved as a group, i.e. as if they were a single component.

- Multiple unions can be defined.

- To create a union of components, select the components then choose the **Create Union from selected components** icon in the Component Placement tools in the Utilities toolbar.

- To remove a component from a union, or to remove the union, choose the **Break Component from Union** icon from the Component Placement tools in the Utilities toolbar. This displays a dialog that lists all components in the union. From here, select the component(s) to be removed from the union. Selecting all components removes the union.

### 14.1.3.2 Component Placement grid

When components are placed or moved, they snap to the Component Placement grid. This grid has an X and a Y value and they are set in the *Board Option* dialog.

## 14.1.4 Interactive Placement commands

There are a number of semi-automated tools that allow you to edit the placement of your PCB design. They are accessed via the **Edit » Align** menu, the **Tools » Component Placement** menu, or the **Alignment** tools in the **Utilities** toolbar. These are described in the following sub-sections.

### 14.1.4.1 Alignment commands

- The Alignment commands (**Edit » Align**) operate on selected objects.
- Use the Spacing commands in the Alignment tools to make the horizontal and vertical spacing between selected components equal, more or less.
- Increasing and decreasing the horizontal (or vertical) spacing for selected components means the horizontal (or vertical) distance between the component reference points is increased (or decreased) by the amount specified in the X (or Y) component placement grid.
- All unlocked components are moved to the closest Component Placement grid point.

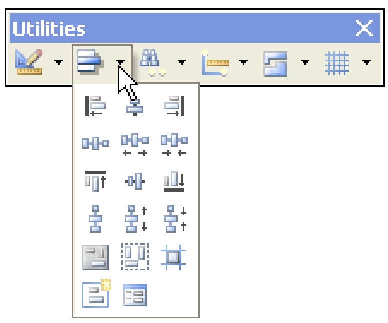



*Figure 2 Align selected objects using the alignment tools.*

### 14.1.4.2 Component Placement commands

Use the interactive component placement commands (**Tools » Component Placement**) to:

| Command | Behavior |
|---|---|
| Arrange Within Room | Components assigned to the nominated room are placed within that room. |
| Arrange Within Rectangle | Selected components are placed within a defined area. |
| Arrange Outside Board | Selected components are moved outside the board area. |
| Reposition Selected Components | Enable Cross Select Mode in the Schematic editor (Tools menu), select multiple schematic components, switch to the PCB editor, then run this command to reposition each component, in the same order they were selected in the schematic. Alternatively, select the components in the PCB panel (with the Select option enabled in the panel), then run the command. |

# 14.2 Re-Annotation and back annotate

Designators can either be assigned based on the component position on the schematic, or their position on the PCB.

- To positionally re-annotate the components on the PCB, select the **Tools » Re-Annotate** command. This displays the *Positional Re-Annotate* dialog, as shown in Figure 3. Select the required direction method and click **OK**.
- Since the PCB component links to its schematic equivalent via a unique identifier, there is no danger in re-annotating multiple times.
- Once you are happy with the re-annotation, update the Schematic with the designator changes using the Synchronizer. To do this, select **Design » Update Schematic**.

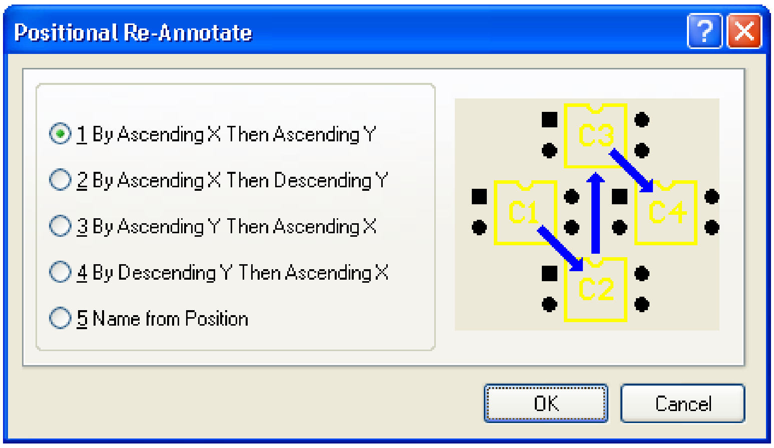



*Figure 3. Positional Re-Annotate dialog*

**Note**: the bottom left corner of the component bounding rectangle is used to determine the component location during re-annotation. The re-annotation command scans for components in a 100mil wide band, stepping in the X or Y direction, according to the selected option.

## 14.2.1 Exercise – Component Placement

In this exercise, you will position the Temperature Sensor components. Use the following image as a guide.

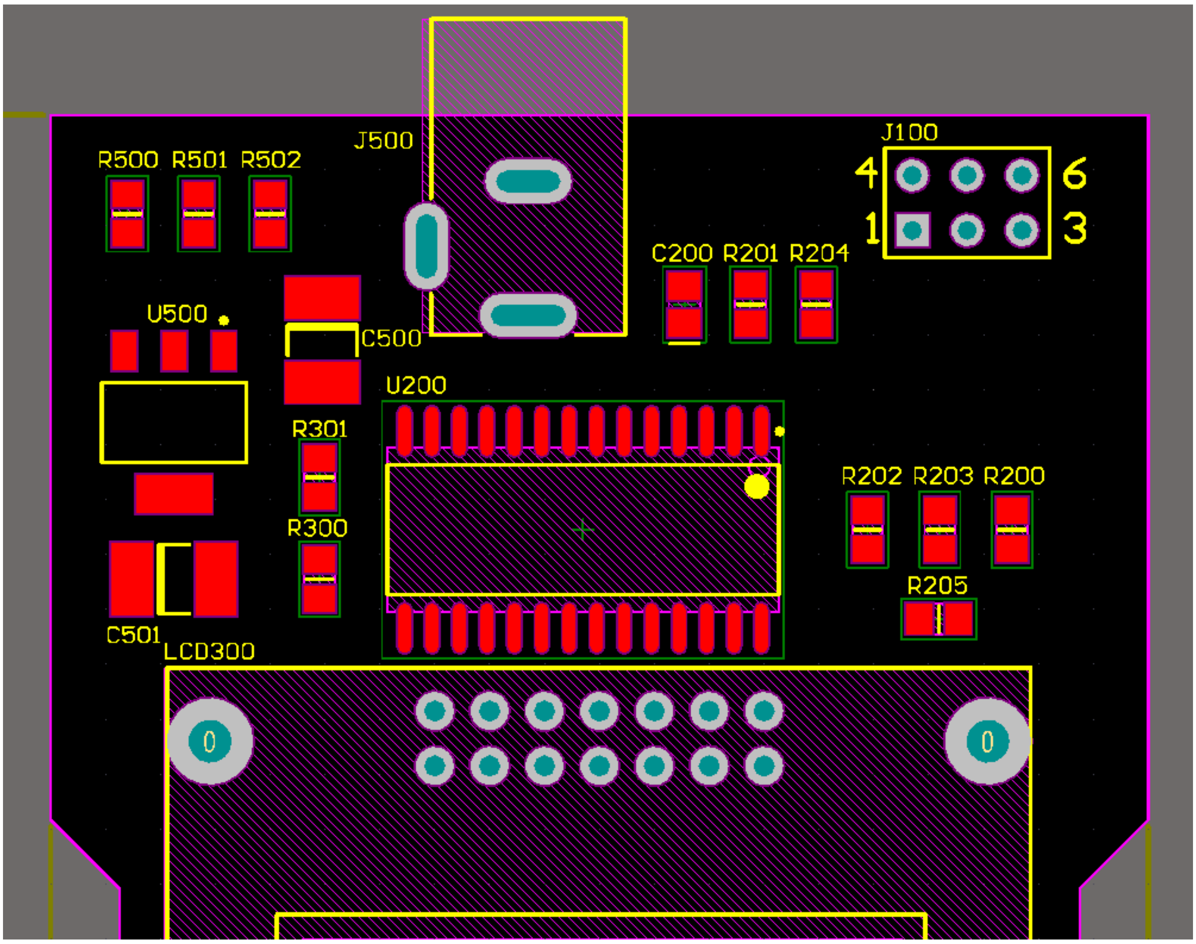



*Figure 4. One possible component placement for the Temperature Sensor board.*

1. The board does not need to be placed exactly as shown, this is only one solution.
2. As you press the spacebar to rotate components, you will notice that the designator remains positioned above the top left of the component. This is controlled by the Designator **Autopostion** option in the *Component* dialog. To manually position a designator, click and drag it to the required location, pressing the spacebar to rotate it if required. To temporarily filter out all objects in the workspace except the designators, type the query `IsDesignator` into the **PCB Filter** panel. You can now move designators without fear of moving anything else in the workspace. To change the size of designator text, press **Ctrl+A** to select all, **F11** to open the **Inspector**, and set the **Text Height** to 30mils and **Text Width** to 4mils. Press **Shift+C** to clear this filter when finished.
3. Each component also has a Comment string, you control the display of this in the Component dialog. To toggle the **Hide** status of all comment strings, enter the Query `IsComment` into the **PCB Filter** panel (confirm that the Select check box is enabled in the panel), then press F11 to open the **Inspector**. The **Inspector** can now be used to edit all selected Comment strings, toggle the state of the **Hide** checkbox to hide/display the Comment strings. Clear the filter when finished.
4. There is a placed copy of the board in the `Backup` folder. You can use this as a reference.
5. Save the board when you have finished but do not route it yet.

TOP SECRET

Altium Designer

# Module 15: Schematic Library Editor

# Module 15: Schematic Library Editor






Module Seq = 15

# 15.1 Introduction to Library Editing

The Creating Library Components section explains the various types of libraries, and the fundamental way in which library documents are created and managed in Altium Designer.

The process of using the Schematic and PCB Library editors will be explored with hands-on exercises showing how to create single and multi-part packages in a schematic library, creating PCB footprints, attaching models, and working with separate library files. It will also introduce Integrated Libraries, a compiled, secure and portable form of library file.

This section will also discuss adding component parameters and provide hands-on exercises on importing pin information from vendor pin lists.

Figure 1. Outlines the workflow to follow when creating component libraries in Altium Designer.

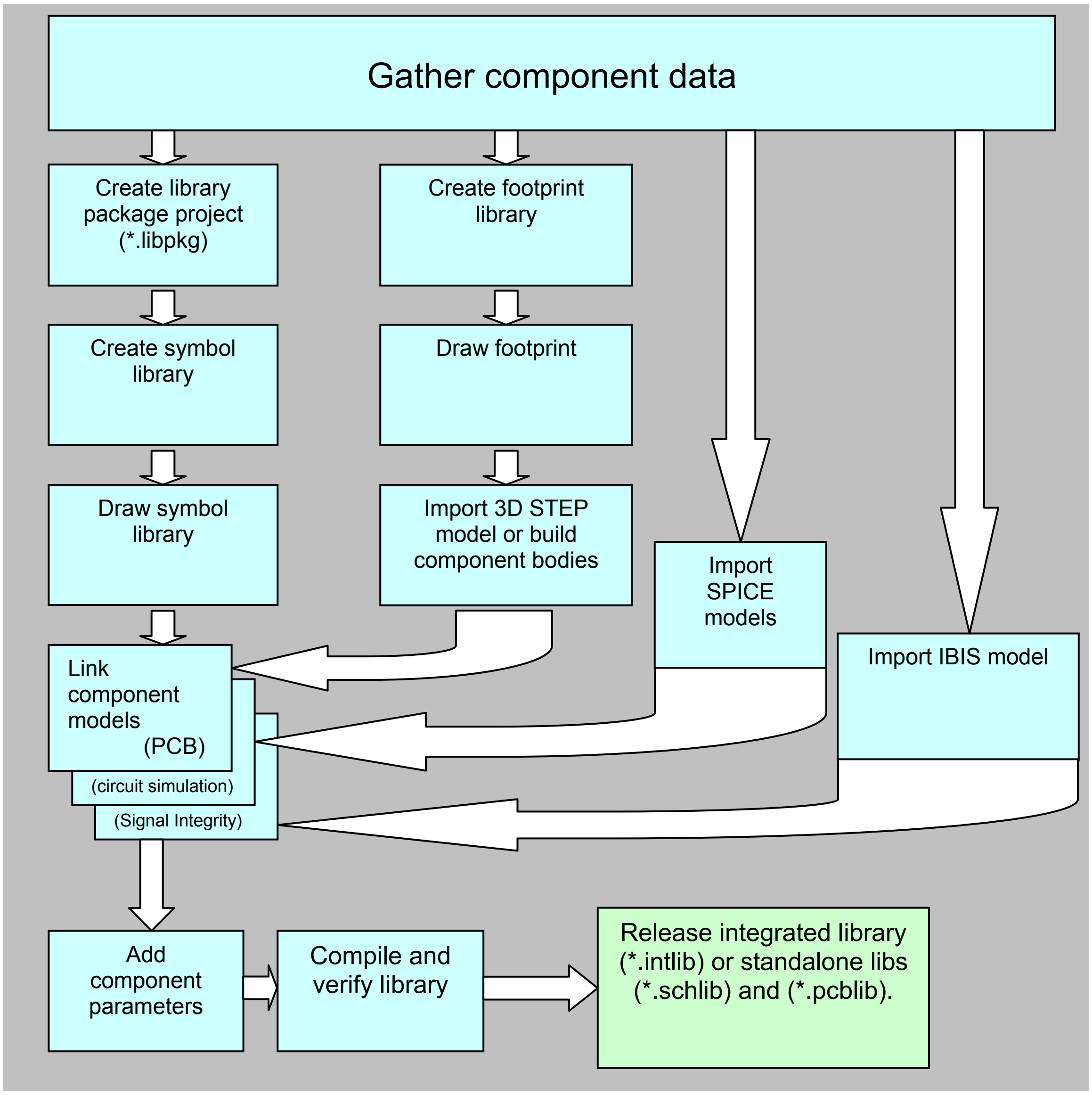



*Figure 1. The Altium Designer component creation workflow.*

# 15.2 Schematic Library Editor

This section covers how to use the Schematic Library Editor and how to create a new component. The Schematic Library Editor is used to:

- Create and modify schematic component symbols
- Attach models to the component
- Add parameters to the component
- Manage component libraries.

The Schematic Library Editor is very similar in operation to the Schematic Editor and shares the same graphical object types (but not the electrical objects). In addition, the Schematic Library Editor has one additional object, the Pin, which is used at points where wires connect to components.

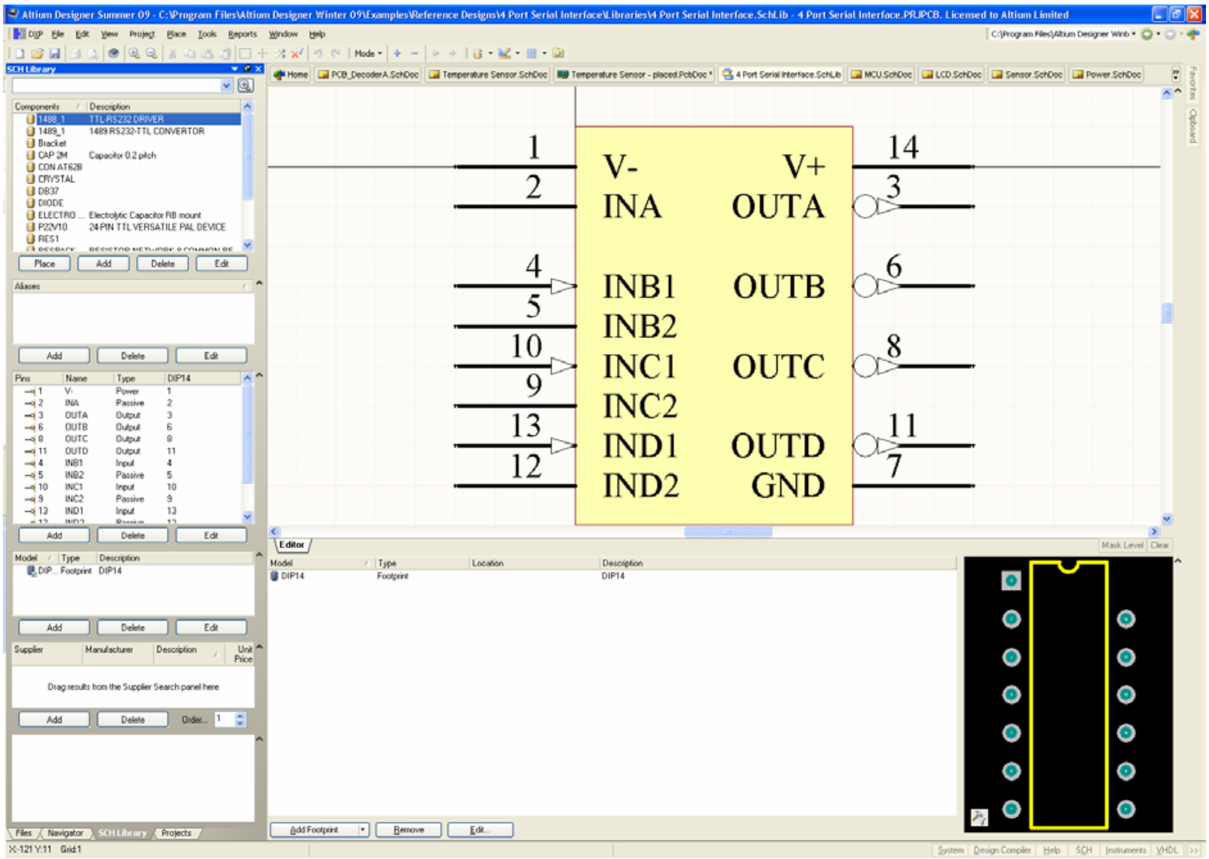


*Figure 2. Schematic Library Editor Workspace*

- **Schematic Libraries (`*.SchLib`)** can be opened for editing using the **File » Open** menu command. Navigate to the folder that the required library is stored in and locate the library, e.g. `C:\Program Files\Altium Designer Summer 09\Examples\Training\PCB Training\Temperature Sensor\Libraries\Temperature Sensor.SchLib` and click on **Open**.
- **Integrated Libraries (`*.IntLib`)** are compiled binary files, which cannot be edited. If you attempt to open an integrated library, it will be de-compiled, i.e. all the source libraries will be extracted and a new Library Package will be created. All the libraries supplied with the software are integrated libraries.

- **Database Libraries (*.DBlib)** is a link to a database (ODBC or ADO based) were references are stored for symbol reference, model linking and parameter information. Each record in the database represents a component, storing all of the parameters, along with links to the models. The record can include links to inventory or other corporate component data. With this approach the schematic component is only used as a symbol, with the models (footprint, 3D Model and simulation model) stored in standard schematic library files, PCB library files, and so on. Components are placed from the database by installing a new DBLib document in the **Libraries** panel, with the DBLib document being configured to reference the component database.
- **Subversion Database Libraries (*.SVNDBLib)** is similar to database libraries however all the linked symbol and footprint data is version controlled within a subversion database. Due to this, each symbol and footprint is stored separately in individual *.schlib and *.pcblib files within the SVN database.

## 15.2.1 Schematic Library Editing Tools

- The Schematic Library Editor has a right-click menu, a **Utilities** toolbar and a **Mode** toolbar, as shown in Figure 3.
- The Utilities toolbar includes a range of standard drawing tools, and a comprehensive set of IEEE symbols.

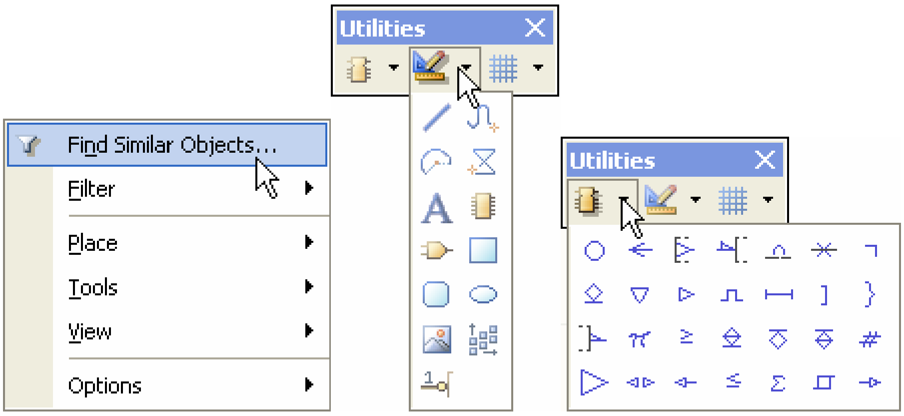



*Figure 3. Library Editor Toolbars and right click command options*

## 15.2.2 Schematic Library Editor Terminology

- **Object** — any individual item that can be placed in the Schematic Library Editor workspace, for example, a pin, line, arc, polygon, IEEE symbol etc.

> **Note:** The IEEE range of symbols can be resized during placement. Press the + and - keys to enlarge and shrink the symbols as you place them.

- **Part** – a collection of graphical objects that represent one part of a multi-part component (e.g. one inverter in a 7404), or a library component in the case of a generic or singly packaged device (e.g. a resistor or an 80486 microprocessor).
- **Part Zero** – this is a special non-visible part available only in multi-part components. Pins added to part zero are automatically added to every part of the component when the component is placed on a schematic. To add a pin to part zero place it on any part, edit it, and set the Part Number attribute in the *Pin Properties* dialog to Zero.
- **Component** – either a single part (e.g. a resistor) or a set of parts that are packaged together (e.g. a 74HCT32).

- **Aliases** – refer to the naming system when a library component has multiple names that share a common component description and graphical image. For example, 74LS04 and 74ACT04 could be aliases of a 7404. Sharing graphical information makes the library more compact. When using database libraries, the use of aliases has become obsolete.
- **Hidden Pins** – these are pins that exist on the component, but do not need to be displayed. Typically, this is done for power pins, which can then be automatically connected to the net specified in the *Pin Properties* dialog. This net does not need to be present on the schematic; one will be created, connecting all hidden pins with the same Connect To net name. The pins will NOT automatically connect if they are visible on the schematic sheet (i.e. un-hidden). Hidden pins can be shown on the schematic sheet by selecting the **Show All Pins** option in the *Component Properties* dialog.
- **Mode** – a component can have up to 255 different display modes. This can be used for things like IEEE component representations, alternate pin arrangements for op-amps, and so on. Use the options in the **Tools » Mode** submenu or the Mode toolbar to add a new mode to a component. The displayed component mode can be changed on the schematic sheet.

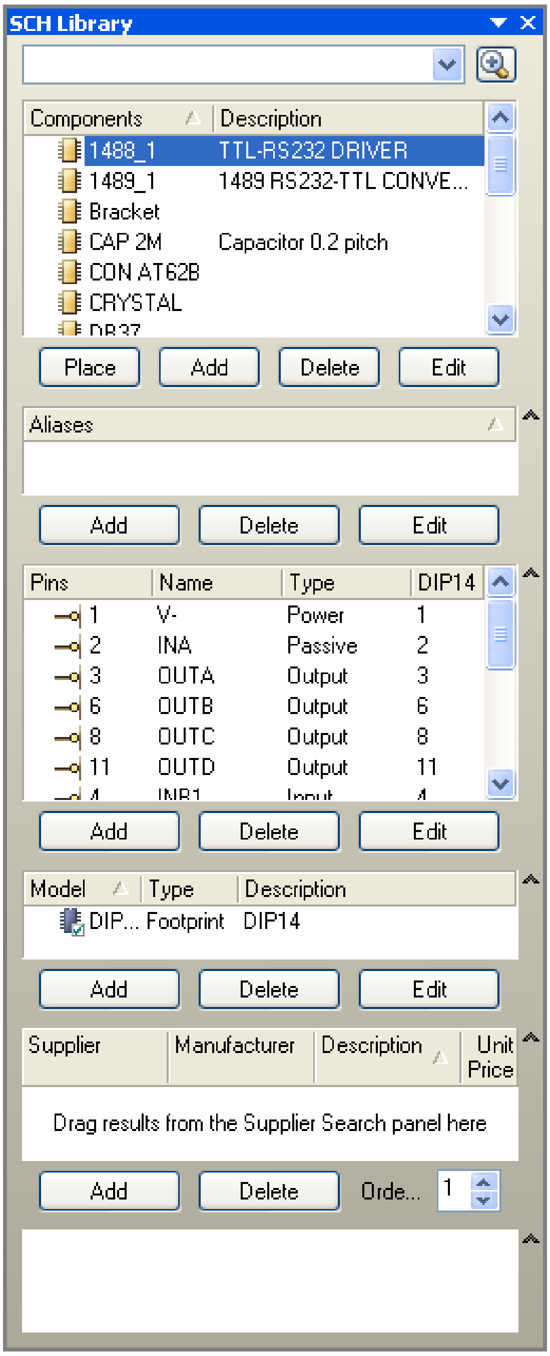


*Figure 4. Schematic Library panel*

### 15.2.2.1 Schematic Library Editor Panel

The Schematic Library Editor panel provides a number of features for working with Schematic components.

- To display the panel, select **SCH » SCH Library** via the panel display buttons at the bottom right of the workspace, when a library is open for editing.
- The buttons below each region of the panel apply to the selected entry in that region.

**Components section**

- Lists all the components in the active library.
- Double-click on a component to open its ***Library Component Properties*** dialog.
- Use the buttons and the options in the right-click menu to manage the library.

**Aliases section**

- This allows you to add alternate names to a component that share the same graphics and description.

**Pins section**

This section lists the pins in the current component.

- Edit individual pins by double-clicking.
- Select **View » Show Hidden Pins** to display all those pins that are defined as hidden. This does not change the actual pin hidden/unhidden status; rather it only displays the hidden pins in the Library Editor.
- When placing multiple pins with incrementing name/designator, press the TAB key after selecting **Place » Pin** from the menus to define the starting value. By default, both the pin number and name will increment.
- Increment behavior can be controlled using the **Auto-Increment during Placement** options in the **Preferences** dialog (the primary value is the pin

number). Enter a negative sign to decrement a value. Enter an alpha value to increment alphabetically. A single alpha followed by numbers increments the leading alpha. If there are multiple alphas, the last character is incremented/decremented.

- The entire set of pins for the current component can also be viewed and edited in the **List** panel. To filter the component to only show pins, right-click in the graphical area and select **Filter » Examples » Pins** from the floating context menu. If the List panel is not currently visible press **Shift+F12** to display it. Note that you can edit multiple pin properties in the **List panel**, and can also copy and paste to and from a spreadsheet using the **Smart Grid** commands in the List panel right-click menu.

> **Note:** If you find that the component pans out of view too often, disable auto pan in the **Schematic – Graphical Editing** page of the *Preferences* dialog. Alternatively, use the **V, F** shortcuts to bring the component back into view.

## 15.2.3 Component Properties

The **Component Properties** dialog is where you add model and parameter information to the component symbol. Double-click on a component name in the **Sch Library panel** to display the dialog.

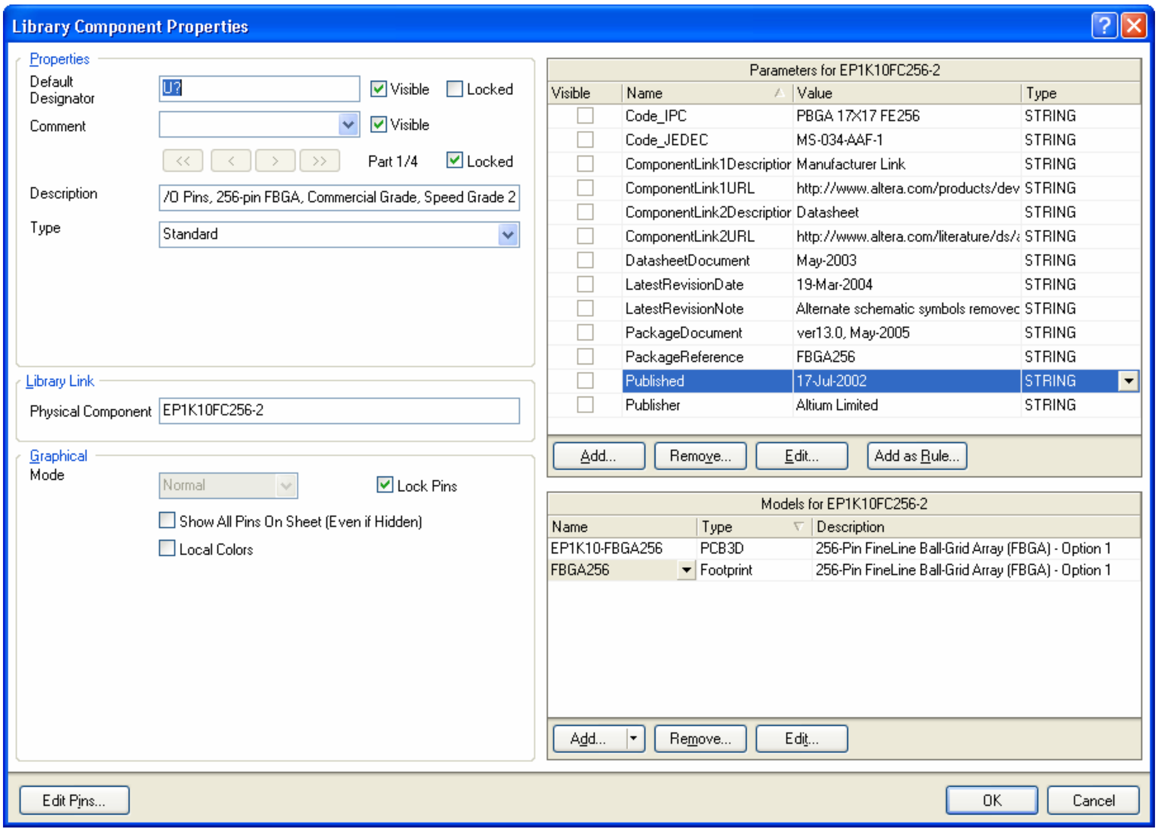



*Figure 5. Component Properties dialog*

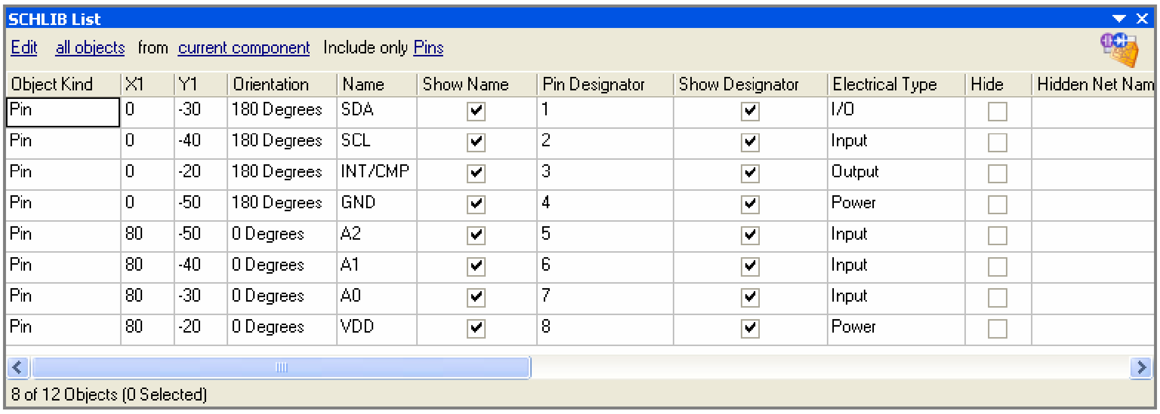
Information that would typically be defined for a component includes:

- **Default Designator** – defines the prefix string to be used with the component designator.
- **Comment** – description of the component. For a component whose definition is fixed, such as a 74HC32, this standard descriptive string would be entered. For a discrete component whose value can change, such as a resistor, the value would be entered. Note that this field supports indirection, which allows you to display the value of any of this component's parameters. Indirection is enabled by entering an equals sign, then the parameter name (spaces are not

supported). If this field is left blank, the component library reference will be entered as the comment when the component is placed, allowing you to define the comment after it has been placed on the schematic.

- **Description** – meaningful description that can be used for searching and in the BOM.
- **Type** – alternate component types are provided for special circumstances. Graphical components do not get synchronized or included in the BOM. Mechanical types only get synchronized if they exist on both the schematic and the PCB and do get included in the BOM. Net Tie components are used for shorting two or more nets on the PCB.
- **Parameters** – any number of parameters can be added either in the Library Editor, or on the schematic sheet. Parameters can be linked to a company database; add a database link document to the project to do this.
- **Models** – various component models can be added, including footprint, simulation, signal integrity, and so on.
- **Lock Pins** – if this option is enabled, you will not be able to edit pins, only the component as a whole entity, when the component is placed on a schematic. Disable this option if you wish to edit the pins and click on the **Edit Pins** button.

**Note:** Use the What's This Help ? for more information about options in the dialog.


SCHLIB List

Edit all objects from current component Include only Pins

| Object Kind | X1 | Y1 | Orientation | Name | Show Name | Pin Designator | Show Designator | Electrical Type | Hide | Hidden Net Nam |
|---|---|---|---|---|---|---|---|---|---|---|
| Pin | 0 | -30 | 180 Degrees | SDA | ☑ | 1 | ☑ | I/O | ☐ | |
| Pin | 0 | -40 | 180 Degrees | SCL | ☑ | 2 | ☑ | Input | ☐ | |
| Pin | 0 | -20 | 180 Degrees | INT/CMP | ☑ | 3 | ☑ | Output | ☐ | |
| Pin | 0 | -50 | 180 Degrees | GND | ☑ | 4 | ☑ | Power | ☐ | |
| Pin | 80 | -50 | 0 Degrees | A2 | ☑ | 5 | ☑ | Input | ☐ | |
| Pin | 80 | -40 | 0 Degrees | A1 | ☑ | 6 | ☑ | Input | ☐ | |
| Pin | 80 | -30 | 0 Degrees | A0 | ☑ | 7 | ☑ | Input | ☐ | |
| Pin | 80 | -20 | 0 Degrees | VDD | ☑ | 8 | ☑ | Power | ☐ | |

8 of 12 Objects (0 Selected)

*Figure 6. Use the List panel to view or edit component pins*

## 15.2.4 Using supplier data

Creating a live link between an Altium Designer component and a Supplier Item has always been a simple process and importing information from an item is equally so. Simply select the parameter(s), data sheet link(s), pricing information, or stock information that you wish to import, in the detailed information section of the **Supplier Search** panel, then drag and drop:

- Anywhere within the main editing area of the Schematic Library Editor – ensuring the source Schematic Library document is the active document in the main design window and the recipient component is focused. You can also add information to a component by dropping onto the component's name (in the **Components** region of the **SCH Library** panel).
- Onto the required component record, in the **Table Browser** tab of the relevant Database Library file or SVN Database Library file – ensuring the DbLib or SVNDbLib file is the active document in the main design window.
- Onto the schematic symbol of a placed component – ensuring the source schematic document is the active document in the main design window.

For parameters, pricing and stock information, the import will proceed in accordance with any defined parameter import options.

The following sections take a look at the import of these different pieces of information.

### 15.2.4.1 Importing Parameters

Consider, as an example, the case where we have a component defined in a Schematic Library, which represents an N-channel MOSFET (a 2N7002). We have already linked a Supplier Item to this component and now need to import the parameters.

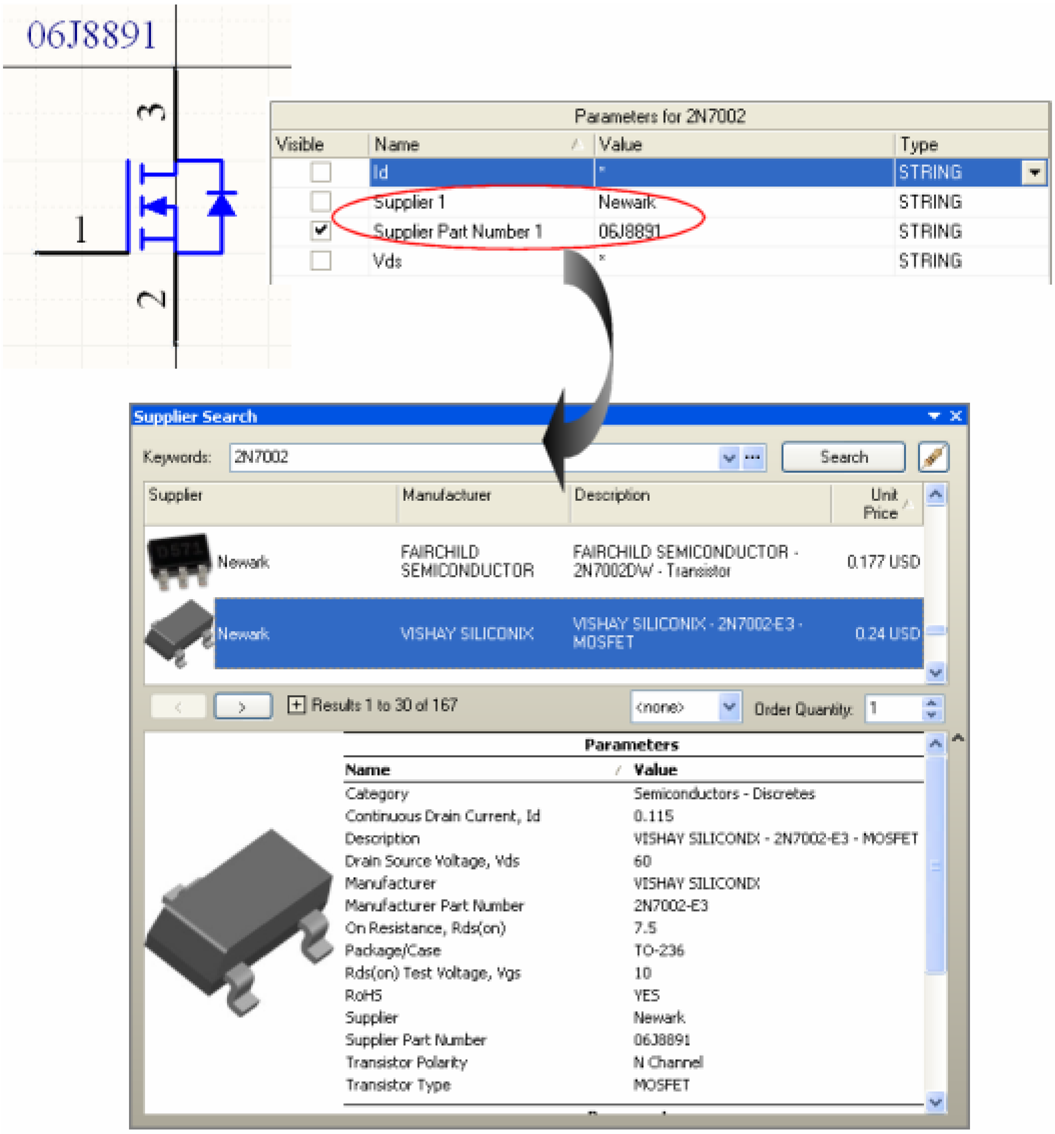


*Figure 7. The example library component and the linked Supplier Item.*

For the purposes of demonstrating the feature, a couple of parameters have already been defined for the component in the library – the Continuous Drain Current (Id) and the Drain Source Voltage (Vds). The Supplier Item names these parameters differently, so let's set up some parameter name mapping to ensure the data comes in to our existing parameters in these two cases.

It's a simple case of adding the two parameters – named as per the Supplier Data area – and using our naming for the **Imported Parameter Name** fields.

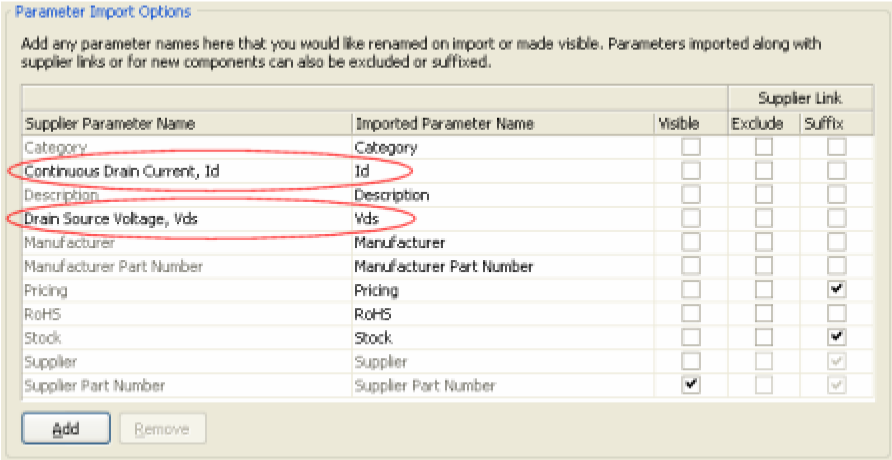



*Figure 8. Define parameter name mappings to ensure the existing parameters for the component get used, rather than adding new parameters for this same data.*

Now the options are set, we need to just select the parameters we want to import, in the detailed **Parameters** section for the Supplier Item. As this Supplier Item is already linked to the component, we don't need to import the *Supplier* or *Supplier Part Number* parameters. Let's import all others. Once selected, just drag the selection onto the component's name – in the **Components** region of the **SCH Library** panel – or within the main editing area for the component itself. That's all there is to it – a quick check in the properties dialog for the component shows that the parameters have been imported, and the two existing parameters have been used to receive data as required!

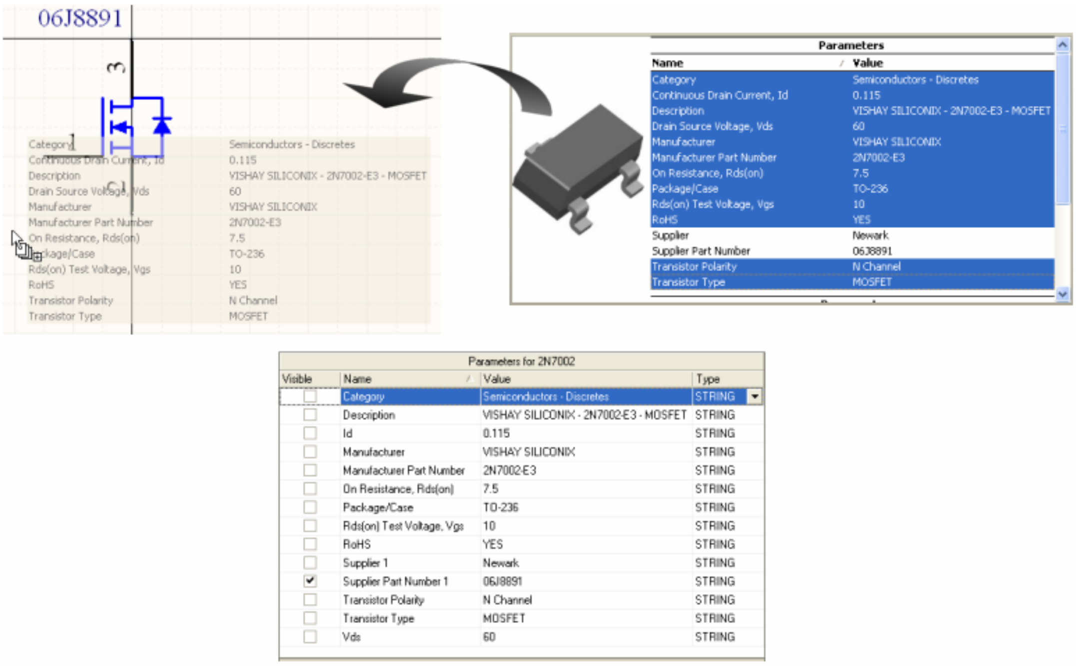



*Figure 9. Parameter import in action!*

**Notes:**

Parameters can be imported without setting up a prior live link to the Supplier Item, however importing the *Supplier* and *Supplier Part Number* parameters in this manual, drag and drop fashion, will not create a live link to the item.

Another way to initiate the import of parameters is to select the required parameter(s) in the detailed Parameters section for the Supplier Item, right click and choose the command from the context menu that appears. If a Schematic Library, Database Library or SVN Database Library is active, the command will appear in the format `Add Parameters To` *`ComponentName`*, where *`ComponentName`* is the currently focused component/component record in the library. If a Schematic document is active, the command will appear in the format `Add Parameters To Part`. Simply click on each placed component to which you want to add the parameters using the cross hair cursor. Right-click or press Esc to exit.

All importable data for a Supplier Item can be imported, along with a new supplier link to that item, in one step. Simply right-click on the Supplier Item entry in the Supplier Search panel and choose the Add Supplier Link And Parameters To command (see Adding a Supplier Link and Parameters Simultaneously).

#### 15.2.4.2 Importing Data Sheet Links

The process for importing a data sheet link from a Supplier Item is the same as that for importing parameters. The only difference is that there are no options to set up beforehand – just click and drag the data sheet link from the detailed **Documents** section for the Supplier Item onto the target component/component record in the active library (SchLib, DbLib, SVNDbLib), or component on the active schematic sheet as required.

The process will add a `ComponentLink` parameter pair, targeting the data sheet. By default, the same value will be entered for both the `ComponentLinknDescription` and `ComponentLinknURL` parameters – the URL for the data sheet. Access the link from the usual **References** sub-menu for the component.

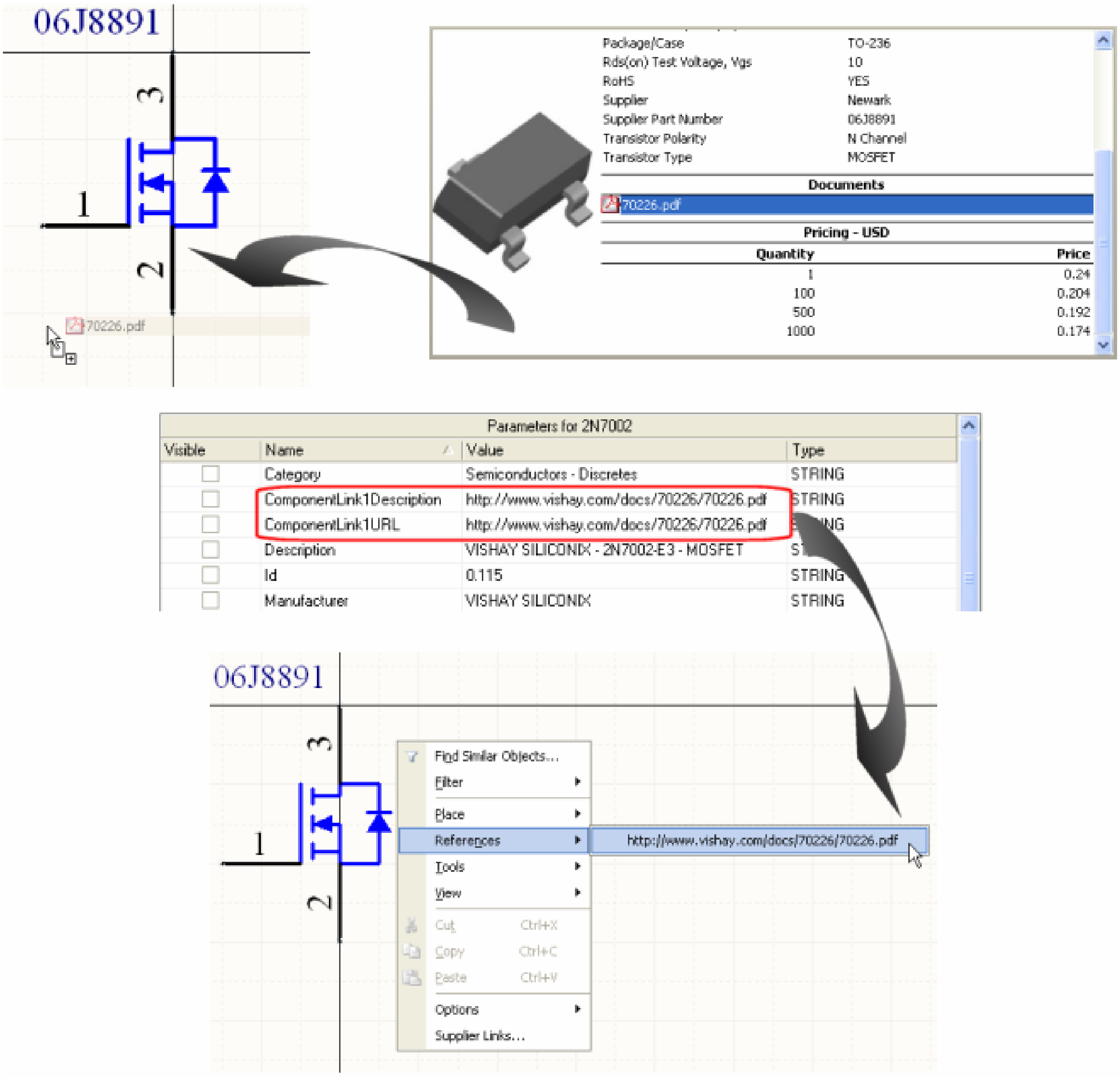


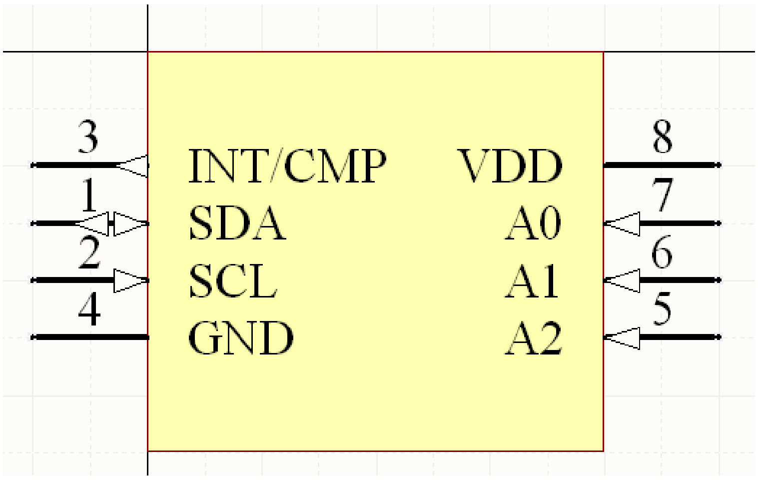

*Figure 10. Add a link to a data sheet for a Supplier Item with a drag of the mouse!*

It can be a good idea to change the value for the `ComponentLinknDescription` parameter to a shorter, more meaningful entry, for better display in the **References** sub-menu.

## 15.2.5 Exercise – Creating a new component symbol

1. We will now create a component – a serial temperature sensor. If it is not open, open the schematic library `\Program Files\Altium Designer Summer 09\Training\PCB Training\Temperature Sensor\Libraries\Temperature Sensor.SchLib`
2. Before creating the component, make sure that this library is in the Temperature Sensor project. If it is not already open, re-open the project created during *Module 4 - Schematic Capture* training session, `\Program Files\Altium Designer Summer 09\Examples\Training\PCB Training\Temperature Sensor\Temperature Sensor.PrjPcb`.
3. To add the library to the project, click and hold on the `Temperature Sensor.SchLib` in the **Projects** panel, then drag and drop it onto the project filename, `Temperature Sensor.PrjPcb`. It will disappear from the Free Documents, instead appearing under the *Libraries* folder icon in the project structure.
4. Right-click on the project name and select **Save Project**.

5. To create a new component Select **Tools » New Component** to create a new component.
6. Enter `TCN75` in the *New Component Name* dialog.
7. When the blank sheet appears, zoom in (PAGE UP) until you can see the grid. Components generally have the top left of the component body located at co-ordinates `0,0` (indicated by the two darker grid lines).
8. Check that the Snap Grid and Visible Grid are set to 10 (**Tools » Document Options**).




**Note:** If the yellow rectangle covers the pin names, use the **Edit » Move » Send to Back** command to move it behind the pins.

*Figure 11. Microchip TCN75 serial temperature sensor*

9. Create the graphical representation for the component as shown in Figure 11. The component body is a **Rectangle**, placed at the origin in the center of the sheet. The origin is indicated by the two darker lines that form a crosshair, zoom in/out to show the crosshair and the gridlines. Start placing the rectangle at the origin, the body is 80 units wide by 70 units high, you can use the coordinates shown on the **Status bar** to guide you.
10. Place the pins for the part. It is important to orient pins so that the 'hot' end is away from the component body. When placing pins, the cursor will be on the 'hot' end of the pin. Press SPACEBAR to rotate the pin or **X** or **Y** to flip it.
11. **Before** placing the first pin, press TAB to edit the pin properties. The *Pin Properties* dialog will open. For each pin, set the Pin Name, Pin Number, Electrical Type as per the table, and set the Pin Length to 20.

| Pin Number | Pin Name | Electrical Type |
|---|---|---|
| 1 | SDA | IO |
| 2 | SCL | Input |
| 3 | INT/CMP | Output |
| 4 | GND | Power |
| 5 | A2 | Input |
| 6 | A1 | Input |
| 7 | A0 | Input |
| 8 | VDD | Power |

**Note:** you can use the auto-increment/decrement feature when placing pins 5, 6 and 7.

**Note:** As well as using the pin properties dialog to edit the pin names, you can also use the List panel to edit the pin properties after they have been placed, as shown in Figure 6.

12. When you have completed drawing the component, set the
    - Designator to `U?`
    - Comment to TCN75

- Description to `Serial temperature sensor`

13. Save the library. You can not place a new component from a library until it has been saved.

**Note:** At the moment this component is really just a symbol, it has no models or parameters – as a minimum it needs a footprint. You will create the footprint for this component in the next section and then come back to the schematic library editor to link it to the symbol.

### 15.2.6 Exercise – Completing the Sensor schematic

At this stage in the training, the structure of your Temperature Sensor project should look like Figure 12. However, the `Sensor.SchDoc` is incomplete. To complete it:

1. Add a new schematic to the project and call it **Sensor.schdoc** if you haven't done so already.
2. Add the Ports, Power Ports and Wiring to finish the schematic, as show in Figure 13.
3. Since the project structure has been modified, you should **Compile** the project and resolve any errors.

**Hint:** use the **Design » Synchronize Sheet Entries and Ports** command to resolve sheet entry-to-port mismatches.

4. Save the Sensor schematic sheet.

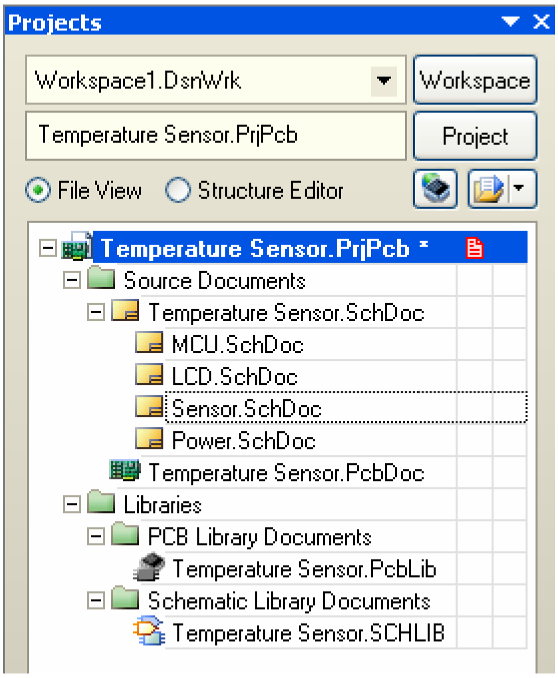



*Figure 12. Project structure after completing the Sensor schematic.*

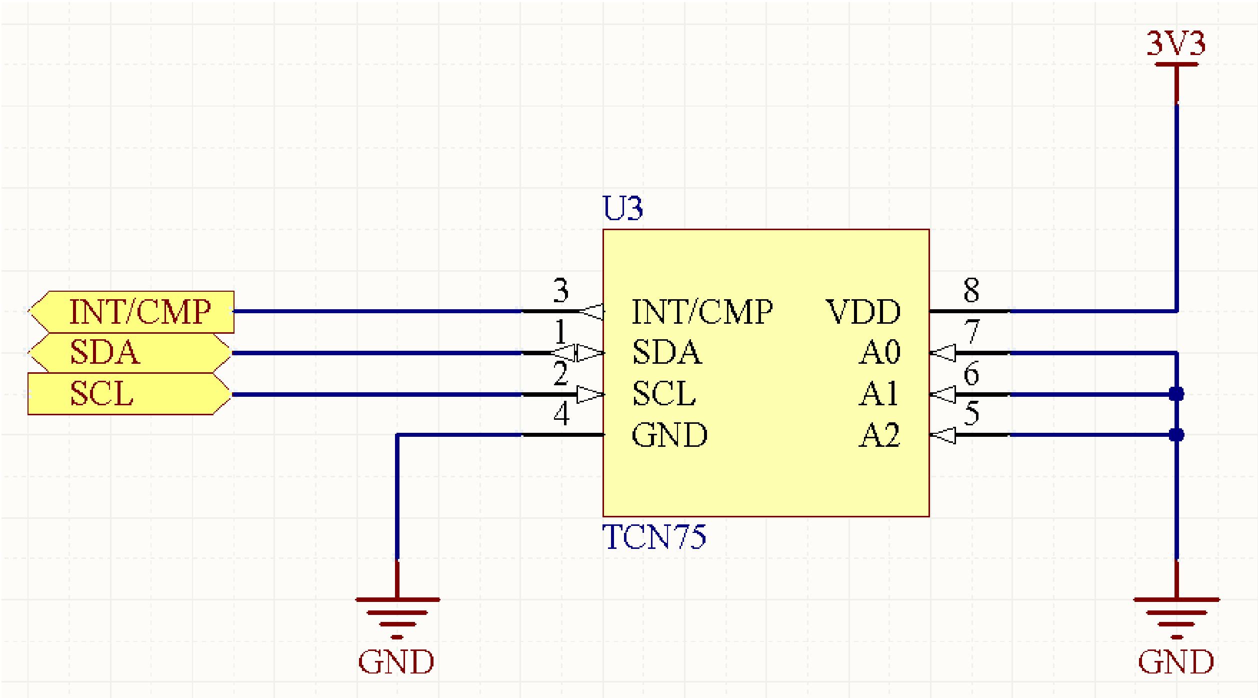
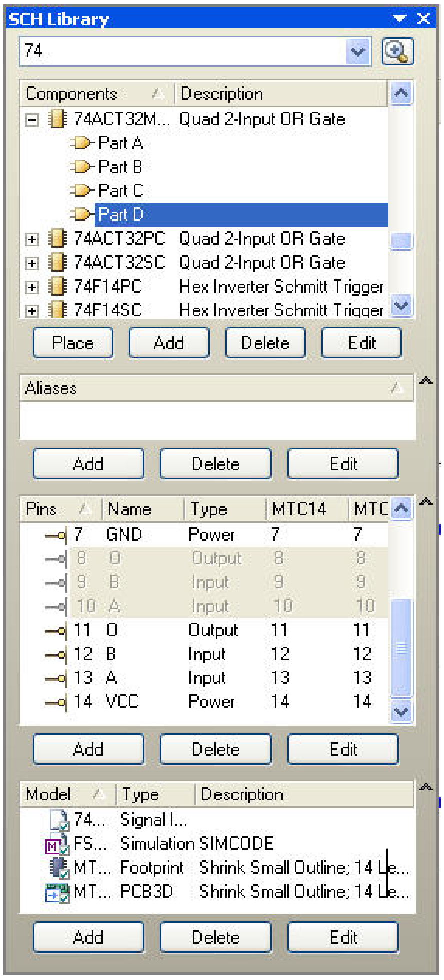


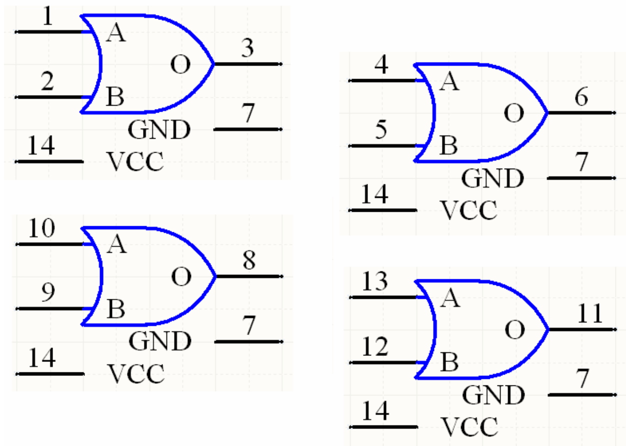

*Figure 13. Wired sensor schematic (Sensor.SchDoc).*

## 15.2.7 Creating a multi-part component

To create a multi-part component:

- First create one part, select all, then copy the part to the clipboard using the **Edit » Copy** menu command.
- Select **Tools » New Part** to add a new part sheet under the same component name.
- Paste the part onto the sheet and update the pin information. Note that the Part field in the panel will now show 2/2, meaning the second of two parts.
- Finally, add hidden pins (typically power pins) to any of the parts. Edit them, enable the **Hide** attribute and set their **Part Number** to zero. If they are to automatically connect to a net, enter the net name in the **Connect To** field.




*The 4 parts of a multipart 74ACT32 component. Note the power pins on each part (hidden pins have been displayed), these exist once, on part zero (a non-visible part). Hidden pins must have the net that they connect to specified in the Pin Properties dialog.*

*Figure 14. Creating a multi-part component.*

TOP SECRET

# Module 16: PCB Library Editor

## Module 16: PCB Library Editor






Module Seq = 16

# 16.1 PCB Library Editor

The PCB Library Editor is used to create and modify PCB component footprints and manage PCB component libraries. The PCB Library Editor also includes a **Component Wizard** that you can guide through the creation of most common PCB component types.

## 16.1.1 The PCB Library workspace

An existing PCB library (*.PcbLib) can be opened using the **File » Open** command, displaying the first footprint in that library, and a list of all footprints in the library in the **PCB Library** panel. Click on the required component in the **Components** list.

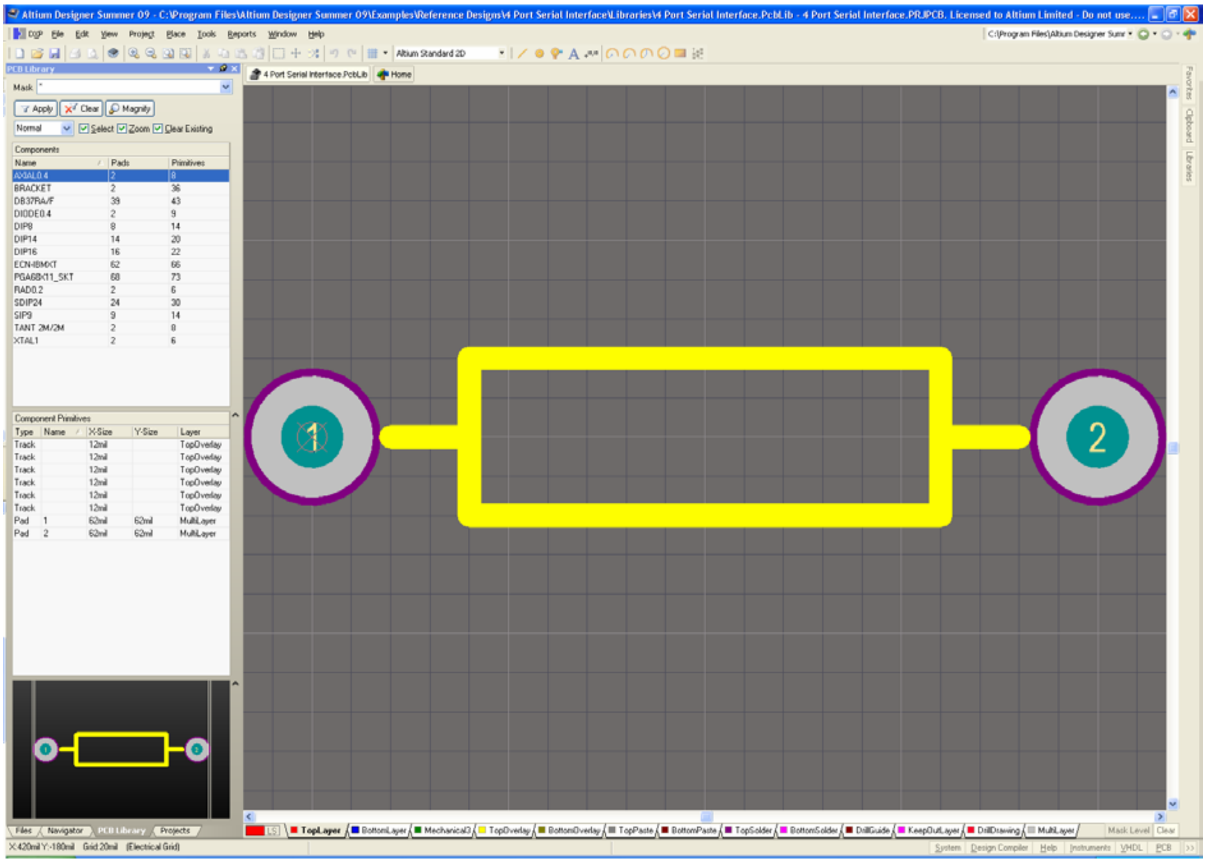


*Figure 1. PCB Library Editor Workspace*

The view commands, primitive objects, layers, selection and focus, grids and general editing functions are all identical to the PCB Editor.

Settings in the **Preferences** dialog and **Board Options** dialog also apply in the PCB Library Editor.

## 16.1.2 PCB Library Editor Panel

The **PCB Library** panel of the PCB Library Editor panel provides a number of features for working with PCB components. These include:

- The **Components** section of the panel lists all the components in the active library.
- Right-click in the **Components** section to display menu options to Create New Components, edit Component Properties, Copy or Paste selected components, or update the component footprints on open PCBs.
- Note that the copy/paste commands in the right-click menu can be used with multiple footprints selected, and support:
  - copying and pasting within a library,
  - copying and pasting from a PCB into a library,
  - copying and pasting between PCB libraries.
- The **Components Primitives** section lists the primitives that belong to the currently selected component. Click on a primitive in the list to highlight it in the design window.
- The way that the chosen primitive is highlighted depends on the options at the top of the panel:
  - Enabling **Mask** will result in only the primitive(s) you click on remaining at normal visibility, all others will be faded. Click the Clear button down the bottom right of the Workspace to remove the filter and restore the display.
  - Enabling **Select** will result in the primitive(s) you click on being selected, ideal if you need to edit them.
- Right-click in the **Component Primitives** section to control which types of primitives are listed in this section.

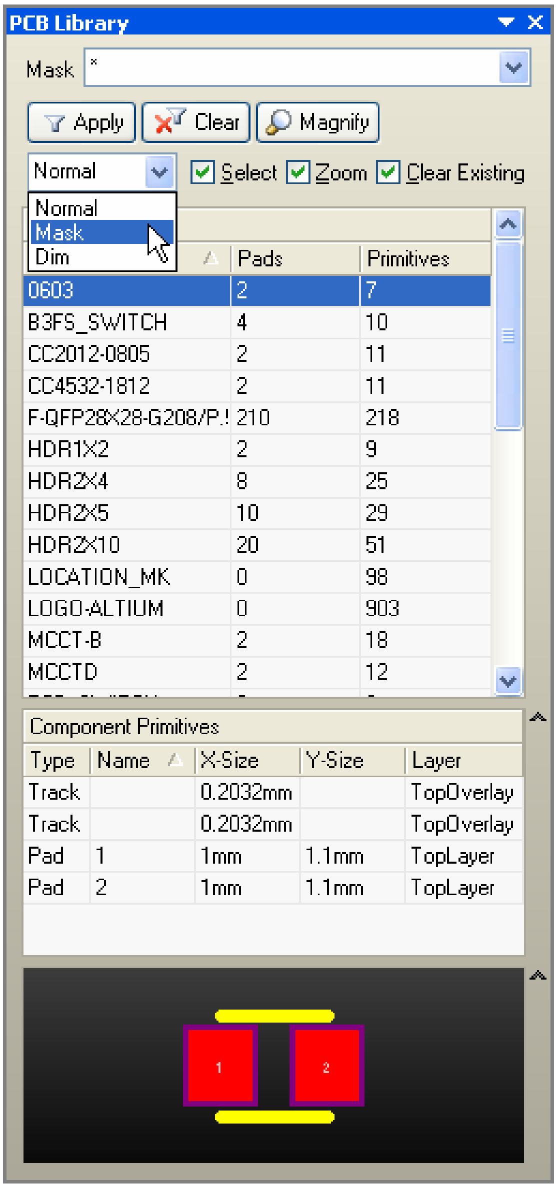



*Figure 2. PCB Library panel*

## 16.1.3 Creating a component using the Component Wizard

The PCB Library Editor includes a **Component Wizard**. This Wizard allows you to select from various package types, fill in appropriate information and it will then build the component footprint for you.

To launch the Component Wizard, right-click on the Components section of the PCB Library Editor panel and select **Component Wizard**, or select the **Tools » New Blank Component**.

## 16.1.4 Creating a component using IPC footprint wizard

The IPC Footprint Wizard creates IPC-compliant component footprints. Rather than working from footprint dimensions, the IPC Footprint Wizard uses dimensional information from the component itself, in accordance with the algorithms released by the IPC.

Available through **Tools » IPC Footprint Wizard** menu when a PCB library is the active document, the new IPC Footprint Wizard creates IPC-compliant component footprints.

In accordance with the IPC standard it also supports three footprint variants, tailored to suit the board density. The wizard supports BGA, BQFP, CFP, CHIP, CQFP, DPAK, LCC, MELF, MOLDED, PLCC, PQFP, QFN, QFN-2ROW, SOIC, SOJ, SOP, SOT143/343, SOT223, SOT23, SOT89, and WIRE WOUND footprints.

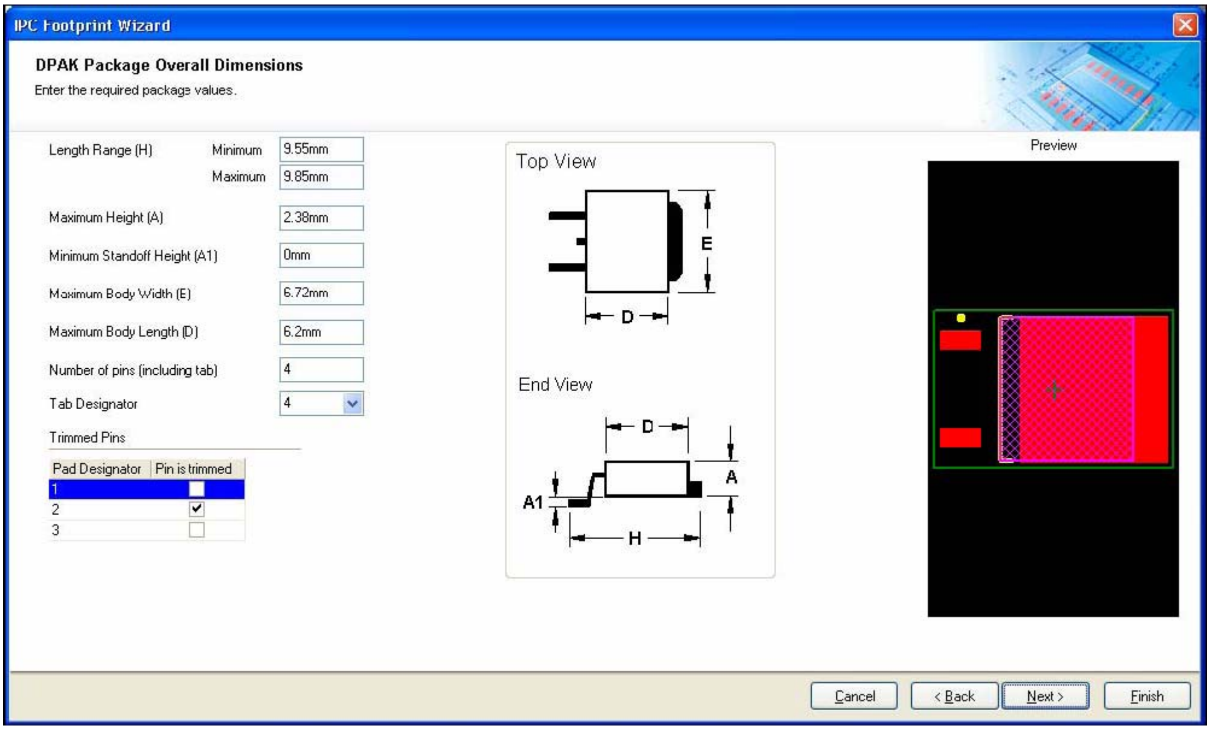



*Figure 3. One of the new supported packages in the IPC footprint Wizard is the DPAK (Transistor Outline).*

Some of the **IPC Footprint Wizard** features include:

- Overall packaging dimensions, pin information, heel spacing, solder fillets and tolerances can be entered and immediately viewed.
- Mechanical dimensions such as Courtyard, Assembly, and Component (3D) Body Information can be entered.
- Wizard is re-entrant and allows reviewing and making adjustments easy. Previews of the footprint are shown at every stage.
- The finish button can be pressed at any stage to generate the currently previewed footprint.

## 16.1.5 Creating a component using IPC Footprints batch generator

Available through **Tools » IPC Footprint Batch Generator** menu when a PCB library is the active document, the **IPC Footprint Batch Generator** makes it possible to quickly generate multiple footprints as well as multiple density levels for a single component from a package input file that contains datasheet package information.

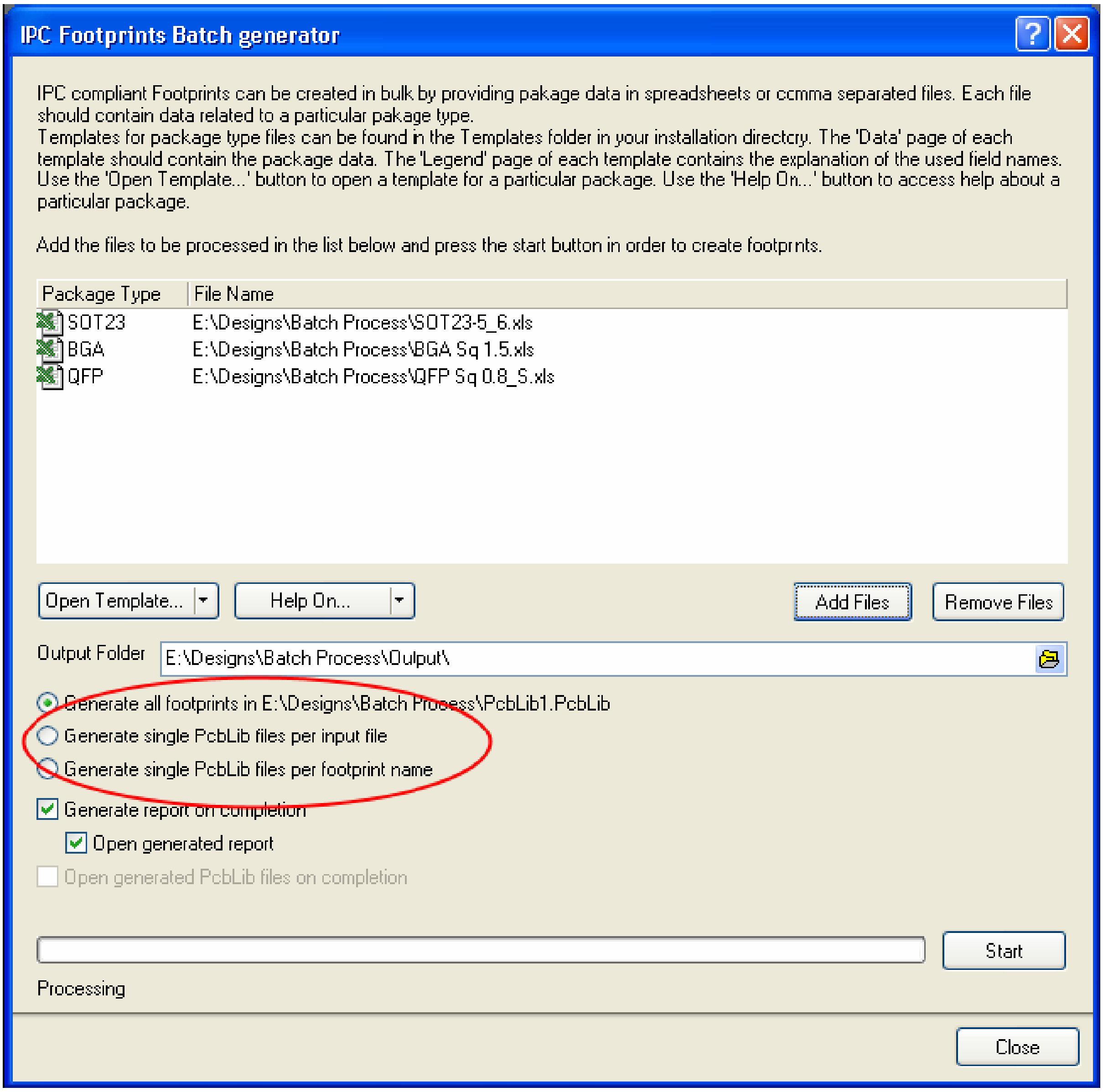


*Figure 4. The IPC Footprint Batch Generator has options that either create all the footprints into the current open .PcbLib, or generate a single .PcbLib file based on either an input file or footprint name.*

Support for the **IPC Footprints Generator** includes:

- Package type blank template files can be opened within the dialog and are provided in the `\Templates` folder of the installation directory. Help for any of the package type templates is also available.
- Package input files need only to contain the information for one or more footprints of a single package type, and can be either an Excel or comma-delimited (CSV) file format.

## 16.1.6 Manually creating a component

Components are created in the PCB Library Editor using the same set of primitive objects available in the PCB Editor. In addition to PCB components, corner markers, phototool targets, mechanical definitions, etc. can be saved as components.

The typical sequence for manually creating a component footprint is:

- Open the desired library in the PCB Library Editor.

- Select the **Tools » New Blank Component** menu command. You will be presented with an empty component footprint workspace, called `PCBComponent_1`. Rename the component by double-clicking on the name in the Components list, select **Component Properties** and enter a new name in the **Component Properties** dialog. Component names can be up to 255 characters.
- Use tracks or other primitive objects to place the component outline on the Silkscreen layer.
- Place the pads according to the component requirements. Prior to placing the first pad, press the TAB key to define all the pad properties. Make sure you set the designator property correctly. Typically, the first pad you place is pin 1, so the set designator to '1' for the first pad. The designator automatically increments.

> **Note:** The 0,0 coordinate is the point where the component is 'held' during placement. Always confirm that it is set to a suitable location. Select **Edit » Set Reference** to change the location.

## 16.1.7 Copying a component

There is often the requirement to copy components, either from one library to another or within the same library. For a single component you can achieve this using the **Edit » Copy Component** command. This command copies the current component, ready for pasting back into a PCB library.

You can also copy/paste multiple component footprints using the commands in the **PCB Library** panel's right-click menu. Select the required component footprints using CTRL+click in the list, then right click and choose **Copy**, then right-click again and choose **Paste X Components** (where X is the number of component footprints you selected).

## 16.1.8 Special strings in the Library Editor

There are two special strings that are active in the Library Editor. These are provided to allow you to control the positioning of the designator (`.Designator`) and the comment (`.Comment`).

Place these in the PCB Library Editor workspace at the location relative to the component where you would like the designator or comment to be placed.

When you use these, you can hide the default designator and comment that are added when the component is placed in the PCB file.

## 16.1.9 Component Rule Check

The **Reports » Component Rule Check** command allows you to check either the current component or the whole library for any of the objects selected in the *Component Rule Check* dialog.

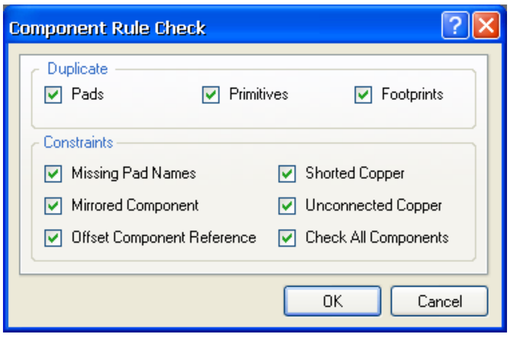
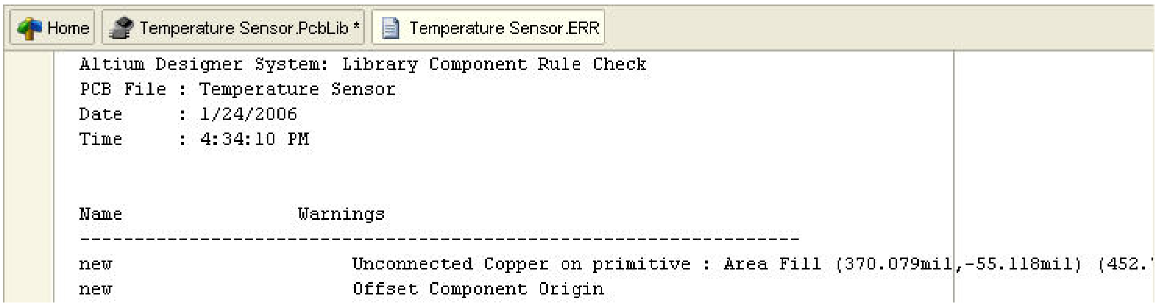



*Figure 5. Component Rule Check dialog*

The results of the component rule check are displayed in a text document.


Home | Temperature Sensor.PcbLib * | Temperature Sensor.ERR

```
Altium Designer System: Library Component Rule Check
PCB File : Temperature Sensor
Date     : 1/24/2006
Time     : 4:34:10 PM


Name                Warnings
-----------------------------------------------------------------
new                      Unconnected Copper on primitive : Area Fill (370.079mil,-55.118mil) (452.
new                      Offset Component Origin
```

*Figure 6. Library Component Rule Check report*

## 16.1.10 Exercise – Creating the component footprint

In this exercise, we will create a new component footprint called `SOIC8`, to use with the Temperature Sensor component you just created, the TCN75.

1. The first thing we need is a footprint library. For this training session we will create the footprint in a project library. If it is not open, open the footprint library `\Program Files\Altium Designer Summer 09\Examples\Training\PCB Training\Temperature Sensor\Libraries\Temperature Sensor.PcbLib`
2. Before creating the footprint, we will make this footprint library part of the Temperature Sensor project. If it is not already open, re-open the project created during the *Environment and Editor Basics* training session, `\Program Files\Altium Designer Summer 09\Examples\Training\PCB Training\Temperature Sensor\Temperature Sensor.PrjPcb.`
3. To add the library to the project, click and hold on the `Temperature Sensor.PcbLib` in the **Projects** panel, then drag and drop it onto the project filename, `Temperature Sensor.PrjPcb`. It will disappear from the Free Documents, instead appearing under the *Libraries* folder icon in the project structure.
4. Right-click on the project name and select **Save Project**.
5. To create the new SOIC8 footprint we will use the IPC Footprint Wizard, select **Tools » IPC Footprint Wizard** to run the Wizard.
6. In the list of pattern types, select Small Outline Integrated Package (SOIC), as shown below in Figure 7.

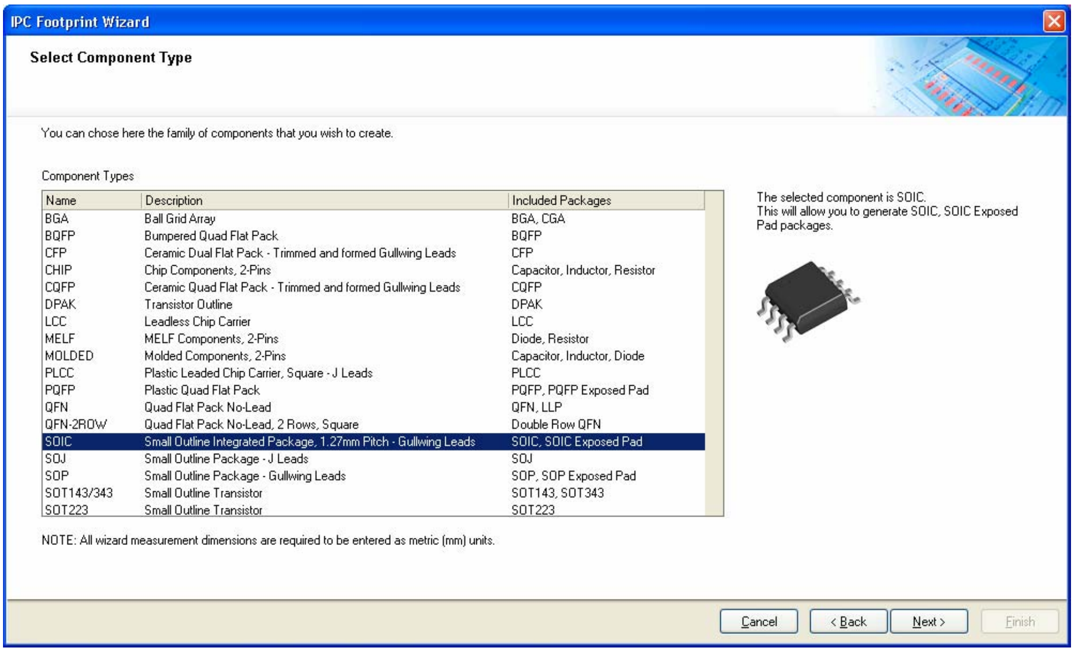



*Figure 7. Choose the footprint type in the IPC Footprint Wizard*

7. Refer to information in Figure 8 for dimensions. Note that it will be created with 8 round-ended pads at this point if we finish the wizard.

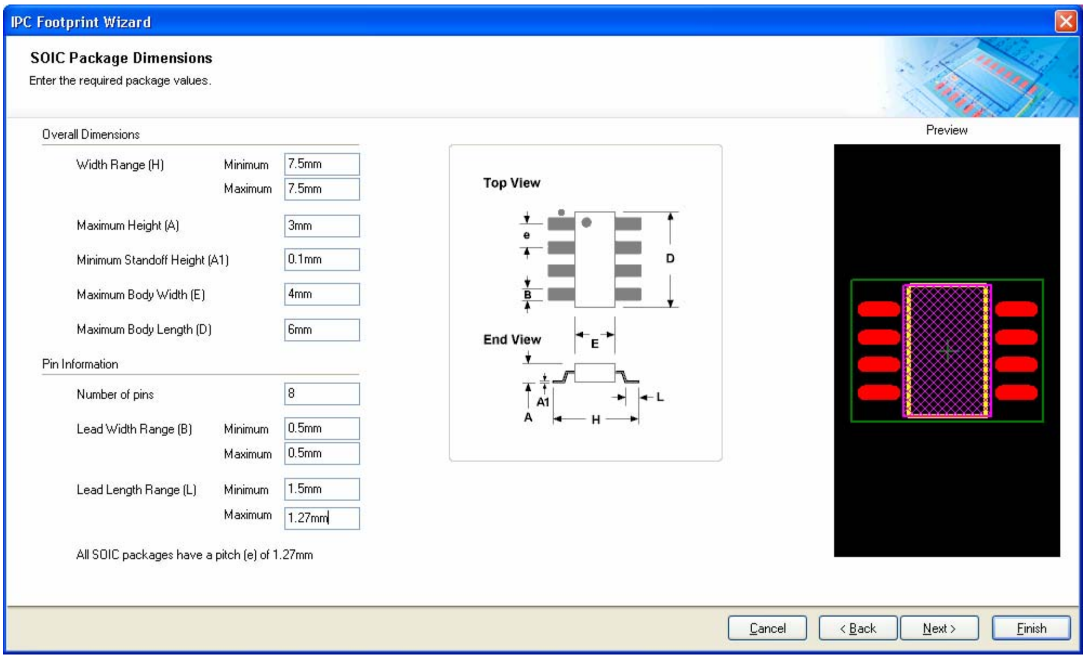



*Figure 8.SOIC package dimensions*

8. Step through the Thermal Pad dimensions, Heel spacing, Solder fillets, Component Tolerances and IPC Tolerances.
9. In the SOIC Footprint Dimensions page change the pad shape to Rectangular.
10. In the SOIC Silkscreen Dimensions page change the silkscreen line width to 0.1mm.
11. Step through the Courtyard, assembly and component board information page.
12. In the footprint Description page change the name to SOIC8 and leave the description as is.

13. In the Footprint Destination page, set to current PcbLib file and set it to *\Program Files\Altium Designer Summer 09\Examples\Training\PCB Training\Temperature Sensor\Libraries\Temperature Sensor.PcbLib*

14. Click on finish to end the wizard and create the new component as per Figure 9.

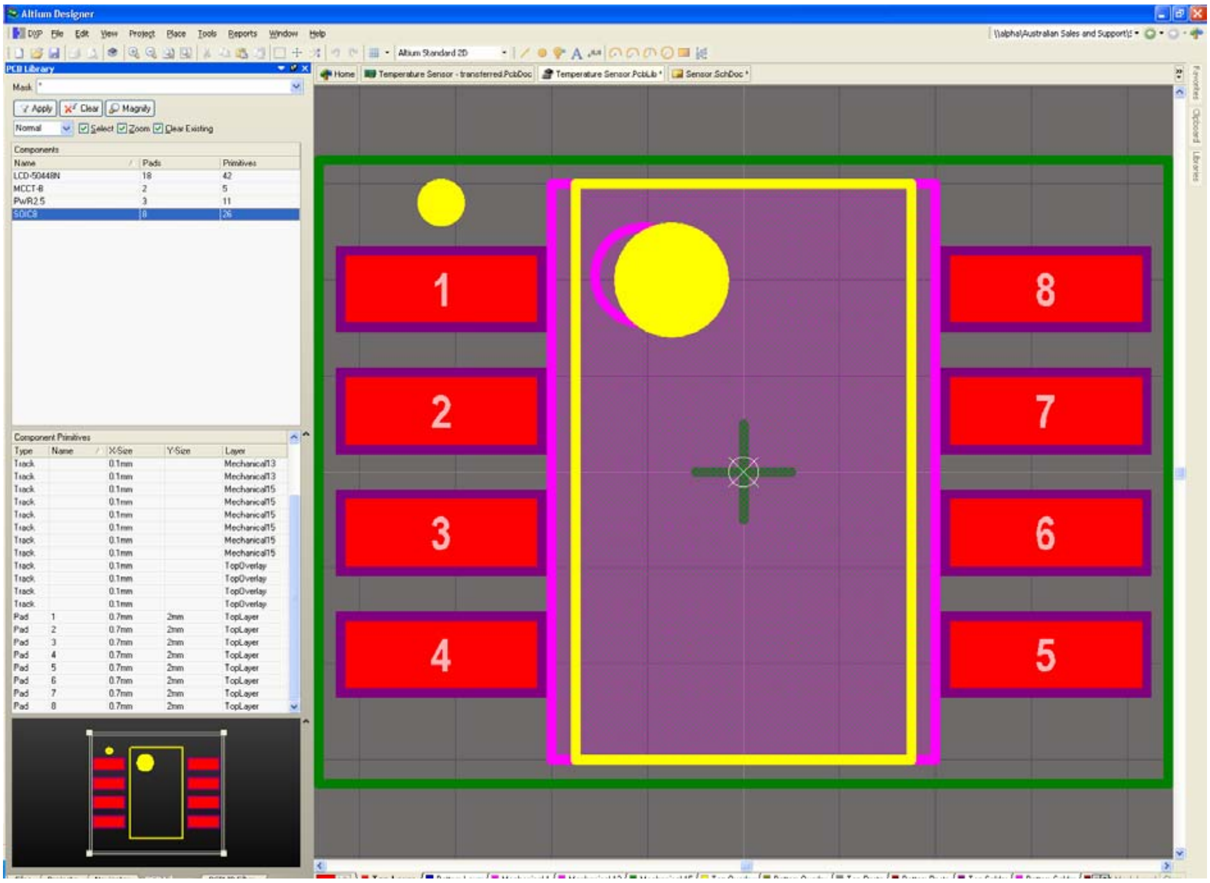


*Figure 9. Finished footprint created in the temperature sensor.pcblib.*

15. Save the library, and the project.

## 16.1.11 Browsing footprint libraries

PCB libraries are accessed through the same panel as schematic libraries – the **Libraries** panel.

- Enable the footprint display mode by clicking the ··· button at the top of the panel and enabling the **Footprints** checkbox.
- Select a library name in the drop down list to choose it and display all the footprints in that library. This can be either an integrated library or a footprint library.
- Footprint libraries that are in the active project, currently installed or found down the search path are available in the panel.
- Click the **Libraries** button at the top panel to install a footprint library.
- Library search paths are defined in the **Search Path** tab of the *Options for Project* dialog.
- To Search for a footprint, first enable the **Footprints** mode, then click the **Search** button.
- Click on a footprint name in the list to display that footprint in the MiniViewer.
- Click on the **Place** button to place the chosen footprint in the workspace, or double-click on the footprint name.

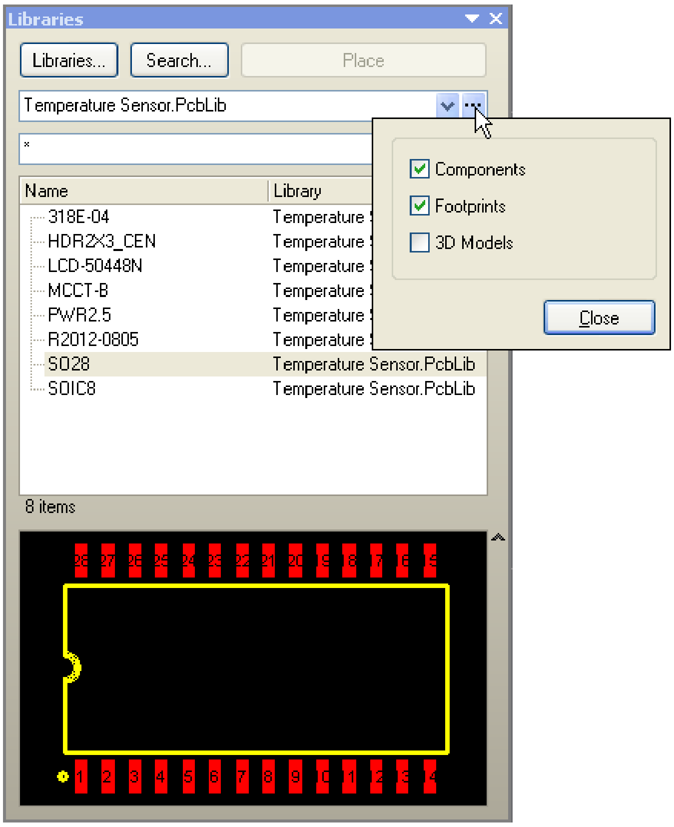



*Figure 10. Libraries panel*

## 16.1.12 Creating footprints with an irregular pad shape

There will be situations where you need to create a footprint with pads that have an irregular shape. This can be done using any of the design objects available in the library editor, but there is an important factor that you must keep in mind.

The software automatically creates solder and paste masks based on the shape of pad objects, if you use pad objects to build up an irregular shape then the matching irregular mask shape will be generated correctly. If you build the irregular shape from other objects, such as lines (tracks), fills, regions or arcs, then you will also need to define any required solder or paste masks by placing suitably enlarged or contracted objects on the solder mask and paste mask layers.

Figure 11 and Figure 12 shows versions of an SOT-89 footprint created by different designers. Figure 11 uses 2 pads to create the large irregular shaped pad in the center, Figure 12 uses a pad and a line (track). Figure 12 would need to have the solder and paste masks defined manually.

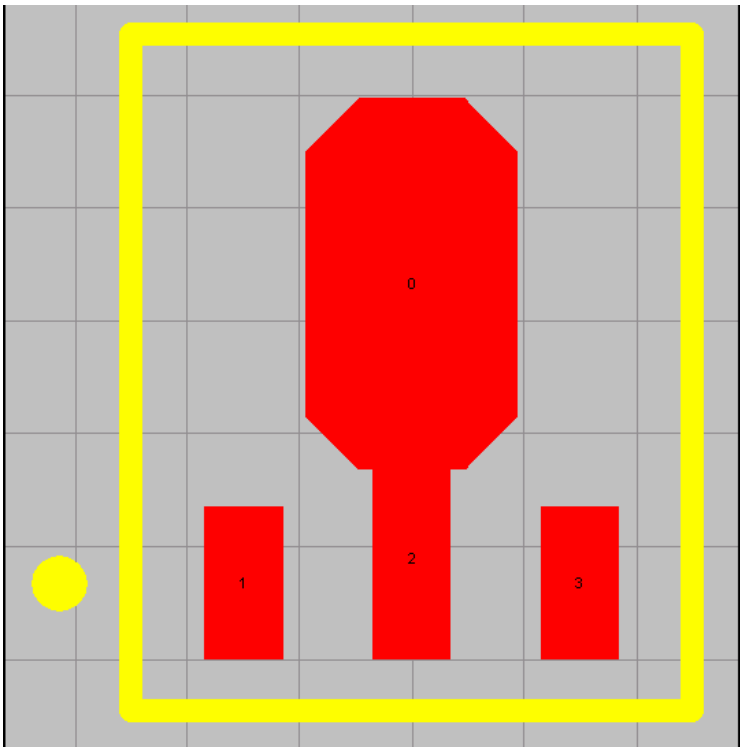



*Figure 11. Using pads to create irregular shapes*

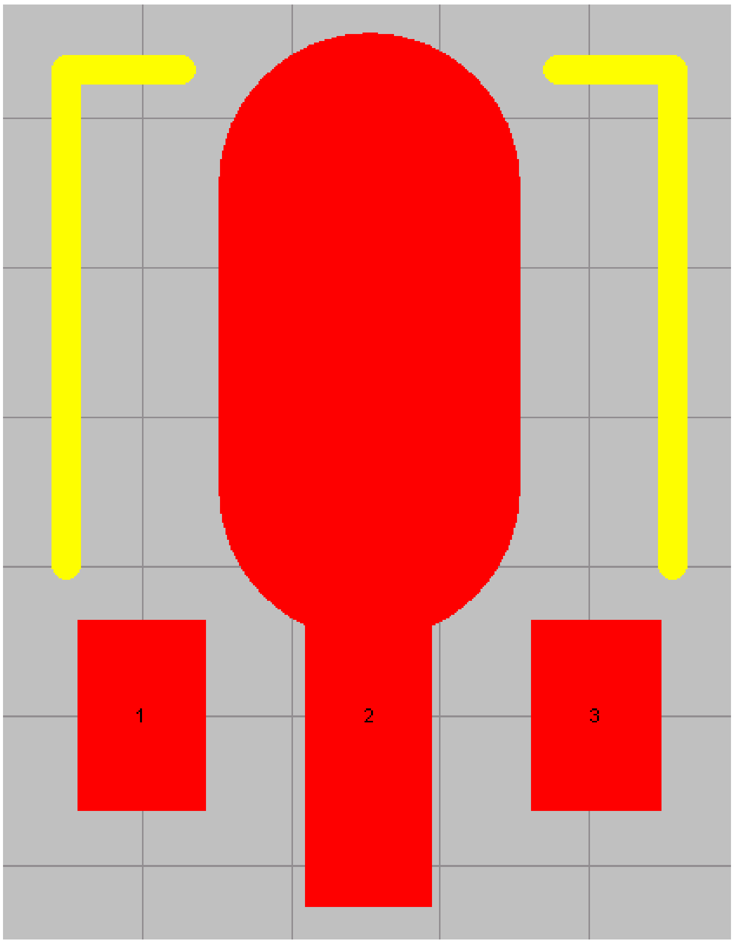



*Figure 12. Using a track to create irregular shapes*

## 16.1.13 Managing components that include routing primitives in their footprint

When you transfer a design, the footprint specified in each component is extracted from the available libraries and placed on the board. Then each pad in the footprint has its net property set to the name of the net connected to that component pin in the schematic. If the footprint includes copper primitives touching the pads, these primitives will not be assigned the net name automatically, and will create a design rule violation. In this case, you will need to perform an update process to assign the net name.

The PCB editor includes a comprehensive net management tool, to launch it select **Design » Netlist » Configure Physical Nets** from the PCB editor menus. Figure 13 shows the **Configure Physical Nets** dialog being used to update the extra primitives detected in the switch footprint shown in Figure 14. Click the **Menu** button for a menu of options, and click the **New Net Name** region to select the net to assign to the unassigned primitives.

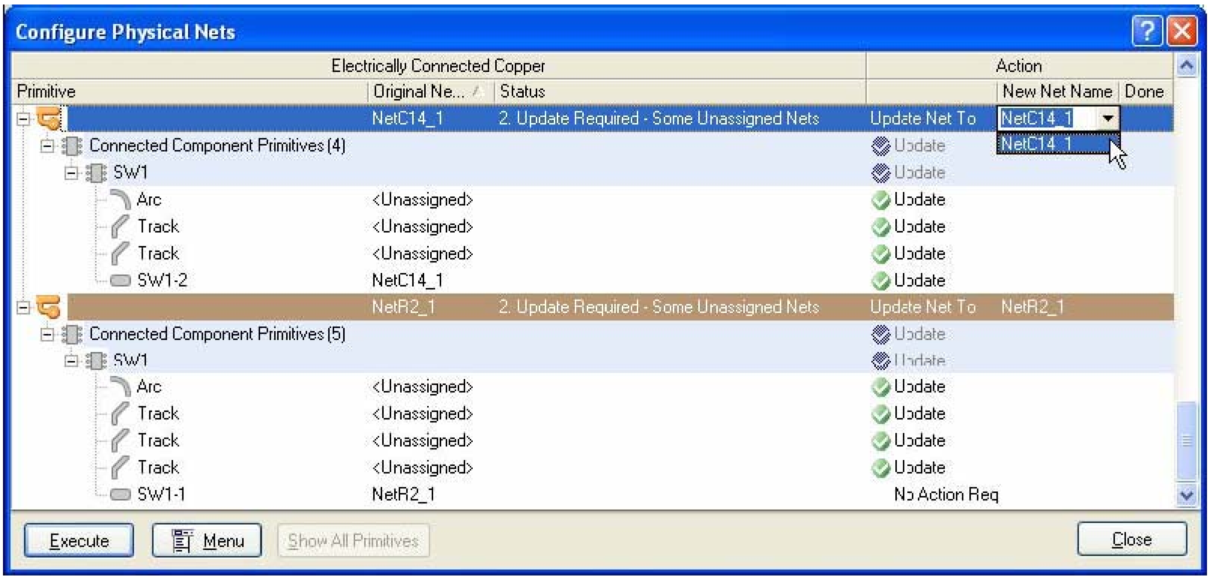



*Figure 13. Update the net name on unnamed footprint primitives in the Configure Physical Nets dialog.*

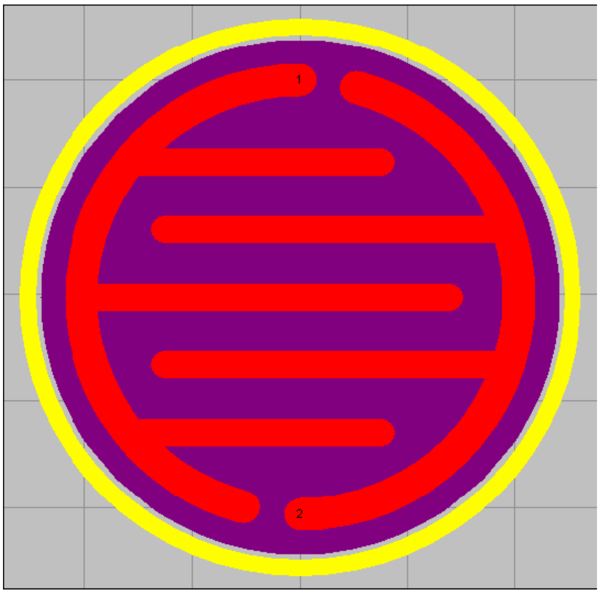


*Figure 14. Printed push button footprint, designed by placing pads, lines and arcs.*

## 16.1.14 Footprints with multiple pads connected to the same pin

The footprint shown in Figure 15, a TO-3 transistor, has multiple pads that connected to the same logical schematic component pin. For this component both of the 2 mounting hole pads have the same designator of '3'.

When the **Design » Update PCB** command is used in the Schematic Editor to transfer design information to the PCB, the resulting synchronization will show the connection lines going to both pads in the PCB Editor, i.e. they are on the same net, as shown in Figure 15. Both of these can be routed.

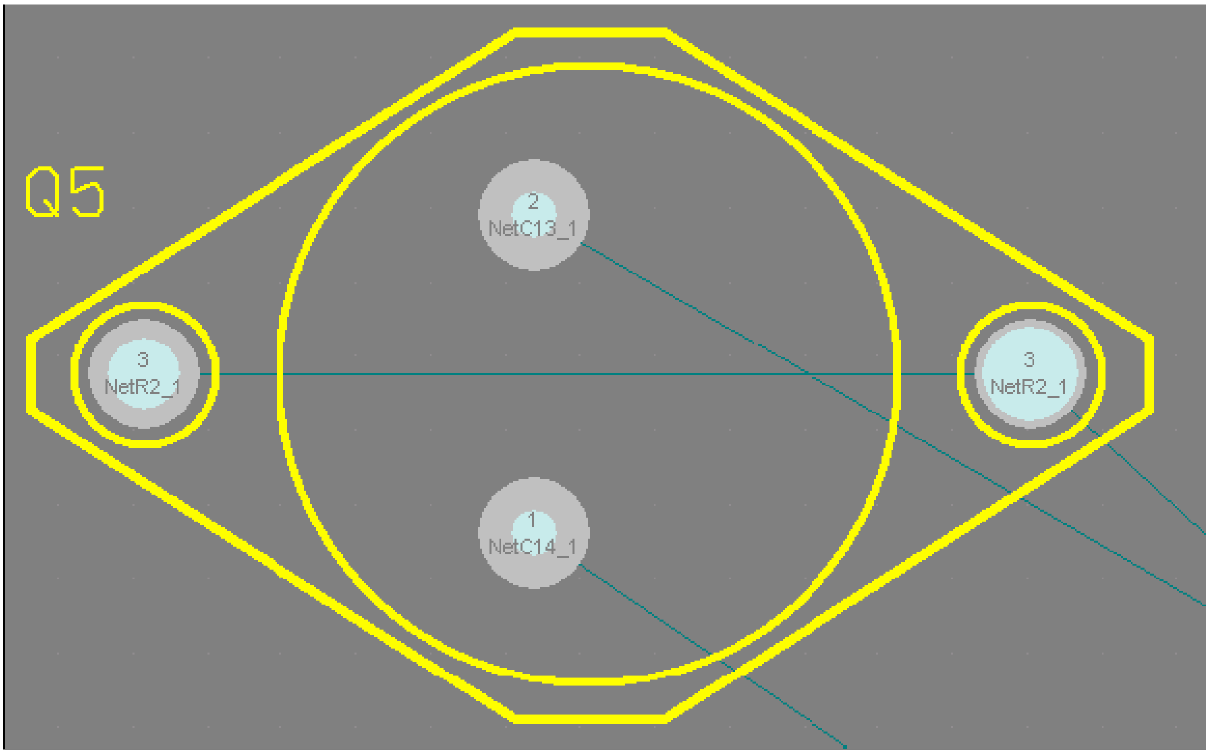



*Figure 15. TO-3 footprint showing two pads with a designator of 3, on the same net*

## 16.1.15 Handling special solder mask requirements

The footprint shown in Figure 14 is the contact set for a push button switch, which is implemented directly in the copper on the surface layer of the PCB.

A rubber switchpad overlay is placed on top of the PCB, with a small captive carbon button that contacts both sets of fingers in the footprint when the button is pressed, creating connectivity.

For this to happen both sets of fingers must not be covered by the solder mask. The circular solder mask opening has been achieved by placing an arc whose width is equal to or greater than the arc radius, resulting in the solid circle shown behind the 2 sets of fingers.

Each set of copper fingers has been defined by an arc, horizontal lines, and a pad. The pads are required to define the points of connectivity.

Note that manually placed solder mask definitions are automatically transferred when the component is placed on the bottom of the board.

## 16.1.16 Other footprint attributes

Solder and paste masks are created automatically at each pad site on the Solder Mask and Paste Mask layers respectively. The shape that is created on the mask layer is the pad shape, expanded or contracted by the amount specified by the Solder Mask and Paste Mask design rules set in the PCB Editor, or specified in the *Pad* dialog.

When you edit a pad you will see the settings for the solder mask and paste mask expansions. While these settings are included to give you localized control of the expansion requirements of a pad, you will not normally need them. Generally it is easier to control the paste mask and solder mask requirements by defining the appropriate design rules in the PCB editor. Using rules you define one rule to set the expansion for all components on the board, then if required you can add other rules that target any specific situations – such as all instances of a specific footprint type used on the board, or a specific pad on a specific component, and so on.

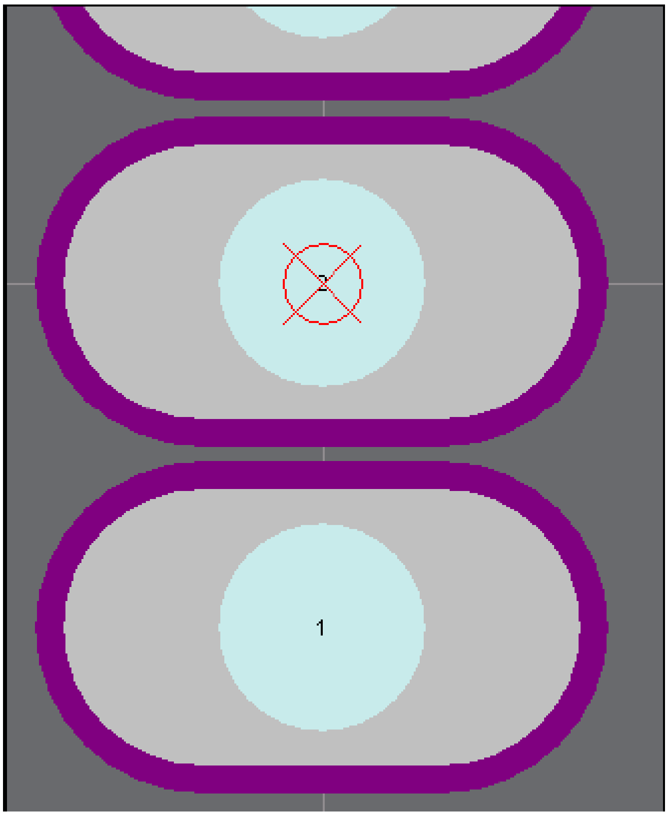



*Figure 16. Pads with the solder mask displayed*

# 16.2 3D dimensional component detail

Given the density and complexity of today's electronic products, today's PCB designer must consider more than the horizontal component clearance requirements, they must also consider height restrictions and component-under-component placement options. There is also the need to transfer the final PCB to a mechanical CAD tool, where a virtual product assembly can verify the complete packaging of the product being developed. Altium Designer includes a number of features to cater for these different situations.

## 16.2.1 Adding height to your PCB footprint

At the simplest level you can add a height attribute to your footprint. To do this double-click on the footprint in the Components list in the **PCB Library** panel to display the **PCB Library Components** dialog and enter the recommended height for the component in the **Height** field.

Height Constraint design rules can then be defined during board design (select **Design » Rules** in the PCB Editor), typically testing for maximum component height in a class of components, or within a room definition.

## 16.2.2 Adding a 3D body to a footprint

For more detailed height requirements, you can add a 3D body object to the footprint. A 3D Body is a polygonal shaped object that can be added to a footprint on any enabled mechanical layer. One or more 3D body objects can be added to define the physical size and shape of a component in both the horizontal and vertical planes. The 3D body objects can then be used by the component clearance design rule check to test for component collisions, and they can also be used by the 3D visualization engine when it renders a 3D representation of the board (**View » Board in 3D** in the PCB Editor).

## 16.2.3 Manually placing 3D body objects

3D body objects can be placed manually in the PCB Library Editor (**Place » 3D body**). They can also be added automatically to footprints in the PCB Library Editor (and to placed footprints in the PCB Editor) using the **Component Body Manager** dialog (**Tools » Manage 3D Bodies for Components on Board…**). Note that can only be placed on a mechanical layer, the current mechanical layer is used if one is active when the command is selected, otherwise the 1$^{st}$ available mechanical layer will be selected in the *3D Body* dialog. The steps involved in setting this up for an example footprint like a DIP14 would include:

1. Select the footprint you wish to add a 3D body to using the PCB library panel.
2. Confirm that a mechanical layer is enabled and is the current layer.
3. Select **Place » Place 3D body** [shortcut P, B]. The *3D body* dialog will open, as shown in Figure 17.

> **Note:** 3D body objects can be created from shapes (extruded rectangular, cylindrical or spherical), or from an imported STEP model. You can also use a combination of both, if needed.

4. Enter overall height and stand off height information, and close the dialog.
5. Click to define the vertices. Note that the placement process for en extruded 3D body is the standard polygonal object placement process, use **Shift+Spacebar** to cycle the corner style, and **Spacebar** to toggle the current placement corner.

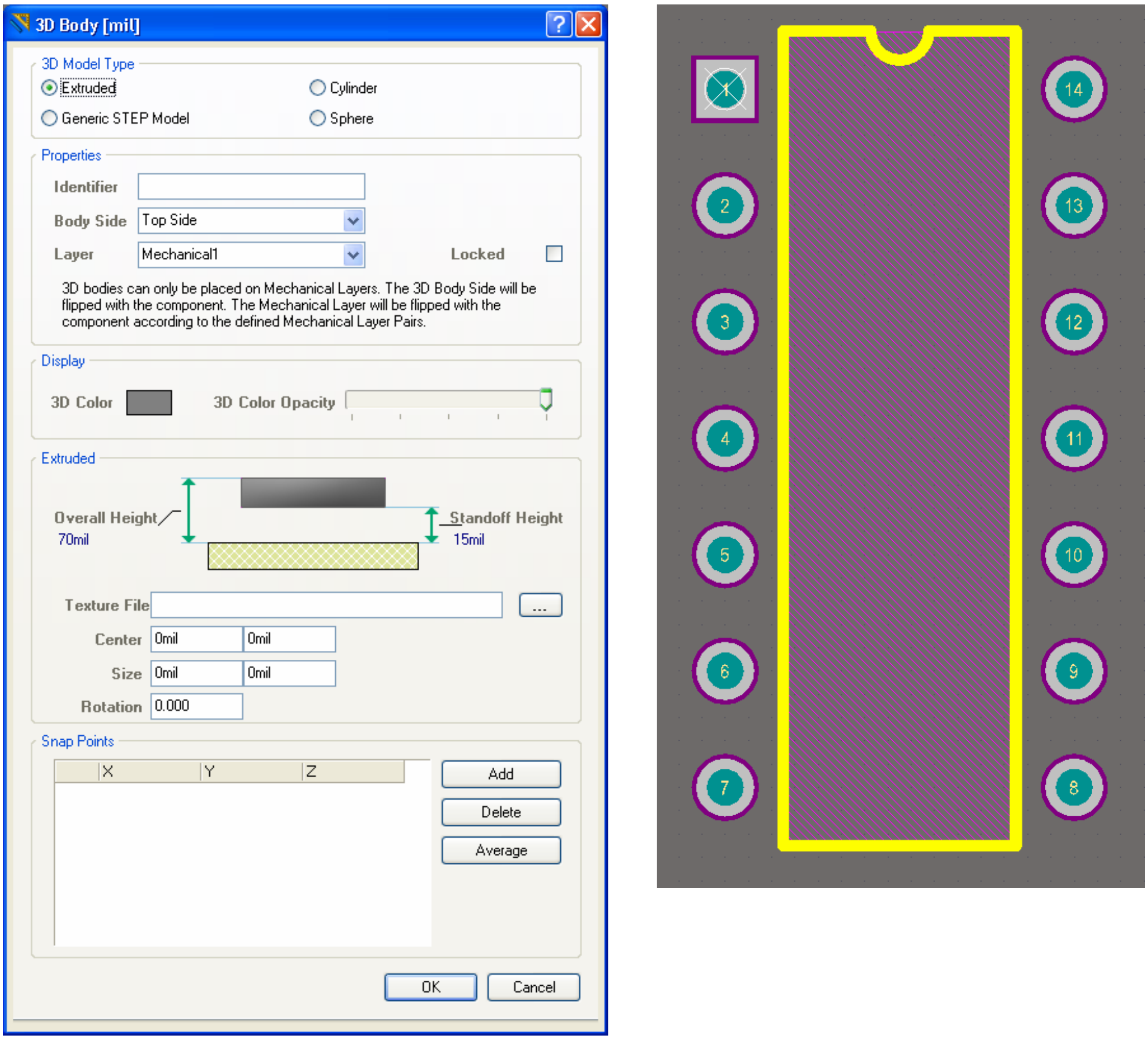



*Figure 17. Define the 3D body overall height and standoff height. The Body after it has been placed on the DIP14.*

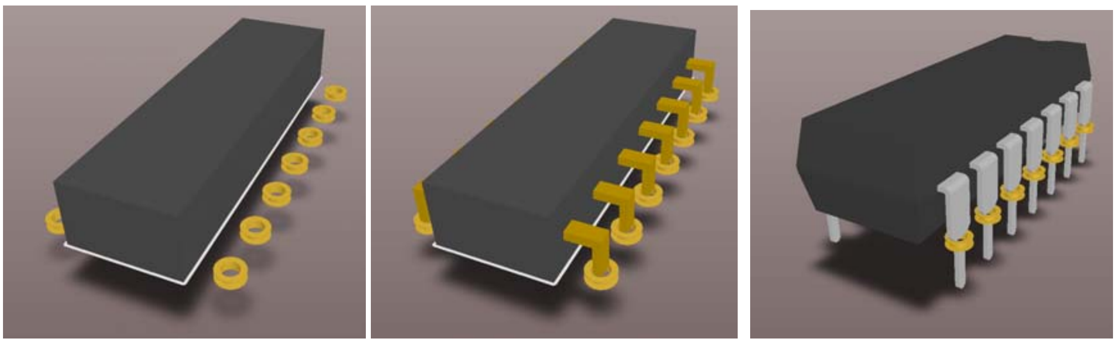


*Figure 18. DIP14 footprint with a single 3D body, then with multiple 3D bodies added to define the pins, then with an imported STEP model for a DIP14.*

## 16.2.4 Using the 3D Body Manager

Using the 3D Body Manager, 3D body objects can be automatically created based on the bounding rectangles or closed polygonal outlines of primitives that already exist in the footprint. The 3D Body Manager can be used for the current footprint, or across the entire PCB library.

In Figure 19 you can see the **3D Body Manager** used to define a 3D body for the transistor footprint, SOIC8. Using this approach is much easier than attempting to define the shape manually, because of the curved edge of the transistor body.

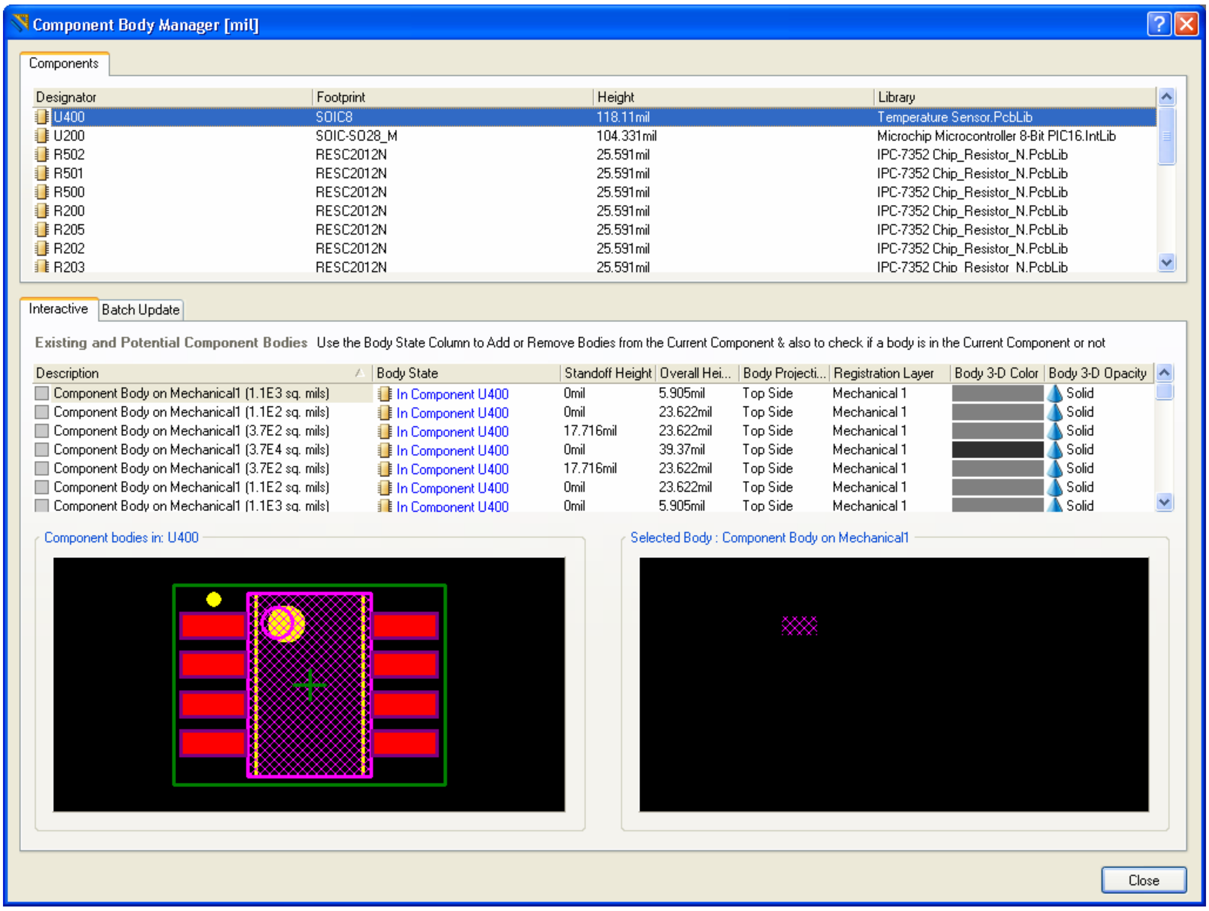



*Figure 19. Use the 3D Body Manager to quickly create body objects based on existing primitives.*

To create a shape that follows the outline defined on the component overlay click the second option that appears in the list, **Polygonal shape created from primitives on TopOverlay**. For this line in the dialog, click on the **Action** button **Add to component_name**), set the **Registration Layer** to the mechanical layer that the body object should be placed on (mechanical layer 3 has been renamed BodyTop in this example), and set the **Overall Height** to a suitable value, for example `200mil`, as shown in Figure 19.

**Note:** Multiple 3D body shapes can be added to build up a 3D body using the dialog in 19. To do this, simply pick and choose which bodies you want from the action column to add. You can also mix this with manually placed bodies to build up a complex 3D model like for example the 16x2 character screen on the Altium NanoBoard, or the Spirit Level board.

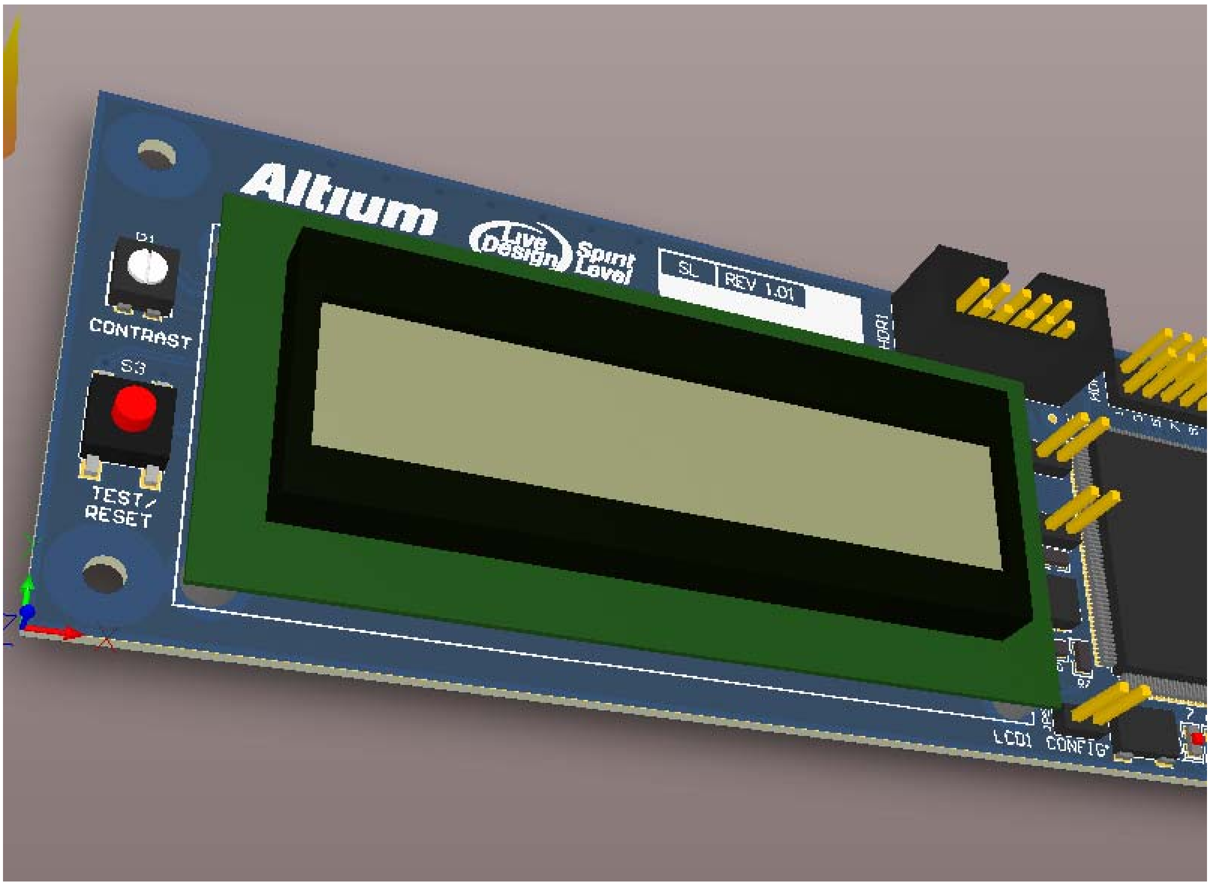



*Figure 20. End result in the 3D view of a complex 3D body created in the PCB library editor.*

TOP SECRET

Altium Designer

# Module 17: Linking Models, Parameters, Library Package and Updates

## Module 17: Linking Models, Parameters, Library Package and Updates






Module Seq = 17

# 17.1 Adding Model and Parameter Detail to a Component

The final stage of preparing the new component is to add model and parameter information.

## 17.1.1 Adding a model

There are a number of ways of adding a models to a component, these include:

- Via the Library Component Properties dialog
- Via the models region at the bottom of the main workspace – click the small down arrow at the bottom right of the workspace if it is not visible.
- Via the Model Manager. The Model Manager allows you to add models to multiple components at the same time, and is ideal for reviewing model assignments across a library.

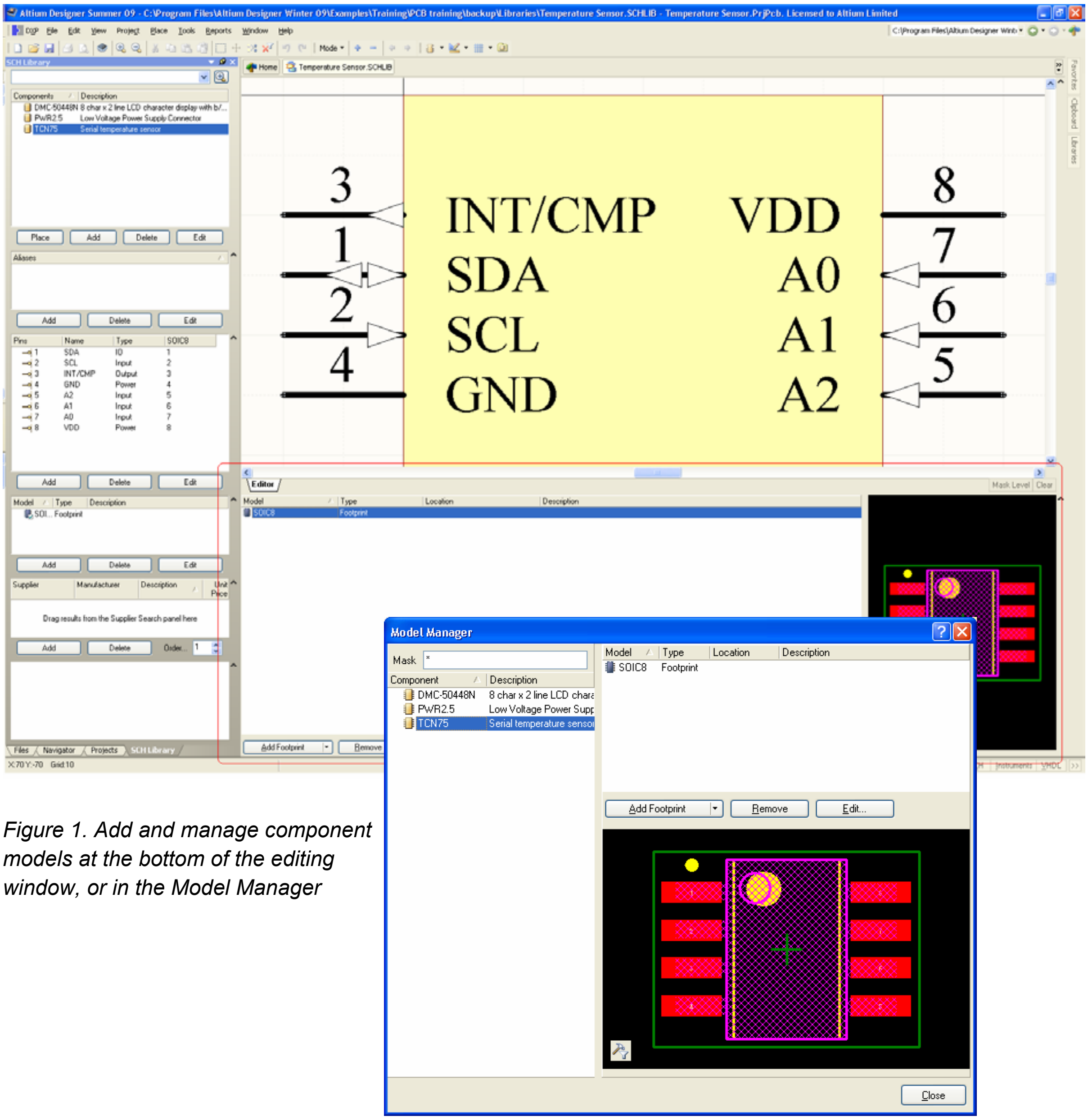



*Figure 1. Add and manage component models at the bottom of the editing window, or in the Model Manager*

## 17.1.2 Exercise – Adding a footprint using the Model Manager

To add a footprint using the Model Manager:

1. If it is not already open, re-open the Temperature Sensor project (`\Program Files\Altium Designer Summer 09\Examples\Training\PCB Training\Temperature Sensor\Temperature Sensor.PrjPcb`).
2. In the **Projects** panel, browse to and open the schematic library, `Temperature Sensor.SchLib`.
3. Select **Tools » Model Manager** in the Schematic Library Editor to open the *Model Manager.* It will show a list of all components in the current library and any models, including Footprint, Simulation, Signal Integrity, and PCB 3D models.
4. In the *Model Manager* dialog click on the TCN75 component that you created, making it the active part.
5. Click **Add Footprint** to launch the *PCB Model* dialog, as shown in Figure 2.

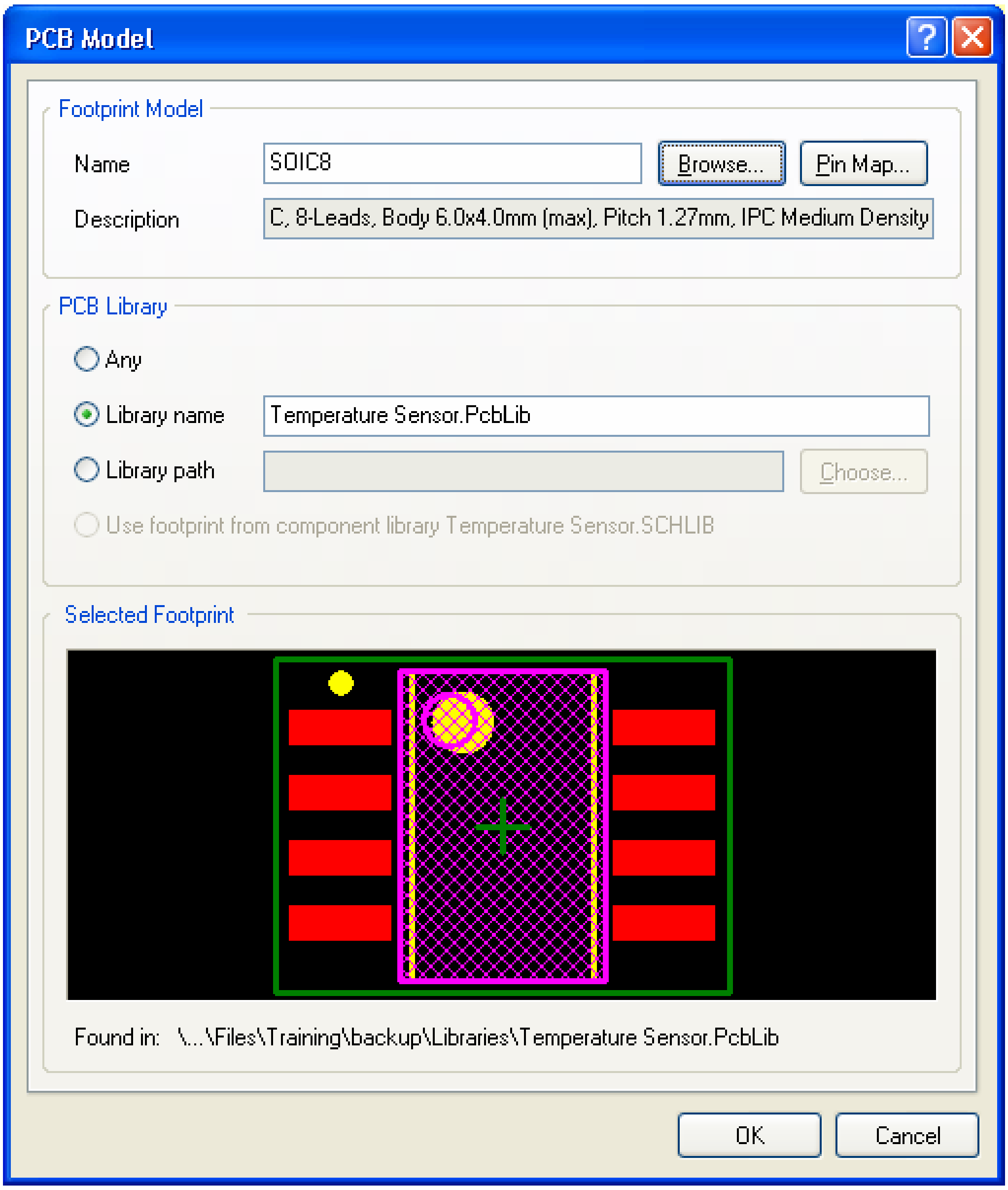



*Figure 2. Adding the footprint model to the component.*

6. If you know the footprint name, and you are confident that is in a currently available footprint library, you can type the name directly into the **Name** field, an image of it will appear if it is located. Otherwise, you can click **Browse** to open the *Browse Libraries* dialog, as shown in Figure 3.

> **Note:** There are different ways you can reference a footprint from the symbol, this is determined in the **PCB Library** region of the *PCB Model* dialog:
>
> **- Any** means find the footprint in Any currently available library,
>
> **- Library Name** means it must come from the specified library,
>
> **- Library Path** means it must come from the specified library in the specified location, and
>
> **- Use from integrated** is set automatically if you have compiled the library into an integrated library.

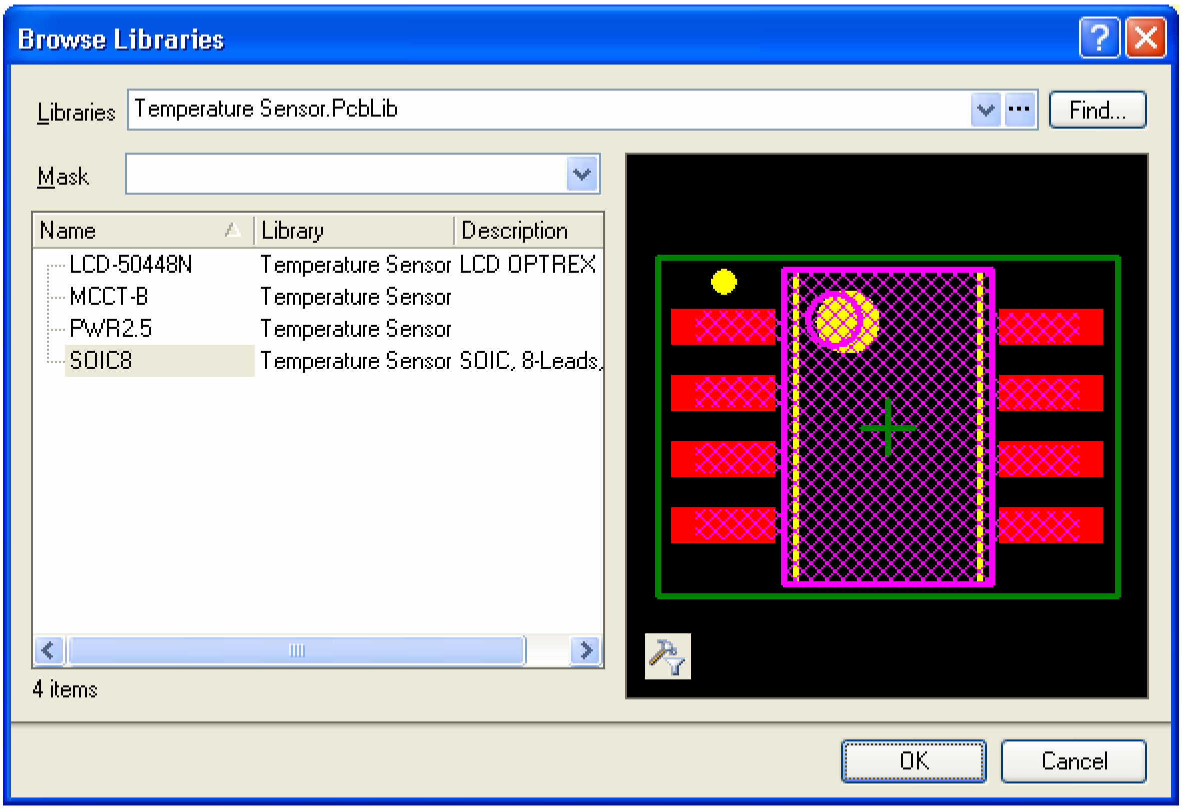



*Figure 3. Use the Browse Libraries dialog to visually select the correct footprint.*

> **Note:** The **Libraries** dropdown at the top of the dialog allows you to choose which library you are currently browsing, from the available footprint libraries. The **Find** button is used to search, this will be demonstrated in the *Schematic Capture* training module.

7. Once you have located your new SOIC8 footprint select it, and it will appear in the *PCB Model* dialog. Click **OK** to close the dialog.

> **Note:** If your footprint was using a different numbering scheme from the pin numbering on the symbol you would need to define the pin-to-pad mapping, click the **Pin Map** button in the *PCB Models* dialog to do this.

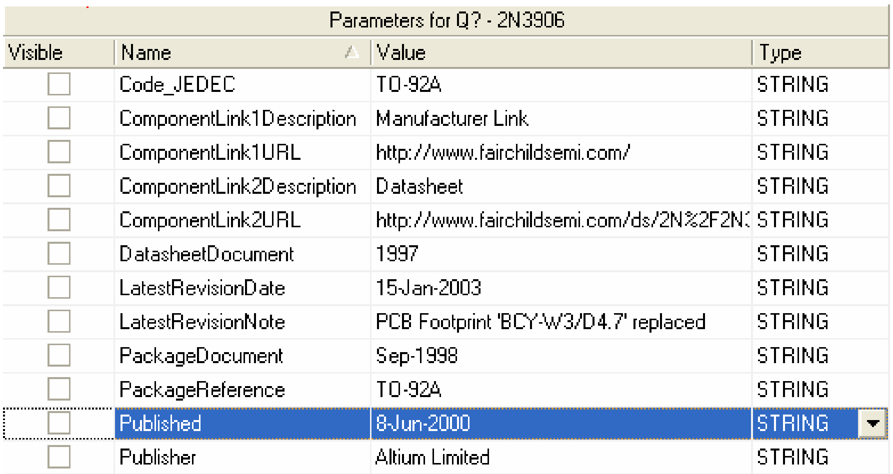
8. Click **Close** to close the **Model Manager**, you have now assigned the SOIC8 footprint to your TCN75 component.

9. Save the library.

## 17.1.3 Component Parameters

Each component that you create can have any number of component parameters, which can be added/edited in the library or on the schematic. Parameters can be used for any purpose you require, including:

- component detail information, such as a voltage rating, component revision, and so on
- company component information, such as stock number or price,
- design reference information, such as special pick and place requirements,
- links to reference information, such as websites and PDFs.

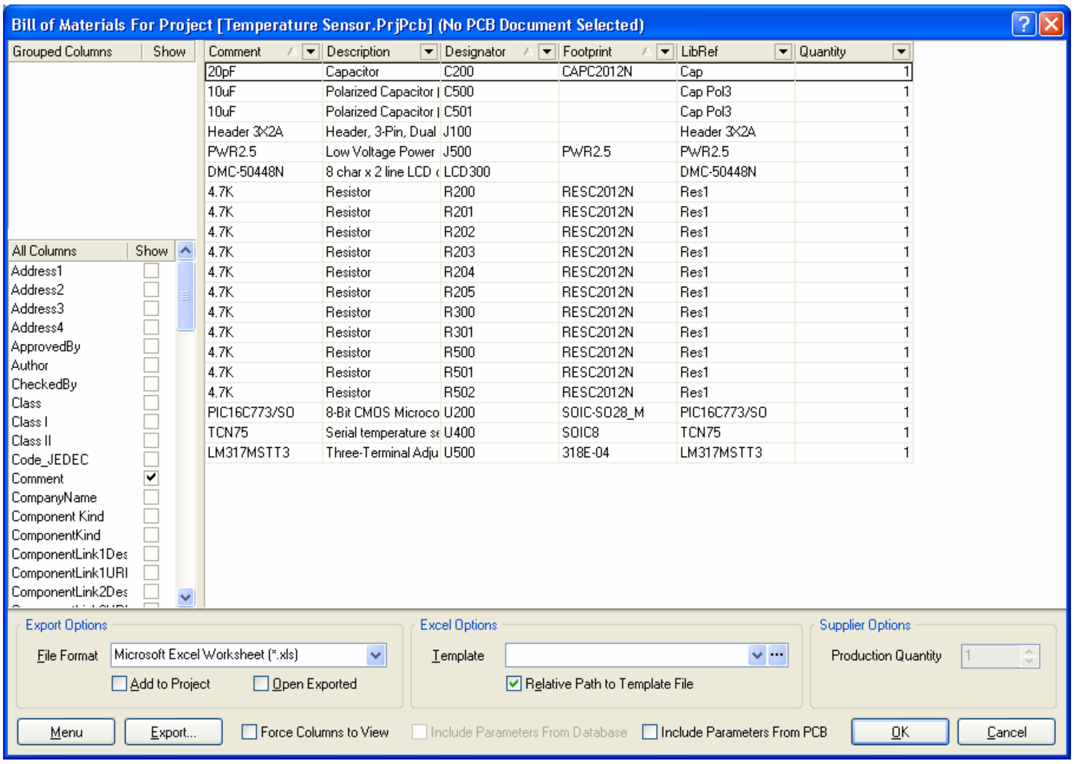


Parameters for Q? - 2N3906

| Visible | Name | Value | Type |
|---|---|---|---|
| ☐ | Code_JEDEC | TO-92A | STRING |
| ☐ | ComponentLink1Description | Manufacturer Link | STRING |
| ☐ | ComponentLink1URL | http://www.fairchildsemi.com/ | STRING |
| ☐ | ComponentLink2Description | Datasheet | STRING |
| ☐ | ComponentLink2URL | http://www.fairchildsemi.com/ds/2N%2F2N: | STRING |
| ☐ | DatasheetDocument | 1997 | STRING |
| ☐ | LatestRevisionDate | 15-Jan-2003 | STRING |
| ☐ | LatestRevisionNote | PCB Footprint 'BCY-W3/D4.7' replaced | STRING |
| ☐ | PackageDocument | Sep-1998 | STRING |
| ☐ | PackageReference | TO-92A | STRING |
| ☐ | Published | 8-Jun-2000 | STRING |
| ☐ | Publisher | Altium Limited | STRING |

*Figure 4. Add parameters to fully describe your components.*

Any component parameters can be included in the Bill of Materials, or any custom report you generate via the Report generation dialog.




| Comment | Description | Designator | Footprint | LibRef | Quantity |
|---|---|---|---|---|---|
| 20pF | Capacitor | C200 | CAPC2012N | Cap | 1 |
| 10uF | Polarized Capacitor ( | C500 | | Cap Pol3 | 1 |
| 10uF | Polarized Capacitor ( | C501 | | Cap Pol3 | 1 |
| Header 3X2A | Header, 3-Pin, Dual | J100 | | Header 3X2A | 1 |
| PWR2.5 | Low Voltage Power | J500 | PWR2.5 | PWR2.5 | 1 |
| DMC-50448N | 8 char x 2 line LCD ( | LCD300 | | DMC-50448N | 1 |
| 4.7K | Resistor | R200 | RESC2012N | Res1 | 1 |
| 4.7K | Resistor | R201 | RESC2012N | Res1 | 1 |
| 4.7K | Resistor | R202 | RESC2012N | Res1 | 1 |
| 4.7K | Resistor | R203 | RESC2012N | Res1 | 1 |
| 4.7K | Resistor | R204 | RESC2012N | Res1 | 1 |
| 4.7K | Resistor | R205 | RESC2012N | Res1 | 1 |
| 4.7K | Resistor | R300 | RESC2012N | Res1 | 1 |
| 4.7K | Resistor | R301 | RESC2012N | Res1 | 1 |
| 4.7K | Resistor | R500 | RESC2012N | Res1 | 1 |
| 4.7K | Resistor | R501 | RESC2012N | Res1 | 1 |
| 4.7K | Resistor | R502 | RESC2012N | Res1 | 1 |
| PIC16C773/SO | 8-Bit CMOS Microco | U200 | SOIC-SO28_M | PIC16C773/SO | 1 |
| TCN75 | Serial temperature se | U400 | SOIC8 | TCN75 | 1 |
| LM317MSTT3 | Three-Terminal Adju | U500 | 318E-04 | LM317MSTT3 | 1 |

*Figure 5. Generate reports that include any component data you require.*

## 17.1.4 Exercise – adding a component parameter

A useful component parameter is to add is a link from the schematic component to a datasheet. A PDF datasheet for the serial temperature sensor component has been included, we will now add a parameter that allows us to open the datasheet for it directly from the schematic.

1. In your *Windows File Explorer*, confirm that the PDF datasheet is available. Browse to the location:

   `C:\Program Files\Altium Designer Summer 09\Examples\Training\PCB Training\Temperature Sensor`

2. In the *Windows File Explorer*, copy this address location (you might need to enable the **Address Bar** via the **View » Toolbars** menu first).
3. Return to Altium Designer, then with your TCN75 component symbol open in the library editor, double click on its name in the **SCH Library** panel to open the *Library Component Properties* dialog.
4. In the **Parameters** region of the dialog, click the **Add** button to add a new parameter, this will open the *Parameter Properties* dialog.
5. In the **Name** field, type `HelpURL`.
6. Click to position the cursor in the **Value** field of the dialog, then Paste in the contents of the Windows clipboard, this should be what you copied from the Address Bar of the Windows File Explorer, namely:

   `C:\Program Files\Altium Designer Summer 09\Examples\Training\PCB Training\Temperature Sensor`

7. Type a backslash ('\') character at the end of the string, then type in the name of the PDF, namely: `TCN75 - 21490b.pdf,` as shown in Figure 6.

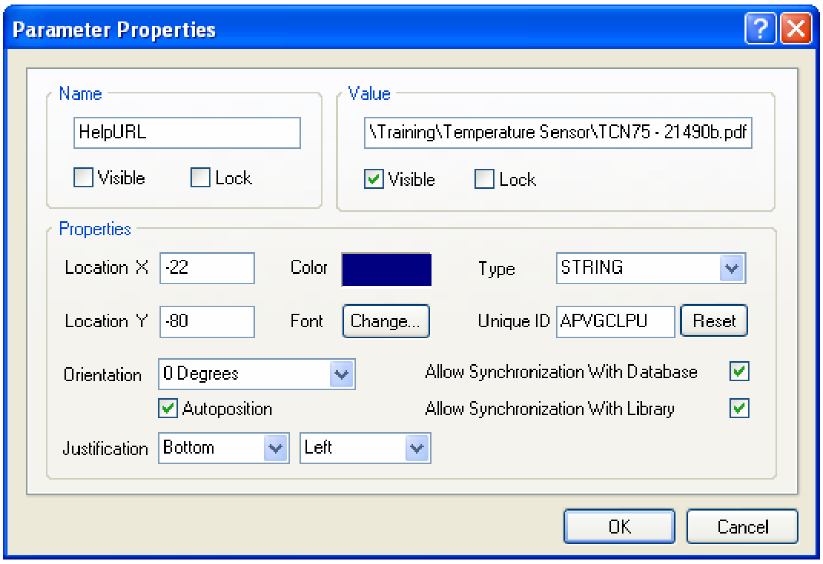



*Figure 6. Add the HelpURL parameter to link the datasheet to the component.*

8. Clear the **Visible** checkbox in the *Parameter Properties* dialog since there is no need to show this string on the schematic.
9. Click **OK** to close the dialog, then click **OK** to close the *Library Component Properties* dialog.
10. Save the library.

**Note:** We'll use this link we have created shortly in 17.2.4 Exercise – Using the new component in schematic

## 17.1.5 Adding and Modifying Component Parameters

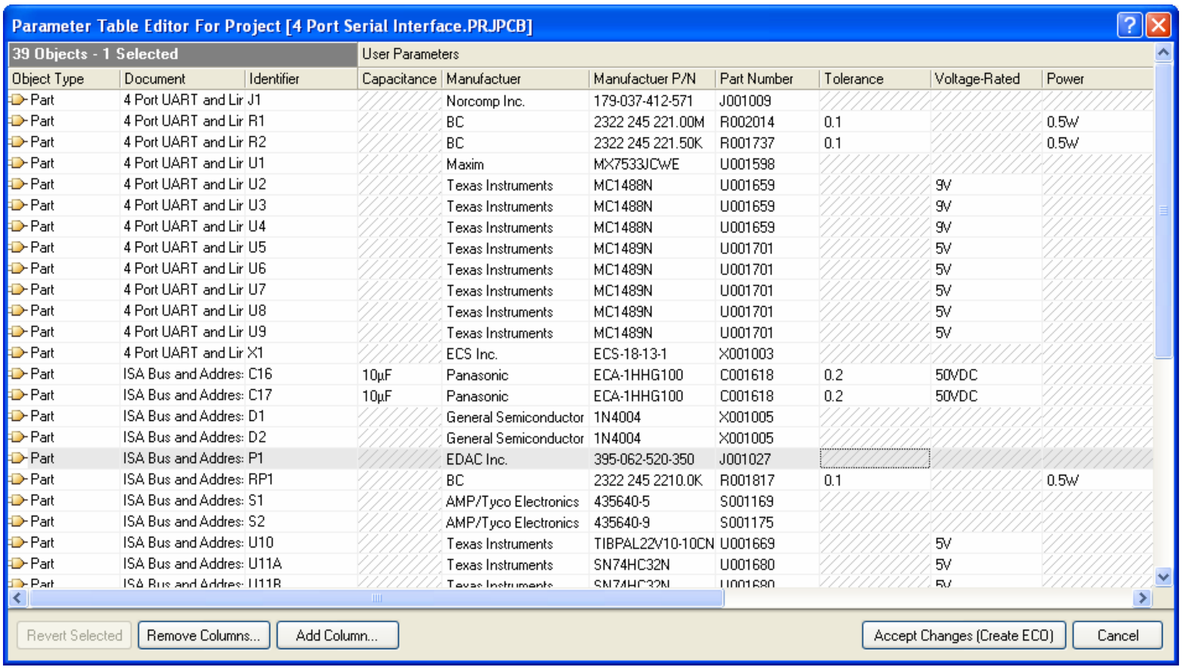
Parameters can be added to an individual component, either in the schematic library editor, or the schematic editor. Adding parameters at the library level ensures consistency in component information and speeds the process of component placement as parameters do not need to be added individually.

Altium Designer also provides a powerful **Parameter Manager** that supports adding parameters globally to a library, schematic, or series of schematics.

To open the **Parameter Manager**:

- In either the Schematic or Schematic Library editors, select **Tools » Parameter Manager** from the menus to launch the *Parameter Editor Options* dialog.
- Enable the **Include Parameters Owned By** check boxes you require, if you are editing components this would typically only be the **Parts** option, then click **OK**.
- The **Parameter Table Editor** will open. Column headings correspond to the parameter names, with the contents of each column corresponds to that parameter's value. Figure 7 shows the Parameter Table Editor.
- Right-clicking in the *Parameter Table Editor* will display options, such as **Add Columns**, **Add Parameter Values**, **Copy**, **Paste**, and so on. Data can also be pasted from standard text and from most spreadsheet applications, such as Microsoft Excel.
- All parameter changes are controlled by an Engineering Change Order (ECO) process that supports the controlled execution of ECO's, including the ability to selectively include and exclude operations as well as generate reports of all changes prior to their being executed.

> **Note:** once you have selected an item (or items) in the Parameter Manager, press F2 to edit the value(s).




| Object Type | Document | Identifier | Capacitance | Manufactuer | Manufactuer P/N | Part Number | Tolerance | Voltage-Rated | Power |
|---|---|---|---|---|---|---|---|---|---|
| Part | 4 Port UART and Lir | J1 | | Norcomp Inc. | 179-037-412-571 | J001009 | | | |
| Part | 4 Port UART and Lir | R1 | | BC | 2322 245 221.00M | R002014 | 0.1 | | 0.5W |
| Part | 4 Port UART and Lir | R2 | | BC | 2322 245 221.50K | R001737 | 0.1 | | 0.5W |
| Part | 4 Port UART and Lir | U1 | | Maxim | MX7533JCWE | U001598 | | | |
| Part | 4 Port UART and Lir | U2 | | Texas Instruments | MC1488N | U001659 | | 9V | |
| Part | 4 Port UART and Lir | U3 | | Texas Instruments | MC1488N | U001659 | | 9V | |
| Part | 4 Port UART and Lir | U4 | | Texas Instruments | MC1488N | U001659 | | 9V | |
| Part | 4 Port UART and Lir | U5 | | Texas Instruments | MC1489N | U001701 | | 5V | |
| Part | 4 Port UART and Lir | U6 | | Texas Instruments | MC1489N | U001701 | | 5V | |
| Part | 4 Port UART and Lir | U7 | | Texas Instruments | MC1489N | U001701 | | 5V | |
| Part | 4 Port UART and Lir | U8 | | Texas Instruments | MC1489N | U001701 | | 5V | |
| Part | 4 Port UART and Lir | U9 | | Texas Instruments | MC1489N | U001701 | | 5V | |
| Part | 4 Port UART and Lir | X1 | | ECS Inc. | ECS-18-13-1 | X001003 | | | |
| Part | ISA Bus and Addres: | C16 | 10µF | Panasonic | ECA-1HHG100 | C001618 | 0.2 | 50VDC | |
| Part | ISA Bus and Addres: | C17 | 10µF | Panasonic | ECA-1HHG100 | C001618 | 0.2 | 50VDC | |
| Part | ISA Bus and Addres: | D1 | | General Semiconductor | 1N4004 | X001005 | | | |
| Part | ISA Bus and Addres: | D2 | | General Semiconductor | 1N4004 | X001005 | | | |
| Part | ISA Bus and Addres: | P1 | | EDAC Inc. | 395-062-520-350 | J001027 | | | |
| Part | ISA Bus and Addres: | RP1 | | BC | 2322 245 2210.0K | R001817 | 0.1 | | 0.5W |
| Part | ISA Bus and Addres: | S1 | | AMP/Tyco Electronics | 435640-5 | S001169 | | | |
| Part | ISA Bus and Addres: | S2 | | AMP/Tyco Electronics | 435640-9 | S001175 | | | |
| Part | ISA Bus and Addres: | U10 | | Texas Instruments | TIBPAL22V10-10CN | U001669 | | 5V | |
| Part | ISA Bus and Addres: | U11A | | Texas Instruments | SN74HC32N | U001680 | | 5V | |
| Part | ISA Bus and Addres: | U11B | | Texas Instruments | SN74HC32N | U001680 | | 5V | |

*Figure 7. Parameter Table Editor*

## 17.1.6 Exercise – Editing Parameters in the Parameter Manager

We'll add a new parameter, called *Lead Time*, to all the components in the Temperature Sensor schematic library.

1. From the `Temperature Sensor.schlib` schematic Library, select **Tools » Parameter Manager** and in the initial dialog, be sure only **Parts** is checked in the **Include Parameters Matched By** section at the top, leaving the other options set to their defaults, the click **OK**.
2. In the *Parameter Table Editor*, right-click and choose **Add Column**.
3. In the **Name** field, enter `Lead Time`, checking the **Add to all objects** option. Leave the **Value** field blank, and click **OK**. A new column will be created with the heading **Lead Time**, at the right hand edge of the dialog (scroll across if necessary).
4. Right click the contents of the column **Lead Time** for the component DMC-50448N and select **Edit**. The field will change to a drop-down with a cursor. Type `14 days` in the field, and press **Enter** to commit the value.
5. Right click on this value of `14 days`, and select **Copy**.
6. Right-click the field **Lead Time** for the component TCN75, and select **Paste**.
7. Select all the cells of the column **Lead Time** (not including the column heading) using either Ctrl+Click, Shift+Click, or dragging to select the contents fields.
8. Right-click on any of the 3 selected cells and select **Edit**. One of the 3 cells will become editable. Type the text `3-5 Days` and press **Enter** on the keyboard. All selected cells will have the edit applied. Using these standard spreadsheet-type editing actions, multiple cells (including columns and rows) can be edited.
9. Click **Accept Changes (Create ECO)** to open the *Engineering Change Order* dialog. This is list of changes that are about to be applied to the library, note that individual changes can be disabled.
10. Click the **Execute Changes** button in the *ECO* dialog to propagate those changes through to the design. The **Report Changes** button will generate a list of change orders, use this if you need to keep a record of changes made to a design file. If you click the **Close** button in the *ECO* dialog the changes will not be applied, so you will be returned to the *Parameter Table Editor* dialog.
11. Save the library.

# 17.2 Component Auditing

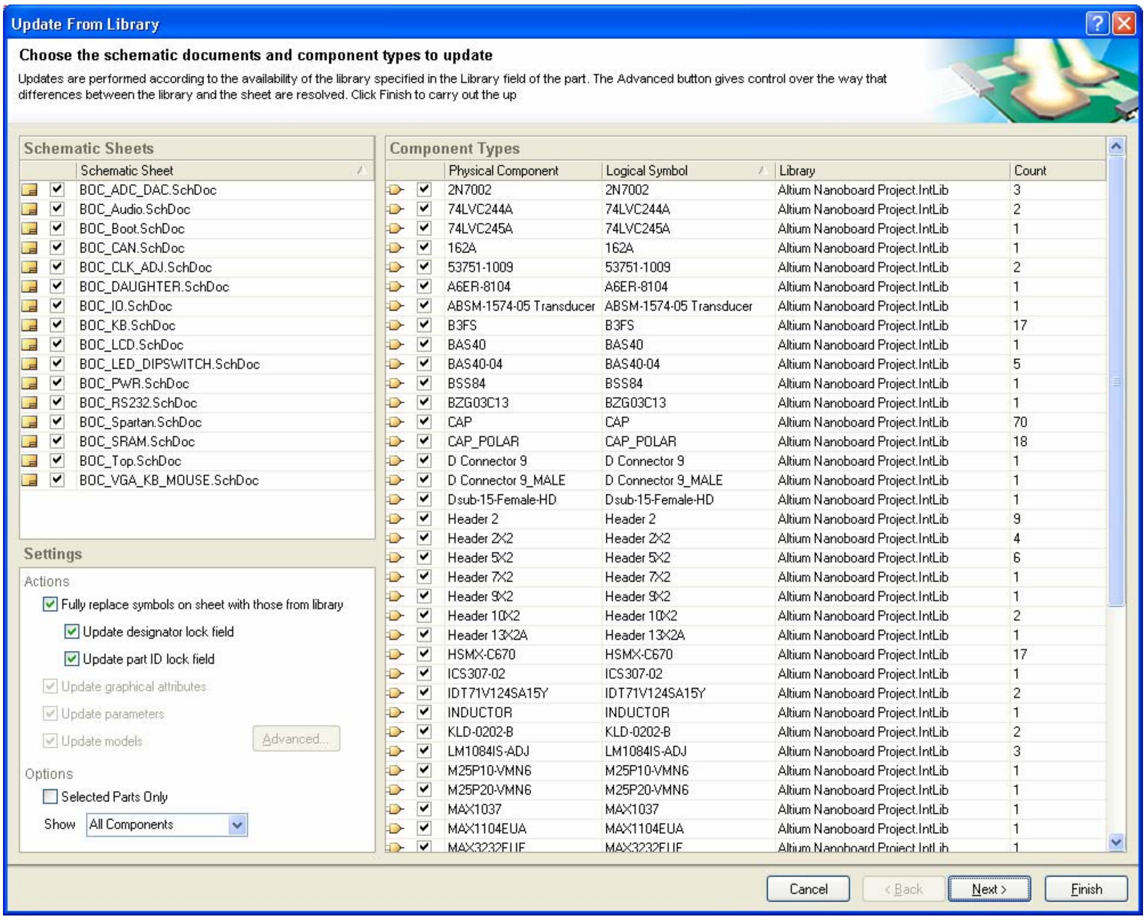
It is standard design practice to create and edit components during the design process. Altium Designer includes updating tools that give complete control over the updating process.

Along with the flexibility of being able to make edits to a given instance of a component or footprint, also comes the risk of making edits that might not be correct or ideal. Altium Designer includes features to help check the schematic symbols and PCB footprints against the libraries they came from, and generate reports that show any differences.

## 17.2.1 Schematic Updating

Schematic components are updated using the *Update from Libraries* wizard in the schematic editor.

- Select **Tools » Update From Libraries**, to open the *Update from Libraries* wizard.
- The default is to do a full update on every component; including its graphical attributes, models and parameters (the designator is not changed).
- To control what attributes of the components are to be updated, uncheck the **Fully replace symbols on sheet with those from library** setting.
- The **Advanced** button provides greater flexibility in terms of how parameter and model difference should be handled.




| Physical Component | Logical Symbol | Library | Count |
| --- | --- | --- | --- |
| 2N7002 | 2N7002 | Altium Nanoboard Project.IntLib | 3 |
| 74LVC244A | 74LVC244A | Altium Nanoboard Project.IntLib | 2 |
| 74LVC245A | 74LVC245A | Altium Nanoboard Project.IntLib | 1 |
| 162A | 162A | Altium Nanoboard Project.IntLib | 1 |
| 53751-1009 | 53751-1009 | Altium Nanoboard Project.IntLib | 2 |
| A6ER-8104 | A6ER-8104 | Altium Nanoboard Project.IntLib | 1 |
| ABSM-1574-05 Transducer | ABSM-1574-05 Transducer | Altium Nanoboard Project.IntLib | 1 |
| B3FS | B3FS | Altium Nanoboard Project.IntLib | 17 |
| BAS40 | BAS40 | Altium Nanoboard Project.IntLib | 1 |
| BAS40-04 | BAS40-04 | Altium Nanoboard Project.IntLib | 5 |
| BSS84 | BSS84 | Altium Nanoboard Project.IntLib | 1 |
| BZG03C13 | BZG03C13 | Altium Nanoboard Project.IntLib | 1 |
| CAP | CAP | Altium Nanoboard Project.IntLib | 70 |
| CAP_POLAR | CAP_POLAR | Altium Nanoboard Project.IntLib | 18 |
| D Connector 9 | D Connector 9 | Altium Nanoboard Project.IntLib | 1 |
| D Connector 9_MALE | D Connector 9_MALE | Altium Nanoboard Project.IntLib | 1 |
| Dsub-15-Female-HD | Dsub-15-Female-HD | Altium Nanoboard Project.IntLib | 1 |
| Header 2 | Header 2 | Altium Nanoboard Project.IntLib | 9 |
| Header 2X2 | Header 2X2 | Altium Nanoboard Project.IntLib | 4 |
| Header 5X2 | Header 5X2 | Altium Nanoboard Project.IntLib | 6 |
| Header 7X2 | Header 7X2 | Altium Nanoboard Project.IntLib | 1 |
| Header 9X2 | Header 9X2 | Altium Nanoboard Project.IntLib | 1 |
| Header 10X2 | Header 10X2 | Altium Nanoboard Project.IntLib | 2 |
| Header 13X2A | Header 13X2A | Altium Nanoboard Project.IntLib | 1 |
| HSMX-C670 | HSMX-C670 | Altium Nanoboard Project.IntLib | 17 |
| ICS307-02 | ICS307-02 | Altium Nanoboard Project.IntLib | 1 |
| IDT71V124SA15Y | IDT71V124SA15Y | Altium Nanoboard Project.IntLib | 2 |
| INDUCTOR | INDUCTOR | Altium Nanoboard Project.IntLib | 1 |
| KLD-0202-B | KLD-0202-B | Altium Nanoboard Project.IntLib | 2 |
| LM1084IS-ADJ | LM1084IS-ADJ | Altium Nanoboard Project.IntLib | 3 |
| M25P10-VMN6 | M25P10-VMN6 | Altium Nanoboard Project.IntLib | 1 |
| M25P20-VMN6 | M25P20-VMN6 | Altium Nanoboard Project.IntLib | 1 |
| MAX1037 | MAX1037 | Altium Nanoboard Project.IntLib | 1 |
| MAX1104EUA | MAX1104EUA | Altium Nanoboard Project.IntLib | 1 |
| MAX3232EUE | MAX3232EUE | Altium Nanoboard Project.IntLib | 1 |



*Figure 8. Update from schematic library to schematic*

- Updates are performed through the ECO system, and a report can be produced if needed.
- The wizard checks each component against the original library it came from, if those libraries are not available, or if you need the update to be done from another library, those components will need to have their **Library Name** attribute disabled, allowing Altium Designer to search all available libraries instead. This attribute is configured in the **Component Properties** dialog.

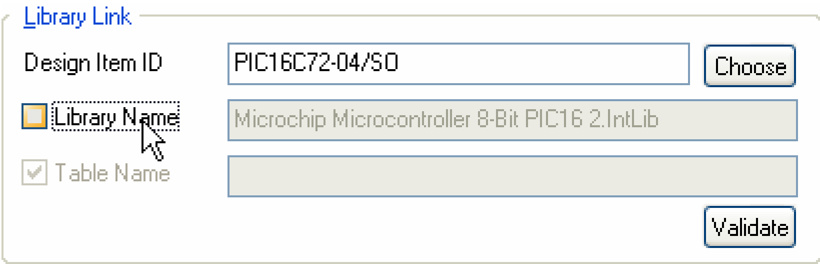



*Figure 9. Disabling the Library Name attribute allows Altium Designer to search all available libraries.*

- If the update process creates more changes than expected, it is probably due to component customization that has been performed after the component was placed. For example, a generic discrete component being given a value after it was placed. Use the Advanced options in the wizard to control parameter updates.

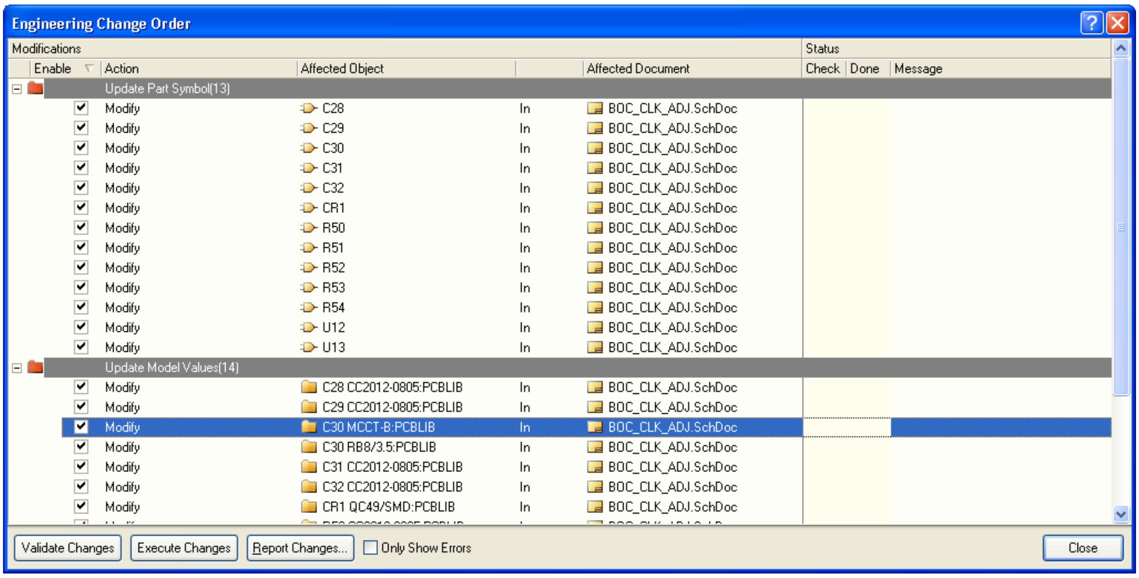



*Figure 10. Engineering change order dialog listing all executable differences found.*

- When the component in the library is unique, for example the resistor's value, footprint and other parameters are fully specified; any detected difference should be carefully checked. When in doubt avoid wholesale updates of all components in the project, instead selectively enable only those components you are currently working on.

> **Note:** It is also possible to update schematic components from the library editor, onto currently open schematic sheets. This is a very basic update feature which does a complete replacement of the component – the only attribute retained from the instance placed on the schematic sheet is the designator value.

> **Note:** To compare components placed from one library with those placed from another, use the Parameter Manager. Pre-select components of interest, then enable the **Selected Objects Only** checkbox in the initial *Parameter Editor Options* dialog that appears when you select **Tools » Parameter Manager**.

## 17.2.2 Updating PCB Footprints

PCB footprints can be updated directly from the PCB library, think of this as *pushing* the footprint from the library onto the board. Footprints can also be updated from the board, using the Update From Libraries tool, which can be thought of as *pulling* the footprints from the libraries onto the board.

- Push-type updating is limited; it can only be done one library at a time, for one footprint or all footprints in that library.
- Pull-type updating is fully configurable; it does a detailed comparison between all primitives in every footprint on the board and the footprints in the library, and can also produce a detailed report. This updating is done by selecting **Tools » Update from PCB Libraries** from the PCB editor menus.
- The PCB footprint libraries need to be available to perform an **Update from PCB libraries**. The library specified in the PCB editor *Component* dialog is the library that the footprint was placed from when the design was initially transferred.

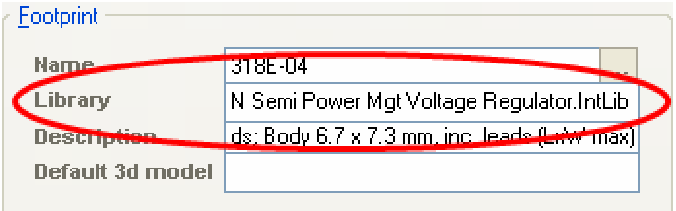



*Figure 11. The source footprint library must be available.*

- Before the update dialog opens an options dialog appears, where you can set the tolerance, and layers to include. The default settings should suit most situations, however if you are working on a legacy design from another design system you may need to adjust the tolerance to avoid showing insignificant differences.

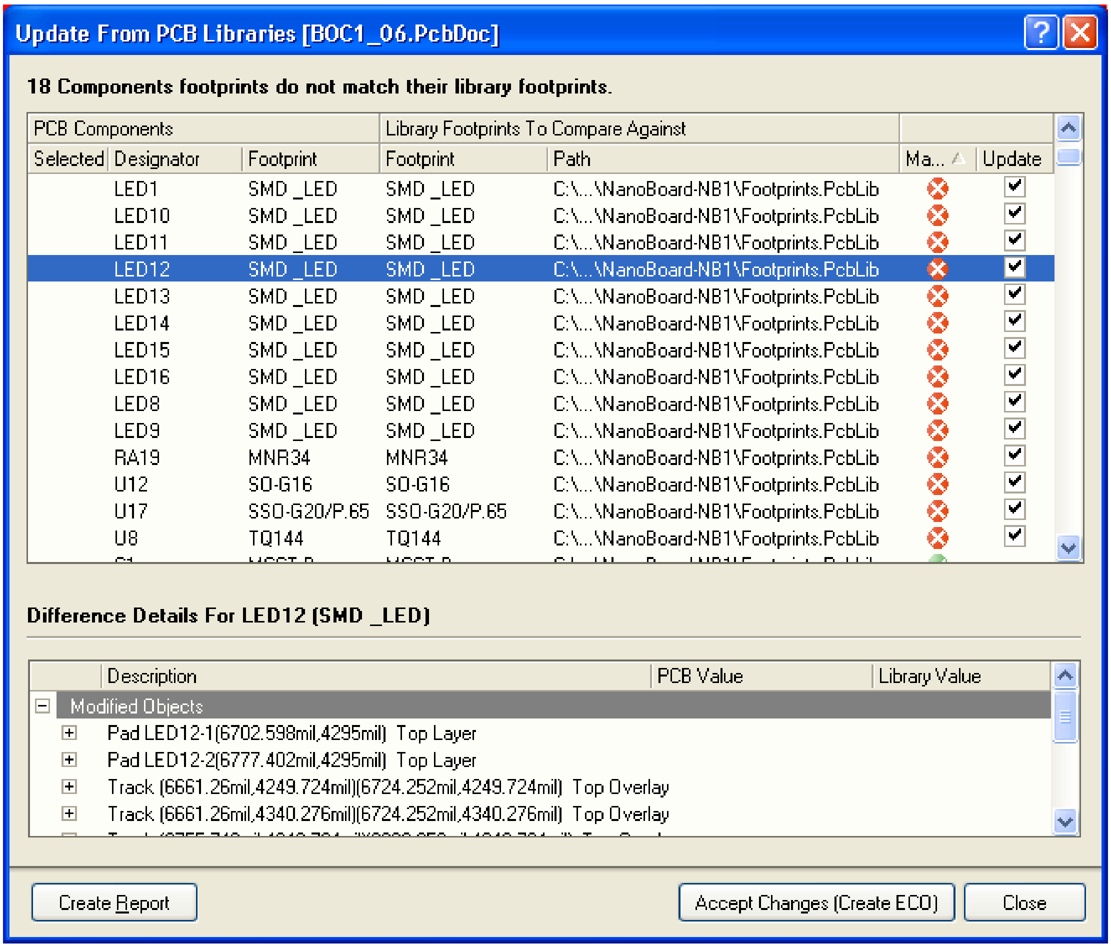



*Figure 12. the Update from PCB libraries dialog details all physical differences.*

- You may not need to include all layers in the comparison. For example, if you have adjusted the widths of the overlay to suit the current design, you can disable the overlay layers.
- The **Create Report** button produces a detailed HTML report.

**Note:** Keep in mind that certain footprint properties, such as solder mask and paste mask expansions, can be defined manually for each footprint (in the library or on the board), or they can be controlled via design rules. Manually defined attributes have the highest priority, so changes made in the library or on the board will override design rule settings for calculated fields such as these.

**Note:** Although it is possible to edit the footprint directly on the board (by unlocking the component primitives), the most common cause of difference is when the footprint has been placed from the wrong library, or has become out of date due to changes to the footprint at the library level. Differences should be considered carefully, and once an update has been performed a Batch DRC should be run to confirm the changes did not affect the integrity of the PCB layout.

## 17.2.3 Other Audit techniques

### 17.2.3.1 Reviewing changes to Documents

The file comparison tools in the Storage Manager provide an excellent way of reviewing changes made to a file.

- Comparisons can be made between the current file and an older version saved by Altium Designer's built-in **Local History** system, or between the current version and an older version stored in your version control system.
- Comparisons can be done between different revisions of the same document or comparing two documents from different projects.

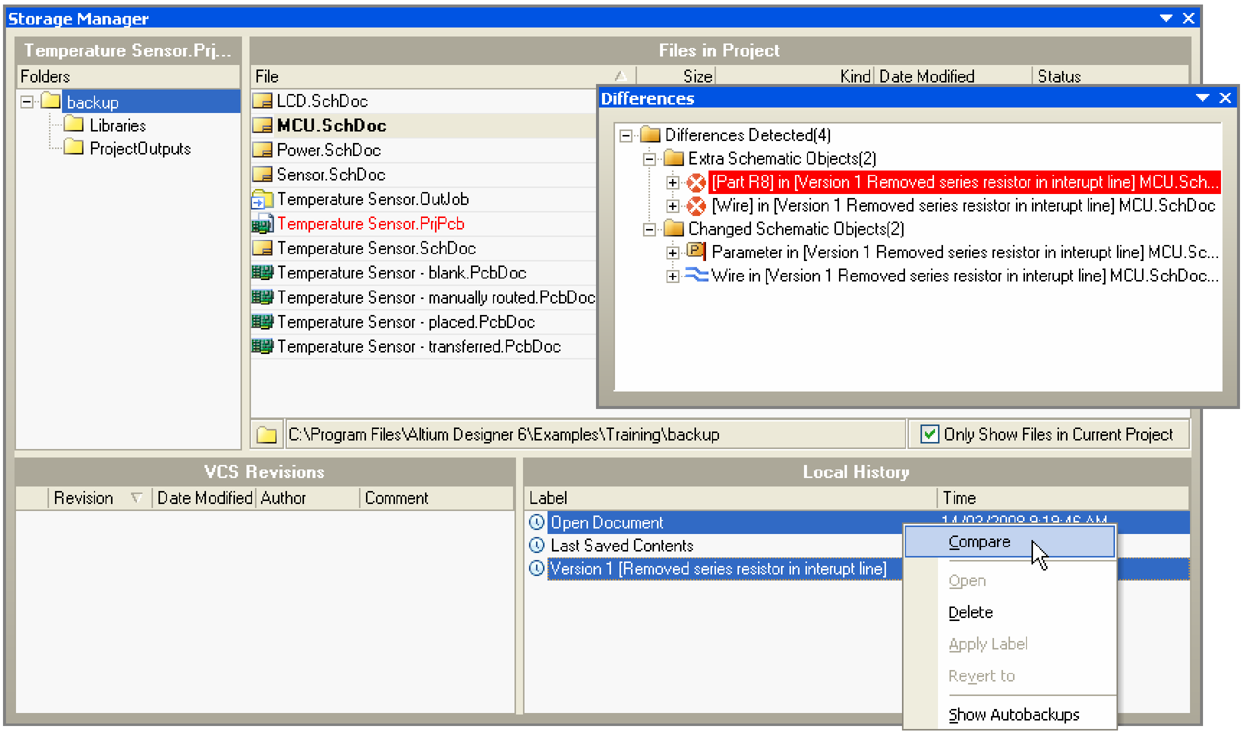
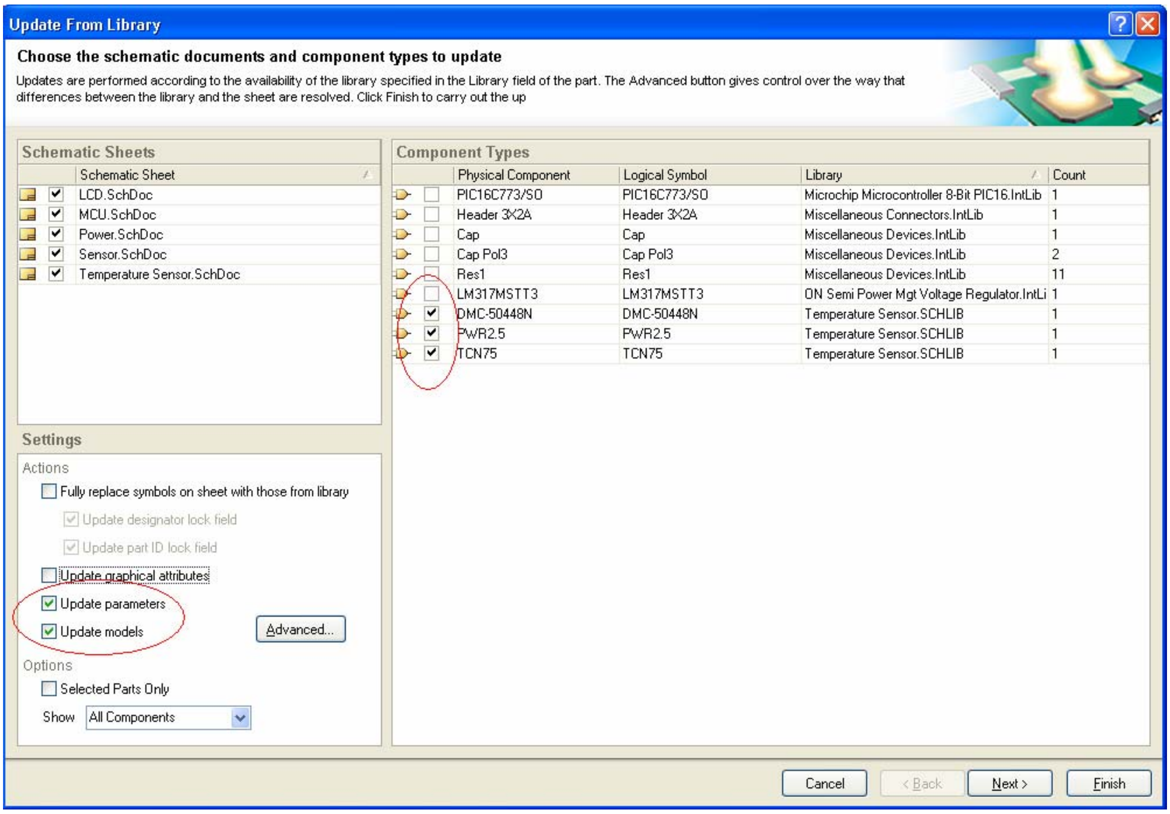


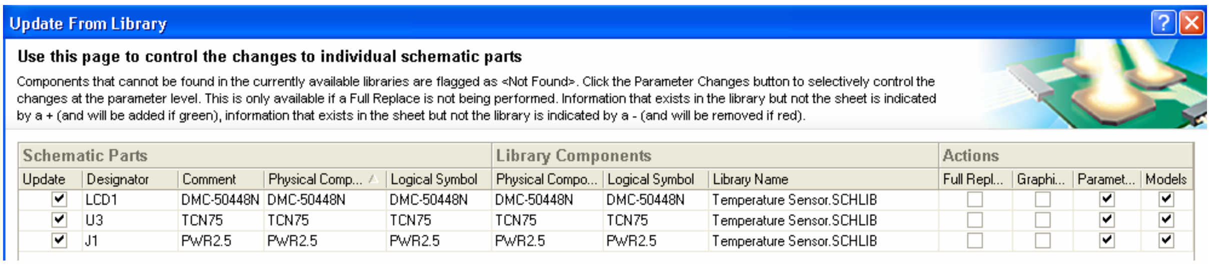

*Figure 13. Using the Storage Manager to compare two different revisions of the same schematic document. The difference panel details all of the differences found, double click to display the affected objects.*

## 17.2.4 Exercise – Using the new component in schematic

1. Open Sensor.schdoc.
2. Select the command **Tools » Update from Libraries**.
3. We are interested in updating not only the TCN75 component, but also the other components from the temperature sensor.schlib, as there have been other parameters and model changes. In the **Update from Library** wizard, set the checkboxes so that only components from the *Temperature sensor.schlib* are updated. Set the update Actions to only update models and parameters, using the checkboxes down the bottom left of the wizard.



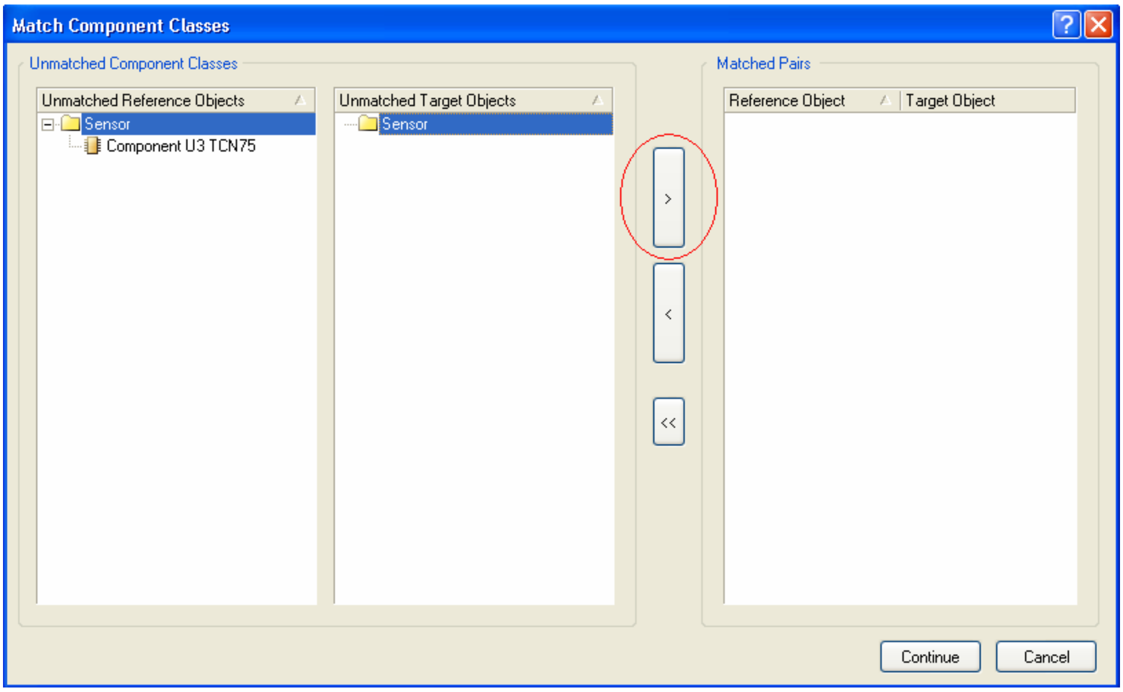

4. Click **Next** and confirm that the components are being updated from the correct library.



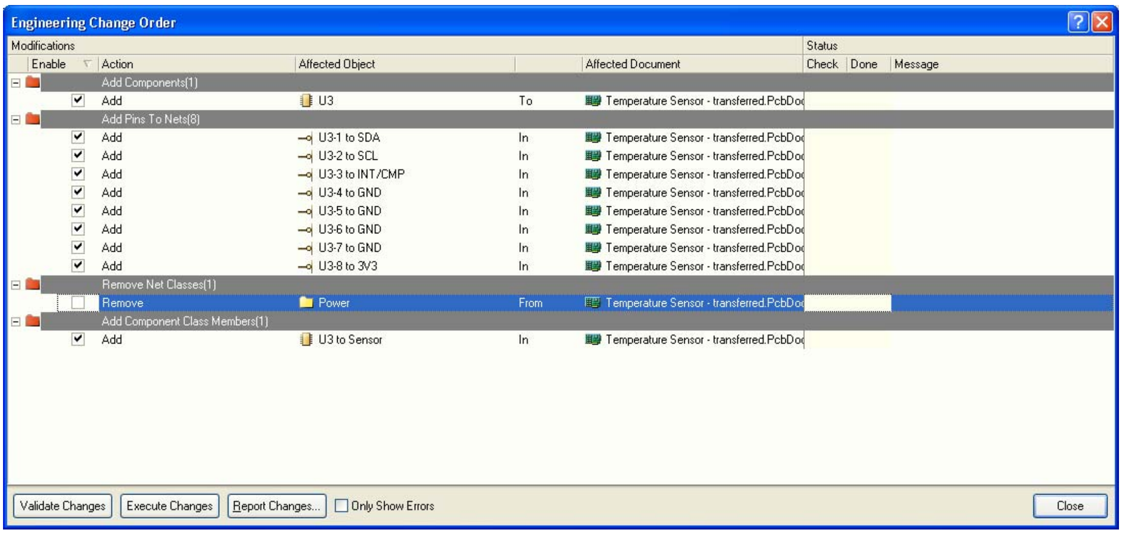

5. Click the **Finish** button to create the ECOs needed to update the selected component model and parameter data, and **Execute** the ECOs to make the changes.
6. Once the update is complete, right-click on the TCN75 symbol, and select **References » Help** from the floating menu. This should open the PDF document that was linked in the Exercise – adding a component parameter. Note that this link can also be pressing **F1** when the cursor is over the TCN75 component.
7. Save all the schematic sheets.
8. Since a new component has been added, we need to update the PCB. Select the **Design » Update PCB** command from any schematic document.

9. Since a new component has been added, it should first detect unmatched component classes, if component classes exist on the board. If the *Match Manually* dialog appears, click **Yes** and continue with step 10. If it does not appear, go straight to step 11.

10. In the *Match Component Classes* dialog, select **Sensor** in both of the **Unmatched** columns, and click the arrow to match them. Click **Continue**.




11. Since a NetClass called **Power** was created in *Module 12 - Design Rules*, you will need to disable the change to **Remove Net Classes (Power)** in the *Engineering Change Order* dialog.




12. **Validate** and **Execute** the remaining changes to add the new SOIC8 footprint to the PCB, close the ECO dialog, and position the SOIC8 on the board. If you find the **Add Pins to Nets** ECOs do not validate, run the Update process again.

> **Note:** To prevent that NetClass being removed by future updates, you can go to the **Comparator** tab in the *Project Options* dialog and disable the checking for **Extra Net Classes**.

## 17.3 3D PCB Components

Given that many boards must fit into unusually shaped and tight-tolerance housings, it is a common requirement that the loaded board be transferred to an M-CAD system.

- With the introduction of 3D visualization in Altium Designer 6 it became possible to display the board in 3D within the PCB editor, and export the board as a STEP format file.
- You can also import STEP models into footprints in the PCB library, creating a complete E-CAD to M-CAD 3D solution.
- Component shapes can be modeled using Altium Designer's 3D body objects, or by importing a STEP format component model. Both types of models export into the board STEP file.
- Alignment tools are included to orient the STEP model with the footprint.

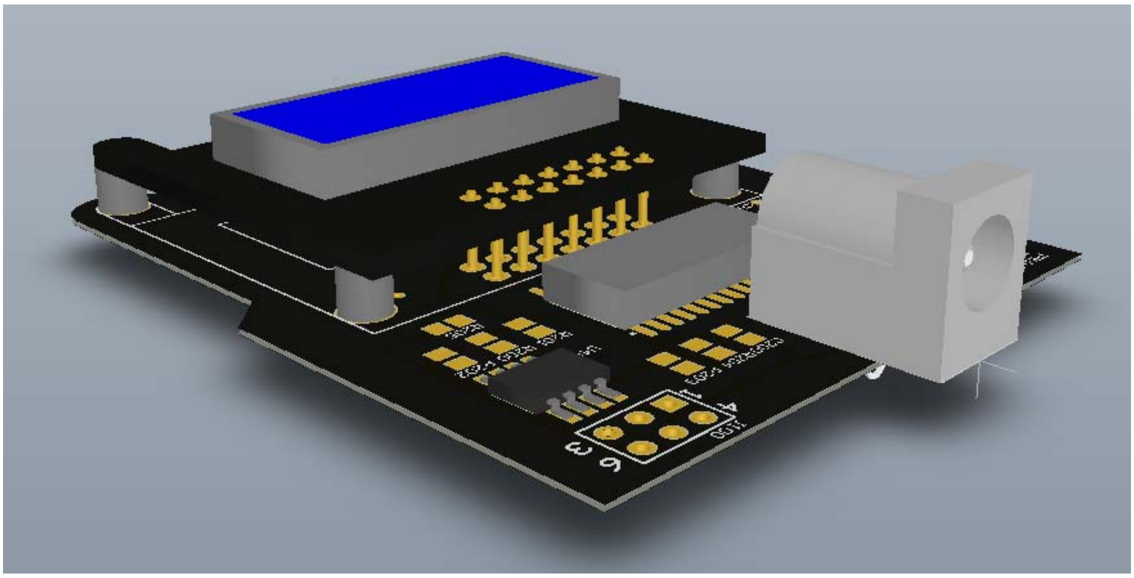


*Figure 14. The completed board can be displayed in a highly realistic 3D view, and then exported.*

| |
|---|
| **Note:** 3D visualization requires a graphics card that supports DirectX 9 (or better), and Shader Model 3 (or better). |

### 17.3.1 Using Altium Designer's 3D Body Objects

3D body objects can be used to model a component's shape. They are ideal when there is no STEP model available, or you do not need a precise shape for a particular component.

- Complex component shapes can be built up by placing multiple body objects onto a mechanical layer. Use the **Place » 3D Body** command in the PCB library editor.
- Shapes can be placed manually, or using the *3D Body Manager*. The *3D Body Manager* (Library editor **Tools** menu) is an ideal way of creating initial shapes, which can then be modified and supplemented by manually placed 3D bodies.
- Use the **2** and **3** shortcut keys to toggle the library editor display between 2D and 3D modes.
- The PCBLib List panel is ideal for reviewing the 3D body objects in a footprint, and editing their height and color settings. This can be done with the footprint displayed in either 2D or 3D, if it is in 3D mode the changes can be examined immediately.

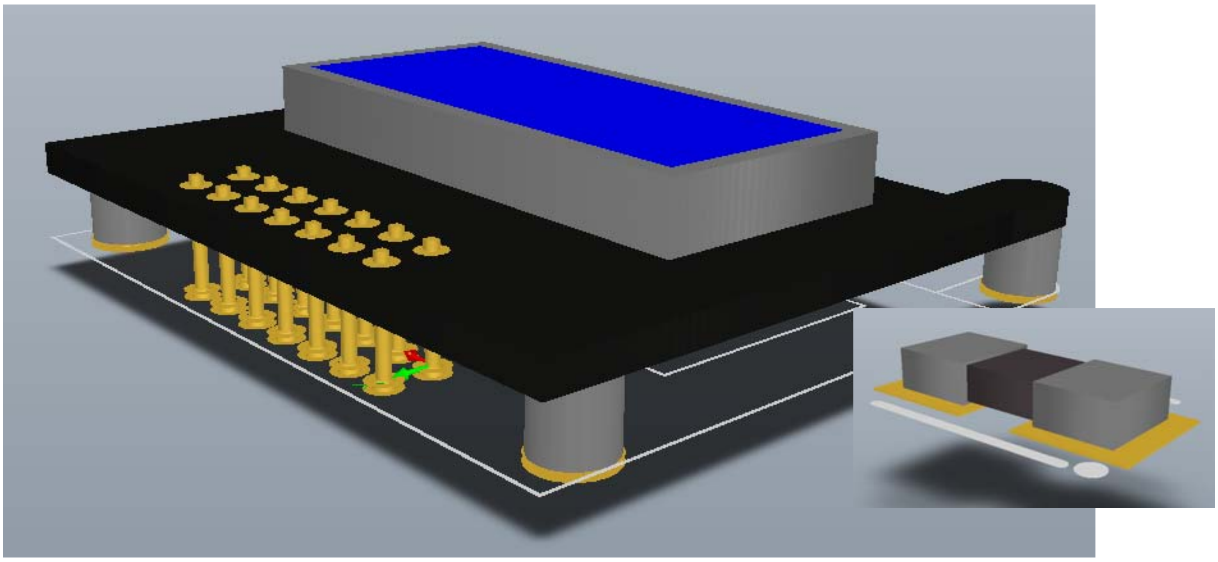


*Figure 15. Component shapes created by placing multiple 3D bodies in the PCB library editor.*

## 17.3.2 Importing a 3D STEP Model

STEP is a popular data exchange format, supported by all the major M-CAD packages. There are different versions of the STEP format, including 203 and 214. Note that the 203 format does not support color information, models in this format will display in Altium Designer in a pale grey color.

- STEP format component models are imported into an Altium Designer 3D body object, using the **Place » 3D Body** from STEP Model command.
- Use the **2** and **3** shortcut keys to toggle the library editor display between 2D and 3D modes.
- You will often find the STEP model has been built using a different orientation than the Altium Designer footprint; there are a number of orientation and alignment tools to help resolve this.
- While they may have different orientations, STEP models often have the same origin as the PCB footprint. Adding a snap point to the model's origin can help aligning them, the easiest way to do this is in the *3D Body* dialog (double click on the Body object containing the imported STEP model), and click the **Add** button.

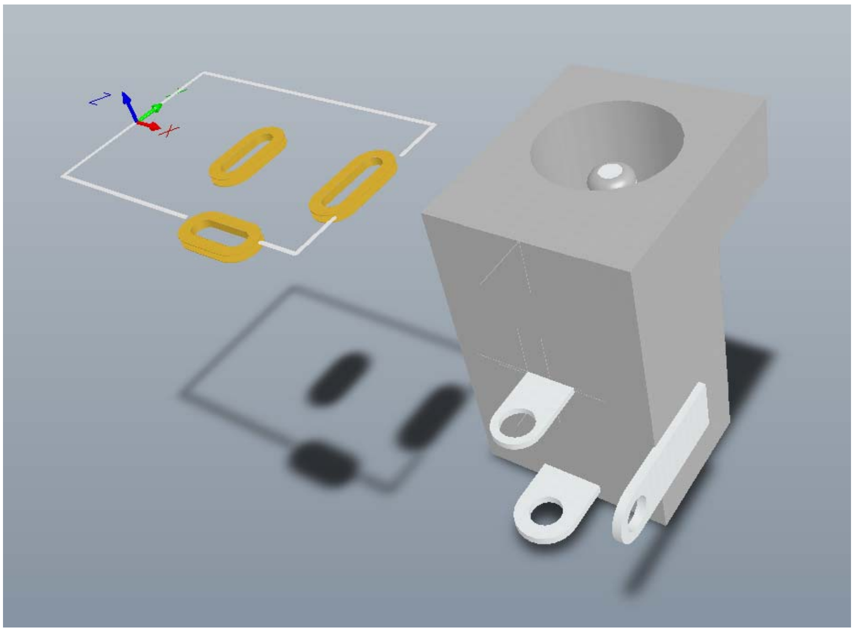


*Figure 16. STEP model imported into a footprint, note that it has a different orientation from the footprint.*

## 17.3.3 Rotating and Positioning the STEP model

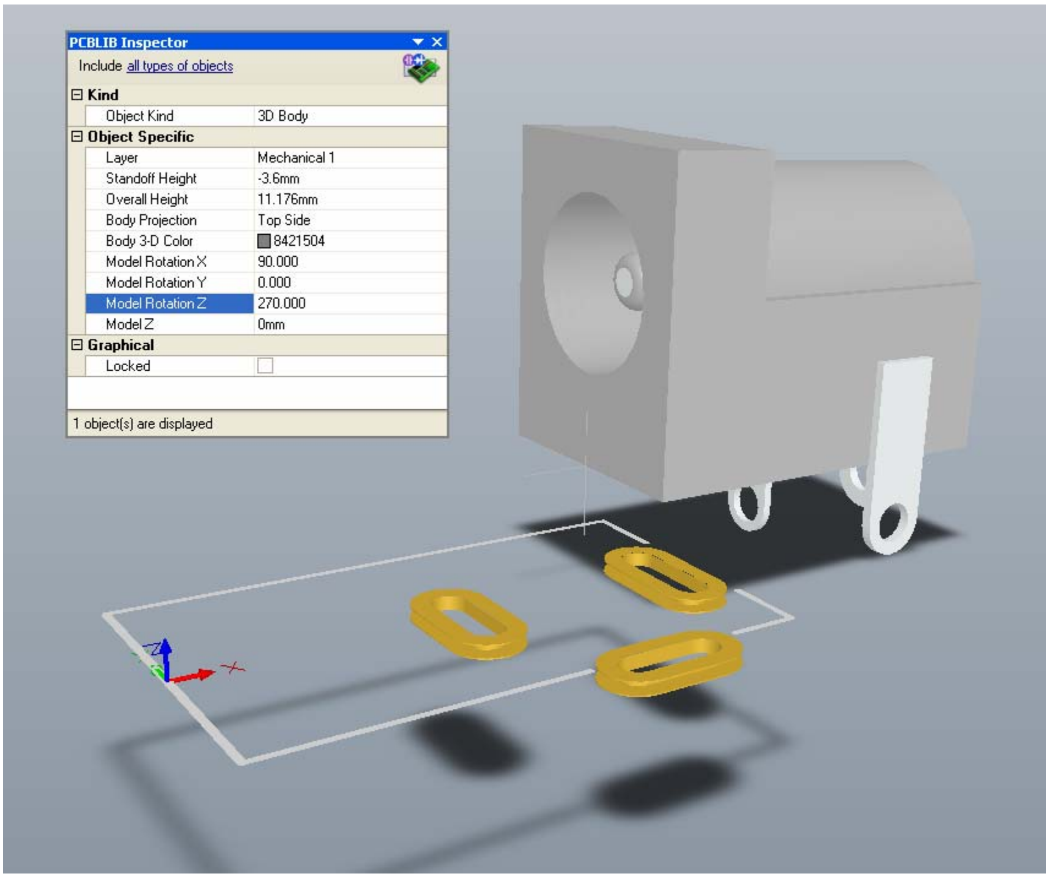



*Figure 17. Use the Inspector to test different rotation values.*

- The Inspector panel is ideal for experimenting with rotation values while the footprint is in 3D view mode.
- Use the **M, M** shortcuts to move the STEP model, click on the model origin as the holding point.
- Use the **J, R** shortcuts to jump to the footprint reference point while the STEP model is being moved, press **Enter** to place it.
- STEP models can also be rotated using the numeric keypad as the model is being moved. Press 2 & 8 for X axis rotation, 4 & 6 for Y axis, and Spacebar & Shift+Spacebar for Z Axis. These shortcuts are detailed in the Shortcuts panel (when the model is being moved).
- Any number of snap points can be added to the model, to facilitate moving and aligning it with the footprint. Use the interactive commands in the **Tools » 3D STEP Body Placement** submenu to do this (available when the footprint is 3D view mode).
- The **Add Snap Point from Vertices** command has 2 modes: use it to add a snap point on the chosen vertex, or press the **Spacebar** to toggle to the midpoint mode, where the snap point is added midway between the 2 vertices you click on. This mode is ideal for adding snap points to the center of component pins. Keep an eye on the Status bar for details.

> **Note:** as well as component manufacturer websites, 3D models can be downloaded from http://www.3dcontentcentral.com/. Try multiple different search strings to get the best results.

## 17.3.4 Exercise – importing a STEP model

In this exercise we will import a STEP model for the power jack that has been downloaded from the 3DContentCentral website.

1. Open the Temperature Sensor.PcbLib, and select the footprint PWR2.5.
2. Make Mechanical 1 the active layer.
3. Select **Place » 3D Body** from the menus, and in the *3D Body* dialog:
   - set the **3D Model Type** to **Generic STEP Model**,
   - then click the **Embed STEP Model** button,
   - then in the *Choose Model* dialog, navigate to the folder `\Altium Designer Summer 09\Examples\Training\PCB Training\Temperature Sensor\Libraries\3D Models`, and select the model `power_jack.step` and click **Open** to close the *Choose Model* dialog
   - then click **OK** to close the *3D Body* dialog.
4. A box will appear floating on the cursor, position it anywhere just off to the right of the footprint.
5. The *3D Body* dialog will re-appear, click **Cancel** to close it.
6. Press the **3** shortcut to display the footprint in 3D mode and use **Shift + Right Mouse Button** to rotate the model slightly.

   **Note:** 3D visualization requires DirectX 9 and Shader Model 3 graphics card support.

7. Press **L** to open the *View Configuration* dialog. In the dialog, enable both the **Show Simple 3D Bodies** and **Show STEP Model** options. Use the **Save As View Configuration** option to save these settings as your own view configuration. This will now be available in the drop down list of configurations at the top of the PCB workspace.
8. The model will have a different orientation to the footprint, as shown in Figure 16. Before attempting to re-orient it, we will add a Snap Point to the model origin. To do this, double click on the upper-most face of the model to open the *3D Body* dialog. Click the **Add** button to add a Snap Point, and **OK** to close the dialog.
9. To re-orient the model using the Inspector panel, press **F11** to open the panel. If the panel is blank click once on the model to select it, loading its attributes into the Inspector.
10. Experiment with different values in the Model Rotation fields (as shown in Figure 17). Keep in mind that you are rotating *around* the axis that you are editing.
11. Once the model has the correct orientation (Model Rotation X=90, Model Rotation Y=0, Model Rotation Z=270), it is time to re-position it on top of the footprint. To do this press **M, M** to run the Move Object command, the cursor will change to blue.
12. We want to pick the model up by the snap point at its origin, which is indicated by the cross hair on the front face. Be careful as you click to select this point, you can pick up a model by any default vertex, or by any user-defined snap point. Since there is a default vertex at the mid-point of every edge, you may find that when you click you are holding the model by the vertex that is just next to the snap point. Zoom in (**Ctrl+roll**) to confirm that you are holding it by the snap point.
13. Once you are moving the model by the snap point, let go of the mouse, press **J, R** on the keyboard to jump to the footprint reference, and press **Enter** on the keyboard to position the model. If you turn the footprint+model over (**Shift + Right Mouse Button**) you will see that the component pins are centered in the footprint pads. Once positioned, press **Esc** or click **Right Mouse Button** to drop out of the Move Object command, the cursor will change from blue back to orange.
14. To update the footprint on an open PCB, right-click on the footprint name in the PCB Library panel, and select **Update PCB With PWR2.5**.

# 17.4 Library Package types

The Schematic PCB libraries have been created, and all the models have been linked to the schematic library components. If required, we are now at the stage were we can link everything together and create a single library. There are different ways this can be done in Altium Designer, each with its own advantages. Possible library options include:

- Separate schematic and PCB libraries
- an integrated library, or IntLib – a single compiled library that is, compact, secure and portable
- A Database library, or DBLib – interfaces to your company component database
- An extension to the database library, called a Subversion database library – interfaces to a component database, and subversion, a free version control system.

For this course we are only going to cover Integrated libraries. For database libraries and SVN database libraries refer to the Advanced PCB course for more information.

## 17.4.1 Integrated Libraries

An integrated library is actually the compiled output from a Library Package. The Library Package is essentially a project file for libraries.

- To create a new library package, select **File » New » Project » Integrated Library** from the menus.
- Source schematic and PCB libraries can be added to the new library package by dragging and dropping them in the Projects panel. Hold Ctrl as you drag and drop if you want to have the source libraries in both the PCB project and the library package.
- To compile the Library Package and create the Integrated Library, right click on the library package in the Projects panel and select **Compile Integrated Library** from the floating menu.
- The compilation process checks that all specified models are available and the linking between symbol and model pins. Check the Messages panel for any compilation Warnings or Errors.
- Even if you choose not to deploy and use the compiled integrated library, compiling is an excellent way of checking the mapping between each symbol and its various models.
- Integrated libraries are also very compact, usually 1/8 to 1/10 the size of the original documents.
- Updating components in a design from Integrated Libraries is faster.

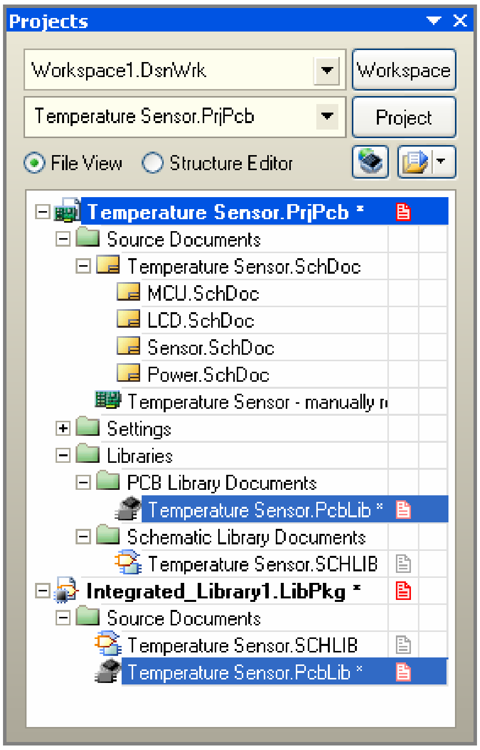



*Figure 18. The Schematic and PCB libraries as source files in the library package.*

- An integrated library can be placed on a network drive and used throughout your company.
- Although the integrated library is compiled, the source libraries can be extracted if necessary. When you attempt to open an Integrated Library (like a file, instead of installing it in the Libraries panel like a library), you will be asked if you wish to extract the sources, as shown in Figure 19.
- Only source schematic and PCB library files are added into the created library package as source files.

- All other model kinds, such as simulation MDL and CKT files, are extracted, you will find them in the same folder as the extracted symbols and footprints. They are included in the project using a Search Path, defined in the Project Options for the library package.

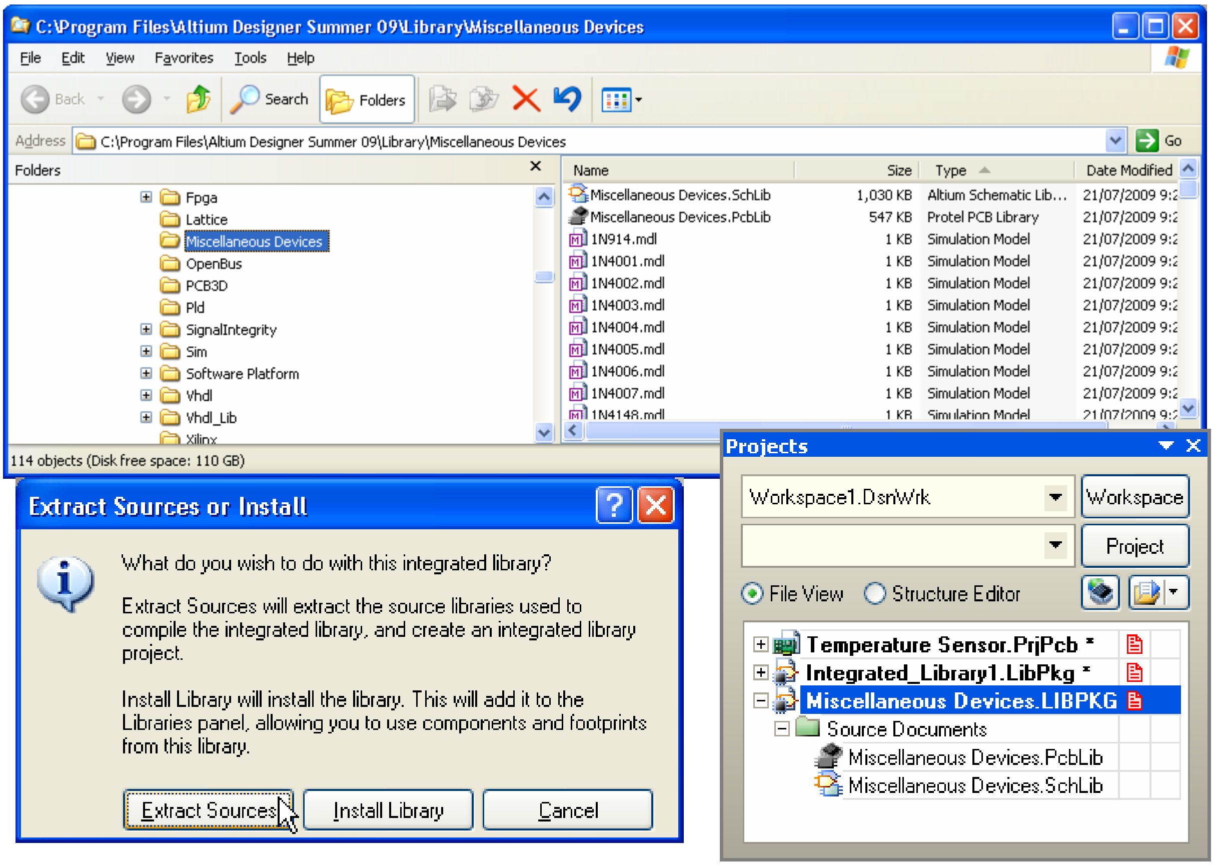



*Figure 19. Result of extracting the contents of the integrated library, Miscellaneous Devices.*

# 17.5 Library Reports

There are several reports available in both the Schematic Library Editor and Schematic Editor that are used to record component, library and project information, such as a Bill of Materials (BOM).

## 17.5.1 Library Editor Reports

There are three reports that can be generated in the Library Editor. All have the syntax `Library_Filename.extension`.

- The **Reports » Component** command generates a `.CMP` file that includes the component name, part count, components in the same group, details of each part and details of all pins. This report can be used to verify that the component has been correctly constructed.
- The **Reports » Library List** command generates a `.REP` file which includes a component count as well as the name and description of each component in the library. This report can be used to create a listing of the components in a library, handy if you need a printed reference of a library.
- The **Reports » Component Rule Check** command opens the *Library Component Rule Check* dialog (Figure 20). These allow you to test for:
  - duplicate component names and pins
  - missing description
  - missing footprint
  - missing default designators
  - missing pin name
  - missing pin number
  - missing pins in sequence.

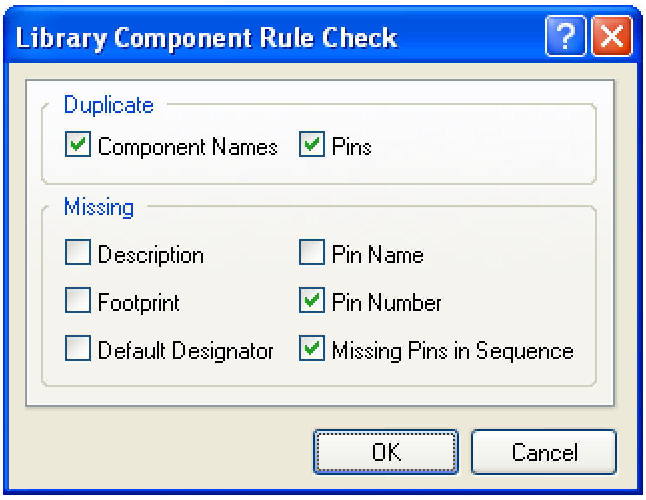



*Figure 20. Library Component Rule Check dialog*

Running this generates a `.ERR` report which reports on all components in the active library. This report can be used to aid in library verification and library management.

- The **Report » Library Report** command presents the Library Report Settings, which allows you to generate either a Microsoft Word® document, or an HTML document. An example report is shown in Figure 21.

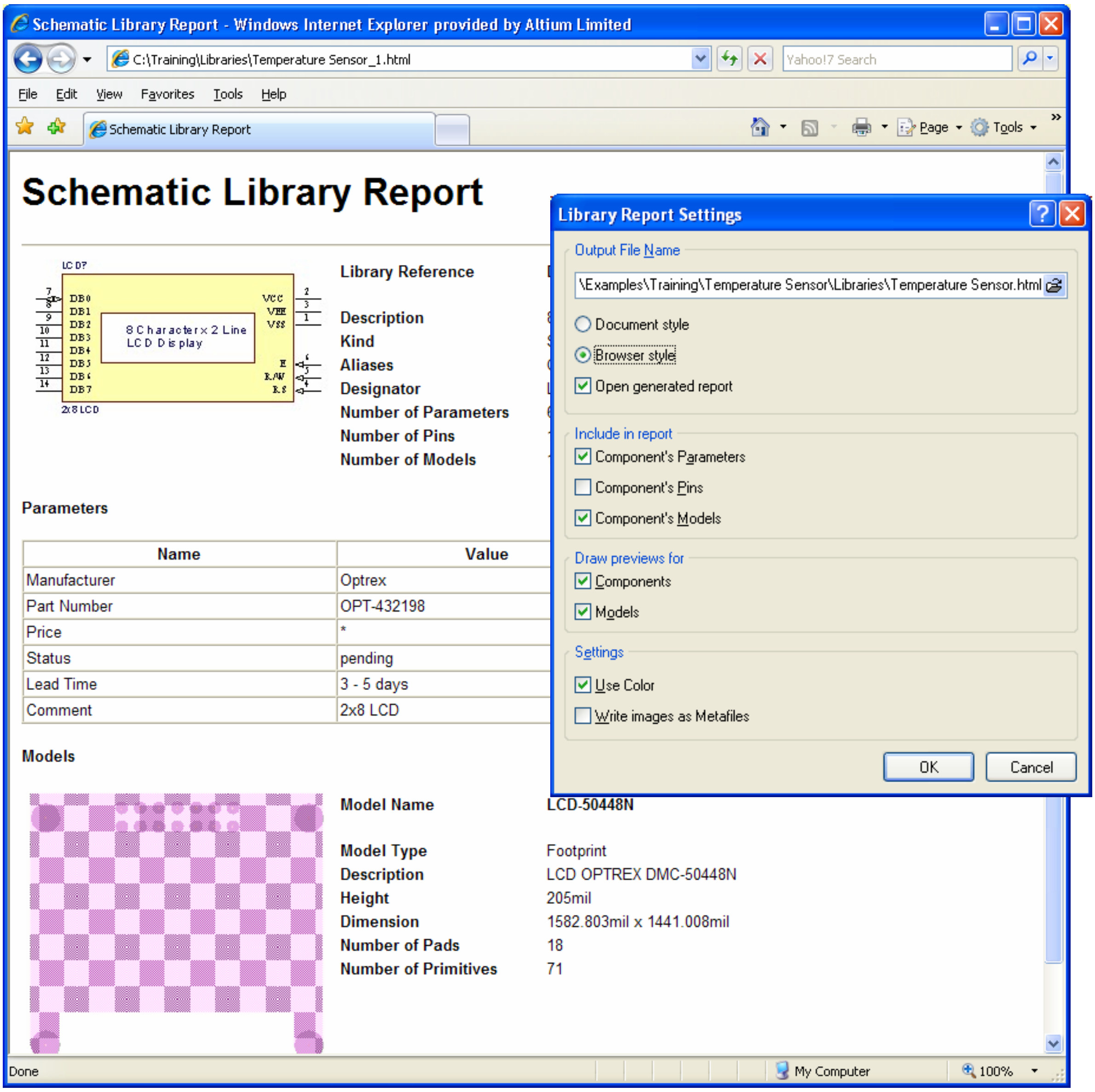



*Figure 21. Library report, and the Library Report Settings dialog*

**Note:** You can also generate a library report from the Libraries panel. Right-click on a component in the panel and select **Library Report** from the context menu.

TOP SECRET

Altium Designer

# Module 18: Routing and Polygons

# Module 18: Routing and Polygons






Module Seq = 18

# 18.1 Routing

## 18.1.1 Interactive routing

Routing is the process of defining connective paths between the nodes in each net.

Altium Designer includes a powerful Interactive Routing engine to help you efficiently route your board. There are two interactive routing commands, both are launched from the **Place** menu.

- **Interactive Routing** – you place track segments to route the selected connection. The routing engine attempts to find a path from the start of the connection (or last click location), to the current cursor location. The path it finds depends on the current routing mode, you can choose between: *Walkaround*, *Push*, *HugNPush* or *Ignore*. When you click, all segments will be placed (except the last one if the look-ahead option is enabled). You can also auto-complete the connection up to the target pad by holding the Ctrl key as you click, if the routing engine can identify a path. Existing routes can also be re-routed by simple placing new segments, with old redundant routing being removed when you finish defining the new route path (if the loop removal option is enabled).
- **Differential Pair Routing** – this command is used to route a pair of nets simultaneously. To do this, the nets must be defined as a differential pair.
- Once you have chosen one of the interactive routing commands, click on a connection line to commence routing that connection. Interactive routing shortcuts can be accessed at any time during routing by pressing the Shift+F1 keys, or by displaying the **Shortcuts** panel.

### 18.1.1.1 Managing connectivity

Once components are placed into a PCB file, *connection lines* display to indicate which pads belong in each net, and must be routed to create the connectivity defined in the schematic.

- Whenever there is an operation on a copper layer that affects connectivity, the PCB Editor analyzes the PCB to determine if any connections have changed. If you have routed a connection (joined 2 pads with track segments on a copper layer), the connection line between those 2 pads is no longer displayed. Also, if a shorter path for any connection is possible because of a routed connection, a shorter connection line is displayed.
- The arrangement or pattern of the connection lines in a net is called the *topology*. The default topology for all nets in a board is Shortest, as determined by the applicable Routing Topology design rule. Because it is shortest, as you move components around the connection lines may jump from one pad in the net to another pad in the net, maintaining the shortest possible length of connection lines for that net.
- You can change the color of the connection lines for a net in the *Edit Net* dialog, double click on the net name in the **PCB** panel to open the dialog.

### 18.1.1.2 Interactive Routing track width

When you select one of the **Interactive Routing** commands and start routing, the track width that you start with is determined by the **PCB Editor – Interactive Routing** settings in the *Preferences* dialog, working in harmony with the applicable Width Constraint design rules.

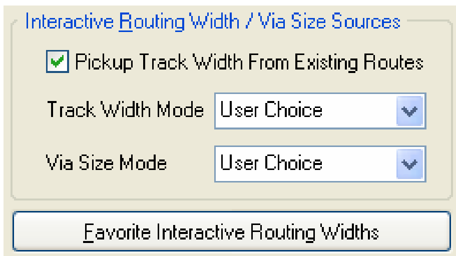



*Figure 1. Interactive routing behavior is determined by these settings.*

While the preferences allow you to change the width as you route, it is always constrained by the applicable rule – if you attempt to change it outside the range defined by the rule it will automatically be clipped back to the rule min or max, whichever is closer.

### Track Width / Via Size Mode

- **User Choice** – With this mode enabled the routing width is selected from the list of favorite widths, press Shift+W while routing to display the list. Use the **Favorite Interactive Routing Widths** button in the preferences dialog to configure the list.
- **Rule Minimum** – With this mode enabled the Minimum size setting in the applicable design rule will be used.
- **Rule Preferred** – With this mode enabled the Preferred size setting in the applicable design rule will be used.
- **Rule Maximum –** With this mode enabled the Maximum size setting in the applicable design rule will be used.

**Note:** You can cycle between the above modes while interactive routing by pressing the **3** (for Track Width) or **4** (for Via Size) shortcut keys, the current setting is indicated on the Status bar.

## 18.1.1.3 Editing during Routing

As well as SHIFT+W to change the track width, there is another level of editing available as you route. Pressing the TAB key will open the *Interactive Routing for Net* dialog (Figure 2), where you can configure many of the interactive routing options, as well as edit the routing width and via size attributes.

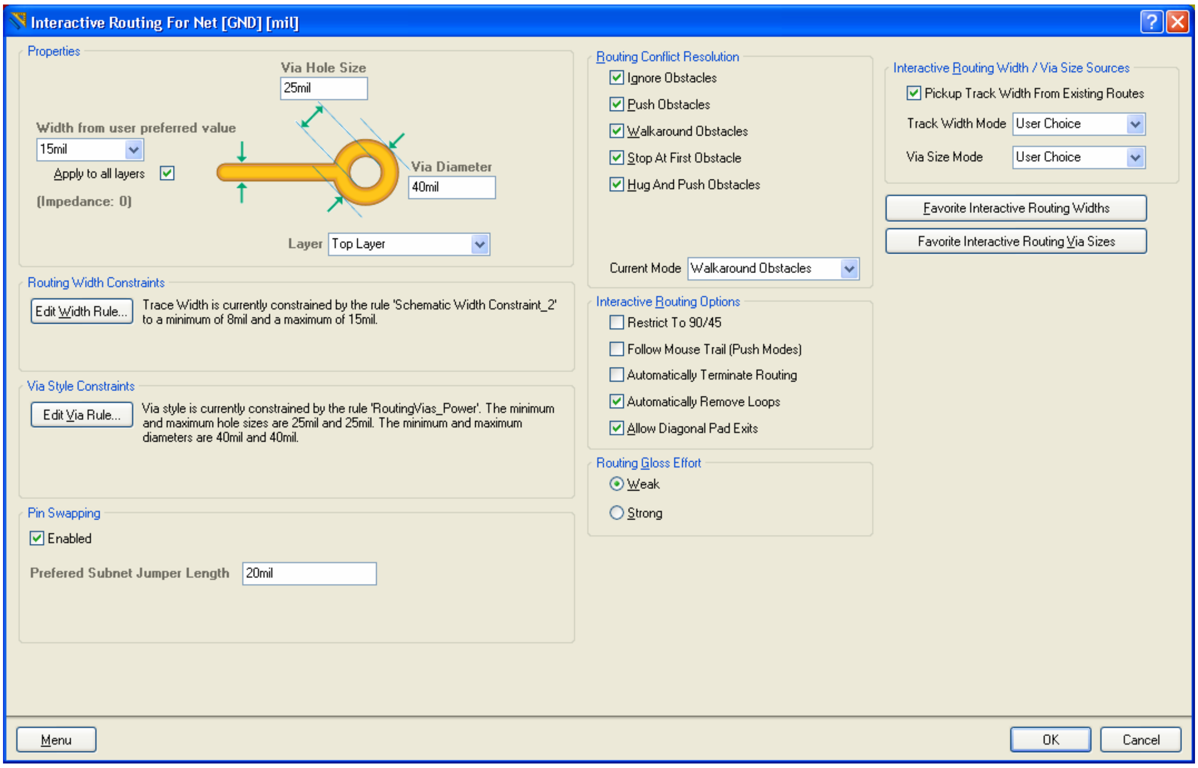



*Figure 2. Interactive Routing dialog*

### 18.1.1.4 Handling conflicts during Interactive Routing

As you route interactively you will be placing track segments amongst other objects that are already on the board. You can control how Altium Designer should handle a potential routing conflict. The conflict resolution mode is set in the **PCB Editor – Interactive Routing** page of the *Preferences* dialog, the applicable settings are shown in **Error! Reference source not found.**.

Conflict resolution modes include:

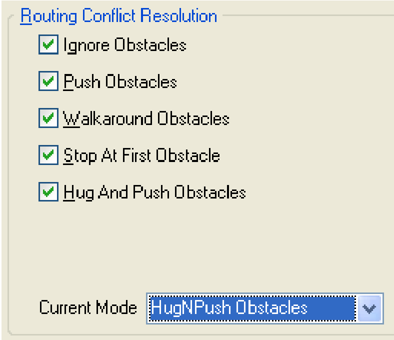



*Figure 3. Define how interactive routing conflicts are handled.*

- **None –** this is the Ignore mode, where conflicts are permitted. You can route over the top of existing objects. Violations are highlighted.
- **Push Conflicting Objects –** in this mode all existing tracks and vias will be pushed to make room for the new route.
- **Walkaround Conflicting Object –** in this mode the new route will *walk around* existing obstacles, or jump them if possible. As you move the cursor, the routing engine continually attempts to find the shortest path from the last click location to the current cursor location – click to define intermediate locations if you don't like the calculated path.
- **Hug And Push Conflicting Object –** in this mode the routing engine will follow existing objects, and only push them when there is insufficient room for the track being routed. In this mode the route path tends to follow the path you draw with the cursor.
- **Stop At First Obstacle –** in this mode the routing engine will stop at the first obstacle that gets in the way.

| **Note:** Press the **Shift+R** shortcut keys to cycle through the different modes while you are routing, keep an eye on the status bar to see which mode you are currently in. |
|---|

### 18.1.1.5 Additional Interactive Routing Options

Altium Designer's routing capabilities have been developed to make the routing process efficient. There are another set of options that go toward that efficiency, which are also set in the **PCB Editor – Interactive Routing** page of the *Preferences* dialog (Figure 4).

These include:

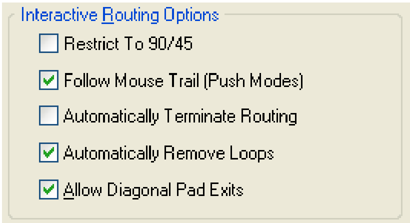



*Figure 4. Additional interactive routing options.*

- **Restrict to 90/45 –** there is a total of 5 possible routing corner modes, cycled through as you press SHIFT+SPACEBAR during interactive routing. Enabling this option will restrict this list to 2, you will only choose between 90 degree or 45 degree corners.
- **Automatically Terminate Routing –** with this option enabled, when you click on the target pad both the current track segment and the look-ahead segment are placed and you are automatically *released* from that route, ready to start on another connection.
- **Automatically Remove Loops –** with this option enabled, loops that are created during manual routing are automatically removed.

| **Note:** Automatic Loop Removal can be disabled on an individual net if you require routing loops in that net. Double-click on the net name in the PCB panel to access the net properties to alter this setting. |
|---|

- **Hug Existing Traces (Walkaround Mode) –** with this enabled walkaround mode still attempts to find the shortest path from the last click to current cursor location, but makes hugging existing objects a higher priority than shortest distance.

### 18.1.1.6 Routing Gloss Effort

**Weak** or **Strong** are the two Options available here. The Routing engine will either optimise or tidy-up routes 'weakly' or 'strongly'. *Glossing* is the term used to describe how much of a 'clean up' the Routing engine should undertake.

### 18.1.1.7 Look-ahead routing

The PCB Editor's interactive routing mode incorporates a *look-ahead* feature that operates as you place tracks during routing. The track segment that is connected to the cursor is a look-ahead segment. The segment between this look-ahead segment and the last-placed segment is the current track that you are placing.

If look-ahead mode is active, you can use the look-ahead segment to work out where you intend to place the next segment and to determine where you wish to terminate the current segment. When you click to place the current segment, its end point will be positioned exactly where you need to commence the next segment. This feature allows you to quickly and accurately place tracks around existing objects and plan where the next track segment can be placed.

As you use the look-ahead segment to guide your routing, you will notice that the track end does not always remain attached to the cursor, it clips as you approach an existing obstacle (if the conflict resolution mode is set to stop at first conflicting object). This feature prevents you from violating any clearance constraints.

**Note:** The look-ahead mode can be toggled off and on while interactively routing by pressing the 1 key. If look-ahead is off each click will place both the 2nd last and the last track segments.

### 18.1.1.8 Working with the Electrical Grid

Whenever you are placing an electrical object, like a track during routing, the Electrical grid is active. An octagonal graphic on the cursor indicates that the Electrical Grid is in operation, pulling the cursor to an existing object on the board. This feature is ideal for routing to off-grid pads. You can inhibit the electrical grid if there is a situation where it is working against you; hold the CTRL key during interactive routing to do this.

**Note: Shift+E** cycles through the three electrical grid modes, including; off, on for current layer, on for all layers. The current state is displayed on the Status bar.

### 18.1.1.9 Changing the routing - automatically remove loops, or drag tracks

Altium Designer has 2 methods for changing existing routing: rerouting using the Interactive Routing command, and dragging track segments.

- **Loop removal** is a feature that automatically removes redundant track segments as you re-route a connection. Using loop removal you can easily re-route existing routing, as soon as you terminate routing any redundant routing is automatically removed. This includes complex routes that pass through many layers, redundant vias are automatically removed along with track segments.
- **Dragging tracks**, you can also drag track segments and preserve the 45 angle to the adjoining track segments. To do this first click to select the segment and the special cursor will indicate the mode (Figure 5). Then click and drag to move the segment. Alternatively, instead of clicking once to select the track segment first, hold the CTRL key as you click and drag on the segment.

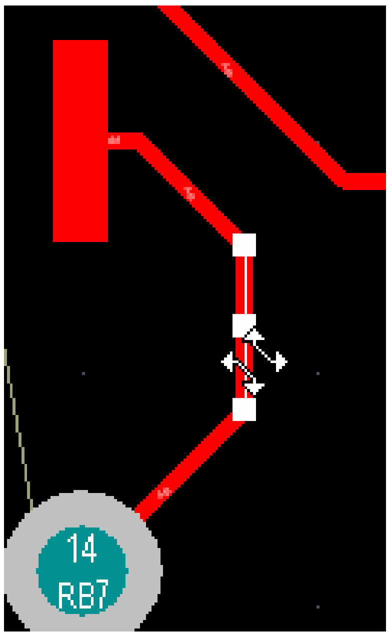



*Figure 5. Note the special cursor, indicating that corner angles will be preserved when the selected segment is dragged.*

- **Dragging Arcs**, you can also drag arc segments and preserve their concurrency. Simply click to select the routing segment(s) you wish to drag – the cursor will change to a quad arrow, see  Figure 76 for more details– and then click and drag to slide to the new location. Alternatively, use the Ctrl+click & drag shortcut to drag without having to select first.

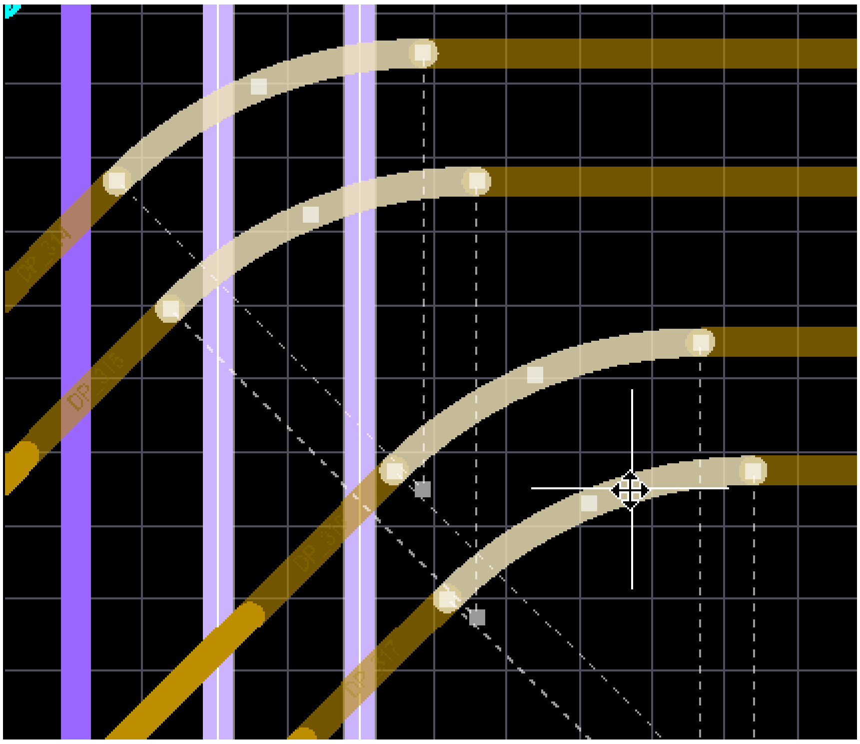


*Figure 6. Note the special quad arrow cursor that appears when dragging arcs.*

## 18.1.2 Exercise – Interactive Routing

In this exercise, you will route all the connections between the LCD module (`LCD1`) and the PIC microcontroller (`U1`).

1. Select **Place » Interactive Routing** and then, starting at the right-hand side of LCD1, route the connections from the LCD1 pads to the U1 pads.
2. Attempt to route one of the power nets.
3. As you are routing the connections, explore the various interactive routing options. Press the **~** key (or **Shift+F1**) to display them.
4. If you are going well, route the rest of the board.

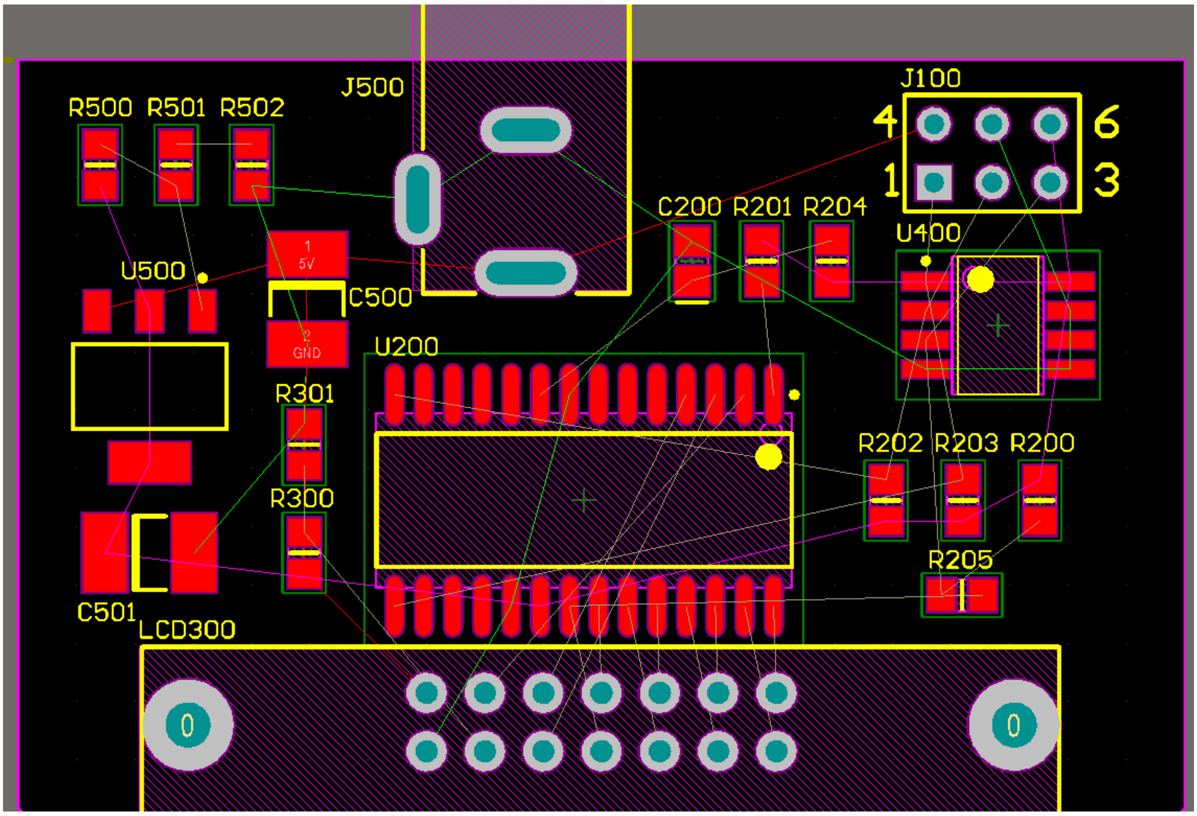



*Figure 7. The placed board, ready to route.*

### Tips for routing

- It can help to change the connection line color for important nets. To do this, double-click on the net name in the PCB panel.
- You can also control which connection lines are displayed by pressing the N shortcut to pop up a display control menu.
- Disabling the display of specific layers, such as the component overlay, can also help. Press the L shortcut to pop up the *View Configurations* dialog.
- Press the * key on the numeric keypad to switch to the next signal layer while routing.
- Press the CTRL+G shortcut keys to display and edit the current snap grid. 5 mils works well for this design.
- For a 2 layer board it is generally advisable to have one layer for predominantly horizontal routing, and the other for predominantly vertical routing.
- Press SPACEBAR during routing to toggle the start-end for the 45 degree track.
- Press SHIFT+SPACEBAR to toggle the corner mode.
- While routing a net, press the SHIFT+R shortcut keys to cycle the conflict resolution modes – keep an eye on the status bar to check the current mode.
- While routing a connection, hold CTRL as you click to automatically complete the routing of that connection.
- To examine the routing of a net, CTRL+Click on the routed net to highlight the entire net. CTRL+CLICK in free space to clear the highlight. Use the **Mask Level** button to control the fading.

## 18.1.3 Differential Pair Interactive Routing

Differential signaling is fast becoming the preferred signaling interface method, driven by the ever increasing signal speeds in electronic products. Altium Designer has excellent support for differential signaling – from defining pairs on the schematic, through to interactive differential pair routing on the PCB.

- Differential pairs are routed as a pair – that is you route two nets simultaneously. To route a differential pair select **Place » Differential Pair Routing** from the menus. You will be prompted to select one of the nets in the pair, click on either to start routing.
- Press the Tilda (~) or **Shift+F1** keys to display a list of differential pair routing shortcuts.

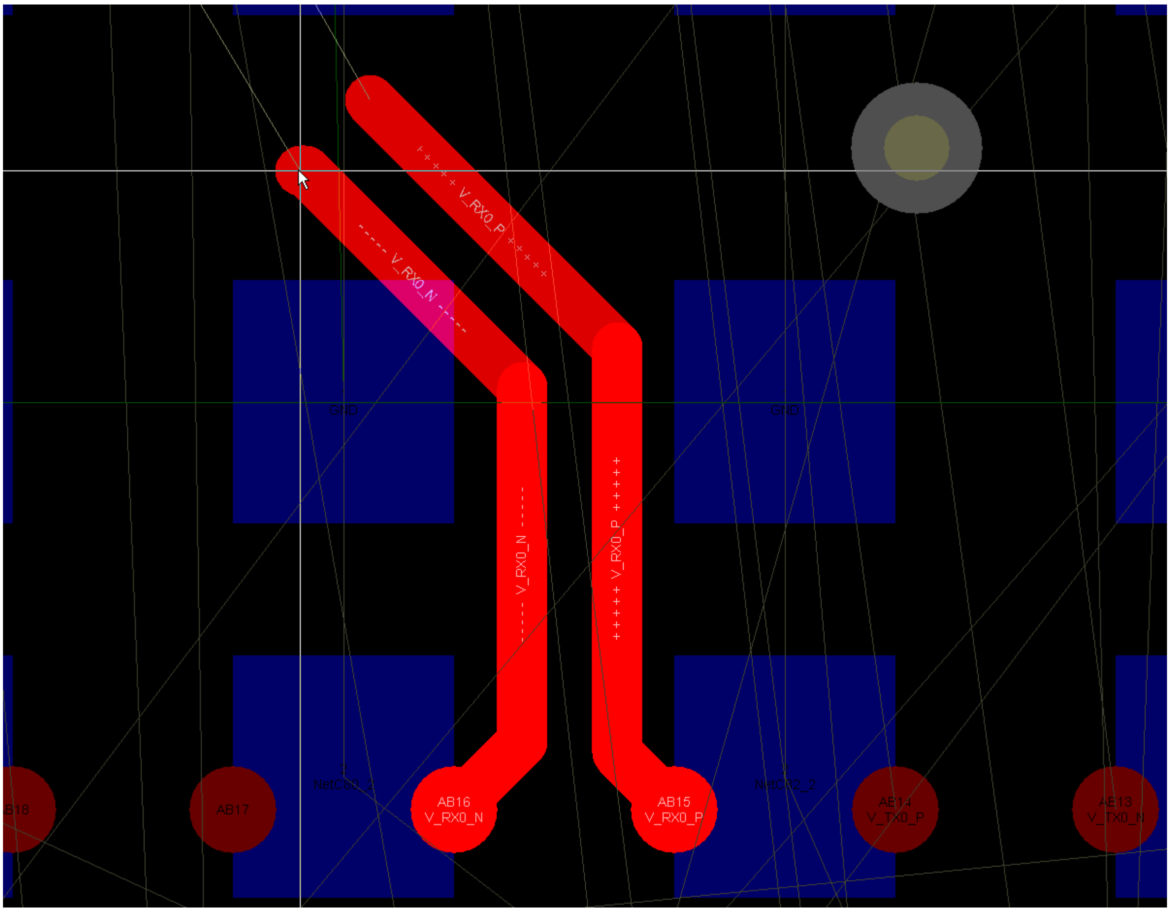



*Figure 8. A differential pair being routed, note that both connections in the pair are routed simultaneously.*

> **Note**: For more information on Altium Designer's differential pair routing capabilities, refer to the application note, *Interactive and Differential Pair Routing*.

## 18.1.4 Multi trace routing

There is two ways of using the multi trace routing, by dragging multiple track ends, or by placing multiple traces.

- The smart drag function allows a group of tracks to be selected and then extended as a single entity. You can use successive drags to continue to add new segments.

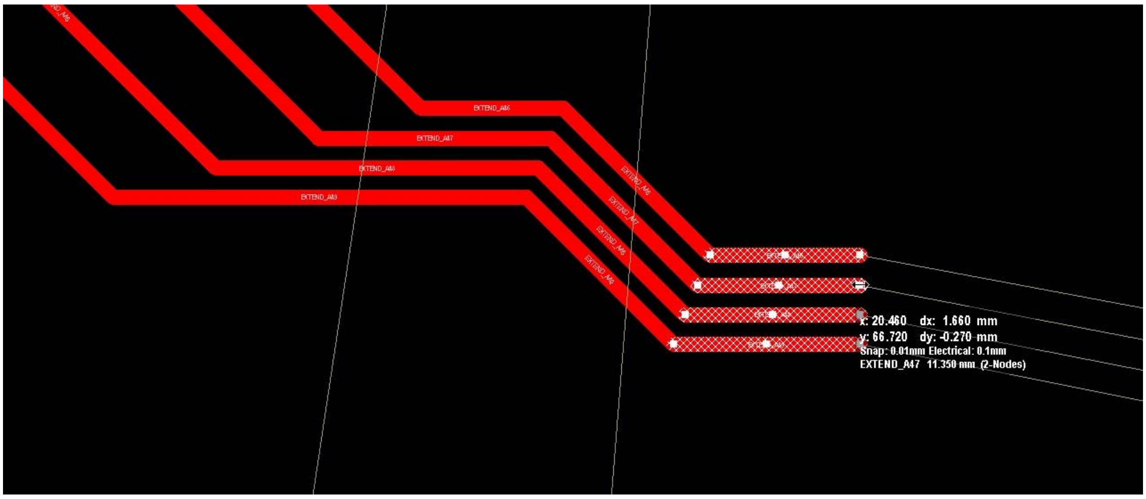



*Figure 9. The smart drag tool can be used to extend selected traces, as shown above.*

- The smart drag tool is a basic tool in that it only works on existing bus routes. The alternate approach is to use the multi trace routing tool, **Place » Interactive Multi-Routing**.
- Using this command you can start with an unrouted component and effectively pull the routing out of the *selected* component pads.
- The multiple traces are then automatically gathered together, as shown in Figure 10. Simply move the cursor around as you place the multiple traces to explore various gather options.

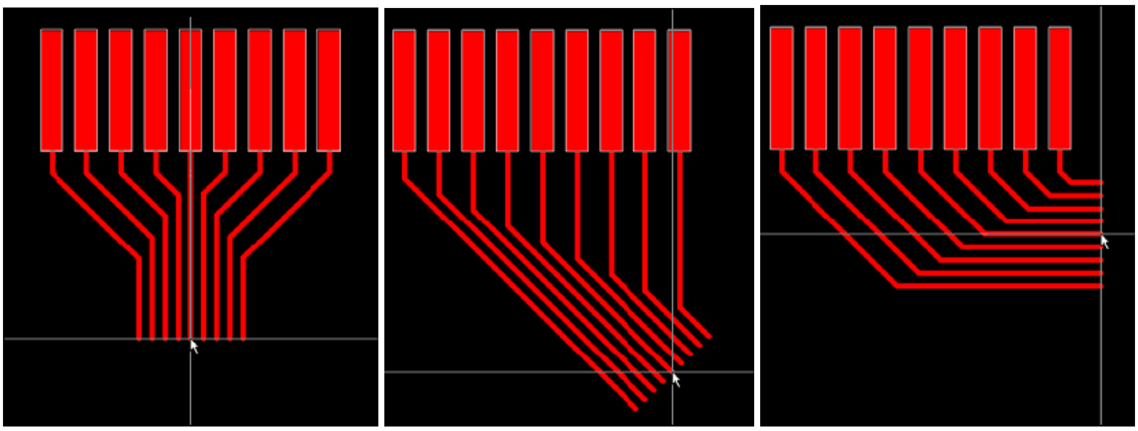


*Figure 10. Use Multiple Traces command to start from selected pads in an unrouted component. Move the cursor to explore gathering options.*

- Rather than selecting component pads one by one, hold the **Ctrl** key as you click and drag a rectangle to select multiple pads. Holding Ctrl limits the selection to the pad objects only, rather than selecting the parent component. This technique also works with the **Select Touching Line** and **Select Touching Rectangle** commands.
- Press the **Tab** key to open the **Bus Routing** dialog, where you set the Bus Spacing (track center to track center separation).

- Alternatively, use the **,** (comma) and **.** (full stop) shortcuts to interactively decrement and increment the bus spacing, in steps of the current snap grid.
- Press the **** (Backslash) to change the end alignment (once the first set of segments has been placed).
- Press the **~** (Tilda) or **Shift+F1** keys for a list of interactive shortcuts.

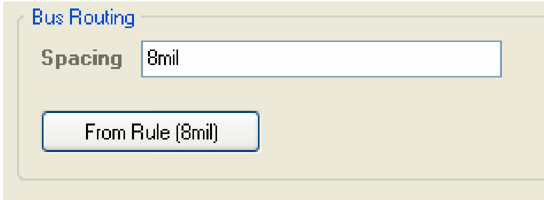



*Figure 11. Setting the bus routing spacing.*

## 18.1.5 Interactive Length Tuning

Matching route lengths is a standard technique for maintaining data integrity in a high-speed digital system, and an essential ingredient of differential pair routing. **Interactive Length Tuning** allows a dynamic means of optimizing and controlling net lengths by allowing variable amplitude patterns to be inserted according to the available space, rules, and obstacles in your design.

- Launched from **Tools » Interactive Length Tuning** menu, tuning can be based on: design rules, properties of the net, or values you enter into a dialog. Once launched, click on the routed net and move the mouse along the route path to add tuning segments.
- The Interactive Length Tuning cursor guides you during the tuning process. The yellow cursor bars indicate the possible minimum and maximum lengths. The green bar indicates the target length, and the sliding indicator shows how close you are to achieving a match, as shown in Figure 12.
- Press Tab during length tuning to open the *Interactive Length Tuning* dialog (Figure 12), where the tuning behavior is configured.
- Target Length can be controlled to meet: design rules, an existing routed net, or manual.
- Three tuning styles available: Mitered with Lines, Mitered with Arcs, and Rounded.
- Option to precisely clip tuning patterns to **Target Length** when Mitered with Lines and Mitered with Arcs styles are used.
- Tuning patterns and properties (such as pitch and amplitude) can be controlled using shortcuts, press Shift+F1 for a list.

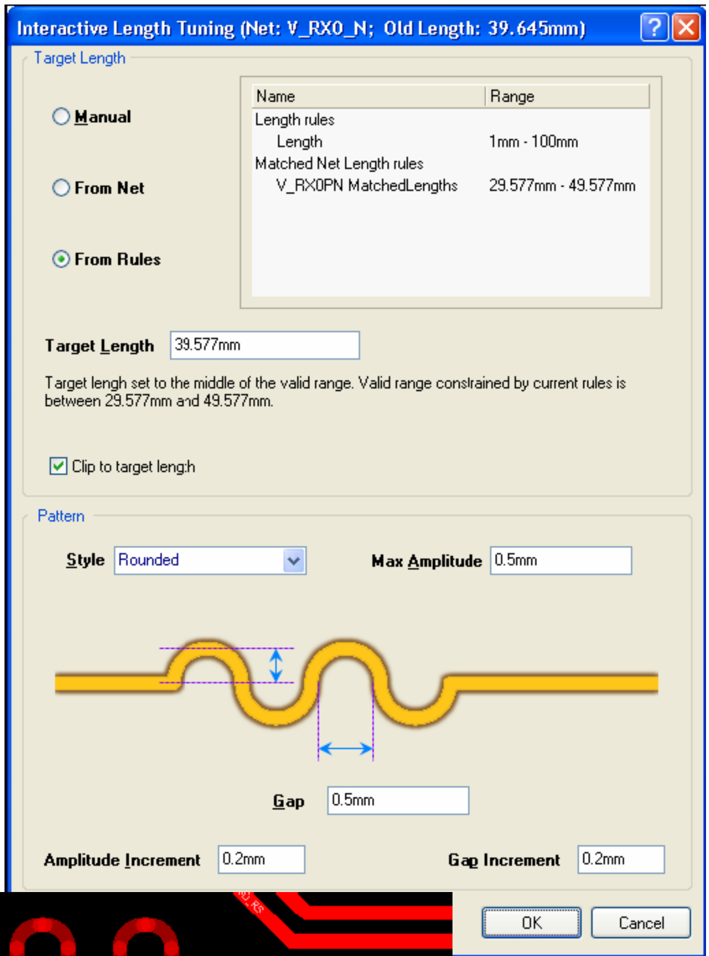




*Figure 12. Tuning parameters are configured in the Interactive Length Tuning dialog. The tuning cursor shows how close the current route length is to the required length.*

# 18.2 Testpoint System

Testing is an important part of the board manufacturing process. After fabrication, the board must be tested to ensure no short or open circuits. Once fully populated with all its components, a board is tested again to ensure signal integrity and device operation. To aide in this process, it is fundamentally beneficial to have a scheme of points on the board – testpoints – which the testing equipment can probe and perform the required tests.

The location of testpoints on a board will depend on factors including the mode of testing and the test equipment used. For example, when performing bare-board fabrication testing, the board is not populated and so all pads and vias are 'fair game' when it comes to assigning testpoints. The locations used for testpoints when performing in-circuit assembly testing however, will almost always be different. As the board is populated, you may no longer have probe access to component pads and certainly no access to pads and vias under a component!

Altium Designer provides a powerful system to handle your testpoint needs and enhance the testability of your boards, allowing you to separately assign testpoints for bare-board fabrication testing and/or in-circuit assembly testing as required. Testpoints can be assigned manually or, in a more streamlined and automated fashion, using the *Testpoint Manager*.

## 18.2.1 Considering Your Testpoint Strategy

Before jumping into the assignation of pads and vias for use as testpoint locations, it is a good idea to step back and think about what is required. The following are just some pointers to consider when defining a strategy to incorporate testpoints into a design:

- When choosing the side of the board that testpoints will be allowed on, consideration should be given to the testing processes and associated fixtures that will be used. For example, will the board be probed from the bottom side only, the top side only, or both sides.
- A testpoint underneath a component (on the same side of the board as the component) is usually used at the bare-board testing stage. This should be taken into consideration when planning testpoint locations for assembled board testing.
- It is advisable to locate all testpoints on one side of the board only, using vias to achieve this if necessary. The reason for this lies in the fact that a dual-head test fixture incurs greater cost than a single-head test fixture.
- The more non-standard and complex your pattern of testpoints, the more costly it will be to configure a fixture with which to test the board. The best philosophy is to develop a methodology that will result in generic testability. A well-honed and adaptable testpoint policy will allow different designs to be tested efficiently and cost-effectively.
- Careful consideration should be given to any via tenting requirements of the design. Tenting a testpoint-designated via will effectively block test probe contact. Even partial tenting using a liquid photoimageable (LPI) solder mask will cause contact problems, as the mask liquid will tend to run away through the via hole. Peelable solder mask may indeed be used to provide temporary tenting of such designated vias, but this can often prove quite costly.
- Consult with your fabrication and assembly houses closely to make sure any specific design parameters are taken into account when specifying testpoints. These could include testpoint-to-testpoint clearances and testpoint-to-component clearances that may be stricter than normal placement and routing clearances.

## 18.2.2 Pad and Via Testpoint Support

Altium Designer provides full support for testpoints, allowing you to specify pads (thru-hole or SMD) and/or vias to be used as testpoint locations in fabrication and/or assembly testing. A Pad or Via is nominated for use as a testpoint by setting its relevant testpoint properties – should it be a fabrication or assembly testpoint, and on which side of the board should it be used as a testpoint. These properties are set from within the Pad or Via properties dialogs.

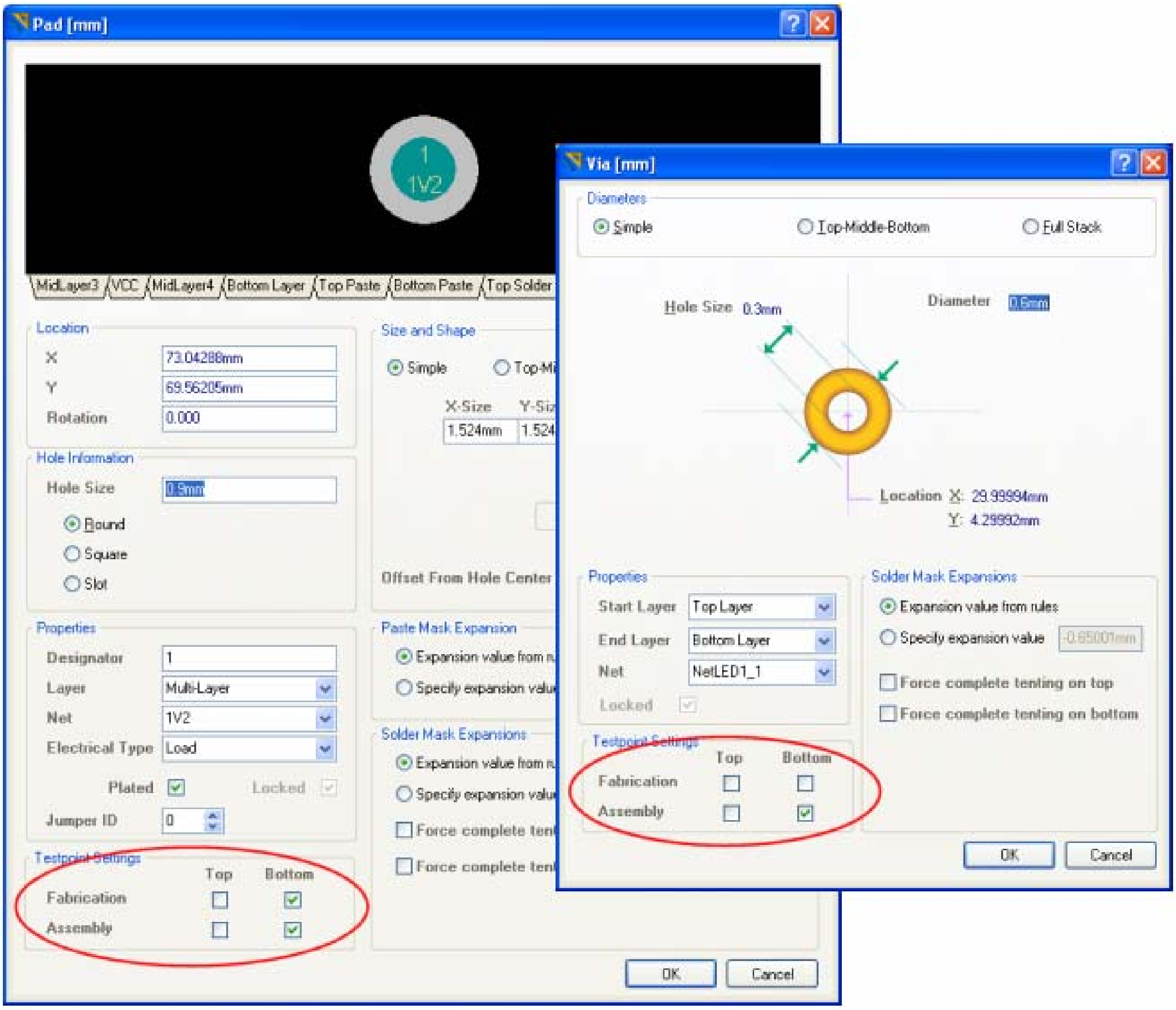


*Figure 13. A pad or via is specified for use as a testpoint through the relevant options in its associated properties dialog.*

You can automatically assign testpoints based on defined design rules and using the *Testpoint Manager*. This automated assignment simply sets the relevant testpoint properties for the pad/via in each case. You of course have the option to manually specify testpoints – in essence, handcrafting at the individual pad/via level – giving you full control over the testpoint scheme employed for your board.

## 18.2.3 Design Rules

The constraints of a PCB design should be thought out and implemented as a well-honed set of design rules. To implement a successful testpoint scheme – where all defined testpoints can be accessed and used as part of the bare-board and/or in-circuit testing, governing constraints must be put in place. To this end, the following rule types are definable as part of the PCB Editor's Design Rules system:

Access and define rules of these types from the *PCB Rules and Constraints Editor* dialog (**Design » Rules**).

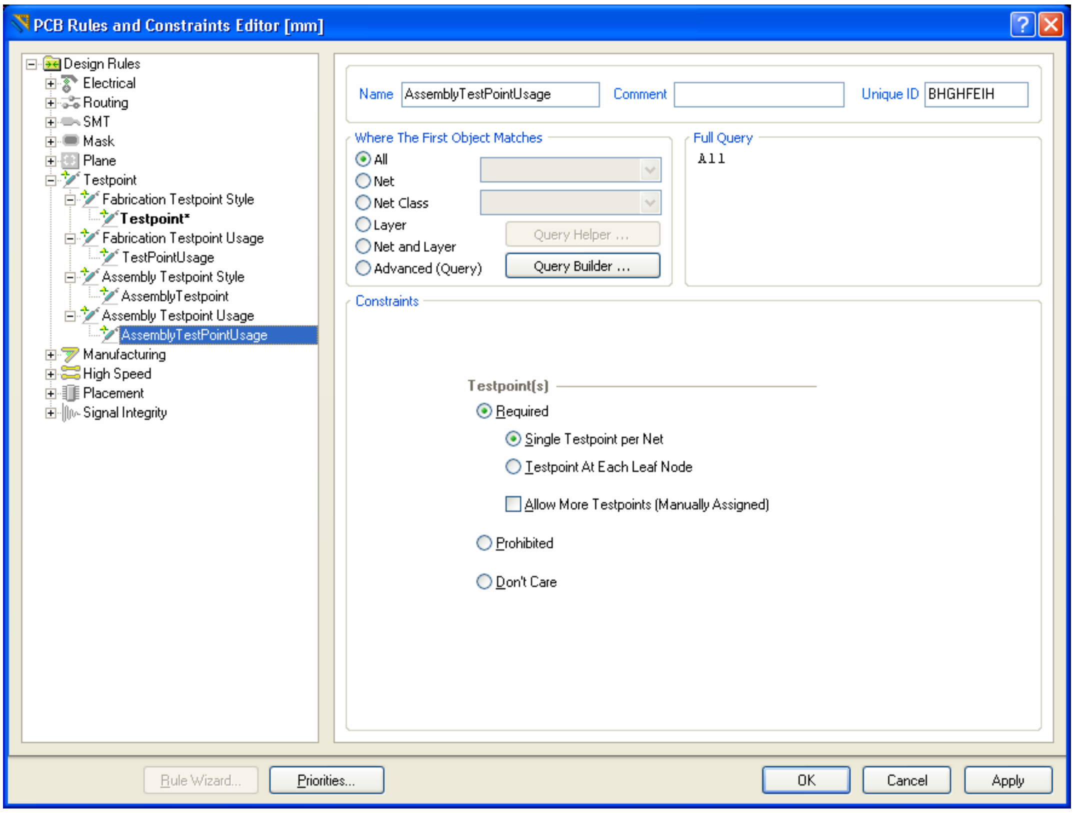


Figure 14.Define separate design rules to constrain which pads and/or vias in the design can be used as Fabrication testpoints and Assembly testpoints, and which nets require testpoints.

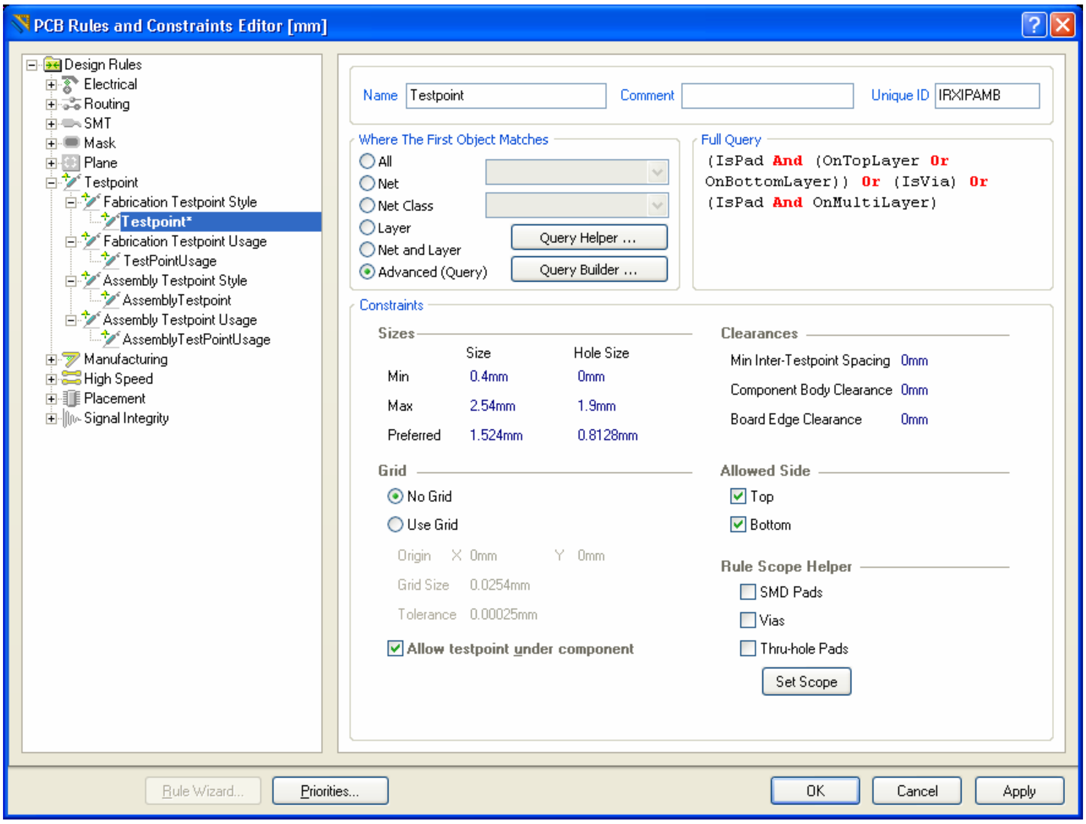


Figure 15. Define separate design rules to constrain which pads and/or vias in the design can be used as Fabrication testpoints and Assembly testpoints, and which nets require testpoints.

The Testpoint Style and Usage rules are identical, in terms of constraints, between the two testing modes (fabrication and assembly). The style rule essentially specifies constraints that a pad or via has to meet in order to be considered for selection as a testpoint location. The usage rule simply specifies which nets require a testpoint. When defining a style rule, the rule scope can be quickly created to target the precise pad and/or via objects for testpoint consideration, using the *Rule Scope Helper*.

The testpoint design rules are used by the *Testpoint Manager*, the Autorouter, Online and Batch DRC processes and also during output generation.

Default Fabrication and Assembly Testpoint Style and Testpoint Usage rules exist. You should check whether these rules meet your board requirements and make changes as necessary.

When opening PCB designs or importing design rules created in a release of the software prior to the Summer 09 release, Testpoint Style rules will become Fabrication Testpoint Style rules and Testpoint Usage rules will become Fabrication Testpoint Usage rules.

## 18.2.4 Managing Testpoints

Assigning testpoints manually can be a painstaking and laborious job at the best of times. Imagine this task on a more complex board, populated with hundreds of components (possibly on both sides of the board) and the process cries out for a more automated method of testpoint assignment. To cater for streamlined management of testpoints in your board designs, Altium Designer equips the PCB Editor with a *Testpoint Manager*.

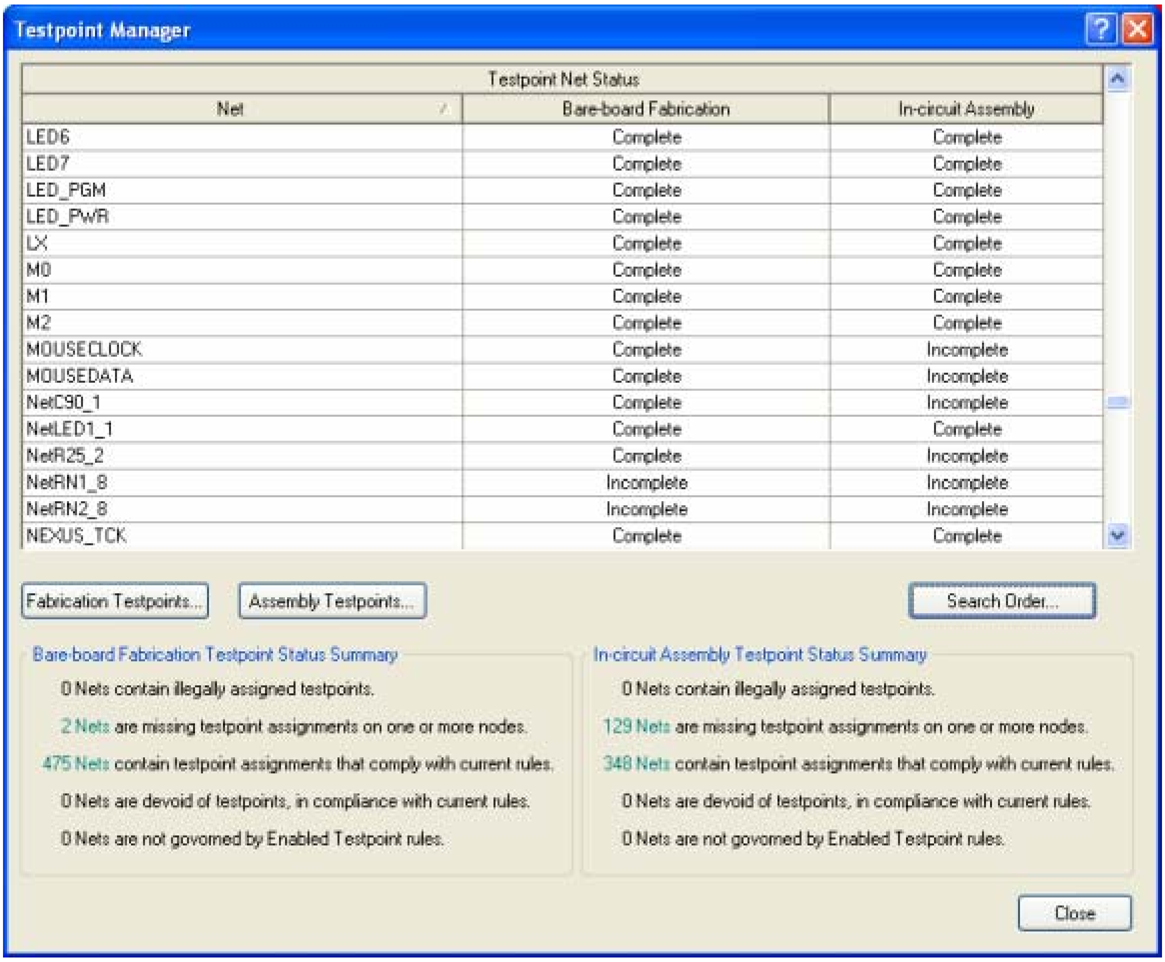



*Figure 16. Manage your fabrication and assembly testpoint requirements quickly and efficiently using the Testpoint Manager.*

Accessed from the PCB Editor's main Tools menu (**Tools » Testpoint Manager**), the *Testpoint Manager* provides controls allowing you to automatically assign and clear testpoints from the one convenient location. A listing of all nets in the design is provided, with status to indicate testpoint coverage – either `Complete` or `Incomplete` – for both bare-board fabrication and in-circuit assembly testing.

Whether assigning testpoints for some or all of the nets in a design, the *Testpoint Manager* follows the style and usage rules defined for fabrication and assembly testpoints. Where rules are defined to use a single testpoint per net, a definable search order of pad/via object types is provided – giving you even finer control over the priority by which such objects are considered.

A full summary of the testpoint status – for both testing modes – is also displayed and this updates with each assignment or clearance action performed.

The *Testpoint Manager* replaces the **Tools » Find and Set Testpoints** and **Tools » Clear All Testpoints** commands found in releases of Altium Designer prior to the Summer 09 release.

### 18.2.5 Checking the Validity of Testpoints

Defined fabrication and assembly testpoint rules are followed as part of the PCB Editor's Design Rule Checking (DRC) facility. Online and/or Batch DRC checking can be enabled for the various rule types from within the *Design Rule Checker* dialog (**Tools » Design Rule Check**).

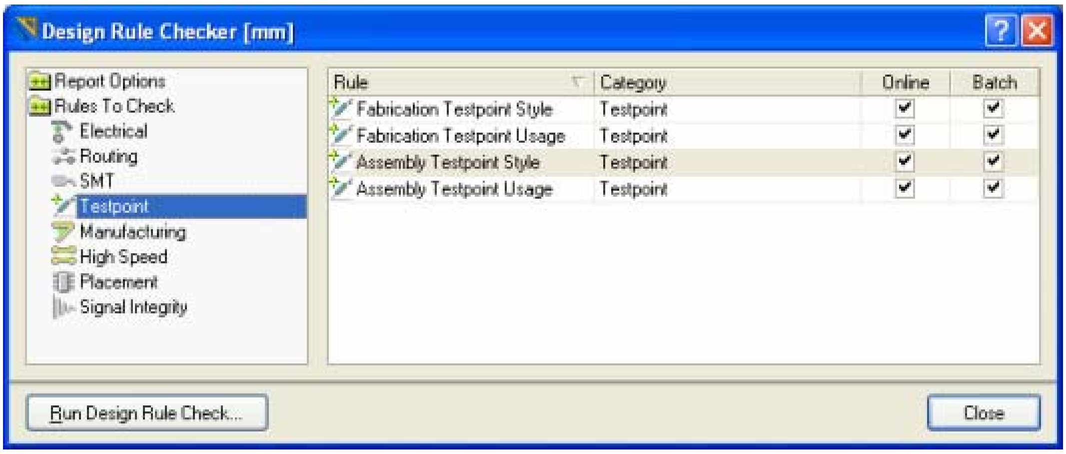



*Figure 17. Include your testpoint design rules as part of the Online or Batch DRC processes.*

# 18.3 Adding and removing teardrops

Teardrops are a common technique for guarding against drill breakout during the board fabrication phase.

- The **Tools » Teardrops** command is used to add or remove tear-dropping from pads and/or vias.
- Options are configured in the **Teardrop Options** dialog (Figure 18).
- To remove teardrops use the **Remove** option in the **Teardrop Options** dialog.
- Teardrop shapes are created by adding additional short track or arc segments.
- Use the **Report** option to identify pads/vias where teardrops could not be added.

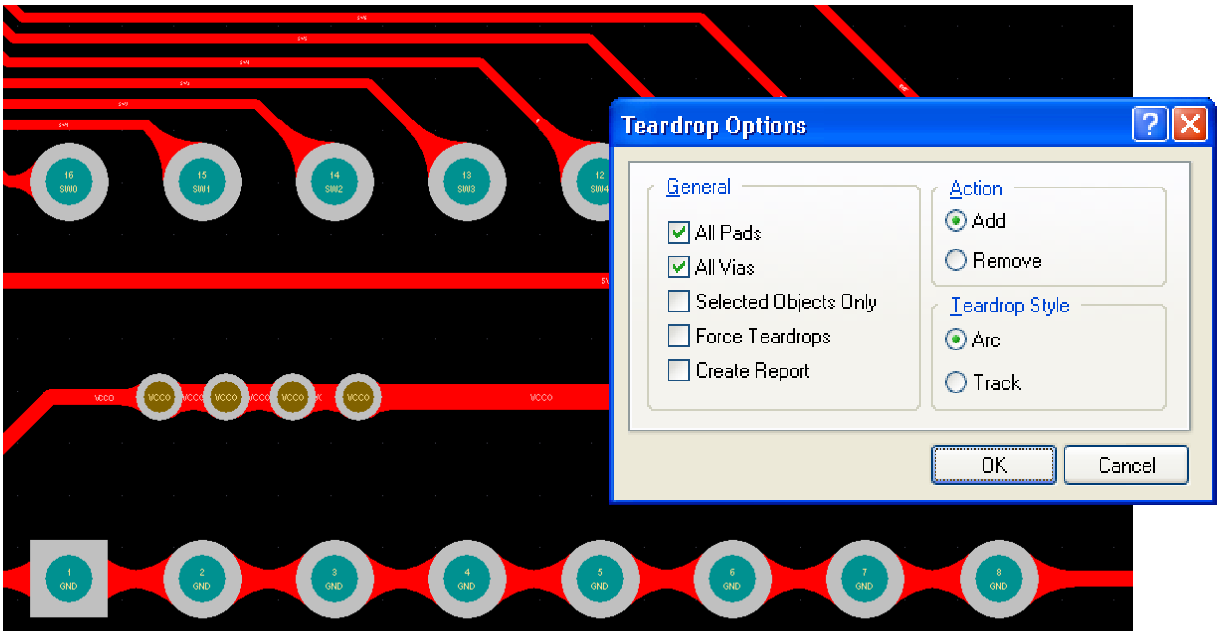



*Figure 18. Teardrops build up the copper at each track entry to pads and vias.*

- Choose the **Selected Objects Only** option to control which pads/vias should have teardrops added.
- The **Force Teardrops** option will apply teardrops to all pads and/or vias, even if it results in a DRC violation.

> **Note:** Arc-style teardrops can create complex shapes at the pads/vias. When polygons are poured over routing with arc-style teardrops, part of the polygon near the teardrop can "break off" (a section of the polygon is missing). If this occurs try a Solid style polygon instead of Hatched, or change the teardrop style from Arc to Track.

# 18.4 Automatic routing

Altium Designer's autorouter is a topological autorouter – it uses topological mapping to find routing paths on the board. The Autorouter adheres to all electrical and routing design rules, except the Routing Corners and Differential Pair design rules.

### 18.4.1.1 Autorouting tips

- The board must include a closed boundary on the Keep Out layer.
- Design rules must be correctly defined for the router to be able to route, it will not route connections that would result in a design rule violation. If there are potential rule conflicts they will be detailed at the top of the *Situs Routing Strategies* dialog. Always check that the rules are appropriately defined before starting the autorouter.
- Routing layer directions must be configured. Default directions are assigned, but these do not take into consideration any existing manual routing, so they should always be checked. Routing layer directions are configured by clicking the **Edit Layer Directions** button in the *Situs Routing Strategies* dialog.
- You can protect pre-routed connections, fan-outs and entire nets by enabling the **Lock all Pre-routes** option in the *Situs Routing Strategies* dialog (**Auto Route » Setup**). This option also protects fan outs and partially routed connections.
- Objects with a net name that are not locked may be moved/ripped up during routing.
- Objects placed on the Keep Out layer create blocks for the router on all layers.
- Signal layer keepout objects create blocks for the router on that signal layer.
- The router does not consider objects on the mechanical layers.
- The router is sensitive to connection lines running at very shallow angles, experiment with the alignment of components to observe this.

### 18.4.1.2 Running the Autorouter

- The Autorouter requires minimal set up. To run the router using a default strategy, select **Auto Route » All** to display the *Situs Routing Strategies* dialog, select a strategy, and click OK to start routing.
- Use the **Auto Route » Stop** command to terminate autorouting.

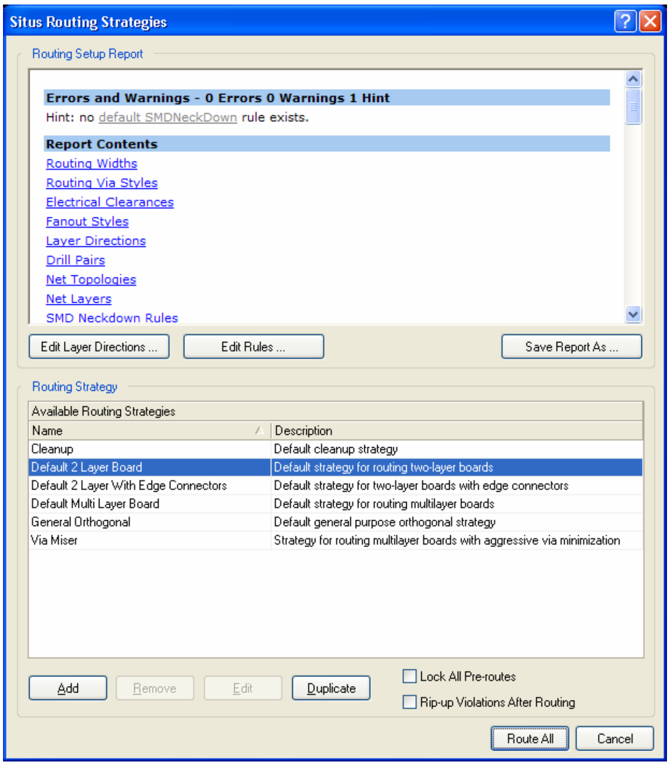



*Figure 19. Autorouter strategy dialog*

### 18.4.1.3 Creating a Custom Routing Strategy

- To create a custom routing strategy select one of the default strategies in the *Routing Strategies* dialog and click **Duplicate**.
- As well as defining the set of routing passes, you can also control the via cost, and the router's tendency to route more diagonally or more orthogonally. If you enable the Orthogonal option in the *Situs Strategy Editor* you should add a Recorner pass to the strategy.

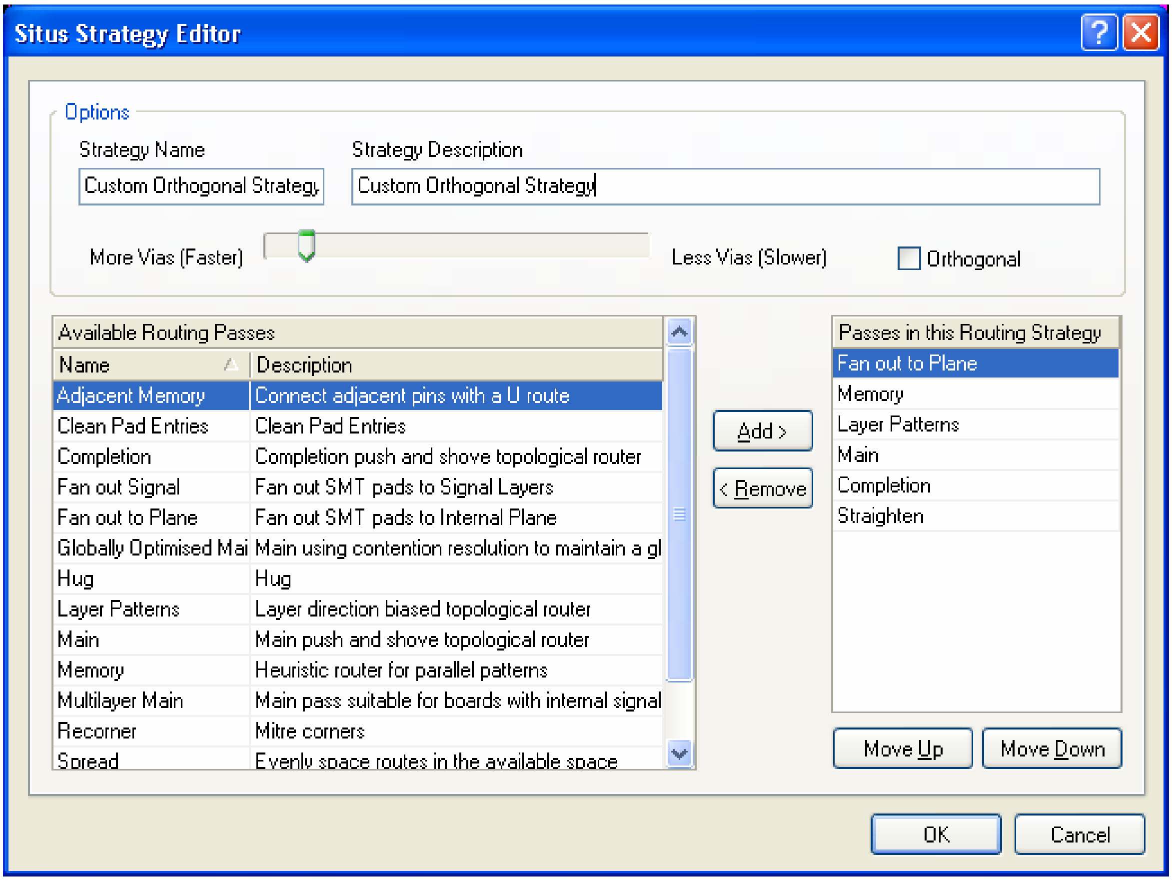



*Figure 20. Custom routing strategy using cheaper vias and orthogonal routing*

## 18.4.2 Exercise – Autorouting

1. Select **Autoroute » All** from the menus.
2. Select the **Default 2 Layer Board** strategy, enable the **Lock All Pre-routes** option if you would like to keep your hand routing, and click the **Route All** button.
3. Examine the routing results. To more easily check each layer, press the **Shift+S** shortcut to toggle to single layer mode, then press the * key to toggle back and forth from Top layer to Bottom layer. To highlight the routing of a particular net hold the **CTRL** key and click on the net. Repeat this where there are no objects under the cursor to clear the highlight. If you have the board in single layer mode, you can enable the **Show All Primitives in Routed Net** checkbox in the *Preferences* dialog to show the routing on all layers.
4. Now unroute (**Tools » Un-route**) so it can be rerouted using a custom strategy. To do this, duplicate the Default 2 Layer Board strategy, set the **More Vias** slider to close to the left end, enable the **Orthogonal** checkbox, and add a **Recorner** pass before the **Straighten** pass (as shown in Figure 20).
5. Autoroute the board with the custom strategy.

6. When you are happy with the routing results, save the board.

### 18.4.3 BGA Escape routing

The BGA escape routing engine will attempt to route each pad out to just beyond the edge of the device – making the remaining routing challenge much easier.

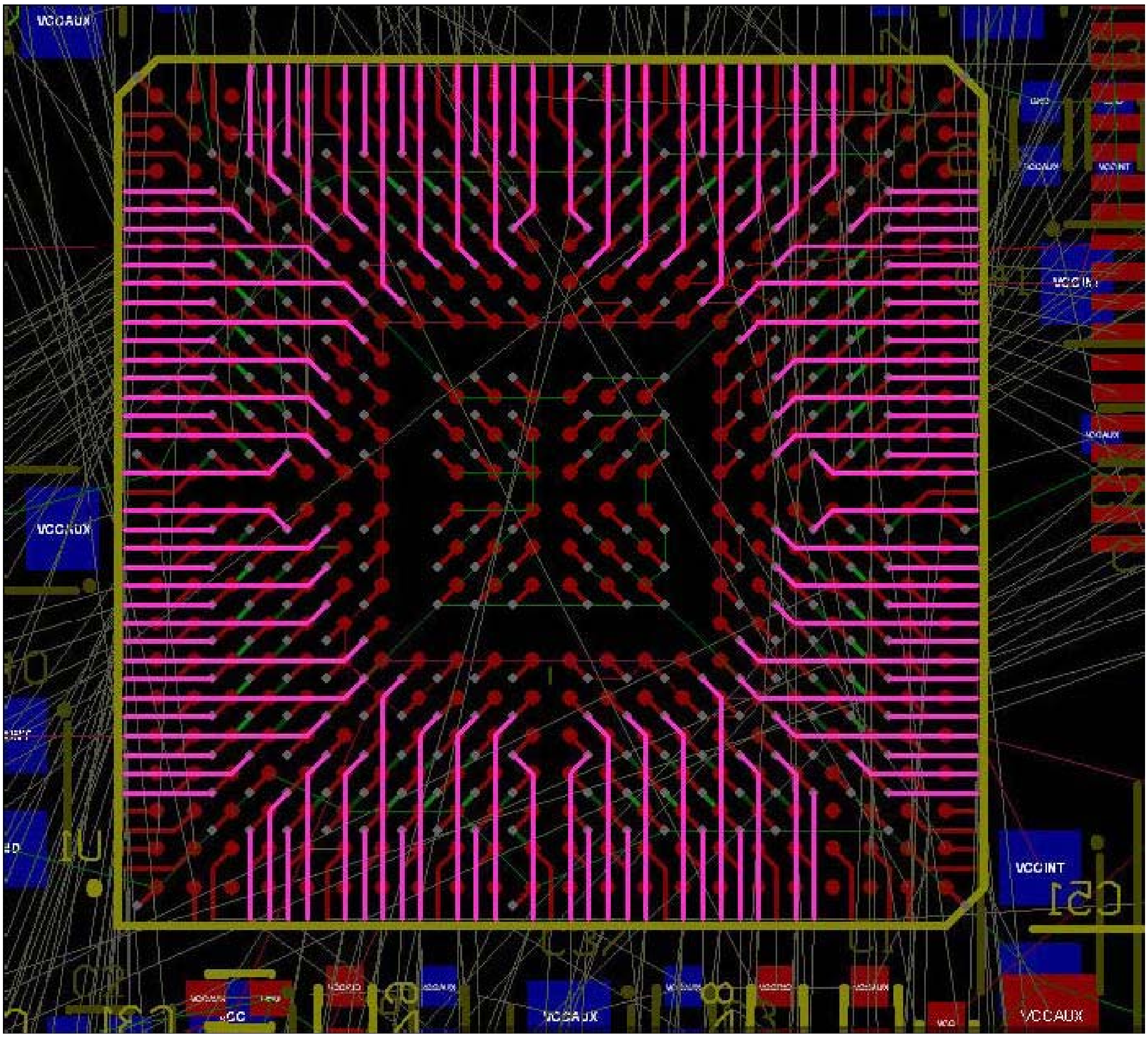


*Figure 21. Note how the escape route feature presents each connected pad as an accessible route outside the edge of the BGA.*

- Figure 21 shows the escape routing from a 1mm pad pitch BGA. Used inner pads are first fanned out using the traditional dog-bone (a short route with a via on the end) to access another layer, and then from the via they are escape routed out just beyond the edge of the device, working through the available routing layers until all pads have been escape routed.
- Right-click on a BGA and select **Component Actions » Fanout Component** from the context menu. The routing will be done in accordance with the applicable design rules (track and via sizes). A report of all pads that could not be escape routed will be generated and opened, click on an entry in the report to cross probe to the PCB and examine that object.

> **Note:** The most common reason escape routing fails is because the vias will not fit between the BGA pads. Ensure that there is a suitable Routing Via Style design rule configured to allow suitably small vias to be used for escape routing.

# 18.5 Polygons and the Polygon Manager

A signal layer polygon is an area of copper which is placed over existing objects, such as tracks and pads, but automatically pours around them, maintaining the specified clearances.

- A polygon can placed to define any enclosed shape.
- A signal layer polygon maintains clearances, defined by the Clearance design rules, from other copper objects.
- A signal layer polygon can be connected to a net.
- A polygon can be Solid or Hatched.
- A Solid polygon is built from Region objects. The advantage of this style of polygon is that there is typically much less data to store in the PCB file, and also less data in the CAM (Gerber or ODB++) files. Also region objects have sharp corners, so the polygon can sometimes better fill the space between other objects.
- A Hatched polygon is built from tracks and arcs. The advantage of this style of polygon is that the CAM processing software does not need to understand polygonal shape definitions.
- Polygons can be placed on other layers, but only pour around other objects on signal layers.
- Polygons can be created from a selected set of primitives, such as lines, as long as they form a closed boundary. Use the command in the **Tools » Convert** submenu.
- Polygons can be shelved, a process that hides them from other design objects, but does not remove them. Shelved polygons can be restored at any time.

## 18.5.1 Placing a polygon

- Place a polygon using the **Place » Polygon Pour** menu command or the toolbar icon. This displays the *Polygon Pour* dialog, where you set up the parameters for the polygon. Note that there are 2 different styles of polygons available:
  - **Solid polygon** – the polygon is constructed from multiple, multi-sided region objects. This style of polygon requires that your fabricator supports polygonal objects in Gerber or ODB++ files (most do). Using these polygons will give much smaller design files.
  - **Hatched polygon** – the polygon is constructed with track segments and arcs.
- Once the parameters are set up, click **OK** and draw the polygon in the workspace. The corner styles for the polygon are the same as those available during routing, press **Shift+Space** to cycle through the corner modes.

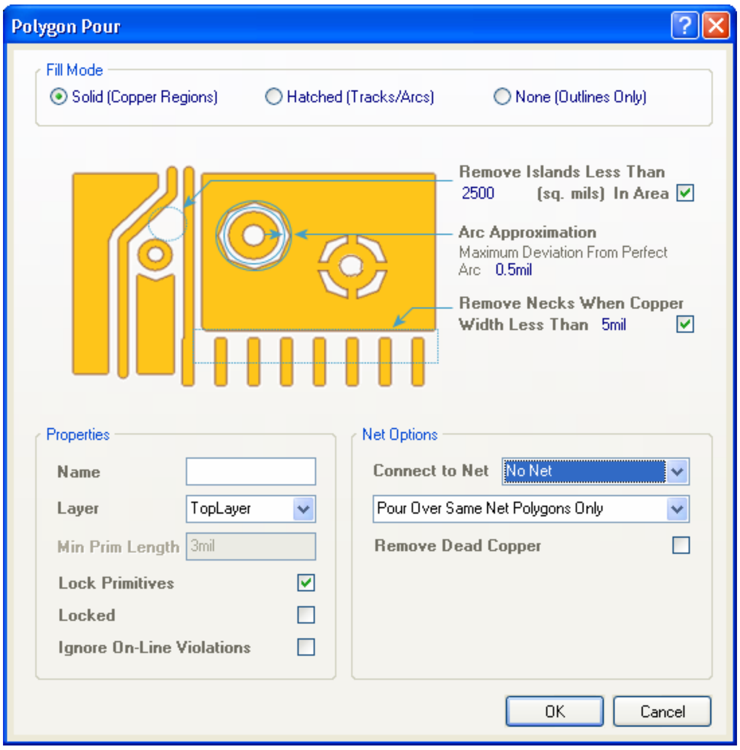



*Figure 22. Polygon Pour dialog*

The parameters for Polygons include:

**Net Options**

- **Connect to Net –** selects the net to be connected to the polygon.
- **Pour Over options –** existing polygons, or existing polygons and existing tracks within the polygon which are part of the net being connected to can be covered by the new polygon.
- **Remove Dead Copper –** removes any part of the polygon that cannot connect to the plane net.

**Properties**

- **Name –** The Name property identifies this polygon. Edit this field to define the name of this polygon. The Name property can be used in the **InNamedPolygon** query to highlight or scope rules, as a member of a polygon class and so on.
- **Layer –** select the signal layer that the polygon is to be placed on.
- **Min Primitive Length** – Tracks or arcs below this setting are not placed when pouring a polygon.
- **Lock Primitives –** if unchecked, individual objects (i.e. tracks or arcs) that make up the plane can be deleted.
- **Locked -** If this option is checked, the polygon is fixed in the workspace and can not be moved by the auto-placer or directly manipulated graphically. If you attempt to manually move the polygon, the warning message "Object is locked, continue?" will pop up, allowing you to move the polygon without unlocking it. The locked attribute remains set after this move. If this option is unchecked, the polygon can be moved directly without confirmation.
- **Ignore On-Line Violations** - Enable this option if you want your PCB document that has polygons, to be scanned for online violations by the Design Rule Checker. This is computationally intensive. Turn this option off to speed up the operations by ignoring polygons especially complicated polygons during the automated design rule checking.

**Plane Settings (Hatched and Outlines Only)**

- **Track Width –** width of tracks that make up the polygon. If Track Width is equal to the Grid Size, the polygon ends up as solid copper. If Grid Size is greater than Track Width, the polygon ends up as hatched.
- **Grid Size –** spacing between tracks that make up the polygon.
- **Surround Pads With**
    - *Octagons* **–** Places a track to form an octagon around pads.
    - *Arc* **–** Places an arc around pads.
- **Hatch Mode**
    - *90-Degree Hatch* **–** Polygon is hatched with horizontal and vertical tracks.
    - *45-Degree Hatch* **–** Polygon is hatched with tracks at 45 degrees and 135 degrees.
    - *Vertical Hatch* **–** Polygon consists of only vertical tracks.
    - *Horizontal Hatch* — Polygon consists of only horizontal tracks.

**Plane Settings (Solid)**

- **Remove Islands** – remove any region that has an area less than specified.
- **Arc Approximation** – solid polygons use short straight edges to surround existing curved shapes (such as pads). This setting defines the maximum allowable amount of deviation.
- **Remove Necks** – narrow necks that have a width less than this amount are removed.

## 18.5.2 Editing a polygon

**Note:** To edit a polygon, first make the layer that the polygon is on the active or current layer.

- To change any of the parameters once a polygon has been placed, double-click on the polygon, or select **Edit » Change** and click on the polygon. When the *Polygon Pour* dialog opens change the settings, click **OK**, and you are will be prompted to re-pour the polygon.

**Moving a polygon**

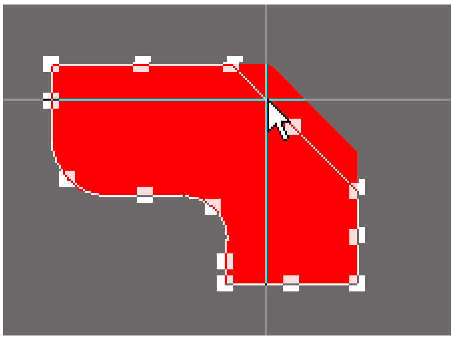
- Move a polygon as you would any other object. Click, hold and move it to the new location. When you release the mouse button, you will be prompted to re-pour the polygon.

**Deleting a polygon**

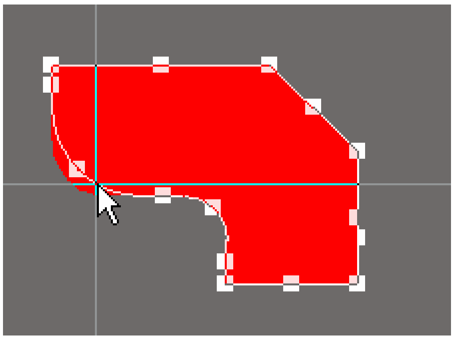
- To delete a polygon, select it, then press Delete on the keyboard.

**Pouring a polygon with a larger clearance**

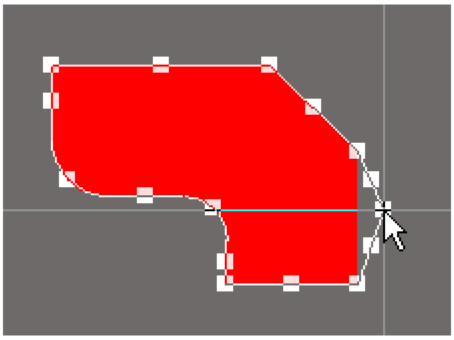
- Often you will want the polygons to have a larger clearance than the standard track to track clearances. This can be achieved by adding a new, higher priority clearance design rule, with one of the object Queries set to **InPolygon**, and the rule clearance set to the required higher value. The **Polygon Manager** can be used to easily create a targeted clearance rule for you.



*Click on the edge away from a vertex to slide that entire edge.*



*Click on an arc to resize it.*



*Click on a midpoint vertex to 'break' the edge.*

*Figure 23. Reshaping a polygon*

### 18.5.2.1 Reshaping a polygon

A polygon can be reshaped at any time. This is done by entering **Move Vertices** mode.

- To do this, right-click on the polygon and select the **Move Vertices** command from the **Polygon Actions** submenu.
- Click on a corner or midpoint vertex and drag it.
- Click on an edge away from the vertex to slide the entire edge.
- To add vertices, click and drag on a midpoint vertex.
- To convert a vertex into an arc, hold down **Shift+A** as you click and drag a vertex. Note that this only works when nearby vertices have not been moved from their original state.
- To delete a vertex, click on the corner vertex beyond the one you want to delete, and press the Delete key on the keyboard. Note that the order of the vertices is determined by the order they were originally placed.
- After dropping out of Move Vertex mode you will be prompted to repour the polygon.

### 18.5.2.2 Polygon cutouts

A polygon cutout, or hole inside a polygon, is actually an object in its own right – a **Region** object with the **Polygon Cutout** property enabled.

- Place a cutout in a polygon using the **Place » Polygon Pour Cutout** command. The standard corner styles are available, use **Shift+Space** to cycle through them.

### 18.5.2.3 Slicing polygons

Often it is easier to cut off part of a polygon, rather than resize it to a new shape.

- Select the **Place » Slice Polygon Pour** command to slice a polygon into two or more separate polygons.
- After launching the command, filtering will be applied to the document, temporarily dimming all objects except polygons. Place line objects through the polygon to define the slice path, using the standard cornering modes. Ensure that the last slice segment is placed beyond the polygon edge when you exit, then right click to drop out of slice mode. The 2 polygons will be re-poured, delete either if necessary.

### 18.5.2.4 Shelving a polygon

If you are modifying a design, perhaps changing components and modifying routing, existing polygons can be shelved, to temporarily remove them from the workspace. Shelved polygons are not deleted from the design, and can be restored at any time.

- To shelve an individual polygon, right-click on it and select **Polygon Actions » Shelve Polygon** from the context menu.
- To shelve specific polygons, select them first and use the **Shelve Selected** command instead.
- To restore shelved polygons, use the **Restore** command in the **Tools » Polygons Pours** submenu.
- Alternatively, use the **Polygon Manager** to selectively Shelve and Restore polygons.

### 18.5.2.5 Converting hatched polygons to solid polygons

If you are updating an existing design, you may wish to convert hatched polygons to solid polygons. This can be done using the **Tools » Polygon Pours » Convert Polygons to Solid** command. Note that older hatched polygons used an edge with a defined width, so during the conversion process you will have the opportunity for the software to expand the polygon by half the original polygon boundary width. Note that this will not affect the polygon clearance, the new polygon will still meet your clearance design rules. This process is done to ensure that the new polygon completely fills the area covered by the old hatched polygon.

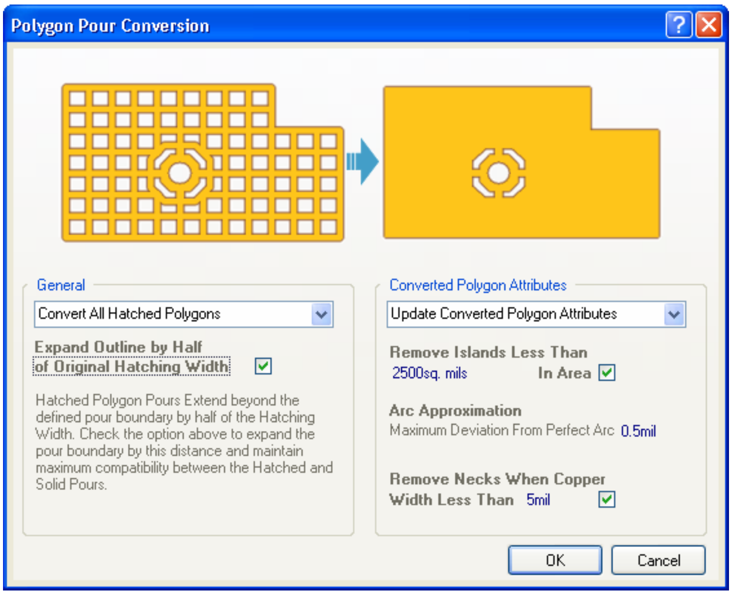



*Figure 24. Polygon hatched to solid conversion dialog.*

## 18.5.3 Managing polygons – the Polygon Manager

Even a small design can include a large number of polygons, to help manage them Altium Designer includes a Polygon Manager.

- Select **Tools » Polygon Pours » Polygon Manager** to open the *Polygon Pour Manager* dialog.
- All polygons are listed in the upper **View/Edit** region of the dialog, click to display a specific polygon in the viewer down the bottom.
- Polygons can be named, use this if you want to target a specific polygon with a design rule.
- The action buttons (**Repour**, **Shelving**, etc) can be used on polygons selected in the **View/Edit** list.
- Use the pour order when there are small polygons completely enclosed within larger polygons, in this situation the smaller polygons must be poured first. The **Auto Generate** button will order the polygons from smallest area to largest area, on a layer-by-layer basis.
- The rule creation buttons build a design rule that targets the polygon(s) currently selected in the **View/Edit** list, select them before clicking the rule creation button.

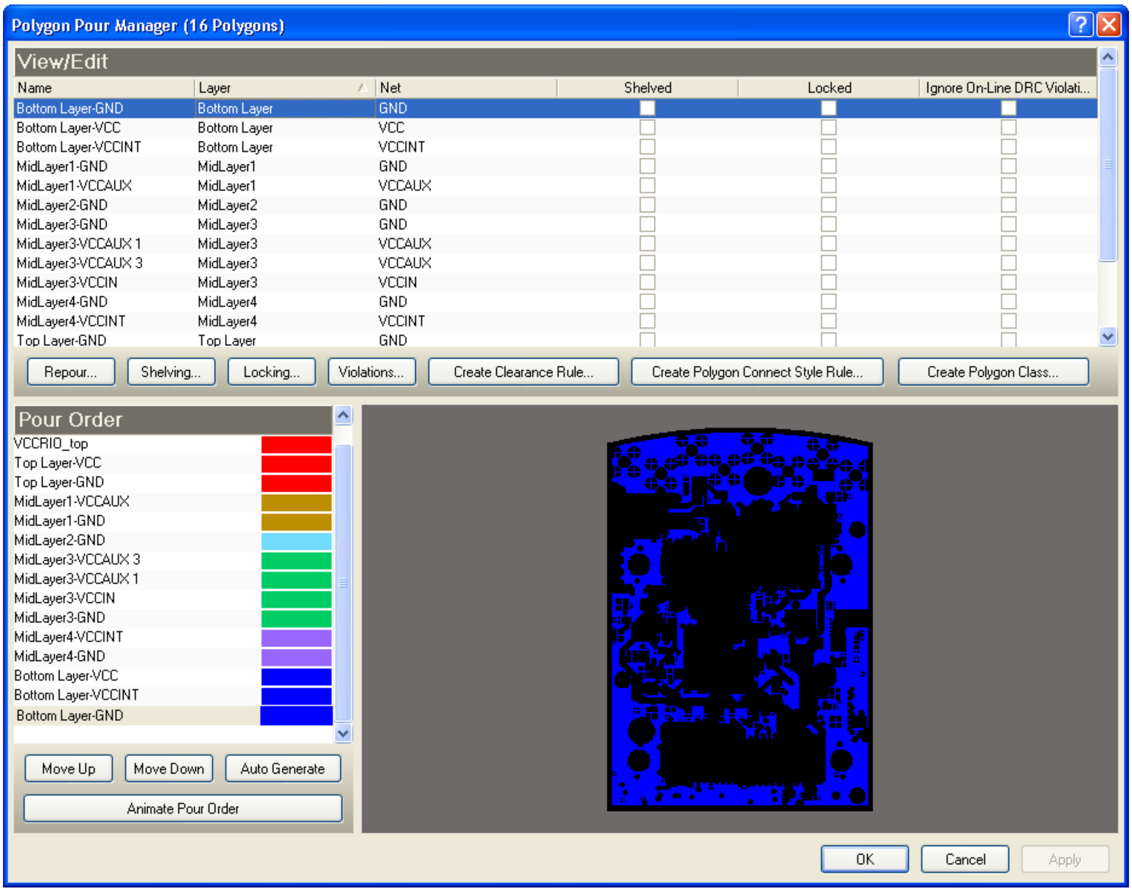



*Figure 25. Use the Polygon Pour Manager to examine and manage all polygons on the board.*

## 18.5.4 Exercise – Working with polygons

In this exercise, you will place a polygon plane on the top layer of the Temperature Sensor PCB.

1. Place a solid polygon on the top layer covering the entire PCB,
   - connected to net GND,
   - named Top Layer-GND,
   - enable the **Pour Over All Same Net Objects** option,
   - enable the **Remove Dead Copper** option.
   - Don't worry about exactly following the board shape, since copper outside the keepout boundary will not be connected to GND, it will be removed.
2. Select **Design » Rules** from menus, and display the **Electrical Clearance** section.
3. Right-click to create a new *Clearance* rule and call it *Polygon to keepout clearance*
4. Set the top query to **InPolygon**
5. Set the bottom query to `OnLayer('Keep-Out Layer')`

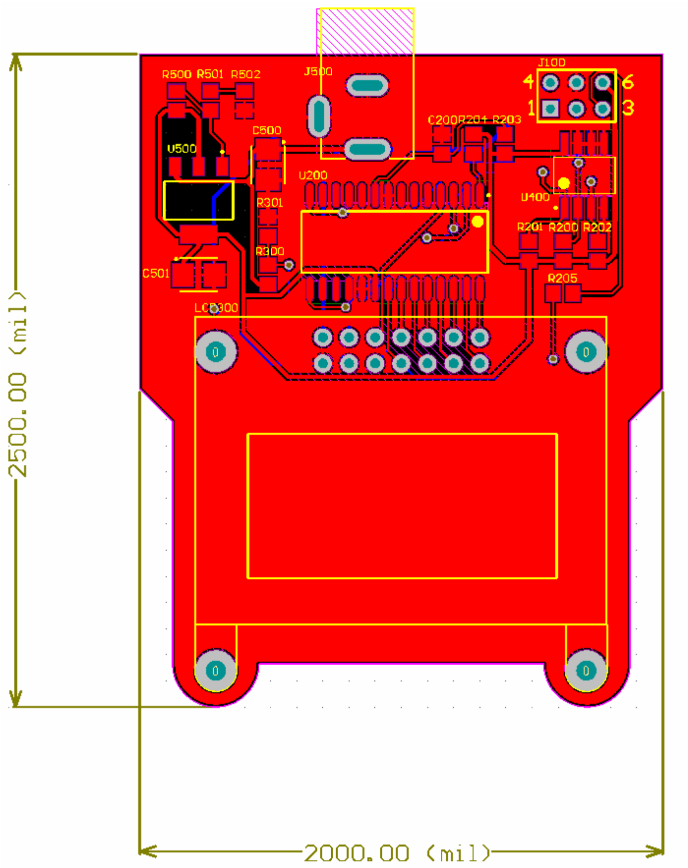



*Figure 26. the Temperature Sensor PCB with a solid polygon.*

6. Set the clearance to **20mil**.
7. Click the **Priorities** button and make sure this new rule has the highest priority.
8. When the Rules dialog is closed the online DRC will run and flag the polygon as a violation, since it does not comply with this new design rule.
9. Repour the polygon so that obeys the new polygon clearance rule.
10. Perform a final design rule check (DRC) to ensure there are no problems with your board. Refer to Module 12 - Design Rules to refresh your memory on checking the design rules.
11. Save the board.

TOP SECRET

Altium Designer

# Module 19: Output Generation and CAM File Editing

## Module 19: Output Generation and CAM File Editing





Module Seq = 19

# 19.1 Bill of Materials

## 19.1.1 Generating a Bill of Materials

Altium Designer includes a powerful report generation engine, which can be used to generate a comprehensive Bill of Materials.

- Any schematic or PCB component property can be included. For components that link to a company database, database fields that were not included in the schematic can also be included in the BOM. Enable the **Show** checkbox to include a component property. To include PCB parameters enable the **Include Parameters from PCB** option, to include additional database parameters enable the **Include Parameters from Database** option.
- The layout and grouping of data in the report is fully customizable, drag and drop columns to re-order them, drag component properties into the **Grouped Columns** region to group by that property.
- Supported output formats include: text, CSV, XLS, HTML and XML.
- Excel (XLS) format output can auto load into your company Excel (XLT) spreadsheet.
- Report customization is remembered when you click **OK** to close the dialog.
- Click **Export** to generate the BOM in the selected **File Format**.
- Project and document parameters can also be included in the BOM. For example, to include the value of the project parameter `DesignRevision` (added in the *Project Options* dialog) in the exported BOM, add the string `Field=DesignRevision` to the Excel template.
- The report generation engine can be accessed via the **Reports » Bill of Materials** menu entry (settings are saved in the project file), or via the Output Job editor (settings are saved in the OutJob file). Choose one approach and use only that approach, since the settings of each are stored separately.

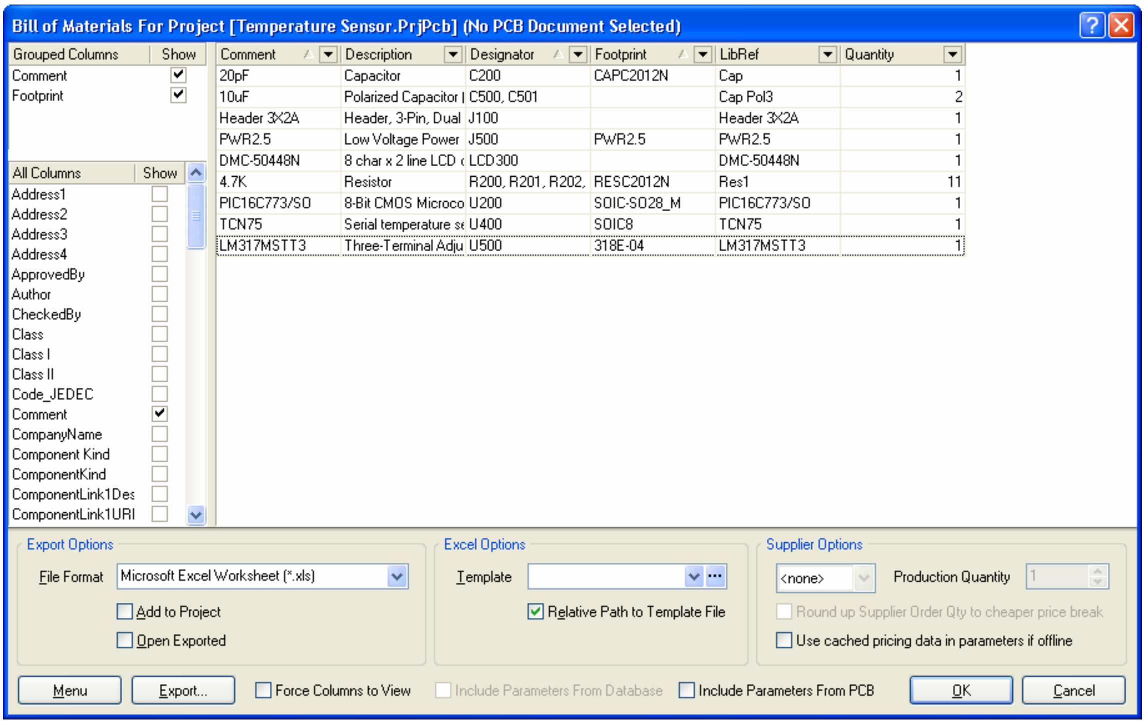



*Figure 1. Configuring the layout of the Bill of Materials, then click Export to generate in the selected file format.*

## 19.1.2 Using databases with the Bill of Materials

Using this method, the Database Link file (*.DBlink) defines linkage between the schematic component and a matched record in a database. The record match is established by key field linking, which can be a single key field (for example a part number), or multiple key fields (by defining a **Where** clause).

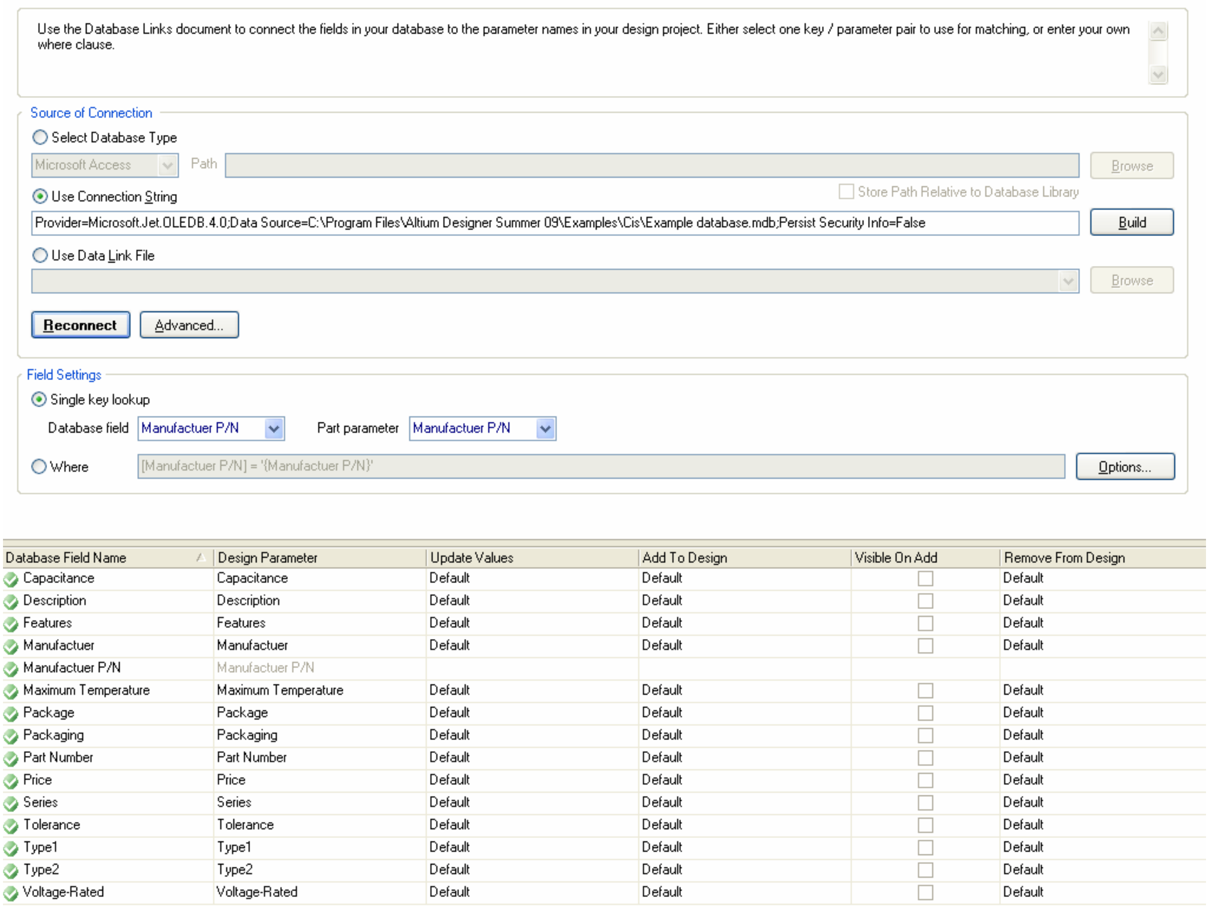
With this method of linking, the model and parameter information for the component must be predefined as part of the Altium Designer library component. The library component must also include the necessary key field information as part of its definition. Once this has been defined you add a database Link document (*.DBlink) to your Library Package or PCB project, then you can synchronize the component information (parameters) with the contents of fields in the database.

Although each physical component defined by each database record does not need to map to a unique Altium Designer library component – many database components can share the same component symbol – this method of linking would typically be used in a "one database record-to-one Altium Designer component" fashion. The unique Altium Designer component can either be an instance placed on a schematic sheet, or a unique component in a component library.

With DBLink-style database linking, you include the Database Link file with the project.

## 19.1.3 Setting up the link to a database

To setup a link you'll need to create a new *.DBlink file by going to **File » New » Database Link File**.




| Database Field Name | Design Parameter | Update Values | Add To Design | Visible On Add | Remove From Design |
|---|---|---|---|---|---|
| Capacitance | Capacitance | Default | Default | | Default |
| Description | Description | Default | Default | | Default |
| Features | Features | Default | Default | | Default |
| Manufactuer | Manufactuer | Default | Default | | Default |
| Manufactuer P/N | Manufactuer P/N | | | | |
| Maximum Temperature | Maximum Temperature | Default | Default | | Default |
| Package | Package | Default | Default | | Default |
| Packaging | Packaging | Default | Default | | Default |
| Part Number | Part Number | Default | Default | | Default |
| Price | Price | Default | Default | | Default |
| Series | Series | Default | Default | | Default |
| Tolerance | Tolerance | Default | Default | | Default |
| Type1 | Type1 | Default | Default | | Default |
| Type2 | Type2 | Default | Default | | Default |
| Voltage-Rated | Voltage-Rated | Default | Default | | Default |

*Figure 2. A Database link file created for a project with all the links setup*

The first step is linking to the database itself.

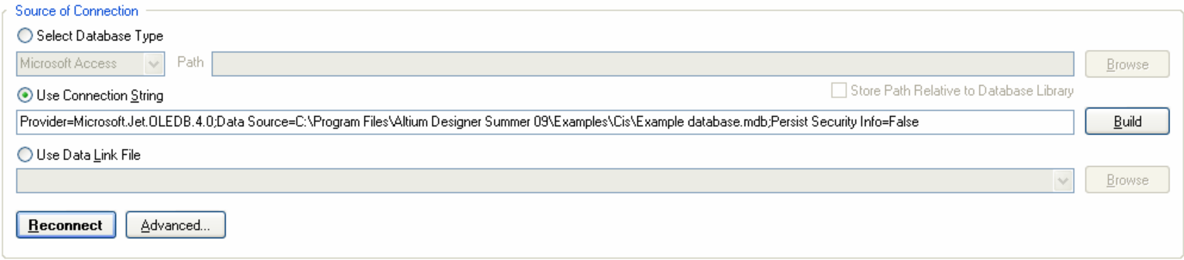



*Figure 3. The database link setup.*

All Different types of databases can be linked to, including Excel, Access, SQL, MYSQL and even Oracle. Any database which provides OLE DB support can be connected to. In Figure 3 an ODBC based connection string has been built using a datalink direct to an access database.

For more details on all different types of connections to databases refer to the document *AP0133 Using Components Directly from Your Company Database.PDF* which you can find in the help directory of the Altium Designer installation, or on the Altium website.

The next step is specifying a matching criteria.

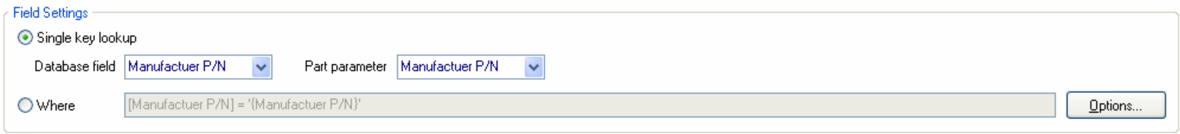
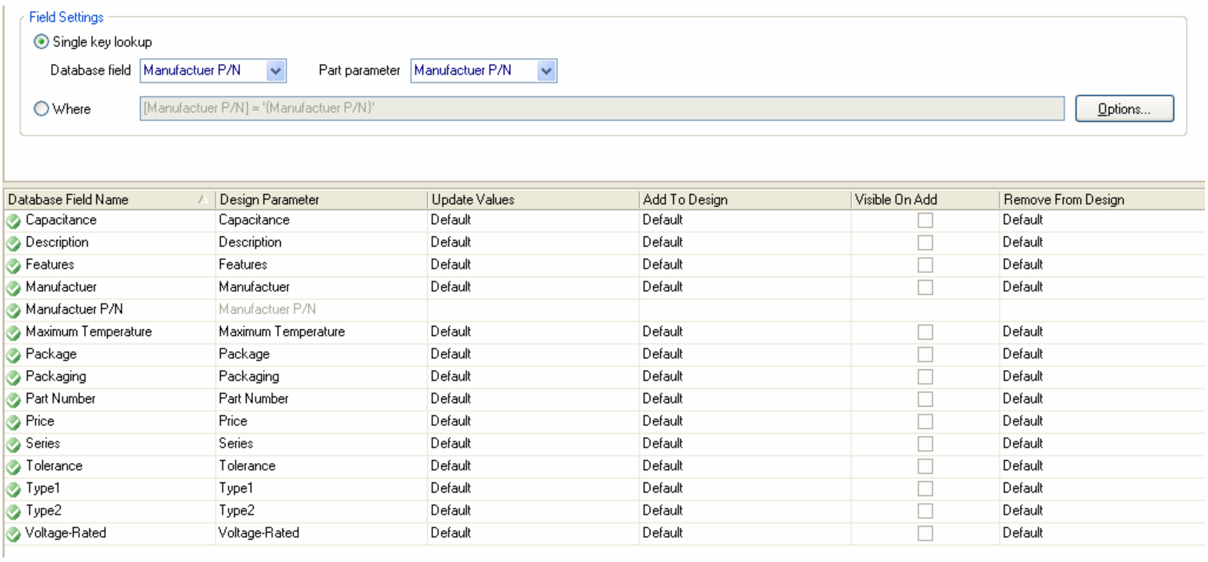



*Figure 4. The matching criteria*

In Figure 4 the criteria is set to Manufacturer P/N for both database field and the part parameter field. This means there needs to exist this parameter in both the database and on a created component were the original component came from, be that a schlib, integrated library or a database library. This field has to be a unique field that exists in both the database and on the components being linked to.

Once the look up key is set, there is also the field mapping to be setup.




| Database Field Name | Design Parameter | Update Values | Add To Design | Visible On Add | Remove From Design |
|---|---|---|---|---|---|
| Capacitance | Capacitance | Default | Default | | Default |
| Description | Description | Default | Default | | Default |
| Features | Features | Default | Default | | Default |
| Manufactuer | Manufactuer | Default | Default | | Default |
| Manufactuer P/N | Manufactuer P/N | | | | |
| Maximum Temperature | Maximum Temperature | Default | Default | | Default |
| Package | Package | Default | Default | | Default |
| Packaging | Packaging | Default | Default | | Default |
| Part Number | Part Number | Default | Default | | Default |
| Price | Price | Default | Default | | Default |
| Series | Series | Default | Default | | Default |
| Tolerance | Tolerance | Default | Default | | Default |
| Type1 | Type1 | Default | Default | | Default |
| Type2 | Type2 | Default | Default | | Default |
| Voltage-Rated | Voltage-Rated | Default | Default | | Default |

*Figure 5. Field Mapping for each of the parameters that exists in the database.*

The first two columns (from the left) on the **Field Mappings** tab allow you to control which information from the database is to be mapped to the component's models and parameters.

The **Database Field Name** column lists all field (column) names in the currently active table of the database. The **Design Parameter** column defines how each corresponding field in the database is to be used – whether it is used to source a schematic component, link-in a particular model, or to be attached to the component as a mapped design parameter.

Initial mapping is performed automatically upon connection to the database, with all database fields mapped.

For fields that you explicitly do not want mapped from the database, set the **Design Parameter** entry to `[None]`. Unmapped database fields are distinguished on the tab by the use of a red cross icon. Mapped database fields are distinguished by a green tick icon.

## 19.1.4 Linking the database to the Bill of Material

Source information for a Bill of Materials (BOM) has, in the past, been taken from the parameter information of the placed components for the design. But that can lead to a lot of information attached to a schematic that is only ever used for the BOM. If you place components from a Database Library the BOM Generator is able to extract any other record information that has not been added as design parameters at the time of placement.

When configuring the Bill of Materials report using the *Report Manager*, simply enable the **Include Parameters from Database** option. This option will only be available if one or more components in your design are linked to an external database. In the parameter listing, the icon is used to distinguish a parameter that exists for one or more placed components in a linked external database.

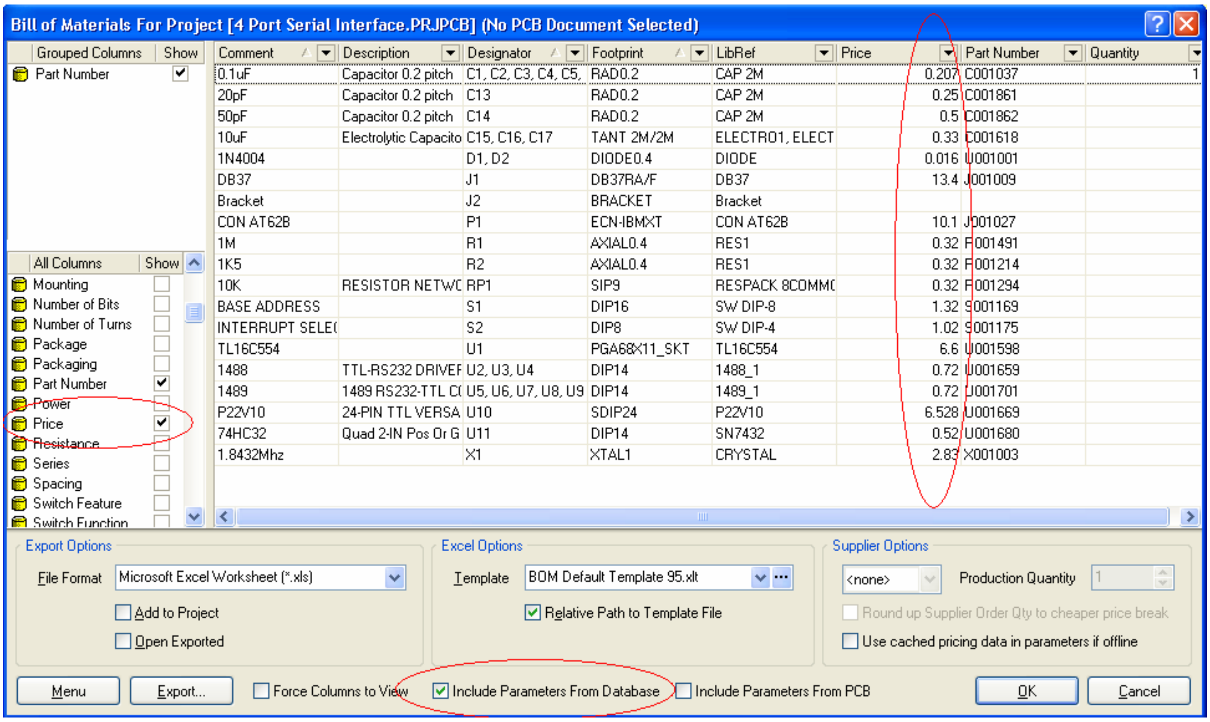



*Figure 6. Include additional component information that exists only in an external database*

## 19.1.5 Adding PCB Information Directly to a BOM

Source information for a Bill of Materials (BOM) can be based on property information taken from the PCB in the event you need to customize and use the report generation for more than a BOM.

An example would be for generation of a pick and place file where every placement machine wants the data (such as X, Y location) in a different column order and in different file formats.

When configuring the Bill of Materials report using the Report Manager dialog, simply enable the **Include Parameters From PCB** option. This option will only be available if there is a PCB document in the project file. In the parameter listing, the icon is used to distinguish a PCB parameter for one or more placed components in the project.

Note that when you have a project with multiple PCBs and you enable the **Include Parameters From PCB** option, the *BOM Report Options* dialog will automatically prompt you to select which PCB to include in the BOM report.

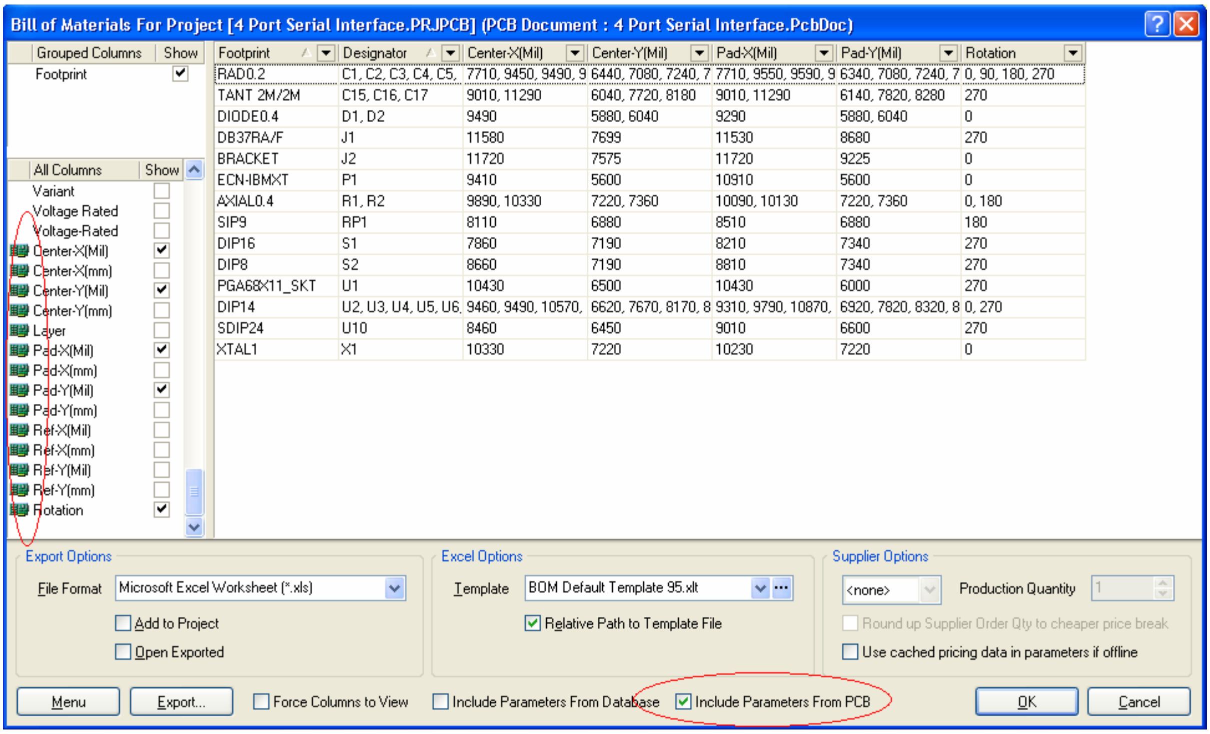


*Figure 7. Include additional component information that exists in a PCB.*

## 19.1.6 Supplier Options

This feature located in the bottom right side of the bill of materials dialog enables the use of linking to direct supplier data that is already embedded in the design from suppliers such as Digikey, Farnell and Newark. In order for this section to function certain parameter need to exist in the design.

- Supplier Unit Price

Once this parameter has been enabled in the left column the supplier options should fully enable.

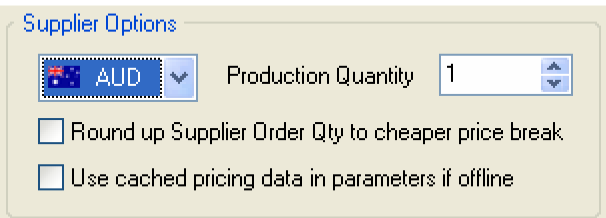



*Figure 8. Supplier options in the bill of materials.*

From here the local currency can be picked. There is two further options to pick, **round up supplier order Qty to cheaper price break**. This tick box should pick the cheapest option for buying in qty. For example if 9 are using on the design, however at 10 there is a price drop, and buying the extra is cheaper than the bill of material should round up to cheaper price point. The second option, **use cached pricing data in the parameters if offline** is useful if the design in progress uses a database library or you are offline, then it'll use the data that is already cached in the design, using the conversion rate done at the time of population of the parameter in the design.

**Note:** The currency data is set in the pricing parameter. If there is a difference between the pricing data currency and local currency a conversion takes place.

## 19.1.7 Exporting the Report

The grid content of the data section can be exported and a report generated by using the **Export** button in the *Report Manager* dialog.

1. Select a File Format from the drop-down list. When exporting the data using the **Export** option from the *Report Manager* dialog, the following file formats are supported:
   - CSV (Comma Delimited) (*.csv)
   - Microsoft Excel Worksheet (*.xls)
   - Tab Delimited Text (*.txt)
   - Web Page (*.htm; *.html)
   - XML Spreadsheet (*.xml).
2. If you want the relevant software application, e.g. Microsoft Excel, to open once the exported file has been saved, make sure the **Open Exported** option is enabled in the *Report Manager* dialog.
3. If you want to have the generated report added to the project after it is created, simply enable the **Add to Project** option in the *Report Manager* dialog.
4. Click on **Export** button in the *Report Manager* dialog and to generate and save the report in the appropriate format.

## 19.1.8 Using Excel Templates

If you want to export your data straight into an Excel template, you can select an existing Excel template or you can use the supplied Excel templates.

1. If the **Microsoft Excel Worksheet (*.xls)** file format is selected, the **Template** field becomes available in the Excel Options region of the *Report Manager* dialog. Enter the required Excel template file (`*.XLT`) directly into this field, or browse for it by clicking the **...** button. The field's drop-down list contains a range of default templates for the installation.

Select the `BOM Default Template.XLT` from the `\Program Files\Altium Designer Summer 09\Templates` folder. The file can be specified with a relative or absolute path using the **Relative Path to Template File** option. For more information about template creation, refer to your Microsoft Excel documentation.

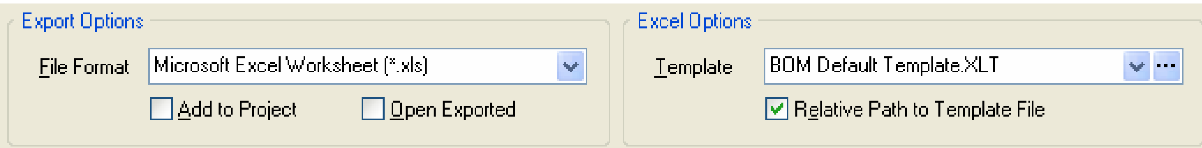



*Figure 9. File format and template data set for Bill of Materials.*

2. If you have the **Open Exported** option selected in the *Report Manager* dialog, the file will open in Excel after export.
3. Click on the **Export** button and nominate a filename and location for your report and click on **Save**. The report opens in Excel, formatted in the nominated Excel template.

4. Click on the Project Information tab to display details about the report.

# Bill of Materials

**4 Port Serial Interface**

**Source Data From:** **4 Port Serial Interface.PRJPCB**
**Project:** **4 Port Serial Interface.PRJPCB**
**Variant:** **None**

Creation Date: 21/07/2009 9:56:28 AM
Print Date: 40015 40015.41425

| Footprint | Comment | LibRef | Designator | Description | Quantity |
|---|---|---|---|---|---|
| RAD0.2 | 0.1uF | CAP 2M | C1, C2, C3, C4, C5, C8, C9, C10, C11, | Capacitor 0.2 pitch | 10 |
| RAD0.2 | 20pF | CAP 2M | C13 | Capacitor 0.2 pitch | 1 |
| RAD0.2 | 50pF | CAP 2M | C14 | Capacitor 0.2 pitch | 1 |
| TANT 2M/2M | 10uF | ELECTRO1, ELE | C15, C16, C17 | Electrolytic Capacitor RB mount | 3 |
| DIODE0.4 | 1N4004 | DIODE | D1, D2 | | 2 |
| DB37RA/F | DB37 | DB37 | J1 | | 1 |
| BRACKET | Bracket | Bracket | J2 | | 1 |
| ECN-IBMXT | CON AT62B | CON AT62B | P1 | | 1 |
| AXIAL0.4 | 1M | RES1 | R1 | | 1 |
| AXIAL0.4 | 1K5 | RES1 | R2 | | 1 |
| SIP9 | 10K | RESPACK 8CON | RP1 | RESISTOR NETWORK 8 COMMON RESIS | 1 |
| DIP16 | BASE ADDRESS | SW DIP-8 | S1 | | 1 |
| DIP8 | INTERRUPT SEL | SW DIP-4 | S2 | | 1 |
| PGA68X11_SI | TL16C554 | TL16C554 | U1 | | 1 |
| DIP14 | 1488 | 1488_1 | U2, U3, U4 | TTL-RS232 DRIVER | 3 |
| DIP14 | 1489 | 1489_1 | U5, U6, U7, U8, U9 | 1489 RS232-TTL CONVERTOR | 5 |
| SDIP24 | P22V10 | P22V10 | U10 | 24-PIN TTL VERSATILE PAL DEVICE | 1 |
| DIP14 | 74HC32 | SN7432 | U11 | Quad 2-IN Pos Or G | 1 |
| XTAL1 | 1.8432Mhz | CRYSTAL | X1 | | 1 |
| | | | | | 37 |

**Approved** **Notes**

*Figure 10. Generated Report in Excel using BOM Default Template.XLT template.*

# 19.2 Output Generation

All output generation settings (print, Gerber, NC drill, ODB++, CAM, report and netlist, etc) can either be:

- Configured and stored as part of the project. If you select print, Gerber, and other outputs from the PCB editor's **File**, **Design** and **Reports** menu these output configurations are stored in the project file.
- Alternatively you can add an Output Job file to the project and store the output setups there. The advantage of an Output Job file is that it supports setting up multiple outputs of any kind. It also allows multiple outputs to be generated in a single operation, and the OutJob file can be copied from one project to another. Any combination of output setups can be included in the OutJob file, and any number of OutJob files can be included in the project. Note that setting made in the OutJob file are completely independent of the settings made in the PCB Editor's menus.

## 19.2.1 Creating a new Output Job file

The Output Job file enables you to define all of your design output configurations - assembly, fabrication, reports, netlists, etc - all in the one convenient and portable file. Each output setup uses a specific data source including the entire project (all schematic sheets), an individual schematic or the PCB.

- Select **File » New » Output Job File** to create a new output job configuration file. A new output job configuration file (`Job1.OutJob`) is created and added to the `Job Files` sub-folder of the focused project in the **Projects** panel. It opens as the active document in the design window and defaults to include all possible output setups.

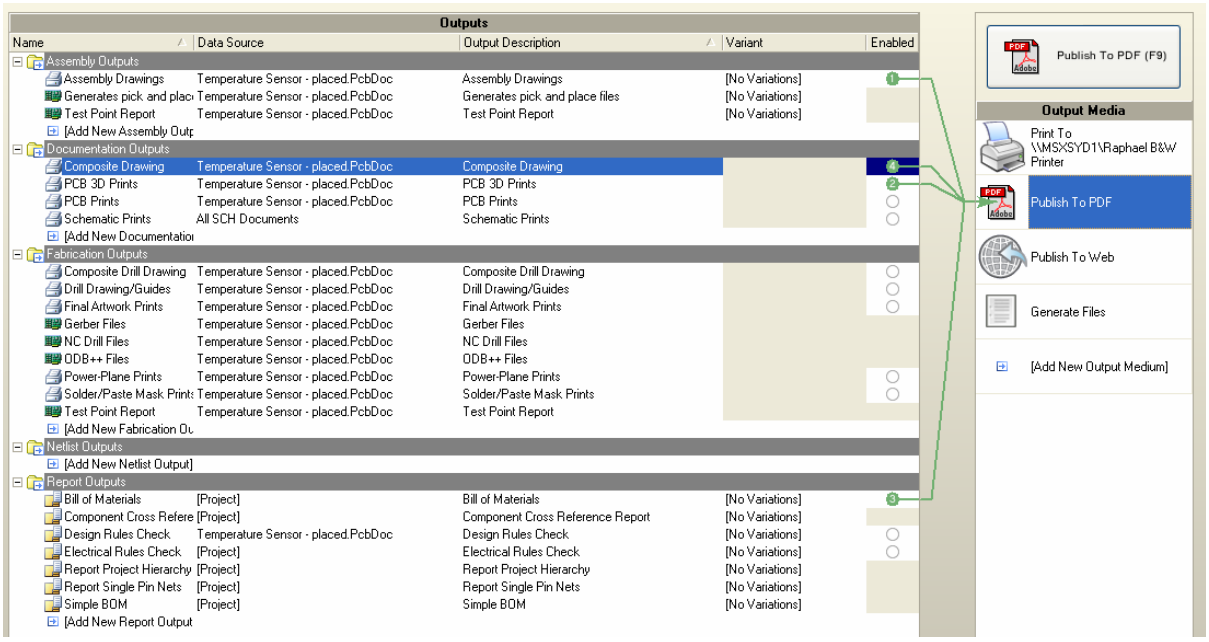


*Figure 11. A Output Job file with four output setups configured to be published into the one PDF fle.*

- Selected setups can be deleted (CTRL+A to select all) and new outputs can be added at any time by clicking on the required **Add New Output**.
- Double-click on an output to configure it in its *Properties* dialog, or right-click for a list of options. The Data Source and Variants columns also have a drop-down list to choose from —

click once to select the item, then click a second time to display the down arrow and then select from the list.

## 19.2.2 Setting up Print job options

- Select a print output from the Output Job file, e.g. Composite Drawing. Double-click to configure this printout option in the *PCB Printout Properties* dialog.

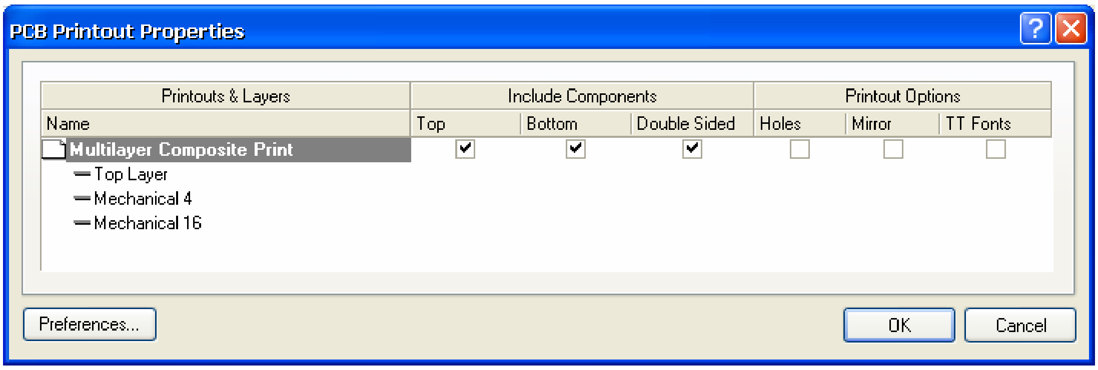



*Figure 12. Printout Properties dialog*

- Click on the **Preferences** button to set the colors and layers to include in the printout.

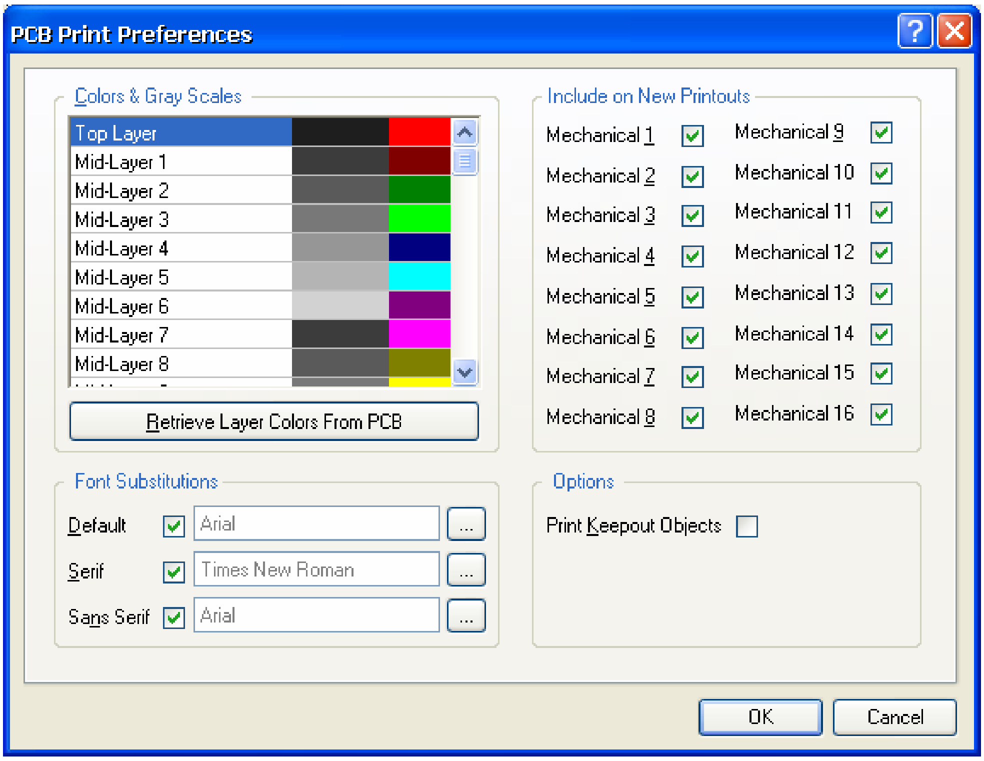



*Figure 13. PCB Print Preferences dialog*

- Right-click on the print option in the Output Job file to configure which printer your output will print to (**Printer Setup**) as the printouts will be sent directly to that printer when you run the output generator.
- Right-click and select **Print Preview** to view your printout. From the preview window you can copy the current Printout preview to the Windows clipboard by right-clicking and selecting **Copy**. You can also save the image as an Enhanced Windows Metafile (`.emf`) by right-clicking and selecting **Export Metafile**.

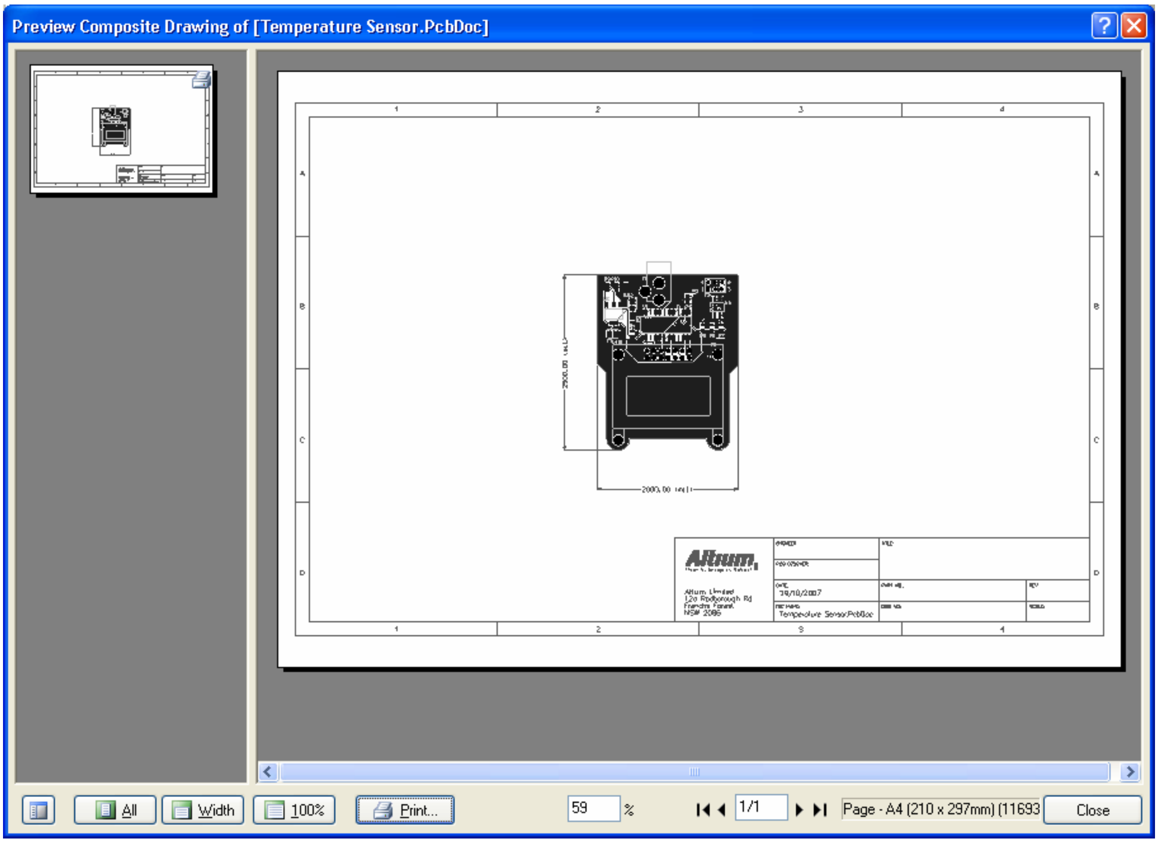



*Figure 14. Print Preview window with all layers displayed.*

- When the printout is configured, you can run it as a batch job (if Batch is enabled) along with all the other setups (**F9**), run the current output generator (SHIFT+F9) or run a selection of output generators (CTRL+SHIFT+F9). These output options are also available in the right-click menu. The printouts are sent to the printer.

### 19.2.3 Creating CAM files

You can setup and create manufacturing output files from the Output Job file, such as:

- Bill of Materials
- Gerber and ODB++ files
- NC Drill files
- Pick and Place files
- Testpoint Report.

The data is output into appropriate documents in a folder within the same folder as your PCB file or in separate folders for each output type as determined in the **Options** tab of the *Options for Project* dialog.

## 19.2.4 Gerber

This option in the Job Output file produces a Photoplotter output in Gerber format. Double-clicking on a **Gerber Files** output displays the *Gerber Setup* dialog. Consult your PCB manufacturer for their preferred settings.

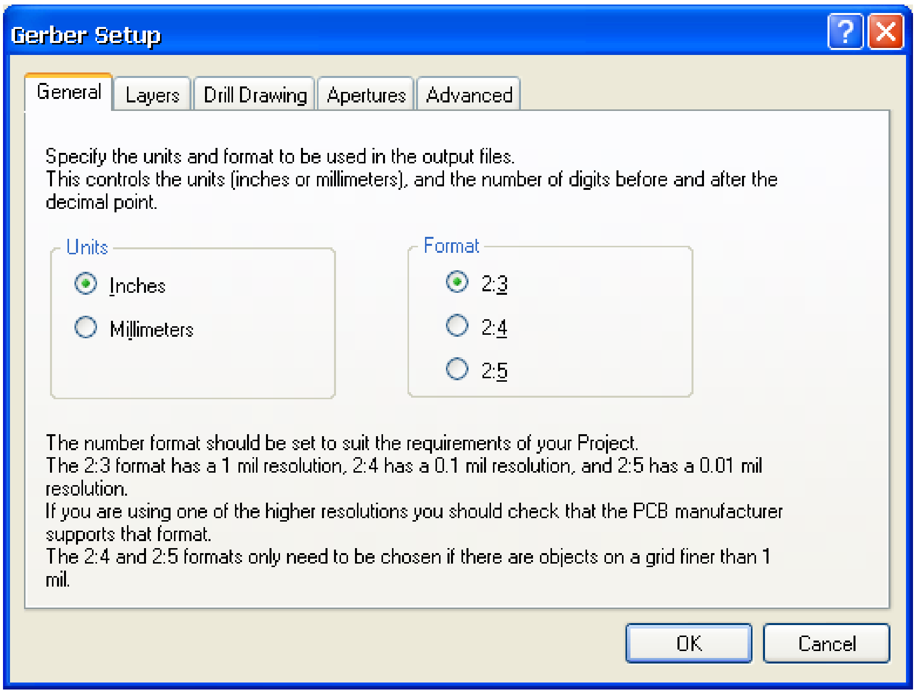



*Figure 15. Gerber Setup dialog*

## 19.2.5 NC Drill

This option produces a NC drill output in an industry standard format. Double-clicking on **NC Drill Files** displays the *NC Drill Setup* dialog. Consult your PCB manufacturer for their preferred settings.

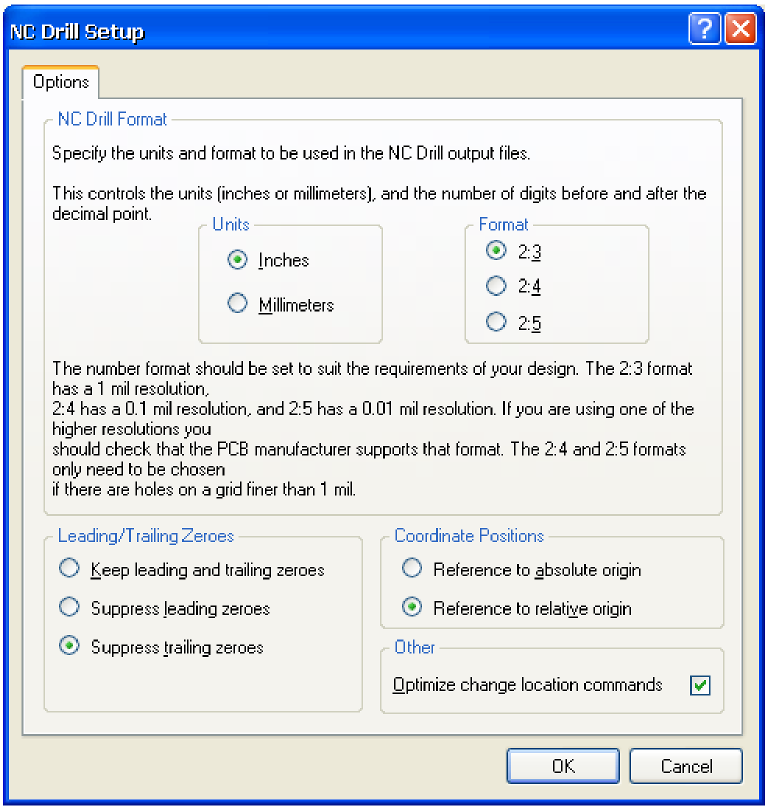



*Figure 16. NC Drill Setup dialog*

## 19.2.6 ODB++ Output

This option produces ODB++ output, ready to load into any ODB++ compliant CAM tool. Double-clicking on **ODB++ Files** displays the *Select Layers to Plot* dialog.

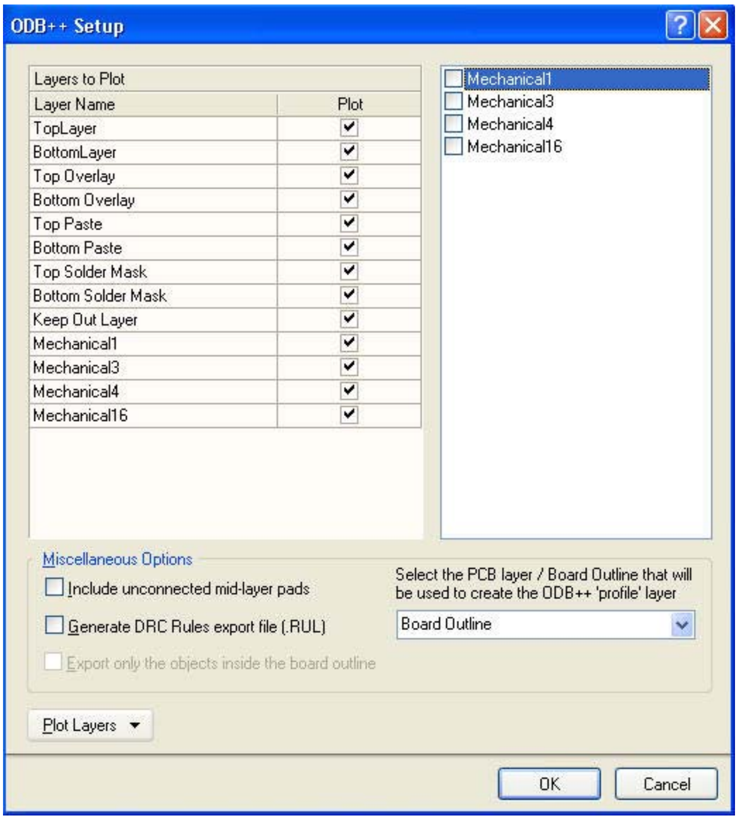
## 19.2.7 Pick and Place

This option produces component data that is used to program a Pick and Place machine. Double-clicking on **Generates Pick & Place Files** displays the *Pick and Place Setup* dialog.

## 19.2.8 Testpoint report

This option produces information on the location and size of Testpoints for use in fabricating test fixtures and programming testers.

Double-clicking on a **Testpoints Reports** displays the *Testpoint Report Setup* dialog.




## 19.2.9 Smart PDFs

Altium Designer has built in PDF generation capabilities. As well as being able to create standard PDF files showing the schematic sheets and the PCB layers, you can also include linked PDF bookmarks to components and nets on the sheets and board.

There is 2 ways of generating PDFs, select **File » SmartPDF** from the schematic or PCB editor menus, or use the **Publish To PDF** option in the Output Job editor.

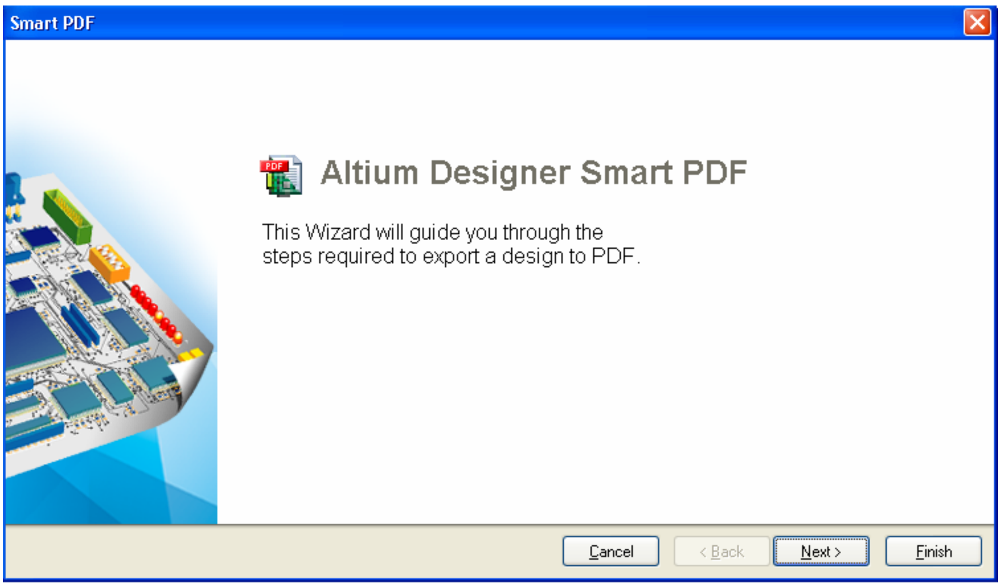



*Figure 18. The SmartPDF wizard*

Using the SmartPDF wizard creates an output job file at the end of the process with all the options that were specified in the wizard. This is a handy tool if you are not too familiar with the output job file and how to modify the setups.

## 19.2.10 Publish To Web

The main role of the new web publisher output medium is to enhance collaboration by making web publication of design outputs (graphical and table based) easy and automatable.
The new web publisher will publish the outputs of Outjob Documents on web hosted media (FTP, S3, WebDav) as well as locally.

### 19.2.10.1 Locating

From the File menu, select **New » Output Job File**.

Click on the **Publish To Web** output medium, and select the outputs you wish to publish by clicking in the **Enabled** column. A green arrow leading to the output medium will appear for each output selected.

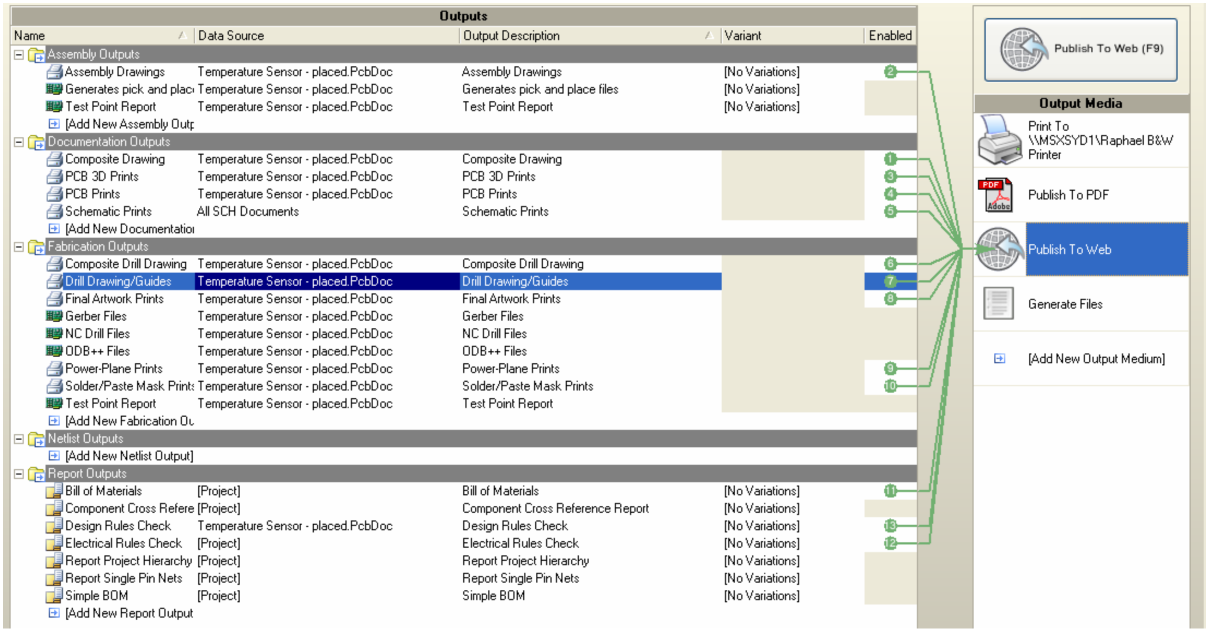


*Figure 19. Publish to web in the output job file*

### 19.2.10.2 Settings Window

Once the desired outputs have been selected, right click on the output medium **Publish To Web** and select **Web Setup**.

The settings window will display and allow you to configure the web output destination.

### 19.2.10.3 Prompt if files already exist

When selected this option will cause the system to confirm with you before overwriting any files in the destination.

### 19.2.10.4 Open after export

When selected this option will cause the system to open the published files in a web browser window.

### 19.2.10.5 Destination Type

Four destination types, each representing a different way of transferring files, are available allowing you to select the method best suited to your environment.

### 19.2.10.6 Destination Type - File System

The file system destination type will publish the web output to a folder of your choosing, either on your local machine, or on a shared network drive.

### 19.2.10.7 Output Folder

This is the folder where the published files should be placed.

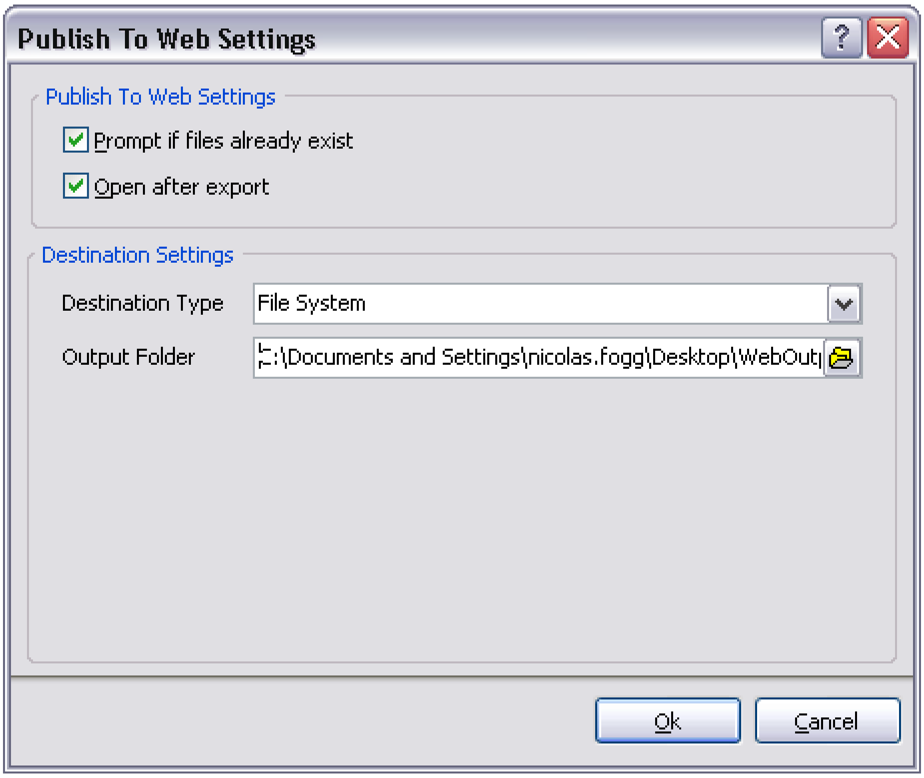



*Figure 20. File system and destination options*

### 19.2.10.8 Destination Type - FTP

The FTP destination type will publish the web output to a server capable of accepting files via the File Transfer Protocol (FTP).

### 19.2.10.9 Server Name

The server name or address used to identify the server on the network.

### 19.2.10.10 User Name

The account on the server with the appropriate upload permissions.

### 19.2.10.11 Password

The password for the account used.

### 19.2.10.12 Output Folder

The folder into which the files should be placed on the server.

### 19.2.10.13 HTTP URL

The URL which can be used to access the folder in a standard web browser. If blank the web browser window will not be opened at the completion of the transfer.

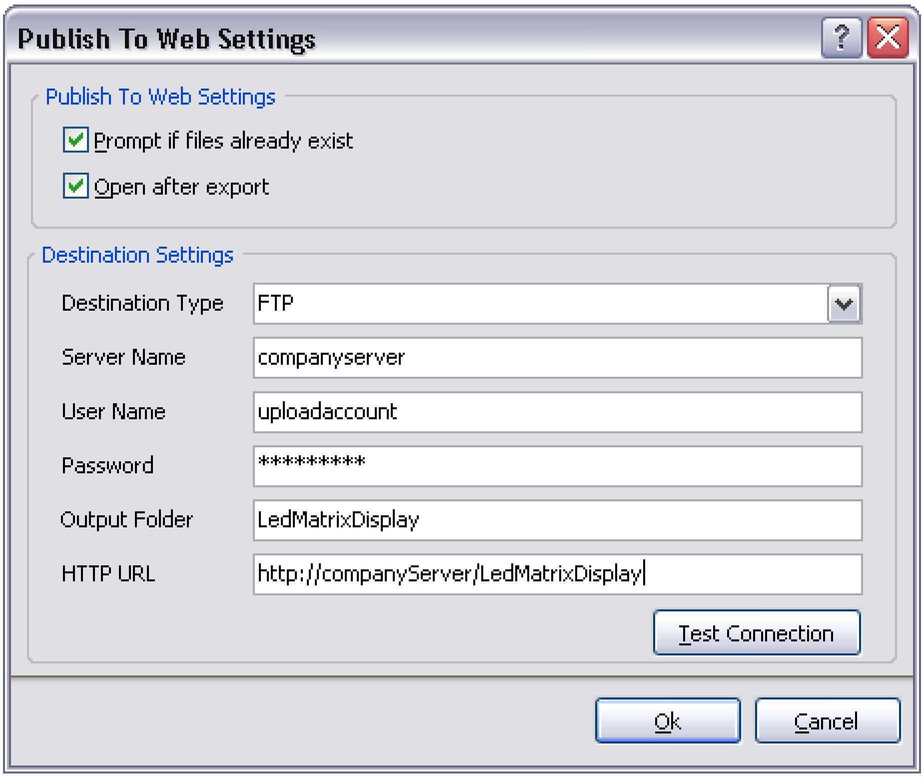



*Figure 21. FTP destination options*

### 19.2.10.14 Destination Type - WebDAV

The WebDAV destination type will publish the web output to a web server capable of accepting files via the WebDAV protocol.

### 19.2.10.15 URL

The full URL of the destination, including both the server name and the output folder.

### 19.2.10.16 User Name

The account on the server with the appropriate upload permissions.

### 19.2.10.17 Password

The password for the account used.

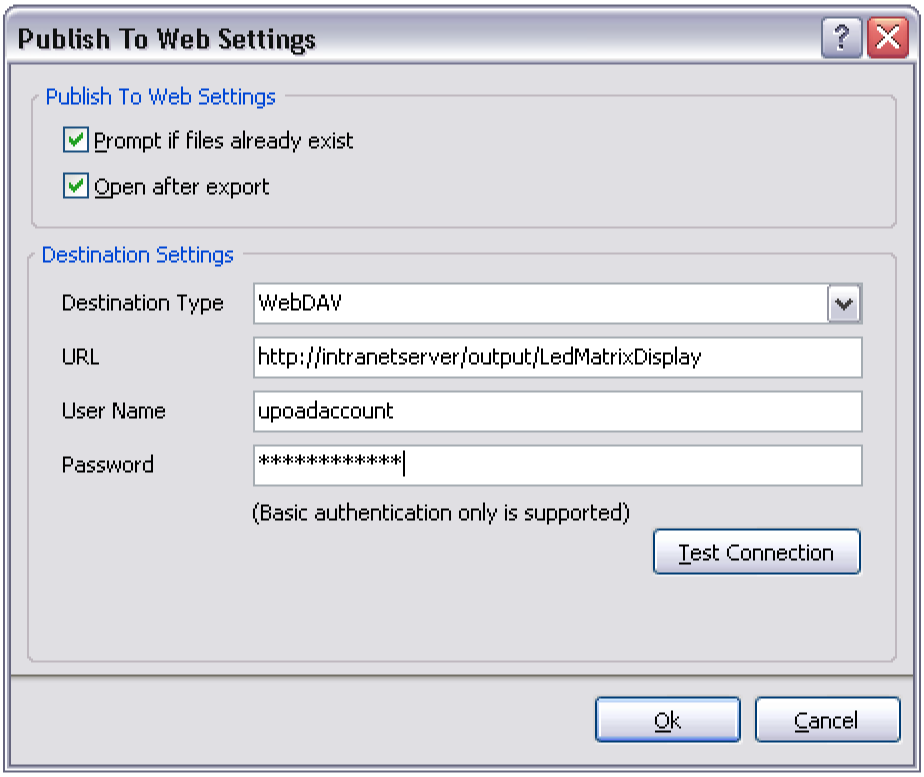


*Figure 22. WebDEV destination options*

### 19.2.10.18 Destination Type - S3

The S3 destination type will publish the web output to Amazon's Simple Storage System (S3). An account with Amazon is required for this destination type.

### 19.2.10.19 Account Name

Your Amazon account name.

### 19.2.10.20 Access Key

Your Amazon access key.

### 19.2.10.21 Secret Key

Your Amazon secret key.

### 19.2.10.22 Bucket

The bucket into which the files should be uploaded.

### 19.2.10.23 Key Prefix

The prefix to add to each file name. A key prefix ending with / will cause the files to be placed in a folder with the key name.

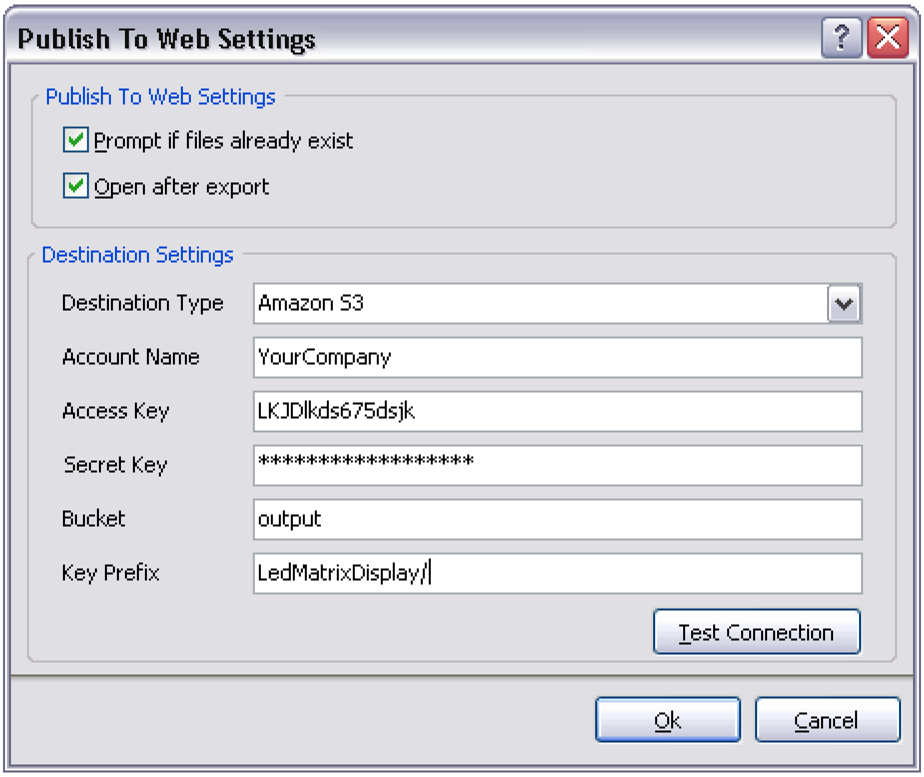



*Figure 23. Amazon S3 destination options*

### 19.2.10.24 Publish

Once the settings have been confirmed you may click the **Publish To Web** button to start the upload. Once the upload has completed a web browser will open and display the published files.

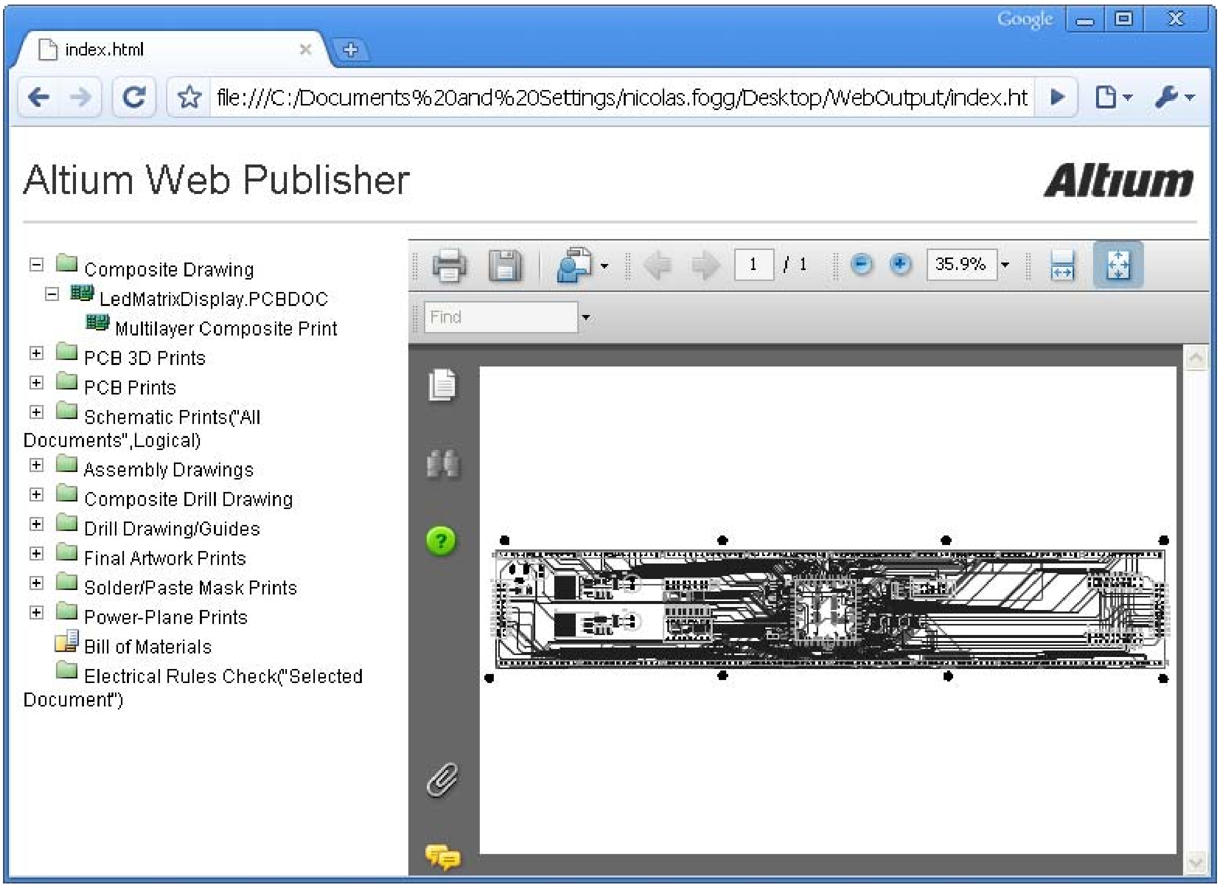



*Figure 24. Final result publish to web*

## 19.2.11 Running the Output Generator

You can run the Output Generator to create your output files and printouts from within the Output Job file itself (right-click menu) or use the **Tools** menu which includes a number of Run options. When the **Run Batch** command is selected (**F9**) all output setups with the Batch checkbox ticked will be generated.

You can also generate output for a selected group of outputs from within the Output Job file by highlighting them and selecting the **Run Selected** command (SHIFT+CTRL+F9).

Fabrication CAM outputs can be set to open automatically in CAMtastic by enabling the relevant options in the *Output Job Options* dialog (**Tools » Output Job Options**).

## 19.2.12 Exercise – adding an OutJob file to the project

1. With the Temperature Sensor project open, select **File » New » Output Job File**.
2. Save the document, naming it as `Temperature Sensor.OutJob`.
3. Select all the output setups (CTRL+A), and press **Delete** to remove them.
4. Add in an Assembly Drawing, Gerber, NC Drill and a Bill of Materials.
5. Click on the Gerber output setup to select it, then right click and configure the output. Make sure layers are ticked, format set correctly, apertures etc… Do the same for the NC drill output.
6. Select **Tools » Output Job Options**.

7. In the *Output Job Options* dialog, enable the Gerber and NC drill output check boxes and close the dialog.
8. Select both the Gerber and NC drill using the ctrl key, right-click and choose **Run Selected** from the menu. The files will be generated, a new CAMtastic document created and the gerber and NC drill documents loaded into it. These can now be checked, panelized, and so on.
9. Once the Gerber is created, go back into the output job file and create a PDF of the assembly drawing using the publish to PDF output type.

# 19.3 CAM Editor

Altium Designer's CAM Editor (CAMtastic) offers a variety of tools, the most basic of which are for viewing and editing CAM data. Once image and drill files have been imported, the Editor can receive instructions determining layer types and stackup, at which point a netlist can be extracted and compared with an IPC netlist generated from the original PCB design software. These netlists will handle not just through-hole components, but blind and buried vias as well. The Editor also offers Design Rule Checking, panelization and NC-Routing (plus milling) tools.

## 19.3.1 Setting up data to use in the CAM Editor

- After importing CAM data into a CAMtastic document, you may notice that there is options grayed-out, like export to PCB and DRC doesn't work.
- The following sections detail steps to be taken in order to get Gerber or ODB++ data ready to use in a CAMtastic document.

## 19.3.2 Import Data

- The starting point for the whole process is the loading of ODB++, or Gerber and NC Drill files, into a new CAMtastic document. If you have an IPC netlist, you should import this as well when importing Gerber and NC Drill data. When importing ODB++, only import an IPC file if the netlist is not in the ODB++ directory. The IPC netlist will allow you to update the extracted nets with their original names, and differentiate between through-hole vias and free (non-component) pads.
- The **File » Import** submenu commands all search for files with certain extensions within a given folder. When importing drill files, for instance, the specified directory will filter out all files except those with `.DR*`, `.ROU`, `.RTE`, `.NC*` and `.TX*` extensions. If you receive dfiles with different extensions than these, you may extend this list on the **CAMtastic – Miscellaneous** page of the *Preferences* dialog (**DXP » Preferences**).

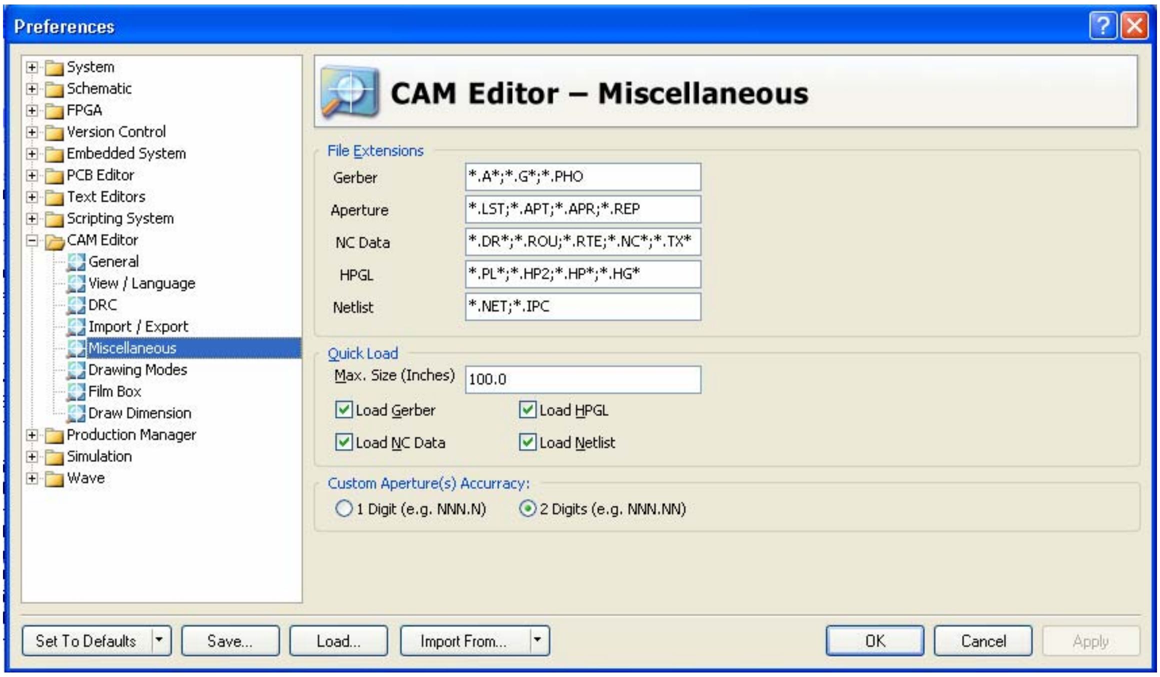



*Figure 25. Miscellaneous options for CAMtastic in the Preferences.*

- From here you can alter any of the default extension lists associated with Gerber, Aperture, HPGL and Netlist files as well.
- The **CAMtastic – Import/Export** page of the *Preferences* dialog lets you change the default import and export settings for Gerber files. For example, if you regularly receive CAM data generated using P-CAD, you might consider changing the coordinate precision format from 2:3 (Altium Designer defaults) to 4:4 (P-CAD defaults). If you don't you will still have the opportunity to change these settings from their default state each time you import a job.

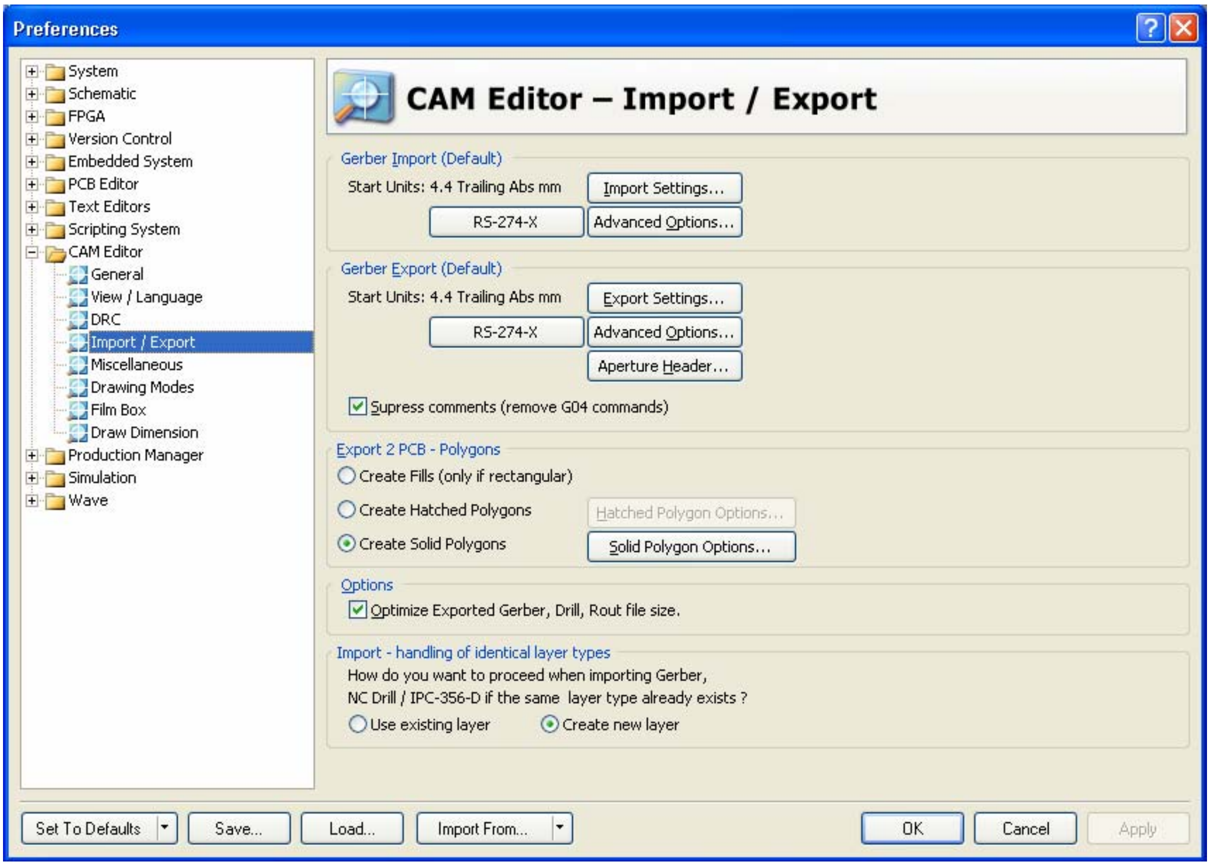
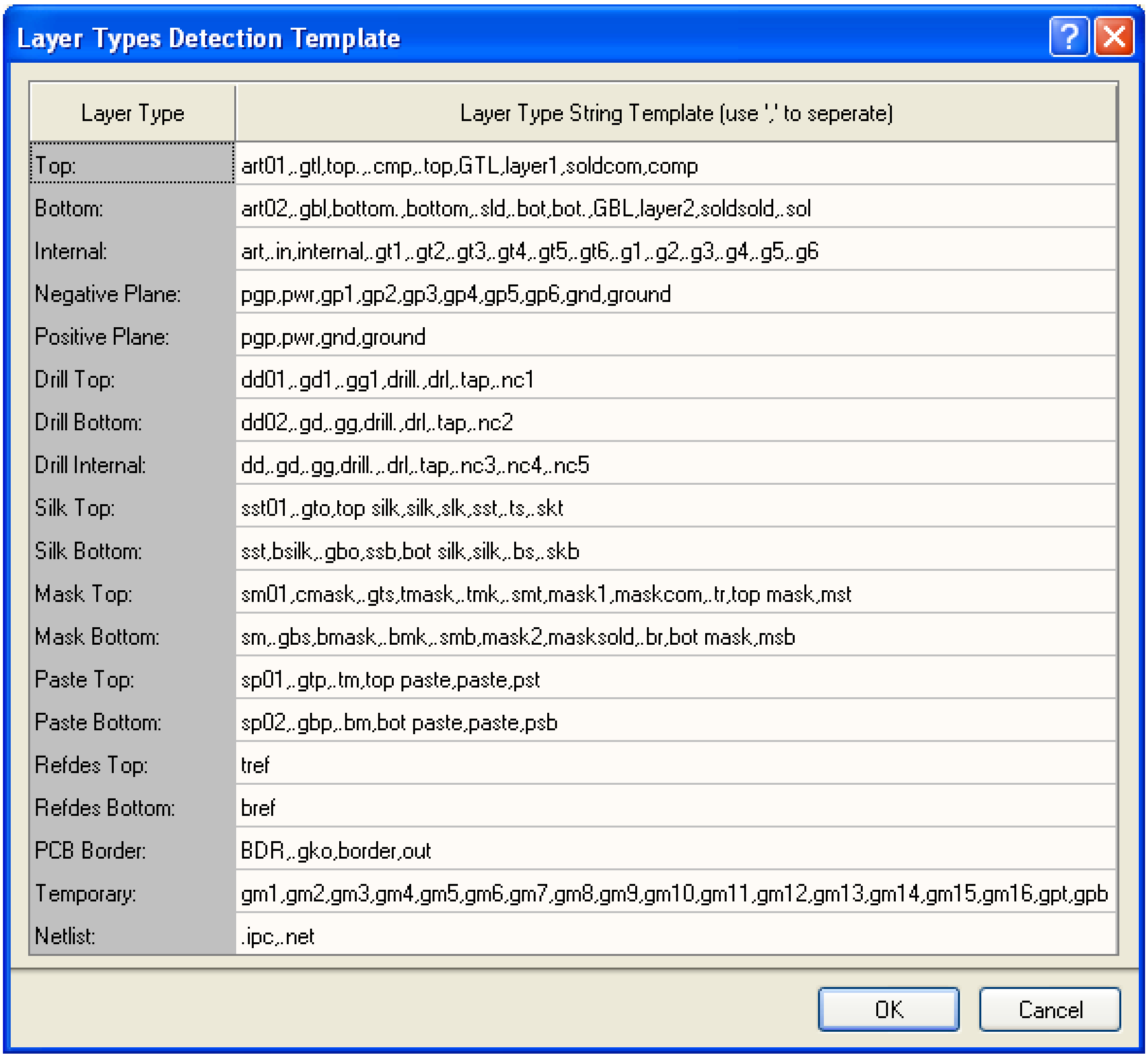



*Figure 26. Import/export options in preferences for CAMtastic.*

## 19.3.3 Layer Type Assignments

- Individual Gerber and Drill files are all assigned to individual layers. In addition, the CAMtastic Editor looks at their names upon import, and tries to assign each one a layer type. It does this according to the information in the *Layer Types Detection Template* dialog (**Tables » Layer Type Detection**). This is a fully-customizable dialog that contains fragments of file names (extensions, keywords or both) that identify individual files as layer types, such as signal, negative plane, border, drill, and temporary layers (mechanical).
- Altium Designer's PCB Editor, for example, differentiates between Gerber layers by their extensions, such as `.gtl` for the top layer and `.gbl` for the bottom. Other design programs might export a `.top` file for a top layer, and a `.sol` file for the bottom (solder) layer. Some of these programs, such as P-CAD, allow users to specify the Gerber output names themselves.
- After importing any group of CAM files, you should open the *Layers Table* dialog and review the type assignments that have taken place automatically.



Layer Types Detection Template

| Layer Type | Layer Type String Template (use ',' to seperate) |
| --- | --- |
| Top: | art01,.gtl,top.,.cmp,.top,GTL,layer1,soldcom,comp |
| Bottom: | art02,.gbl,bottom.,bottom,.sld,.bot,bot.,GBL,layer2,soldsold,.sol |
| Internal: | art,.in,internal,.gt1,.gt2,.gt3,.gt4,.gt5,.gt6,.g1,.g2,.g3,.g4,.g5,.g6 |
| Negative Plane: | pgp,pwr,gp1,gp2,gp3,gp4,gp5,gp6,gnd,ground |
| Positive Plane: | pgp,pwr,gnd,ground |
| Drill Top: | dd01,.gd1,.gg1,drill.,drl,.tap,.nc1 |
| Drill Bottom: | dd02,.gd,.gg,drill.,drl,.tap,.nc2 |
| Drill Internal: | dd,.gd,.gg,drill.,.drl,.tap,.nc3,.nc4,.nc5 |
| Silk Top: | sst01,.gto,top silk,silk,slk,sst,.ts,.skt |
| Silk Bottom: | sst,bsilk,.gbo,ssb,bot silk,silk,.bs,.skb |
| Mask Top: | sm01,cmask,.gts,tmask,.tmk,.smt,mask1,maskcom,.tr,top mask,mst |
| Mask Bottom: | sm,.gbs,bmask,.bmk,.smb,mask2,masksold,.br,bot mask,msb |
| Paste Top: | sp01,.gtp,.tm,top paste,paste,pst |
| Paste Bottom: | sp02,.gbp,.bm,bot paste,paste,psb |
| Refdes Top: | tref |
| Refdes Bottom: | bref |
| PCB Border: | BDR,.gko,border,out |
| Temporary: | gm1,gm2,gm3,gm4,gm5,gm6,gm7,gm8,gm9,gm10,gm11,gm12,gm13,gm14,gm15,gm16,gpt,gpb |
| Netlist: | .ipc,.net |

OK Cancel

*Figure 27. Layer types detection setup.*

- You may open the drop-down list for the type and modify any assignment. For example, you will notice that all drill files are automatically set to type `Drill Top`. You might want to distinguish between top, bottom and internal drill sets at this point. This is not required, however. Later on we'll discuss pairing layers to accommodate boards containing blind and/or buried vias.
- If you customarily receive Gerber or drill files that are not automatically assigned to a layer type correctly, you should look over the file name fany distinguishing characteristics whereby the CAMtastic Editor might identify them, and add them to the *Layer Types Detection Template* dialog.
- ODB++ does not need to use the *Layer Types Detection Template* dialog because all the information needed to identify each layer is stored in the matrix file. However, it is good practice to check the layer type assignments after the loading process ends.

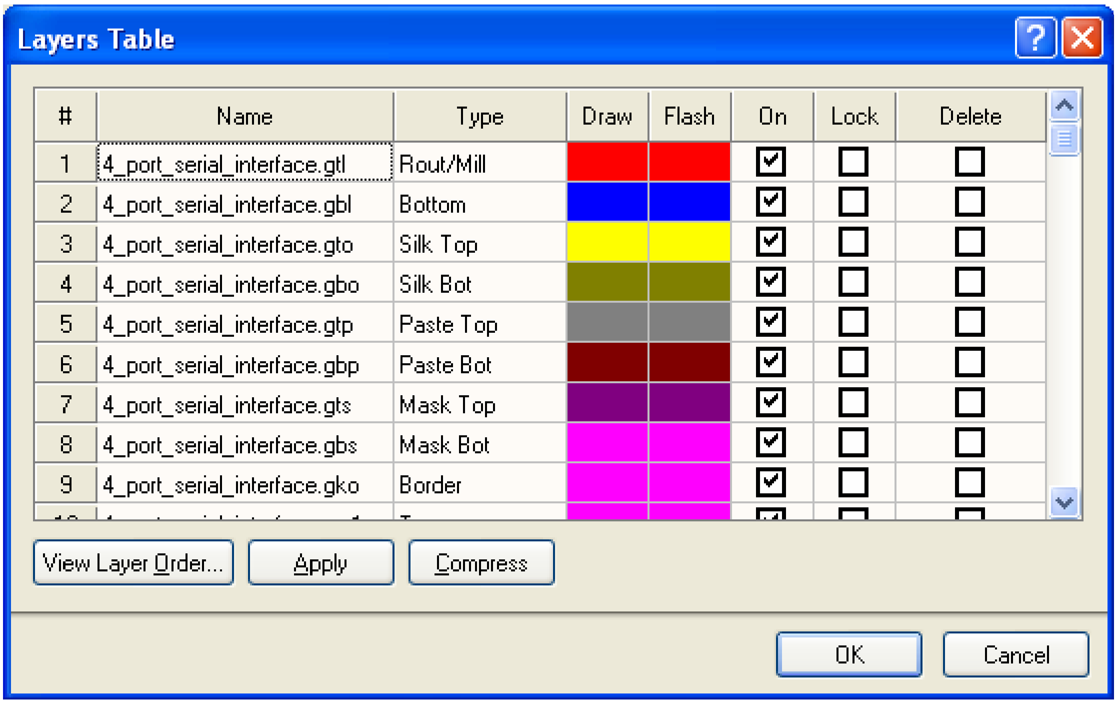



*Figure 28.Layers table assigning Gerber layers to Camtastic layers.*

## 19.3.4 Board Stackup and Drill Sets

- Layers that have been assigned as signal types (top, bottom or internal) or planes will now appear in the Create/Update Layers Order dialog (Tables » Layers Order). In most cases, the CAMtastic Editor will be able to determine the stackup from the provided CAM data, but in some cases all that will be provided is the logical order (the order in which the layers were imported). It is your responsibility to make sure that the information in the Layer Physical Order column is correct, with number one being the top layer. Be careful not to assign the same order number to different layers. Once you have specified a valid stackup, the physical order column will take precedence over the logical list when this dialog is reopened and you will see the layers listed in their proposed stackup.

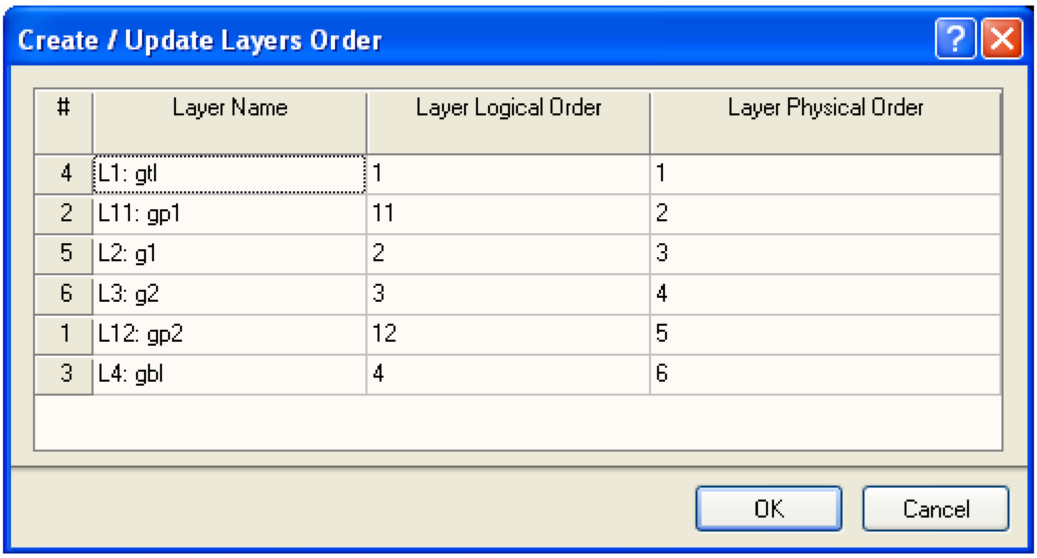



*Figure 29. Logically layer order setup.*

- This same physical order will be reflected in the *Create/Update Layers Sets* dialog (**Tables » Layers Sets**), where you match drill files to layer sets. Consider a 6-layer board (four-signal, two-plane) with both blind and buried vias. The design might contain four separate drill files:
  - one for the Top-Bottom pair (thru-holes)
  - one for the Top-InternalPlane1 pair
  - one for the MidLayer1-MidLayer2 pair
  - one for the Bottom-InternalPlane2 pair.
- Remember that you are determining a drill set, not just outermost pairs. For blind/buried drill layers, all layers that are drilled need to be in the layers set. For through hole drill layers, the start and end layers should suffice.

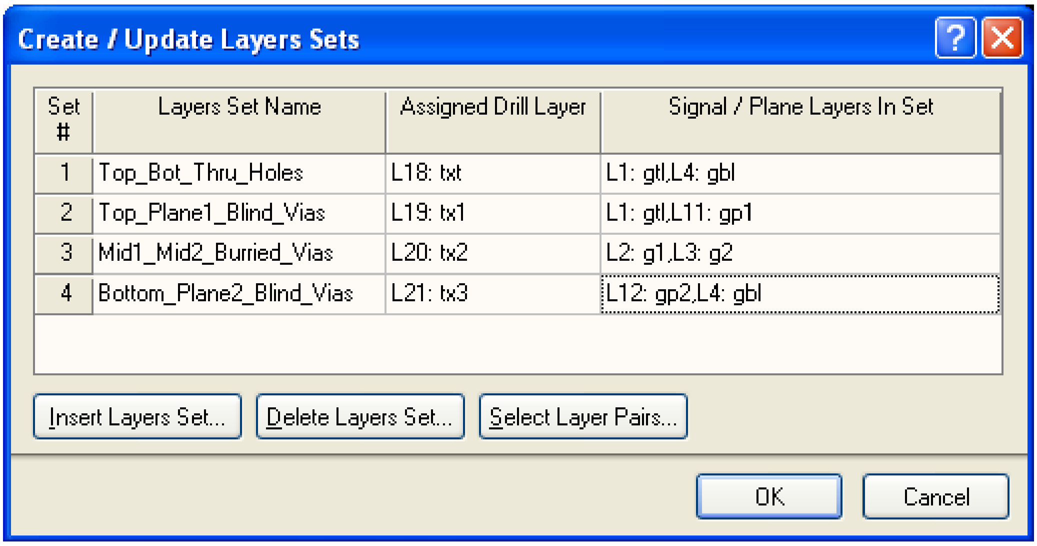



*Figure 30. Drill setup for thru hole and blind and buried setups.*

## 19.3.5 Netlist Extract

- This step (**Tools » Netlist » Extract**) relies on the accuracy of the steps preceding it. You can't do it at all unless you've got a layer type assignment for every layer in your board (you should set mechanical layers to `Temporary`).
- The CAMtastic Editor will trace connectivity from one layer to another, consulting the NC drill files to make layer to layer connections. For this reason, the layer stackup you defined in the *Create/Update Layers Order* dialog, and the drill pairs you selected in the *Create/Update Layers Sets* dialog, are critical precursors to extracting an accurate netlist from the CAM data.
- If you have included an IPC netlist file with your imported Gerber and NC Drill files, you may restore the original net names (**Tools » Netlist » Rename Nets**) and differentiate between through-hole vias and free pads in your new PCB file. But much more importantly, an IPC netlist gives you a reference with which the extracted netlist can be compared (**Tools » Netlist » Compare**).
- For IPC netlists generated from Altium Designer's PCB Editor, you will notice that the CAMtastic Editor's compare function will usually find a series of "missing nets" without names. This is because the PCB Editor's IPC format includes single-pin nets – the CAMtastic Editor, on the other hand, disregards pads without any copper connections as far as the netlist is concerned. In fact, its DRC has an option to remove non-functional pads.

## 19.3.6 Running a DRC

- Now we can run a DRC (Design Rule Check) to verify there are no violations in your CAM file that will affect the fabrication. You are presented with 18 rules. The values for these rules will be retained from your previous CAMtastic sessions, unless you loaded your CAM data from a Protel output that included a .RUL file. In any case, the values of these rules may be modified before running the DRC.

### 19.3.6.1 Setting up the DRC

1. Select Analysis » PCB Design Check/Fix. The PCB Design Check/Fix dialog displays.

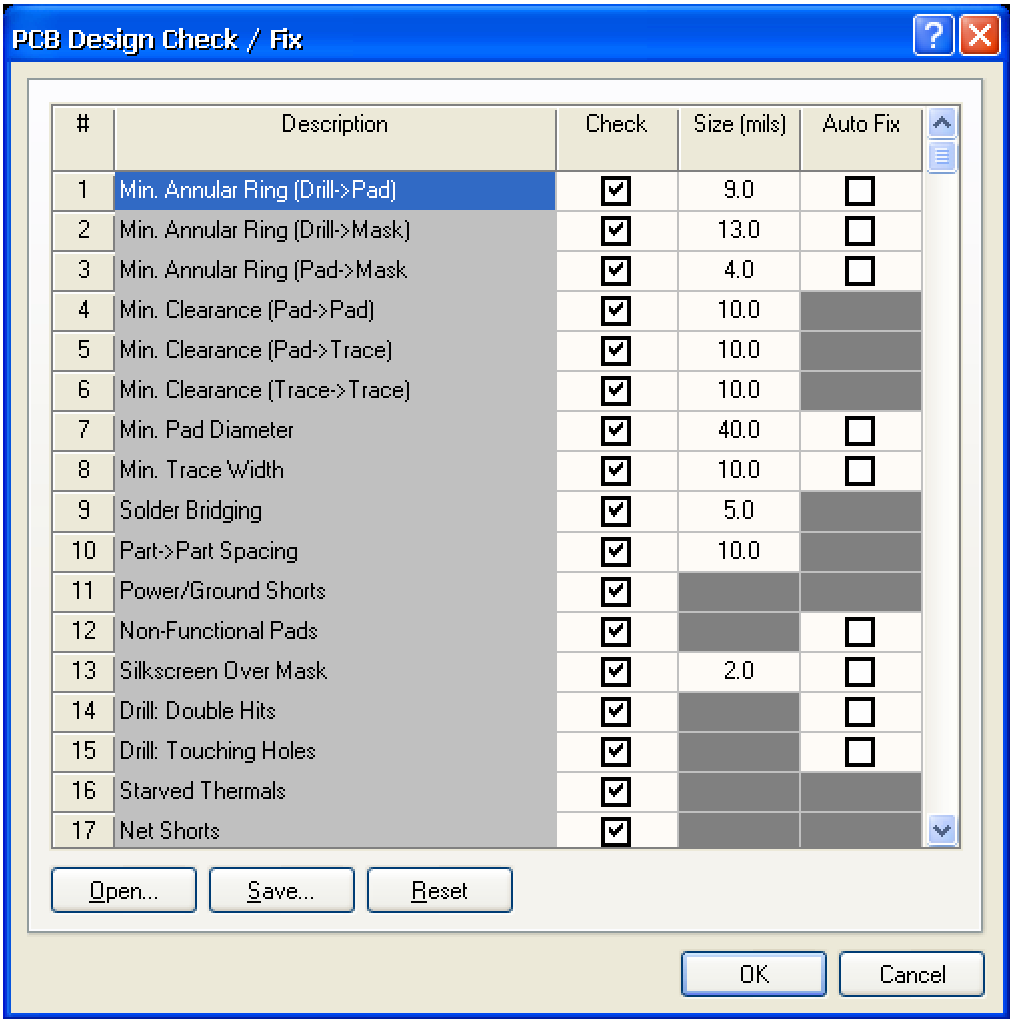



*Figure 31. DRC setup in Camtastic.*

2. From this dialog, you can change relevant size values, if required, or enable the Auto Fix option, if available, so CAMtastic will attempt to fix any violations found. We will first run the DRC without Auto Fix enabled to review the number of violations and then with an Auto Fix option enabled. Type in the sizes as displayed in the *PCB Design Check/Fix* dialog above. Enable all the **Check** column boxes. You can click on the Check header to toggle all options on or off.

3. Once you have set up the DRC, you can save the DRC settings to a `.DRC` file by clicking on **Save**. Use the **Open** button to reload saved `.DRC` files.

4. Click **OK** to run the DRC. The DRC runs and any violations display in the *CAMDXP* dialog.

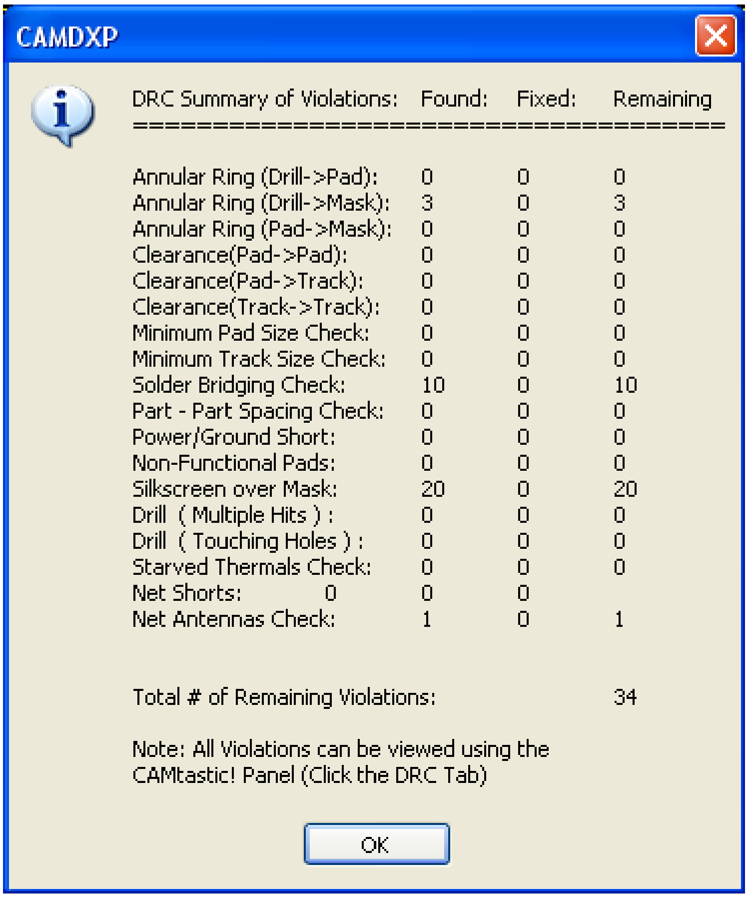



*Figure 32. Results of a DRC check in Camtastic.*

5. Click OK to close the dialog and click on the Drc tab in the CAMtastic panel to view more details about each of the violations.

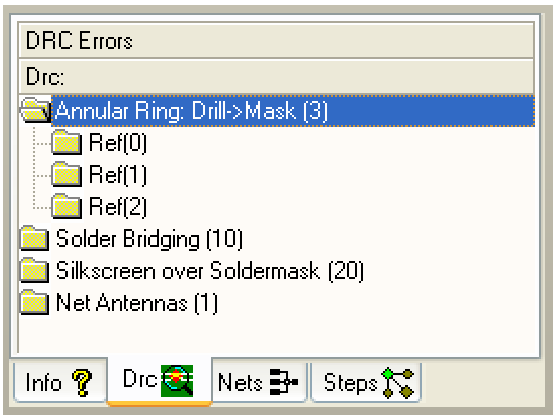



*Figure 33. Navigation of DRC violation in the Camtastic Panel.*

6. Double-click on a violation error folder in the Drc tab of the CAMtastic panel, e.g. Silkscreen over Solder mask, to view the individual error subfolders. Click on a subfolder, e.g. Ref (13), to zoom in on and highlight the offending object/s in the design window.

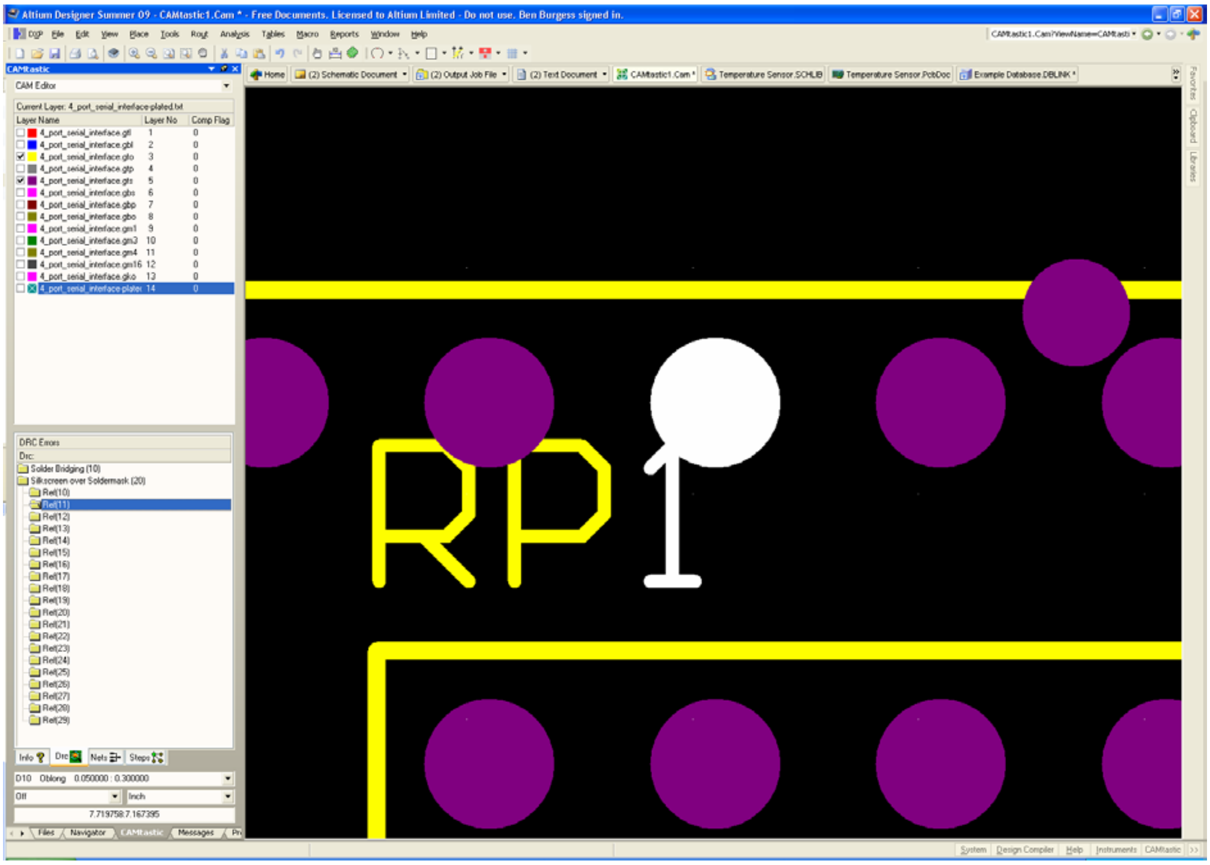


*Figure 34. A Silkscreen over soldermask problem found in CAMtastic.*

### 19.3.6.2 Querying a violation for further information

You can find out more information about the possible reason for errors by querying the object/s involved in the violation.

1. If the CAMtastic panel is active, press **Shift+F5** to make the design window active, or click in the workspace. You can now use the zoom and pan commands.
2. Press Q for Query (or select Analysis » Query » Object) and the cursor changes to a pointing hand. Click on the object you wish to find more information about. The information about the selected object is displayed in the Info tab of the CAMtastic panel. At the bottom of the Info Query section, all the DRC errors are listed that relate to the queried object. Click on these errors to zoom into those related violations.
3. You may also wish to measure distances between objects if there are clearance issues. Select a measuring option, such as Point to Point or Object to Object from the Analysis » Measure submenu and click on the points or objects you want to measure. The measurements are displayed in the Info tab of the CAMtastic panel.

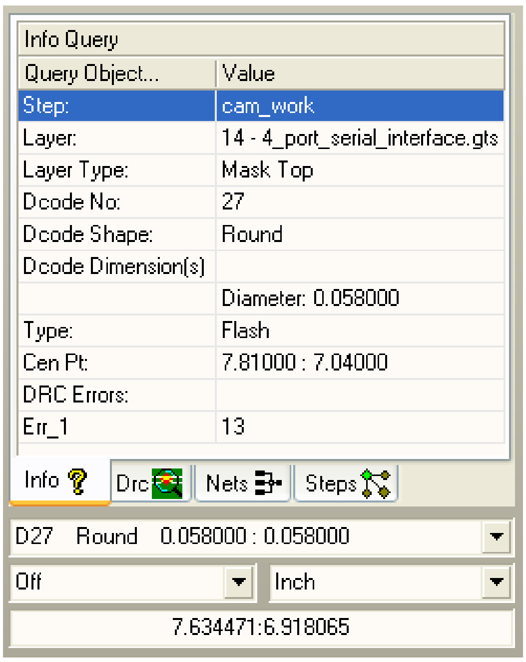



*Figure 35. Query results in CAMtastic.*

## 19.3.7 Using Auto Fix

1. In the PCB Design Check/Fix dialog (**Analysis » PCB Design Check/Fix**), enable all items in the Check column. Enable the Auto Fix checkbox for the Silkscreen over Solder Mask design rule only. Click OK to rerun the DRC. The CAMDXP dialog displays the results.
2. You will notice that number of violations has been reduced as the Auto Fix option has resolved the Silkscreen over Solder Mask errors. Auto Fix removes enough of the overlaid object to clear the violations.

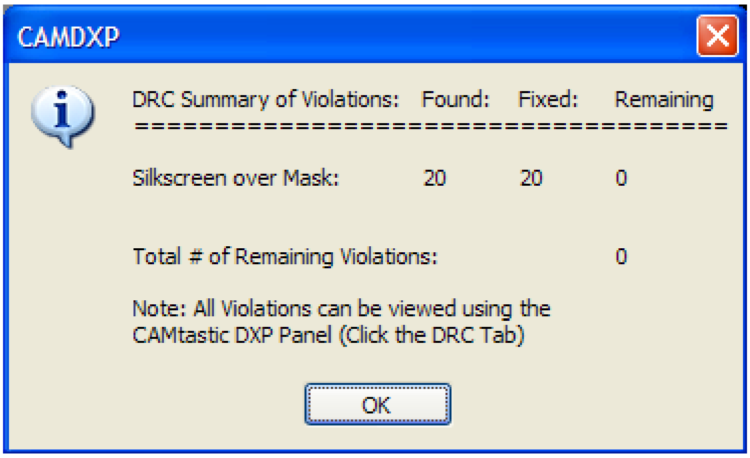
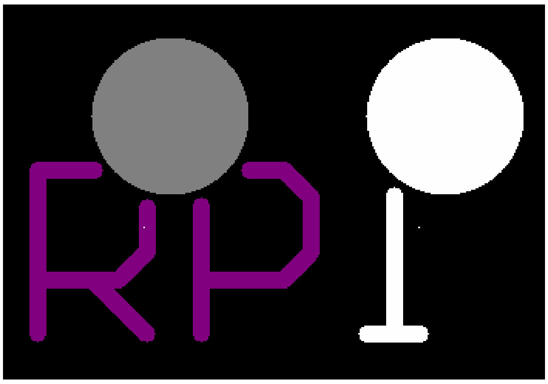



*Figure 36. Summary of the fixes made in CAMtastic.*

3. Check what has been achieved by using the Auto Fix option. Double-click on the Silkscreen over Solder mask violation error folder in the **Drc** tab of the **CAMtastic** panel, then click on the Ref(13) subfolder to display and highlight the auto fixed objects in the design window.
4. Remember you can always use **Edit » Undo** (shortcut Ctrl+Z) to reverse any auto fixes.



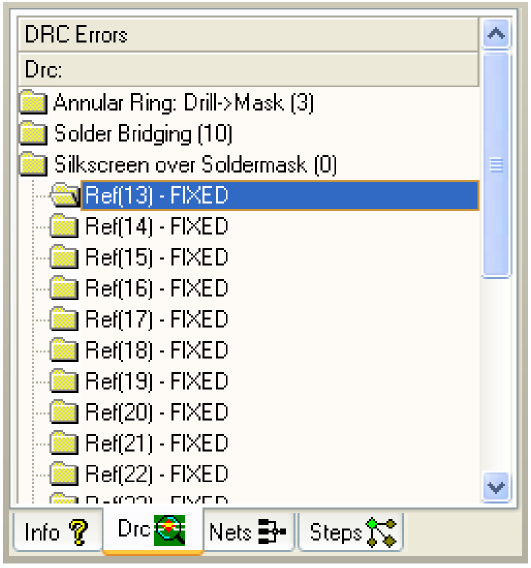



*Figure 37. Fixed problems on the result of the fix on the left.*

### 19.3.7.1 Using Auto Fix in the CAMtastic Panel

You can also use the Auto Fix option (where applicable) from the right-click menu when in the DRC tab of the CAMtastic panel. This allows you to fix individual errors as well as entire DRC types.

To auto fix all the Silkscreen over Solder Mask errors, for example, from the DRC tab, right-click on the Silkscreen over Solder Mask violations folder in DRC tab of the CAMtastic panel and select **Fix All - Silkscreen over Solder Mask errors**. All violations in that folder are fixed.

To auto fix individual errors, right-click on an individual error Ref folder and select **Fix DRC Error**. The error is fixed.

## 19.3.8 Checking Remaining Violations

Check the remaining violations and resolve any other errors as necessary. The table below explains the reasons for the remaining violations.

| DRC Violations | Notes |
| --- | --- |
| Annular Ring (Drill>Mask) | The three violations are for annular rings (drill to mask) that have a negative value, which is not interpreted by the DRC. Investigation using the Analysis » Query » Object command (shortcut Q) will show that the drill holes are larger than the pads and have a diameter of 140 mil and the masks are 150 mil in diameter. The drill to mask expansion of 5 mil (half of the difference between the drill hole and mask diameters) is still better than the fab house minimum of 4 mil (pad to mask), so these errors can be ignored. |
| Solder Bridging | Solder bridging checks the Top and Bottom layers against the Mask Top and Mask Bottom layers for the distance from the edge of the mask opening to any objects under the mask but on a different net from the pad exposed by the mask opening. DRC flags a violation if the nets are different and the objects are not correctly covered by the mask.<br><br>In this example, the solder bridging occurs where the routing to the end connector comes in contact with the mask. Since there are many different nets under the mask, errors are generated. Visual inspection shows that this is OK, so the violations could be ignored. |
| Net Antennas | A rogue flash has been detected. This DRC looks for track ends not terminating on a track end, pad or via. This trace can be deleted by selecting Edit » Clear and then selecting the flash, right-click to Clear and press ESC to end the command. Notify the PCB designer of this alteration. |

*Table 1. Summary of DRC violations and what they mean.*

The data in the design has now been verified and the files are ready for the next stage of fabrication.

TOP SECRET

Altium Designer

# Module 20: Interfacing to 3D Mechanical CAD

# Module 20: Interfacing to 3D Mechanical CAD





Module Seq = 20

# 20.1 Interfacing to 3D Mechanical CAD

Altium Designer has strong support for interfacing to mechanical CAD design tools, including 3D visualization within the PCB editor, and 2D and 3D export. The completed PCB can be exported as a 3D STEP format file, or a 2D description can be exported as IDF.

You can also import 3D STEP objects, such as components and housings, into the PCB library and PCB editors, and also import 2D IDF and DXF data.

## 20.1.1 Viewing a Board in 3D

- To view the board in 3D, press the **3** shortcut key.

> **Note:** Viewing the board in 3D requires a graphics card that supports DirectX 9 or better, and Shader Model 3 or better. If you are unsure if your graphics card meets these specifications, click the [Test DirectX...] button in the **PCB Editor – Display** page of the *Preferences* dialog.

- By default the board will display as a blank, or unloaded board. Press the **L** shortcut to display the *View Configurations* dialog, where you can configure the color scheme, and also enable the display of 3D bodies. Note that components can only be displayed if the footprint includes 3D body objects, or an imported STEP format component description.
- Press the **2** shortcut to change back to the traditional 2D display mode.

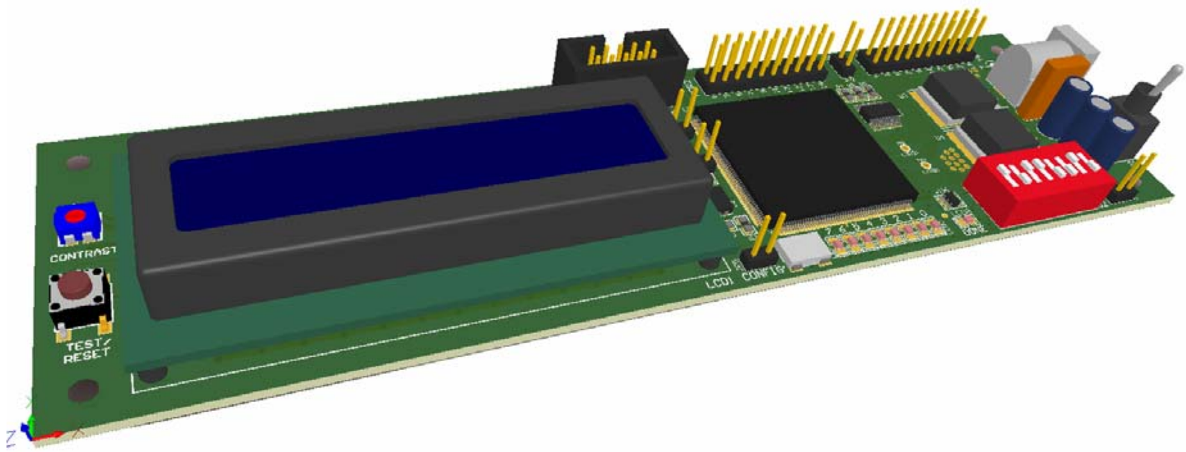



*Figure 1. 3D view of Altium's Spiritlevel design.*

- To rotate the 3D view of the board, hold the Shift key to display the spherical rotation cursor, then click and hold the right mouse button and drag the board around to a new position.
- Altium Designer also supports the Space Navigator, a 3 dimensional navigation device developed specifically for 3D CAD work. Visit [www.3dconnexion.com](http://www.3dconnexion.com) for more information.

## 20.1.2 Exporting a 3D description of the board to an MCAD package

The best way to pass a 3 dimensional description of your board to your MCAD software is to export it from Altium Designer as a STEP file.

- Although STEP (**St**andard for the **E**xchange of **P**roduct model data – ISO10303) is a standard data interchange format, not all CAD packages produce the same STEP format files, so you may need to experiment with the export options to see what gives the best result.
- To export the board to a STEP format file, select **File » Save As** from the menus, and select **Export STEP** from the **Save as Type** list. The *STEP Export Options* dialog will appear, where you can configure what is exported, and how components are exported. Exporting holes will increase the file size, exclude these if necessary.

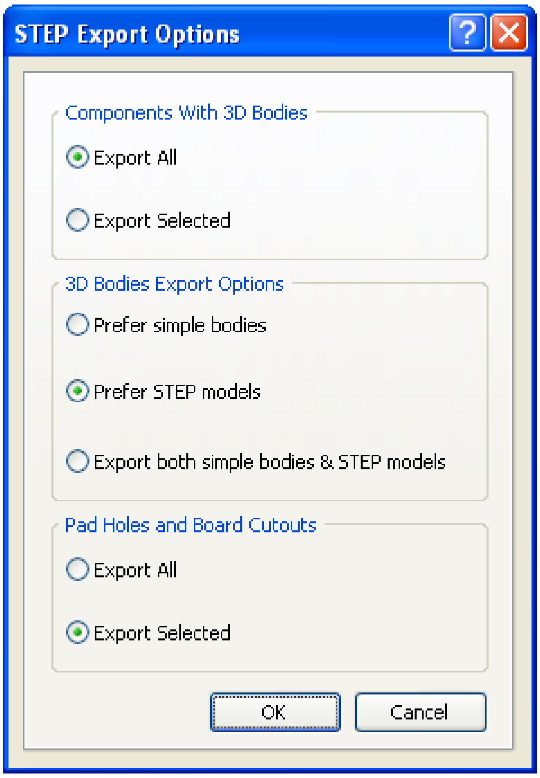



*Figure 2. The STEP Export options dialog*

> **Note:** For information on creating 3D component definitions from imported STEP models or Altium Designer 3d bodies, ready for use in your board, refer to the training module *Linking Models, Parameters, Library Package and Updates*.

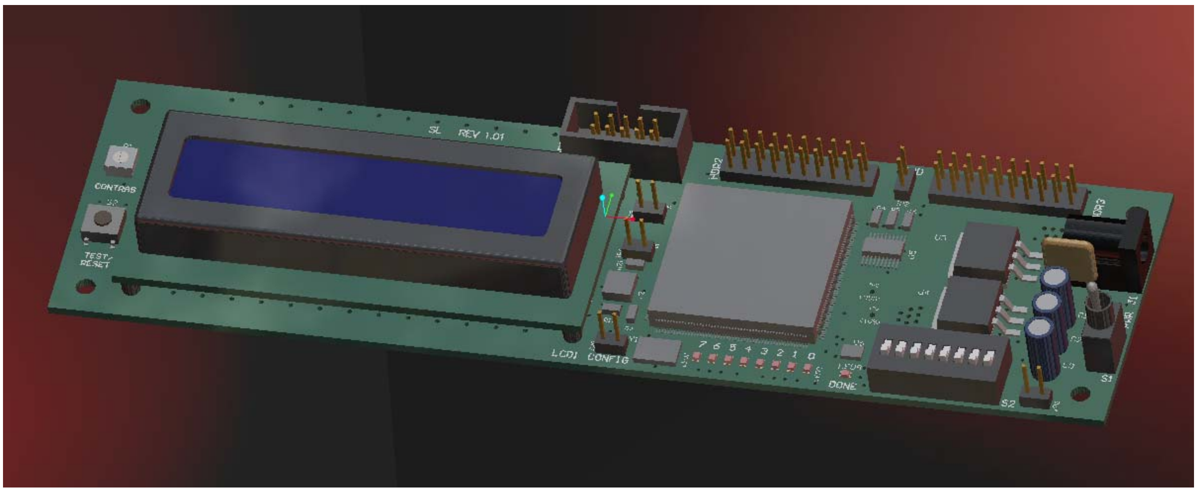


*Figure 3. The STEP fie imported into Pro Engineer Wildfire*

## 20.1.3 Checking external mechanical objects against the board

A common development requirement for a compact, complex electronic product is to be able to test the fit of the board inside its housing, before committing to manufacturing the board or the tooling for the housing.

- Altium Designer supports importing of any 3D model, such as a case or a bracket, directly into the PCB editor. To do this, switch to 3D view mode, then select **Place » 3D Model** from the menus.
- Orient the model using the Spacebar and arrow keys on the numeric keypad, or use the **PCB Inspector** panel to test different rotation settings.
- When you click and hold to move the model into position, you will be holding it by the nearest vertex in the model. You can also add your own snap points, either via the *3D body* dialog (double click on the imported model to open it), or via the **Tools » 3D Body Placement » Add Snap Points From Vertices** command.
- Set the **PCB** Panel to **3D Models** mode, select the model, and use the **Highlighted Models** drop down to change the transparency level

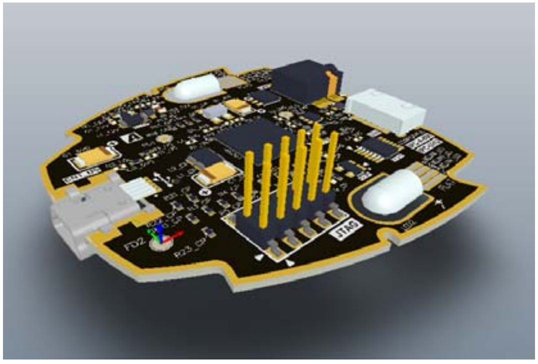
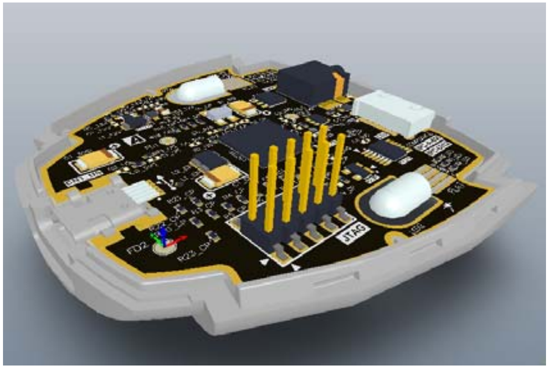



*Figure 4. Board only in 3D view mode, then shown inside bottom half of enclosure.*

*Figure 5. Note how connector is currently interfering with enclosure (DRC violations disabled).*

## 20.1.4 IDF import/export

Altium Designer supports both import and export of IDF v2.0 and v3.0.

- Two files are generated when exporting to IDF, *.BRD and *.PRO. The BRD file is the board file and the PRO file is the library information.
- The exported BRD file contains the board outline and component position information.
- The PRO file details the names of the components and some basic component shape information. If component bodies are used this information is a lot more usable. The ideal solution is to have a 3D representation of the component in your MCAD package, which is placed from the library information.
- To export the current board, select **File » Save As** and set the **Save As Type** option to **Export SDRC-IDF Brd Files**.

**Note:** Different CAD packages use different file extensions to Altium Designer, for example Pro Engineer uses:
*.emp - IDF library information (also called profiles files)
*.emn - IDF board files (also called neutral files)

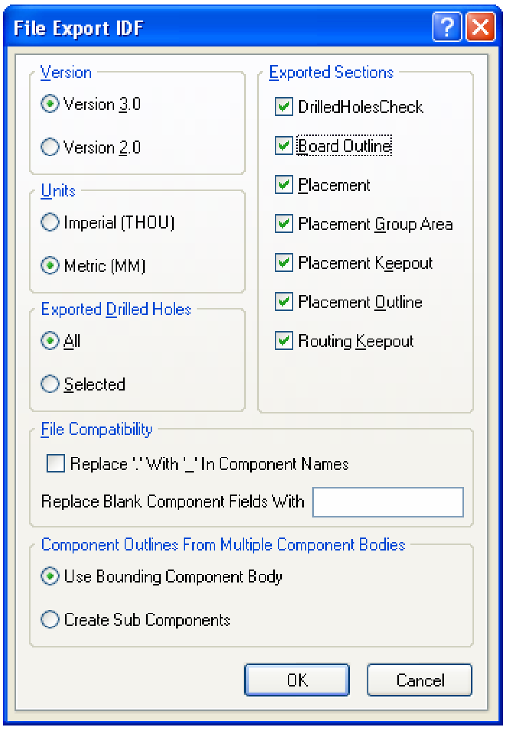



*Figure 6. Export to IDF dialog*

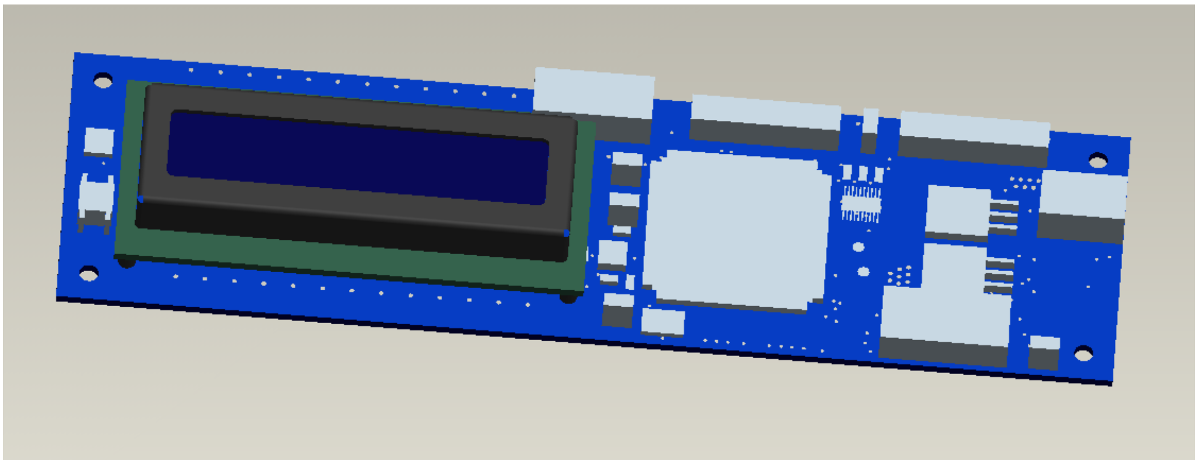


*Figure 7. Imported IDF with a 3D component for the LCD screen drawn from a Pro Engineer library. The rest are built from Altium Designer component bodies.*

TOP SECRET

Altium Designer

# Module 21: Circuit Simulation

# Module 21: Circuit Simulation





Module Seq = 21

# 21.1 Circuit Simulation

The Altium Designer-based Circuit Simulator is a true mixed-signal simulator, meaning that it can analyze circuits that include both analog and digital devices.

The Simulator uses an enhanced version of the event-driven XSpice, developed by the Georgia Tech Research Institute (GTRI), which itself is based on Berkeley's SPICE3 code. It is fully SPICE3f5 compatible, as well as providing support for a range of PSpice® device models.

## 21.1.1 Model Types

The models supported by the Simulator can be effectively grouped into the following categories.

### 21.1.1.1 SPICE3f5 analog models

These are predefined analog device models that are built-in to SPICE. They cover the various common analog component types, such as resistors, capacitors and inductors, as well as voltage and current sources, transmission lines and switches. The five most common semiconductor devices are also modeled - diodes, BJTs, JFETs, MESFETs and MOSFETs.

A large number of model files (*.mdl) are also included, that define the behavior of specific instances of these devices.

### 21.1.1.2 PSPICE analog models

These are predefined analog device models that are built-in to PSpice. To support these models, changes have been made to the general form for the corresponding SPICE3f5 device and/or additional parameter support has been added for use in a linked model file.

**Note:** These models are not listed separately in this reference. PSpice support information is included as part of the information for the relevant SPICE3f5 device model.

### 21.1.1.3 XSPICE analog models

These are predefined analog device code models that are built-in to XSpice. Code models allow the specification of complex, non-ideal device characteristics, without the need to develop long-winded sub-circuit definitions that can adversely affect Simulator speed performance. The supplied models cover special functions such as gain, hysteresis, voltage and current limiting and definitions of sdomain transfer functions.

The SPICE prefix for these models is A.

### 21.1.1.4 Sub-Circuit models

These are models for more complex devices, such as operational amplifiers, timers, crystals, etc, that have been described using the hierarchical sub-circuit syntax.

A sub-circuit consists of SPICE elements that are defined and referenced in a fashion similar to device models. There is no limit on the size or complexity of sub-circuits and sub-circuits can call other subcircuits. Each sub-circuit is defined in a sub-circuit file (*.ckt).

The SPICE prefix for theses models is X.

#### 21.1.1.5 Digital models

These are digital device models that have been created using the Digital SimCode™ language. This is a special descriptive language that allows digital devices to be simulated using an extended version of the event-driven XSpice. It is a form of the standard XSpice code model.

Source SimCode model definitions are stored in an ASCII text file (*.txt). Compiled SimCode models are stored in a compiled model file (*.scb). Multiple device models can be placed in the same file, with each reference by means of a special "func=" parameter.

The SPICE prefix for theses models is A.

Digital SimCode is a proprietary language - devices created with it are not compatible with other simulators, nor are digital components created for other simulators compatible with the Altium Designer-based mixed-signal Simulator.

### 21.1.2 Exercise/Demo

1. Select **File** » **New** » **PCB Project** from the menus.
2. Rename the new project file (with a .PrjPCB extension) by selecting **File** » **Save Project As**. Navigate to a location where you would like to store the project on your hard disk, type the name **Filter.PrjPCB** in the File Name field and click on Save.
3. Select **File** » **New** » **Schematic**. A blank schematic sheet named sheet1.SchDoc displays in the design window of the Schematic Editor and the schematic sheet is now listed under Source Documents beneath the project name in the Projects panel.
4. Rename the new schematic file by selecting **File** » **Save As**.
5. Now we can create the Filter circuit shown below in figure 86. Before we can run a simulation, the schematic must contain components with SIM models attached, voltage sources to power the filter, an excitation source, a ground reference for the simulations and some net labels on the points of the circuit where we wish to view waveforms.

**Note:** This project is available in the Altium Designer circuit simulation examples.

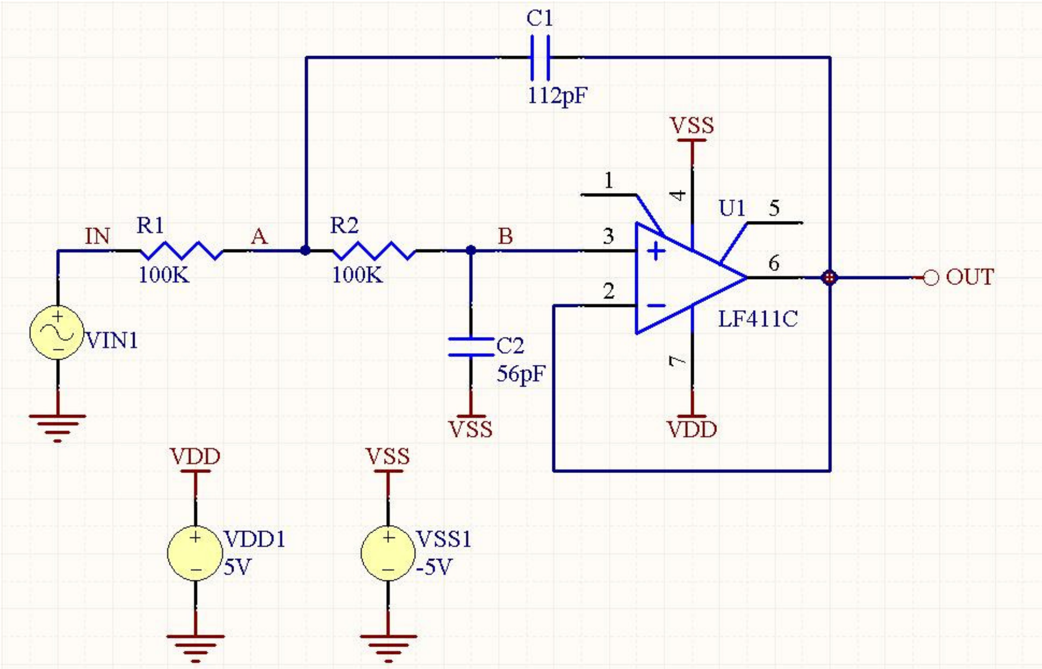



*Figure 1. Circuit to simulate*

**Note:** The Altium simulator supports **Spice 3F5 models and PSPICE models**. SPICE models used for circuit simulation (.ckt and .mdl files) are located within the integrated libraries in the `C:\Program Files\Altium Designer Summer 08\Library` folder. You must use the correct file extension for each model type.

**Tip:** A circuit for simulation must always have a path to ground. If it doesn't you'll receive iteration limit reached error. To fix this provide a path to ground or modified the advanced options value for RSHUNT to a figure of around 2-4Meg. Remember that the simulator assumes Capacitors are perfect, i.e. no leakage.

6. Click on the **Search** button in the **Libraries** panel.

7. To search for all LF411 type components, in all supplied libraries, enter LF411 in the upper part of the *Libraries Search* dialog. Set the **Scope** to **Libraries on Path**, and set the **Path** to `C:\Program Files\Altium Designer Summer 08\Library`, with the **Include Subdirectories** option enabled. Note that this will find all LF411 type components, you can check if the chosen component includes a Spice model by selecting it in the **Libraries** panel, and noting if there is a simulation model detailed in the **Model Name** region of the panel.

**Note:** to search for only components that include a simulation model, you would have used the search string `(Name like '*LF411*')` **`and`** `HasModel('SIM','*',true)`

8. You may want to use a simulation model in your design other than the one already supplied with the component in its integrated library. If you download a model file from a manufacturer's website, it is recommended that you copy it into your design's project folder.

For the sake of this example, we will remove and re-add the SPICE model, LF411C.ckt.

9. Copy LF411C.ckt and paste this file into the folder where your project files reside using windows Explorer. If you have created the circuit from scratch, there is a copy of the model file in `C:\Program Files\Altium Designer Summer 08\Examples\Circuit Simulation\Filter`.

10. Add the model file to the project by selecting the project name (**Filter.PrjPCB**) in the Projects panel, right-click and select **Add Existing to Project**. Choose the model file and click Open.

11. Double-click on the op amp to open its *Component Properties* dialog. Delete the existing SIM model in the Models section by selecting it and clicking on **Remove** and confirming the deletion.
12. Click on **Add** in the Models section to display the *Add New Model* dialog.
13. Select **Simulation** from the Model Type drop-down list and click **OK**. The *SIM Model-General/Generic Editor* dialog displays.
14. Select **Spice Subcircuit** from the **Model Sub-Kind** list, the Spice Prefix will automatically set to X. Note that the dialog name has changed to reflect the Model Sub-Kind.
15. Click the **Browse** button to open the *Browse Libraries* dialog. Because you have added the model file to the project it will automatically be available, select the LF411C.ckt model from the list and click **OK** to close the dialog.

**Note:** The Model Name detailed in the Model Kind tab is not the name of the file, it is actually matched against the name found in the .SUBCKT statement in the model file. CKT files can contain multiple .SUBCKT sections so be careful the correct name is specified.

16. The final step is to configure the pin mapping from the model to the schematic symbol (generic numbering is used in Spice model files, not physical pin numbers). Click on the **Port Map** tab of the *SIM Model-General / Spice Subcircuit* dialog to show the current mapping, and the **Model File** tab to show the generic numbers used in the .SUBCKT statement in the model file.

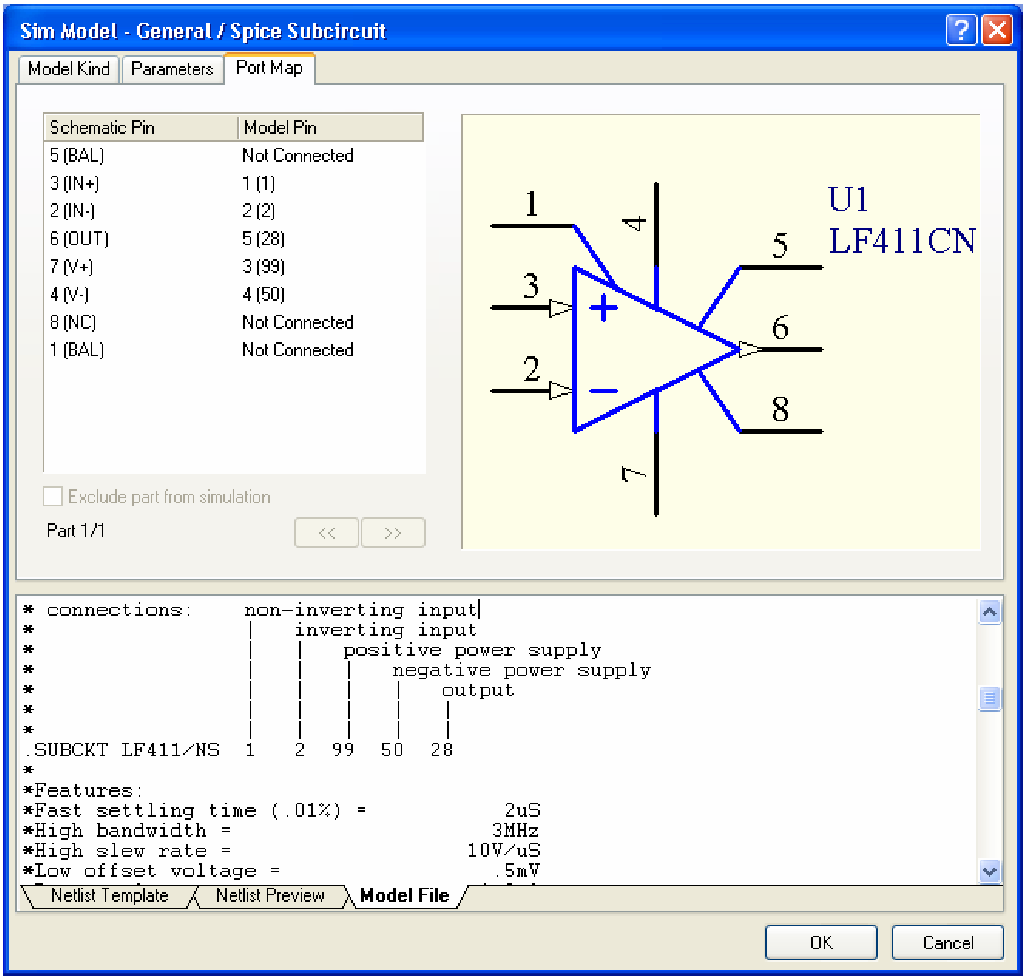



*Figure 2. Port Map information for sub-circuit simulation model*

17. Modify the Model Pin mapping by selecting the matching pins from the Model Pin drop-down lists, as shown in Figure 2. Close the dialogs when finished.

## 21.1.3 Adding the voltage sources

We can add the voltage sources needed to power the design when simulating.

1. We will place the VDD power source first. Search for the component VSRC in the **Libraries** panel and then add the simulation sources.IntLib library to the Available Libraries list. Note that there are other simulation libraries available in the \Project Files\Altium2004\Library\Simulation folder.
2. As you place the power source, press **TAB** to edit its properties. Click on the **SIM model VSRC** in the Models list in the Component Properties dialog and click on **Edit**. In the Sim Model-Voltage Source/DC Source dialog, check the Model Kind is set to **Voltage Source** and the Model Sub-Kind to **DC Source**.
3. Click on the **Parameters tab** to set up the voltage value required. Type **5V** in the Value field and tick its Component Parameter box. This will automatically create the parameter 'Value' for you in the Component Properties dialog. Leave the other fields set to **zero**.

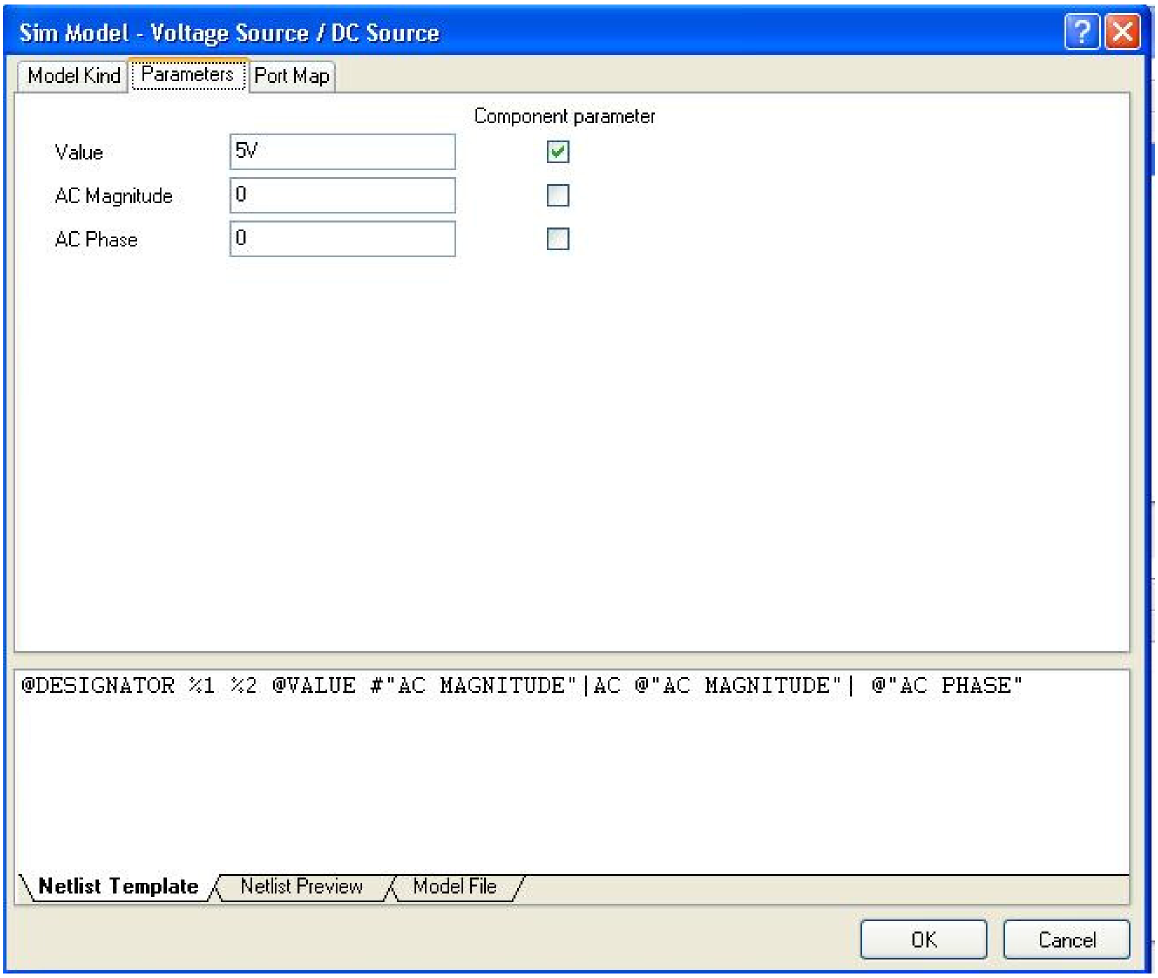
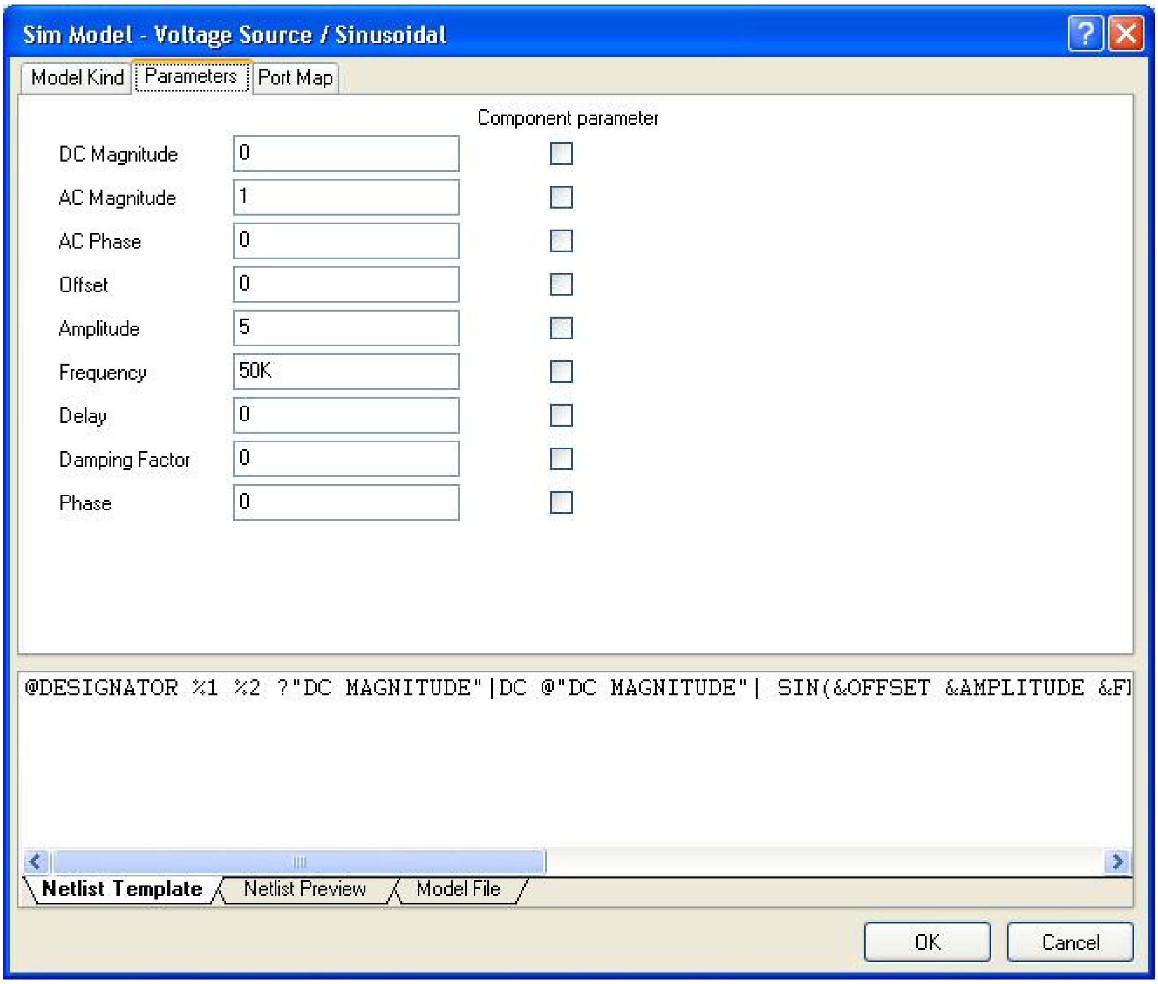



*Figure 3. Parameter information for the simulation source model*

4. Now place the VSS power source, remembering to set the model file parameter Value to –5V.
5. Finally add the Sinusoidal (excitation) Voltage Source, VSIN, also available from the Simulation Sources. IntLib. Press **TAB** to edit its properties before placing and change the

frequency from 1 KHz to 50 KHz for this example. In the Component Properties dialog, clock on the SIM model VSIN in the models list and click on **Edit**.

6. Click on the **Parameters tab** to set up the voltage value required. Type in the parameter values as shown in the Sim Model-Voltage Source / Sinusoidal dialog.




7. **Save** your schematic.

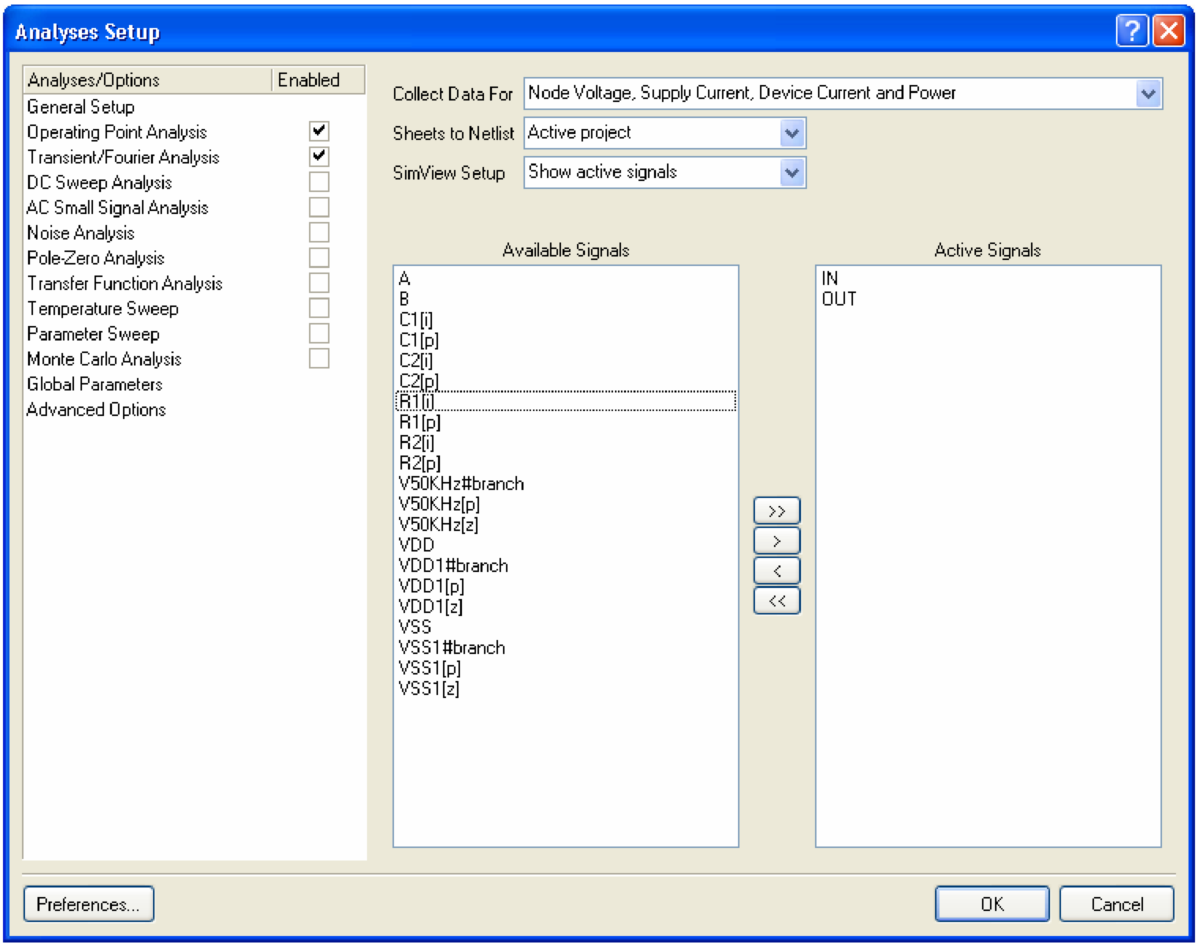
**Note:** Using values like 4k7 for a resistor or 5v5 does not work in Altium Designer. The values must be 4.7k and 5.5v respectively to function correctly.

## 21.1.4 Setting up the analyses

Altium Designer allows you to run an array of circuit simulations directly from a schematic. In this example, we will simulate the output waveforms produced by our Filter circuit.

The simulation can be set up and run using the Simulate menu command or by clicking on the appropriate button on the Mixed Sim toolbar.

1. With ***Filter.SchDoc*** open in the Schematic Editor, select **Design » Simulate » Mix Sim** to display the General Setup page of the Analyses Setup dialog.
2. First, we will set up the nodes in the circuit that we want to observe. In the **Collect Data For field**, select **Node Voltage, Supply Current, Device Current and Power** from the list. This option defines what type of data you want calculated during the simulation run, i.e. it saves data for the voltage at each node, the current in each supply and the current and power in each device, Set the **SimView Setup** to **Show Active Signals**.
3. In the **Available Signals** field, double-click on the **IN and OUT** signal names. As you double-click on each one, it will move to the **Active Signals** field.




4. Each individual analysis type is configured on a separate page of the Analyses Setup dialog. Click on the analysis name to activate the corresponding setup page.

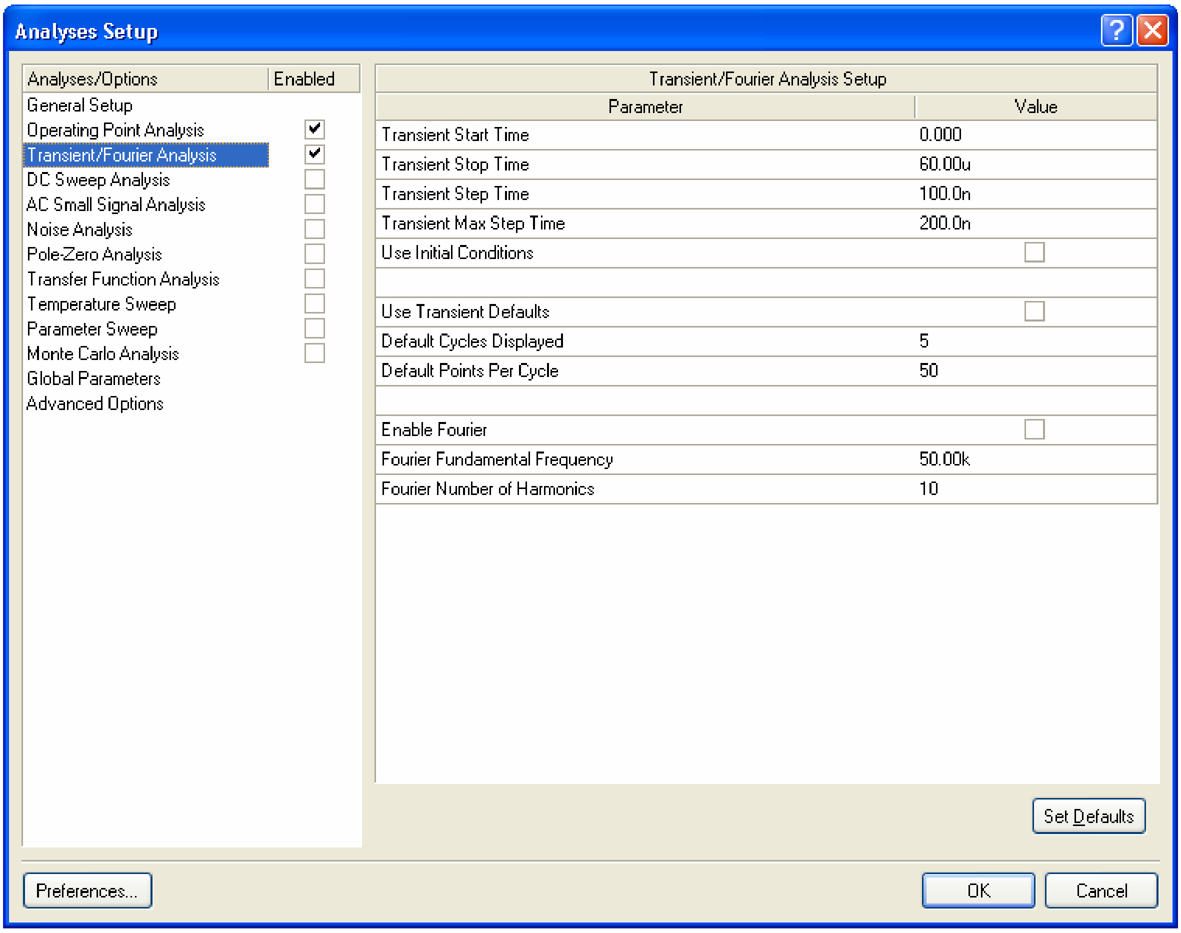
## 21.1.5 Setting up a Transient/Fourier analysis

A Transient/Fourier analysis generates output like that normally shown on an oscilloscope, computing the transient output variables (voltage, current or power) as a function of time over the user-specified time interval.

An Operating Point analysis is automatically performed prior to a Transient analysis to determine the DC bias of the circuit.

To view six cycles of the 50 KHz excitation voltage, we'll need to set up to view a 60u portion of the waveform.

1. Make sure the **Transient/Fourier** checkbox is enabled (ticked) in the Analyses Setup dialog for this analysis.




2. Check that the **Use Transient Defaults** option is disabled, so that the Transient Analysis parameters can be modified.

3. To specify a 60u simulation window, set the **Transient Stop Time** field to 60u.

4. Now set the **Transient Step Time** field to 100n, indicating that the simulation should calculate a point every 100ns.

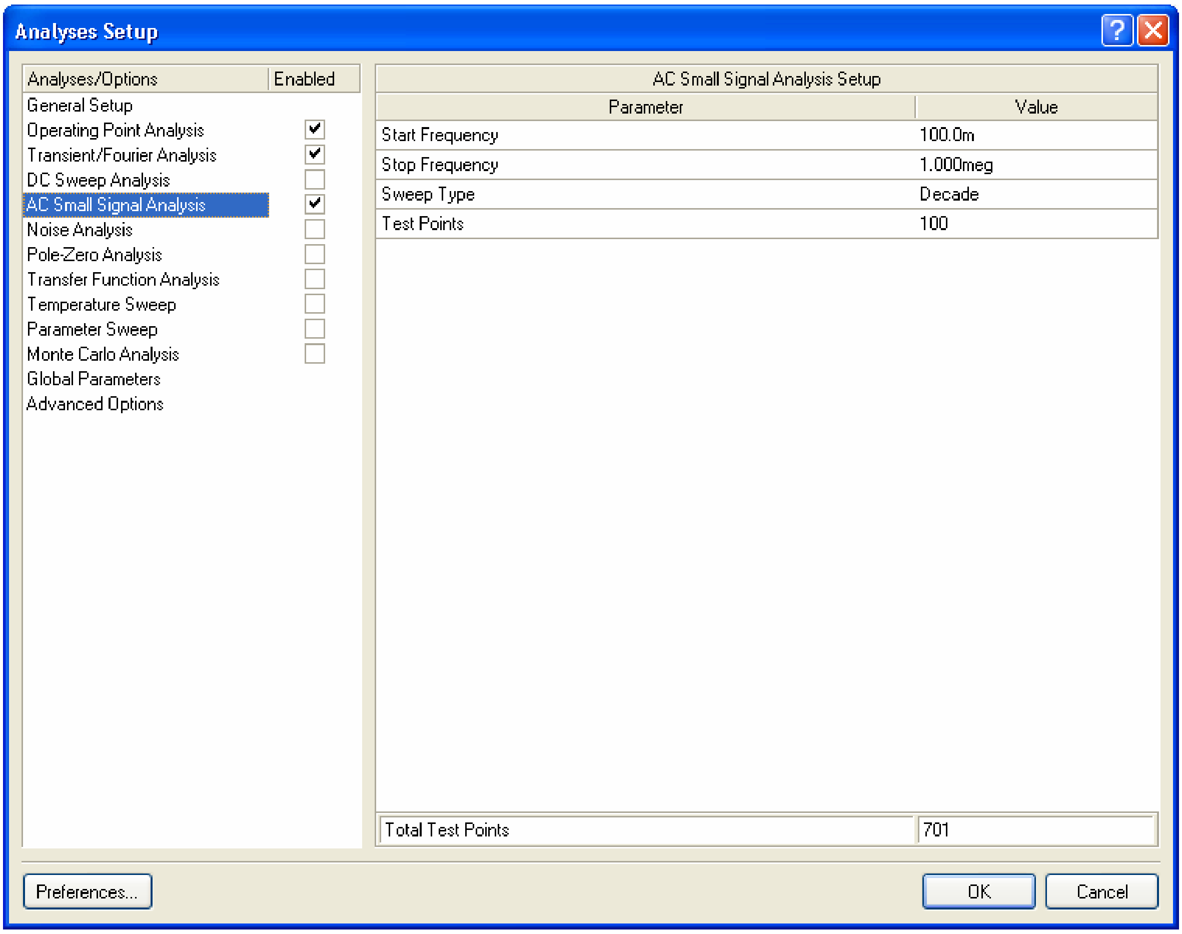
5. During simulation, the actual time step is varied automatically to achieve convergence and the required accuracy. The **Maximum Step** field limits the variation of the timestep size, so set the **Transient Max Step Time** to 200n.

## 21.1.6 Setting up an AC Small Signal analysis

An AC analysis generates output that shows the frequency response of the circuit, calculating the small-signal AC output variables as a function of frequency. The desired output of an AC small-signal analysis is usually a transfer function, e.g. voltage gain.

The schematic must contain at least one AC source with a value entered for the AC Magnitude parameter in its SIM model. We have already set up the parameters of our sinusoidal excitation source (VSIN) to contain the AC Magnitude value, frequency and amplitude.

1. Make sure the **AC Small Signal Analysis** checkbox is enabled in the Analyses Setup dialog.




2. Enter the parameter values as shown above.
3. When this analysis is run, an Operating Point analysis is run first to determine the DC bias of the circuit. The signal source is then replaced with a fixed amplitude sine wave generator and the circuit is analyzed over the specified frequency range, stepping in increments defined by the values in the **Test Points** and **Sweep Type** fields.

## 21.1.7 Running the simulation

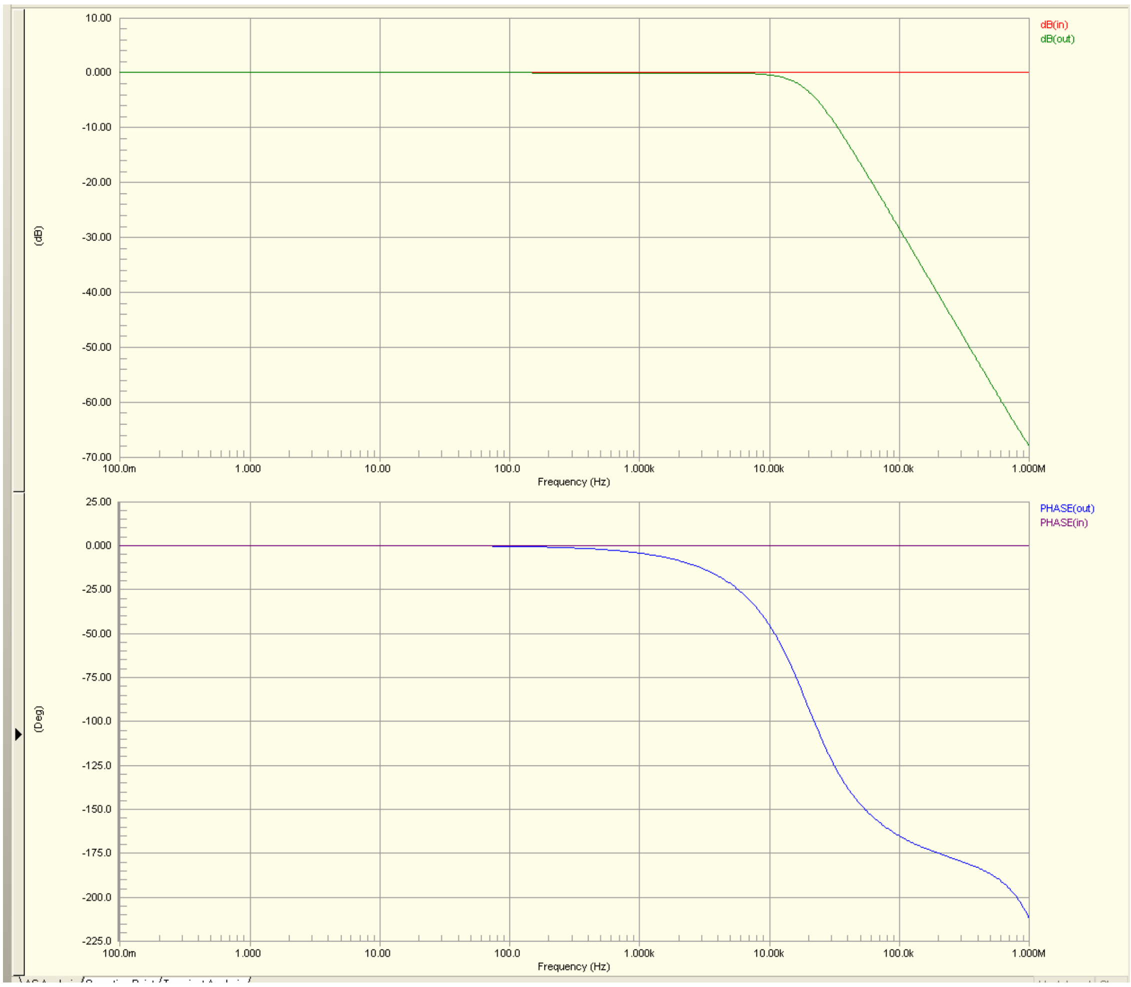
You are now ready to run the enabled analyses. Note that you can also run a simulation without entering the Analyses Setup dialog every time by clicking on the **Run Simulation** button on the Mixed Sim toolbar.

1. Click the OK button at the bottom of the Analyses Setup dialog to run the simulation.
2. If there are no errors in the circuit, a SPICE netlist (*.nsx) is created and passed to the simulator. Note that the netlist will be regenerated each time a simulation is run.
3. The simulation begins and a simulation data file (*.sdf) will open. The results of each analysis are shown as a separate chart in the SimData Editor's Waveform Analysis window. The Operating Point analysis is performed first to determine the DC bias of the circuit.
4. When the simulation is finished, you should see outputs waveform.

### 21.1.7.1 Creating a Bode plot

A Bode plot consists of two curves- the log of gain and phase-as functions of the log of frequency. The gain is decibels (dB) and the phase are plotted linearly along the y axis on a chart that has several cycles of a log scale on the x axis. Each cycle represents a factor of ten in frequency. We can create a Bode plot using the waveform functions available in the Edit Waveform dialog.

1. In the AC Analysis tab of the Waveform Analyzer window, we will display dB(in), dB(out), PHASE(in) and PHASE(out).
2. In the Waveform Analyzer window, click on net in. Right-click on the selected net and select Edit Wave (or select **Wave » Edit Wave**). The Edit Waveform dialog displays. The selected wave appears in the Expression field.
3. Select the function from the Complex Functions list, i.e. Magnitude (dB). Click on Create to see the waveform dB(net_name) on the plot.
4. To create dB(out), right-click on the plot and select Add Wave to Plot. The Add Wave to Plot dialog displays which works in the same way as the Edit Waveform dialog.
5. Repeat step 4 to create PHASE(in) and PHASE(out) on the second plot by selecting the Phase(Deg) complex function in the Add Wave to Plot dialog.




6. These waveforms could be displayed on different Y axes, if required, by selecting **Add to new Y** axis in the Edit Waveform dialog. Note that if you remove the new Y axis, all waves that are plotted to this axis are removed as well as any measurement cursors attached to these waves. There is no Undo functionality. If you added a new Y axis, it would appear as shown below.

### 21.1.7.2 Using the measurement cursors

Now we can determine the 3dB point using the measurement cursors.

1. Click on the net name **DB(out)** to select the wave in the Waveform Analyzer window.
2. Right-click and select **Cursor A**. Position cursor A in the low pass section by dragging the marker.
3. Right-click and select **Cursor B**. Position cursor B at a point such that **B-A=-3** in the Measurement section of the Sim Data panel.
4. Read off the X value of cursor B in the Measurement Cursors section of the **Sim Data** panel and you will find the 3dB point = 20kHz.

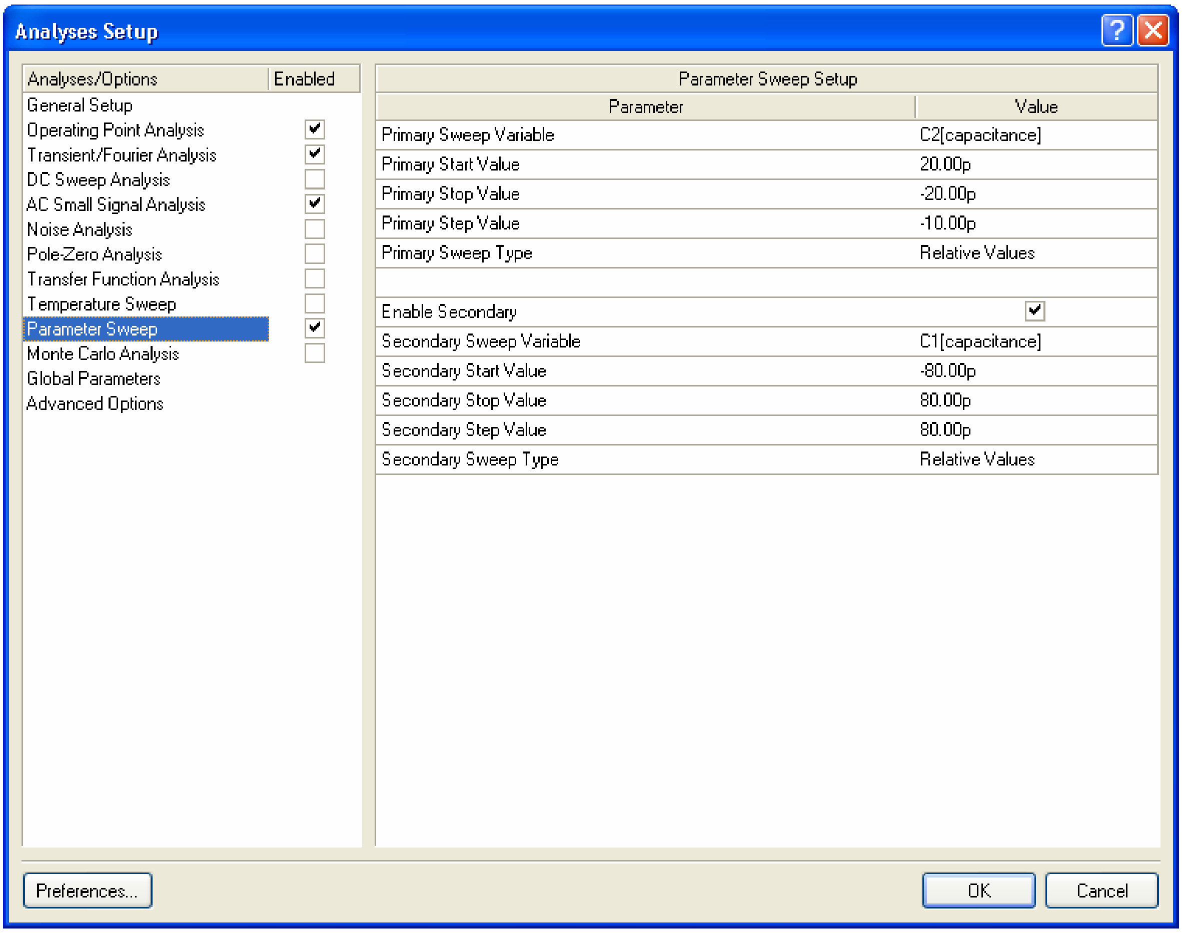
## 21.1.8 Running a Parameter Sweep Analysis

Now that we have set up and run some analyses, let's setup a parameter sweep to see the effect of varying some of the capacitor/resistor values on the frequency response.

A Parameter Sweep analysis allows you to sweep the value of a device in defined increments, over a specified range. AC, DC or Transient analyses must also be enabled in order to perform a Parameter Sweep analysis as the simulator performs multiple passes of each enabled analyses.

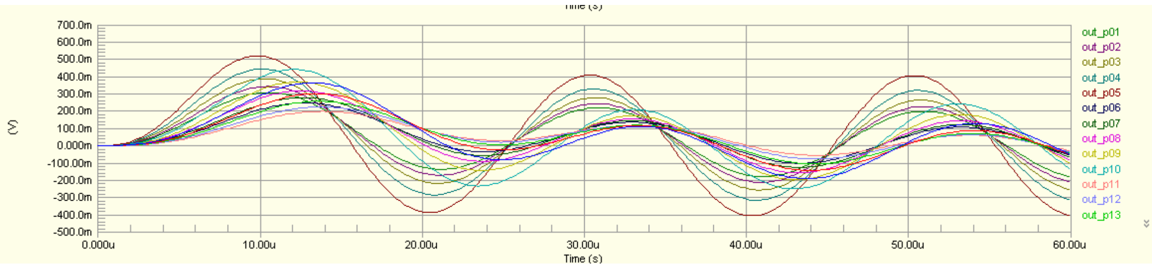
This analysis can vary basic component values and model parameters; however subcircuit data is not varied during the analysis.

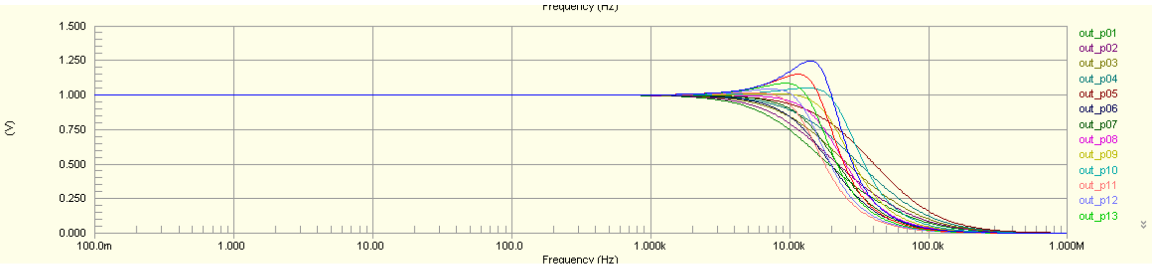
1. Click on the Filter.SchDoc tab to make the schematic available in the design window. Select **Design » Simulate » Mixed Sim**. Make sure **Show Active Signals** is selected as the SIMView Setup in the General Setup page of the Analyses Setup dialog.
2. Click on **Parameter Sweep** in the Analyses/Options section of the Analyses Setup dialog to enable this analysis type.
3. Enter the parameter to be swept in the Primary Sweep Variable field. In this example, we will primary sweep parameter C2, so select it from the drop-down list.
4. To define the range of values for the sweep, set the Primary Start Value to 20p and Primary Stop Value to –20p. Set the Primary Step Value to define step increments to –10p.
5. Set the Primary Sweep Type option to **Relative Values**. The values entered in the Primary Start Value, Primary Stop Value and Primary Step Value fields will be added to the parameter's existing or default value.
6. Click on **Enable Secondary** to add secondary sweep values. When a Secondary parameter is defined, the Primary parameter is swept for each value of the Secondary parameter. Set up a secondary sweep for C1 with a Secondary Start Value of –80p and Secondary Stop Value to 80p. Set the Primary Step Value to define step increments to 80p. Set the Secondary Sweep Type to **Relatives Values**.




7. Click **OK** to run the simulation. Each primary sweep appears as a waveform with the notation<net_name_P<sweep_number>, e.g. out_p01, in a new plot in the AC Analysis and Transient Analysis tabs of the Waveform Analyzer window.

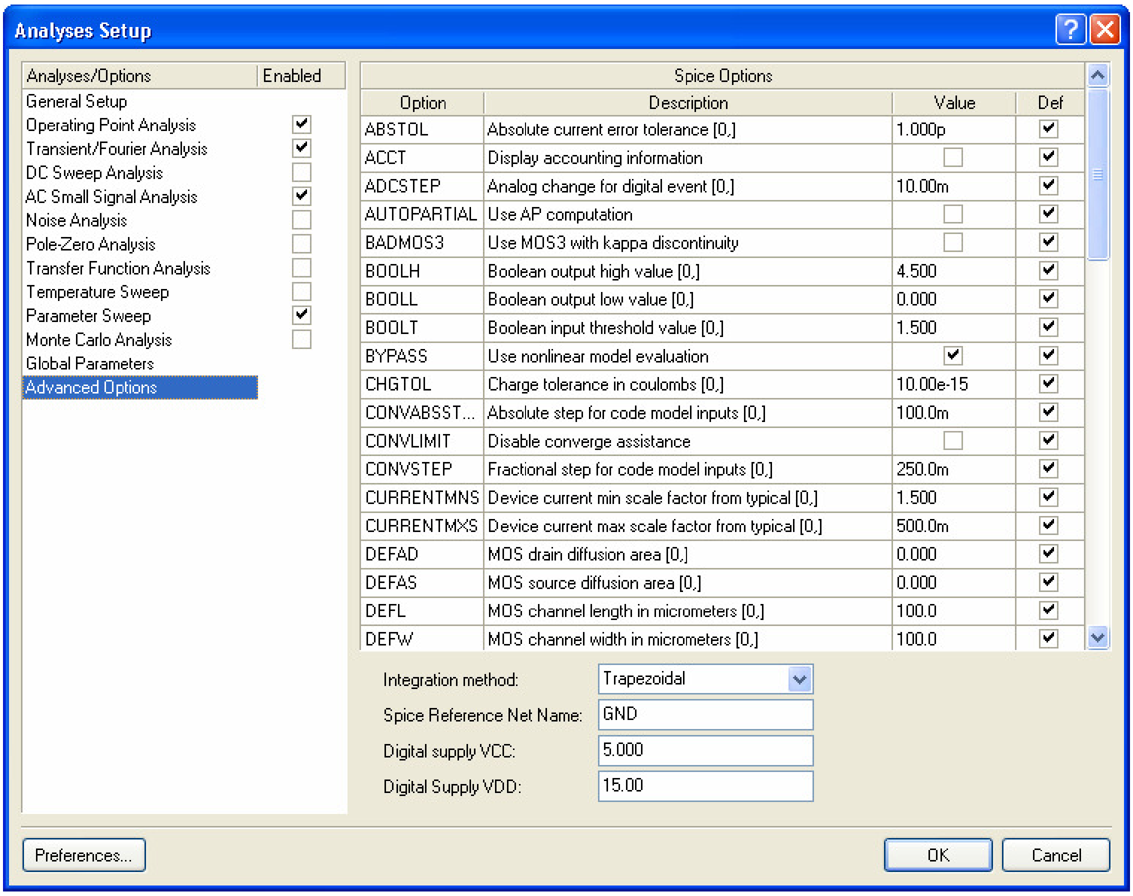
8. Click on a sweep parameter name to display more information, e.g. clicking out_po1 displays the sweep information underneath the plot.



## 21.1.9 Using Advanced Options

The Advanced Options page in the Analyses Setup dialog contains a list of internal SPICE options that can be set to effect the simulation calculations, such as error tolerances and iteration limits.




To change the value of a SPICE option, e.g. the iteration value for ITL1:

1. Select the variable, e.g. ITL1. Type in the new value in the Value field, or choose the value by clicking on the scroll arrows that appear.
2. Press Enter, or click in another field, and the Default (Def) option is then disabled.

**Note:** You could also choose a different integration method from this page, for example, if you have a circuit design with unexpected high frequency oscillations, you could change the standard integration method from Trapezoidal to Gear. The Trapezoidal method is relatively fast and accurate, but tends to oscillate under certain conditions. The Gear methods require longer simulation times but tend to be more stable. Theoretically, the higher the Gear order, the more accurate the results but the simulation time increases

## 21.1.10 Using a SPICE netlist for simulation

You can also perform a simulation directly from a SPICE netlist, allowing you to use the DXP simulator in conjunction with other schematic capture tools. To do this:

1. Include the netlist in a project file (*.PrjPcb). This allows you to save setup information between simulation sessions. Setup information is stored in the project file in the same manner as for a project containing schematics. Note that a project file is automatically added when you run simulation for the first time from P-CAD. If the netlist is opened as a free document, setup information will not be stored between sessions of DXP.
2. The DXP simulation engine requires a netlist that includes the component information, design connectivity, the model data, as well as the setup information. If there is no simulation setup information already contained in the netlist when you select **Simulate » Setup**, a new netlist is created named <original filename>_tmp.nsx. This file contains setup information from the project file plus the netlist information from the original .nsx file.

If there is simulation information present in the netlist, then the simulation is run directly with no modifications to the netlist. This is useful for advanced users who want to be able to modify settings directly in he netlist. Note that you cannot add or configure setup information using the Analyses Setup dialog if the netlist already includes any setup information.

3. Once you have added a netlist, which has no setup information to a project, you can configure the analyses by selecting **Simulate » Setup**.
4. To run a simulation, select **Simulate » Run** from the menus. The simulation waveforms will display in a .sdf document.
5. Rename the netlist file so it is not overwritten next time you run a simulation.

TOP SECRET

Altium Designer

# Module 22: Signal Integrity

# Module 22: Signal Integrity





Module Seq = 22

# 22.1 Signal Integrity

To help designers cope with the added complexity of transmission line analysis on a PCB, EDA vendors are starting to include signal integrity analysis software as part of the standard board design toolset.

- Unlike circuit simulation, signal integrity analysis does not concern itself with the functional operation of the circuit – components are only modeled in terms of the I/O characteristics of their pins, without regard to the component's function.
- Connections between component pins are modeled using transmission line techniques that factor in the length of the trace, the characteristic impedance of the trace at the stimulus frequency, and the termination characteristics at each end of the connection.
- Traditionally, signal integrity tools are designed to work with a fully routed board. While this gives accurate results because each individual trace length is known, it does mean that the analysis is performed quite late in the design cycle. One of the most common signal integrity issues that arises is signal degradation resulting from reflection caused by mismatched impedances of the pins at either end of a connection. The solution to this problem usually involves the addition of a termination resistor or R/C network to match the termination impedances and minimize reflections – a task which requires modifying the source schematic. Altium Designer supports schematic-level signal integrity analysis, to help preempt potential impedance miss-matches before the board layout stage.

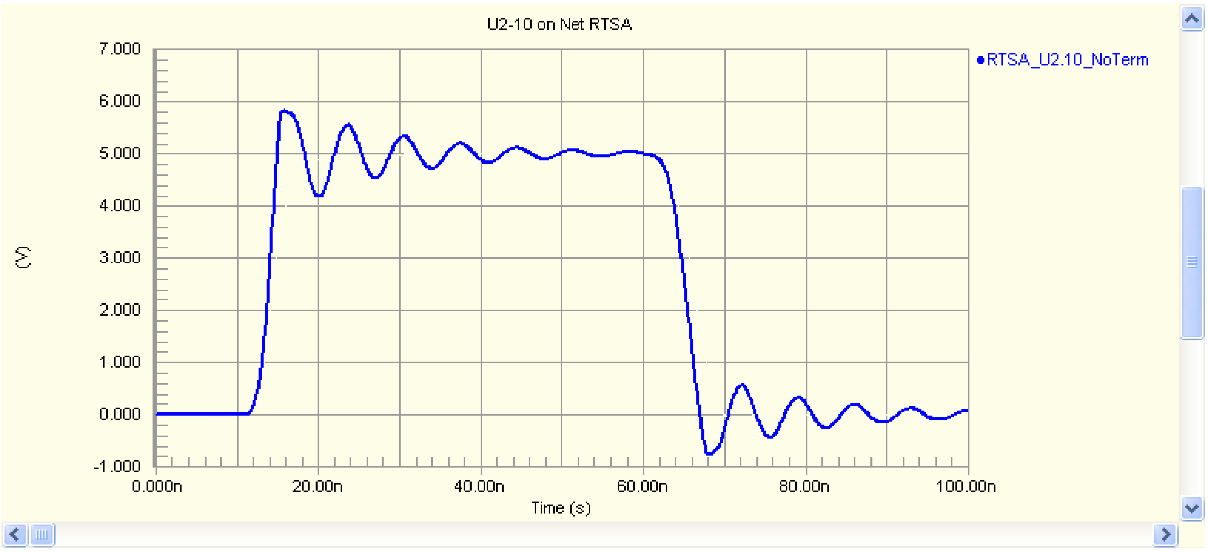



*Figure 1. Output from a reflection simulation for a problem net on a board.*

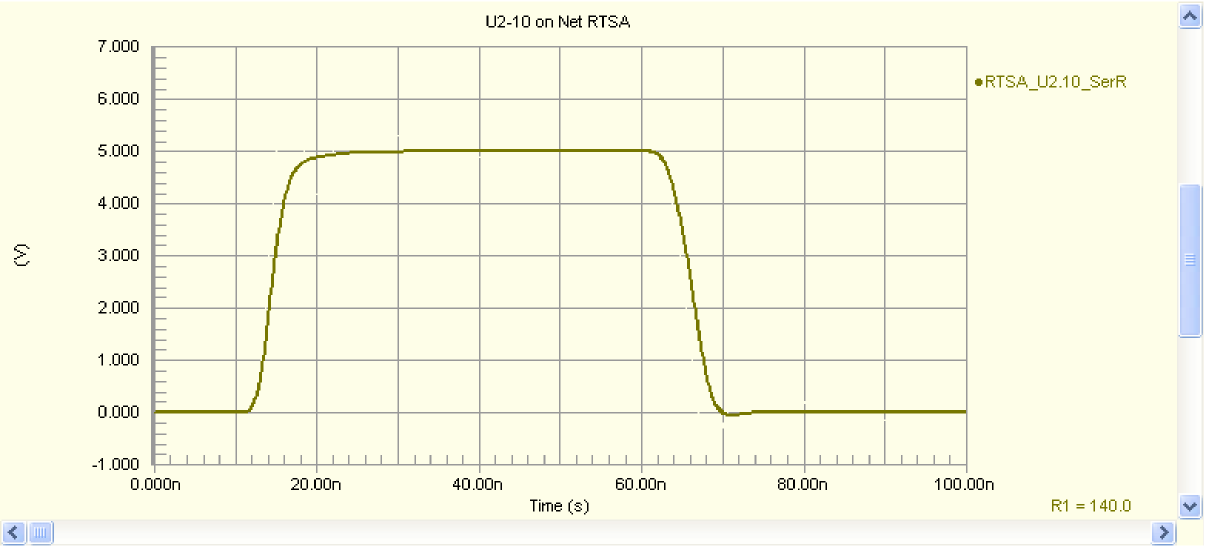



*Figure 2. Same net after the insertion of an appropriate series termination resistor.*

- Using signal integrity simulation to uncover problems in a completed board design before any prototyping is undertaken can reduce the number of prototype iterations needed to complete a project. However, the addition of new components late in the board design stage can be a major problem, particularly on dense boards. The rework involved can be time consuming, partially negating the time saved in the prototype/test phase.
- For this module we'll run through an exercise that takes you through the steps of setting up, finding and fixing signal integrity problems on a design using an FPGA, SDRAM and a discrete processor.

# 22.2 Checking Signal Integrity on an FPGA design

This tutorial relates to the following example project:

`\Program Files\Altium Designer Summer 08\Examples\Signal Integrity\NBP-28\NBP-28.PrjPcb`

This example is based on the daughterboard, NBP28. This daughterboard includes a Xilinx Spartan 3, a Sharp LH79520 incorporating an ARM 7 processor, SRAM and Flash RAM.

In this example we are going to answer the question *how hard can I drive the signals D[31..0] before ringing and crosstalk will prevent correct operation?* Or alternatively, *what is the optimum slew and drive settings to use for FPGA pins driving signals D[31..0]?*

## 22.2.1 Setting Up

Before we do any analysis we need to make sure that the components connected to the signals of interest all have the correct models set up.

These components are listed below.

- U1 FPGA - Xilinx XC3S1000-4FG456C
- U5 PROCESSOR - Sharp LH79520
- U7, U9 SRAM - Micron MT48LC16M16A2FG-75
- U8 FLASH RAM - AMD AM29LV640DU90RWHI

U1 is the FPGA. Suitable pin models for all FPGAs supported by Altium Designer are included with the software. The actual pin model that is automatically chosen depends on the final program state of the device, and the settings for pin IO standard, slew rate and drive strength. Part of this example is to determine the optimum settings for these.

## 22.2.2 Importing IBIS Models

You will need to import the IBIS models for the other ICs listed above in order to have the pin models added to the SI pin model libraries in your installation of Altium Designer. The required IBIS models have been downloaded from the vendor websites and can be found in the `\ibis models` folder in the example project directory. The IBIS model is imported by editing the schematic component.

Locate U5 on the schematics, and:

1. Double click to open the **Component Properties** dialog.
2. Edit the existing Signal Integrity model, the **Signal Integrity Model** dialog will open.
3. Click the Import IBIS button.
4. Locate and select the IBIS model file for the device.
5. Select the correct component if a choice is given in the upper region of the **IBIS Converter** dialog.
6. Click OK to exit the dialogs. You will be informed that the model data has been written to the libraries, and that the model has been assigned. A report file will open, listing which pin model has been assigned to each device pin.

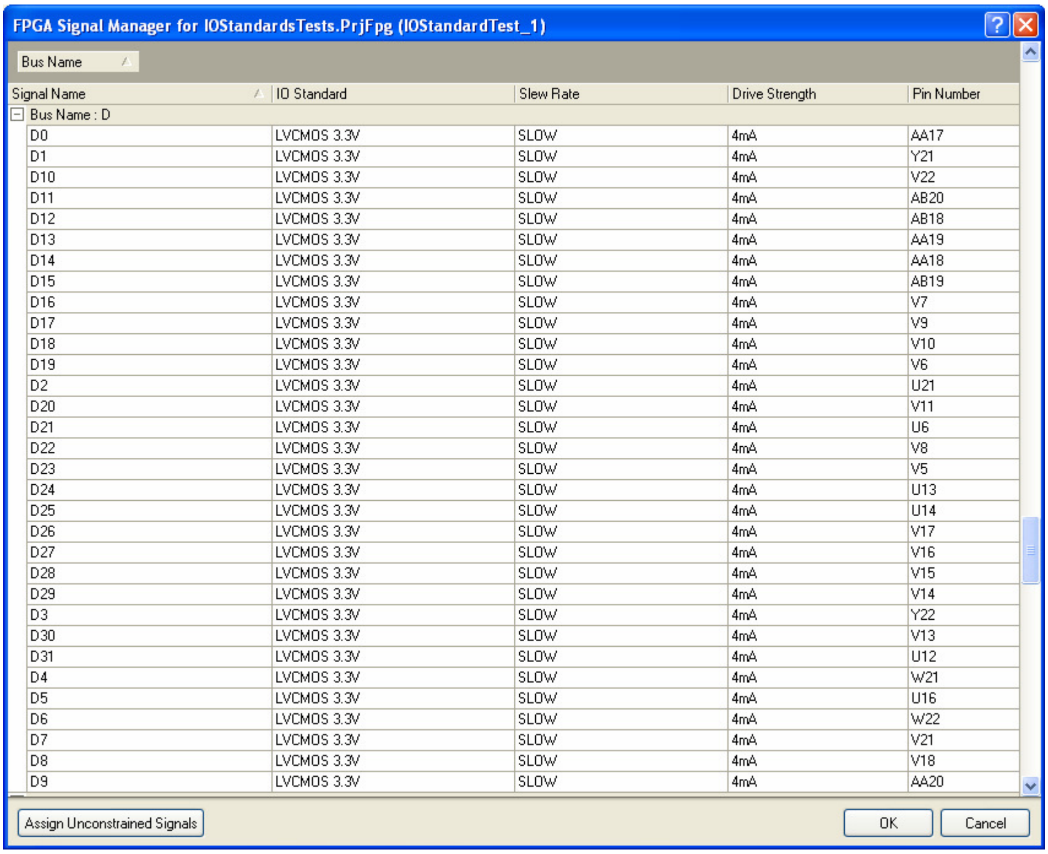
7. Now repeat this process for U7, U8 and U9. Note that we are assuming that the IBIS models from the manufacturer accurately model the pin characteristics. As with any simulation, the models used are extremely important to accuracy. A simulation is only as accurate as the models used!

> **Note:** For the SRAM, there are 3 possible IBIS model files. For this example, we will use `y16a.ibs`.

- Another essential requirement for successful SI simulation is to set up supply nets design rules for each of the supply nets (**Design » Rules**), so that the Signal Integrity Analyzer will handle them correctly. It is particularly important to set up supply net rules for the plane layers. These design rules have already been configured in the example board.
- The layer stack for the PCB must also be defined correctly. The Signal Integrity Analyzer requires continuous power planes. Split planes are not supported, so the net that is assigned to the plane is used. If they are not present, they are assumed, so it is far better to add them and set them up appropriately. The thickness for all layers, cores and prepreg must also be set correctly for the board. These properties, as well as dielectric values, can be defined in the *Layer Stack Manager* dialog (**Design » Layer Stack Manager**). All such settings have been defined already for the example board.

## 22.2.3 Initial IO Standard, Drive and Slew settings

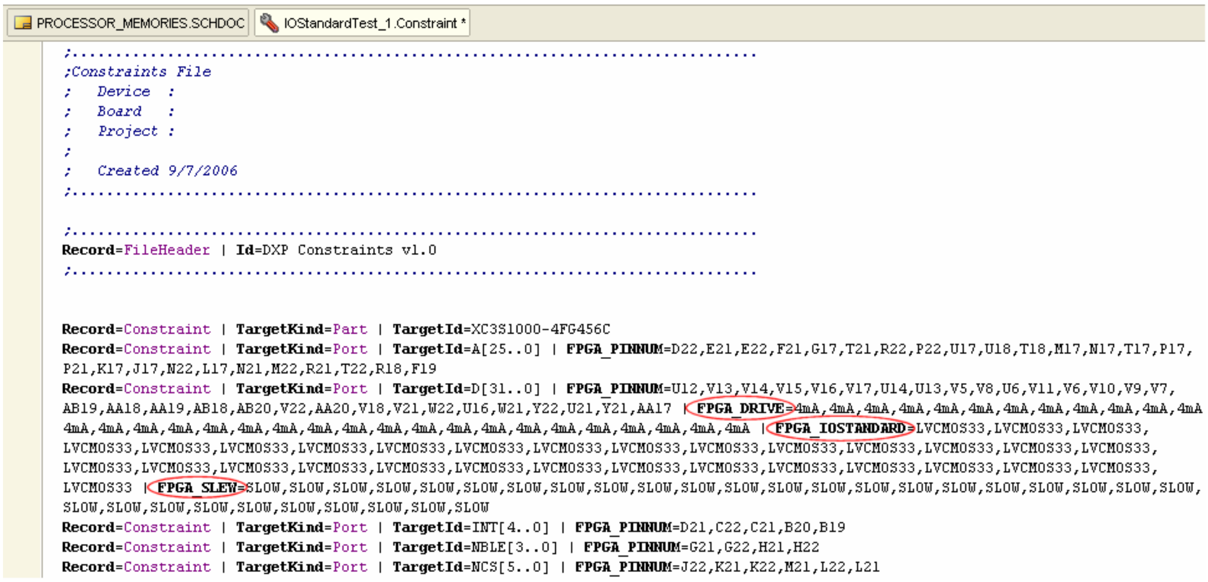
The electrical properties for pins of the physical FPGA device are defined in the FPGA Signal Manager dialog Figure 3. Access this dialog from the schematic or PCB document using the Tools » FPGA Signal Manager menu command.




| Signal Name | IO Standard | Slew Rate | Drive Strength | Pin Number |
|---|---|---|---|---|
| Bus Name : D | | | | |
| D0 | LVCMOS 3.3V | SLOW | 4mA | AA17 |
| D1 | LVCMOS 3.3V | SLOW | 4mA | Y21 |
| D10 | LVCMOS 3.3V | SLOW | 4mA | V22 |
| D11 | LVCMOS 3.3V | SLOW | 4mA | AB20 |
| D12 | LVCMOS 3.3V | SLOW | 4mA | AB18 |
| D13 | LVCMOS 3.3V | SLOW | 4mA | AA19 |
| D14 | LVCMOS 3.3V | SLOW | 4mA | AA18 |
| D15 | LVCMOS 3.3V | SLOW | 4mA | AB19 |
| D16 | LVCMOS 3.3V | SLOW | 4mA | V7 |
| D17 | LVCMOS 3.3V | SLOW | 4mA | V9 |
| D18 | LVCMOS 3.3V | SLOW | 4mA | V10 |
| D19 | LVCMOS 3.3V | SLOW | 4mA | V6 |
| D2 | LVCMOS 3.3V | SLOW | 4mA | U21 |
| D20 | LVCMOS 3.3V | SLOW | 4mA | V11 |
| D21 | LVCMOS 3.3V | SLOW | 4mA | U6 |
| D22 | LVCMOS 3.3V | SLOW | 4mA | V8 |
| D23 | LVCMOS 3.3V | SLOW | 4mA | V5 |
| D24 | LVCMOS 3.3V | SLOW | 4mA | U13 |
| D25 | LVCMOS 3.3V | SLOW | 4mA | U14 |
| D26 | LVCMOS 3.3V | SLOW | 4mA | V17 |
| D27 | LVCMOS 3.3V | SLOW | 4mA | V16 |
| D28 | LVCMOS 3.3V | SLOW | 4mA | V15 |
| D29 | LVCMOS 3.3V | SLOW | 4mA | V14 |
| D3 | LVCMOS 3.3V | SLOW | 4mA | Y22 |
| D30 | LVCMOS 3.3V | SLOW | 4mA | V13 |
| D31 | LVCMOS 3.3V | SLOW | 4mA | U12 |
| D4 | LVCMOS 3.3V | SLOW | 4mA | W21 |
| D5 | LVCMOS 3.3V | SLOW | 4mA | U16 |
| D6 | LVCMOS 3.3V | SLOW | 4mA | W22 |
| D7 | LVCMOS 3.3V | SLOW | 4mA | V21 |
| D8 | LVCMOS 3.3V | SLOW | 4mA | V18 |
| D9 | LVCMOS 3.3V | SLOW | 4mA | AA20 |

*Figure 3. Controlling FPGA pin electrical characteristics using the FPGA Signal Manager.*

- For each pin, the dialog can be used to set the IO Standard, the Slew Rate and the Drive Strength. The available slew rates and drive strengths will depend on the current IO Standard that the pin is set to.
- From Figure 3, you can see that the default IO Standard for the data pins has been set to LVCMOS 3.3V. This is the voltage level expected for the signals by the processor and memories in the design.
- The Slew Rate for each data pin has been set to SLOW and the Drive Strength to 4mA.
- The defined signal information is stored in the corresponding constraint file (`IOStandardTest_1.Constraint`) for the FPGA project (`IOStandardsTests.PrjFpg`), with the following parameters added to the corresponding constraint record for the D[31..0] port:
  - FPGA_DRIVE
  - FPGA_IOSTANDARD
  - FPGA_SLEW.



PROCESSOR_MEMORIES.SCHDOC | IOStandardTest_1.Constraint *

```
;..............................................................................
;Constraints File
;   Device  :
;   Board   :
;   Project :
;
;   Created 9/7/2006
;..............................................................................

;..............................................................................
Record=FileHeader | Id=DXP Constraints v1.0
;..............................................................................


Record=Constraint | TargetKind=Part | TargetId=XC3S1000-4FG456C
Record=Constraint | TargetKind=Port | TargetId=A[25..0] | FPGA_PINNUM=D22,E21,E22,F21,G17,T21,R22,P22,U17,U18,T18,M17,N17,T17,P17,
P21,K17,J17,N22,L17,N21,M22,R21,T22,R18,F19
Record=Constraint | TargetKind=Port | TargetId=D[31..0] | FPGA_PINNUM=U12,V13,V14,V15,V16,V17,U14,U13,V5,V8,U6,V11,V6,V10,V9,V7,
AB19,AA18,AA19,AB18,AB20,V22,AA20,V18,V21,W22,U16,W21,Y22,U21,Y21,AA17 | FPGA_DRIVE=4mA,4mA,4mA,4mA,4mA,4mA,4mA,4mA,4mA,4mA,4mA,4mA
4mA,4mA,4mA,4mA,4mA,4mA,4mA,4mA,4mA,4mA,4mA,4mA,4mA,4mA,4mA,4mA,4mA,4mA,4mA,4mA | FPGA_IOSTANDARD=LVCMOS33,LVCMOS33,LVCMOS33,
LVCMOS33,LVCMOS33,LVCMOS33,LVCMOS33,LVCMOS33,LVCMOS33,LVCMOS33,LVCMOS33,LVCMOS33,LVCMOS33,LVCMOS33,LVCMOS33,LVCMOS33,LVCMOS33,
LVCMOS33,LVCMOS33,LVCMOS33,LVCMOS33,LVCMOS33,LVCMOS33,LVCMOS33,LVCMOS33,LVCMOS33,LVCMOS33,LVCMOS33,LVCMOS33,LVCMOS33,LVCMOS33,
LVCMOS33 | FPGA_SLEW=SLOW,SLOW,SLOW,SLOW,SLOW,SLOW,SLOW,SLOW,SLOW,SLOW,SLOW,SLOW,SLOW,SLOW,SLOW,SLOW,SLOW,SLOW,SLOW,SLOW,SLOW,SLOW,
SLOW,SLOW,SLOW,SLOW,SLOW,SLOW,SLOW,SLOW,SLOW,SLOW
Record=Constraint | TargetKind=Port | TargetId=INT[4..0] | FPGA_PINNUM=D21,C22,C21,B20,B19
Record=Constraint | TargetKind=Port | TargetId=NBLE[3..0] | FPGA_PINNUM=G21,G22,H21,H22
Record=Constraint | TargetKind=Port | TargetId=NCS[5..0] | FPGA_PINNUM=J22,K21,K22,M21,L22,L21
```

*Figure 4. Electrical characteristics added to the relevant constraint file.*

## 22.2.4 Simulating the Reflecting Characteristics

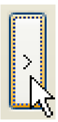
Now that things are setup we can simulate the reflection characteristics of these signals.

Firstly, to analyze the reflection characteristics:

1. Open one of the project documents (either a schematic or the PCB)
2. Run Tools » Signal Integrity
3. Click the **Continue** button when the **Errors or Warnings Found** dialog pops up. This dialog opens because there are components that the Signal Integrity Analyzer cannot correctly guess a suitable model for, in this case it is a number of test points – we can ignore these as they are not part of the nets we are interested in.

**Note:** The design will be analyzed and the **Signal Integrity** panel will open, listing all the nets in the design. What has happened is that a fast analysis has been performed on *all* nets in the design, called a screening analysis, with the results being listed down the left side of the **Signal Integrity** panel. Screening is used to quickly identify potential problem nets, which can then be analyzed in more detail. To analyze a net in more detail it is taken over to the right hand side of the panel, where a reflection or cross talk analysis can be performed. Note that a number of nets have a status of Not Analyzed, typically this is because these nets include a connector pin.

As part of the screening analysis results, you will see that failed nets have red backgrounds to their cells. For Overshoot and Undershoot conditions, the Signal Integrity engine has built-in thresholds that it can test against. These are configured in the *Set Screening Analysis Tolerances* dialog. Access this dialog by clicking on the **Menu** button and choosing the **Set Tolerances** command.



4. Locate the net D0, and click the Take Over button  at the top of the panel to add D0 to the right hand section of the panel.
5. Make sure that the direction of the FPGA pin on this net is set to Bi/Out, and that all other pins are set to Bi/In. Note that the FPGA is component U1. To change this, right click in the pins on each pin that needs to be changed, and choose "Toggle In/Out" to change the status. The reason for this is that we want the FPGA pin to be driving the net for the signal integrity analysis.

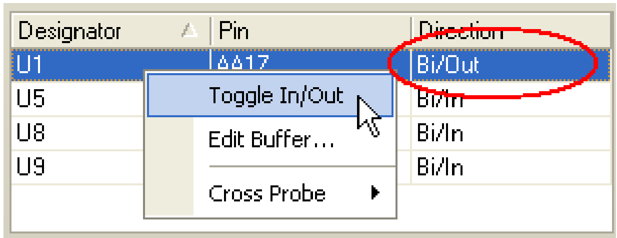



*Figure 5.Set the FPGA pin to Bi/Out, and the rest of the pins in the net to Bi/In.*

6. Click the **Reflections** button at the bottom of the panel to run a Reflection Analysis on the net D0. The resulting waveform will appear, as shown in Figure 6.

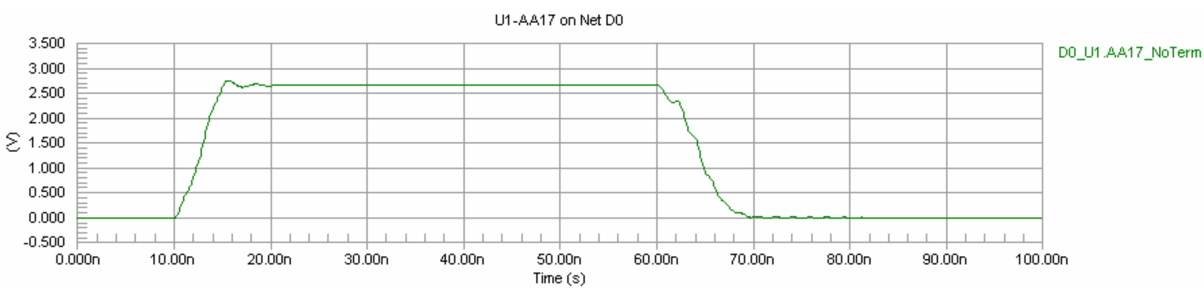



*Figure 6. Waveform for D0 at the FPGA pin, with Slew = SLOW and Drive = 4mA.*

7. Locate the signal D0 in the *FPGA Signal Manager* dialog and set the Slew Rate to the FAST option. Click **OK** and then **Execute** the subsequent engineering change order that is generated. The associated constraint file will be updated with the new setting and opened as the active document. Save and close this file.
8. Display the Signal Integrity panel again (you can click the Signal Integrity button down the bottom of the workspace if you closed the panel), and click the **Reanalyze Design** button.
9. Ensure that the FPGA pin is still set to be the output pin on the net (Bi/Out). Then run a reflection analysis on D0 again and note that there is now some ringing on this signal (Figure 7).

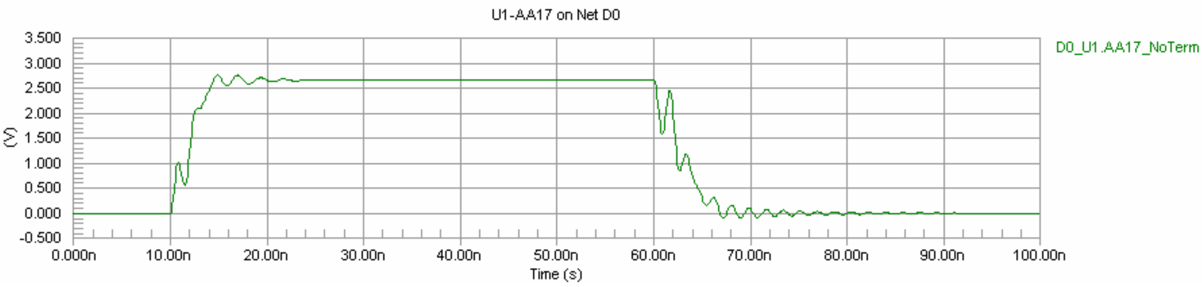



*Figure 7. Waveform for D0 at the FPGA pin, with Slew = FAST and Drive = 4mA.*

So just how much ringing can we allow on these signals in order to avoid glitches on the input pins? Consulting the datasheets for components U5, U7 and U9 reveals the following characteristics:

- Maximum VIL = 0.8
- Min VIH = 2.0

In the NBP-28.sdf waveform document you will find Charts for a range of possible Slew and Drive Strength settings. There is also a chart named "Comparison" with all the different waveforms combined for comparison. To scroll through the different waves, select a wave name and use the mouse scroll wheel or up down arrows.

> **Note:** The naming of the waveform in the comparison follows the following format: <signal>_<comp designator>.<pin>_<slew>_<drive> For example D0_U1.AA17_F_6mA is the waveform for signal D0, component U1, pin AA17, Fast slew rate, 6mA drive strength.

The comparison graph was generated using the **File** » **Export** » **Chart** command to create a CSV file for each of the charts. The wave names were changed inside this file to follow the above convention, and then the CSV files were all imported into a new chart.

From browsing these waveforms we can see that most of the FAST slew rate waveforms are not acceptable as they have excessive ringing present on the signals. With the slew rate set to SLOW, 6mA drive strength gives the cleanest looking waveform (Figure 8).

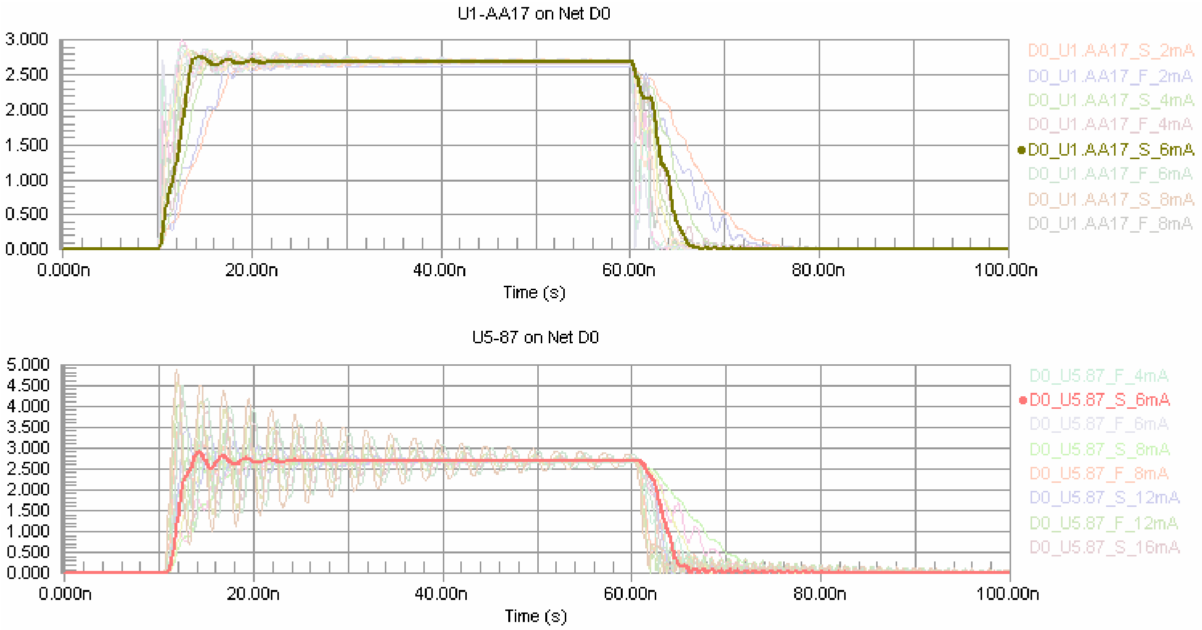



*Figure 8. Waveform for D0 at the FPGA and Processor pins, with Slew = SLOW and Drive = 6mA.*

A drive strength of 8mA would probably be acceptable although there is some ringing (Figure 9).

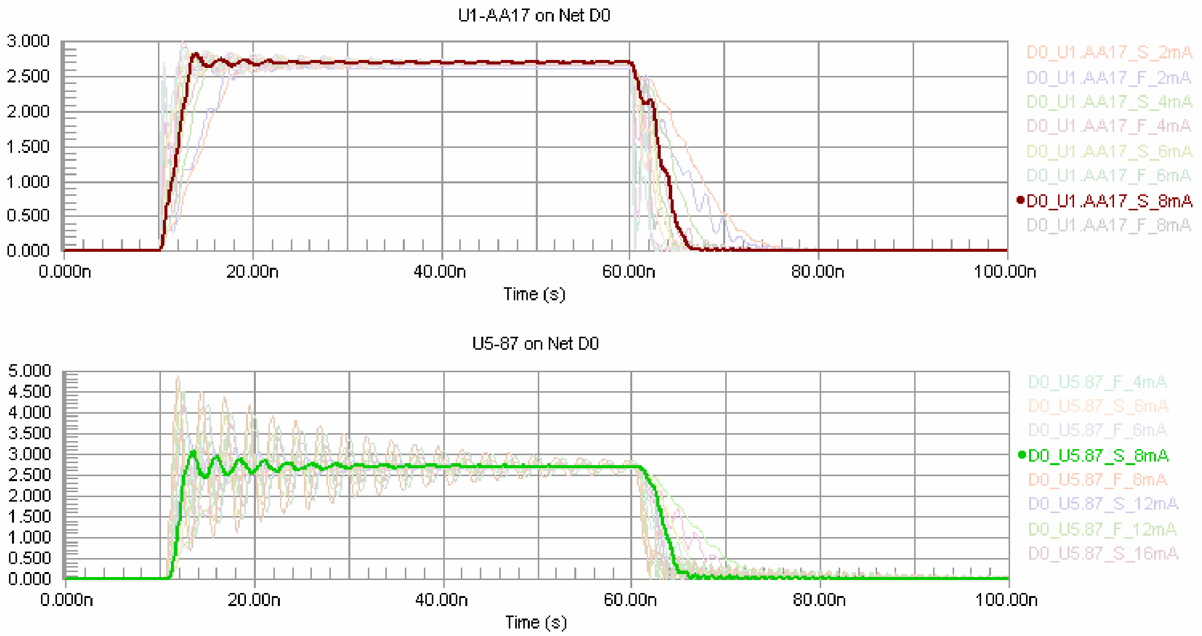



*Figure 9.* Waveform for D0 at the FPGA and Processor pins, with Slew = SLOW and Drive = 8mA.

Any drive strengths above this generate large amounts of ringing, particularly on U5, pin 87 (Figure 10).

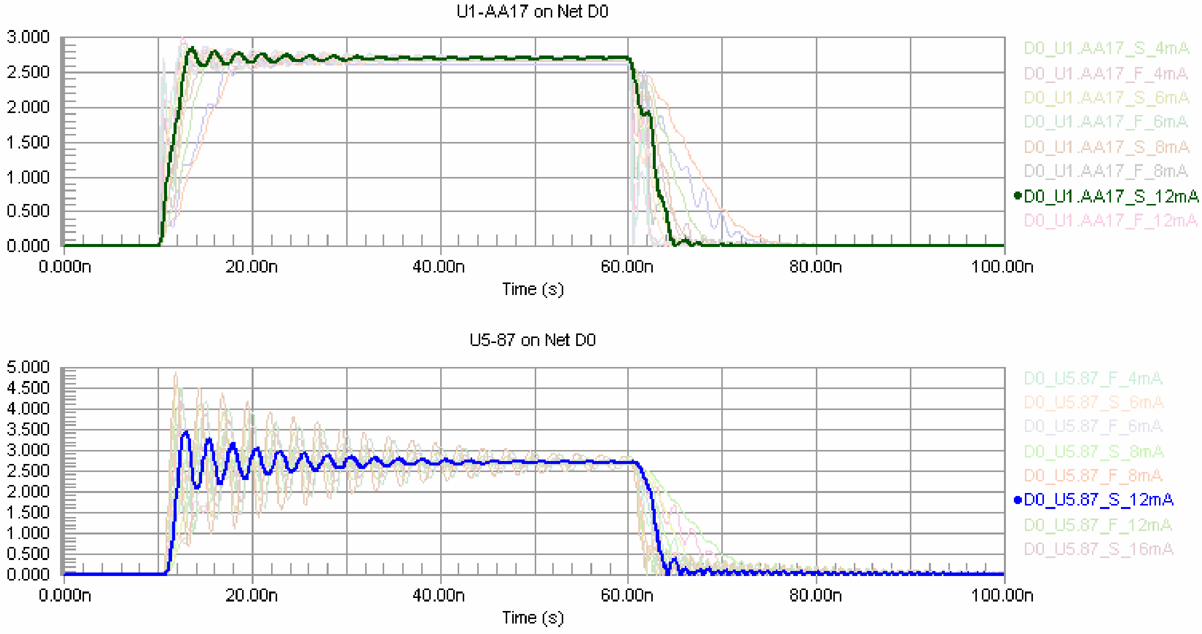



*Figure 10.* Waveform for D0 at the FPGA and Processor pins, with Slew = SLOW and Drive = 12mA.

You can see from the waveforms that the slew rate and drive strengths can have a huge effect on how clean the signals are. By modifying the slew and drive characteristics we can select values that provide a suitably fast and clean response.

Given the above findings we would select SLOW slew rate and 6mA drive strength for the FPGA pins. We would definitely not want to drive it any harder than 8mA or use the FAST slew rate setting without adding terminations on the PCB.

Selecting SLOW Slew and 6mA drive strength for all the FPGA signals D[31..0] in the signal manager we can now verify that this gives satisfactory waveforms for all these signals.

1. Make these changes in the *FPGA Signal Manager* dialog.
2. Re-analyze the design with the Signal Integrity tool.
3. Take across signal D[31..0]
4. Remember to check the i/o direction of the pins, make sure that for each signal the FPGA pins are set to Bi/Out and all other pins are Bi/In
5. Click the **Reflections** button. This generates a chart of reflection waveforms for each of the selected signals.

Examining the generated waveforms shows that 6mA drive strength gives acceptable waveforms for all these signals.

If we wanted to drive the pins harder than this then it would be necessary to place terminations on the PCB. The termination advisor is a useful tool for deciding upon a type of termination to use, and it can be used to sweep values of termination components across a range.

Using the termination advisor:

1. Using the *FPGA Signal Manager* dialog, set the drive current for D0 to 24mA.
2. Re-analyze the design with the Signal Integrity tool
3. Take across signal D0.
4. Make sure that the FPGA pin AA17 is set to Bi/Out, and all other pins are set to input.
5. Enable all the different termination types.
6. Un-check the **Perform Sweep** checkbox.

7. Click the **Reflections** button, and waves will be generated for each termination type.
8. From the results we can see that a serial resistor looks like it will fix the ringing.
9. In the Signal Integrity panel, disable all terminations except for Serial Resistor.
10. Check the **Perform Sweep** option.
11. Click the **Reflections** button.
12. Click on the name of the first wave in the legend, and use the mouse wheel to scroll through the waves for different termination resistance values. It would appear that a value around 47 Ohms would work well, so lets perform a reflection simulation with this value.
13. Uncheck the **Perform Sweep** option.
14. Change the value of the Serial Resistor to 47.
15. Run the reflection analysis.

As you can see, this can be a useful tool for selecting terminations in order to solve SI issues. The values used for each sweep are shown in the waveform hint for each wave. In the above example we can see that a value of 47 ohms for the serial resistor should provide a good termination if we needed to use the high drive current option for the FPGA output pin. In `NBP-28.sdf` there is a chart named Comparison2. This contains a comparison of the 24mA drive current terminated with a 47 Ohm serial resistor with the 6mA drive current waveforms (Figure 11).

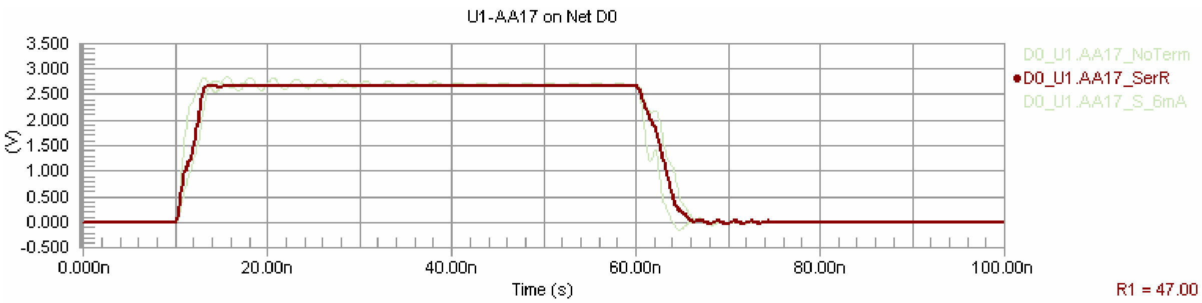



*Figure 11.* Waveform for D0 at the FPGA pin, with Slew = SLOW, Drive = 24mA and 47Ohm Series Resistor.

There is not much difference between these signals, so unless we needed the higher drive current we are best off sticking to the 6mA and avoiding the need to add termination resistors.

## 22.2.5 Checking for Crosstalk

Finally we will look at how to examine the crosstalk between signals. Given that we were able to choose drive and slew settings that minimized ringing on the signals, crosstalk problems should have already been minimized.

There are essentially two stages in performing a crosstalk analysis. First, we must identify the nets for which crosstalk may pose a problem. Then we must take over the nets which we want to more closely examine, and run a detailed crosstalk analysis.

## 22.2.6 Finding coupled Nets

The Signal Integrity panel includes a feature that enables you to quickly identify which nets are considered to be coupled, to the selected net(s) you choose in the screening analysis results area of the panel. This feature – Find Coupled Nets – is ideal for working out which nets may be susceptible to crosstalk. It essentially analyzes the PCB and identifies traces that run parallel to each other according to coupling options defined on the Configuration tab of the Signal Integrity Preferences dialog (Figure 12). Access this dialog by clicking on the Menu button in the panel, and choosing Preferences from the resulting menu.

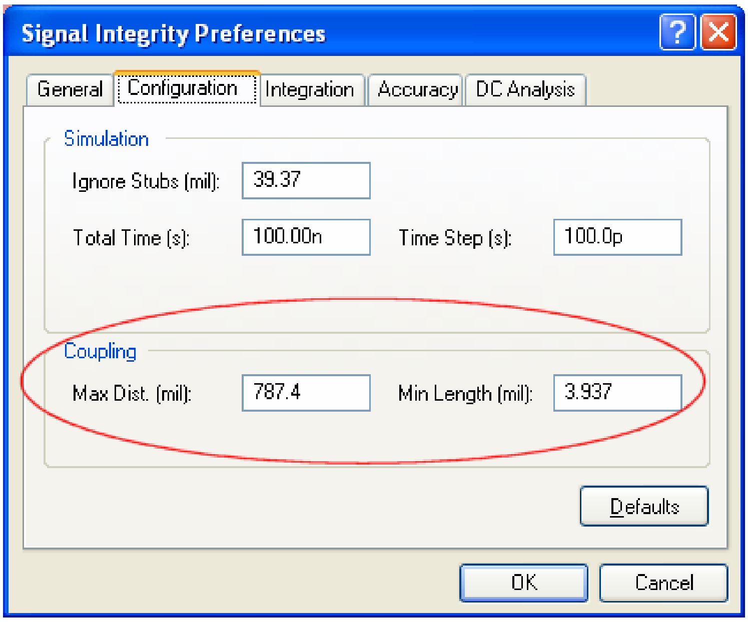



*Figure 12. Set criteria to identify coupled nets.*

Two options constitute the coupling criteria:

- **Max Dist**. – this option defines the proximity for neighboring nets. Only nets within the specified distance of the selected base net will be considered for coupling purposes.
- **Min Length** – this option specifies the minimum length that traces of nets must run parallel to that of the selected base net, in order to be considered coupled.

As a general rule, the closer the traces the more likely crosstalk is to occur. Similarly, the longer the parallel sections of trace, the more crosstalk will occur.

**Note:** When running a crosstalk analysis, the typical number of nets taken across would be three – a net and its two immediate neighbors.

To illustrate the power of this feature, let us find all nets coupled to the net D5 (our base net). The default setting for the Max Dist is a little high and would easily find all nets in the D[31..0] bus, as well as many others. Instead, we will vary Max Dist – using 20mil and 10mil values – and observe the resulting nets that are deemed to be coupled. In addition we will set the Min Length to be 50mil (i.e. any nets not having traces running parallel to D5 for at least this distance will not be considered coupled).

To find coupled nets with **Max Dist** = 20mil, **Min Length** = 50mil:

1. Change the **Max Dist** coupling option to 20mil
2. Change the **Min Length** coupling option to 50mil
3. Select net D5 in the screening analysis region of the **Signal Integrity** panel
4. Right-click and choose **Find Coupled Nets**
5. Observe that the following data nets – D1, D4, D6 and D7 – are all found to be coupled to D5 and are selected ready to take across for a more detailed crosstalk analysis (**Error! Reference source not found.**). Net A17 is also found to be coupled, but as we are interested only in the data lines, this should be deselected prior to taking the nets across.

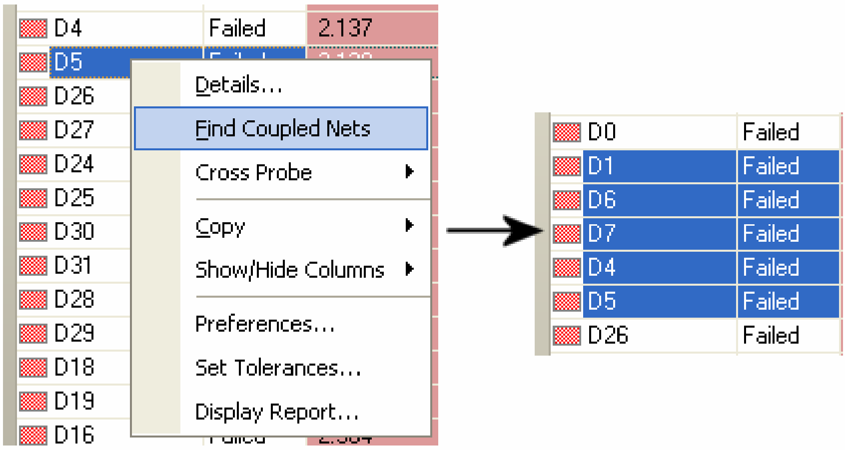



*Figure 13. Nets coupled to D5 (Max Dist = 20mil, Min Length = 50mil).*

To find coupled nets with **Max Dist** = 10mil, **Min Length** = 50mil:

1. Change the **Max Dist** coupling option to 10mil
2. Change the **Min Length** coupling option to 50mil
3. Select net D5 in the screening analysis region of the **Signal Integrity** panel
4. Right-click and choose **Find Coupled Nets**
5. Observe that the following data nets – D6 and D7 – are all found to be coupled to D5 and are selected ready to take across for a more detailed crosstalk analysis (**Error! Reference source not found.**).

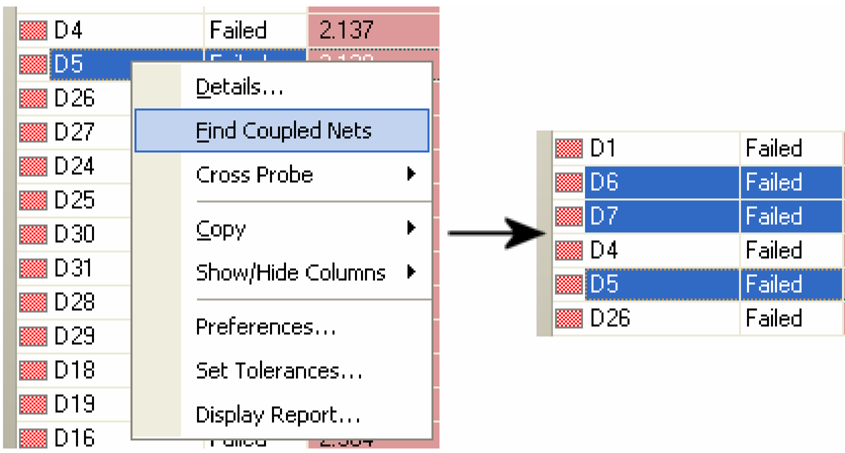



*Figure 14. Nets coupled to D5 (Max Dist = 10mil, Min Length = 50mil).*

## 22.2.7 Performing Crosstalk Analysis

In the previous section, we used the Find Coupled Nets feature to identify nets coupled to the net D5, and that may suffer from the effects of crosstalk. We will now perform crosstalk analysis to assess the level of crosstalk involved when the net D5 is set to Aggressor and is driven at different signal strengths. Note that although several coupled nets were found by the Find Couple Nets feature, we will concentrate on just examining D5 and D6, since these two nets run parallel to each other for much of their length.

1. Set the drive current of D5 to 6mA.
2. Analyze the design using the Signal Integrity tool.
3. Take over signals D5 and D6.
4. Make sure that all the FPGA pin directions are set to Bi/Out.
5. Right Click on D5 in the list of nets and select "Set Aggressor". This means that during crosstalk analysis we will be driving this net and simulating its effect on the other "victim" nets.
6. Click the **Crosstalk** button to run a crosstalk analysis.
7. In the generated chart we can see that up to about 125mV is induced in D6. (see chart "Crosstalk Analysis 6mA"). This maximum occurs at pin 79 of component U5 (Figure 15). As anticipated it should not be enough to cause any problems.

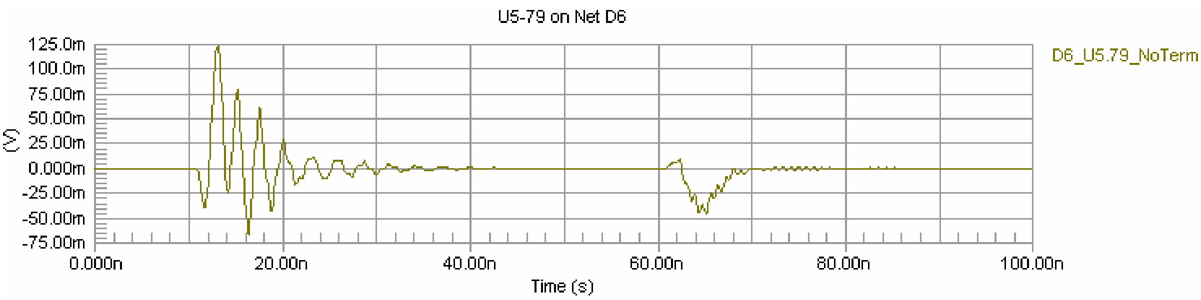



*Figure 15. Crosstalk induced in D6 by D5 (D5 driven at 6mA – max crosstalk on pin 79 of U5).*

If we repeat the above steps but with 24mA drive strength for D5 we see that the simulation waveforms show up to about 500mV is induced in the coupled nets (see chart "Crosstalk Analysis 24mA"). This maximum is again found on pin 79 of component U5 (Figure 16). This is quite significant and would certainly be likely to cause problems on the signal D6.

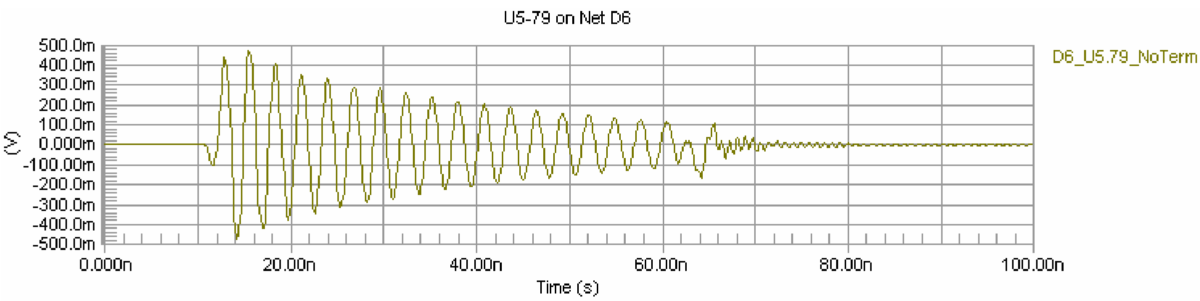



*Figure 16.* Crosstalk induced in D6 by D5 (D5 driven at 24mA – max crosstalk on pin 79 of U5).

You can experiment further with other slew and drive settings in the *FPGA Signal Manager* dialog and see the effect on crosstalk.

## 22.2.8 Summing Up

In conclusion we have used the Signal Integrity tool to find the optimum drive and slew settings for the FPGA pins D[31..0]. In this case we chose a SLOW slew rate and 6mA drive strength. We have also used the termination advisor to choose appropriate terminations if the higher drive current is required. We have also looked at how to assess crosstalk between signals.

For detailed information on the **Signal Integrity** panel, press **F1** while the panel has focus. Use the available link in the upper region of the **Knowledge Center** panel to access the relevant topic for this panel, within the *Altium Designer Panels Reference*.

For conceptual information regarding what is required to address possible signal integrity issues that may arise in a design, refer to the article *Putting Signal Integrity in its Place*.

For further tutorial-based information on Signal Integrity – including pre- and post-layout analysis – refer to the *Performing Signal Integrity Analyses* tutorial.

For further information on working with generated waveforms, refer to the application note *Working with the Sim Data Editor*.

TOP SECRET

Altium Designer

# Module 23: Customization and the Scripting System

# Module 23: Customization and  the Scripting System





Module Seq = 23

# 23.1 Customizing toolbars, menus & shortcut keys

All methods of command selection can be customized, including menus, toolbars and shortcut key menus. These are often referred to as *resources* in Altium Designer.

## 23.1.1 Customizing resources

- Resources are customized via the DXP System menu, or by right-clicking on a menu or toolbar and selecting **Customize**.
- Figure 1 shows the *Customizing Schematic Editor* dialog. When you select **Customize** with a schematic as the active document, this dialog opens ready to customize the resources for that editor. Customization options include adding, deleting or re-ordering menu entries and toolbar buttons, and adding new shortcut key definitions.

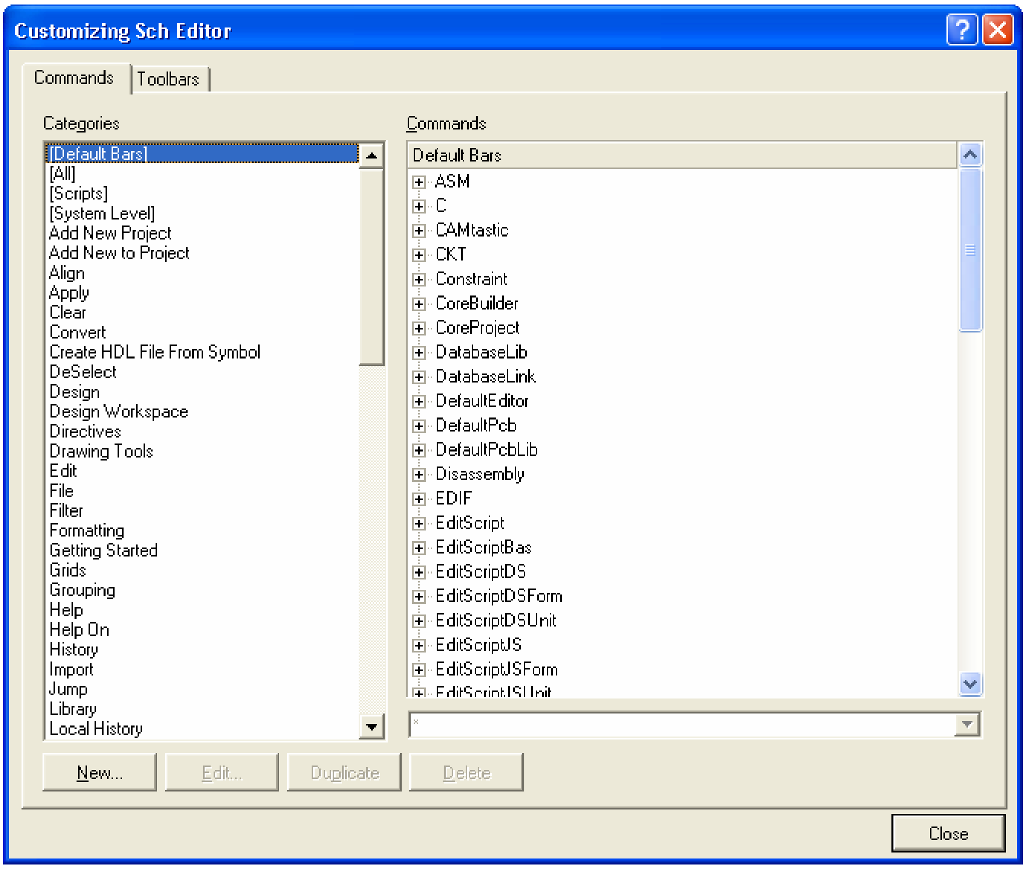



*Figure 1. Toolbar Properties dialog*

#### 23.1.1.1 Adding a command to a menu and toolbar

- The **Commands** tab of the *Customizing* dialog gives access to all the commands available to this editor.
- There are essentially two ways of accessing a command:
  - selecting **Default Bars**, then using the tree-like structure on the right, or
  - choosing a flat list of commands, either **All** commands in one list, or clicking on a menu name on the left to access a command in that menu.

- When the required command has been located, click and drag it to the required toolbar or menu, then release in the required location.
- When the *Customizing* dialog is open, menu entries and toolbar buttons can be:
  - moved, by clicking and dragging
  - copied, by holding **Ctrl** while you click and drag
  - edited, by double-clicking.
- When the *Customizing* dialog is open, separators can be:
  - added to a menu by clicking and dragging a menu entry down slightly from the previous entry to add a separator in between
  - removed by dragging the entry that follows the separator up and releasing on top of the separator.
  - Use the same techniques to add/remove a separator from a toolbar.
- When the *Customizing* dialog is not open, hold Ctrl as you click on a menu entry or toolbar button to directly access the *Edit Command* dialog for that command.

#### 23.1.1.2 Bars – the menu bar and toolbars

- Toolbars and the main menu are all classified as bars. Set any bar to be the main menu in the **Bars** tab of the *Customizing* dialog.
- When you create a new toolbar in the **Bars** tab of the *Customizing* dialog, the blank bar appears just to the right of the main menu bar.
- Alternate menu bars can be created and kept as a toolbar, then switched to be the menu bar when required.

#### 23.1.1.3 Shortcut keys

- Shortcuts are defined as part of the command. To examine all shortcuts, click on **All** in the *Customizing* dialog, then click on the Shortcut heading in the Commands section of the dialog on the right to sort by shortcut key.
- When the *Customizing* dialog is open, a Shortcut menu appears on the toolbar at the top of the workspace.
- Only one set of shortcuts can be defined for each editor.

### 23.1.2 Behind the scenes - processes and parameters

- Underlying every command in the DXP environment is a process. Each DXP server presents its functionality to the environment as a set of processes.
- Many processes support parameters, where each parameter is used to control the behavior of the process.
- Commands, which are edited in the *Customizing* dialog, are pre-packed combinations of a process + required parameters + menu caption + shortcut keys.

#### 23.1.2.1 Using parameters

- Adding parameters can further customize the operation of any process.
- An example of the use of parameters is the **Digital Objects** tools, available on the **Utilities** toolbar in the Schematic Editor (**View » Toolbars » Utilities** to control the display of the toolbar**)**.

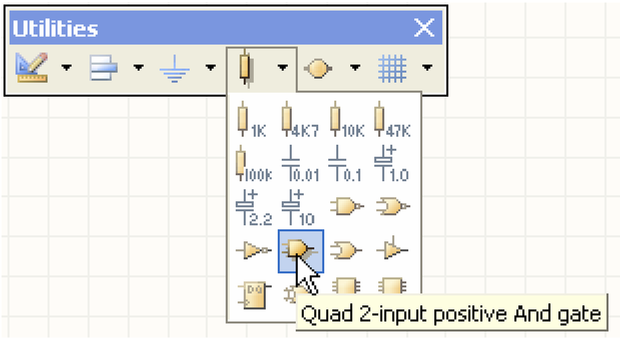



*Figure 2. Digital Objects tools, accessed on the Utilities bar*

All the buttons on this toolbar use the process:
`IntegratedLibrary:PlaceLibraryComponent`

it is the parameters that specify which part is placed, e.g.
`LibReference=SN74F08D | Library=Texas Instruments\TI Logic Gate 2.IntLib | Orientation=0`

**Note:** Multiple parameters are separated by a pipe symbol (|).

## 23.1.3 Exercises — Customizing resources

### 23.1.3.1 Adding a command to a toolbar

In this exercise, we will add the **Find** » **Text** command to the Schematic Editor's Main toolbar.

1. While in a Schematic document, right-click on the main menu (or a toolbar) and select **Customize** from the floating menu that appears. The *Customizing* dialog will appear.
2. The **Find Text** command is already available in the menus, so rather than finding it in the *Customizing* dialog, we will simply copy the command from a menu to the toolbar.
3. Click once on **Edit** menu, then click once on the **Find Text** command. It will be highlighted with a black box.
4. Holding the CTRL key, click and hold on the **Find Text** command and drag it up to the main toolbar, dropping it before the **Cut** button, as shown in Figure 3.
5. Close the *Customizing* dialog, then click the new button to confirm that it works, the *Find Text* dialog should open.

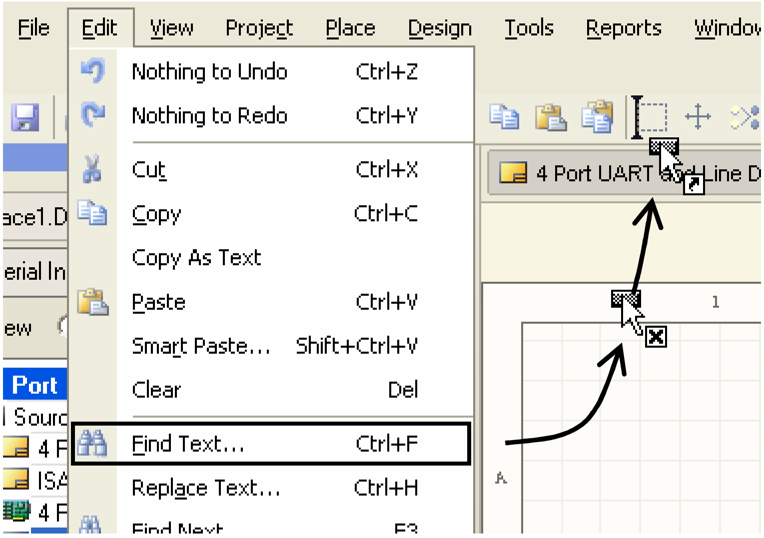



*Figure 3. Copying a command from a menu to a toolbar*

### 23.1.3.2 Adding an item to the main menu or right–click menu

In this exercise, you will add the Deselect All command to the right-click menu of the Schematic Editor. Menu items that appears in the **Right Mouse Click** menu, **Options** popup menu (press the O shortcut key) or **Filter** popup menu (press the Y shortcut key) are listed under the **Help** » **Popups** menu.

1. While in a Schematic document, right-click on the main menu (or a toolbar) and select **Customize** from the floating menu that appears. The *Customizing* dialog will appear.

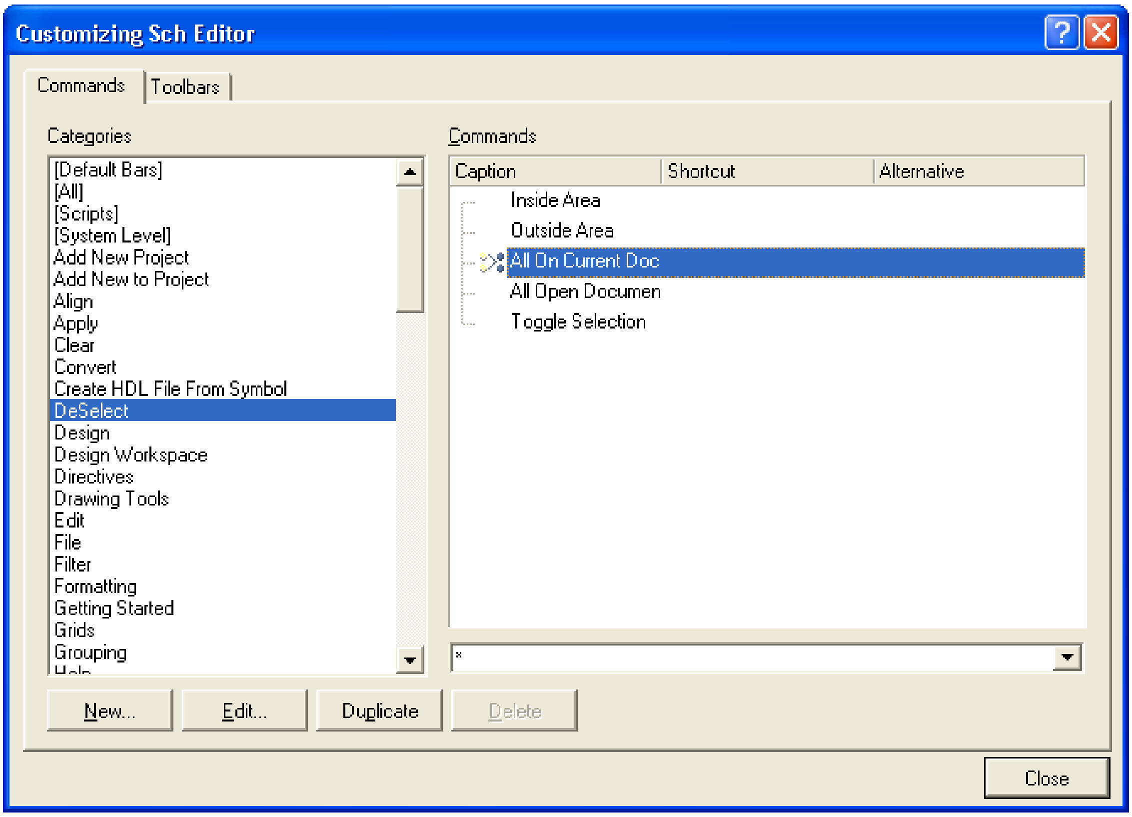



*Figure 4. Customizing dialog with Right Mouse Click commands displayed*

2. In the dialog, select **DeSelect** in the **Categories** list, then in the **Commands** list on the right locate the **All on Current Document** command.
3. Click and hold on this command and drag it up to the **Help** menu. Once it opens, drag down to **Popups**, then down to **Right Mouse Click**, then drop the command below the **Clear Filter** menu entry.
4. Before closing the menu we will edit the caption that appears in the menu. To do this, double-click on the new menu entry to open the *Edit Command* dialog.
5. In the Edit Command dialog, edit the caption to read **De&Select All**. Note the location of the ampersand character (&). This defines the letter that will act as the accelerator key. The letter `S` has been chosen because the letters `D` and `A` are already assigned in this menu. You are free to reassign any of the accelerator keys that are used in the menu.

> **Note:** Resource customizations are stored in the file DXP.RCS, which is located in the `C:\Documents and Settings\<your logon name>\Application Data\AltiumDesignerSummer08` folder.

## 23.1.4 Creating a new menu, toolbar or shortcut key menu

Creating a new menu bar or toolbar is similar to editing one. The procedure is outlined below.

Select the **Customize** command from the **DXP System** menu (to the left of the **File** menu). This displays *Customize Editor* dialog shown in Figure 5.

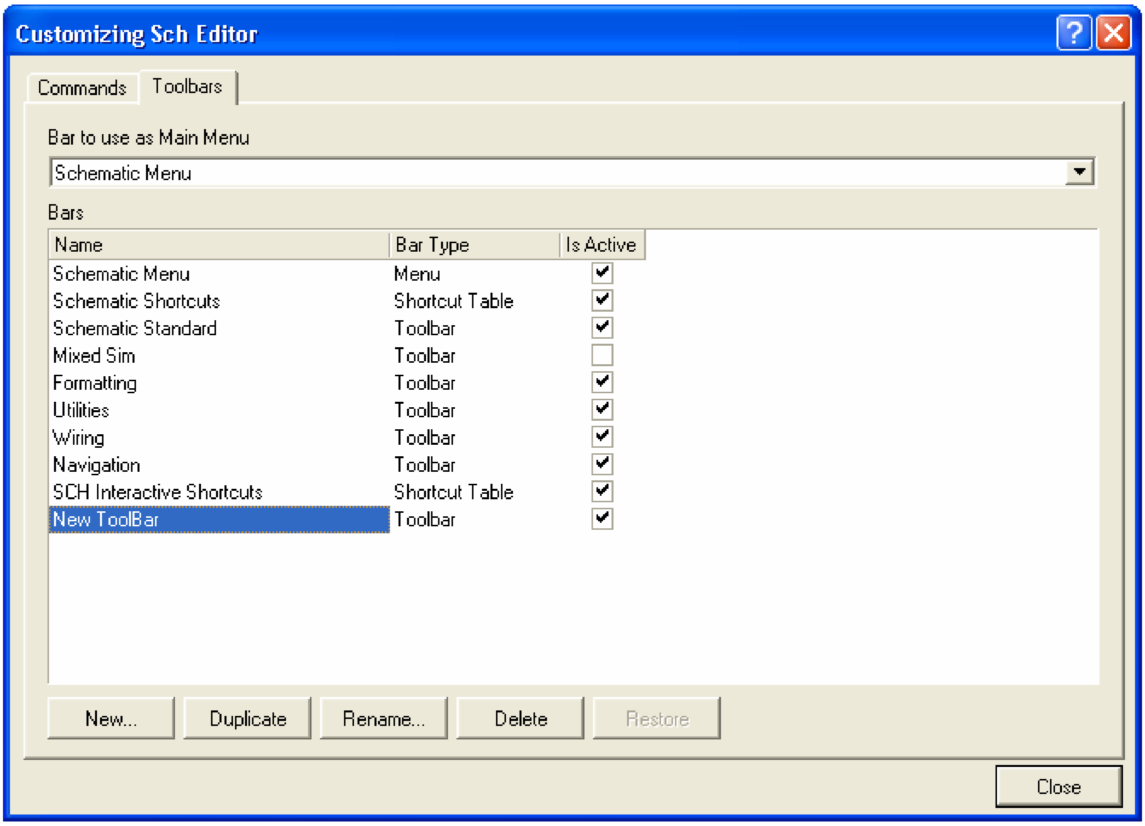



*Figure 5. Bars tab of the Customizing dialog*

The **Bars** tab can be used to create a new toolbar, control the display of toolbars and select which bar will be the menu bar. Only one menu can be active at any one time but any toolbar can be selected to be the menu bar. To set a new bar to be the menu bar, change the **Bar to Use as Main Menu** drop down.

### 23.1.4.1 Exercise — Creating a new toolbar

1. While the Schematic Editor is active, select the Customize command from the DXP menu to display the Customizing dialog.
2. Click on the Toolbars tab and click New. A new toolbar will appear in the list. Click Rename and rename it as My Toolbar, then enable the Is Active check box to display it.
3. Locate the new blank bar, if the menu and toolbars are in the default locations it will be to the right of the Help menu, and drag it so it is floating in the workspace.
4. Finally, add some buttons to your new toolbar using the steps detailed in exercise 23.1.3.1 Adding a command to a toolbar.

## 23.1.5 Adding a script to a menu item

You have the ability to assign a script to a server menu, toolbar or hot key in Altium Designer which makes it possible for you to run the script over a current PCB document for example. You will need to specify the full path to the script project and specify which script unit and procedure to execute the script.

There are two parameters in this case: the `ProjectName` and the `ProcName`. For the `ProcName` parameter, you need to specify the script filename and the main procedure in this script. So the format is as follows: `ProcName = ScriptFileName>ProcedureName`. Note the **GreaterThan** symbol used between the script file name and the procedure name.

1. Double click on the PCB menu and the *Customizing PCB Editor* dialog appears.
2. Click on the **New** button of the *Customizing PCB Editor* dialog.
3. Choose **ScriptingSystem:RunScript** process in the **Process:** field of the *Customizing PCB Editor* dialog.
4. Enter `ProjectName = C:\Program Files\Altium Designer Summer 08\Examples\Scripts\DelphiScript Scripts\General\HelloWorld.PrjScr | ProcName = HelloWorldDialog>RunHelloWorld` text in the **Parameters:** field of the *Customizing PCB Editor* dialog.

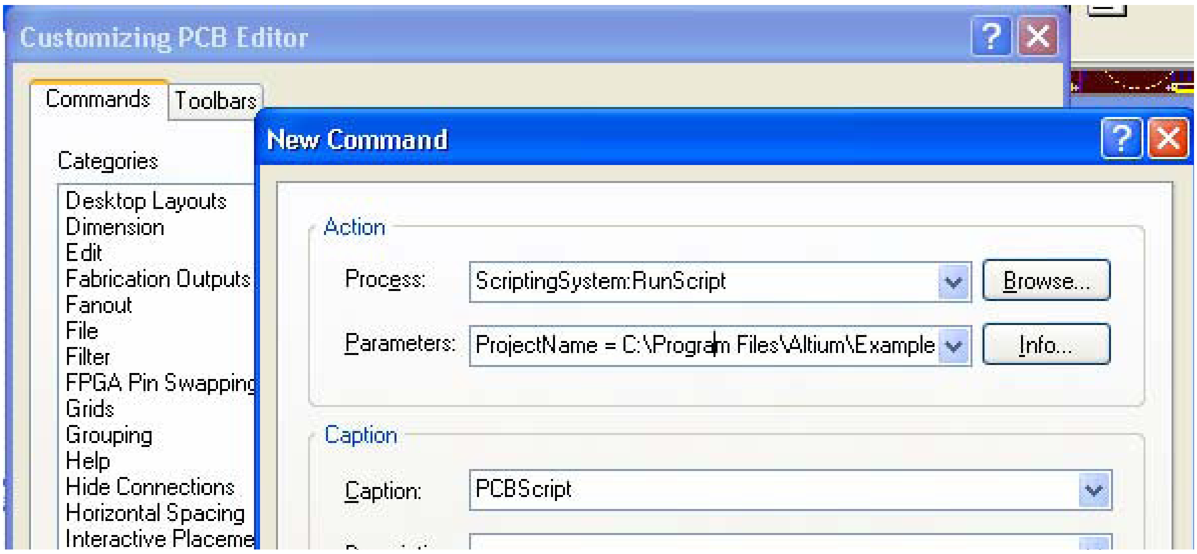



*Figure 6. Creating a menu item for a script*

5. You will need to give a name to this new command and assign a new icon if you wish. In this case, the name is `PCBScript` in the **Caption:** field of this dialog. The new commands appear in the **[Custom]** category of the **Categories** list. Click on the **[Custom]** entry from the **Categories** list. The **PCBScript** command appears in the **Commands** list of this dialog.
6. You then need to drag the new **PCBScript** command onto the **PCB** menu from the *Customizing PCB Editor* dialog. The command appears on the menu. You can then click on this new command and the HelloWorld form appears.